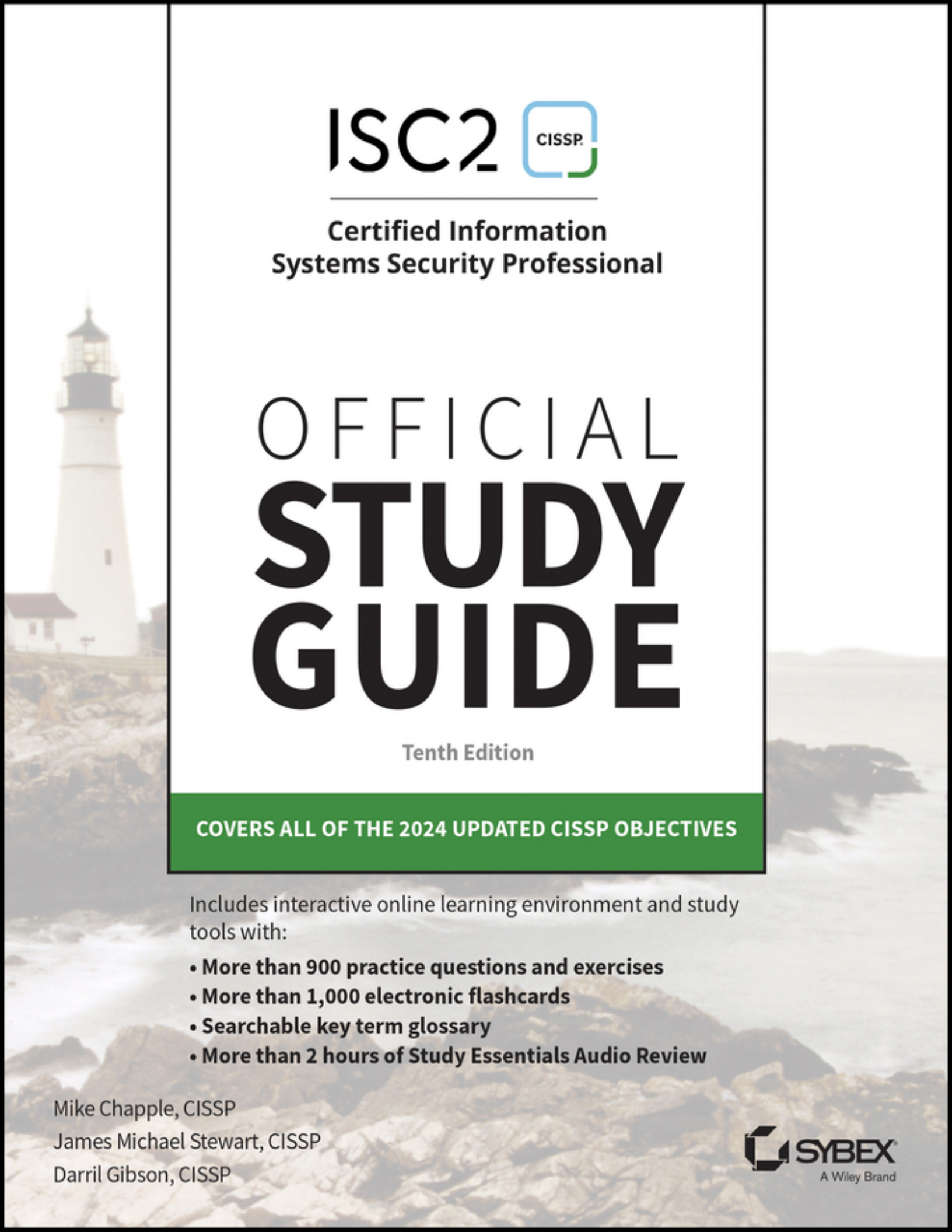


ISC2 CISSP

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Darril Gibson, CISSP


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To Darril Gibson, my friend and co-author of many years. You made a tremendous impact on the cybersecurity field and we will be eternally grateful for your contributions.

—Mike Chapple

To Cathy, I continue to love threading the zigzaggednesses of life with you.

—James Michael Stewart

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—Mike and Michael

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—Mike Chapple

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—James Michael Stewart

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Darril Gibson, CISSP (1958–2022), was the CEO of YCDA, LLC and regularly wrote and consulted on a wide variety of technical and security topics and held numerous other certifications, including MCSE, MCDBA, MCSD, MCITP, ITIL v3, and Security+. He authored or coauthored more than 30 books, including multiple prior editions of the *CISSP Study Guide*. Darril was greatly respected in the cybersecurity, training, and education fields and will be missed.

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Introduction

The *ISC2® CISSP® Certified Information Systems Security Professional Official Study Guide, Tenth Edition*, offers you a solid foundation for the Certified Information Systems Security Professional (CISSP) exam. By purchasing this book, you've shown a willingness to learn and a desire to develop the skills you need to achieve this certification. This introduction provides you with a basic overview of this book and the CISSP exam.

This book is designed for readers and students who want to study for the CISSP certification exam. If your goal is to become a certified security professional, then the CISSP certification and this CISSP Study Guide are for you. The purpose of this book is to adequately prepare you to take the CISSP exam.



The information presented here in this Introduction was based on the publicly available documentation from ISC2 as of April 15, 2024. However, these details and exam parameters are subject to change at any time based upon ISC2 operational decisions. Please consult isc2.org to confirm, verify, or learn about updated exam specifics.

Before you dive into this book, you need to have accomplished a few tasks on your own. You need to have a general understanding of IT and of security. You should have the necessary five years of cumulative full-time work experience (or four years if you have a college degree) in two or more of the eight domains covered by the CISSP exam. Part-time work and internship experience is also acceptable with conditions; see www.isc2.org/certifications/cissp/cissp-experience-requirements. If you are qualified to take the CISSP exam according to ISC2, then you are sufficiently prepared to use this book to study for it. For more information on ISC2, see the next section.

Alternatively, ISC2 allows for a one-year reduction of the five-year experience requirement if you have earned one of the approved certifications from the ISC2 prerequisite pathway. As of Q1 2024, the qualified certifications are:

- AWS Certified Security - Specialty
- Certified in Governance, Risk and Compliance (CGRC)
- Certified Cloud Security Professional (CCSP)
- Certified Computer Examiner (CCE)
- Certified Ethical Hacker v8 or higher
- Certified Information Security Manager (CISM)
- Certified Information Systems Auditor (CISA)
- Certified Internal Auditor (CIA)
- Certified Protection Professional (CPP) from ASIS
- Certified in Risk and Information Systems Control (CRISC)
- Certified Secure Software Life cycle Professional (CSSLP)
- Certified Wireless Security Professional (CWSP)
- Cisco Certified CyberOps Associate/Professional
- Cisco Certified Internetwork Expert (CCIE) Security
- Cisco Certified Network Associate Security (CCNA Security)
- Cisco Certified Network Professional Security (CCNP Security)
- CIW Web Security Professional
- CIW Web Security Specialist
- CompTIA Advanced Security Practitioner (CASP+)
- CompTIA CySA+
- CompTIA Security+
- Computer Hacking Forensic Investigator (CHFI)
- CSA Certificate of Cloud Security Knowledge (CCSK)

- EC-Council Certified Security Specialist (ECSS)
- EC-Council Certified SOC Analyst (CSA)
- GIAC Certified Enterprise Defender (GCED)
- GIAC Certified Incident Handler (GCIH)
- GIAC Certified Intrusion Analyst (GCIA)
- GIAC Cyber Threat Intelligence (GCTI)
- GIAC Global Industrial Cyber Security Professional (GICSP)
- GIAC Information Security Fundamentals (GISF)
- GIAC Information Security Professional (GISP)
- GIAC Security Essentials Certificate (GSEC)
- GIAC Security Leadership Certification (GSLC)
- GIAC Strategic Planning, Policy, and Leadership (GSTRT)
- GIAC Systems and Network Auditor (GSNA)
- HealthCare Information Security and Privacy Practitioner (HCISPP)
- Information Security Management Systems Lead Auditor (IRCA)
- Information Security Management Systems Principal Auditor (IRCA)
- Juniper Networks Certified Internet Expert (JNCIE-SEC)
- Microsoft Identity and Access Management
- Microsoft Security Operations Analyst
- Microsoft Certified Cybersecurity Architect
- Offensive Security Certified Professional/Expert (OSCP/E)
- Systems Security Certified Practitioner (SSCP)

For the complete and current list of qualifying certifications, visit www.isc2.org/certifications/cissp/cissp-experience-requirements.



You can use only one of the experience reduction measures, either a college degree or a certification, not both.

ISC2 offers an entry program known as an Associate of ISC2. This program allows someone without any or enough experience to qualify as a CISSP applicant to take the CISSP exam anyway and then obtain experience afterward. Associates are granted six years to obtain five years of security experience. Only after providing proof of such experience, usually by means of endorsement (discussed later), can the individual be awarded the full CISSP certification.

If you are just getting started on your journey to CISSP certification and do not yet have the work experience, then our book can still be a useful tool in your preparation for the exam. However, you may find that some of the topics covered assume knowledge that you don't have. For those topics, you may need to do some additional research using other materials, and then return to this book to continue learning about the CISSP topics.

ISC2

The CISSP exam is governed by the International Information System Security Certification Consortium ISC2. ISC2 is a global nonprofit organization. It has the mission of “ISC2 strengthens the influence, diversity and vitality of the field through advocacy, expertise and workforce empowerment that accelerates cyber safety and cybersecurity in an interconnected world.”

ISC2 is operated by a board of directors elected from the ranks of its certified practitioners.

ISC2 supports and provides a wide variety of certifications, including CISSP, ISSAP, ISSMP, ISSEP, SSCP, CC^{SM}, CCSP, CGRC^{SM}, and CSSLP. These certifications are designed to verify the knowledge and skills of IT security professionals across all industries. You can obtain more information about ISC2 and its other certifications from its website at isc2.org.

The CISSP credential is for security professionals “with the knowledge, skills and abilities to lead an organization's information security program.”

Topical Domains

The CISSP certification covers material from the eight topical domains. These eight domains are as follows:

- Domain 1: Security and Risk Management
- Domain 2: Asset Security
- Domain 3: Security Architecture and Engineering
- Domain 4: Communication and Network Security
- Domain 5: Identity and Access Management (IAM)
- Domain 6: Security Assessment and Testing
- Domain 7: Security Operations
- Domain 8: Software Development Security

These eight domains provide a vendor-independent overview of a common security framework. This framework is the basis for a discussion on security practices that can be supported in all types of organizations worldwide.

Prequalifications

ISC2 has defined the qualification requirements you must meet to become a CISSP. First, you must be a practicing security professional with at least five years’ work experience or with four years’ experience and a recent IT or IS degree or an approved security certification (as mentioned previously). Professional experience is defined as security work performed (with or without pay) within two or more of the eight CISSP domains.

Second, you must agree to adhere to a formal code of ethics. The ISC2 Code of Ethics is a set of guidelines ISC2 wants all certification candidates to follow to maintain professionalism in the field of information systems security. You can find the ISC2 Code of Ethics at isc2.org/ethics.

Overview of the CISSP Exam

The CISSP exam focuses on security from an overview perspective; it deals more with theory and concept than implementation and procedure. It is very broad but not very deep. To successfully complete this exam, you'll need to be familiar with every domain but not necessarily be a master of each domain.

The CISSP exam is in an adaptive format that ISC2 calls CISSP CAT (Computerized Adaptive Testing). For complete details of this form of exam presentation, please see www.isc2.org/certifications/CISSP/CISSP-CAT.

The CISSP CAT exam has a minimum of 100 questions and a maximum of 150. Not all items (i.e., questions) presented count toward your proficiency level, competency requirements, or passing status. There are 25 unscored questions that are called *pre-test or unscored items* by ISC2, whereas the scored questions are called *operational items*. The questions are not labeled on the exam as to whether they are scored (i.e., operational items) or unscored (i.e., pre-test questions). Test candidates will receive 25 unscored items on their exam, regardless of whether they achieve a passing rank at question 100 or see all of the 150 questions. However, an exam's pass/fail report is determined by only the last 75 operational items answered by the test candidate.

The CISSP CAT grants a maximum of three (3) hours to take the exam. If you run out of time before achieving a passing rank, you will automatically fail.

The CISSP CAT does not allow you to return to a previous question to change your answer. Your answer selection is final once you leave a question by submitting your answer selection.

To pass the CISSP CAT exam, you must score 700 out of a possible 1000 points, within the last 75 operational items (i.e., questions). If you do not achieve the minimum passing score after submitting your answer to question 150, then you fail. If you run out of time, then you fail.

If you do not pass the CISSP exam on your first attempt, you are allowed to retake the CISSP exam under the following conditions:

- You can take the CISSP exam a maximum of four times per 12-month period. (Note that on the CISSP CAT FAQ the limit is defined as 3 times per a 12-month period.)
- You must wait 30 days after your first attempt before trying a second time.
- You must wait an additional 60 days after your second attempt before trying a third time.
- You must wait an additional 90 days after your third or subsequent attempts before trying again.

The exam retake policy may be updated; you can read the most current version of the official policy here: www.isc2.org/Exams/After-Your-Exam.

You will need to pay full price for each additional exam attempt. However, ISC2 offers promotions from time to time that may allow you to retake an exam at no additional cost. This promotion has been called "Peace of Mind Protection," but could be renamed. It is limited to first-time test-takers only and has time restrictions. Be sure to read the fine print before acting on any such promotional offers.

The CISSP CAT exam is available in English, Chinese, German, Japanese, and Spanish.

For more details and the most up-to-date information on the CISSP exam direct from ISC2, please visit www.isc2.org/Certifications/CISSP and download the CISSP Ultimate Guide and visit www.isc2.org/certifications/cissp/cissp-certification-exam-outline to download the CISSP Exam Outline. You might also find useful information on the ISC2 Insights blog at www.isc2.org/Insights.



The total number of questions you may see on the exam, the total number of questions that count toward your score, the means and methods of scoring, and the time limit for the test are things that ISC2 reevaluates and changes from time to time. The best advice for preparing is to always recheck the ISC2 website for up to date exam specifications and policies.

CISSP Exam Question Types

Most of the questions on the CISSP exam are four-option, multiple-choice questions with a single correct answer. Some are straightforward, such as asking you to select a definition. Some are a bit more involved, asking you to select the appropriate concept or best practice. And some questions present you with a scenario or situation and ask you to select the best response.

You must select the one correct or best answer and mark it. In some cases, the correct answer will be obvious to you. In other cases, several answers may seem correct. In these instances, you must choose the best answer for the question asked. Watch for general, specific, universal, superset, and subset answer selections. In other cases, none of the answers will seem correct. In these instances, you'll need to select the least incorrect answer.

Some multiple-choice questions may require that you select more than one answer; if so, these will state what is necessary to provide a complete answer.

In addition to the standard multiple-choice question format, the exam may include a few advanced question formats, which ISC2 calls *advanced innovative questions*. These include drag-and-drop questions and hotspot questions. These types of questions require you to place topics or concepts in order of operations, in priority preference, or in relation to proper positioning for the needed solution. Specifically, the drag-and-drop questions require the test taker to move labels or icons to mark items on an image. The hotspot questions require the test-taker to pinpoint a location on an image with a crosshair marker. These question concepts are easy to work

with and understand, but be careful about your accuracy when dropping or marking.



ISC2 introduced the advanced innovative questions in 2014. They maintained a page describing these questions until 2017. While they still use this phrase to reference the question concepts, they no longer provide an explanation or examples of these questions on their website.

Advice on Taking the Exam

The CISSP exam consists of two key elements. First, you need to know the material from the eight domains. Second, you must have good test-taking skills. You have a maximum of 3 hours to achieve a passing standard with the potential to see up to 150 questions. Thus, you will have on average just over a minute for each question, so it is important to work quickly, without rushing, but also without wasting time.

Question skipping is not allowed on the CISSP CAT exam. You cannot return to view or change a previous question, and you're also not allowed to jump around, so one way or another, you have to come up with your best answer on each question as it is presented to you. If you don't know how to answer a question, then we recommend that you attempt to eliminate as many answer options as possible before making a guess. Then you can make educated guesses from a reduced set of options to increase your chance of getting a question correct. Since you have to answer every question presented, and you might not know the answer to some questions, you should develop a guessing strategy to select an answer promptly to minimize further wasting time.

Also note that ISC2 does not disclose if there is partial credit given for multiple-part questions if you get only some of the elements correct. So, pay attention to questions with checkboxes, and be sure to select as many items as necessary to properly address the question.

You will be provided with a dry-erase board and a marker to jot down thoughts and make notes. But nothing written on that board will be used to alter your score. That board must be returned to the test administrator prior to departing the test facility.

To maximize your test-taking activities, here are some general guidelines:

- Read each question carefully, then read the answer options, and then reread the question.
- Eliminate wrong answers before selecting the correct one.
- Watch for double negatives.
- Pay attention to universal terms, such as always or never.
- Look for relationships between answer options, such as similes, antonyms, sets, groups, parent/child, category/example, etc.
- Be sure you understand what the question is asking.

Manage your time. You can take breaks during your test, but this will consume some of your test time. You might consider bringing a drink and snacks, but your food and drink will be stored for you away from the testing area, and that break time will count against your test time limit. Be sure to bring any medications or other essential items, but leave all things electronic at home or in your car. You should avoid wearing anything on your wrists, including watches, fitness trackers, and jewelry. You are not allowed to bring any form of noise-canceling headsets or earbuds, although you can use foam earplugs. We also recommend wearing comfortable clothes and taking a light jacket with you (some testing locations are a bit chilly).

You may want to review the ISC2 CISSP Glossary document at `www.isc2.org/certifications/cissp/cissp-student-glossary`.

Finally, ISC2 exam policies are subject to change. Please be sure to check `isc2.org` for the current policies before you register and take the exam.

Study and Exam Preparation Tips

We recommend planning for a month or so of nightly intensive study for the CISSP exam. Here are some suggestions to maximize your learning time; you can modify them as necessary based on your own learning habits:

- Take one or two evenings to read each chapter in this book.
- Read and understand the Study Essentials for each chapter.
- Complete the written labs from each chapter.
- Answer all the review questions for each chapter.
- Be sure to research each question that you get wrong in order to learn what you didn't know.
- Review ISC2's Exam Outline to make sure you understand each listed item.
- Use the flashcards included with the online study tools to reinforce your understanding of concepts.
- Take the 4 full-length bonus practice exams provided in the online test engine.



We recommend spending about half of your study time reading and reviewing concepts and the other half taking practice exams. Students have reported that the more time they spent taking practice exams, the better they retained test topics. In addition to the practice tests with this Study Guide, Sybex also publishes *ISC2 CISSP Certified Information Systems Security Professional Official Practice Tests, 4th Edition* (ISBN: 978-1-394-25507-8). It contains 100 or more practice questions for each domain and four additional full-sized practice exams. Like this Study Guide, it also comes with an online version of the questions.

Completing the Certification Process

Once you have been informed that you successfully passed the CISSP certification, there is one final step before you are actually awarded the CISSP certification. That final step is known as *endorsement*. Basically, this involves getting someone who holds any ISC2 certification in good standing and is familiar with your work history to endorse you. Endorsement is the evaluation of your prerequisite qualifications (i.e., work experience) and the recommendation to ISC2 to award you the certification. Once you pass the CISSP exam, you will receive an email with instructions. However, you can review the endorsement application process at isc2.org/Endorsement. This URL is also where you initiate the endorsement process. You will need to know the ISC2 membership number of the person who will endorse you.

If you registered for CISSP, then you must complete endorsement within nine months of your exam. If you registered for Associate of ISC2, then you have six years from your exam data to complete endorsement. Once ISC2 accepts your endorsement, the certification process will be completed and you will be sent a welcome packet with confirmation of the certification achieved. You should also receive an email confirmation of the endorsement process's completion and another when the certification is awarded to you.

Once you have achieved your CISSP certification, you must maintain it. You will need to earn 120 Continuing Professional Education (CPE) credits by your third-year anniversary. For details on earning and reporting CPEs, please consult the ISC2 Continuing Professional Education (CPE) Handbook (www.isc2.org/-/media/Project/ISC2/Main/Media/documents/members/CPE-Handbook-2023.pdf) and the CPE Opportunities page (www.isc2.org/members/cpe-opportunities). You will also be required to pay an annual maintenance fee (AMF) upon earning your certification and at each annual anniversary. For details on the AMF, please see the ISC2 CPE Handbook, www.isc2.org/Policies-Procedures/AMFs-Overview, and www.isc2.org/Policies-Procedures/Member-Policies.

The Elements of This Study Guide

Each chapter includes common elements to help you focus your studies and test your knowledge. Here are descriptions of those elements:

Tips and Notes Throughout each chapter you will see inserted statements that you should pay additional attention to. These items are often focused details related to the chapter section or related important material.

Summaries The summary is a brief review of the chapter to sum up what was covered.

Study Essentials The Study Essentials highlight topics that could appear on the exam in some form. This section reinforces significant concepts that are key to understanding the concepts and topics of the chapter. The Study Essentials point out specific knowledge you want to retain from a chapter.

Written Labs Each chapter includes written labs that synthesize various concepts and topics that appear in the chapter. These raise questions that are designed to help you put together various pieces you've encountered individually in the chapter and assemble them to propose or describe potential security strategies or solutions. We highly encourage you to write out your answers before viewing our suggested solutions in Appendix B.

Chapter Review Questions Each chapter includes practice questions that have been designed to measure your knowledge of key ideas that were discussed in the chapter. After you finish each chapter, answer the questions; if some of your answers are incorrect, it's an indication that you need to spend some more time studying the corresponding topics. The answers to the practice questions can be found in Appendix A.

Interactive Online Learning Environment and Test Bank

Studying the material in the *ISC2 CISSP: Certified Information Systems Security Professional Official Study Guide, Tenth Edition* is an important part of preparing for the Certified Information Systems Security Professional (CISSP) certification exam, but we provide additional tools to help you prepare. The online Test Bank will help you understand the types of questions that will appear on the certification exam.

The sample tests in the Test Bank include all the questions in each chapter as well as the questions from the Assessment test in this Introduction section. In addition, there are four bonus practice exams that you can use to evaluate your understanding and identify areas that may require additional study. These four additional practice exams include 125 questions each and cover the breadth of domain topics in a similar percentage ratio as the real exam. They can be used as real exam simulations to evaluate your preparedness.

The online flashcards will push the limits of what you should know for the certification exam. The questions are provided in digital format. Each online flashcard has one question and one correct answer.

The downloadable PDF glossary is a searchable list of key terms from this exam guide that you should know for the CISSP certification exam.

A downloadble audio review is available where Mike Chapple reads aloud the Study Essentials from each chapter. You can listen to the audio review to keep your knowledge skills sharps as you go about your day. Its another means to sneak in a few more minutes of study time.

To start using these to study for the exam, go to www.wiley.com/go/sybextestprep, register your book to receive your unique PIN, and then once you have the PIN, return to www.wiley.com/go/sybextestprep and register a new account or add this book to an existing account.



Like all exams, the CISSP certification from ISC2 is updated periodically and may eventually be retired or replaced. At some point after ISC2 is no longer offering this exam, the old editions of our books and online tools will be retired. If you have purchased this book after the exam was retired, or are attempting to register in the Sybex online learning environment after the exam was retired, please know that we make no guarantees that this exam's online Sybex tools will be available once the exam is no longer available.

Study Guide Exam Objectives

This table provides the extent, by percentage, to which each domain is represented on the actual examination.

Domain	% of exam
Domain 1: Security and Risk Management	16%
Domain 2: Asset Security	10%
Domain 3: Security Architecture and Engineering	13%
Domain 4: Communication and Network Security	13%
Domain 5: Identity and Access Management (IAM)	13%
Domain 6: Security Assessment and Testing	12%
Domain 7: Security Operations	13%
Domain 8: Software Development Security	10%
Total	100%

Objective Map

This book is designed to cover each of the eight CISSP Exam Outline domains in sufficient depth to provide you with a clear understanding of the material. The main body of this book consists of 21 chapters. Here is a complete CISSP Exam Outline mapping each objective item to its location in this book's chapters.

Domain	Description	Chapter
1.0	**Security and Risk Management**	
1.1	Understand, adhere to, and promote professional ethics	19
1.1.1	ISC2 Code of Professional Ethics	19
1.1.2	Organizational code of ethics	19
1.2	Understand and apply security concepts	1
1.2.1	Confidentiality, integrity, and availability, authenticity, and nonrepudiation (5 Pillars of Information Security)	1

Domain	Description	Chapter
1.3	Evaluate and apply security governance principles	1
1.3.1	Alignment of the security function to business strategy, goals, mission, and objectives	1
1.3.2	Organizational processes (e.g., acquisitions, divestitures, governance committees)	1
1.3.3	Organizational roles and responsibilities	1
1.3.4	Security control frameworks (e.g., International Organization for Standardization (ISO), National Institute of Standards and Technology (NIST), Control Objectives for Information and Related Technology (COBIT), Sherwood Applied Business Security Architecture (SABSA), Payment Card Industry (PCI), Federal Risk and Authorization Management Program (FedRAMP))	1
1.3.5	Due care/due diligence	1
1.4	Understand legal, regulatory, and compliance issues that pertain to information security in a holistic context	4
1.4.1	Cybercrimes and data breaches	4
1.4.2	Licensing and Intellectual Property requirements	4
1.4.3	Import/export controls	4
1.4.4	Transborder data flow	4
1.4.5	Issues related to privacy (e.g., General Data Protection Regulation (GDPR), California Consumer Privacy Act, Personal Information Protection Law, Protection of Personal Information Act)	4
1.4.6	Contractual, legal, industry standards, and regulatory requirements	4
1.5	Understand requirements for investigation types (i.e., administrative, criminal, civil, regulatory, industry standards)	19

Domain	Description	Chapter
1.6	Develop, document, and implement security policy, standards, procedures, and guidelines	1
1.7	Identify, analyze, assess, prioritize, and implement Business Continuity (BC) requirements	3
1.7.1	Business Impact Analysis (BIA)	3
1.7.2	External dependencies	3
1.8	Contribute to and enforce personnel security policies and procedures	2
1.8.1	Candidate screening and hiring	2
1.8.2	Employment agreements and policy driven requirements	2
1.8.3	Onboarding, transfers, and termination processes	2
1.8.4	Vendor, consultant, and contractor agreements and controls	2
1.9	Understand and apply risk management concepts	2
1.9.1	Threat and vulnerability identification	2
1.9.2	Risk analysis, assessment, and scope	2
1.9.3	Risk response and treatment (e.g., cybersecurity insurance)	2
1.9.4	Applicable types of controls (e.g., preventive, detection, corrective)	2
1.9.5	Control assessments (e.g., security and privacy)	2
1.9.6	Continuous monitoring and measurement	2
1.9.7	Reporting (e.g., internal, external)	2
1.9.8	Continuous improvement (e.g., risk maturity modeling)	2

Domain	Description	Chapter
1.9.9	Risk frameworks (e.g., International Organization for Standardization (ISO), National Institute of Standards and Technology (NIST), Control Objectives for Information and Related Technology (COBIT), Sherwood Applied Business Security Architecture (SABSA), Payment Card Industry (PCI))	2
1.10	Understand and apply threat modeling concepts and methodologies	1
1.11	Apply supply chain risk management (SCRM) concepts	1
1.11.1	Risks associated with the acquisition of products and services from suppliers and providers (e.g., product tampering, counterfeits, implants)	1
1.11.2	Risk mitigations (e.g., third-party assessment and monitoring, minimum security requirements, service level requirements, silicon root of trust, physically unclonable function, software bill of materials)	1
1.12	Establish and maintain a security awareness, education, and training program	2
1.12.1	Methods and techniques to increase awareness and training (e.g., social engineering, phishing, security champions, gamification)	2
1.12.2	Periodic content reviews to include emerging technologies and trends (e.g., cryptocurrency, artificial intelligence (Al), blockchain)	2
1.12.3	Program effectiveness evaluation	2
2.0	**Asset Security**	
2.1	Identify and classify information and assets	5
2.1.1	Data classification	5
2.1.2	Asset classification	5
2.2	Establish information and asset handling requirements	5
2.3	Provision information and assets securely	16

Domain	Description	Chapter
2.3.1	Information and asset ownership	16
2.3.2	Asset inventory (e.g., tangible, intangible)	16
2.3.3	Asset management	16
2.4	Manage data lifecycle	5
2.4.1	Data roles (i.e., owners, controllers, custodians, processors, users/subjects)	5
2.4.2	Data collection	5
2.4.3	Data location	5
2.4.4	Data maintenance	5
2.4.5	Data retention	5
2.4.6	Data remanence	5
2.4.7	Data destruction	5
2.5	Ensure appropriate asset retention (e.g., End of Life (EOL), End of Support)	5
2.6	Determine data security controls and compliance requirements	5
2.6.1	Data states (e.g., in use, in transit, at rest)	5
2.6.2	Scoping and tailoring	5
2.6.3	Standards selection	5
2.6.4	Data protection methods (e.g., Digital Rights Management (DRM) data loss prevention (DLP), cloud access security broker (CASB))	5
3.0	**Security Architecture and Engineering**	
3.1	Research, implement, and manage engineering processes using secure design principles	1, 8, 9, 16
3.1.1	Threat modeling	1
3.1.2	Least privilege	16
3.1.3	Defense in depth	1
3.1.4	Secure defaults	8

Domain	Description	Chapter
3.1.5	Fail securely	8
3.1.6	Segregation of Duties (SoD)	16
3.1.7	Keep it simple and small	8
3.1.8	Zero trust or trust but verify	8
3.1.9	Privacy by design	8
3.1.10	Shared responsibility	9
3.1.11	Secure access service edge	8
3.2	Understand the fundamental concepts of security models (e.g., Biba, Star Model, Bell-LaPadula)	8
3.3	Select controls based upon systems security requirements	8
3.4	Understand security capabilities of Information Systems (IS) (e.g., memory protection, Trusted Platform Module (TPM), encryption/decryption)	8
3.5	Assess and mitigate the vulnerabilities of security architectures, designs, and solution elements	6, 7, 9, 16, 20
3.5.1	Client-based systems	9
3.5.2	Server-based systems	9
3.5.3	Database systems	20
3.5.4	Cryptographic systems	6, 7
3.5.5	Industrial Control Systems (ICS)	9
3.5.6	Cloud-based systems (e.g., Software as a Service (SaaS), Infrastructure as a Service (IaaS), Platform as a Service (PaaS))	16
3.5.7	Distributed systems	9
3.5.8	Internet of Things (IoT)	9
3.5.9	Microservices (e.g., application programming interface (API))	9
3.5.10	Containerization	9

Domain	Description	Chapter
3.5.11	Serverless	16
3.5.12	Embedded systems	9
3.5.13	High-Performance Computing systems	9
3.5.14	Edge computing systems	9
3.5.15	Virtualized systems	9
3.6	Select and determine cryptographic solutions	6, 7
3.6.1	Cryptographic life cycle (e.g., keys, algorithm selection)	6, 7
3.6.2	Cryptographic methods (e.g., symmetric, asymmetric, elliptic curves, quantum)	6, 7
3.6.3	Public Key Infrastructure (PKI) (e.g., quantum key distribution)	7
3.6.4	Key management practices (e.g., rotation)	7
3.6.5	Digital signatures and digital certificates (e.g., non-repudiation, integrity)	7
3.7	Understand methods of cryptanalytic attacks	7, 14, 21
3.7.1	Brute force	7
3.7.2	Ciphertext only	7
3.7.3	Known plaintext	7
3.7.4	Frequency analysis	7
3.7.5	Chosen ciphertext	7
3.7.6	Implementation attacks	7
3.7.7	Side-channel	7
3.7.8	Fault injection	7
3.7.9	Timing	7
3.7.10	Man-in-the-Middle (MITM)	7
3.7.11	Pass the hash	14
3.7.12	Kerberos exploitation	14

Domain	Description	Chapter
3.7.13	Ransomware	21
3.8	Apply security principles to site and facility design	10
3.9	Design site and facility security controls	10
3.9.1	Wiring closets/intermediate distribution frame	10
3.9.2	Server rooms/data centers	10
3.9.3	Media storage facilities	10
3.9.4	Evidence storage	10
3.9.5	Restricted and work area security	10
3.9.6	Utilities and Heating, Ventilation, and Air Conditioning (HVAC)	10
3.9.7	Environmental issues (e.g., natural disasters, man-made)	10
3.9.8	Fire prevention, detection, and suppression	10
3.9.9	Power (e.g., redundant, backup)	10
3.10	Manage the information system lifecycle	8
3.10.1	Stakeholders needs and requirements	8
3.10.2	Requirements analysis	8
3.10.3	Architectural design	8
3.10.4	Development /implementation	8
3.10.5	Integration	8
3.10.6	Verification and validation	8
3.10.7	Transition/deployment	8
3.10.8	Operations and maintenance/sustainment	8
3.10.9	Retirement/disposal	8
4.0	**Communication and Network Security**	
4.1	Apply secure design principles in network architectures	11, 12
4.1.1	Open System Interconnection (OSI) and Transmission Control Protocol/Internet Protocol (TCP/IP) models	11

Domain	Description	Chapter
4.1.2	Internet Protocol (IP) version 4 and 6 (IPv6) (e.g., unicast, broadcast, multicast, anycast)	11
4.1.3	Secure protocols (e.g., Internet Protocol Security (IPSec), Secure Shell (SSH), Secure Sockets Layer (SSL)]/Transport Layer Security (TLS))	11
4.1.4	Implications of multilayer protocols	11
4.1.5	Converged protocols (e.g., Internet Small Computer Systems Interface (iSCSI), Voice over Internet Protocol (VoIP), InfiniBand over Ethernet, Compute Express Link)	11
4.1.6	Transport architecture (e.g., topology, data/control/management plane, cut-through/store-and-forward)	11
4.1.7	Performance metrics (e.g., bandwidth, latency, jitter, throughput, signal-to-noise ratio)	12
4.1.8	Traffic flows (e.g., north-south, east-west)	11
4.1.9	Physical segmentation (e.g., in-band, out-of-band, air-gapped)	11
4.1.10	Logical segmentation (e.g., virtual local area networks (VLANs), virtual private networks (VPNs), virtual routing and forwarding, virtual domain)	11
4.1.11	Micro-segmentation (e.g., network overlays/encapsulation; distributed firewalls, routers, intrusion detection system (IDS)/intrusion prevention system (IPS), zero trust)	11
4.1.12	Edge networks (e.g., ingress/egress, peering)	11
4.1.13	Wireless networks (e.g., Bluetooth, Wi-Fi, Zigbee, satellite)	11
4.1.14	Cellular/mobile networks (e.g., 4G, 5G)	11
4.1.15	Content distribution networks (CDN)	11

Domain	Description	Chapter
4.1.16	Software defined networks (SDN), (e.g., application programming interface (API), Software-Defined Wide-Area Network, network functions virtualization)	11
4.1.17	Virtual Private Cloud (VPC)	11
4.1.18	Monitoring and management (e.g., network observability, traffic flow/shaping, capacity management, fault detection and handling)	12
4.2	Secure network components	11
4.2.1	Operation of infrastructure (e.g., redundant power, warranty, support)	11
4.2.2	Transmission media (e g. physical security of media, signal propagation quality)	11
4.2.3	Network Access Control (NAC) systems (e.g., physical, and virtual solutions)	11
4.2.4	Endpoint security (e.g., host-based)	11
4.3	Implement secure communication channels according to design	12
4.3.1	Voice, video, and collaboration (e.g., conferencing, Zoom rooms)	12
4.3.2	Remote access (e.g., network administrative functions)	12
4.3.3	Data communications (e.g., backhaul networks satellite)	12
4.3.4	Third-party connectivity (e.g., telecom providers, hardware support)	12
5.0	**Identity and Access Management (IAM)**	
5.1	Control physical and logical access to assets	13
5.1.1	Information	13
5.1.2	Systems	13
5.1.3	Devices	13
5.1.4	Facilities	13

Domain	Description	Chapter
5.1.5	Applications	13
5.1.6	Services	13
5.2	Design identification and authentication strategy (e.g., people, devices, and services)	13
5.2.1	Groups and Roles	13
5.2.2	Authentication, Authorization and Accounting (AAA) (e.g., multi-factor authentication (MFA), password-less authentication)	13
5.2.3	Session management	13
5.2.4	Registration, proofing, and establishment of identity	13
5.2.5	Federated Identity Management (FIM)	13
5.2.6	Credential management systems (e.g., Password vault)	13
5.2.7	Single sign-on (SSO)	13
5.2.8	Just-ln-Time	13
5.3	Federated identity with a third-party service	13
5.3.1	On-premise	13
5.3.2	Cloud	13
5.3.3	Hybrid	13
5.4	Implement and manage authorization mechanisms	14
5.4.1	Role-based access control (RBAC)	14
5.4.2	Rule based access control	14
5.4.3	Mandatory access control (MAC)	14
5.4.4	Discretionary access control (DAC)	14
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Assessment Test

1. Which of the following types of access control seeks to discover evidence of unwanted, unauthorized, or illicit behavior or activity?
    A. Preventive
    B. Deterrent
    C. Detection
    D. Corrective
2. Define and detail the aspects of password selection that distinguish good password choices from ultimately poor password choices.
    A. Is difficult to guess or unpredictable
    B. Meets minimum length requirements

C. Meets specific complexity requirements

D. All of the above

3. Some adversaries use DoS attacks as their primary weapon to harm targets, whereas others may use them as weapons of last resort when all other attempts to intrude on a target fail. Which of the following is most likely to detect DoS attacks?

   A. Host-based IDS

   B. Network-based IDS

   C. Vulnerability scanner

   D. Penetration testing

4. Unfortunately, attackers have many options of attacks to perform against their targets. Which of the following is considered a denial-of-service (DoS) attack?

   A. Pretending to be a technical manager over the phone and asking a receptionist to change their password

   B. While surfing the web, sending to a web server a malformed URL that causes the system to consume 100 percent of the CPU

   C. Intercepting network traffic by copying the packets as they pass through a specific subnet

   D. Sending message packets to a recipient who did not request them, simply to be annoying

5. Hardware networking devices operate within the protocol stack just like protocols themselves. Thus, hardware networking devices can be associated with an OSI model layer related to the protocols they manage or control. At which layer of the OSI model does a router operate?

   A. Network Layer

   B. Layer 1

   C. Transport Layer

   D. Layer 5

6. Which type of firewall automatically adjusts its filtering rules based on the content and context of the traffic of existing sessions? (Choose all that apply.)

   A. Static packet filtering

   B. Application-level gateway

   C. Circuit-level gateway

   D. Stateful inspection firewall

7. A VPN can be a significant security improvement for many communication links. A VPN can be established over which of the following?

   A. Wireless LAN connection

   B. Remote access dial-up connection

   C. WAN link

   D. All of the above

8. Adversaries will use any and all means to harm their targets. This includes mixing attack concepts together to make a more effective campaign. What type of malware uses social engineering to trick a victim into installing it?

   A. Virus

   B. Worm

   C. Trojan horse

   D. Logic bomb

9. Security is established by understanding the assets of an organization that need protection and understanding the threats that could cause harm to those assets. Then, controls are selected that provide protection for the CIA Triad of the assets at risk. The CIA Triad consists of what elements?

   A. Contiguousness, interoperable, arranged

   B. Authentication, authorization, accountability

C. Capable, available, integral

D. Availability, confidentiality, integrity

10. The security concept of AAA services describes the elements that are necessary to establish subject accountability. Which of the following is not a required component in the support of accountability?

    A. Logging

    B. Privacy

    C. Identification verification

    D. Authorization

11. Collusion is when two or more people work together to commit a crime or violate a company policy. Which of the following is not a defense against collusion?

    A. Separation of duties

    B. Restricted job responsibilities

    C. Group shared user accounts

    D. Job rotation

12. A data custodian is responsible for securing resources after _____ has assigned the resource a security label.

    A. Senior management

    B. The data owner

    C. An auditor

    D. Security staff

13. In what phase of the Capability Maturity Model for Software (SW-CMM) are quantitative measures used to gain a detailed understanding of the software development process?

    A. Repeatable

    B. Defined

    C. Managed

D. Optimizing

14. Which one of the following is a layer of the protection ring model design concept that is not normally implemented?

    A. Ring 0

    B. Ring 1

    C. Ring 3

    D. Ring 4

15. TCP operates at the Transport Layer and is a connection-oriented protocol. It uses a special process to establish a session each time a communication takes place. What is the last phase of the TCP three-way handshake sequence?

    A. SYN flagged packet

    B. ACK flagged packet

    C. FIN flagged packet

    D. SYN/ACK flagged packet

16. The lack of secure coding practices has enabled an uncountable number of software vulnerabilities that attackers have discovered and exploited. Which one of the following vulnerabilities would be best countered by adequate parameter checking?

    A. Time-of-check to time-of-use

    B. Buffer overflow

    C. SYN flood

    D. Distributed denial of service (DDoS)

17. Computers are based on binary mathematics. All computer functions are derived from the basic set of Boolean operations. What is the value of the logical operation shown here?

```
X:     0 1 1 0 1 0
Y:     0 0 1 1 0 1
_____________________
X ⊕ Y:   ?
```

A. 0 1 0 1 1 1

B. 0 0 1 0 0 0

C. 0 1 1 1 1 1

D. 1 0 0 1 0 1

18. Which of the following are considered standard data type classifications used in either a government/military or a private sector organization? (Choose all that apply.)

    A. Public

    B. Healthy

    C. Private

    D. Inside only

    E. Sensitive

    F. Proprietary

    G. Essential

    H. Certified

    I. Critical

    J. Confidential

    K. For Your Eyes Only

19. The General Data Protection Regulation (GDPR) has defined several roles in relation to the protection and management of personally identifiable information (PII). Which of the following statements is true?

    A. A data processor is the entity assigned specific responsibility for a data asset in order to ensure its protection for use by the organization.

    B. A data custodian is the entity that performs operations on data.

    C. A data controller is the entity that makes decisions about the data they are collecting.

D. A data owner is the entity assigned or delegated the day-to-day responsibility of proper storage and transport as well as protecting data, assets, and other organizational objects.

20. If Renee receives a digitally signed message from Mike, what key does she use to verify that the message truly came from Mike?

    A. Renee's public key

    B. Renee's private key

    C. Mike's public key

    D. Mike's private key

21. A systems administrator is setting up a new data management system. It will be gathering data from numerous locations across the network, even from remote offsite locations. The data will be moved to a centralized facility, where it will be stored on a massive RAID array. The data will be encrypted on the storage system using AES-256, and most files will be signed as well. The location of this data warehouse is secured so that only authorized personnel can enter the room and all digital access is limited to a set of security administrators. Which of the following describes the data?

    A. The data is encrypted in transit.

    B. The data is encrypted in processing.

    C. The data is redundantly stored.

    D. The data is encrypted at rest.

22. The _____ is the entity assigned specific responsibility for a data asset in order to ensure its protection for use by the organization.

    A. Data owner

    B. Data controller

    C. Data processor

    D. Data custodian

23. A security auditor is seeking evidence of how sensitive documents made their way out of the organization and onto a public document

distribution site. It is suspected that an insider exfiltrated the data over a network connection to an external server, but this is only a guess. Which of the following would be useful in determining whether this suspicion is accurate? (Choose two.)

A. NAC

B. DLP alerts

C. Syslog

D. Log analysis

E. Malware scanner reports

F. Integrity monitoring

24. A new Wireless Access Point (WAP) is being installed to add wireless connectivity to the company network. The configuration policy indicates that WPA3 is to be used and thus only newer or updated endpoint devices can connect. The policy also states that ENT authentication will not be implemented. What authentication mechanism can be implemented in this situation?

A. IEEE 802.1X

B. IEEE 802.1q

C. Simultaneous authentication of equals (SAE)

D. EAP-FAST

25. When securing a mobile device, what types of authentication can be used that depend on the user's physical attributes? (Choose all that apply.)

A. Fingerprint

B. TOTP (time-based one-time password)

C. Voice

D. SMS (short message service)

E. Retina or iris

F. Gait

G. Phone call

H. Facial recognition

I. Smartcard

J. Password

26. A recently acquired piece of equipment is not working properly. Your organization does not have a trained repair technician on staff, so you have to bring in an outside expert. What type of account should be issued to a trusted third-party repair technician?

A. Guest account

B. Privileged account

C. Service account

D. User account

27. Security should be designed and integrated into the organization as a means to support and maintain the business objectives. However, the only way to know if the implemented security is sufficient is to test it. Which of the following is a procedure designed to test and perhaps bypass a system's security controls?

A. Logging usage data

B. War dialing

C. Penetration testing

D. Deploying secured desktop workstations

28. Security needs to be designed to support the business objectives, but it also needs to be legally defensible. To defend the security of an organization, a log of events and activities must be created. Auditing is a required factor to sustain and enforce what?

A. Accountability

B. Confidentiality

C. Accessibility

D. Redundancy

29. Risk assessment is a process by which the assets, threats, probabilities, and likelihoods are evaluated in order to establish criticality prioritization. What is the formula used to compute the ALE?

    A. ALE = AV * EF * ARO

    B. ALE = ARO * EF

    C. ALE = AV * ARO

    D. ALE = EF * ARO

30. Incident response plans, business continuity plans, and disaster recovery plans are crafted when implementing business-level redundancy. These plans are derived from the information obtained when performing a business impact assessment (BIA). What is the first step of the BIA process?

    A. Identification of priorities

    B. Likelihood assessment

    C. Risk identification

    D. Resource prioritization

31. Many events can threaten the operation, existence, and stability of an organization. Some of those threats are human caused, whereas others are from natural events. Which of the following represent natural events that can pose a threat or risk to an organization?

    A. Earthquake

    B. Flood

    C. Tornado

    D. All of the above

32. What kind of recovery facility enables an organization to resume operations as quickly as possible, if not immediately, upon failure of the primary facility?

    A. Hot site

    B. Warm site

C. Cold site

D. All of the above

33. During an account review, an auditor provided the following report:

User	Last Login Length	Last Password Change
Bob	4 hours	87 days
Sue	3 hours	38 days
John	1 hour	935 days
Kesha	3 hours	49 days

The security manager reviews the account policies of the organization and takes note of the following requirements:

- Passwords must be at least 12 characters long.
- Passwords must include at least one example of three different character types.
- Passwords must be changed every 180 days.
- Passwords cannot be reused.

Which of the following security controls should be corrected to enforce the password policy?

A. Minimum password length

B. Account lockout

C. Password history and minimum age

D. Password maximum age

34. Any evidence to be used in a court proceeding must abide by the Rules of Evidence to be admissible. What type of evidence refers to written documents that are brought into court to prove a fact?

A. Best evidence

B. Parol evidence

C. Documentary evidence

D. Testimonial evidence

35. DevOps manager John is concerned with the CEO's plan to minimize his department and outsource code development to a foreign programming group. John has a meeting scheduled with the board of directors to encourage them to retain code development in house due to several concerns. Which of the following should John include in his presentation? (Choose all that apply.)

    A. Code from third parties will need to be manually reviewed for function and security.

    B. If the third party goes out of business, existing code may need to be abandoned.

    C. Third-party code development is always more expensive.

    D. A software escrow agreement should be established.

36. When TLS is being used to secure web communications, what URL prefix appears in the web browser address bar to signal this fact?

    A. `SHTTP://`

    B. `TLS://`

    C. `FTPS://`

    D. `HTTPS://`

37. A new update has been released by the vendor of an important software product that is an essential element of a critical business task. The chief security officer (CSO) indicates that the new software version needs to be tested and evaluated in a virtual lab, which has a cloned simulation of many of the company's production systems. Furthermore, the results of this evaluation must be reviewed before a decision is made as to whether the software update should be installed and, if so, when to install it. What security principle is the CSO demonstrating?

    A. Business continuity planning (BCP)

    B. Onboarding

    C. Change management

    D. Static analysis

38. What type of token device produces new time-derived passwords on a specific time interval that can be used only a single time when attempting to authenticate?

    A. HOTP

    B. HMAC

    C. SAML

    D. TOTP

39. Your organization is moving a significant portion of their data processing from an on-premises solution to the cloud. When evaluating a cloud service provider (CSP), which of the following is the most important security concern?

    A. Data retention policy

    B. Number of customers

    C. Hardware used to support VMs

    D. Whether they offer MaaS, IDaaS, and SaaS

40. Most software vulnerabilities exist because of a lack of secure or defensive coding practices used by the developers. Which of the following is considered a secure coding technique? (Choose all that apply.)

    A. Using immutable systems

    B. Using stored procedures

    C. Using code signing

    D. Using server-side validation

    E. Optimizing file sizes

    F. Using third-party software libraries

Answers to Assessment Test

1. C. Detection access controls are used to discover (and document) unwanted or unauthorized activity. Preventive access controls block

the ability to perform unwanted activity. Deterrent access controls attempt to persuade the perpetrator not to perform unwanted activity. Corrective access controls restore a system to normal function in the event of a failure or system interruption.

2. D. Strong password choices are difficult to guess, unpredictable, and of specified minimum lengths to ensure that password entries cannot be computationally determined. They may be randomly generated and use any of the alphabetic, numeric, and allowed special characters; they should never be written down or shared; they should not be stored in publicly accessible or generally readable locations; and they shouldn't be transmitted in the clear.
3. B. Network-based IDSs are usually able to detect the initiation of an attack or the ongoing attempts to perpetrate an attack (including denial of service, or DoS). They are, however, unable to provide information about whether an attack was successful or which specific systems, user accounts, files, or applications were affected. Host-based IDSs have some difficulty with detecting and tracking down DoS attacks. Vulnerability scanners don't detect DoS attacks; they test for possible vulnerabilities. Penetration testing may cause a DoS or test for DoS vulnerabilities, but it is not a detection tool.
4. B. Not all instances of DoS are the result of a malicious attack. Errors in coding OSs, services, and applications have resulted in DoS conditions. Some examples of this include a process failing to release control of the CPU or a service consuming system resources out of proportion to the service requests it is handling. Social engineering (i.e., pretending to be a technical manager) and sniffing (i.e., intercepting network traffic) are typically not considered DoS attacks. Sending message packets to a recipient who did not request them simply to be annoying may be a type of social engineering, and it is definitely spam, but unless the volume of the messages is significant, it does not warrant the label of DoS.
5. A. Routers function at Layer 3, the Network Layer. Layer 1, the Physical Layer, is where repeaters and hubs operate, not routers. Network devices usually do not operate at the transport layer alone, but across layers, such as firewalls, proxies, and load balancers. Layer

5, the Session Layer, does not actually exist in a modern TCP/IP network, and thus no hardware directly operates at this layer, but its functions are performed by TCP in the Transport Layer, Layer 4, when sessions are in use.

6. B, D. Stateful inspection firewalls (aka dynamic packet-filtering firewalls) enable the real-time modification of the filtering rules based on traffic content and context. An application-level gateway is a type of stateful firewall. Static packet filtering and circuit-level firewalls are both stateless and thus do not consider the context when applying filtering rules.
7. D. A virtual private network (VPN) link can be established over any network communication connection. This could be a typical LAN cable connection, a wireless LAN connection, a remote access dial-up connection, a WAN link, or an internet connection used by a client for access to the office LAN.
8. C. A Trojan horse is a form of malware that uses social engineering tactics to trick a victim into installing it—the trick is to make the victim believe that the only thing they have downloaded or obtained is the host file, when in fact it has a malicious hidden payload. Viruses and logic bombs do not typically use social engineering as an element in their means of infecting a system. A worm sometimes is designed to take advantage of social engineering, such as when the worm is an executable email attachment and the message tricks the victim into opening it. However, not all worms are designed this way—this is a core design concept of a Trojan horse.
9. D. The components of the CIA Triad are confidentiality, integrity, and availability. The other options are not the terms that define the CIA Triad, although they are security concepts that need to be evaluated when establishing a security infrastructure.
10. B. Privacy is not necessary to provide accountability. The required elements of accountability, as defined in AAA services, are as follows: identification (which is sometimes considered an element of authentication, a silent first step of AAA services, or represented by IAAA), authentication (i.e., identification verification), authorization

(i.e., access control), auditing (i.e., logging and monitoring), and accounting.

11. C. Group shared user accounts allow for multiple people to log in under a single user account. This allows collusion because it prevents individual accountability. Separation of duties, restricted job responsibilities, and job rotation help establish individual accountability and control access (especially to privileged capabilities), which in turn limits or restricts collusion.

12. B. The data owner must first assign a security label to a resource before the data custodian can secure the resource appropriately. Senior management is ultimately responsible for the success or failure of a security endeavor. An auditor is responsible for reviewing and verifying that the security policy is properly implemented, that the derived security solutions are adequate, and that user events are in compliance with security policy. The security staff is responsible for designing, implementing, and managing the security infrastructure once approved by senior management.

13. C. The Managed phase (level 4) of the SW-CMM involves the use of quantitative development metrics. The Software Engineering Institute (SEI) defines the key process areas for this level as Quantitative Process Management and Software Quality Management. The Repeatable phase (level 2) is where basic life cycle processes are introduced. The Defined phase (level 3) is where developers operate according to a set of formal, documented development processes. The Optimizing phase (level 5) is where a process of continuous improvement is achieved.

14. B. Rings 1 and 2 contain device drivers but are not normally implemented in practice, since they are often collapsed into Ring 0. Ring 0 always contains the security kernel. Ring 3 contains user applications. Ring 4 does not exist in the design concept, but it may exist in customized implementations.

15. B. The SYN flagged packet is first sent from the initiating host to the destination host. The destination host then responds with a SYN/ACK flagged packet. The initiating host sends an ACK flagged packet, and the connection is then established. The FIN flagged packet is not used

in the TCP three-way handshake to establish a session; it is used in the session teardown process.

16. B. Parameter checking (i.e., confirming input is within reasonable boundaries) is used to prevent the possibility of buffer overflow attacks. Time-of-check to time-of-use (TOCTOU) attacks are not directly addressed by parameter checking or input filtering; defensive coding practices are needed to eliminate or reduce this issue. SYN flood attacks are a type of DoS, which is not fully protected against with just improved coding practices. A DDoS is also not prohibited by just improved coding practices such as parameter checking. For any type of DoS, adequate filtering and processing capacity are the most effective security responses.

17. A. The ⊕ symbol represents the XOR function and returns a true value when only one of the input values is true. If both values are false or both values are true, the output of the XOR function is false. Option B is the result if these two values were combined using the AND (the ∧ symbol) function, which returns a value of true if the two values are both true. Option C is the result if these two values were combined using the OR (the ∨ symbol) function, which returns a value of true if either input values is true. Option D is the result if only the X value was subjected to the NOR (the ~ symbol) function, which reverses the value of an input.

18. A, C, E, F, I, J. There are six standard data type classifications used in either a government/military or a private sector organization in this list of options: public, private, sensitive, proprietary, critical, and confidential. The other options (healthy, inside only, essential, certified, and for your eyes only) are incorrect since they are not typical or standard classifications. Note: There is a “For internal use only” classification, but the option here is “Inside only,” which while it may express the same sentiment/intention, is not the exact classification label.

19. C. The correct statement is regarding the data controller. The other statements are incorrect. The correct versions of those statements are as follows. A data owner is the entity assigned specific responsibility for a data asset in order to ensure its protection for use by the

organization. A data processor is the entity that performs operations on data. A data custodian is the entity assigned or delegated the day-to-day responsibility for proper storage and transport as well as protecting data, assets, and other organizational objects.

20. C. Any recipient can use Mike's public key to verify the authenticity of the digital signature. Renee's (the recipient's) public key is not used in this scenario. However, it could be used to create a digital envelope to protect a symmetric session encryption key sent from Mike to Renee. Renee's (the recipient's) private key is not used in this scenario. However, it could be used if Renee becomes a sender to send Mike a digitally signed message. Mike's (the sender's) private key was used to encrypt the hash of the data to be sent to Renee, and this is what creates the digital signature.
21. D. In this scenario, the data is encrypted at rest with AES-256. There is no mention of encryption for transfer or processing. The data is not stored redundantly, since it is being moved, not copied, to the central data warehouse, and there is no mention of a backup.
22. A. The data owner is the person(s) (or entity) assigned specific responsibility for a data asset in order to ensure its protection for use by the organization. The data controller is the entity that makes decisions about the data they are collecting. A data processor is the entity that performs operations on data on behalf of a data controller. A data custodian is a subject who has been assigned or delegated the day-to-day responsibility for proper storage and transport as well as protecting data, assets, and other organizational objects.
23. B, D. In this scenario, the data loss prevention (DLP) alerts and log analysis are the only options that would potentially include useful information in regard to an insider exfiltrating the sensitive documents. The other options are incorrect because they do not provide relevant information. Network access control (NAC) is a security mechanism to prevent rogue devices and ensure authorized systems meet minimum security configuration requirements. Syslog is a logging service used to maintain centralized real-time copies of active log files. Malware scanner reports are not relevant here since there is no suspicious or malicious code being used but only access abuses and unauthorized

file distribution. Integrity monitoring is also not relevant to this situation, since there is no indication that the documents were altered, just that they were released to the public.

24. C. WPA3 supports ENT (Enterprise Wi-Fi authentication, aka IEEE 802.1X) and SAE authentication. Simultaneous authentication of equals (SAE) still uses a password, but it no longer encrypts and sends that password across the connection to perform authentication. Instead, SAE performs a zero-knowledge proof process known as Dragonfly Key Exchange, which is itself a derivative of Diffie–Hellman. IEEE 802.1X defines port-based network access control that ensures that clients can't communicate with a resource until proper authentication has taken place. It's based on Extensible Authentication Protocol (EAP) from Point-to-Point Protocol (PPP). However, this is the technology behind the label of ENT; thus, it is not an option in this scenario. IEEE 802.1q defines the use of virtual local area network (VLAN) tags and thus is not relevant to Wi-Fi authentication. Flexible Authentication via Secure Tunneling (EAP-FAST) is a Cisco protocol proposed to replace Lightweight Extensible Authentication Protocol (LEAP), which is now obsolete, thanks to the development of WPA2, and is not supported in WPA3 either.
25. A, C, E, H. Biometrics are authentication factors that are based on a user's physical attributes; they include fingerprints, voice, retina, iris, and facial recognition. Gait is a form of biometrics, but it is not appropriate for use as authentication on a mobile device; it is used from a stationary position to monitor people walking toward or past a security point. The other options are valid authentication factors, but they are not biometrics.
26. B. A repair technician typically requires more than a normal level of access to perform their duties, so a privileged account for even a trusted third-party technician is appropriate. A guest account or user (normal, limited) account is insufficient for this scenario. A service account is to be used by an application or background service, not a repair technician or other user.
27. C. Penetration testing is the attempt to bypass security controls to test overall system security. Logging usage data is a type of auditing and is

useful in the authentication, authorization, accounting (AAA) service process in order to hold subjects accountable for their actions. However, it is not a means to test security. War dialing is an attempt to locate modems and fax machines by dialing phone numbers. This process is sometimes still used by penetration testers and adversaries to find targets to attack, but it is not an actual attack or stress test itself. Deploying secured desktop workstations is a security response to the results of a penetration test, not a security testing method.

28. A. Auditing is a required factor to sustain and enforce accountability. Auditing is one of the elements of the AAA services concept of identification, authentication, authorizations, auditing, and accounting. Confidentiality is a core security element of the CIA Triad, but it is not dependent on auditing. Accessibility is the assurance that locations and systems are able to be used by the widest range of people/users possible. Redundancy is the implementation of alternatives, backup options, and recovery measures and methods to avoid single points of failure to ensure that downtime is minimized while maintaining availability.

29. A. The annualized loss expectancy (ALE) is computed as the product of the asset value (AV) times the exposure factor (EF) times the annualized rate of occurrence (ARO). This is the longer form of the formula ALE = SLE * ARO, since SLE = AV * EF. The other formulas displayed here do not accurately reflect this calculation, since they are not valid or typical risk formulas.

30. A. Identification of priorities is the first step of the business impact assessment process. Likelihood assessment is the third step or phase of BIA. Risk identification is the second step of BIA. Resource prioritization is the last step of BIA.

31. D. Natural events that can threaten organizations include earthquakes, floods, hurricanes, tornadoes, wildfires, and other acts of nature. Thus options A, B, and C are correct because they are natural and not human caused.

32. A. Hot sites provide backup facilities maintained in constant working order and fully capable of taking over business operations. Warm sites consist of preconfigured hardware and software to run the business,

neither of which possesses the vital business information. Cold sites are simply facilities designed with power and environmental support systems but no configured hardware, software, or services. Disaster recovery services can facilitate and implement any of these sites on behalf of a company.

33. D. The issue revealed by the audit report is that one account has a password that is older than the requirements allow for; thus, correcting the password maximum age security setting should resolve this. There is no information in regard to password length, lockout, or password reuse in the audit report, so these options are not of concern in this situation.

34. C. Written documents brought into court to prove the facts of a case are referred to as documentary evidence. Best evidence is a form of documentary evidence, but specifically it is the original document rather than a copy or description. Parol evidence is based on a rule stating that when an agreement between parties is put into written form, the written document is assumed to contain all the terms of the agreement, and no verbal agreements may modify the written agreement. Testimonial evidence consists of the testimony of a witness's experience, either verbal testimony in court or written testimony in a recorded deposition.

35. A, B. If your organization depends on custom-developed software or software products produced through outsourced code development, then the risks of that arrangement need to be evaluated and mitigated. First, the quality and security of the code needs to be assessed. Second, if the third-party development group goes out of business, can you continue to operate with the code as is? You may need to abandon the existing code to switch to a new development group. It is not true that third-party code development is always more expensive; it is often less expensive. A software escrow agreement (SEA) is not an issue that John would want to bring up as a reason to keep development in-house, since a SEA is a means to reduce the risk of a third-party developer group ceasing to exist.

36. D. `HTTPS://` is the correct prefix for the use of HTTP (Hypertext Transfer Protocol) over TLS (Transport Layer Security). This was the

same prefix when SSL (Secure Sockets Layer) was used to encrypt HTTP, but SSL has been deprecated. `SHTTP://` is for Secure HTTP, which was SSH, but SHTTP is also deprecated. `TLS://` is an invalid prefix. `FTPS://` is a valid prefix that can be used in some web browsers, and it uses TLS to encrypt the connection, but it is for securing FTP file exchange rather than web communications.

37. C. The CSO in this scenario is demonstrating the need to follow the security principle of change management. Change management usually involves extensive planning, testing, logging, auditing, and monitoring of activities related to security controls and mechanisms. This scenario is not describing a BCP event. A BCP event would involve the evaluation of threats to business processes and then the creation of response scenarios to address those issues. This scenario is not describing onboarding. Onboarding is the process of integrating a new element (such as an employee or device) into an existing system of security infrastructure. Although loosely similar to change management, onboarding focuses more on ensuring compliance with existing security policies by the new member, rather than testing updates for an existing member. Static analysis is used to evaluate source code as a part of a secure development environment. Static analysis may be used as an evaluation tool in change management, but it is a tool, not the principle of security referenced in this scenario.

38. D. The two main types of token devices are TOTP and HOTP. Time-based one-time password (TOTP) tokens or synchronous dynamic password tokens are devices or applications that generate passwords at fixed time intervals, such as every 60 seconds. Thus, TOTP produces new time-derived passwords on a specific time interval that can be used only a single time when attempting to authenticate. HMAC-based one-time password (HOTP) tokens or asynchronous dynamic password tokens are devices or applications that generate passwords not based on fixed time intervals but instead based on a nonrepeating one-way function, such as a hash or hash-based message authentication code (HMAC—a type of hash that uses a symmetric key in the hashing process) operation. HMAC is a hashing function, not a means to authenticate. Security Assertions Markup Language (SAML) is used to

create authentication federation (i.e., sharing) links; it is not itself a means to authenticate.

39. A. The most important security concern from this list of options in relation to a CSP is the data retention policy. The data retention policy defines what information or data is being collected by the CSP, how long it will be kept, how it is destroyed, why it is kept, and who can access it. The number of customers and what hardware is used are not significant security concerns in comparison to data retention. Whether the CSP offers MaaS, IDaaS, and SaaS is not as important as data retention, especially if these are not services your organization needs or wants. One of the keys to answering this question is to consider the range of CSP options, including software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service (IaaS), and the type of organizations that are technically CSP SaaS but that we don't often think of as such (examples include Facebook, Google, and Amazon). These organizations absolutely have access to customer/user data, and thus, their data retention policies are of utmost concern (at least compared to the other options provided).

40. B, C, D. Programmers need to adopt secure coding practices, which include using stored procedures, code signing, and server-side validation. A stored procedure is a subroutine or software module that can be called on or accessed by applications interacting with a relational database management system (RDBMS). Code signing is the activity of crafting a digital signature of a software program in order to confirm that it was not changed and who it is from. Server-side data validation is suited for protecting a system against input submitted by a malicious user. Using immutable systems is not a secure coding technique; instead, an immutable system is a server or software product that, once configured and deployed, is never altered in place. File size optimization may be efficient but is not necessarily a secure coding technique. Using third-party software libraries may reduce workload to minimize the amount of new code to author, but third-party software libraries are a risk because they can introduce vulnerabilities, especially when closed source libraries are used. Thus, use of third-party software libraries is not a secure coding technique

unless the security posture of the externally sourced code is verified, which was not mentioned as an answer option.

Chapter 1
Security Governance Through Principles and Policies

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 1.0: Security and Risk Management**
  - 1.2 Understand and apply security concepts
    - 1.2.1 Confidentiality, integrity, and availability, authenticity, and nonrepudiation (5 Pillars of Information Security)
  - 1.3 Evaluate and apply security governance principles
    - 1.3.1 Alignment of the security function to business strategy, goals, mission, and objectives
    - 1.3.2 Organizational processes (e.g., acquisitions, divestitures, governance committees)
    - 1.3.3 Organizational roles and responsibilities
    - 1.3.4 Security control frameworks (e.g., International Organization for Standardization (ISO), National Institute of Standards and Technology (NIST), Control Objectives for Information and Related Technology (COBIT), Sherwood Applied Business Security Architecture (SABSA), Payment Card Industry (PCI), Federal Risk and Authorization Management Program (FedRAMP))
    - 1.3.5 Due care/due diligence
  - 1.6 Develop, document, and implement security policy, standards, procedures, and guidelines
  - 1.10 Understand and apply threat modeling concepts and methodologies
  - 1.11 Apply supply chain risk management (SCRM) concepts
    - 1.11.1 Risks associated with the acquisition of products and services from suppliers and providers (e.g., product tampering, counterfeits, implants)
    - 1.11.2 Risk mitigations (e.g., third-party assessment and monitoring, minimum security requirements, service level

requirements, silicon root of trust, physically unclonable function, software bill of materials)

- ✓ **Domain 3.0 Security Architecture and Engineering**
  - 3.1 Research, implement, and manage engineering processes using secure design principles
    - 3.1.1 Threat modeling
    - 3.1.3 Defense in depth

The Security and Risk Management domain encompasses many of the foundational elements of security solutions. Additional elements of this domain are discussed in various chapters:

- Chapter 2, “Personnel Security and Risk Management Concepts”
- Chapter 3, “Business Continuity Planning”
- Chapter 4, “Laws, Regulations, and Compliance”
- Chapter 19, “Investigations and Ethics”

Please review all these chapters to have a complete perspective on the topics of this domain.

Security 101

We often hear how important security is, but we don't always understand why. Security is essential because it helps to ensure that an organization can continue to exist and operate despite any attempts to steal its data or compromise its physical or logical elements. Security is an element of business management rather than only an IT concern. Furthermore, IT and security are different. *Information technology (IT)* or even *information systems (IS)* is the hardware and software that support the operations or functions of a business. Security is the business management tool that ensures the reliable and protected operation of IT/IS. Security exists to support the organization's objectives, mission, and goals.

Generally, a security framework that provides a starting point for implementing security should be adopted. Once security is initiated, fine-tuning that security is accomplished through continuous evaluation and stress testing. There are three common types of security evaluation: risk assessment, vulnerability assessment, and penetration testing (these are covered in detail in Chapter 2 and Chapter 15, "Security Assessment and Testing"). *Risk assessment* is identifying assets, threats, and vulnerabilities to calculate risk. Once risk is understood, it is used to guide the improvement of the existing security infrastructure. *Vulnerability assessment* uses automated tools to locate known security weaknesses, which can be addressed by adding more defenses or adjusting the current protections. *Penetration testing* uses trusted teams to stress-test the security infrastructure to find issues that may not be discovered by the prior two means and to find those concerns before an adversary takes advantage of them.

Security should be cost-effective. Organizations do not have infinite budgets and, thus, must allocate their funds appropriately. Additionally, an organizational budget includes a percentage of monies dedicated to security, just as most other business tasks and processes require capital, not to mention payments to employees, insurance, retirement, etc. You should select security controls that provide the most significant protection for the lowest resource cost.

Security should be legally defensible. The laws of your jurisdiction are the backstop of organizational security. When someone intrudes into your environment and breaches security, especially when such activities are illegal, prosecution in court may be the only available response for compensation or closure. Also, many decisions made by an organization will have legal liability issues. If required to defend a security action in the courtroom, legally supported security will go a long way toward protecting your organization from facing significant fines, penalties, or charges of negligence.

Security is a journey, not a finish line. It is not a process that will ever be concluded. It is impossible to fully secure something because security issues are always changing. Our deployed technology is changing with the passage of time, by users' activities, and by adversaries discovering flaws and developing exploits. The defenses that were sufficient yesterday may

not be sufficient tomorrow. As new vulnerabilities are discovered, new means of attack are crafted, and new exploits are built, we have to respond by reassessing our security infrastructure and responding appropriately.

Understand and Apply Security Concepts

Security management concepts and principles are inherent elements in a security policy and solution deployment. They define the basic parameters needed for a secure environment. They also define the goals and objectives that both policy designers and system implementers must achieve to create a secure solution.

The 5 Pillars of Information Security are confidentiality, integrity, availability, authenticity, and nonrepudiation. The first three of these, namely confidentiality, integrity, and availability, are so commonly discussed as a group they have been labeled with their own phrase, the *CIA Triad*. The elements of the CIA Triad are often perceived as the primary goals and objectives of a security infrastructure (see Figure 1.1).

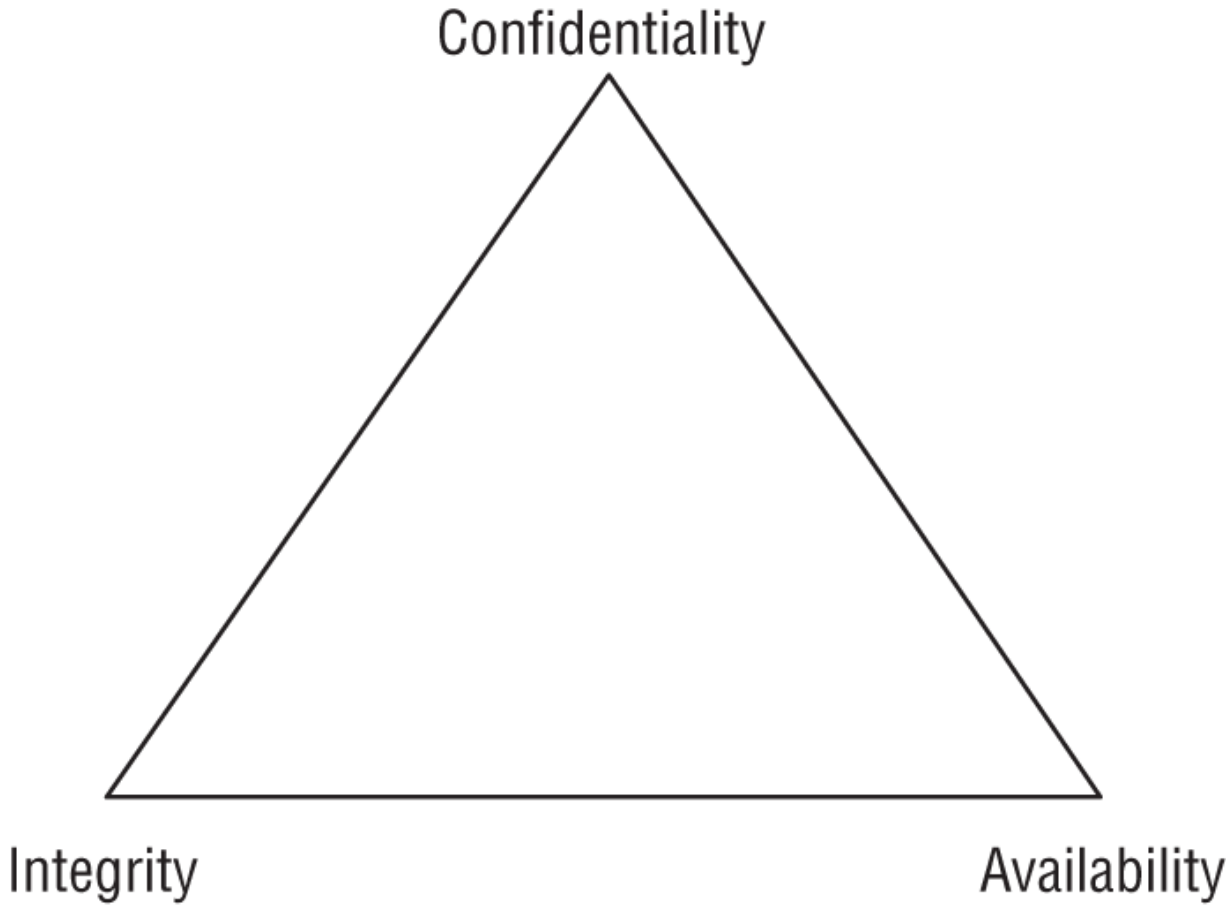


FIGURE 1.1 The CIA Triad

Security controls are typically evaluated on how well they address these three core information security tenets. Vulnerabilities and risks are also evaluated based on the threat they pose against one or more of the CIA Triad principles.

Confidentiality

The first principle of the CIA Triad is confidentiality. *Confidentiality* is the concept of the measures used to ensure the protection of the secrecy of data, objects, or resources. The goal of confidentiality protection is to prevent or minimize unauthorized access to data. Confidentiality protections prevent disclosure while protecting authorized access.

Violations of confidentiality are not limited to directed intentional attacks. Many instances of unauthorized disclosure of sensitive or confidential information are the result of human error, oversight, or ineptitude.

Confidentiality violations can result from the actions of an end user or a system administrator. They can also occur because of an oversight in a security policy or a misconfigured security control.

Numerous countermeasures can help ensure confidentiality against possible threats. These include encryption, network traffic padding, strict access control, rigorous authentication procedures, data classification, and extensive personnel training.

Concepts, conditions, and aspects of confidentiality include the following:

Sensitivity *Sensitivity* refers to the quality of information that could cause harm or damage if disclosed.

Discretion *Discretion* is a decision where an operator can influence or control disclosure to minimize harm or damage.

Criticality The level to which information is mission critical is its measure of *criticality*. The higher the level of criticality, the more likely the need to maintain the confidentiality of the information.

Concealment *Concealment* is the act of hiding or preventing disclosure. Concealment is often viewed as a means of cover, obfuscation, or distraction. A related concept to concealment is *security through obscurity*, which attempts to gain protection through hiding, silence, or secrecy.

Secrecy *Secrecy* is the act of keeping something a secret or preventing the disclosure of information.

Privacy *Privacy* refers to keeping information confidential that is personally identifiable or that might cause harm, embarrassment, or disgrace to someone if revealed.

Seclusion *Seclusion* involves storing something in an out-of-the-way location, likely with strict access controls.

Isolation *Isolation* is the act of keeping something separated from others.

Organizations should evaluate the nuances of confidentiality they wish to enforce. Tools and technology that implement one form of confidentiality

might not support or allow other forms.

Integrity

Integrity is the concept of protecting the reliability and correctness of data. Integrity protection prevents unauthorized alterations of data. Properly implemented integrity protection provides a means for authorized changes while protecting against intended and malicious unauthorized activities (such as viruses and intrusions) and mistakes made by authorized users (such as accidents or oversights).

Integrity can be examined from three perspectives:

- Preventing unauthorized subjects from making modifications
- Preventing authorized subjects from making unauthorized modifications, such as mistakes
- Maintaining the internal and external consistency of objects so that their data is a correct and true reflection of the real world and any relationship with any other object is valid, consistent, and verifiable

For integrity to be maintained on a system, controls must be in place to restrict access to data, objects, and resources. Maintaining and validating object integrity across storage, transport, and processing requires numerous variations of controls and oversight.

Numerous attacks focus on the violation of integrity. These include viruses, logic bombs, unauthorized access, errors in coding and applications, malicious modification, intentional replacement, and system backdoors.

Human error, oversight, or ineptitude accounts for many instances of unauthorized alteration of sensitive information. They can also occur because of an oversight in a security policy or a misconfigured security control.

Numerous countermeasures can ensure integrity against possible threats. These include strict access control, rigorous authentication procedures, intrusion detection systems, object/data encryption, hash verifications (see Chapter 6, “Cryptography and Symmetric Key Algorithms,” and Chapter 7,

“PKI and Cryptographic Applications”), interface restrictions, input/function checks, and extensive personnel training.

Concepts, conditions, and aspects of integrity include the following:

- *Accuracy*: Being correct and precise
- *Truthfulness*: Being a true reflection of reality
- *Validity*: Being factually or logically sound
- *Accountability*: Being responsible or obligated for actions and results
- *Responsibility*: Being in charge or having control over something or someone
- *Completeness*: Having all necessary components or parts
- *Comprehensiveness*: Being complete in scope; the full inclusion of all needed elements

Availability

Availability means authorized subjects are granted timely and uninterrupted access to objects. Often, availability protection controls support sufficient bandwidth and timeliness of processing as deemed necessary by the organization or situation. Availability includes efficient, uninterrupted access to objects and prevention of denial-of-service (DoS) attacks. Availability also implies that the supporting infrastructure—including network services, communications, and access control mechanisms—is functional and allows authorized users to gain access.

For availability to be maintained on a system, controls must be in place to ensure authorized access and an acceptable level of performance, to quickly handle interruptions, provide for redundancy, maintain reliable backups, and prevent data loss or destruction.

There are numerous threats to availability. These include device failure, software errors, and environmental issues (heat, static electricity, flooding, power loss, and so on). Some forms of attack focus on the violation of availability, including DoS attacks, object destruction, and communication interruptions.

Many availability breaches are caused by human error, oversight, or ineptitude. They can also occur because of an oversight in a security policy or a misconfigured security control.

Numerous countermeasures can ensure availability against possible threats. These include designing intermediary delivery systems properly, using access controls effectively, monitoring performance and network traffic, using firewalls and routers to prevent DoS attacks, implementing redundancy for critical systems, and maintaining and testing backup systems. Most security policies, as well as business continuity planning (BCP), focus on the use of fault tolerance features at the various levels of access/storage/security (that is, disk, server, or site) with the goal of eliminating single points of failure to maintain the availability of critical systems.

Availability depends on both integrity and confidentiality. Without integrity and confidentiality, availability cannot be maintained.

Concepts, conditions, and aspects of availability include the following:

- *Usability*: The state of being easy to use or learn or being able to be understood and controlled by a subject
- *Accessibility*: The assurance that the widest range of subjects can interact with a resource regardless of their capabilities or limitations
- *Timeliness*: Being prompt, on time, within a reasonable time frame, or providing a low-latency response

DAD, Overprotection, Authenticity, Nonrepudiation, and AAA Services

In addition to the CIA Triad, you need to consider a plethora of other security-related concepts and principles when designing a security policy and deploying a security solution. These include the DAD Triad, the risks of overprotection, authenticity, nonrepudiation, and AAA services.

One interesting security concept is the opposite of the CIA Triad, which is the DAD Triad. Disclosure, alteration, and destruction make up the *DAD Triad*. The DAD Triad represents the failures of security protections in the CIA Triad. It may be useful to recognize what to look for when a security

mechanism fails. Disclosure occurs when sensitive or confidential material is accessed by unauthorized entities. It is a violation of confidentiality. Alteration occurs when data is either maliciously or accidentally changed. It is a violation of integrity. Destruction occurs when a resource is damaged or made inaccessible to authorized users (technically, we usually call the latter denial of service [DoS]). Destruction is a violation of availability.

It may also be worthwhile to know that too much security can be its own problem. Overprotecting confidentiality can result in a restriction of availability. Overprotecting integrity can result in a restriction of availability. Overproviding availability can result in a loss of confidentiality and integrity.

Authenticity is the security concept that data is authentic or genuine and originates from its alleged source. This is related to integrity but more closely related to verifying that it is from a claimed origin. When data has authenticity, the recipient can have a high level of confidence that the data is from whom it claims to be and did not change in transit (or storage).

Nonrepudiation ensures that the subject of an activity or who caused an event cannot deny that the event occurred. Nonrepudiation prevents a subject from claiming not to have sent a message, not to have performed an action, or not to have been the cause of an event. It is made possible through identification, authentication, authorization, auditing, and accounting. Nonrepudiation can be established using digital certificates, session identifiers, transaction logs, and numerous other transactional and access control mechanisms. A system built without proper enforcement of nonrepudiation does not provide verification that a specific entity performed a certain action. Nonrepudiation is an essential part of accounting. A suspect cannot be held accountable if they can repudiate the claim against them.

AAA services are a core security mechanism of all security environments. The three As in this abbreviation refer to authentication, authorization, and accounting (or sometimes auditing). However, what is not as clear is that although there are three letters in the acronym, it actually refers to five elements: identification, authentication, authorization, auditing, and accounting. These five elements represent the following processes of security:

Identification *Identification* is claiming to be an identity when attempting to access a secured area or system.

Authentication *Authentication* is proving that you are that claimed identity.

Authorization *Authorization* defines the permissions (i.e., allow/grant and/or deny) of a resource and object access for a specific identity or subject.

Auditing *Auditing* is recording a log of the events and activities related to the system and subjects.

Accounting *Accounting* (aka *accountability*) is reviewing log files to check for compliance and violations in order to hold subjects accountable for their actions, especially violations of organizational security policy.

Although AAA is typically referenced in relation to authentication systems, it is actually a foundational concept for security. Missing any of these five elements can result in an incomplete security mechanism. The following sections discuss identification, authentication, authorization, auditing, and accounting (see Figure 1.2).

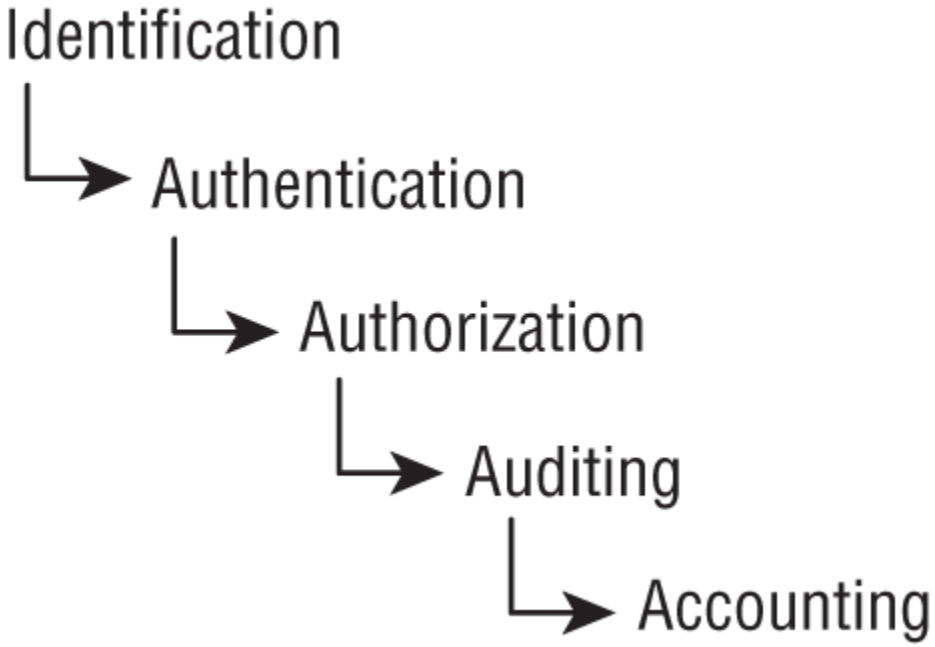


FIGURE 1.2 The five elements of AAA services

Identification

A subject must perform identification to start the process of authentication, authorization, and accounting (AAA). Providing an identity can involve typing in a username; swiping a smartcard; waving a proximity device; speaking a phrase; or positioning your face, hand, or finger for a camera or scanning device. Without an identity, a system has no way to correlate an authentication factor with the subject.

Once a subject has been identified (that is, once the subject's identity has been recognized and verified), the identity is accountable for any further actions by that subject. IT systems track activity by identities, not by the subjects themselves. A computer doesn't know one individual from another, but it does know that your user account is different from all other user accounts. Simply claiming an identity does not imply access, authorization, or authority. The identity must be proven before use is allowed or access is granted. That process is authentication.

Authentication

The process of verifying whether a claimed identity is valid is authentication. Authentication requires the subject to provide additional information that corresponds to the identity they are claiming. The most common form of authentication is using a password. Authentication verifies the identity of the subject by comparing one or more factors against the database of valid identities (that is, user accounts). The capability of the subject and system to maintain the secrecy of the authentication factors for identities directly reflects the level of security of that system.

Identification and authentication are often used together as a single two-step process. Providing an identity is the first step, and providing the authentication factors is the second step. Without both, a subject cannot gain access to a system—neither element alone is useful in terms of security. In some systems, it may seem as if you are providing only one element but gaining access, such as when keying in an ID code or a PIN. However, in these cases, either the identification is handled by another means, such as physical location, or authentication is assumed by your ability to access the system physically. Both identification and

authentication take place, but you might not be as aware of them as when you manually type in both a username and a password.

Each authentication technique or factor has its unique benefits and drawbacks. Thus, it is important to evaluate each mechanism in light of the environment in which it will be deployed to determine viability. We discuss authentication at length in Chapter 13, “Managing Identity and Authentication.”

Authorization

Once a subject is authenticated, access must be authorized. The process of authorization ensures that the requested activity or access to an object is possible, given the rights and privileges assigned to the authenticated identity. In most cases, the system evaluates the subject, the object, and the assigned permissions related to the intended activity. If the specific action is allowed, the subject is authorized. If the specific action is not allowed, the subject is not authorized.

Keep in mind that just because a subject has been identified and authenticated does not mean they have been authorized to perform any function or access all resources within the controlled environment. Identification and authentication are all-or-nothing aspects of access control. Authorization has a wide range of variations between all or nothing for each object within the environment. A user may be able to read a file but not delete it, print a document but not alter the print queue, or log on to a system but not access any resources. Authorization is discussed in Chapter 13.

Auditing

Auditing is the programmatic means by which a subject's actions are tracked and recorded to hold the subject accountable for their actions while authenticated on a system through the documentation or recording of subject activities. It is also the process of detecting unauthorized or abnormal activities on a system. Auditing is recording the activities of a subject and its objects and the activities of application and system functions. Log files provide an audit trail for re-creating the history of an event, intrusion, or system failure. Auditing is needed to detect malicious actions by subjects, attempted intrusions, and system failures. Auditing is

also necessary to reconstruct timelines of compromise events, provide evidence for prosecution, and produce problem reports and analyses. Auditing is usually a native feature of operating systems and most applications and services. Thus, configuring the system to record information about specific types of events is fairly straightforward.



Monitoring is part of what is needed for audits, and audit logs are part of a monitoring system, but the two terms have different meanings. Monitoring is a type of watching or oversight, whereas auditing is recording the information into a record or file. It is possible to monitor without auditing, but you can't audit without some form of monitoring.

Accounting

An organization's security policy can be properly enforced only if accounting is maintained. In other words, you can maintain security only if subjects are held accountable for their actions. Effective accounting relies on the capability to prove a subject's identity and track their activities. Accountability is established by linking an individual to the activities of an online identity through the security services and mechanisms of auditing, authorization, authentication, and identification. Thus, individual accountability is ultimately dependent on the strength of these processes. Without a strong authentication process, there is doubt that the person associated with a specific user account was the actual entity controlling that user account when the undesired action took place.

To have viable accountability, you must be able to support your security decisions and their implementation in a court of law. If you are unable to legally support your security efforts, then you will be unlikely to be able to hold an individual accountable for actions linked to a user account. With only a password as authentication, there is significant room for doubt. Passwords are the least secure form of authentication, with dozens of different methods available to compromise them. However, with the use of multifactor authentication (MFA), such as a password, smartcard, and fingerprint scan in combination, there is very little possibility that any other

individual could have compromised the authentication process in order to impersonate the person responsible for the user account.

Protection Mechanisms

Another aspect of understanding and applying security controls is the concept of protection mechanisms or protection controls. Not all security controls must have them, but many controls offer their protection through the use of these mechanisms. Some common examples of these mechanisms are defense in depth, abstraction, data hiding, and using encryption.

Defense in Depth

Defense in depth, also known as *layering*, is the use of multiple controls in a series. No one control can protect against all possible threats. Using a multilayered solution allows for numerous different controls to guard against whatever threats come to pass. When security solutions are designed in layers, a single failed control should not result in the exposure of systems or data.

Using layers in a series rather than in parallel is important. Performing security restrictions in a series means linearly enforcing one after the other. Only through a series configuration will each attack be scanned, evaluated, or mitigated by every security control. In a series configuration, failure of a single security control does not render the entire solution ineffective. If security controls were implemented in parallel, a threat could pass through a single checkpoint that did not address its particular malicious activity.

Serial configurations are very narrow but deep, whereas parallel configurations are very wide but shallow. Parallel systems are useful in distributed computing applications, but parallelism is not often a useful concept in the realm of security.

Within the context of defense in depth, the terms levels, multilevel, and layers are often used. Additionally, there are numerous other terms that also relate to this concept, including classifications, zones, realms, compartments, silos, segmentations, lattice structures, and protection rings. You will see these terms used often throughout this book. When you see them, think about the concept of defense in depth in relation to the context of where the term is used.



Defense in breadth or *diversity of defense* is also an important related concept to defense in depth. It can be problematic if elements of several security layers are from the same vendor or share common code, since a vulnerability could affect numerous layers simultaneously. Using a range of security products from varied vendors significantly reduces or avoids the risk of a single exploit compromising several layers at once.

Abstraction

Abstraction is used for efficiency. Similar elements are put into groups, classes, or roles that are collectively assigned security controls, restrictions, or permissions. Abstraction simplifies security by enabling you to assign security controls to a group of objects collected by type or function. Thus, the concept of abstraction is used when classifying objects or assigning roles to subjects.

Abstraction is one of the fundamental principles behind the field known as object-oriented programming (OOP). In OOP, the unknown environment doctrine states that users of an object (or operating system component) don't necessarily need to know the details of how the object works; they just need to know the proper syntax for using the object and the type of data that will be returned as a result (that is, how to send input and receive output). This is very much what's involved in mediated access to data or services, such as when user-mode applications use system calls to request administrator-mode services or data (and such mediated access requests may be granted or denied depending on the requester's credentials and permissions) rather than obtaining direct, unmediated access. (See the “Protection Rings” section of Chapter 9, “Security Vulnerabilities, Threats, and Countermeasures,” for more on the topic of mediated access.)

Another way in which abstraction applies to security is the introduction of object groups, sometimes called classes, where access controls and operation rights are assigned to groups of objects rather than on a per-object basis. This approach allows security administrators to define and name

groups easily (the names are often related to job roles or responsibilities) and helps make the administration of rights and privileges easier (when you add an object to a class, you confer rights and privileges rather than having to manage rights and privileges for each object separately).

Data Hiding

Data hiding is preventing data from being discovered or accessed by a subject by positioning the data in a logical storage compartment that is not accessible to nor seen by the subject. This means the subject cannot see or access the data, not just that it is unseen. Data hiding includes keeping a database from being accessed by unauthorized visitors and restricting a subject at a lower classification level from accessing data at a higher classification level. Preventing an application from accessing hardware directly is also a form of data hiding. Data hiding is often a key element in security controls as well as in programming. Steganography is an example of data hiding (see Chapter 7).

Data hiding is a vital characteristic in multilevel secure systems. It ensures that data existing at one level of security is not visible to processes running at different security levels. From a security perspective, data hiding relies on placing objects in security containers different from those that subjects occupy to hide object details from those without the need to know about them or the means to access them.

The term *security through obscurity* may seem relevant here. However, that concept is different. Data hiding is intentionally positioning data so that it is not viewable or accessible to an unauthorized subject, whereas security through obscurity is the idea of not informing a subject about an object being present and thus hoping that the subject will not discover the object. In other words, in security through obscurity, the subject could access the data if they find it. It is digital hide and seek. Security through obscurity does not actually implement any form of protection. It is instead an attempt to hope something important is not discovered by keeping knowledge of it a secret. An example of security though obscurity is when a programmer is aware of a flaw in their software code, but they release the product anyway hoping that no one discovers the issue and exploits it.

Encryption

Encryption is the science of hiding the meaning or intent of a communication from unintended recipients. Encryption can take many forms and should be applied to every type of electronic communication and storage. Encryption is discussed at length in Chapters 6 and 7.

Security Boundaries

A *security boundary* is the line of intersection between areas, subnets, or environments with different security requirements or needs. A security boundary exists between high-security and low-security areas, such as between a LAN (local area network) and the Internet. Recognizing the security boundaries on your network and in the physical world is essential to establishing reliable security barriers. Once you identify a security boundary, you must deploy mechanisms to control the flow of information across that boundary.

Divisions between security areas can take many forms. For example, objects may have different classifications. Each classification defines which subjects can perform functions on which objects. The distinction between classifications is a security boundary.

Security boundaries also exist between the physical environment and the logical environment. To provide logical security, you must provide security mechanisms different from those used to provide physical security. Both must be present to provide a complete security structure, and both must be addressed in a security policy. However, they are different and must be assessed as separate elements of a security solution.

Security boundaries, such as a perimeter between protected and unprotected areas, should always be clearly defined. In a security policy, it's important to state the point at which control ends or begins and to identify that point in both the physical and logical environments. Logical security boundaries are where electronic communications interface with devices or services for which your organization is legally responsible. In most cases, that interface is clearly marked, and unauthorized subjects are informed that they do not have access, and that attempts to gain access will result in prosecution.

The security perimeter in the physical environment often reflects the security perimeter of the logical environment. In most cases, the area for

which the organization is legally responsible determines the reach of a security policy in the physical realm. This can be the walls of an office, the walls of a building, or the fence around a campus. In secured environments, warning signs are posted indicating that unauthorized access is prohibited and that attempts to gain access will be thwarted and result in prosecution.

When transforming a security policy into actual controls, you must consider each environment and security boundary separately. Simply deduce what available security mechanisms would provide the most reasonable, cost-effective, and efficient solution for a specific environment and situation. However, all security mechanisms must be weighed against the value of the objects they are to protect. Deploying countermeasures that cost more than the value of the protected objects is unwarranted.

Evaluate and Apply Security Governance Principles

Security governance is the collection of practices related to supporting, evaluating, defining, and directing an organization's security efforts. Optimally, security governance is performed by a board of directors or governance committee, but smaller organizations may have the chief executive officer (CEO) or chief information security officer (CISO) perform the activities of security governance. Security governance seeks to compare the security processes and infrastructure used within the organization with knowledge and insight obtained from external sources. This is why a board of directors often consists of people from various backgrounds and industries. The board members can bring their varied experience and wisdom to guide the improvement of the organization they oversee.

Security governance principles are closely related to and often intertwined with corporate and IT governance. The goals of these three governance agendas are often the same or interrelated, such as maintaining business processes while striving toward growth and resiliency.

Some aspects of governance are imposed on organizations due to legislative and regulatory compliance needs, whereas industry guidelines or license requirements impose others. All forms of governance, including security

governance, must be assessed and verified from time to time. Various requirements for auditing and validation may be present due to government regulations or industry best practices. This is especially problematic when laws in different countries differ or, in fact, conflict. The organization as a whole should be given the direction, guidance, and tools to provide sufficient oversight and management to address threats and risks, with a focus on eliminating downtime and keeping potential loss or damage to a minimum.

As you can tell, the definitions of security governance are often rather stilted and high-level. Ultimately, security governance is the implementation of a security solution and a management method that are tightly interconnected. Security governance directly oversees and gets involved in all levels of security. Security is not and should not be treated as an IT issue only. Instead, security affects every aspect of an organization. Security is a business operations issue. Security is an organizational process, not just something the IT geeks do behind the scenes. Using the term *security governance* is an attempt to emphasize this point by indicating that security needs to be managed and governed throughout the organization, not just in the IT department.

There are numerous security frameworks and governance guidelines, including the National Institute of Standards and Technology (NIST) SP 800-53 and NIST SP 800-100. Although the NIST guidance is focused on government and military use, it can be adopted and adapted by other types of organizations as well. Many organizations adopt security frameworks in an effort to standardize and organize what can become a complex and bewilderingly messy activity, namely, attempting to implement reasonable security governance.

Third-Party Governance

Third-party governance is the system of external entity oversight that law, regulation, industry standards, contractual obligation, or licensing requirements may mandate. The actual method of governance may vary, but it generally involves an outside investigator or auditor. A governing body might designate these auditors or might be consultants hired by the target organization.

Another aspect of third-party governance is the application of security oversight to third parties that your organization relies on. Many organizations choose to outsource various aspects of their business operations. Outsourced operations can include security guards, maintenance, technical support, and accounting services. These parties must comply with the primary organization's security stance. Otherwise, they present additional risks and vulnerabilities to the primary organization.

Third-party governance focuses on verifying compliance with stated security objectives, requirements, regulations, and contractual obligations. On-site assessments can provide firsthand exposure to the security mechanisms employed at a location. Those performing on-site assessments or audits must follow auditing protocols (such as Control Objectives for Information and Related Technologies [COBIT]) and have a specific checklist of requirements to investigate.

In the auditing and assessment process, both the target and the governing body should participate in full and open document exchange and review. An organization needs to know the full details of all requirements it must comply with. The organization should submit security policy and self-assessment reports back to the governing body. This open document exchange ensures that all parties involved agree about all the issues of concern. It reduces the chances of unknown requirements or unrealistic expectations. Document exchange does not end with the transmission of paperwork or electronic files. Instead, it leads to the process of documentation review.

See Chapter 12, “Secure Communications and Network Attacks,” for a discussion of third-party connectivity.

Documentation Review

Documentation review is the process of reading the exchanged materials and verifying them against standards and expectations. The documentation review is typically performed before any on-site inspection takes place. If the exchanged documentation is sufficient and meets expectations (or at least requirements), then an on-site review will be able to focus on compliance with the stated documentation. However, if the documentation is incomplete, inaccurate, or otherwise insufficient, the on-site review is

postponed until the documentation can be updated and corrected. This step is important because if the documentation is not in compliance, the location will likely not be in compliance either.

In many situations, especially those related to government or military agencies or contractors, failing to provide sufficient documentation to meet requirements of third-party governance can result in a loss of or a voiding of *authorization to operate (ATO)*. Complete and sufficient documentation can often maintain existing ATOs or provide a temporary ATO (TATO). However, once an ATO is lost or revoked, complete documentation and on-site review showing full compliance are usually necessary to reestablish the ATO.

A portion of the documentation review is the logical and practical investigation of the business processes and organizational policies in light of standards, frameworks, and contractual obligations. This review ensures that the stated and implemented business tasks, systems, and methodologies are practical, efficient, and cost-effective, and most of all (at least in relation to security governance) that they support the goal of security through the reduction of vulnerabilities and the avoidance, reduction, or mitigation of risk. Managing, assessing, and addressing risk are all methods and techniques involved in performing process/policy review.

Manage the Security Function

The *security function* is the aspect of operating a business that focuses on the task of evaluating and improving security over time. To manage the security function, an organization must implement proper and sufficient security governance.

The act of performing a risk assessment to drive the security policy is the clearest and most direct example of management of the security function. The process of risk assessment is discussed in Chapter 2.

Security must be measurable. Measurable security means that the various aspects of the security mechanisms function, provide a clear benefit, and have one or more metrics that can be recorded and analyzed. Similar to performance metrics, security metrics are measurements of performance, function, operation, action, and so on as related to the operation of a

security feature. When a countermeasure or safeguard is implemented, security metrics should show a reduction in unwanted occurrences or an increase in the detection of attempts. The act of measuring and evaluating security metrics is the practice of assessing the completeness and effectiveness of the security program. This should also include measuring it against common security guidelines and tracking the success of its controls. Tracking and assessing security metrics is part of effective security governance.

Managing the security function includes the development and implementation of information security strategies. Most of the content of this book, addresses the various aspects of the development and implementation of information security strategies.

Alignment of Security Function to Business Strategy, Goals, Mission, and Objectives

Security management planning ensures the proper creation, implementation, and enforcement of a *security policy*. Security management planning aligns the security functions to the strategy, goals, mission, and objectives of the organization. This includes designing and implementing security based on business cases, budget restrictions, or scarcity of resources. A *business case* is usually a documented argument or stated position in order to define a need to make a decision or take some form of action. To make a business case is to demonstrate a business-specific need to alter an existing process or choose an approach to a business task. A business case is often made to justify the start of a new project, especially a project related to security. Money and resources, such as people, technology, and space, are limited in most organizations. Due to resource limitations like these, the maximum benefit needs to be obtained from any endeavor.

A *top-down approach* is one of the most effective ways to tackle security management planning. Upper or senior management is responsible for initiating and defining policies for the organization. Security policies provide direction for all levels of the organization's hierarchy. Middle management's responsibility is to flesh out the security policy into standards, baselines, guidelines, and procedures. The operational managers or security professionals must then implement the configurations prescribed

in the security management documentation. Finally, the end users must comply with all the security policies of the organization.



The opposite of the top-down approach is the bottom-up approach. In a *bottom-up approach* environment, the IT staff makes security decisions directly without input from senior management. The bottom-up approach is rarely used in organizations and is considered problematic in the IT industry.

Security management is a responsibility of upper management, not of the IT staff, and is considered an issue of business operations rather than IT administration. The team or department responsible for security within an organization should be autonomous. The *information security (InfoSec) team* should be led by a designated *chief information security officer (CISO)* who reports directly to senior management, such as the chief information officer (CIO), the chief executive officer (CEO), or the board of directors. Placing the autonomy of the CISO and the CISO's team outside the typical hierarchical structure in an organization can improve security management across the entire organization. It also helps avoid cross-department and internal political issues. The term *chief security officer (CSO)* is sometimes used as an alternative to CISO, but in many organizations, the CSO position is a subposition under the CISO that focuses on physical security. Another potential term for the CISO is *information security officer (ISO)*, but this also can be used as a subposition under the CISO.



The *chief information officer (CIO)* focuses on ensuring information is used effectively to accomplish business objectives. The *chief technical officer (CTO)* focuses on ensuring that equipment and software work properly to support the business functions.

Elements of security management planning include defining security roles; prescribing how security will be managed, who will be responsible for

security, and how security will be tested for effectiveness; developing security policies; performing risk analysis; and requiring security education for employees. These efforts are guided through the development of management plans.

The best security plan is useless without one key factor: approval by *senior management*. Without senior management's approval of and commitment to the security policy, the policy will not succeed. It is the responsibility of the policy development team to educate senior management sufficiently so managers understand the risks, liabilities, and exposures that remain even after security measures prescribed in the policy are deployed. Developing and implementing a security policy is evidence of due diligence and due care on the part of senior management. If a company does not practice due diligence and due care, managers can be held liable for negligence and held accountable for both asset and financial losses.

A security management planning team should develop three types of plans, as shown in Figure 1.3:

Strategic Plan A *strategic plan* is a long-term plan that is fairly stable. It defines the organization's security purpose. It defines the security function and aligns it with the goals, mission, and objectives of the organization. It's useful for about five years if it is maintained and updated annually. The strategic plan also serves as the planning horizon. Long-term goals and visions for the future are discussed in a strategic plan. A strategic plan should include a risk assessment.

Tactical Plan The *tactical plan* is a midterm plan developed to provide more details on accomplishing the goals set forth in the strategic plan or can be crafted ad hoc based on unpredicted events. A tactical plan is typically useful for about a year and often prescribes and schedules the tasks necessary to accomplish organizational goals. Some examples of tactical plans are project plans, acquisition plans, hiring plans, budget plans, maintenance plans, support plans, and system development plans.

Operational Plan An *operational plan* is a short-term, highly detailed plan based on strategic and tactical plans. It is valid or useful only for a short time. Operational plans must be updated often (such as

monthly or quarterly) to retain compliance with tactical plans. Operational plans spell out how to accomplish the various goals of the organization. They include resource allotments, budgetary requirements, staffing assignments, scheduling, and step-by-step or implementation procedures. Operational plans include details on how the implementation processes are in compliance with the organization's security policy. Examples of operational plans are training plans, system deployment plans, and product design plans.

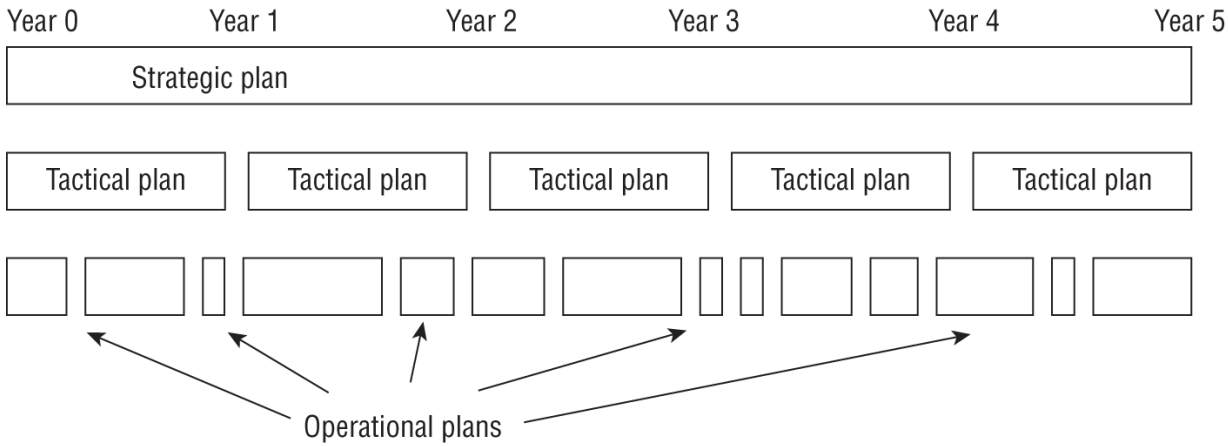


FIGURE 1.3 Strategic, tactical, and operational plan timeline comparison

Security is a continuous process. Thus, the activity of security management planning may have a definitive initiation point, but its tasks and work are never fully accomplished or complete. Effective security plans focus attention on specific and achievable objectives, anticipate change and potential problems, and serve as a basis for decision-making for the entire organization. Security documentation should be concrete, well-defined, and clearly stated. For a security plan to be effective, it must be developed, maintained, and actually used.

Organizational Processes

Security governance should address every aspect of an organization, including the organizational processes of acquisitions, divestitures, and governance committees. Acquisitions and mergers place an organization at an increased level of risk. Such risks include inappropriate information disclosure, data loss, downtime, or failure to achieve sufficient return on investment (ROI). In addition to all the typical business and financial

aspects of mergers and acquisitions, a healthy dose of security oversight and increased scrutiny is often essential to reduce the likelihood of losses during such a period of transformation.

Similarly, divestiture or any form of asset or employee reduction is another time period of increased risk and, thus, increased need for focused security governance. Assets need to be sanitized to prevent data leakage. Storage media should be removed and destroyed because media sanitization techniques do not guarantee against data remnant recovery. Employees released from duty need to be debriefed. This process is often called an exit interview. This process usually involves reviewing any nondisclosure agreements and any other binding contracts or agreements that will continue after employment has ceased.

When acquisitions and mergers are made without security considerations, the risks inherent in those obtained products remain throughout their deployment life span. Minimizing inherent threats in acquired elements will reduce security management costs and likely reduce security violations.

It is important to evaluate the risks associated with hardware, software, and services. Products and solutions that have resilient integrated security are often more expensive than those that fail to have a security foundation. However, this additional initial expense is often a much more cost-effective expenditure than addressing security deficiencies over the life of a poorly designed product. Thus, when considering the cost of a merger/acquisition, it is important to consider the total cost of ownership over the life of the product's deployment rather than just initial purchase and implementation.

Acquisitions do not relate exclusively to hardware and software. Outsourcing, contracting with suppliers, and engaging consultants are also elements of acquisition. Integrating security assessments when working with external entities is just as important as ensuring a product was designed with security in mind.

In many cases, ongoing security monitoring, management, and assessment may be required. This could be an industry best practice or a regulation. Such assessment and monitoring might be performed by the organization internally or may require the use of external auditors. When engaging third-party assessment and monitoring services, keep in mind that the external entity needs to show security-mindedness in their business operations. If an

external organization is unable to manage their own internal operations on a secure basis, how can they provide reliable security management functions for yours?

When evaluating a third party for your security integration, consider the following processes:

On-Site Assessment Visit the site of the organization to interview personnel and observe their operating habits.

Document Exchange and Review Investigate the means by which datasets and documentation are exchanged and the formal processes by which they perform assessments and reviews. This focuses on the means and processes.

Process/Policy Review Request copies of their security policies, processes/procedures, and documentation of incidents and responses for review. This focuses on the written policies.

Third-Party Audit Having an independent third-party auditor, as defined by the American Institute of Certified Public Accountants (AICPA), can provide an unbiased review of an entity's security infrastructure, based on System and Organization Controls (SOC) reports. See Chapter 15 for details on SOC reports.

For all acquisitions, establish minimum security requirements. These should be modeled after your existing security policy. The security requirements for new hardware, software, or services should always meet or exceed the security of your existing infrastructure. When working with an external service, be sure to review any service-level agreement (SLA) to ensure that security is a prescribed component of the contracted services. When that external provider is crafting software or providing a service (such as a cloud provider), then a service-level requirement (SLR) may need to be defined.

An SLR is a statement of the expectations of service and performance from the product or service of a vendor. Often, an SLR is provided by the customer/client prior to the establishment of the SLA (which should incorporate the elements of the SLR if the vendor expects the customer to sign the agreement).

Two additional examples of organizational processes that are essential to strong security governance are change control/change management (see Chapter 16, “Managing Security Operations”) and data classification (see Chapter 5, “Protecting Security of Assets”).

Organizational Roles and Responsibilities

A *security role* is an individual's part in the overall scheme of security implementation and administration within an organization. Security roles are not necessarily prescribed in job descriptions because they are not always distinct or static. Familiarity with security roles will help in establishing a communications and support structure within an organization. This structure will enable the deployment and enforcement of the security policy. This section focuses on general-purpose security roles for managing an overall security infrastructure. See Chapter 5 for roles related specifically to data management.

The following are the common security roles present in a typical secured environment:

Senior Manager The organizational owner (*senior manager*) role is assigned to the person who is ultimately responsible for the security maintained by an organization and who should be most concerned about the protection of its assets. The senior manager must sign off on all security policy issues. There is no effective security policy if the senior management does not authorize and support it. The senior manager is the person who will be held liable for the overall success or failure of a security solution and is responsible for exercising due diligence and due care in establishing security for an organization. Even though senior managers are ultimately responsible for security, they rarely implement security solutions. In most cases, that responsibility is delegated to security professionals within the organization.

Security Professional The *security professional*, *information security (InfoSec) officer*, or *cyber incident response team (CIRT)* role is assigned to a trained and experienced network, systems, and security engineer who is responsible for following the directives mandated by senior management. The security professional has the functional

responsibility for security, including writing the security policy and implementing it. The role of a security professional may be labeled as an IS/IT role, but its focus is on protection more than function. The security professional role is often filled by a team that is responsible for designing and implementing security solutions based on the approved security policy. Security professionals are not decision makers; they are implementers. All decisions must be left to the senior manager.

Asset Owner The *asset owner* role is assigned to the person who is responsible for classifying information for placement and protection within the security solution. The asset owner is typically a high-level manager who is ultimately responsible for asset protection. However, the asset owner usually delegates the responsibility of the actual data management tasks to a custodian.

Custodian The *custodian* role is assigned to the person who is responsible for the tasks of implementing the prescribed protection defined by the security policy and senior management. The custodian performs all activities necessary to provide adequate protection for the CIA Triad (confidentiality, integrity, and availability) of data and to fulfill the requirements and responsibilities delegated by upper management. These activities can include performing and testing backups, validating data integrity, deploying security solutions, and managing data storage based on classification.

User The *user* (*end user* or *operator*) role is assigned to any person who has access to the secured system. A user's access is tied to their work tasks and is limited so that they have only enough access to perform the tasks necessary for their job position (the principle of least privilege). Users are responsible for understanding and upholding the security policy of an organization by following prescribed operational procedures and operating within defined security parameters.

Auditor An *auditor* is responsible for reviewing and verifying that the security policy is properly implemented and the derived security solutions are adequate. The auditor produces compliance and effectiveness reports that are reviewed by the senior manager. Issues

discovered through these reports are transformed into new directives assigned by the senior manager to security professionals or custodians.

All of these roles serve an important function within a secured environment. They are useful for identifying liability and responsibility as well as for identifying the hierarchical management and delegation scheme.

Security Control Frameworks

One of the first and most important security planning steps is to consider the overall *security control framework* or structure of the security solution desired by the organization. Security control frameworks, often referred to as security frameworks or cybersecurity frameworks, are structured sets of guidelines, standards, best practices, and controls designed to help organizations effectively manage and enhance their information security and cybersecurity posture. These frameworks provide a systematic and comprehensive approach to identifying, implementing, and monitoring security controls and measures to protect an organization's data, systems, networks, and sensitive information. There are numerous organizations that produce and maintain security control frameworks.

International Organization for Standardization (ISO)

The *International Organization for Standardization (ISO)* is a worldwide standards-setting group of representatives from various national standards organizations. ISO defines standards for industrial and commercial equipment, software, protocols, and management, among others. It issues six main products: International Standards, Technical Reports, Technical Specifications, Publicly Available Specifications, Technical Corrigenda, and Guides. ISO standards are widely accepted across many industries and have even been adopted as requirements or laws by various governments. For more information on ISO, please visit `ISO.org`. Specifically, the ISO/IEC 27000 family group is an international security standard that can be the basis for implementing organizational security and related management practices. (The International Electrotechnical Commission [IEC] is an international standards organization that prepares and publishes international standards for all electrical, electronic, and related

technologies. ISO and IEC often work together in establishing worldwide standards.)

National Institute of Standards and Technology (NIST)

The *National Institute of Standards and Technology (NIST)* is a U.S. federal agency that operates under the umbrella of the U.S. Department of Commerce. NIST's mission is to promote and maintain measurement standards, as well as advance technology and innovation. It plays a pivotal role in developing and promoting standards and best practices, especially in the areas of science and technology. NIST is responsible for establishing and maintaining various standards, including those related to computer security. One of the most well-known publications from NIST is the NIST Special Publication (SP) 800-53 “Security and Privacy Controls for Information Systems and Organizations,” which outlines a comprehensive set of security controls and guidelines for information systems used by U.S. federal agencies. This publication is widely used as a reference for implementing information security practices, especially within government organizations and in various sectors that handle sensitive data. NIST also established the *Risk Management Framework (RMF)* and *Cybersecurity Framework (CSF)* (both covered in Chapter 2).



The *Center for Internet Security (CIS)* provides OS, application, and hardware security configuration guides at `www.cisecurity.org/cis-benchmarks`. These are not considered security control frameworks, but they are often used in conjunction with them.

Control Objectives for Information and Related Technologies (COBIT)

Control Objectives for Information and Related Technologies (COBIT) is a documented set of best IT security practices crafted by ISACA. (Previously spelled out as Information Systems Audit and Control Association, however the organization no longer uses the full name only the acronym as their name.) It prescribes goals and requirements for security controls and

encourages the mapping of IT security ideals to business objectives. COBIT is based on six key principles for the governance and management of enterprise IT:

- Provide Stakeholder Value
- Holistic Approach
- Dynamic Governance System
- Governance Distinct from Management
- Tailored to Enterprise Needs
- End-to-End Governance System

COBIT is used not only to plan the IT security of an organization but also as a guideline for auditors (www.isaca.org/resources/cobit).

Sherwood Applied Business Security Architecture (SABSA)

Sherwood Applied Business Security Architecture (SABSA) is a framework and methodology for developing risk-driven enterprise security and information assurance architectures. It is known for its holistic and business-focused approach to security architecture.

Key aspects of SABSA include:

- *Risk-focused:* SABSA places a strong emphasis on identifying and managing security risks within the context of the business. It aims to align security measures with an organization's specific risks and objectives.
- *Business-driven:* SABSA promotes the idea that security should be integrated into an organization's business processes and goals. It helps organizations understand how security supports and enables business activities.
- *Layered approach:* SABSA uses a layered architectural model to address security concerns at various levels, from strategic planning down to operational security controls. These layers include the business context, information domain, systems, technology, and physical security.

- *Framework and methodology:* SABSA provides a structured framework for developing security architectures and a comprehensive methodology for designing, implementing, and managing security solutions.
- *Certification:* SABSA offers a certification program that allows security professionals to become certified in SABSA methodologies and practices.

SABSA is often used in large organizations and enterprises where a robust and business-aligned approach to security architecture is required. It helps organizations create security architectures that are not only effective in addressing security risks but that are also closely tied to the organization's strategic goals and objectives.

Payment Card Industry Data Security Standard (PCI DSS)

Payment Card Industry Data Security Standard (PCI DSS) is a set of security standards and requirements designed to ensure the protection of sensitive credit card and debit card information. PCI DSS was established by major credit card companies to enhance the security of payment card transactions and to protect cardholder data.

Key components of PCI DSS include:

- *Data security:* PCI DSS sets guidelines for the secure handling of payment card data, including cardholder names, primary account numbers (PANs), expiration dates, and card verification values (CVVs).
- *Network security:* PCI DSS mandates the implementation of robust network security practices, including firewalls, encryption, and access controls, to protect cardholder data during transmission.
- *Access control:* PCI DSS requires organizations to restrict access to cardholder data on a need-to-know basis. Access should be limited to authorized personnel only.
- *Regular monitoring and testing:* Continuous monitoring and regular security testing are necessary to identify and address vulnerabilities in systems and applications that process cardholder data.

- *Information security policies:* Organizations must develop and maintain comprehensive security policies and procedures to guide employees in secure practices related to payment card data.
- *Vulnerability management:* This involves the timely identification and remediation of security vulnerabilities to protect against potential threats.
- *Physical security:* PCI DSS also includes requirements for the physical security of cardholder data, including restricted access to servers, storage, and point-of-sale (POS) devices.
- *Incident response:* Having an incident response plan is essential to respond promptly and effectively to security incidents and data breaches.
- *Compliance audits:* Organizations that handle payment card data are required to undergo regular PCI DSS compliance audits. These audits are conducted by independent qualified security assessors (QSAs) or internal security assessors (ISAs) who are certified to assess compliance. The goal of these audits is to determine whether the organization complies with the PCI DSS requirements.

Compliance with PCI DSS is mandatory for any entity that processes payment card transactions, including merchants, service providers, and financial institutions. Failure to comply with PCI DSS can lead to fines, loss of card processing privileges, and reputational damage.

PCI DSS is typically updated periodically to address evolving security threats and technology changes, so organizations subject to its requirements must stay current with the latest version and maintain compliance.

Federal Risk and Authorization Management Program (FedRAMP)

The Federal Risk and Authorization Management Program (FedRAMP) is a U.S. government-wide program designed to standardize the security assessment, authorization, and continuous monitoring processes for cloud products and services used by federal agencies. Its primary goal is to ensure that cloud services meet stringent security requirements and can be used by

U.S. government organizations to process, store, and transmit sensitive and classified information.

Key elements of FedRAMP include:

- *Security standardization:* FedRAMP establishes a set of security controls, baselines, and requirements that cloud service providers (CSPs) must adhere to when offering cloud solutions to federal agencies. These requirements are based on NIST SP 800-53, which outlines security controls for federal information systems.
- *Authorization process:* CSPs seeking to offer their cloud services to federal agencies must go through a rigorous authorization process. This process involves a comprehensive security assessment, documentation, and evaluation by a third-party assessment organization.
- *Continuous monitoring:* Once authorized, CSPs are required to maintain ongoing security monitoring and reporting to ensure that their services continue to meet the established security standards and remain secure throughout their life cycle.
- *Reuse of authorizations:* FedRAMP encourages the reuse of security authorizations across federal agencies. When a CSP receives a FedRAMP authorization, other agencies can reuse that authorization rather than conducting their own assessments, streamlining the procurement process.
- *Collaboration:* FedRAMP fosters collaboration between federal agencies, CSPs, and third-party assessors. It aims to create a more efficient and standardized approach to cloud security while reducing duplication of effort.
- *Three impact levels:* FedRAMP has three impact levels (low, moderate, and high) to account for different levels of sensitivity and classification of federal data. The required security controls and assessment processes vary based on the impact level.
- *Compliance framework:* FedRAMP provides a framework that ensures the security of cloud services and helps federal agencies make informed decisions when selecting and implementing cloud solutions.

FedRAMP plays a critical role in securing federal government data and systems by ensuring that cloud services meet rigorous security standards. It also simplifies the procurement and adoption of cloud solutions for federal agencies by providing a standardized and transparent process for assessing and authorizing cloud services for government use.

Information Technology Infrastructure Library (ITIL)

Information Technology Infrastructure Library (ITIL) (`itlibrary.org`), initially crafted by the British government, is a set of recommended best practices for the optimization of IT services to support business growth, transformation, and change. ITIL focuses on understanding how IT and security need to be integrated with and aligned to the objectives of an organization. ITIL and operational processes are often used as a starting point for the crafting of a customized IT security solution within an established infrastructure.



There are many specialized security control frameworks, such as the SWIFT security control framework. The SWIFT security control framework refers to the set of security measures, guidelines, and best practices established by SWIFT (Society for Worldwide Interbank Financial Telecommunication) to ensure the security, trust, and integrity of financial messaging and transactions within the global financial network.

Due Diligence and Due Care

Why is planning to plan security so important? One reason is the requirement for *due diligence* and *due care*. Due diligence is establishing a plan, policy, and process to protect the interests of an organization. Due care is practicing the individual activities that maintain the due diligence effort. For example, due diligence is developing a formalized security structure containing a security policy, standards, baselines, guidelines, and procedures. Due care is the continued application of this security structure onto the IT infrastructure of an organization. Operational security is the

ongoing maintenance of continued due diligence and due care by all responsible parties within an organization. Due diligence is knowing what should be done and planning for it; due care is doing the right action at the right time.

Due diligence is also used as a detection mechanism, referred to as “do detect.” The idea is that while due care (aka “do correct”) activities are being performed, due diligence is used to oversee and confirm that the proper actions are being taken and that a record of such actions is being created. This is an extension of the planning concept based on continued oversight while performing tasks properly. Additionally, as conditions change (whether new threats, risks, or business tasks), due diligence is the adjustment of prior plans to take into account new conditions and concerns. Once updated, due care implements the revised plans.

In today's business environment, prudence is mandatory. Showing due diligence and due care is the only way to disprove negligence in an occurrence of loss. Senior management must show due care and due diligence to reduce their culpability and liability when a loss occurs.

Security Policy, Standards, Procedures, and Guidelines

For most organizations, maintaining security is an essential part of ongoing business. To reduce the likelihood of a security failure, implementing security has been formalized with a hierarchical organization of documentation. Developing and implementing documented security policies, standards, procedures, and guidelines produces a solid and reliable security infrastructure.

Security Policies

The top tier of the formalization is known as a security policy. A *security policy* is a document that defines the scope of security needed by the organization and discusses the assets that require protection and the extent to which security solutions should go to provide the necessary protection. The security policy is an overview or generalization of an organization's security needs. It defines the strategic security objectives, vision, and goals

and outlines the security framework of an organization. The security policy is used to assign responsibilities, define roles, specify audit requirements, outline enforcement processes, indicate compliance requirements, and define acceptable risk levels. This document is often used as proof that senior management has exercised due diligence in protecting itself against intrusion, attack, and disaster. Security policies are compulsory.

Many organizations employ several types of security policies to define or outline their overall security strategy. An organizational security policy focuses on issues relevant to every aspect of an organization. An issue-specific security policy focuses on a specific network service, department, function, or other aspect that is distinct from the organization as a whole. A system-specific security policy focuses on individual systems or types of systems and prescribes approved hardware and software, outlines methods for locking down a system, and even mandates firewall or other specific security controls.

From the security policies flow many other documents or sub-elements necessary for a complete security solution. Policies are broad overviews, whereas standards, baselines, guidelines, and procedures include more specific, detailed information on the actual security solution. Standards are the next level below security policies.

Security Standards, Baselines, and Guidelines

Once the main security policies are set, the remaining security documentation can be crafted under the guidance of those policies. *Standards* define compulsory requirements for the homogenous use of hardware, software, technology, and security controls. They provide a course of action by which technology and procedures are uniformly implemented throughout an organization.

A *baseline* defines a minimum level of security that every system throughout the organization must meet. A baseline is a more operationally focused form of a standard. All systems not complying with the baseline should be taken out of production until they can be brought up to the baseline. The baseline establishes a common foundational secure state on which all additional and more stringent security measures can be built.

Baselines are usually system-specific and often refer to an industry or government standard.

Guidelines are the next element of the formalized security policy structure. A *guideline* offers recommendations on how standards and baselines are implemented and serves as an operational guide for both security professionals and users. Guidelines are flexible, so they can be customized for each unique system or condition and can be used in the creation of new procedures. They state which security mechanisms should be deployed instead of prescribing a specific product or control and detailing configuration settings. They outline methodologies, include suggested actions, and are not compulsory.

Security Procedures

Procedures are the final element of the formalized security policy structure. A *procedure* or *standard operating procedure (SOP)* is a detailed, step-by-step how-to document that describes the exact actions necessary to implement a specific security mechanism, control, or solution. A procedure could discuss the entire system deployment operation or focus on a single product or aspect. They must be updated as the hardware and software of a system evolve. The purpose of a procedure is to ensure the integrity of business processes through standardization and consistency of results.

Keeping these various security documents as separate entities provides these benefits:

- Not all users need to know the security standards, baselines, guidelines, and procedures for all security classification levels.
- When changes occur, it is easier to update and redistribute only the affected material rather than updating a monolithic policy and redistributing it throughout the organization.

Many organizations struggle just to define the foundational parameters of their security, much less detail every single aspect of their day-to-day activities. However, in theory, a detailed and complete security policy supports real-world security in a directed, efficient, and specific manner. Once the security policy documentation is reasonably complete, it can be

used to guide decisions, train new users, respond to problems, and predict trends for future expansion.

Threat Modeling

Threat modeling is the security process where potential threats are identified, categorized, and analyzed. Threat modeling can be performed as a proactive measure during design and development or as a reactive measure once a product has been deployed. In either case, the process identifies the potential harm, the probability of occurrence, the priority of concern, and the means to eradicate or reduce the threat.

Threat modeling isn't meant to be a single event. Instead, it's meant to be initiated early in the design process of a system and continue throughout its life cycle. For example, Microsoft uses a security development life cycle (SDL) which includes a range of procedures aimed at bolstering security assurance and compliance prerequisites. (See www.microsoft.com/en-us/securityengineering/sdl). This SDL aids developers in creating software that is more secure by diminishing the quantity and seriousness of software vulnerabilities, all the while trimming development expenses.

A *defensive approach* to threat modeling takes place during the early stages of systems development, specifically during initial design and specifications establishment. This method is based on predicting threats and designing in specific defenses during the coding and crafting process. In most cases, integrated security solutions are more cost-effective and more successful than those shoehorned in later. While not a formal term, this concept could be considered a *proactive approach* to threat management.



Unfortunately, not all threats can be predicted during the design phase, so a *reactive approach* to threat management is still needed to address unforeseen issues. This concept is often called *threat hunting* or may be referred to as an *adversarial approach*. Threat hunting is the activity of looking for existing evidence of a compromise once symptoms or an IoC (indication of compromise) of an exploit become known. Threat modeling looks for zero-day exploits before harm is experienced, whereas threat hunting uses IoC information to find harm that has already occurred.

An adversarial approach to threat modeling takes place after a product has been created and deployed. This deployment could be in a test or laboratory environment or in the general marketplace. This technique of threat hunting is the core concept behind ethical hacking, penetration testing, source code review, and fuzz testing. Although these processes are often useful in finding flaws and threats, they, unfortunately, result in additional effort in coding to add new countermeasures, typically released as patches. This results in less effective security improvements (over defensive threat modeling) at the cost of potentially reducing functionality and user-friendliness.

Fuzz testing is a specialized dynamic testing technique that provides many different types of input to software to stress its limits and find previously undetected flaws. See Chapter 15 for more on fuzz testing.

Identifying Threats

There's an almost infinite possibility of threats, so it's important to use a structured approach to accurately identify relevant threats. For example, some organizations use one or more of the following three approaches:

Focused on Assets This method uses asset valuation results and attempts to identify threats to valuable assets.

Focused on Attackers Some organizations are able to identify potential attackers and can identify the threats they represent based on

the attackers' motivations, goals, or tactics, techniques, and procedures (TTPs).

Focused on Software If an organization develops software, it can consider potential threats against the software.

It's common to pair threats with vulnerabilities to identify threats that can exploit assets and represent significant risks to the organization. The ultimate goal of threat modeling is to prioritize the potential threats against an organization's valuable assets.

When attempting to inventory and categorize threats, it is often helpful to use a guide or reference. Microsoft developed a threat categorization scheme known as the STRIDE threat model. *STRIDE* is an acronym standing for the following:

- *Spoofing:* An attack with the goal of gaining access to a target system through the use of a falsified identity. When an attacker spoofs their identity as a valid or authorized entity, they are often able to bypass filters and blockades against unauthorized access.
- *Tampering:* Any action resulting in unauthorized changes or manipulation of data, whether in transit or in storage.
- *Repudiation:* The ability of a user or attacker to deny having performed an action or activity by maintaining plausible deniability. Repudiation attacks can also result in innocent third parties being blamed for security violations.
- *Information disclosure:* The revelation or distribution of private, confidential, or controlled information to external or unauthorized entities.
- *Denial of service (DoS):* An attack that attempts to prevent authorized use of a resource. This can be done through flaw exploitation, connection overloading, or traffic flooding.
- *Elevation of privilege:* An attack where a limited user account is transformed into an account with greater privileges, powers, and access.

Process for Attack Simulation and Threat Analysis (PASTA) is a seven-stage threat modeling methodology. PASTA is a risk-centric approach that aims at selecting or developing countermeasures in relation to the value of the assets to be protected. The following are the seven steps of PASTA:

- Stage I: Definition of the Objectives (DO)
- Stage II: Definition of the Technical Scope (DTS)
- Stage III: Application Decomposition and Analysis (ADA)
- Stage IV: Threat Analysis (TA)
- Stage V: Weakness and Vulnerability Analysis (WVA)
- Stage VI: Attack Modeling & Simulation (AMS)
- Stage VII: Risk Analysis & Management (RAM)

Each stage of PASTA has specific objectives to achieve and deliverables to produce in order to complete the stage.

Visual, Agile, and Simple Threat (VAST) is a threat modeling concept that integrates threat and risk management into an Agile programming environment on a scalable basis (see Chapter 20, "Software Development Security," regarding Agile).

These are just a few in the vast array of threat modeling concepts and methodologies available from community groups, commercial entities, government agencies, and international associations.

Be Alert for Individual Threats

Competition is often a key part of business growth, but overly adversarial competition can increase the threat level from individuals. In addition to criminal hackers and disgruntled employees, adversaries, contractors, employees, and even trusted partners can be a threat to an organization if relationships go sour.

Potential threats to your business are broad and varied. A company faces threats from nature, technology, and people. Always consider the best and

worst possible outcomes of your organization's activities, decisions, and interactions. Identifying threats is the first step toward designing defenses to help reduce or eliminate downtime, compromise, and loss.

Determining and Diagramming Potential Attacks

The next step in threat modeling is to determine the potential attack concepts that could be realized. This is often accomplished through the creation of a diagram of the elements involved in a transaction, along with indications of data flow and privilege boundaries. Figure 1.4 shows each major component of a system, the boundaries between security zones, and the potential flow or movement of information and data.

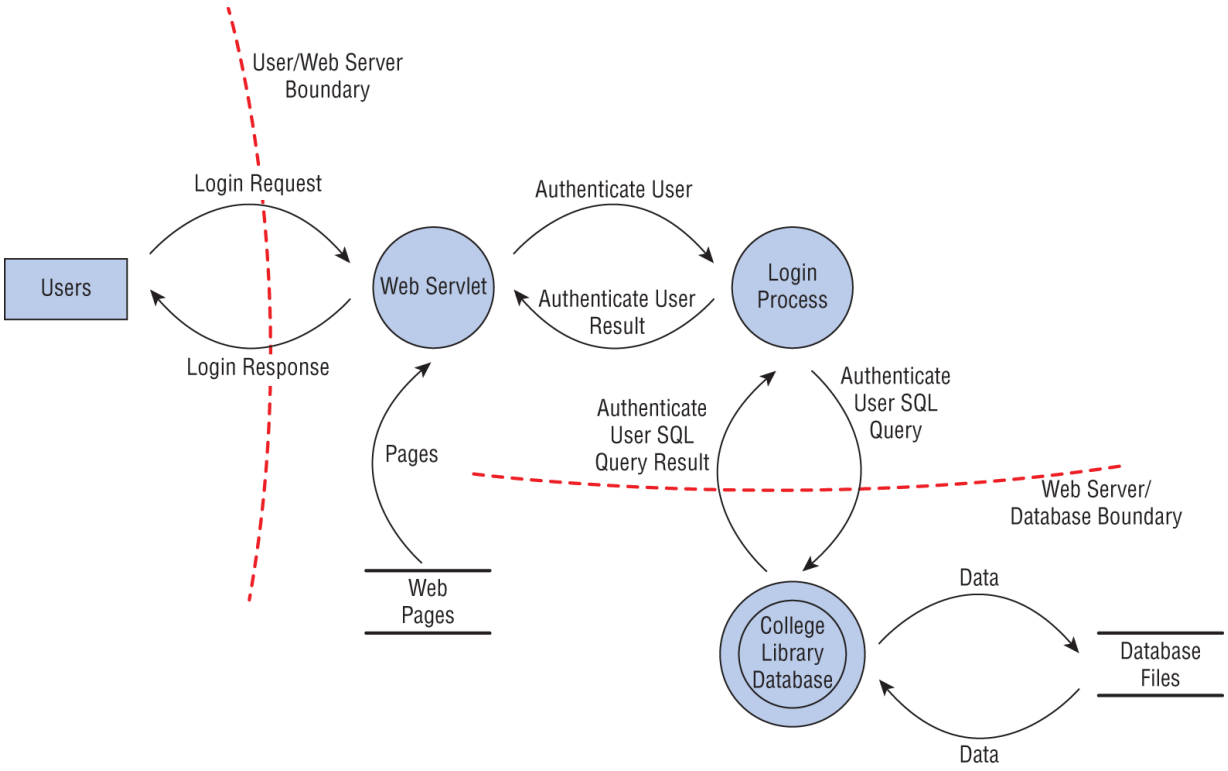


FIGURE 1.4 An example of diagramming to reveal threat concerns

This is a high-level overview and not a detailed evaluation of the coding logic. However, for more complex systems, multiple diagrams may need to be created at various focus points and at varying levels of detail magnification.

Once a diagram has been crafted, identify all of the technologies involved. Next, identify attacks that could be targeted at each element of the diagram.

Keep in mind that all forms of attacks should be considered, including logical/technical, physical, and social. This process will quickly lead you into the next phase of threat modeling: reduction analysis.

Performing Reduction Analysis

The next step in threat modeling is to perform a reduction analysis. *Reduction analysis* is also known as *decomposing* the application, system, or environment. The purpose of this task is to gain a greater understanding of the logic of the product, its internal components, as well as its interactions with external elements. Whether an application, a system, or an entire environment, it needs to be divided into smaller containers or compartments. Those might be subroutines, modules, or objects if you're focusing on software, computers, or operating systems; they might be protocols if you're focusing on systems or networks; or they might be departments, tasks, and networks if you're focusing on an entire business infrastructure. Each identified element should be evaluated in order to understand inputs, processing, security, data management, storage, and outputs.

In the decomposition process, you must identify five key concepts:

Trust Boundaries Any location where the level of trust or security changes

Dataflow Paths The movement of data between locations

Input Points Locations where external input is received

Privileged Operations Any activity that requires greater privileges than a standard user account or process typically required to make system changes or alter security

Details about Security Stance and Approach The declaration of the security policy, security foundations, and security assumptions

Breaking down a system into its constituent parts makes it much easier to identify the essential components of each element as well as take notice of vulnerabilities and points of attack. The more you understand exactly how a program, system, or environment operates, the easier it is to identify threats to it.

Once threats are identified, they should be fully documented by defining the means, target, and consequences of a threat. Consider including the techniques required to implement an exploitation and list potential countermeasures and safeguards.

Prioritization and Response

After documentation, the next step is to rank or rate the threats. This can be accomplished using a wide range of techniques, such as Probability × Damage Potential ranking, high/medium/low rating, or the DREAD system.

The ranking technique of *Probability* × *Damage Potential* produces a risk severity number on a scale of 1 to 100, with 100 being the most severe risk possible. Each of the two initial values can be assigned numbers between 1 and 10, with 1 being the lowest and 10 the highest. These rankings can be somewhat arbitrary and subjective, but since the same person or team will be assigning the numbers for their own organization, it should still result in assessment values that are accurate on a relative basis.

The high/medium/low (1/2/3 or green/yellow/red) rating process is even simpler. It creates a basic risk matrix or heat map (Figure 1.5). As with any means of risk assessment, the purpose is to help establish criticality prioritization. When using a risk matrix, each threat can be assigned a probability and a damage potential level. Then, when these two values are compared, the result is a combined value somewhere in the nine squares. Those threats in the HH (high probability/high damage potential) area are of the highest priority and concern, whereas those in the LL (low probability/low damage potential) area are of the least priority and concern.

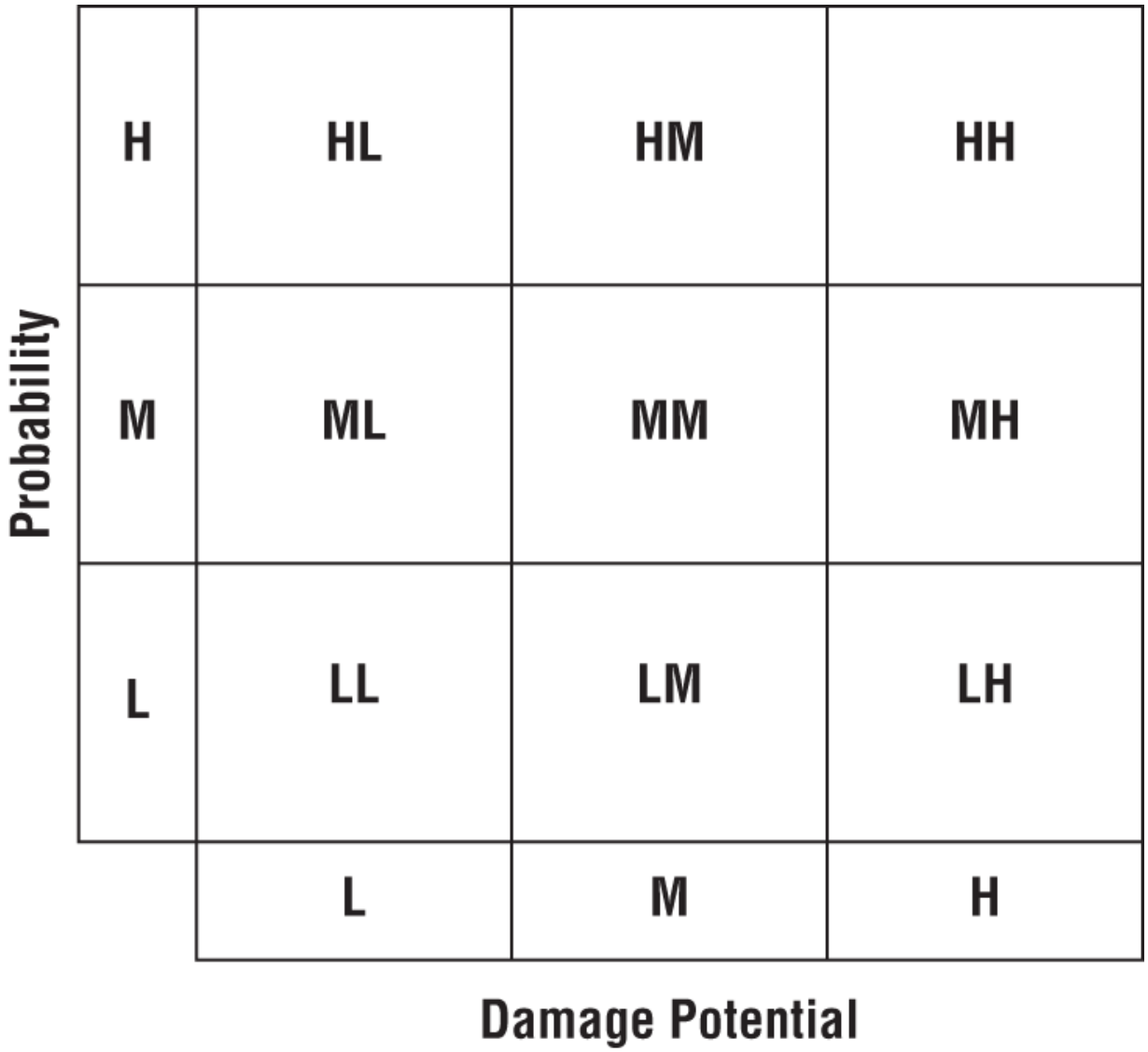


FIGURE 1.5 A risk matrix or risk heat map

The *Damage, Reproducibility, Exploitability, Affected Users, and Discoverability (DREAD)* rating system is designed to provide a flexible rating solution that is based on the answers to five main questions about each threat:

Damage (potential) How severe is the damage likely to be if the threat is realized?

Reproducibility How complicated is it for attackers to reproduce the exploit?

Exploitability How hard is it to perform the attack?

Affected Users How many users are likely to be affected by the attack (as a percentage)?

Discoverability How hard is it for an attacker to discover the weakness?

Once threat priorities are set, responses to those threats need to be determined. Technologies and processes to remediate threats should be considered and weighted according to their cost and effectiveness. Response options should include making adjustments to software architecture, altering operations and processes, and implementing defensive and detection components.

This process is similar to the risk assessment process discussed in Chapter 2. The difference is that threats are the focus of threat modeling, whereas assets are often the focus of risk assessment.

Supply Chain Risk Management

Applying risk-based management concepts to the supply chain is a means to ensure a more robust and successful security strategy in organizations of all sizes. A *supply chain* is the concept that most computers, devices, networks, systems, and even cloud services are not built by a single entity. In fact, most of the companies we know of as computer and equipment manufacturers generally perform the final assembly rather than manufacture all the individual components. Often, the CPU, memory, drive controllers, hard drives, SSDs, and video cards are created by other third-party vendors. Even these commodity vendors are unlikely to have mined their own metals, processed the oil for plastics, or etched the silicon of their chips. Thus, any finished system has a long and complex history, known as its supply chain, that enabled it to come into existence.

Supply chain risk management (SCRM) is the means to ensure that all of the vendors or links in the supply chain are reliable, trustworthy, reputable organizations that disclose their practices and security requirements to their business partners (although not necessarily to the public). SCRM should be evaluated for every organizational acquisition of products and services from third-party suppliers and providers.

Each link in a supply chain should be responsible and accountable to the next link in the chain. Each handoff should be properly organized, documented, managed, and audited. The goal of a secure supply chain is to ensure that the finished product is of sufficient quality, meets performance and operational goals, and provides stated security mechanisms. And, at no point in the process was any element subjected to unauthorized or malicious manipulation or sabotage.

When evaluating organizational risk, consider external factors that can affect the organization, especially related to company stability and resource availability. The supply chain can be a threat vector, where materials, software, hardware, or data are being obtained from a supposedly trusted source, but the supply chain behind that source could have been compromised and the asset poisoned or modified.

An organization's supply chain should be assessed to determine what risks it places on the organization. Is the organization operating on a just-in-time basis where materials are delivered just before or just as they are needed by manufacturing? If there is any delay in delivery, does the organization have access to any surplus or buffer of materials that can be used to maintain production while the supply chain operations are reconstituted?

Most organizations rely on products manufactured by other entities. Most of those products are produced as part of a long and complex supply chain. Attacks on that supply chain could result in flawed or less reliable products or could allow for remote access or listening mechanisms to be embedded into otherwise functioning equipment. Supply chain attacks include product tampering, counterfeits, and even implants.

Supply chain attacks present a risk that can be challenging to address. An organization may elect to inspect all equipment in order to reduce the chance of modified devices going into production networks. However, with miniaturization, it may be nearly impossible to discover an extra chip placed on a device's mainboard. Also, the manipulation may be through firmware or software instead of hardware. Organizations can choose to source products from trusted and reputable vendors or attempt to use vendors who manufacture most of their products domestically.

In many cases, ongoing security monitoring, management, and assessment may be required. This could be an industry best practice or a regulation.

Such assessment and monitoring of a supply chain may be performed by the primary or end-of-chain organization or may require the use of external auditors. When engaging third-party assessment and monitoring services, keep in mind that each element of the supply chain entity needs to show security-mindedness in their business operations. If an organization is unable to manage their own operations on a secure basis, how can they provide reliable security management functions to the supply chain?

When possible, establish minimum security requirements for each entity in a supply chain. The security requirements for new hardware, software, or services should always meet or exceed the security expected in the final product. This often requires a detailed review of SLAs, contracts, and actual performance. This is to ensure that security is a prescribed component of the contracted services. When a supply chain component provider is crafting software or providing a service (such as a cloud provider), then a service-level requirement (SLR) may need to be defined. Often, an SLR is provided by the customer/client prior to the establishment of the SLA (which should incorporate the elements of the SLR if the vendor expects the customer to sign the agreement).

SCRM may require the integration of numerous security mechanisms, including silicon root of trust, physically unclonable functions (PUFs), and/or a software bill of materials (SBOM).

A *silicon root of trust (RoT)*, also known as a hardware root of trust, is a foundational and tamper-resistant component within a computer's hardware that provides a secure starting point for establishing trust and security in a system. The primary purpose of a silicon RoT is to ensure the integrity, authenticity, and confidentiality of the system's boot process and software. A silicon RoT can be an essential element of an SCRM plan.

Key characteristics of a silicon RoT include:

- *Tamper resistance:* The silicon RoT is typically implemented in a way that makes it extremely difficult to tamper with or compromise. This may involve using dedicated hardware security modules, secure enclaves, or other techniques to protect it from physical and software attacks.

- *Secure boot:* A silicon RoT supports a secure boot process. It verifies the integrity of the firmware, bootloader, and operating system during the boot sequence to ensure that no unauthorized or malicious code is executed.
- *Cryptographic operations:* The silicon RoT typically has built-in cryptographic capabilities, allowing it to perform operations such as digital signatures, encryption, and decryption. This is essential for securing data, establishing secure communications, and authenticating the system.
- *Remote attestation:* Many silicon RoTs support remote attestation, which enables a remote entity to verify the trustworthiness of a system. This is crucial for cloud computing and IoT devices.

Silicon roots of trust are fundamental in building secure systems, particularly in environments where the integrity of the hardware and software is critical, such as data centers, cloud computing, and Internet of Things (IoT) devices. They provide a solid foundation for security by ensuring that the system starts in a known, trusted state and can maintain that trust throughout its operation.

A *physically unclonable function (PUF)* is a specialized physical electronic component or function that generates a unique, unpredictable digital identifier based on the inherent physical properties of the component. PUFs are used to provide a hardware-based security feature by creating a unique fingerprint for electronic devices or integrated circuits. This fingerprint can be used for device authentication, encryption keys, or other security-related purposes. PUFs have gained prominence in the field of hardware security and have applications in various domains, including IoT devices, hardware-based cryptography, secure boot processes, and authentication of integrated circuits. PUF components prevent counterfeits or implants along a supply chain, therefore establishing a more secure SCRM.

A *software bill of materials (SBOM)* is a structured and comprehensive inventory or list of all the software components and dependencies that make up a software application or system. An SBOM provides detailed information about the various software components used in a system, including their versions, sources, and relationships. The primary purpose of

an SBOM is to enhance software transparency, security, compliance, and management.

SBOMs play a vital role in software security, particularly in identifying and addressing vulnerabilities and risks associated with the software components. Security teams can use SBOMs to track and address known vulnerabilities and apply patches or updates as needed. In the context of software supply chain management, SBOMs help organizations track the origins and sources of software components, ensuring that they come from trusted and secure sources.

SBOMs are becoming increasingly important as software ecosystems grow in complexity, and organizations rely on a multitude of software components from various sources. They aid in managing software supply chains, improving security, and ensuring compliance with legal and regulatory requirements. In some cases, SBOMs may also be used to provide information to users, customers, and stakeholders about the software components used in a particular product or system.

Numerous security elements can be incorporated into an SCRM plan. Not all of these may be relevant to every organization. However, most organizations would benefit from integrating some or all of these security features into their existing supply chain management processes.

Summary

Security governance, management concepts, and principles are inherent elements in a security policy and in solution deployment. They define the basic parameters needed for a secure environment. They also define the goals and objectives that both policy designers and system implementers must achieve in order to create a secure solution.

The primary goals and objectives of security are contained within the CIA Triad: confidentiality, integrity, and availability. Confidentiality is the principle that objects are not disclosed to unauthorized subjects. Integrity is the principle that objects retain their veracity and are intentionally modified only by authorized subjects. Availability is the principle that authorized subjects are granted timely and uninterrupted access to objects.

Other security-related concepts and principles that should be considered and addressed when designing a security policy and deploying a security solution are identification, authentication, authorization, accounting, auditing, nonrepudiation, defense in depth, abstraction, data hiding, and encryption.

Security roles determine who is responsible for the security of an organization's assets. Common roles include senior manager, security professionals, asset owners, custodians, users, and auditors.

A formalized security policy structure consists of policies, standards, baselines, guidelines, and procedures. These individual documents are elements essential to the design and implementation of security in any environment. To be effective, the approach to security management must be a top-down approach.

Security control frameworks are structured sets of guidelines, standards, best practices, and controls designed to help organizations effectively manage and enhance their information security and cybersecurity posture. Entities and examples of SCFs include ISO, NIST, COBIT, SABSA, PCI DSS, FedRAMP, and ITIL.

Threat modeling is the security process where potential threats are identified, categorized, and analyzed. Threat modeling can be performed as a proactive measure during design and development or as a reactive measure once a product has been deployed. In either case, the process identifies the potential harm, the probability of occurrence, the priority of concern, and the means to eradicate or reduce the threat.

Integrating cybersecurity risk management with supply chain, acquisition strategies, and business practices is a means to ensure a more robust and successful security strategy in organizations of all sizes. When purchases are made without security considerations, the risks inherent in those products remain throughout their deployment life span.

Study Essentials

Understand the CIA Triad elements of confidentiality, integrity, and availability. Confidentiality is the principle that objects are not disclosed to unauthorized subjects. Integrity is the principle that objects retain their

veracity and are intentionally modified only by authorized subjects. Availability is the principle that authorized subjects are granted timely and uninterrupted access to objects.

Know the elements of AAA services. AAA services focus on identification, authentication, authorization, auditing, and accounting.

Be able to explain how identification works. Identification is when a subject professes an identity and accounting is initiated. A subject must provide an identity to a system to start the process of authentication, authorization, and accounting.

Understand the process of authentication. Authentication is the process of verifying or testing that a claimed identity is valid. Authentication requires information from the subject that must exactly correspond to the identity indicated.

Know how authorization fits into a security plan. Once a subject is authenticated, its access must be authorized. The process of authorization ensures that the requested activity or object access is possible given the rights and privileges assigned to the authenticated identity.

Be able to explain the auditing process. Auditing is the programmatic means by which subjects are held accountable for their actions while authenticated on a system through the documentation or recording of subject activities.

Understand the importance of accounting. Security can be maintained only if subjects are held accountable for their actions. Effective accounting relies on the capability to prove a subject's identity and track their activities.

Be able to explain the concept of abstraction. Abstraction is used to collect similar elements into groups, classes, or roles that are assigned security controls, restrictions, or permissions as a collective. It adds efficiency to carrying out a security plan.

Know about security boundaries. A security boundary is the line of intersection between any two areas, subnets, or environments that have different security requirements or needs.

Understand security governance. Security governance is the collection of practices related to supporting, defining, and directing the security efforts

of an organization.

Know about third-party governance. Third-party governance is the system of external entity oversight that may be mandated by law, regulation, industry standards, contractual obligation, or licensing requirements. The actual method of governance may vary, but it generally involves an outside investigator or auditor.

Understand documentation review. Documentation review is the process of reading the exchanged materials and verifying them against standards and expectations. In many situations, especially those related to government or military agencies or contractors, failing to provide sufficient documentation to meet requirements of third-party governance can result in a loss of or a voiding of authorization to operate (ATO).

Understand the alignment of security function to business strategy, goals, mission, and objectives. Security management planning ensures the proper creation, implementation, and enforcement of a *security policy*. Security management planning aligns the security functions to the strategy, goals, mission, and objectives of the organization. This includes designing and implementing security based on business cases, budget restrictions, or scarcity of resources.

Know what a business case is. A business case is usually a documented argument or stated position in order to define a need to make a decision or take some form of action. To make a business case is to demonstrate a business-specific need to alter an existing process or choose an approach to a business task. A business case is often made to justify the start of a new project, especially a project related to security.

Understand security management planning. Security management is based on three types of plans: strategic, tactical, and operational. A strategic plan is a long-term plan that is fairly stable. It defines the organization's goals, mission, and objectives. The tactical plan is a midterm plan developed to provide more details on accomplishing the goals set forth in the strategic plan. Operational plans are short-term and highly detailed plans based on strategic and tactical plans.

Know the elements of a formalized security policy structure. To create a comprehensive security plan, you need the following items in place:

security policy, standards, baselines, guidelines, and procedures.

Understand key security roles. The primary security roles are senior manager, security professional, asset owner, custodian, user, and auditor.

Understand due diligence and due care. Due diligence is establishing a plan, policy, and process to protect the interests of an organization. Due care is practicing the individual activities that maintain the due diligence effort. Due diligence is knowing what should be done and planning for it; due care is doing the right action at the right time.

Know the basics of threat modeling. Threat modeling is the security process where potential threats are identified, categorized, and analyzed. Threat modeling can be performed as a proactive measure during design and development or as a reactive measure once a product has been deployed. Key concepts include assets/attackers/software, STRIDE, PASTA, VAST, diagramming, reduction/decomposing, and DREAD.

Understand supply chain risk management (SCRM) concepts. SCRM is a means to ensure that all the vendors or links in the supply chain are reliable, trustworthy, reputable organizations that disclose their practices and security requirements to their business partners. SCRM includes evaluating risks associated with hardware, software, and services; performing third-party assessment and monitoring; establishing minimum security requirements; and enforcing service-level requirements.

Written Lab

1. Discuss and describe the CIA Triad.
2. What are the requirements to hold a person accountable for the actions of their user account?
3. Name the six primary security roles as defined by ISC2 for CISSP.
4. What are the four components of a complete organizational security policy and their basic purpose?

Review Questions

1. Confidentiality, integrity, and availability are typically viewed as the primary goals and objectives of a security infrastructure. Which of the following is not considered a violation of confidentiality?

    A. Stealing passwords using a keystroke logging tool

    B. Eavesdropping on wireless network communications

    C. Hardware destruction caused by arson

    D. Social engineering that tricks a user into providing personal information to a false website

2. Security governance requires a clear understanding of the objectives of the organization as the core concepts of security. Which of the following contains the primary goals and objectives of security?

    A. A network's border perimeter

    B. The CIA Triad

    C. AAA services

    D. Ensuring that subject activities are recorded

3. Jamie recently discovered an attack taking place against his organization that prevented employees from accessing critical records. What element of the CIA Triad was violated?

    A. Identification

    B. Availability

    C. Encryption

    D. Layering

4. Optimally, security governance is performed by a board of directors, but smaller organizations may simply have the CEO or CISO perform the activities of security governance. Which of the following is true about security governance?

    A. Security governance ensures that the requested activity or access to an object is possible, given the rights and privileges assigned to the authenticated identity.

B. Security governance is used for efficiency. Similar elements are put into groups, classes, or roles that are assigned security controls, restrictions, or permissions as a collective.

C. Security governance is a documented set of best IT security practices that prescribes goals and requirements for security controls and encourages the mapping of IT security ideals to business objectives.

D. Security governance seeks to compare the security processes and infrastructure used within the organization with knowledge and insight obtained from external sources.

5. You have been tasked with crafting a long-term security plan that is fairly stable. It needs to define the organization's security purpose. It also needs to define the security function and align it with the goals, mission, and objectives of the organization. What are you being asked to create?

   A. Tactical plan

   B. Operational plan

   C. Strategic plan

   D. Rollback plan

6. Annaliese's organization is undergoing a period of increased business activity where they are conducting a large number of mergers and acquisitions. She is concerned about the risks associated with those activities. Which of the following are examples of those risks? (Choose all that apply.)

   A. Inappropriate information disclosure

   B. Increased worker compliance

   C. Data loss

   D. Downtime

   E. Additional insight into the motivations of inside attackers

   F. Failure to achieve a sufficient return on investment (ROI)

7. Which security control framework is a set of security standards and requirements designed to ensure the protection of sensitive credit card and debit card information?

    A. ITIL

    B. ISO 27000

    C. PCI DSS

    D. CSF

8. A security role is the part an individual plays in the overall scheme of security implementation and administration within an organization. What is the security role that has the functional responsibility for security, including writing the security policy and implementing it?

    A. Senior management

    B. Security professional

    C. Custodian

    D. Auditor

9. Control Objectives for Information and Related Technologies (COBIT) is a documented set of best IT security practices crafted by ISACA. It prescribes goals and requirements for security controls and encourages the mapping of IT security ideals to business objectives. COBIT is based on six key principles for the governance and management of enterprise IT. Which of the following are among these key principles? (Choose all that apply.)

    A. Holistic Approach

    B. End-to-End Governance System

    C. Provide Stakeholder Value

    D. Maintaining Authenticity and Accountability

    E. Dynamic Governance System

10. In today's business environment, prudence is mandatory. Showing due diligence and due care is the only way to disprove negligence in an

occurrence of loss. Which of the following are true statements? (Choose all that apply.)

A. Due diligence is establishing a plan, policy, and process to protect the interests of an organization.

B. Due care is developing a formalized security structure containing a security policy, standards, baselines, guidelines, and procedures.

C. Due diligence is the continued application of a security structure onto the IT infrastructure of an organization.

D. Due care is practicing the individual activities that maintain the security effort.

E. Due care is knowing what should be done and planning for it.

F. Due diligence is doing the right action at the right time.

11. Security documentation is an essential element of a successful security program. Understanding the components is an early step in crafting the security documentation. Match the following components to their respective definitions.

    1. Policy
    2. Standard
    3. Procedure
    4. Guideline

    I. A detailed, step-by-step how-to document that describes the exact actions necessary to implement a specific security mechanism, control, or solution.

    II. A document that defines the scope of security needed by the organization and discusses the assets that require protection and the extent to which security solutions should go to provide the necessary protection.

    III. A minimum level of security that every system throughout the organization must meet.

    IV. Offers recommendations on how security requirements are implemented and serves as an operational guide for both security

professionals and users.

V. Defines compulsory requirements for the homogenous use of hardware, software, technology, and security controls.

A. 1 – I; 2 – IV; 3 – II; 4 – V

B. 1 – II; 2 – V; 3 – I; 4 – IV

C. 1 – IV; 2 – II; 3 – V; 4 – I

D. 1 – V; 2 – I; 3 – IV; 4 – III

12. STRIDE is often used in relation to assessing threats against applications or operating systems. When confidential documents are exposed to unauthorized entities, which element of STRIDE is used to reference that violation?

A. S

B. T

C. R

D. I

E. D

F. E

13. A development team is working on a new project. During the early stages of systems development, the team considers the vulnerabilities, threats, and risks of their solution and integrates protections against unwanted outcomes. What concept of threat modeling is this?

A. Threat hunting

B. Proactive approach

C. Qualitative approach

D. Adversarial approach

14. Supply chain risk management (SCRM) is a means to ensure that all the vendors or links in the supply chain are reliable, trustworthy, reputable organizations. Which of the following are true statements? (Choose all that apply.)

A. Each link in the supply chain should be responsible and accountable to the next link in the chain.

B. Commodity vendors are unlikely to have mined their own metals, processed the oil for plastics, or etched the silicon of their chips.

C. If the final product derived from a supply chain meets expectations and functional requirements, it is assured to not have unauthorized elements.

D. Failing to properly secure a supply chain can result in flawed or less reliable products, or even embedded listing or remote control mechanisms.

15. Your organization has become concerned with risks associated with the supply chain of their retail products. Fortunately, all coding for their custom product is done in-house. However, a thorough audit of a recently completed product revealed that a listening mechanism was integrated into the solution somewhere along the supply chain. The identified risk is associated with what product component in this scenario?

    A. Software

    B. Services

    C. Data

    D. Hardware

16. Cathy's employer has asked her to perform a documentation review of the policies and procedures of a third-party supplier. This supplier is just the final link in a software supply chain. Their components are being used as a key element of an online service operated for high-end customers. Cathy discovers several serious issues with the vendor, such as failing to require encryption for all communications and not requiring multifactor authentication on management interfaces. What should Cathy do in response to this finding?

    A. Write up a report and submit it to the CIO.

    B. Void the ATO of the vendor.

    C. Require that the vendor review their terms and conditions.

D. Have the vendor sign an NDA.

17. Whenever your organization works with a third party, its supply chain risk management (SCRM) processes should be applied. One of the common requirements is the establishment of minimum security requirements for the third party. What should these requirements be based on?

    A. Existing security policy

    B. Third-party audit

    C. On-site assessment

    D. Vulnerability scan results

18. It's common to pair threats with vulnerabilities to identify threats that can exploit assets and represent significant risks to the organization. The ultimate goal of threat modeling is to prioritize the potential threats against an organization's valuable assets. Which of the following is a risk-centric threat-modeling approach that aims at selecting or developing countermeasures in relation to the value of the assets to be protected?

    A. VAST

    B. DREAD

    C. PASTA

    D. STRIDE

19. The next step after threat modeling is reduction analysis. Reduction analysis is also known as decomposing the application, system, or environment. The purpose of this task is to gain a greater understanding of the logic of the product, its internal components, as well as its interactions with external elements. Which of the following are key components to identify when performing decomposition? (Choose all that apply.)

    A. Patch or update versions

    B. Trust boundaries

    C. Dataflow paths

D. Open vs. closed source code use

E. Input points

F. Privileged operations

G. Details about security stance and approach

20. Defense in depth is the use of multiple controls in a series. No one control can protect against all possible threats. Using a multilayered solution allows for numerous different controls to guard against whatever threats come to pass. Which of the following are terms that relate to or are based on defense in depth? (Choose all that apply.)

    A. Layering

    B. Classifications

    C. Zones

    D. Realms

    E. Compartments

    F. Silos

    G. Segmentations

    H. Lattice structure

    I. Protection rings

Chapter 2
Personnel Security and Risk Management Concepts

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 1.0: Security and Risk Management**
  - 1.8 Contribute to and enforce personnel security policies and procedures
    - 1.8.1 Candidate screening and hiring
    - 1.8.2 Employment agreements and policy driven requirements
    - 1.8.3 Onboarding, transfers, and termination processes
    - 1.8.4 Vendor, consultant, and contractor agreements and controls
  - 1.9 Understand and apply risk management concepts
    - 1.9.1 Threat and vulnerability identification
    - 1.9.2 Risk analysis, assessment, and scope
    - 1.9.3 Risk response and treatment (e.g., cybersecurity insurance)
    - 1.9.4 Applicable types of controls (e.g., preventive, detection, corrective)
    - 1.9.5 Control assessments (e.g., security and privacy)
    - 1.9.6 Continuous monitoring and measurement
    - 1.9.7 Reporting (e.g., internal, external)
    - 1.9.8 Continuous improvement (e.g., risk maturity modeling)
    - 1.9.9 Risk frameworks (e.g., International Organization for Standardization (ISO), National Institute of Standards and Technology (NIST), Control Objectives for Information and Related Technology (COBIT), Sherwood Applied Business Security Architecture (SABSA), Payment Card Industry (PCI))

- 1.12 Establish and maintain a security awareness, education, and training program
  - 1.12.1 Methods and techniques to present awareness and training (e.g., social engineering, phishing, security champions, gamification)
  - 1.12.2 Periodic content reviews to include emerging technologies and trends (e.g., cryptocurrency, artificial intelligence (Al), blockchain)
  - 1.12.3 Program effectiveness evaluation

Additional elements of this domain are discussed in various chapters:

- Chapter 1, “Security Governance Through Principles and Policies”
- Chapter 3, “Business Continuity Planning”
- Chapter 4, “Laws, Regulations, and Compliance”

Please review all of these chapters to have a complete perspective on the topics of this domain.

Personnel Security Policies and Procedures

Humans are often considered the weakest element in any security solution. No matter what physical or logical controls are deployed, humans can discover ways to avoid, circumvent, subvert, or disable them. Thus, it is important to consider your users' humanity when designing and deploying security solutions for your environment. To understand and apply security governance, you must address the potentially weakest link in your security chain—people.

However, people can also become a key security asset when they are properly trained and motivated to protect not only themselves but the security of the organization. It is important to not treat personnel as a problem to be solved, but as people who can become valued partners in a security endeavor.

Issues, problems, and compromises related to humans occur at all stages of a security solution development. This is because humans are involved throughout any solution's development, deployment, and ongoing management. Therefore, you must evaluate the effect users, designers, programmers, developers, managers, vendors, consultants, and implementers have on the process.

Job Descriptions and Responsibilities

Hiring new staff typically involves several distinct steps: creating a *job description* or *position description*, setting a classification for the job, screening employment candidates, and hiring and training someone best suited for the job. Without a job description, there is no consensus on what type of individual should be hired. Any job description for any position within an organization should address relevant security issues, such as whether the position requires handling sensitive material or access to classified information. In effect, the job description defines the roles to which an employee needs to be assigned to perform their work tasks. Job roles typically align to a rank or level of privilege, whereas job descriptions map to specifically assigned responsibilities and tasks.

Job responsibilities are the specific work tasks an employee is required to perform regularly. Employees require access to various objects, resources, and services depending on their responsibilities. Job responsibilities should be detailed in a job description. Thus, a list of job responsibilities guides the assignment of access rights, permissions, and privileges. On a secured network, users must be granted access privileges for those elements related to their work tasks.

Job descriptions are not used exclusively for the hiring process; they should be maintained throughout the organization's life. Only through detailed job descriptions can a comparison be made between what a person should be responsible for and what they actually are responsible for. Managers should audit privilege assignments to ensure that workers do not obtain access that is not strictly required for them to accomplish their work tasks.

Candidate Screening and Hiring

Employment *candidate screening* for a specific position is based on the sensitivity and classification defined by the job description. Thus, the thoroughness of the screening process should reflect the security of the position to be filled.

Employment candidate screening, background checks, reference checks, education verification, and security clearance validation are essential elements in proving that a candidate is adequate, qualified, and trustworthy for a secured position. *Background checks* include obtaining a candidate's work and educational history; checking references; verifying education; interviewing colleagues; checking police and government records for arrests or illegal activities; verifying identity through fingerprints, driver's license, and/or birth certificate; and holding a personal interview. Depending on the job position, this process could also include skill challenges, drug testing, credit checks, checking driving records, and personality testing/evaluation.

Performing online background checks and reviewing the social networking accounts of applicants has become standard practice for many organizations. If a potential employee has posted inappropriate materials online, then they are not as promising a candidate as those who did not. A general picture of a person's attitude, intelligence, loyalty, common sense, diligence, honesty, respect, consistency, and adherence to social norms and/or corporate culture can be gleaned quickly by viewing a person's online identity. However, being fully aware of the legal restrictions against discrimination is essential. Various countries have vastly different freedoms or limitations on background checks, especially criminal history research. Always confirm with your legal department before evaluating an applicant's online persona.

During the initial applicant review process, the human resources (HR) staff are looking to confirm that a candidate is appropriately qualified for a job, but they are also on the lookout for issues that would disqualify the applicant.

Interviewing qualified applicants is the next filter to eliminate those not suited for the job or the organization. When conducting interviews, it is important to have a standardized interview process in order to treat each candidate fairly. Although some aspects of an interview are subjective and

based on the interplay of personalities of the candidates and the interviewer, the decision whether or not to hire someone needs to be legally defensible.

Onboarding: Employment Agreements and Policy-Driven Requirements

Once a qualified but not-disqualified candidate is found and interviewed, they can be offered the job. If accepted, the new hire will need to be integrated into the organization. This process is known as *onboarding*. As with all tasks within a security-focused organization, the process of onboarding should be guided by policy-driven requirements.

Onboarding is the process of adding new employees to the organization, having them review and sign employment agreements and policies, be introduced to managers and coworkers, and be trained in employee operations and logistics. Onboarding can also include organizational socialization and orientation. This is the process by which new employees are trained in order to be properly prepared for performing their job responsibilities. It can include training, job skill acquisition, and behavioral adaptation in an effort to integrate employees efficiently into existing organizational culture, processes, and procedures. Well-designed onboarding can result in higher levels of job satisfaction, higher levels of productivity, faster integration with existing workers, a rise in organizational loyalty, stress reduction, and a decreased resignation rate.

A new employee will often be provided with a computer/network user account. This is accomplished through the *identity and access management (IAM)* system of an organization, which will provision the account and assign necessary privileges and access. The onboarding process is also used when an employee's role or position changes or when that person is awarded additional levels of privilege or access.

To maintain security, access should be assigned according to the principle of least privilege. The *principle of least privilege* states that users should be granted the minimum amount of access necessary for them to complete their required work tasks or job responsibilities. True application of this principle requires low-level granular control over all resources and functions. Further discussion of least privilege is in Chapter 16, “Managing Security Operations.”

When a new employee is hired, they should sign an employment agreement. Such a document outlines the rules and restrictions of the organization, the security policy, details of the job description, violations and consequences, and the minimum or probationary length of time the position is to be filled by the employee. These items might be separate documents, such as an acceptable use policy (AUP). In such a case, the employment agreement is used to verify that the employment candidate has read and understood the associated documentation and signed their agreement to adhere to the necessary policies related to their prospective job position.



An acceptable use policy (AUP) defines what is and what is not an acceptable activity, practice, or use for company equipment and resources. The AUP is specifically designed to assign security roles within the organization as well as prescribe the responsibilities tied to those roles. This policy defines a level of acceptable performance and expectations of behavior and activity. Failure to comply with the policy may result in job action warnings, penalties, or termination.

In addition to employment agreements, there may be other security-related documentation and policy-driven requirements that must be addressed. One common document is a *nondisclosure agreement (NDA)*. An NDA is used to protect confidential information within an organization from being disclosed by a current or former employee. Violations of an NDA are often met with strict penalties. Throughout a worker's employment, they may be asked to sign additional NDAs as their job responsibilities change and they need to access new sensitive, proprietary, or confidential assets. When an employee leaves the organization, they should be reminded of their legal obligation to maintain silence on all items covered by any signed NDAs. In fact, they may be required to re-sign the NDA upon departure as a means to legally confirm that they are fully aware of their legally recognized obligation to maintain trade secrets and other confidential information.

There are several forms of NDA to be aware of. A unilateral NDA, also known as a one-way NDA, is used when one party needs to share sensitive information with another party while retaining control and protection over

that information. A bilateral NDA (aka mutual NDA or two-way NDA) is a legally binding contract between two parties, often individuals or organizations, where both parties agree to protect each other's confidential information. A multilateral NDA is a legal contract involving three or more parties, each of whom agrees to protect and keep confidential the sensitive information shared by the other parties.

Another potential element of an employment contract is a non-compete agreement (NCA). A non-compete agreement, often called a non-compete clause or covenant not to compete (CNC), is a legal contract between an employer and an employee, a business and an independent contractor, or between business partners. The primary purpose of a non-compete agreement is to restrict the ability of one party (usually the employee or business partner) from engaging in competitive activities or working for a competing entity, typically within a specific geographical area and for a defined duration, after the termination of their relationship with the other party. Non-compete agreements are common in various industries to protect a business's interests, trade secrets, and client relationships. However, their enforceability can vary by jurisdiction, and there are often legal restrictions on the extent to which non-compete agreements can be applied, as they must strike a reasonable balance between protecting a business's legitimate interests and an individual's right to pursue their livelihood. Before entering into or enforcing a non-compete agreement, seeking legal counsel to ensure that it complies with applicable laws and regulations in a specific jurisdiction is advisable.

Employee Oversight

Throughout the employment lifetime of personnel, managers should regularly review or audit the job descriptions, work tasks, privileges, and responsibilities for every staff member. It is common for work tasks and privileges to drift over time. Drifting job responsibilities or privilege creep can also result in security violations. Excess privileges held by a worker represent an increased risk to the organization. That risk includes the greater chance for mistakes to damage asset confidentiality, integrity, and availability (CIA) outside of the worker's actual responsibilities, greater ability for a disgruntled worker to cause harm on purpose, and greater ability for an attack that takes over a worker's account to cause harm.

Reviewing and then adjusting user capabilities to realign with the principle of least privilege is a risk-reduction strategy.

For some organizations, mostly those in the financial industry, a key part of this review process is enforcing mandatory vacations. *Mandatory vacations* are used as a peer review process. This process requires a worker to be away from the office and without remote access for one to two weeks per year. While the worker is on "vacation," a different worker (i.e., an auditor) performs the work duties (potentially with the same account), which makes it easier to verify the work tasks and privileges of workers while attempting to detect abuse, fraud, or negligence. This does not mean that passwords are shared. Mandatory vacation auditing could be implemented by redefining the target worker's account's password for the auditor, then redefining the password again when the worker returns. Or another account could be created for the auditor with the same access, permissions, and privileges as the worker's.

Other user and worker management and evaluation techniques include separation of duties, job rotation, and cross-training. These concepts are discussed in Chapter 16.

When several people work together to perpetrate a crime, it's called *collusion*. Employing the principles of separation of duties, restricted job responsibilities, mandatory vacations, job rotation, and cross-training reduces the likelihood that a coworker will be willing to collaborate on an illegal or abusive scheme because of the higher risk of detection, reporting, or whistleblowing. Collusion and other privilege abuses can also be reduced through strict monitoring of special privileges and privileged accounts, such as those of an administrator, root, and others.

For many job positions that are considered sensitive or critical, especially in medical, financial, government, and military organizations, periodic revaluation of employees may be needed. This could be a process that is just as thorough as the original background check and investigation performed when the individual was hired, or it may require performing only a few specific checks to confirm consistency in the person's qualifications as well as researching for any new information regarding disqualifications.

User behavior analytics (UBA) and *user and entity behavior analytics (UEBA)* are the concepts of analyzing the behavior of users, subjects,

visitors, customers, and so forth for some specific goal or purpose. The *E* in UEBA extends the analysis to include *entity* activities (i.e., devices, systems, networks, and applications) that take place but that are not necessarily directly linked or tied to a user's specific actions, but that can still correlate to a vulnerability, reconnaissance, intrusion, breach, or exploit occurrence. Information collected from UBA/UEBA monitoring can be used to improve personnel security policies, procedures, training, and related security oversight programs.

Offboarding, Transfers, and Termination Processes

Offboarding is the reverse of the onboarding process. *Offboarding* is the removal of an employee's identity from the IAM system once that person has left the organization. But offboarding can also be an element used when an *employee transfers* to a new position at the same organization, especially when shifting between departments, facilities, or geographic locations. Personnel transfers may be treated as a termination/rehire rather than a personnel move. This depends on the organization's policies and the means they have determined to best manage this change. Some of the elements that go into making the decision as to which procedure to use include whether the same user account will be retained, if their clearance will be adjusted, if their new work responsibilities are similar to the previous position, and if a “clean slate” account is required for auditing purposes in the new job position.

When a full offboarding is going to occur, whether as part of a termination/rehire transfer, a retirement, or a termination, this can include disabling and/or deleting the user account, revoking certificates, canceling access codes, and terminating other specifically granted privileges. It is common to deactivate the accounts of prior employees in order to retain the identity for auditing purposes for a few months. After the allotted time, if no incidents are discovered regarding the former employee's account, it can be deleted from the IAM completely. If the account is deleted prematurely, any logged events that are of a security concern no longer point to an actual account and thus can make tracking down further evidence of violations more complicated.



An internal employee transfer should not be used to move a problem employee into a different department rather than firing them. Consider the overall 5 Pillars of Information Security and benefit to the organization; if a person is not acceptable as an employee in one department, is it realistic to assume they would be in another? Rather than passing around the problem, the better option is to terminate the problematic employee, especially if direct training and coaching do not provide a resolution.

The offboarding process may also include informing security guards and other physical facility and property access management personnel to disallow entry to the former employee in the future.

The procedures for onboarding and offboarding should be clearly documented to ensure consistency of application and compliance with regulations or contractual obligations. Disclosure of these policies may need to be a standard element of the hiring process.

Numerous issues must be addressed when an employee must be terminated or offboarded. A strong relationship between the security department and HR is essential to maintain control and minimize risks during termination.

Terminations are typically unpleasant processes for all involved. However, they might be elevated to a neutral experience when well planned and scripted. The intent of a termination policy is to reduce the risk associated with employee termination while treating the person with respect. The termination meeting should take place with at least one witness, preferably a higher-level manager and/or a security guard. Once the employee has been informed of their release, they should be reminded of the liabilities and restrictions placed on the former employee based on the employment agreement, NDAs, and any other security-related documentation. During this meeting, all organization-specific identification, access, or security badges as well as devices, cards, keys, and access tokens should be collected (see Figure 2.1). The termination of an employee should be handled in a private and respectful manner. However, this does not mean that precautions should not be taken.

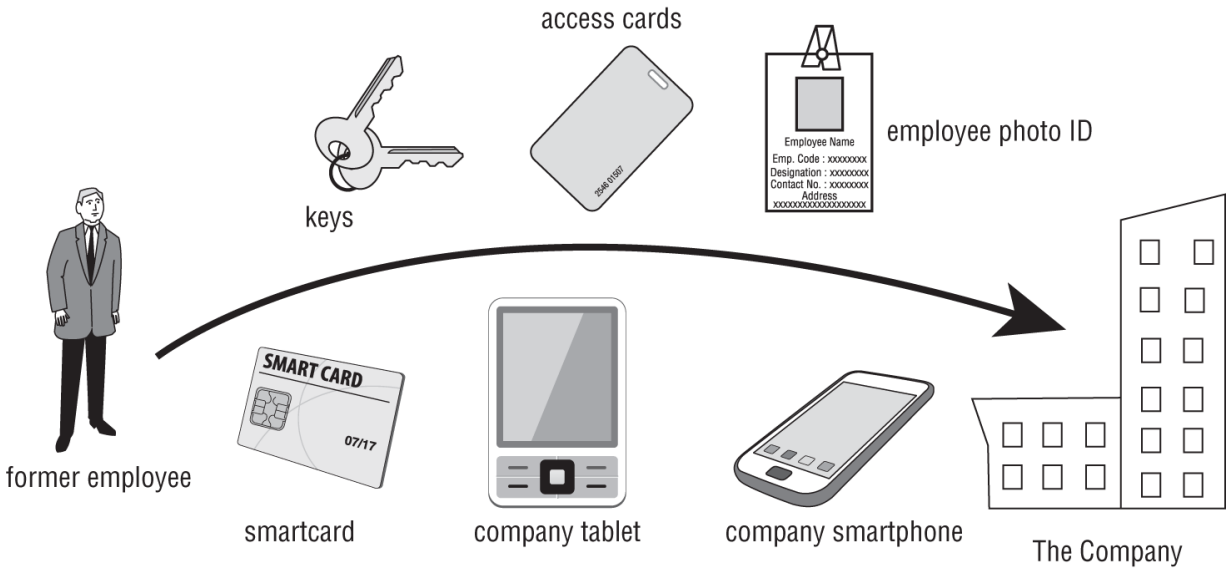


FIGURE 2.1 Former employees must return all company property.

For nonvoluntary terminations where there is a perceived risk of a confrontation, the termination process may need to be abrupt and attended by security guards. Any need to resolve HR issues, retrieve company equipment, review NDAs, and so forth can be handled afterward through an attorney.

For terminations that are expected to be professional and for voluntary separations (such as quitting, retiring, or taking extended leave), an additional process may be called an exit interview. An *exit interview* is normally done by an HR person who specializes in those interviews with the idea of learning from the employee's experience. The purpose of an exit interview is to understand why the employee is leaving, what their perspective is of the organization (its personnel, culture, process, etc.), and what they suggest could be done to improve conditions for current and future employees. Information learned from an exit interview may assist the organization with retaining employees through employment improvements and process/policy changes.

Whether an abrupt termination process is used or a cordial process was concluded, the now former employee should be escorted off the premises and not allowed to return to their work area without an escort for any reason.

The following list includes some other security issues that should be handled as soon as possible:

- Remove or deactivate the employee's user account at the same time as or just before they are notified of being terminated.
- Make sure the employee returns any organizational equipment or supplies from their vehicle or home.
- Arrange for a security department member to accompany the released employee while they gather their personal belongings from the work area.
- Inform all security personnel and anyone else who watches or monitors any entrance point to ensure that the former employee does not attempt to reenter the building without an escort.

Firing: Timing Is Everything

Firing an employee has become a complex process. That's why you need a well-designed termination process. However, it must be followed correctly every time. Unfortunately, this doesn't always happen. You might have heard of some fiasco caused by a botched termination procedure. Common examples include performing any of the following before the employee is officially informed of their termination (thus giving the employee prior warning of their termination):

- The IT department is requesting the return of a mobile device
- Deactivating a network user account
- Blocking a person's personal identification number (PIN) or smartcard for building entrance
- Revoking a parking pass
- Distributing a revised company organizational chart
- Positioning a new employee in their cubicle or workspace
- Allowing layoff information to be leaked to the media

Vendor, Consultant, and Contractor Agreements and Controls

Vendor, consultant, and contractor controls are used to define the levels of performance, expectation, compensation, and consequences for entities, persons, or organizations that are external to the primary organization.

Multiparty risk exists when several entities or organizations are involved in a project. The risk or threats are often due to the variations of objectives, expectations, timelines, budgets, and security priorities of those involved. Risk management is the processes of identifying risks, assessing those risks, then selecting responses to risks that need mitigation. The risk response strategies implemented by one party may in fact cause additional risks against or from another party. Often a risk management governing body

must be established to oversee the multiparty project and enforce consistent security parameters for the member entities, at least as their interactions relate to the project.

Using service-level agreements (SLAs) is a means to ensure that organizations providing services maintain an appropriate level of service agreed on by the service provider, vendor, or contractor and the customer organization. You'd be wise to put SLAs in place for any data circuits, applications, information processing systems, databases, or other critical components that are vital to your organization's continued viability. SLAs are important when using any type of third-party service provider, including cloud services. SLAs also commonly include financial and other contractual remedies that kick in if the agreement is not maintained. For example, if a critical circuit is down for more than 15 minutes, the service provider might agree to waive all charges on that circuit for one week.

SLAs and vendor, consultant, and contractor controls are an important part of risk reduction and risk avoidance. By clearly defining the expectations and penalties for external parties, everyone involved knows what is expected of them and what the consequences are in the event of a failure to meet those expectations. Although it may be very cost-effective to use outside providers for a variety of business functions or services, it does increase potential risk by expanding the potential attack surface and range of vulnerabilities. SLAs should include a focus on protecting and improving security in addition to ensuring quality and timely services at a reasonable price. Some SLAs are set and cannot be adjusted, whereas, with others, you may have significant influence over their content. You should ensure that an SLA supports the tenets of your security policy and infrastructure rather than being in conflict with them, which could introduce weak points, vulnerabilities, or exceptions.

Outsourcing is the term often used to describe the use of an external third party, such as a vendor, consultant, or contractor, rather than performing the task or operation in-house. Outsourcing can be used as a risk response option known as transference or assignment (see the “Risk Responses” section later in this chapter). However, though the risk of operating a function internally is transferred to a third party, other risks are taken on by using a third party. This aspect needs to be evaluated as to whether it is a

benefit or a consequence of the SLA. For more on service-level agreements (SLAs), see Chapter 16.

Vendors, consultants, and contractors also represent an increase in the risk of trade secret theft or espionage. Outsiders often lack organizational loyalty that internal employees typically have; thus, the temptation to take advantage of intellectual property access opportunities may seem easier or less of an internal conflict to a perpetrator. For more on espionage, see Chapter 17, "Preventing and Responding to Incidents."

Some organizations may benefit from a *vendor management system (VMS)*. A VMS is a software solution that assists with managing and procuring staffing services, hardware, software, and other needed products and services. A VMS can offer ordering convenience, order distribution, training, consolidated billing, and more. In regard to security, a VMS can potentially keep communications and contracts confidential, require encrypted and authenticated transactions, and maintain a detailed activity log of events related to vendors and suppliers. A VMS is particularly valuable for organizations that work with a large number of vendors and require a centralized and systematic approach to vendor relationship management. A VMS helps enhance vendor performance, reduce costs, manage risks, and maintain compliance while fostering collaborative and productive relationships with external partners.



Compliance is the act of conforming to or adhering to rules, policies, regulations, standards, or requirements. Compliance is an important concern of security governance.

Understand and Apply Risk Management Concepts

Risk management is a detailed process of identifying factors that could damage or disclose assets, evaluating those factors in light of asset value and countermeasure cost, and implementing cost-effective solutions for mitigating or reducing risk. The overall process of risk management is used

to develop and implement information security strategies that support the mission of the organization. The result of performing risk management for the first time is the skeleton of a security policy. Subsequent risk management events are used to improve and sustain an organization's security infrastructure over time as internal and external conditions change.

The primary goal of risk management is to reduce risk to an acceptable level. What that level actually is depends on the organization, the value of its assets, the size of its budget, and many other factors. One organization might consider something to be an acceptable risk, whereas another organization might consider the very same thing to be an unreasonably high level of risk. It is impossible to design and deploy a totally risk-free environment; however, significant risk reduction is possible, often with modest effort.

Risks to an IT infrastructure are not all computer-based. In fact, many risks come from non-IT sources. It is important to consider all possible risks when performing risk evaluation, including accidents, natural disasters, financial threats, civil unrest, pandemics, physical threats, technical exploitations, and social engineering attacks. Failing to evaluate and respond to all forms of risk properly will leave a company vulnerable.

Risk management is composed of two primary elements:

- *Risk assessment* or *risk analysis* is the examination of an environment for risks, evaluating each threat event as to its likelihood of occurring and the severity of the damage it would cause if it did occur, and assessing the cost of various countermeasures for each risk. This results in a sorted criticality prioritization of risks. From there, risk response takes over.
- *Risk response* involves evaluating countermeasures, safeguards, and security controls using a cost/benefit analysis; adjusting findings based on other conditions, concerns, priorities, and resources; and providing a proposal of response options in a report to senior management. Based on management decisions and guidance, the selected responses can be implemented into the IT infrastructure and integrated into the security policy documentation. This activity is also known as risk

reduction or risk mitigation, which is the overall goal of risk management.

A concept related to risk management is risk awareness. *Risk awareness* is the effort to increase the knowledge of risks within an organization. This includes understanding the value of assets, inventorying the existing threats that can harm those assets, and the responses selected and implemented to address the identified risk. Risk awareness helps to inform an organization about the importance of abiding by security policies and the consequences of security failures.

Risk Terminology and Concepts

Risk management employs a vast terminology that must be clearly understood. This section defines and discusses all the important risk-related terminology:

Asset An *asset* is anything used in a business process or task. If an organization relies on a person, place, or thing, whether tangible or intangible, then it is an asset.

Asset Valuation *Asset valuation* is the value assigned to an asset based on a number of factors, including importance to the organization, use in critical processes, actual cost, and nonmonetary expenses/costs (such as time, attention, productivity, and research and development). When performing a math-based risk evaluation (i.e., quantitative; see the "Quantitative Risk Analysis" section later in this chapter), a dollar figure is assigned as the asset value (AV).

Threats Any potential occurrence that may cause an undesirable or unwanted outcome for an organization or for a specific asset is a *threat*. Threats are any action or inaction that could cause damage, destruction, alteration, loss, or disclosure of assets or that could block access to or prevent maintenance of assets. They can be intentional or accidental. They can originate from inside or outside. You can loosely think of a threat as a weapon that could cause harm to a target.

Threat Agent/Actors *Threat agents* or *threat actors* intentionally exploit vulnerabilities. Threat agents are usually people, but they could

also be programs, hardware, or systems. Threat agents wield threats to cause harm to targets. Aka attacker, adversary, or bad guy.

Threat Events *Threat events* are accidental occurrences and intentional exploitations of vulnerabilities. They can also be natural or person-made. Threat events include fire, earthquake, flood, system failure, human error (due to a lack of training or ignorance), and power outage.

Threat Vector A *threat vector* or *attack vector* is the path or means by which an attack or threat agent can gain access to a target to cause harm. Threat vectors can include email, web surfing, external drives, Wi-Fi networks, physical access, mobile devices, cloud, social media, supply chain, removable media, and commercial software.

Vulnerability The weakness in an asset or the absence or the weakness of a safeguard or countermeasure is a *vulnerability*. In other words, a vulnerability is a flaw, loophole, oversight, error, limitation, frailty, or susceptibility that enables a threat to cause harm.

Exposure *Exposure* is being susceptible to asset loss because of a threat; there is the possibility that a vulnerability can or will be exploited by a threat agent or event. Exposure doesn't mean that a realized threat (an event that results in loss) is actually occurring, just that there is the potential for harm to occur. The quantitative risk analysis value of the exposure factor (EF) is derived from this concept.

Risk

Risk is the possibility or likelihood that a threat will exploit a vulnerability to cause harm to an asset and the severity of damage that could result. The more likely it is that a threat event will occur, the greater the risk. The greater the amount of harm that could result if a threat is realized, the greater the risk. Every instance of exposure is a risk. When written as a conceptual formula, risk can be defined as follows:

risk = threat * vulnerability

or

risk = probability of harm * severity of harm

Safeguards A *safeguard*, *security control*, protection mechanism, or *countermeasure* is anything that removes or reduces a vulnerability or protects against one or more specific threats. This concept is also known as a risk response. A safeguard is any action or product that reduces risk through the elimination or lessening of a threat or a vulnerability. Safeguards are the means by which risk is mitigated or resolved. It is important to remember that a safeguard need not involve purchasing a new product; reconfiguring existing elements and removing elements from the infrastructure are also valid safeguards or risk responses.

Attack An *attack* is intentionally exploiting a vulnerability by a threat agent to cause damage, loss, or disclosure of assets, whether or not the attempt is successful. An attack can also be viewed as any violation or failure to adhere to an organization's security policy. A malicious event does not have to succeed in violating security to be considered an attack.

Breach A *breach*, intrusion, or penetration is when a security mechanism is bypassed or thwarted by a threat agent. A breach is a successful attack.

Hazard

A *hazard* refers to a potential source or situation that has the capability to cause harm, loss, damage, injury, or adverse consequences to an organization, its assets, individuals, or the environment.

Some of these risk terms and elements are clearly related, as shown in Figure 2.2. Threats exploit vulnerabilities, which results in exposure. Exposure is a risk, and risk is mitigated by safeguards. Safeguards protect assets that are endangered by threats.

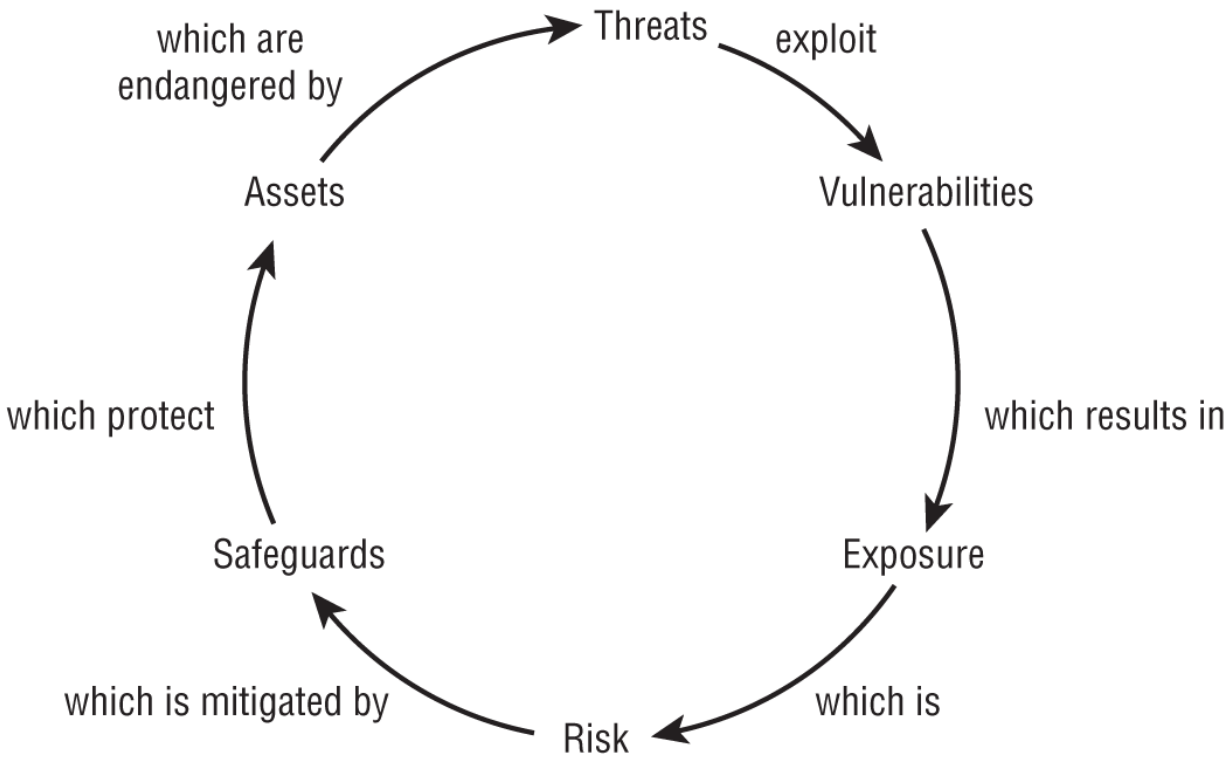


FIGURE 2.2 The cyclical relationships of risk elements

There are many approaches to risk assessment. Some are initiated by evaluating threats, whereas others focus first on assets. Whether a risk assessment starts with inventorying threats, then looks for assets that could be harmed, or starts with inventorying assets, then looks for threats that could cause harm, both approaches result in asset-threat pairings that then need to be risk evaluated. Both approaches have merit, and organizations should shift or alternate their risk assessment processes between these methods. When focusing first on threats, a broader range of harmful issues may be considered, without being limited to the context of the assets. But this may result in the collection of information about threats that the organization does not need to worry about as they don't have the assets or vulnerabilities that the threat focuses on. When focusing first on assets, the entirety of organizational resources can be discovered without being limited to the context of the threat list. But this may result in spending time evaluating assets of very low value and low risk (which would or will be defined as acceptable risk), which may increase the overall time involved in risk assessment.

Risk perspectives, also known as risk management perspectives or approaches, are different lenses through which organizations and individuals can view and address risks. Each perspective emphasizes certain aspects of risk and can guide decision-making, risk assessment, and mitigation strategies. There are innumerable options of risk perspective, including asset, outcome, vulnerability, threat, financial, strategic, operational, compliance, legal, reputational, supply chain, third-party, and workforce. Each risk perspective offers a unique way to approach risk management and provides insights into different aspects of risk. An effective risk management strategy may incorporate elements of multiple perspectives to comprehensively assess and address risks based on their impact, likelihood, and the organization's specific objectives.

The general idea of a threat-based risk assessment was discussed in Chapter 1, “Security Governance Through Principles and Policies”—that is, threat modeling. The discussion of risk assessment in this chapter will focus on an asset-based risk assessment approach.

Asset Valuation

An asset-based or asset-initiated risk analysis starts with inventorying all organizational assets. Once that inventory is complete, a valuation needs to be assigned to each asset. The evaluation or appraisal of each asset helps establish its importance or criticality to the business operations. If an asset has no value, there is no need to provide protection for it. A primary goal of risk analysis is to ensure that only cost-effective safeguards are deployed. It makes no sense to spend $100,000 protecting an asset that is worth only $1,000. Therefore, the value of an asset directly affects and guides the level of safeguards and security deployed to protect it. As a rule, the annual costs of safeguards should not exceed the potential annual cost of asset value loss.

When the cost of an asset is evaluated, there are many aspects to consider. The goal of asset valuation is to assign to an asset a specific dollar value that encompasses tangible costs as well as intangible ones. Determining the exact value of an asset is often difficult, if not impossible, but nevertheless, a specific value must be established in order to perform quantitative mathematical calculations. (Note that the discussion of qualitative versus quantitative risk analysis later in this chapter may clarify this issue; see the

“Risk Assessment/Analysis” section.) Improperly assigning value to assets can result in failing to protect an asset or implementing financially infeasible safeguards properly. The following list includes tangible and intangible issues that contribute to the valuation of assets:

- Purchase cost
- Development cost
- Administrative or management cost
- Maintenance or upkeep cost
- Cost of acquiring an asset
- Cost to protect or sustain an asset
- Value to owners and users
- Value to competitors
- Intellectual property or equity value
- Market valuation (sustainable price)
- Replacement cost
- Productivity enhancement or degradation
- Operational costs of asset presence and loss
- Liability of asset loss
- Usefulness
- Relationship to research and development

Assigning or determining the value of assets to an organization can fulfill numerous requirements by:

- Serving as the foundation for performing a cost/benefit analysis of asset protection when performing safeguard selection
- Serving as a means for evaluating the cost-effectiveness of safeguards and countermeasures

- Providing values for insurance purposes and establishing an overall net worth or net value for the organization
- Helping senior management understand exactly what is at risk within the organization
- Preventing negligence of due care/due diligence and encouraging compliance with legal requirements, industry regulations, and internal security policies

If threat-based or threat-initiated risk analysis is being performed, asset valuation occurs after the organization discovers threats and identifies vulnerable assets to those threats.

Identify Threats and Vulnerabilities

An essential part of risk management is identifying and examining threats. This involves creating an exhaustive list of all possible threats for the organization's identified assets. The list should include threat agents as well as threat events. Keep in mind that threats can come from anywhere. Threats to IT are not limited to IT sources or concepts. When compiling a list of threats, be sure to consider threats from a wide range of sources.

For an expansive and formal list of threat examples, concepts, and categories, consult National Institute of Standards and Technology (NIST) Special Publication (SP) 800-30r1 Appendix D, “Threat sources,” and Appendix E, “Threat events.” For coverage of threat modeling, see Chapter 1.

In most cases, a team rather than a single individual should perform risk assessment and analysis. Also, the team members should be from various departments within the organization. It is not usually a requirement that all team members be security professionals or even network/system administrators. The team's diversity based on the organization's demographics will help to broadly identify and address a wider range of threats and risks.

The Consultant Cavalry

Risk assessment is a highly involved, detailed, complex, and lengthy process. Often risk analysis cannot be properly handled by existing employees because of the size, scope, or liability of the risk; thus, many organizations bring in risk management consultants to perform this work. This provides a high level of expertise, does not bog down employees, and can be a more reliable measurement of real-world risk. However, even risk management consultants do not perform risk assessment and analysis on paper only; they typically employ risk assessment software. This software streamlines the overall task, provides more reliable results, and produces standardized reports that are acceptable to insurance companies, boards of directors, and so on.

Risk Assessment/Analysis

Risk management is primarily the responsibility of upper management. However, upper management typically assigns risk analyses and risk response modeling tasks to a team from the IT and security departments. The results of the risk assessment team will be submitted as a proposal to upper management. Upper management will make the final decisions as to which responses are implemented by the organization.

It is the responsibility of upper management to initiate and support risk analysis and assessment by defining the scope and purpose of the endeavor. All risk assessments, results, decisions, actions, and outcomes must be understood and approved by upper management as an element in providing prudent due care/due diligence.

“Prudent actions” and “reasonable actions” are terms used in legal and ethical contexts to describe different standards of behavior or decision-making, especially related to risk management. While these standards share some similarities, they have distinct meanings. Prudent actions refer to actions or decisions that are marked by a high degree of caution, care, and foresight. They are characterized by careful consideration of potential risks, a focus on preventing harm, and a commitment to acting in a manner that is consistent with established best practices or industry standards. Prudent

actions often involve taking additional precautions beyond what might be considered “reasonable” to ensure the highest level of safety and protection. Prudence implies a proactive and diligent approach to decision-making. In a legal context, acting prudently may serve to protect an individual or organization from liability in cases where a higher standard of care is expected.

Reasonable actions refer to actions or decisions that are in line with what a person of ordinary prudence and judgment would do in similar circumstances. These actions are based on the idea of acting in a manner that is sensible, rational, and consistent with societal norms and expectations. Reasonable actions are a standard often used in legal and ethical contexts to assess whether an individual's behavior or decisions meet a minimum threshold of acceptability. In a legal context, “acting reasonably” often serves as a benchmark to determine whether someone has met their duty of care, particularly in cases where negligence or liability is in question.

Therefore, prudent actions are characterized by an above-average level of caution and diligence, whereas reasonable actions are aligned with common expectations and societal norms. The choice between these standards may depend on the specific circumstances, legal requirements, and ethical considerations, as well as the degree of care and caution expected in a given situation.

All IT systems have risk. All organizations have risk. Every task performed by a worker has risk. There is no way to eliminate 100 percent of all risks. Instead, upper management must decide which risks are acceptable and which are not. Determining which risks are acceptable requires detailed and complex asset and risk assessments, as well as a thorough understanding of the organization's budget, internal expertise and experience, business conditions, and many other internal and external factors. What is deemed acceptable to one organization may not be viewed the same way by another. For example, you might think that losing $100 is a significant loss and impact to your monthly personal budget, but the wealthy might not even realize if they lost or wasted hundreds or thousands of dollars. Risk is personal, or at least specific to an organization based on its assets, its threats, its threat agents/actors, and its risk tolerance.

Scope refers to the extent or boundaries of a risk management process, project, or assessment. It defines what is included and what is excluded in the risk management efforts. Determining the scope is a critical step in effectively managing and addressing risks, as it helps organizations focus their resources and efforts on the most relevant areas. Establishing a well-defined scope is essential for effective risk management because it helps organizations prioritize their efforts, allocate resources efficiently, and ensure that risks are adequately assessed and mitigated in the areas that matter most. A clear and well-communicated scope also reduces ambiguity and ensures that all relevant parties are on the same page regarding the objectives and boundaries of the risk management process.

Once an inventory of threats and assets (or assets and threats) is developed based on a defined scope, each asset-threat pairing must be individually evaluated and its related risk calculated or assessed. There are two primary risk assessment methodologies: quantitative and qualitative. *Quantitative risk analysis* assigns real dollar figures to the loss of an asset and is based on mathematical calculations. *Qualitative risk analysis* assigns subjective and intangible values to the loss of an asset and takes into account perspectives, feelings, intuition, preferences, ideas, and gut reactions. Both methods are necessary for a complete perspective on organizational risk. Most environments employ a hybrid of both risk assessment methodologies in order to gain a balanced view of their security concerns.

The goal of risk assessment is to identify risks (based on asset-threat pairings) and rank them in order of criticality. This risk criticality prioritization is needed in order to guide the organization in optimizing the use of their limited resources on protections against identified risks, from the most significant to those just above the risk acceptance threshold.

The two risk assessment approaches (quantitative and qualitative) can be seen as distinct and separate concepts or endpoints on a sliding scale. As discussed in Chapter 1, a basic probability versus potential damage 3×3 matrix relies on an innate understanding of the assets and threats and relies on a judgment call of the risk analyst to decide whether the likelihood and severity are low, medium, or high. This is likely the simplest form of qualitative assessment. It requires minimum time and effort. However, if it fails to provide the needed clarity or distinction of criticality prioritization, then a more in-depth approach should be undertaken. A 5×5 matrix or even

larger could be used. However, each increase in matrix size requires more knowledge, more research, and more time to assign a level to probability and severity properly. At some point, the evaluation shifts from being mostly subjective qualitative to more substantial quantitative.

Another perspective on the two risk assessment approaches is that a qualitative mechanism can be used first to determine whether a detailed and resource/time-expensive quantitative mechanism is necessary. An organization can also perform both approaches and use them to adjust or modify each other; for example, qualitative results can be used to fine-tune quantitative priorities.

Qualitative Risk Analysis

Qualitative risk analysis is more scenario-based, perception-based, or gut reaction-based than it is mathematically-based. Rather than assigning exact dollar figures to possible losses, you rank threats on a relative scale to evaluate their risks, costs, and effects. Since a purely quantitative risk assessment is not possible, balancing the results of a quantitative analysis is essential. The method of combining quantitative and qualitative analysis into a final assessment of organizational risk is known as *hybrid assessment* or *hybrid analysis*. The process of performing qualitative risk analysis involves judgment, intuition, and experience. You can use many techniques to perform qualitative risk analysis:

- Brainstorming
- Storyboarding
- Focus groups
- Surveys
- Questionnaires
- Checklists
- One-on-one meetings
- Interviews
- Scenarios
- Delphi technique

Determining which mechanism to employ is based on the culture of the organization and the types of risks and assets involved. It is common for several methods to be employed simultaneously and their results compared and contrasted in the final risk analysis report to upper management. Two of these that you need to be more aware of are scenarios and the Delphi technique.

Scenarios

The basic process for all these mechanisms involves the creation of scenarios. A scenario is a written description of a single major threat. The description focuses on how a threat would be instigated and what effects its occurrence could have on the organization, the IT infrastructure, and specific assets. Generally, the scenarios are limited to one page of text to keep them manageable. For each scenario, several safeguards are described that would completely or partially protect against the major threat discussed in the scenario. The analysis participants then assign to the scenario a threat level, a loss potential, and the advantages of each safeguard. These assignments can be simple—such as High, Medium, and Low, or a basic number scale of 1 to 10—or they can be detailed essay responses. The responses from all participants are then compiled into a single report that is presented to upper management. For examples of reference ratings and levels, please see Tables D-3, D-4, D-5, D-6, and E-4 in NIST SP 800-30 Rev.1: csrc.nist.gov/pubs/sp/800/30/r1/final.

The usefulness and validity of a qualitative risk analysis improves as the number and diversity of the participants in the evaluation increases. Whenever possible, include one or more people from each level of the organizational hierarchy, from upper management to end users. It is also important to include a cross-section from each major department, division, office, or branch.

Delphi Technique

The Delphi technique is probably the primary mechanism on the previous list that is not immediately recognizable and understood. The *Delphi technique* is simply an anonymous feedback-and-response process used to enable a group to reach an anonymous consensus. Its primary purpose is to elicit honest and uninfluenced responses from all participants, while

minimizing the influence of bias and discrimination. The participants are usually gathered in a single meeting room. To each request for feedback, each participant writes down their response on paper or through digital messaging services anonymously. The results are compiled and presented to the group for evaluation. The process is repeated until a consensus is reached. The goal or purpose of the Delphi technique is to facilitate the evaluation of ideas, concepts, and solutions on their own merit without the discrimination that often occurs based on who the idea comes from.

Quantitative Risk Analysis

The quantitative method results in concrete probability indications or a numeric indication of relative risk potential. That means the end result is a report that has dollar figures for levels of risk, potential loss, cost of countermeasures, and value of safeguards. This report is usually fairly easy to understand, especially for anyone with knowledge of spreadsheets and budget reports. Think of quantitative analysis as the act of assigning a quantity to risk—in other words, placing a dollar figure on each asset and threat impact. However, a purely quantitative analysis is not sufficient—not all elements and aspects of the analysis can be accurately quantified because some are qualitative, subjective, or intangible.

The process of quantitative risk analysis starts with asset valuation and threat identification (which can be performed in any order). This results in asset-threat pairings that need to have estimations of harm potential/severity and frequency/likelihood assigned or determined. This information is then used to calculate various cost functions that are used to evaluate safeguards.

The major steps or phases in quantitative risk analysis are as follows (see Figure 2.3, with terms and concepts defined after this list of steps):

1. Inventory assets and assign a value (asset value [AV]).
2. Research each asset and produce a list of all possible threats to each individual asset. This results in asset-threat pairings.
3. For each asset-threat pairing, calculate the exposure factor (EF).
4. Calculate the single loss expectancy (SLE) for each asset-threat pairing.

5. Perform a threat analysis to calculate the likelihood of each threat being realized within a single year—that is, the annualized rate of occurrence (ARO).
6. Derive the overall loss potential per threat by calculating the annualized loss expectancy (ALE).
7. Research countermeasures for each threat and then calculate the changes to ARO, EF, and ALE based on an applied countermeasure.
8. Perform a cost/benefit analysis of each countermeasure for each threat for each asset. Select the most appropriate response to each threat.

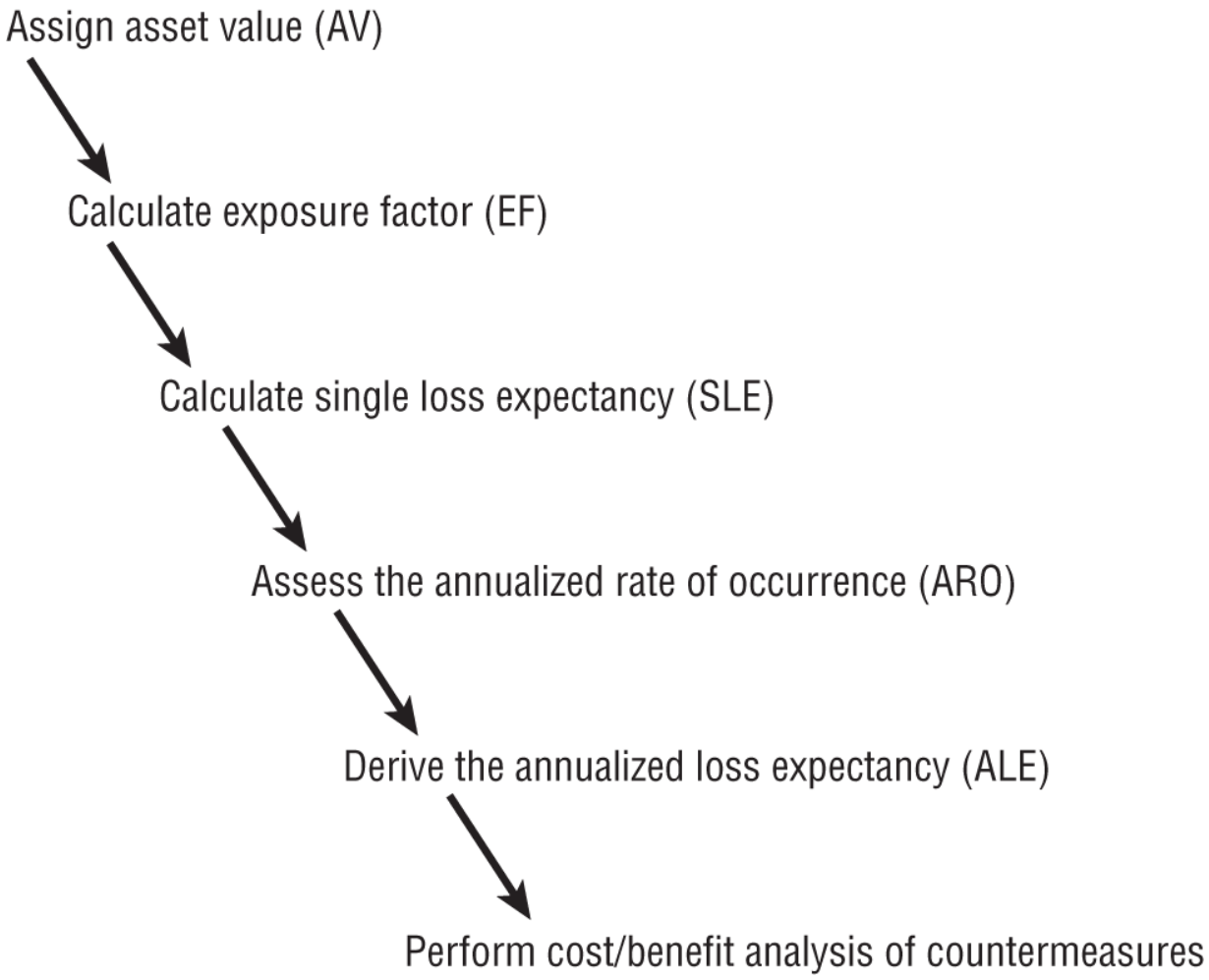


FIGURE 2.3 The six major elements of quantitative risk analysis

The cost functions associated with quantitative risk analysis include the following:

Exposure Factor The *exposure factor (EF)* represents the percentage of loss that an organization would experience if a specific asset were violated by a realized risk. The EF can also be called the *loss potential*. In most cases, a realized risk does not result in the total loss of an asset. The EF simply indicates the expected overall asset value loss because of a single realized risk. The EF is usually small for assets that are easily replaceable, such as hardware. It can be very large for assets that are irreplaceable or proprietary, such as product designs or a database of customers. The EF is expressed as a percentage. The EF is determined by using historical internal data, performing statistical analysis, consulting public or subscription risk ledgers/registers, working with consultants, or using a risk management software solution.

Single-Loss Expectancy The *single-loss expectancy (SLE)* is the potential loss associated with a single realized threat against a specific asset. It indicates the potential amount of loss an organization would or could experience if an asset were harmed by a specific threat occurring.

The SLE is calculated using the following formula:

SLE = asset value (AV) * exposure factor (EF)

or more simply:

SLE = AV * EF

The SLE is expressed in a dollar value. For example, if an asset is valued at $200,000 and it has an EF of 45 percent for a specific threat, then the SLE of the threat for that asset is $90,000. It is not always necessary to calculate an SLE, as the ALE is the most commonly needed value in determining criticality prioritization. Thus, sometimes, during risk calculation, SLE may be skipped entirely.

Annualized Rate of Occurrence The *annualized rate of occurrence (ARO)* is the expected frequency with which a specific threat or risk will occur (that is, become realized) within a single year. The ARO can range from a value of 0.0 (zero), indicating that the threat or risk will never be realized, to a very large number, indicating that the threat or risk occurs often. Calculating the ARO can be complicated. It can be

derived by reviewing historical internal data, performing statistical analysis, consulting public or subscription risk ledgers/registers, working with consultants, or using a risk management software solution. The ARO for some threats or risks is calculated by multiplying the likelihood of a single occurrence by the number of users who could initiate the threat. ARO is also known as a probability determination. Here's an example: the ARO of an earthquake in Tulsa may be .00001, whereas the ARO of an earthquake in San Francisco may be .03 (for a 6.7+ magnitude), or you can compare the ARO of an earthquake in Tulsa of .00001 to the ARO of an email virus in an office in Tulsa of 10,000,000.

Annualized Loss Expectancy

The *annualized loss expectancy (ALE)* is the possible yearly loss of all instances of a specific realized threat against a specific asset. The ALE is calculated using the following formula:

ALE = single loss expectancy (SLE) * annualized rate of occurrence (ARO)

or

ALE = asset value (AV) * exposure factor (EF) * annualized rate of occurrence (ARO)

or more simply:

ALE = SLE * ARO

or

ALE = AV * EF * ARO

For example, if the SLE of an asset is $90,000 and the ARO for a specific threat (such as total power loss) is .5, then the ALE is $45,000. If the ARO for a specific threat (such as a compromised user account) is 15 for the same asset, then the ALE would be $1,350,000.

The task of calculating EF, SLE, ARO, and ALE for every asset and every threat/risk is a daunting one. Fortunately, quantitative risk assessment software tools can simplify and automate much of this process. These tools produce an asset inventory with valuations and then, using predefined

AROs along with some customizing options (industry, geography, IT components, and so on), produce risk analysis reports.

Once an ALE is calculated for each asset-threat pairing, then the entire collection should be sorted from largest ALE to smallest. Although the actual number of the ALE is not an absolute number (it is an amalgamation of intangible and tangible value multiplied by a future prediction of loss multiplied by a future prediction of likelihood), it does have relative value. The largest ALE is the biggest problem the organization is facing and, thus, the first risk to be addressed in risk response.

The "Cost vs. Benefit of Security Controls" section, later in this chapter, discusses the various formulas associated with quantitative risk analysis that you should be familiar with.

Both the quantitative and qualitative risk analysis mechanisms offer useful results. However, each technique involves a unique method of evaluating the same set of assets and risks. Prudent due care requires that both methods be employed in order to obtain a balanced perspective on risk. Table 2.1 describes the benefits and disadvantages of these two systems.

TABLE 2.1 Comparison of quantitative and qualitative risk analysis

Characteristic	Qualitative	Quantitative
Employs math functions	No	Yes
Uses cost/benefit analysis	May	Yes
Requires estimation	Yes	Some
Supports automation	No	Yes
Involves a high volume of information	No	Yes
Is objective	Less so	More so
Relies substantially on opinion	Yes	No
Requires significant time and effort	Sometimes	Yes
Offers useful and meaningful results	Yes	Yes

At this point, the risk management process shifts from risk assessment to risk response. Risk assessment is used to identify the risks and set criticality priorities, and then risk response is used to determine the best defense for

each identified risk. However, identified risks need to be prioritized before any response strategies can be selected or implemented.

Prioritization in risk management is the process of systematically ranking and organizing risks based on their significance, potential impact, likelihood, or other relevant criteria. The objective of prioritization is to identify and focus on the most critical or high-priority risks so that limited resources, time, and attention can be allocated effectively to address them. Prioritization is a fundamental step in the risk management process, helping organizations make informed decisions about risk mitigation and risk response strategies. Risk prioritization (or criticality prioritization) can be as simple as ordering risks from worst to least unfavorable or sorting ALEs from largest to smallest. However, complex and integrated risk analysis methods may integrate qualitative and quantitative elements together, making the prioritization process a less than simple process.

Risk Responses

Whether a quantitative or qualitative risk assessment was performed, there are many elements of risk response that apply equally to both approaches. Once the risk analysis is complete, management must address each specific risk. There are several possible responses to risk:

- Mitigation or reduction
- Assignment or transfer
- Deterrence
- Avoidance
- Acceptance
- Reject or ignore

These risk responses are all related to an organization's risk appetite and risk tolerance. *Risk appetite* is the total amount of risk that an organization is willing to shoulder in aggregate across all assets. *Risk capacity* is the level of risk an organization is able to shoulder. An organization's desired risk appetite may be greater than its actual capacity. *Risk tolerance* is the amount or level of risk that an organization will accept per individual asset-

threat pair. This is often related to a risk target, which is the preferred level of risk for a specific asset-threat pairing. A *risk limit* is the maximum level of risk above the risk target that will be tolerated before further risk management actions are taken.

You need to know the following information about the possible risk responses:

Risk Mitigation *Reducing risk*, or *risk mitigation*, is the implementation of safeguards, security controls, and countermeasures to reduce and/or eliminate vulnerabilities or block threats. Deploying encryption and using firewalls are common examples of risk mitigation or reduction. Elimination of an individual risk can sometimes be achieved, but typically, some risk remains even after mitigation or reduction efforts.

Risk Assignment *Assigning risk* or *transferring risk* is the placement of the responsibility of loss due to a risk onto another entity or organization. Purchasing cybersecurity insurance or traditional insurance and outsourcing are common forms of assigning or transferring risk. Aka assignment of risk and transference of risk.

Risk Deterrence *Risk deterrence* is the process of implementing deterrents to would-be violators of security and policy. The goal is to convince a threat agent not to attack. Some examples include implementing auditing, security cameras, and warning banners; using security guards; and making it known that the organization is willing to cooperate with authorities and prosecute those who participate in cybercrime.

Risk Avoidance *Risk avoidance* is the process of selecting alternate options or activities that have less associated risk than the default, common, expedient, or cheap option. For example, choosing to fly to a destination instead of driving to it is a form of risk avoidance. Another example is to locate a business in Arizona instead of Florida to avoid hurricanes. The risk is avoided by eliminating the risk cause. A business leader terminating a business endeavor because it does not align with organizational objectives and that has a high risk-versus-reward ratio is also an example of risk avoidance.

Risk Acceptance *Accepting risk*, or acceptance of risk, is the result after a cost/benefit analysis shows countermeasure costs would outweigh the possible cost of loss due to a risk. It also means that management has agreed to accept the consequences and the loss if the risk is realized. In most cases, accepting risk requires a clearly written statement that indicates why a safeguard was not implemented, who is responsible for the decision, and who will be responsible for the loss if the risk is realized, usually in the form of a document signed by senior management.

Risk Rejection An unacceptable possible response to risk is to *reject risk* or *ignore risk*. Denying that a risk exists and hoping that it will never be realized are not valid or prudent due care/due diligence responses to risk. Rejecting or ignoring risk may be considered negligence in court.

Inherent risk is the level of natural, native, or default risk in an environment, system, or product before any risk management efforts are performed. Inherent risk can exist due to the supply chain, developer operations, design and architecture of a system, or an organization's knowledge and skill base. Inherent risk is also known as *initial risk* or *starting risk*. This is the risk that is identified by the risk assessment process.

Once safeguards, security controls, and countermeasures are implemented, the risk that remains is known as residual risk. *Residual risk* consists of threats to specific assets against which upper management chooses not to implement a response. In other words, residual risk is the risk that management has chosen to accept rather than mitigate. In most cases, the presence of residual risk indicates that the cost/benefit analysis showed that the available safeguards were not cost-effective deterrents.

Total risk is the amount of risk an organization would face if no safeguards were implemented. A conceptual formula for total risk is as follows:

threats and vulnerabilities and asset value = total risk

The difference between total risk and residual risk is known as the controls gap. The *controls gap* is the amount of risk that is reduced by implementing

safeguards. A conceptual formula for residual risk is as follows:

total risk – controls gap = residual risk

As with risk management in general, handling risk is not a onetime process. Instead, security must be continually maintained and reaffirmed. In fact, repeating the risk assessment and risk response processes is a necessary function to assess the completeness and effectiveness of the security program over time. Additionally, it helps locate deficiencies and areas where change has occurred. Since security changes over time, reassessing on a periodic basis is essential to maintaining reasonable security.

Control risk is the risk that is introduced by the introduction of the countermeasure to an environment. Most safeguards, security controls, and countermeasures are themselves some sort of technology. No technology is perfect, and no security is perfect, so some vulnerability exists in regard to the control itself. Although a control may reduce the risk of a threat to an asset, it may also introduce a new risk of a threat that can compromise the control itself. Thus, risk assessment and response must be an iterative operation that looks back on itself to make continuous improvements.

Cybersecurity Insurance

Cybersecurity insurance, also known as cyber insurance or cyber risk insurance, is a type of insurance policy that provides coverage and financial protection to organizations or individuals in the event of cyber-related incidents, data breaches, or cyberattacks. This form of insurance is designed to help mitigate the financial and legal consequences of cybersecurity breaches, which can result in data loss, financial loss, legal liabilities, and reputational damage. It is a form of risk assignment response.

Key features and aspects of cybersecurity insurance include:

- *Coverage for data breaches.* Cybersecurity insurance typically covers the costs associated with data breaches, including expenses related to notifying affected individuals, credit monitoring services, and the costs of investigating and mitigating the breach.
- *Financial loss protection.* It provides coverage for financial losses resulting from cyberattacks, such as theft of funds, fraudulent

transactions, and extortion payments demanded by cybercriminals.

- *Legal liabilities.* Cyber insurance can cover legal expenses and liability costs associated with cybersecurity incidents, including lawsuits, regulatory fines, and penalties for noncompliance with data protection regulations.
- *Reputation management.* Some policies include coverage for expenses related to reputation management and public relations efforts to rebuild trust with customers and stakeholders after a data breach.
- *Business interruption.* Cyber insurance may offer coverage for losses related to business interruption caused by a cyber incident. This can include income loss due to system downtime.
- *Ransomware protection.* Many policies include coverage for ransomware attacks and may cover ransom payments, investigation costs, and remediation.
- *Forensic services.* Insurers often provide access to cybersecurity experts and forensic services to investigate and assess the extent of a breach.
- *Incident response.* Cyber insurance may include support for incident response planning and coordination, helping organizations navigate the complexities of managing a cyber incident.
- *Regulatory compliance.* Policies may cover expenses related to regulatory compliance and fines, particularly in cases where data protection laws have been violated.
- *Third-party liability.* Coverage may extend to liabilities arising from third parties, such as vendors, partners, and customers, who are affected by a cybersecurity incident involving the insured organization.

Cybersecurity insurance is often an essential component of an organization's risk management strategy, especially as cyberthreats continue to evolve and pose significant financial and operational risks. It helps businesses and individuals transfer some of the financial risks associated with cybersecurity incidents to insurance providers, reducing the potential financial burden and providing peace of mind in the face of cyberthreats. However, it's crucial for policyholders to carefully review the terms,

coverage limits, and conditions of their cyber insurance policies to ensure they meet their specific needs and risk profile.

Cost vs. Benefit of Security Controls

Often, additional calculations are involved in risk response when a quantitative risk assessment is performed. These relate to the mathematical evaluation of the cost/benefit of a safeguard. For each identified risk in criticality priority order, safeguards are considered in regard to their potential loss reduction and benefit potential. For each asset-threat pairing (i.e., identified risk), an inventory of potential and available safeguards must be made. This may include investigating the marketplace, consulting with experts, and reviewing security frameworks, regulations, and guidelines. Once a list of safeguards is obtained or produced for each risk, those safeguards should be evaluated as to their benefit and their cost relative to the asset-threat pair. This is the cost/benefit evaluation of safeguards.

Legal and in Compliance

Every organization needs to verify that its operations and policies are legal and in compliance with its stated security policies, industry obligations, contracts, and regulations. Auditing is necessary for compliance testing, also called compliance checking. Verification that a system complies with laws, regulations, baselines, guidelines, standards, best practices, contracts, and policies is an important part of maintaining security in any environment. Compliance testing ensures that all necessary and required elements of a security solution are properly deployed and functioning as expected. These are all important considerations when selecting risk response strategies.

Safeguards, security controls, and countermeasures will primarily reduce risk through a reduction in the potential rate of compromise (i.e., ARO). However, some safeguards will also reduce the amount or severity of damage (i.e., EF). For those safeguards that only reduce the ARO, the amount of loss of a single realized event (i.e., SLE) is the same with or without the safeguard. But for those safeguards that also reduce the EF, any

single realized event will cause less damage than if the safeguard was not present. Either way, a reduction of the ARO and potentially a reduction of the EF will result in a smaller ALE with the safeguard than without. Thus, this potential ALE with the safeguard should be calculated (ALE = AV * EF * ARO). We can then consider the original asset-threat pair risk ALE as ALE1 (or ALE pre-safeguard) and the safeguard-specific ALE as ALE2 (or ALE post-safeguard). An ALE2 should be calculated for each potential safeguard for each asset-threat pair. The best of all possible safeguards would reduce the ARO to 0, although this is extremely unlikely.

Any safeguard that is selected to be deployed will cost the organization something. It might not be purchase cost; it could be costs in terms of productivity loss, retraining, changes in business processes, or other opportunity costs. An estimation of the yearly costs for the safeguard to be present in the organization is needed. This estimation can be called the *annual cost of the safeguard (ACS)*. Several common factors affect ACS:

- Cost of purchase, development, and licensing
- Cost of implementation and customization
- Cost of annual operation, maintenance, administration, and so on
- Cost of annual repairs and upgrades
- Productivity improvement or loss
- Changes to environment
- Cost of testing and evaluation

The value of the asset to be protected determines the maximum expenditures for protection mechanisms. Security should be cost-effective, and thus it is not prudent to spend more (in terms of cash or resources) protecting an asset than its value to the organization. If the cost of the countermeasure is greater than the value of the asset (i.e., the cost of the risk), that safeguard should not be considered a reasonable option. Also, if the ACS is greater than the ALE1 (i.e., the potential annual loss of an asset due to a threat), then the safeguard is not a cost-effective solution. If no safeguard options are cost-effective, then accepting the risk may be the only remaining option.

Once you know the potential annual cost of a safeguard, you can then evaluate the benefit of that safeguard if applied to an infrastructure. The final computation in this process is the *cost/benefit calculation*, or *cost/benefit analysis*. This calculation is used to determine whether a safeguard actually improves security without costing too much. To determine whether the safeguard is financially equitable, use the following formula:

> [ALE pre-safeguard – ALE post-safeguard] – annual cost of safeguard (ACS) = value of the safeguard to the company

If the result is negative, the safeguard is not a financially responsible choice. If the result is positive, then that value is the annual savings your organization may reap by deploying the safeguard because the rate of occurrence is not a guarantee of occurrence. If multiple safeguards seem to have a positive cost/benefit result, then the safeguard with the largest benefit is the most cost-effective option.

The annual savings or loss from a safeguard should not be the only consideration when evaluating safeguards. You should also consider the issues of legal responsibility and prudent due care/due diligence. In some cases, it makes more sense to lose money in the deployment of a safeguard than to risk legal liability in the event of an asset disclosure or loss.

In review, to perform the cost/benefit analysis of a safeguard, you must calculate the following three elements:

- The pre-safeguard ALE for an asset-threat pairing
- The potential post-safeguard ALE for an asset-threat pairing
- The ACS (annual cost of the safeguard)

With those elements, you can finally obtain a value for the cost/benefit formula for this specific safeguard against a specific risk against a specific asset:

> (pre-safeguard ALE – post-safeguard ALE) – ACS

or, even more simply:

(ALE1 – ALE2) – ACS

The countermeasure with the greatest resulting value from this cost/benefit formula makes the most economic sense to deploy against the specific asset-threat pairing.

It is important to realize that with all the calculations used in the quantitative risk assessment process (Table 2.2), the end values are used for prioritization and selection. The values themselves do not truly reflect real-world losses or costs due to security breaches. This should be obvious because of the level of guesswork, statistical analysis, and probability predictions required in the process.

TABLE 2.2 Quantitative risk analysis formulas

Concept	**Formula or meaning**
Asset value (AV)	$
Exposure factor (EF)	% Loss
Single loss expectancy (SLE)	SLE = AV * EF
Annualized rate of occurrence (ARO)	# / year
Annualized loss expectancy (ALE)	ALE = SLE * ARO or ALE = AV * EF * ARO
Annual cost of the safeguard (ACS)	$ / year
Value or benefit of a safeguard (i.e., cost/benefit equation)	(ALE1 – ALE2) – ACS

Once you have calculated a cost/benefit for each safeguard for each asset-threat pair, you must then sort these values. In most cases, the cost/benefit with the highest value is the best safeguard to implement for that specific risk against a specific asset. But as with all things in the real world, this is only one part of the decision-making process. Although very important and often the primary guiding factor, it is not the sole element of data. Other items include actual cost, security budget, compatibility with existing systems, skill/knowledge base of IT staff, availability of products, political issues, partnerships, market trends, fads, marketing, contracts, and favoritism. As part of senior management or even the IT staff, it is your responsibility to either obtain or use all available data and information to

make the best security decision for your organization. For further discussion of safeguard, security control, and countermeasure selection issues, see the "Countermeasure Selection and Implementation" section later in this chapter.

Most organizations have a limited and all-too-finite budget to work with. Thus, obtaining the best security for the cost is an essential part of security management. To effectively manage the security function, you must assess the budget, the benefit and performance metrics, and the necessary resources of each security control. Only after a thorough evaluation can you determine which controls are essential and beneficial not only to security, but also to your bottom line. Generally, it is not an acceptable excuse that the reason the organization did not protect against an unacceptable threat or risk was solely because of a lack of funds. The entirety of safeguard selections needs to be considered in relation to the current budget. Compromise or adjustments of priorities may be necessary in order to reduce overall risk to an acceptable level with available resources. Keep in mind that organizational security should be based on a business case, be legally justifiable, and be reasonably in line with security frameworks, regulations, and best practices.

Countermeasure Selection and Implementation

Selecting a countermeasure, safeguard, or control (short for *security control*) within the realm of risk management relies heavily on the cost/benefit analysis results. However, you should consider several other factors when assessing the value or pertinence of a security control:

- The cost of the countermeasure should be less than the value of the asset.
- The cost of the countermeasure should be less than the benefit of the countermeasure.
- The result of the applied countermeasure should make the cost of an attack greater for the perpetrator than the derived benefit from an attack.
- The countermeasure should provide a solution to a real and identified problem. (Don't install countermeasures just because they are

available, are advertised, or sound appealing.)

- The benefit of the countermeasure should not be dependent on its secrecy. Any viable countermeasure can withstand public disclosure and scrutiny and thus maintain protection even when known.
- The benefit of the countermeasure should be testable and verifiable.
- The countermeasure should provide consistent and uniform protection across all users, systems, protocols, and so on.
- The countermeasure should have few or no dependencies to reduce cascade failures.
- The countermeasure should require minimal human intervention after initial deployment and configuration.
- The countermeasure should be tamperproof.
- The countermeasure should have overrides accessible to privileged operators only.
- The countermeasure should provide fail-safe and/or fail-secure options.

Keep in mind that security should be designed to support and enable business tasks and functions. Thus, countermeasures and safeguards need to be evaluated in the context of a business process. If there is no clear business case for a safeguard, it is probably not an effective security option.

Security controls, countermeasures, and safeguards can be implemented administratively, logically/technically, or physically. These three categories of security mechanisms should be implemented in a conceptual layered defense-in-depth manner in order to provide maximum benefit (see Figure 2.4). This idea is based on the concept that policies (part of administrative controls) drive all aspects of security and thus form the initial protection layer around assets. Next, logical and technical controls provide protection against logical attacks and exploits. Then, the physical controls provide protection against real-world physical attacks against the facility and devices.

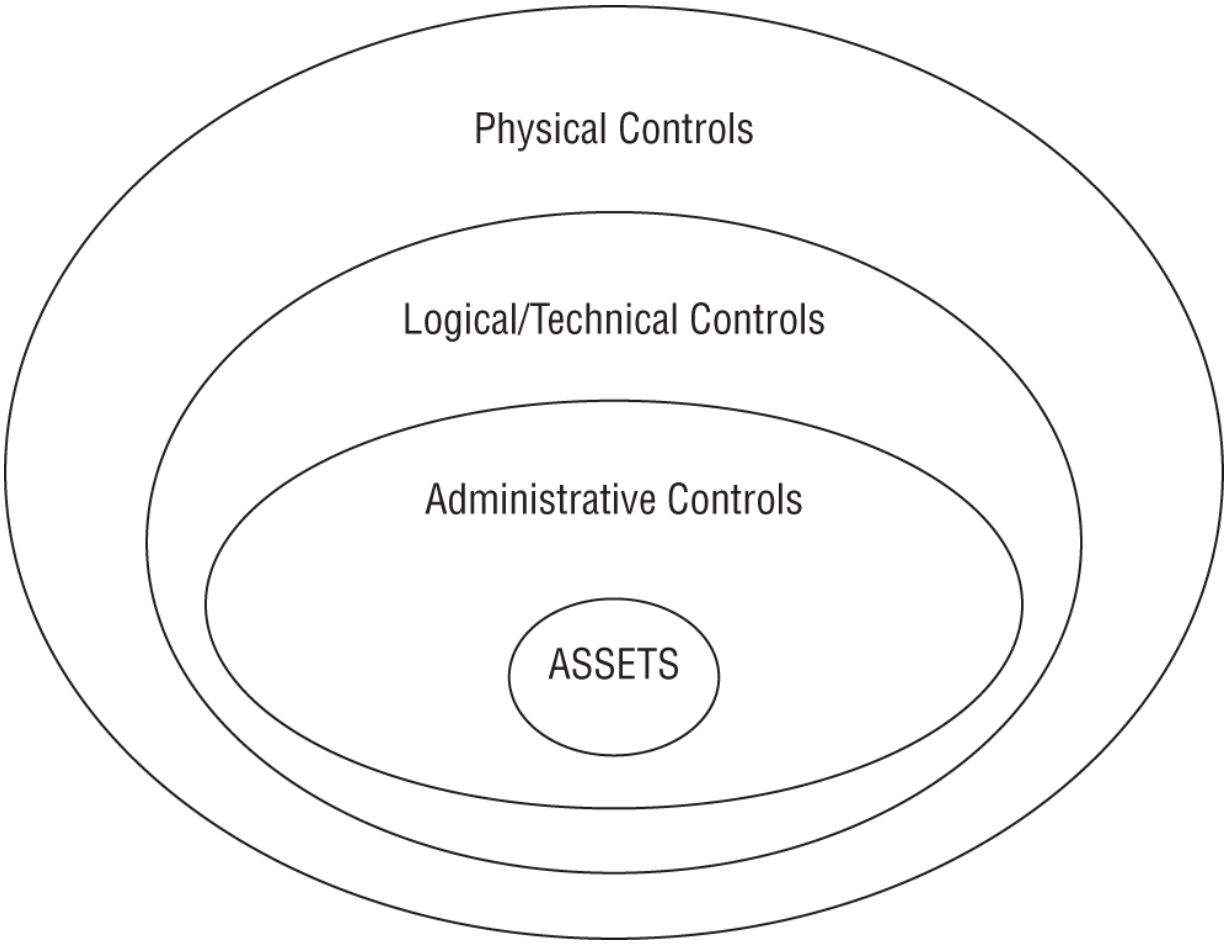


FIGURE 2.4 The categories of security controls in a defense-in-depth implementation

Administrative

The category of *administrative controls* includes the policies and procedures defined by an organization's security policy and other regulations or requirements. They are sometimes referred to as *management controls*, *managerial controls*, or *procedural controls*. These controls focus on personnel oversight and business practices. Examples of administrative controls include policies, procedures, hiring practices, background checks, data classifications and labeling, security awareness and training efforts, reports and reviews, work supervision, personnel controls, and testing.

Technical or Logical

The category of *technical controls* or *logical controls* involves the hardware or software mechanisms used to manage access and provide protection for

IT resources and systems. Examples of logical or technical controls include authentication methods (such as passwords, smartcards, and biometrics), encryption, constrained interfaces, access control lists, protocols, firewalls, routers, and intrusion detection systems (IDSs).

Physical

Physical controls are security mechanisms focused on providing protection to the facility and real-world objects. Examples of physical controls include guards, fences, motion detectors, locked doors, sealed windows, lights, cable protection, laptop locks, badges, swipe cards, guard dogs, video cameras, access control vestibules, and alarms.

Applicable Types of Controls

The term *security control* refers to a broad range of controls that perform such tasks as ensuring that only authorized users can log on and preventing unauthorized users from gaining access to resources. Controls mitigate a wide variety of information security risks.

Whenever possible, you want to prevent any type of security problem or incident. Of course, this isn't always possible, and unwanted events occur. When they do, you want to detect the events as soon as possible. And once you detect an event, you want to correct it.

As you read the control descriptions, notice that some are listed as examples of more than one access control type. For example, a fence (or perimeter-defining device) placed around a building can be a preventive control (physically barring someone from gaining access to a building compound) and/or a deterrent control (discouraging someone from trying to gain access).

Preventive

A *preventive control* (aka *preventative control*) is deployed to thwart or stop unwanted or unauthorized activity from occurring. Examples of preventive controls include fences, locks, authentication, access control vestibules, alarm systems, separation of duties, job rotation, data loss prevention (DLP), penetration testing, access control methods, encryption, auditing,

security policies, security-awareness training, antimalware software, firewalls, and intrusion prevention systems (IPSs).



Keep in mind that there are no perfect security mechanisms or controls. They all have issues that can allow a threat agent to still cause harm. Controls may have vulnerabilities, can be turned off, may be avoided, can be overloaded, may be bypassed, can be tricked by impersonation, may have backdoors, can be misconfigured, or have other issues. Thus, this known imperfection of individual security controls is addressed by using a defense-in-depth strategy.

Detection

A *detection control* (aka *detective control*) is deployed to discover or detect unwanted or unauthorized activity. Detection controls operate after the fact and can discover the activity only after it has occurred. Examples of detection controls include security guards, motion detectors, recording and reviewing of events captured by security cameras or CCTV, job rotation, mandatory vacations, audit trails, honeypots or honeynets, intrusion detection systems (IDSs), violation reports, supervision and review of users, and incident investigations.

Corrective

A *corrective control* modifies the environment to return systems to normal after an unwanted or unauthorized activity has occurred. It attempts to correct any problems resulting from a security incident. Corrective controls can be simple, such as terminating malicious activity or rebooting a system. They also include antimalware solutions that can remove or quarantine a virus, backup and restore plans to ensure that lost data can be restored, and intrusion prevention systems (IPSs) that can modify the environment to stop an attack in progress. The control is deployed to repair or restore resources, functions, and capabilities after a violation of security policies. Examples include installing a spring on a door so that it will close and relock, and using file integrity–checking tools, such as `sigverif` from Windows, which

will replace corrupted boot files upon each boot event to protect the stability and security of the booted OS.

Recovery

Recovery controls are an extension of corrective controls but have more advanced or complex abilities. A recovery control attempts to repair or restore resources, functions, and capabilities after a security policy violation. Recovery controls typically address more significant damaging events compared to corrective controls, especially when security violations may have occurred. Examples of recovery controls include backups and restores, fault-tolerant drive systems, system imaging, server clustering, antimalware software, and database or virtual machine shadowing. In relation to business continuity and disaster recovery, recovery controls can include hot, warm, and cold sites; alternate processing facilities; service bureaus; reciprocal agreements; cloud providers; rolling mobile operating centers; and multisite solutions.

Deterrent

A *deterrent control* is deployed to discourage security policy violations. Deterrent and preventive controls are similar, but deterrent controls often depend on individuals being convinced not to take an unwanted action. Some examples include policies, security-awareness training, locks, fences, security badges, guards, access control vestibules, and security cameras.

Directive

A *directive control* is deployed to direct, confine, or control the actions of subjects to force or encourage compliance with security policies. Examples of directive controls include security policy requirements or criteria, posted notifications, guidance from a security guard, escape route exit signs, monitoring, supervision, and procedures.

Compensating

A *compensating control* (aka *compensation control*) is deployed to provide various options to other existing controls to aid in the enforcement and support of security policies. They can be any controls used in addition to, or in place of, another control. They can be a means to improve the

effectiveness of a primary control or as the alternate or failover option in the event of a primary control failure. For example, if a preventive control fails to stop the deletion of a file, a backup can be a compensating control, allowing for the restoration of that file. Here's another example: if a building's fire prevention and suppression systems fail and the building is damaged by fire so that it is not inhabitable, a compensating control would be having a disaster recovery plan (DRP) with an alternate processing site available to support work operations.

Security Control Assessment

A *security control assessment (SCA)* is the formal evaluation of a security infrastructure's individual mechanisms against a baseline or reliability expectation. The SCA can be performed in addition to or independently of a full security evaluation, such as a penetration test or vulnerability assessment.

The goals of an SCA are to ensure the effectiveness of the security mechanisms, evaluate the quality and thoroughness of the risk management processes of the organization, and produce a report of the relative strengths and weaknesses of the deployed security infrastructure. The results of an SCA may confirm that a security mechanism has sustained its previous level of verified effectiveness or that action must be taken to address a deficient security control. In addition to verifying the reliability of security controls, an assessment should consider whether security controls affect privacy. Some controls may improve privacy protection, whereas others may in fact cause a breach of privacy. The privacy aspect of a security control should be evaluated in light of regulations, contractual obligations, and the organization's privacy policy/promise.

Generally, an SCA is a process implemented by federal agencies based on NIST SP 800-53 Rev. 5, titled “Security and Privacy Controls for Information Systems and Organizations.” However, though defined as a government process, the concept of evaluating the reliability and effectiveness of security controls should be adopted by every organization that is committed to sustaining a successful security endeavor.

Monitoring and Measurement

Security controls should provide benefits that can be continuously monitored and measured. If a security control's benefits cannot be quantified, evaluated, or compared, then it does not actually provide any security. A security control may provide native or internal monitoring, or external monitoring may be required. You should take this into consideration when making initial countermeasure selections.

Measuring the effectiveness of a countermeasure is not always an absolute value. Many countermeasures offer degrees of improvement rather than specific hard numbers as to the number of breaches prevented or attack attempts thwarted. Often to obtain countermeasure success or failure measurements, monitoring and recording of events both prior to and after safeguard installation are necessary. Benefits can only be accurately measured if the starting point (i.e., the normal point or initial risk level) is known. Part of the cost/benefit equation takes countermeasure monitoring and measurement into account. Just because a security control provides some level of increased security does not necessarily mean that the benefit gained is cost-effective. A significant improvement in security should be identified to clearly justify the expense of a new countermeasure deployment.

Risk Reporting and Documentation

Risk reporting is a key task to perform at the conclusion of a risk analysis. Risk reporting involves the production of a risk report and a presentation of that report to the interested/relevant parties. For many organizations, risk reporting is an internal concern only, whereas other organizations may have regulations that mandate third-party or public reporting of their risk findings. A risk report should be accurate, timely, comprehensive of the entire organization, clear and precise to support decision-making, and updated on a regular basis.

Internal and external reporting in risk management are processes through which organizations communicate information about their risk-related activities, assessments, and strategies to different stakeholders. These two forms of reporting serve distinct purposes and audiences:

- *Internal reporting* in risk management is primarily intended for an organization's internal stakeholders, including executives,

management, employees, and relevant departments. The primary purpose is to support informed decision-making, risk mitigation, and the overall management of risks within the organization. Internal reports typically contain detailed information about the organization's risk assessments, risk exposures, risk control measures, and the effectiveness of risk management strategies. These reports may include risk registers, risk heat maps, key risk indicators (KRIs), and the results of risk assessments.

- *External reporting* in risk management is intended for external stakeholders, including regulatory bodies, shareholders, investors, creditors, customers, and the general public. The primary purpose is to provide transparency and disclosure of an organization's risk profile, risk exposure, and risk management practices to external parties. External reports typically focus on high-level information about the organization's risk exposure, its policies and practices for risk management, and the potential impact of risks on the organization's financial health and operations. These reports may include annual reports, financial statements, disclosures in compliance with accounting standards, and regulatory filings.

It's important for organizations to maintain a clear distinction between internal and external reporting in risk management. Balancing these two forms of reporting is essential for effective risk management and maintaining transparency and trust with both internal and external audiences.

A *risk register* or *risk log* is a document that inventories all the identified risks to an organization or system or within an individual project. A risk register is used to record and track the activities of risk management, including the following:

- Identifying risks
- Evaluating the severity of and prioritizing those risks
- Prescribing responses to reduce or eliminate the risks
- Tracking the progress of risk mitigation

A risk register can serve as a project management document to track completion of risk response activities as well as a historical record of risk management over time. The contents of a risk register could be shared with others to facilitate a more realistic evaluation of real-world threats and risks through the amalgamation of risk management activities by other organizations.

A *risk matrix* or *risk heat map* is a form of risk assessment that is performed on a basic graph or chart. It is sometimes labeled as a qualitative risk assessment. The simplest form of a risk matrix is a 3×3 grid comparing probability and damage potential. This was covered in Chapter 1.

Continuous Improvement

Risk analysis is performed to provide upper management with the details necessary to decide which risks should be mitigated, which should be transferred, which should be deterred, which should be avoided, and which should be accepted. The result is a cost/benefit comparison between the expected cost of asset loss and the cost of deploying safeguards against threats and vulnerabilities. Risk analysis identifies risks, quantifies the impact of threats, and aids in budgeting for security. It helps integrate the needs and objectives of the security policy with the organization's business goals and intentions. The risk analysis/risk assessment is a “point-in-time” metric. Threats and vulnerabilities constantly change, and the risk assessment needs to be redone periodically in order to support continuous improvement.

Security is always changing. Thus, any implemented security solution requires updates and changes over time. If a continuous improvement path is not provided by a selected countermeasure, it should be replaced with one that offers scalable improvements to security.

An *enterprise risk management (ERM)* program can be evaluated using the *Risk Maturity Model (RMM)*. An RMM assesses the key indicators and activities of a mature, sustainable, and repeatable risk management process. There are several RMM systems, each prescribing various means to achieve greater risk management capability. They generally relate the assessment of risk maturity against a five-level model (similar to that of the Capability

Maturity Model [CMM]; see Chapter 20, "Software Development Security"). The typical RMM levels are as follows:

1. *Ad hoc.* A chaotic starting point from which all organizations initiate risk management.
2. *Preliminary.* Loose attempts are made to follow risk management processes, but each department may perform risk assessment uniquely.
3. *Defined.* A common or standardized risk framework is adopted organization-wide.
4. *Integrated.* Risk management operations are integrated into business processes, metrics are used to gather effectiveness data, and risk is considered an element in business strategy decisions.
5. *Optimized.* Risk management focuses on achieving objectives rather than just reacting to external threats, increased strategic planning is geared toward business success rather than just avoiding incidents, and lessons learned are reintegrated into the risk management process.

To learn more about RMM, see "Developing a Generic Risk Maturity Model (GRMM) for Evaluating Risk Management in Construction Projects." This is an interesting study of numerous RMM systems and the attempt to derive a generic RMM from the common elements.

Legacy Risk

An often-overlooked area of risk is that of legacy devices, which may be EOL and/or EOS/EOSL:

- *End of life (EOL)* is the point at which a manufacturer no longer produces a product. Service and support may continue for a period of time after EOL, but no new versions will be made available for sale or distribution. An EOL product should be scheduled for replacement before it fails or reaches end of support (EOS) or end of service life (EOSL). EOL is sometimes perceived or used as the equivalent of EOSL.
- *End of service-life (EOSL)* or *end of support (EOS)* are those systems that are no longer receiving updates and support from the vendor. If an

organization continues to use an EOSL system, then the risk of compromise is high because any future exploitation will never be patched or fixed. It is of utmost importance to move off EOSL systems in order to maintain a secure environment. It might not seem initially cost-effective or practical to move away from a solution that still works just because the vendor has terminated support. However, the security management efforts you will expend will likely far exceed the cost of developing and deploying a modern system–based replacement. For example, Windows 10 will reach its EOSL on October 14, 2025. Microsoft recommends moving on to Windows 11 by that deadline.

Risk Frameworks

A *risk framework* is a guideline or recipe for how risk is to be assessed, resolved, and monitored. NIST established the Risk Management Framework (RMF) and the Cybersecurity Framework (CSF). These are both U.S. government guides for establishing and maintaining security, but the CSF is designed for critical infrastructure and commercial organizations, whereas the RMF establishes mandatory requirements for federal agencies. RMF was established in 2010, and the CSF was established in 2014.



Exam Outline objective 1.9.9 includes a list of risk frameworks. All of these risk frameworks are also listed in objective 1.3.4 as security control frameworks. It is common for a security control framework to be either directly used as a risk framework or have a subset of elements that are a risk framework. The following concepts are covered in Chapter 1 as security control frameworks, but they are also relevant to the concepts here in Chapter 2 as risk frameworks:

- International Organization for Standardization (ISO)
- National Institute of Standards and Technology (NIST)
- Control Objectives for Information and Related Technologies (COBIT)
- Sherwood Applied Business Security Architecture (SABSA)
- Payment Card Industry (PCI)

The *Cybersecurity Framework (CSF)* 2.0 (released in early 2024) is based on a framework core that consists of six functions:

- *Identify.* Understand and catalog assets, risks, and vulnerabilities.
- *Protect.* Implement safeguards to protect assets and data.
- *Detect.* Develop and deploy mechanisms for identifying and detecting security incidents.
- *Respond.* Define strategies and processes for responding to and mitigating cybersecurity incidents.
- *Recover.* Develop and implement strategies for recovery and resilience after a cybersecurity incident.
- *Govern.* Establish, communicate, and oversee roles, responsibilities, and policies that ensure a proactive and adaptive approach to cybersecurity.

The CSF is not a checklist or procedure—it is a prescription of operational activities that are to be performed on an ongoing basis for the support and improvement of security over time. The CSF is more of an improvement system rather than its own specific risk management process or security infrastructure. The CSF provides a structured approach for organizations to assess, develop, and enhance their cybersecurity posture and resilience against cyberthreats.

The *Risk Management Framework (RMF)*, defined by NIST in SP 800-37 Rev. 2, is a structured and comprehensive framework used by the U.S. federal government and other organizations to manage and mitigate information security and cybersecurity risks associated with their information systems and networks. RMF establishes mandatory security requirements for U.S. federal agencies. The RMF has seven phases (six of which are used cyclically) (see Figure 2.5):

Prepare to execute the RMF from an organization- and system-level perspective by establishing a context and priorities for managing security and privacy risk.

Categorize the system and the information processed, stored, and transmitted by the system based on an analysis of the impact of loss.

Select an initial set of controls for the system and tailor the controls as needed to reduce risk to an acceptable level based on an assessment of risk.

Implement the controls and describe how the controls are employed within the system and its environment of operation.

Assess the controls to determine if the controls are implemented correctly, operating as intended, and producing the desired outcomes with respect to satisfying the security and privacy requirements.

Authorize the system or common controls based on a determination that the risk to organizational operations and assets, individuals, other organizations, and the Nation is acceptable.

Monitor the system and the associated controls on an ongoing basis to include assessing control effectiveness, documenting changes to the system and environment of operation, conducting risk assessments and impact analyses, and reporting the security and privacy posture of the system.

[From NIST SP 800-37 Rev. 2]

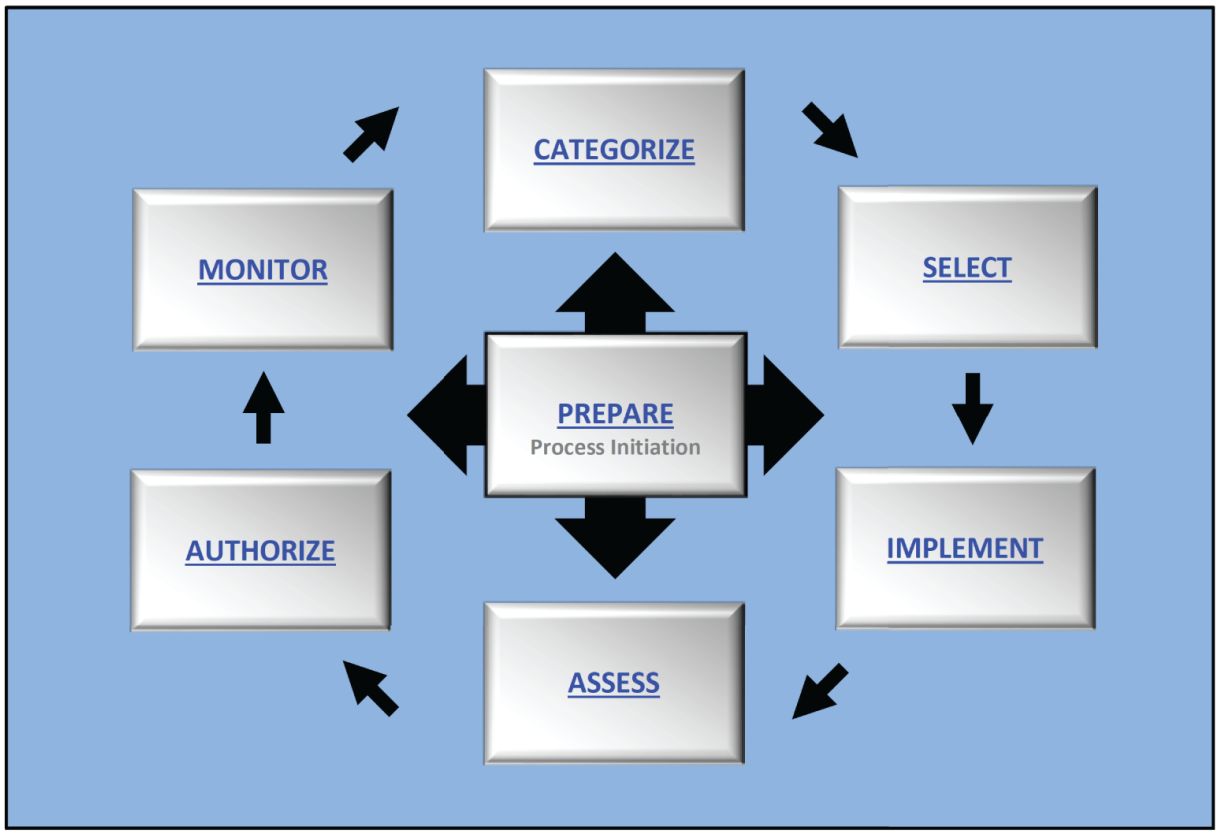


FIGURE 2.5 The elements of the risk management framework (RMF) (from NIST SP 800-37 Rev. 2, Figure 2)

The later six phases are to be performed in order and repeatedly throughout the life of the organization. RMF is intended as a risk management process to identify and respond to threats. Use of the RMF will result in the establishment of a security infrastructure and a process for ongoing improvement of the secured environment.

There is significantly more detail about RMF in the official NIST publication; we encourage you to review this publication in its entirety for a complete perspective on the RMF.

Another important risk framework or guide to risk management is the ISO/IEC 31000 document "Risk Management — Guidelines." This is a high-level overview of the idea of risk management that many will benefit from reading. This ISO guideline is intended to be useful to any type of organization, whether government or private sector. Another related guideline is ISO/IEC 27005, "Information Security, Cybersecurity and Privacy Protection: Guidance on Managing Information Security Risks."



A companion guide, ISO/IEC 31004 “Risk Management — Guidance for the Implementation of ISO 31000” (www.iso.org/standard/56610.html), might also be of interest. However, ISO 31004 has been withdrawn and had not been replaced as of Q1 2024.

While the NIST RMF is a common focus of the CISSP, you might want to review other risk management frameworks for use in the real world. Please consider the following for future research:

- The Committee of Sponsoring Organizations (COSO) of the Treadway Commission's Enterprise Risk Management — Integrated Framework
- ISACA's Risk IT Framework
- Operationally Critical Threat, Asset, and Vulnerability Evaluation (OCTAVE)
- Factor Analysis of Information Risk (FAIR)
- Threat Assessment and Remediation Analysis (TARA)

Understanding that there are a number of well-recognized frameworks and that selecting one that fits your organization's requirements and style is important.

Social Engineering

Social engineering is a form of attack that exploits human nature and human behavior. People are a weak link in security because they can make mistakes, be fooled into causing harm, or intentionally violate company security. Social engineering attacks exploit human characteristics such as a basic trust in others, a desire to provide assistance, or a propensity to show off. It is important to consider the risks that personnel represent to your organization and implement security strategies to minimize and handle those risks.

Social engineering attacks take two primary forms: convincing someone to perform an unauthorized operation or convincing someone to reveal confidential information. In just about every case, the social engineering attacker tries to convince the victim to perform some activity or reveal a piece of information that they shouldn't. The result of a successful attack is information leakage or the attacker being granted logical or physical access to a secure environment.

Here are some example scenarios of common social engineering attacks:

- A website claims to offer free temporary access to its products and services, but it requires web browser and/or firewall alterations in order to download the access software. These alterations may reduce the security protections or encourage the victim to install malicious browser helper objects (also known as plug-ins, extensions, or add-ons).
- The help desk receives a call from someone claiming to be a department manager who is currently involved in a sales meeting in another city. The caller claims to have forgotten their password and needs it to be reset so that they can log in remotely to download an essential presentation.
- Someone who looks like a repair technician claims a service call was received for a malfunctioning device in the building. The “technician” is sure the unit can be accessed from inside your office work area and asks to be given access to repair the system.
- If a worker receives a communication from someone asking to talk with a coworker by name, and there is no such person currently or previously working for the organization, this could be a ruse to either reveal the names of actual employees or convince you to “provide assistance” because the caller has incorrect information.
- When a contact on a discussion forum asks personal questions, such as your education, history, and interests, they could be focused on learning the answers to password reset questions.

Some of these examples may also be legitimate and benign occurrences, but you can see how they could mask the motives and purposes of an attacker.

Social engineers attempt to mask and hide their true intentions by crafting their attacks to seem as normal and typical as possible.

Whenever a security breach occurs, an investigation should be performed to determine what was affected and whether the attack is ongoing. Personnel should be retrained to detect and avoid similar social engineering attacks in the future. Although social engineering attacks primarily focus on people, the results of an attack can be a disclosure of private or confidential materials, physical damage to a facility, or remote access to an IT environment. Therefore, any attempted or successful social engineering breach should be thoroughly investigated and responded to.

Methods to protect against social engineering include the following:

- Training personnel about social engineering attacks and how to recognize common signs
- Requiring authentication when performing activities for personnel over the phone
- Defining restricted information that is never communicated over the phone or through plaintext communications such as standard email
- Always verifying the credentials of a repair person and verifying that a real service call was placed by authorized personnel
- Never following the instructions of an email without verifying the information with at least two independent and trusted sources
- Always erring on the side of caution when dealing with anyone you don't know or recognize, whether in person, over the phone, or over the Internet/network

If several workers report the same odd event, such as a call or email, an investigation should look into what the contact was about, who initiated it, and what the intention or purpose was.

The most important defense against social engineering attacks is user education and awareness training. A healthy dose of paranoia and suspicion will help users detect or notice more social engineering attack attempts than without such preparation. Training should include role-playing and walking through numerous examples of the various forms of social engineering

attacks. However, keep in mind that attackers are constantly altering their approaches and improving their means of attack. So, keeping current with newly discovered means of social engineering attacks is also necessary to defend against this human-focused threat.

Users should receive training when they first enter an organization, and they should receive periodic refresher training, even if it's just an email from the administrator or training officer reminding them of the threats.

Social Engineering Principles

Social engineering works so well because we're human. The principles of social engineering attacks are designed to focus on various aspects of human nature and take advantage of them. Although not every target succumbs to every attack, most of us are vulnerable to one or more of the following common social engineering principles.

Authority

Authority is an effective technique because most people are likely to respond to authority with obedience. The trick is to convince the target that the attacker is someone with valid internal or external authority. Some attackers claim their authority verbally, and others assume authority by wearing a costume or uniform.

An example is an email sent using the spoofed email of the CEO in which workers are informed that they must visit a specific universal resource locator (URL)/universal resource indicator (URI) to fill out an important HR document. This method works when the victims blindly follow instructions that claim to be from a person of authority.

Intimidation

Intimidation can sometimes be seen as a derivative of the authority principle. Intimidation uses authority, confidence, or even the threat of harm to motivate someone to follow orders or instructions. Often, intimidation is focused on exploiting uncertainty in a situation where a clear directive of operation or response isn't defined.

An example is expanding on a previous CEO and HR document email to include a statement claiming that employees will face a penalty if they do

not fill out the form promptly. The penalty could be a loss of casual Friday, exclusion from Taco Tuesday, a reduction in pay, or even termination.

Consensus

Consensus or social proof is the act of taking advantage of a person's natural tendency to mimic what others are doing or are perceived as having done in the past. For example, bartenders often seed their tip jar with money to make it seem as if previous patrons were appreciative of the service. As a social engineering principle, the attacker attempts to convince the victim that a particular action or response is necessary to be consistent with social norms or previous occurrences.

An example is an attacker claiming that a worker who is currently out of the office promised a large discount on a purchase and that the transaction must occur now with you as the salesperson.

Scarcity

Scarcity is a technique used to convince someone that an object has a higher value based on the object's scarcity. This could relate to the existence of only a few items produced or limited opportunities, or that the majority of stock is sold and only a few items remain.

An example is an attacker claiming that there are only two tickets left to your favorite team's final game, and it would be a shame if someone else enjoyed the game rather than you. If you don't grab them now, the opportunity will be lost. This principle is often associated with the principle of urgency.

Familiarity

Familiarity or liking, as a social engineering principle, attempts to exploit a person's native trust in that which is familiar. The attacker often tries to appear to have a common contact or relationship with the target, such as mutual friends or experiences, or uses a facade to take on the identity of another company or person. If the target believes a message is from a known entity, such as a friend or their bank, they're much more likely to trust the content and even act or respond.

An example is an attacker using a vishing attack while falsifying the caller ID as their doctor's office.

Trust

Trust as a social engineering principle involves an attacker working to develop a relationship with a victim. This may take seconds or months, but eventually, the attacker attempts to use the value of the relationship (the victim's trust in the attacker) to convince the victim to reveal information or perform an action that violates company security.

An example is an attacker approaching you as you walk along the street, when they appear to pick up a $100 bill from the ground. The attacker asks you to hold the money while they ask around to find someone who lost it. When they return, the attacker says that since the two of you were close when the money was found, you two should split it. They ask if you have change to split the found money. Since the attacker had you hold the money while they went around to find the person who lost it, this might have caused you to have trust in this stranger so that you are willing to take cash out of your wallet and give it to them. But you won't realize until later that the $100 was counterfeit and you've been robbed.

Urgency

Urgency often dovetails with scarcity, because the need to act quickly increases as scarcity indicates a greater risk of missing out. Urgency is often used as a method to get a quick response from a target before they have time to carefully consider or refuse compliance.

An example is an attacker using an invoice scam through business email compromise (BEC) to convince you to pay an invoice immediately because either an essential business service is about to be cut off or the company will be reported to a collection agency.

Eliciting Information

Eliciting information is the activity of gathering or collecting information from systems or people. In the context of social engineering, it is used as a research method in order to craft a more effective pretext. A *pretext* is a false statement crafted to sound believable in order to convince you to act

or respond in favor of the attacker. Any and all of the social engineering techniques covered in this chapter can be used both as a weapon to harm the target victim and as a means to obtain more information (or access). Thus, social engineering is a tool of both reconnaissance and attack. Data gathered via social engineering can be used to support a physical or logical/technical attack.

Any means or method by which a social engineer can gather information from the target is eliciting information. Any fact, truth, or detail that can be collected, gathered, or gleaned from the target can be used to form a more complete and believable pretext or false story, which in turn may increase the chance of success of the next level or stage of an attack.

Consider that many cyberattacks are similar to actual warfare attacks. The more the attacker knows about the targeted enemy, the more effectively a plan of attack can be crafted.

Defending against eliciting information events generally involves the same precautions as those used against social engineering. Those include classifying information, controlling the movement of sensitive data, watching for attempted abuses, training personnel, and reporting any suspicious activity to the security team.

Prepending

Prepending is the adding of a term, expression, or phrase to the beginning or header of some other communication. Often, prepending is used in order to further refine or establish the pretext of a social engineering attack, such as spam, hoaxes, and phishing. An attacker can precede the subject of an attack message with RE: or FW: (which indicates “in regard to” and “forwarded,” respectively) to make the receiver think the communication is the continuance of a previous conversation rather than the first contact of an attack. Other often-used prepending terms are EXTERNAL, PRIVATE, and INTERNAL.

Prepending attacks can also be used to fool filters, such as spam filters, antimalware, firewalls, and intrusion detection systems (IDSs). This could be accomplished with SAFE, FILTERED, AUTHORIZED, VERIFIED, CONFIRMED, or APPROVED, among others. It might even be possible to

interject alternate email header values, such as “X-Spam-Category: LEGIT” or “X-Spam-Condition: SAFE,” which could fool spam and abuse filters.

Phishing

Phishing is a form of social engineering attack focused on stealing credentials or identity information from any potential target. It is derived from “fishing” for information. Phishing can be waged in numerous ways using a variety of communication media, including email and the web; in face-to-face interactions or over the phone; and even through more traditional communication mediums, such as the post office or couriered packages.

Attackers send phishing emails indiscriminately as spam, without knowing who will get them but in the hope that some users will respond. Phishing emails sometimes inform the user of a bogus problem and say that if the user doesn't take action, the company will lock the user's account. The From email address is often spoofed to look legitimate, but the Reply To email address is an account controlled by the attacker. Sophisticated attacks include a link to a bogus website that looks legitimate, but that captures credentials and passes them to the attacker.

Sometimes, the goal of phishing is to install malware on user systems. The message may include an infected file attachment or a link to a website that installs a malicious drive-by download without the user's knowledge.



A drive-by download is a type of malware that installs itself without the user's knowledge when the user visits a website. Drive-by downloads take advantage of vulnerabilities in browsers or plug-ins.

To defend against phishing attacks, end users should be trained to do the following:

- Be suspicious of unexpected email messages or email messages from unknown senders.

- Never open unexpected email attachments.
- Never share sensitive information via email.
- Avoid clicking any link received via email, instant messaging, or a social network message.

If a message claims to be from a known source, such as a website commonly visited, the user should visit the supposed site by using a preestablished bookmark or by searching for the site by name. If, after accessing their account on the site, a duplicate message does not appear in the online messaging or alert system, the original message is likely an attack or a fake. Any such false communications should be reported to the targeted organization, and then the message should be deleted. If the attack relates to your organization or employer, it should be reported to the security team there as well.

Organizations should consider the consequences and increased risk that granting workers access to personal email and social networks through company systems pose. Some companies have elected to block access to personal Internet communications while using company equipment or through company-controlled network connections. This reduces the risk to the organization even if an individual succumbs to a phishing attack on their own.

A *phishing simulation* is a tool used to evaluate the ability of employees to resist or fall for a phishing campaign. A security manager or penetration tester crafts a phishing attack so that any clicks by victims are redirected to a notification that the phishing message was a simulation and they may need to attend additional training to avoid falling for a real attack.

Smishing

Short Message Service (SMS) phishing or *smishing* (spam over instant messaging [SPIM]) is a social engineering attack that occurs over or through standard text messaging services. There are several smishing threats to watch out for, including these:

- Text messages asking for a response or reply. In some cases, replies could trigger a cramming event. Cramming is when a false or

unauthorized charge is placed onto your mobile service plan.

- Text messages could include a hyperlink/URI/URL to a phishing or scam website or trigger the installation of malicious code.
- Text messages could contain pretexts to get you involved in a conversation.
- Text messages could include phone numbers. Always research a phone number before calling it, especially from an unknown source. There are phone numbers with the same structure as local or domestic numbers, but that may actually be long distance and not included in your calling service or plan, and calling them could cause a connection charge and a high per-minute toll charge.

Although smishing refers to SMS-based attacks, it can sometimes be used to refer to similar attacks occurring through Multimedia Messaging Service (MMS), Rich Communication Services (RCS), Google Chat, Android Messages (i.e., SMS), Facebook Messenger, WeChat, Apple/iPhone iMessage, WhatsApp, Slack, Discord, Microsoft Teams, and so on.

Vishing

Vishing (i.e., voiced-based phishing) or SpIT (spam over internet telephony) is phishing done over any telephony or voice communication system. This includes traditional phone lines, voice-over-IP (VoIP) services, and mobile phones. Most of the social engineers waging vishing campaigns use VoIP technology to support their attacks. VoIP allows the attacker to be located anywhere in the world, make free phone calls to victims, and be able to falsify or spoof their origin caller ID.

Vishing calls can display a caller ID or phone number from any source the attacker thinks might cause the victim to answer the call. Some attackers just duplicate your area code and prefix in order to trick the victim into thinking the call is from a neighbor or other local entity. Vishing is simply another form of phishing attack. Vishing involves the pretexting of the displayed caller ID and the story the attacker spouts. Always assume caller ID is false or at least incorrect.

Spear Phishing

Spear phishing is a more targeted form of phishing where the message is crafted and directed specifically to a group of individuals. Often, attackers use a stolen customer database to send false messages crafted to seem like a communication from the compromised business but with falsified source addresses and incorrect URI/URLs. The hope of the attacker is that someone who already has an online/digital relationship with an organization is more likely to fall for the false communication.

All of the concepts and defenses discussed in the previous section, "Phishing," apply to spear phishing.

Spear phishing can also be crafted to seem as if it originated from a CEO or other top office in an organization. This version of spear phishing is called *business email compromise (BEC)*. BEC is often focused on convincing members of accounting or financial departments to transfer funds or pay invoices based on instructions seeming to originate from a boss, manager, or executive. BEC has defrauded organizations of billions of dollars in the last few years. BEC is also known as *CEO fraud* or *CEO spoofing*.

As with most forms of social engineering, defenses for spear phishing require the following:

- Labeling information, data, and assets with their value, importance, or sensitivity
- Training personnel on proper handling of those assets based on their labels
- Requesting clarification or confirmation on any actions that seem abnormal, off-process, or otherwise overly risky to the organization

Some abusive concepts to watch out for are requests to pay bills or invoices using prepaid gift cards, changes to wiring details (especially at the last minute), or requests to purchase products that are atypical for the requester and that are needed in a rush. When seeking to confirm a suspected BEC, do not use the same communication medium that the BEC used. Make a phone call, go to their office, text-message their cell phone, or use the company-approved internal messaging service. Establishing a second "out-of-band" contact with the requester will further confirm whether the message is legitimate or false.

Whaling

Whaling is a form of spear phishing that targets specific high-value individuals (by title, by industry, from media coverage, and so forth), such as the CEO or other C-level executives, administrators, or high-net-worth clients. Whaling attacks require significantly more research, planning, and development on the part of the attackers in order to fool the victim. That is because these high-level personnel are often well aware that they are a high-value target.

Spam

Spam is any type of email that is undesired and/or unsolicited. But spam is not just unwanted advertisements; it can also include malicious content and attack vectors as well. Spam is often used as the carrier of social engineering attacks.

Spam is a problem for numerous reasons:

- Some spam carries malicious code such as viruses, logic bombs, ransomware, or Trojan horses.
- Some spam carries social engineering attacks (also known as hoax messages).
- Unwanted email wastes your time while you sort through it, looking for legitimate messages.
- Spam wastes internet resources: storage capacity, computing cycles, and throughput.

The primary countermeasure against spam is an email spam filter. These email filters can examine the header, subject, and contents of a message to look for keywords or phrases that identify it as a known type of spam, and then take the appropriate actions to discard, quarantine, or block the message.

Antispam software is a variation on the theme of antimalware software. It specifically monitors email communications for spam and other forms of unwanted email in order to stop hoaxes, identity theft, waste of resources, and possible distribution of malicious software. Antispam software can

often be installed on email servers to protect an entire organization as well as on local client systems for supplemental filtering by the user.

In addition to client application or client-side spam filters, there are enterprise spam tools, including Sender Policy Framework (SPF), Domain Keys Identified Mail (DKIM), and Domain Message Authentication Reporting and Conformance (DMARC) (see Chapter 12, “Secure Communications and Network Attacks”).

Another important issue to address when managing spam is spoofed email. A *spoofed email* is a message that has a fake or falsified source address. DMARC is used to filter spoofed messages.

Spam is most commonly associated with email, but spam also exists in instant messaging (IM), SMS, USENET (Network News Transfer Protocol [NNTP]), social media apps, and web content (such as threaded discussions, forums, comments, and blogs). Failing to block spam allows it to waste resources, consume bandwidth, distract workers from productive activities, and potentially expose users and systems to malware.

Shoulder Surfing

Shoulder surfing is often a physical world or in-person form of social engineering. Shoulder surfing occurs when someone is able to watch a user's keyboard or view their display. Often, shoulder surfing is stopped by dividing worker groups by sensitivity levels and limiting access to certain areas of the building by using locked doors. Additionally, users should not orient their displays to be visible through windows (from outside) or walkways/doorways (for internal issues). And they should not work on sensitive data while in a public space. Password fields should mask characters as they are typed. Another defense against shoulder surfing is the use of screen filters, which limit the field of view to mostly a perpendicular orientation.

Invoice Scams

Invoice scams are social engineering attacks that often attempt to steal funds from an organization or individuals through the presentation of a false invoice, often followed by strong inducements to pay. Attackers often try to target members of financial departments or accounting groups. Some

invoice scams are actually spear phishing scams in disguise. It is also possible for a social engineer to use an invoice scam approach over a voice connection.

This attack is similar to some forms of the BEC concept. In fact, some invoice scams are combined with BEC so that the invoice sent to an accounting worker is seemingly sent from the CEO. This intertwining of attack elements adds more legitimacy to the invoice, thus potentially convincing the target to pay the invoice.

To protect against invoice scams, workers must be informed of the proper channels through which they should receive invoices and the means by which to confirm that any invoices are actually valid. One such method is the use of assigned purchase order numbers from authorized work orders, which must then be included on all related invoices. When invoices arrive, they should be compared against the expected bills based on approved acquisitions. Any invoice that is not expected or otherwise abnormal should trigger a face-to-face discussion with the supervisor or other financial executive.

Separation of duties should exist between workers who place orders for products and services and those who pay invoices. These two groups should also have a third group that audits and governs their activities. All potential acquisitions should be reviewed and approved by a supervisor, and then notice of the acquisition should be sent to the accounts payable department by that supervisor.

Discovery of any fraudulent invoices must be reported to the authorities. Digital transmission and postal delivery of invoice scams are considered a crime of fraud and potential theft. The sending of false invoices through the U.S. Postal Service may be considered postal fraud as well.

Hoax

A *hoax* is a form of social engineering designed to convince targets to perform an action that will cause problems or reduce their IT security. A hoax can be an email that proclaims some imminent threat is spreading across the Internet and that you must perform certain tasks in order to protect yourself. The hoax often claims that taking no action will result in harm. Victims may be instructed to delete files, change configuration

settings, or install fraudulent security software, which results in a compromised OS, a nonbooting OS, or a reduction in their security defenses. Additionally, hoax emails often encourage the victim to forward the message to all their contacts in order to “spread the word.” Hoax messages are often spoofed without a verifiable origin.

Whenever you encounter a potential hoax or just are concerned that a claimed threat is real, do the research. A couple of great places to check for hoax information or to look up your suspected hoax message are `snopes.com` and `phishtank.com`.

Impersonation and Masquerading

Impersonation is the act of taking on the identity of someone else. This can take place in person, over the phone, through email, by logging into someone's account, or through any other means of communication. Impersonation can also be known as *masquerading*, spoofing, and even identity fraud. In some circumstances, impersonation is defined as a more sophisticated and complex attack, whereas masquerading is amateurish and simpler. This distinction is emphasized in the difference between renting an Elvis costume (i.e., masquerading) for a party versus being a career Elvis impersonator.

Defenses against physical location impersonation can include the use of access badges and security guards, and requiring the presentation and verification of ID at all entrances. If nontypical personnel are to visit a facility, the visit should be prearranged and the security guards provided with reasonable and confirmed notice that a nonemployee will be visiting. The organization from which the visitor hails should provide identification details, including a photo ID. When the person arrives, their identity should be compared against the provided credentials. In most secure environments, visitors are not allowed to roam free. Instead, an escort must accompany the visitor for their entire time within the company's security perimeter.

Tailgating and Piggybacking

Tailgating occurs when an unauthorized entity gains access to a facility under the authorization of a valid worker but without their knowledge. This attack can occur when a worker uses their valid credentials to unlock and

open a door, then walks into the building as the door closes, granting the attacker the opportunity to stop the door from closing and to sneak in without the victim realizing. Tailgating is an attack that does not depend on the consent of the victim—just their obliviousness to what occurs behind them as they walk into a building.

Each and every time a user unlocks or opens a door, they should ensure that it is closed and locked before walking away. This action alone eliminates tailgating, but it does require that workers change their behavior. There is also social pressure to hold open a door for someone who is walking up behind you, but this courtesy should not be extended to include secure entry points, even if you think you know the person walking up behind.

Company policy should be focused on changing user behavior toward more security, but realize that working against human nature is very hard. Therefore, other means of enforcing tailgating protections should be implemented. These can include the use of access control vestibules (previously known as mantraps), security cameras, and security guards. Security cameras act as a deterrent more than a prevention, but having a recording of tailgating events can help track down the perpetrators as well as pinpoint the workers who need more security training. A security guard can watch over an entrance to ensure that only valid personnel are let through a security checkpoint.

A problem similar to tailgating is piggybacking. *Piggybacking* occurs when an unauthorized entity gains access to a facility under the authorization of a valid worker by tricking the victim into providing consent. This could happen when the intruder feigns the need for assistance by holding a large box or lots of paperwork and asks someone to “hold the door.” The goal of the intruder is to distract the victim while the attacker gains access in order to prevent the victim from realizing that the attacker did not provide their own credentials. This ploy depends on the good nature of most people to believe the pretext, especially when the intruder seems to have “dressed the part.”

When someone asks for assistance in holding open a secured door, users should ask for proof of authorization or offer to swipe the person's access card on their behalf. Or, the worker should redirect the person to the main entrance controlled by security guards or call over a security guard to

handle the situation. Also, the use of access control vestibules, turnstiles, and security cameras is useful in response to piggybacking. These controls reduce the chance of an outsider bluffing their way into your secured areas.

Baiting

When direct physical entry isn't possible or attempts fail, adversaries may use a baiting technique to deposit malware onto internal systems. Baiting is when the attacker drops USB sticks, optical discs, or even wallets in a location where a worker is likely to encounter it. The hope is the worker will plug the USB drive or insert the disc into a work computer where the malware will auto-infect the system. The wallet often has a note in it with a URL or IP address along with credentials. The hope is the victim will visit the site from a work computer and be infected by a drive-by-download event or be tricked by a phishing site.

Dumpster Diving

Dumpster diving is the act of digging through trash, discarded equipment, or abandoned locations in order to obtain information about a target organization or individual. Typical collected items include old calendars, calling lists, handwritten meeting notes, discarded forms, product boxes, user manuals, sticky notes, printed reports, or the test sheet from a printer. Just about anything that is of any minor internal value or sensitivity is a treasure to be discovered through dumpster diving. The materials gathered via dumpster diving can be used to craft a more believable pretext.

To prevent dumpster diving, or at least reduce its value to an attacker, all documents should be shredded and/or incinerated before being discarded. Additionally, no storage media should ever be discarded in the trash; use a secure disposal technique or service. Secure storage media disposal often includes incineration, shredding, or chipping.

Identity Fraud

Identity fraud and *identity theft* are terms that are often used interchangeably. In fact, the U.S. Department of Justice (DoJ) states that

“Identity theft and identity fraud are terms used to refer to all types of crime in which someone wrongfully obtains and uses another person's personal data in some way that involves fraud or deception, typically for economic gain.” Identity fraud and identity theft can be both the purpose of a social engineering attack (i.e., to steal PII) as well as a tool used to further the success of a social engineering attack.

However, it is important to recognize that while we can use the terms as synonyms (especially in casual conversation), there is more value to be gained by understanding how they are different.

Identity theft is the act of stealing someone's identity. Specifically, this can refer to the initial act of information gathering or elicitation where usernames, emails, passwords, answers to secret questions, credit card numbers, Social Security numbers, healthcare services numbers, and other related and relevant facts are stolen or otherwise obtained by the attacker. So, the first definition of identity theft is the actual theft of the credentials and information for someone's accounts or financial positions.

A second definition of identity theft is when those stolen credentials and details are used to take over someone's account. This could include logging into their account on an online service; making false charges to their credit card, ATM card, or debit card; writing false checks against their checking account; or opening a new line of credit in the victim's name using their Social Security number. When an attacker steals and uses a victim's credentials, this is known as *credential hijacking*.

This second definition of identity theft is also very similar to the definition of identity fraud. Fraud is when you claim something that is false to be true. Identity fraud is when you falsely claim to be someone else through the use of stolen information from the victim. Identity fraud is criminal impersonation or intentional deception for personal or financial gain. Examples of identity fraud include taking employment under someone else's Social Security number, initiating phone service or utilities in someone else's name, or using someone else's health insurance to gain medical services.

You can consider identity theft and identity fraud to be a form of spoofing. *Spoofing* is any action to hide a valid identity, often by taking on the identity of something else. In addition to the concept of human-focused

spoofing (i.e., identity fraud), spoofing is a common tactic for malicious hackers against technology. Attackers often spoof email addresses, IP addresses, media access control (MAC) addresses, Address Resolution Protocol (ARP) communications, Wi-Fi networks, websites, mobile phone apps, and more. These and other spoofing-related topics are covered elsewhere in this book.

Identity theft and identity fraud are also related to impersonation. *Impersonation* is the act of taking on someone's identity. This might be accomplished by logging into their account with stolen credentials or claiming to be someone else when on the phone. These and other impersonation concepts were covered earlier in the “Impersonation and Masquerading” section.

As a current or future victim of identity theft/fraud, you should take actions to reduce your vulnerability, increase the chance of detecting such attacks, and improve your defenses against this type of injustice.

Typosquatting

Typosquatting is a practice employed to capture and redirect traffic when a user mistypes the domain name or IP address of an intended resource. This is a social engineering attack that takes advantage of a person's potential to mistype a fully qualified domain name (FQDN) or address. A malicious site squatter predicts URL typos and then registers those domain names to direct traffic to their own site. This can be done for competition or for malicious intent. The variations used for typosquatting include common misspellings (such as `googel.com`), typing errors (such as `gooogle.com`), variations on a name or word (for example, plurality, as in `googles.com`), and different top-level domains (TLDs) (such as `google.edu`).

URL hijacking can also refer to the practice of displaying a link or advertisement that looks like that of a well-known product, service, or site but, when clicked, redirects the user to an alternate location, service, or product. This may be accomplished by posting sites and pages and exploiting search engine optimization (SEO) to cause your content to occur higher in search results, or through the use of adware that replaces legitimate ads and links with those leading to alternate or malicious locations.

Clickjacking is a means to redirect a user's click or selection on a web page to an alternate, often malicious target instead of the intended and desired location. This can be accomplished through several techniques. Some alter the code of the original web page in order to include a script that will automatically replace the valid URL with an alternate URL at the moment the mouse click or selection occurs. Another means is to add an invisible or hidden overlay, frame, or image map over the displayed page. The user sees the original page, but any mouse click or selection will be captured by the floating frame and redirected to the malicious target. Clickjacking can be used to perform phishing attacks, hijacking, and Adversary-in-the-Middle ([AitM], aka on-path, previously known as Man-in-the-Middle [MitM]) attacks.

Influence Campaigns

Influence campaigns are social engineering attacks that attempt to guide, adjust, or change public opinion. Although such attacks might be undertaken by attackers against individuals or organizations, most influence campaigns seem to be waged by nation-states against their real or perceived foreign enemies.

Influence campaigns are linked to the distribution of false or misleading content, including:

- *Disinformation.* Intentionally false or misleading information spread with the purpose of deceiving or manipulating people. It is often used as a tool for political, ideological, or malicious agendas.
- *Misinformation.* Inaccurate or misleading information that is spread without malicious intent. It can be the result of errors, misunderstandings, or the unintentional sharing of false information.
- *Propaganda.* A systematic effort to spread ideas, information, or opinions, often of a biased or misleading nature, to promote a particular cause, political viewpoint, or ideology. It aims to shape public perception and behavior.
- *False information.* Any information that is factually incorrect or inaccurate. It can be created or spread unintentionally or intentionally and may or may not have a specific agenda.

- *“Fake news.”* A term used to describe deliberately fabricated news stories or hoaxes presented as genuine journalism. It often serves to misinform or deceive readers, and it may be politically motivated or created for profit. It can also be used to label genuine journalism as false.
- *Doxing.* Short for “document tracing” or “dropping documents.” Involves researching and publishing private or personally identifiable information about an individual, such as their real name, address, contact details, or other sensitive data, often with malicious intent, such as harassment or public shaming. Doxing can also refer to the release of false and fabricated information. Doxing can also be against organizations.

Misleading, incomplete, crafted, and altered information can be used as part of an influence campaign to adjust the perception of readers and viewers to the concepts, thoughts, and ideologies of the influencer. These tactics have been used by invaders for centuries to turn a population against their own government. In the current digital information age, influence campaigns are easier to wage than ever before and some of the perpetrators are domestic. Modern influence campaigns don't need to rely on the distribution of printed materials but can digitally transmit the propaganda directly to the targets.

Hybrid Warfare

Nations no longer limit their attacks against their real or perceived enemies using traditional, kinetic weaponry. Now they combine classical military strategy with modern capabilities, including social engineering, digital influence campaigns, psychological warfare efforts, political tactics, and cyberwarfare capabilities. This is known as *hybrid warfare*. Some entities use the term *nonlinear warfare* or *irregular warfare* to refer to this concept.

It is important to realize that nations will use whatever tools or weapons are available to them when they feel threatened or decide they must strike first. With the use of hybrid warfare tactics, there is far greater risk to every individual than in battles of the past. Now with cyberwar and influence campaigns, every person can be targeted and potentially harmed. Keep in

mind that harm is not just physical in hybrid warfare; it can also damage reputation, finances, digital infrastructure, and relationships.

For a more thorough look at hybrid warfare, read the U.S. Government Accountability Office's “Hybrid Warfare” report.



“Cyberwarfare: Origins, Motivations and What You Can Do in Response” is a helpful paper you can find at www.globalknowledge.com/us-en/resources/resource-library/white-papers/cyberwarfare-origins-motivations-and-what-you-can-do-in-response.

Social Media

Social media has become a weapon in the hands of nation-states as they wage elements of hybrid warfare against their targets. In the last decade, we have seen evidence of several nations, including our own, participating in social media–based influence campaigns. You should realize that you cannot just assume that the content you see on a social network is accurate, valid, or complete. Even when quoted by your friends, when referenced in popular media, when seemingly in line with your own expectations, you have to be skeptical of everything that reaches you through your digital communication devices. The use and abuse of social media by adversaries, foreign and domestic, brings the social engineering attack concept to a whole new level.



A great resource for learning how not to fall for false information distributed through the Internet is the “Navigating Digital Information” series presented by the YouTube channel CrashCourse: www.youtube.com/playlist?list=PL8dPuuaLjXtN07XYqqWSKpPrtNDiCHTzU.

Workers can easily waste time and system resources by interacting with social media when that task is not part of their job description. The company's acceptable user policy (AUP) should indicate that workers need to focus on work while at work rather than spending time on personal or non-work-related tasks.

Social media can be a means by which workers intentionally or accidentally distribute internal, confidential, proprietary, or PII data to outsiders. This may be accomplished by typing in messages or participating in chats in which they reveal confidential information. This can also be accomplished by distributing or publishing sensitive documents. Responses to social media issues can include blocking access to social media sites by adding IP blocks to firewalls and resolution filters to Domain Name System (DNS) queries. Violating workers need to be reprimanded or even terminated.

Establish and Maintain a Security Awareness, Education, and Training Program

The successful implementation of a security solution requires changes in user behavior. These changes primarily consist of alterations in normal work activities to comply with the standards, guidelines, and procedures mandated by the security policy. *Behavior modification* involves some level of learning on the part of the user. To develop and manage security education, training, and awareness, all relevant items of knowledge transference must be clearly identified and programs of presentation, exposure, synergy, and implementation crafted.

Awareness

A prerequisite to security training is *awareness*. The goal of creating awareness is to bring security to the forefront and make it a recognized entity for users. Awareness establishes a common baseline or foundation of security understanding across the entire organization and focuses on key or basic topics and issues related to security that all employees must understand. Awareness is not exclusively created through a classroom type of presentation but also through the work environment reminders such as posters, newsletter articles, and screen savers.



Instructor-led awareness, training, and education provide the best opportunity for real-time feedback from attendees.

Awareness establishes a minimum standard common denominator or foundation of security understanding. All personnel should be fully aware of their security responsibilities and liabilities. They should be trained to know what to do and what not to do.

The issues that users must be aware of include avoiding waste, fraud, and unauthorized activities. All members of an organization, from senior management to temporary interns, need the same level of awareness. The awareness program in an organization should be tied in with its security policy, incident-handling plan, business continuity, and disaster recovery procedures. For an awareness-building program to be effective, it must be fresh, creative, and updated often. The awareness program should also be tied to an understanding of how the corporate culture will affect and impact security for individuals as well as the organization as a whole. If employees do not see enforcement of security policies and standards among the C-level executives, especially at the awareness level, then they may not feel obligated to abide by them either.

Training

Training is teaching employees to perform their work tasks and to comply with the security policy. Training is typically hosted by an organization and is targeted to groups of employees with similar job functions. All new employees require some level of training so they will be able to comply with all standards, guidelines, and procedures mandated by the security policy. Training is an ongoing activity that must be sustained throughout the lifetime of the organization for every employee. It is considered an administrative security control.

Methods and techniques to present awareness and training should be revised and improved over time to maximize benefits. This will require that training metrics be collected and evaluated. Improved awareness and training programs may include post-learning testing as well as monitoring

for job consistency improvements and reductions in downtime, security incidents, or mistakes. This can be considered a program effectiveness evaluation.

Awareness and training are often provided in-house. That means these teaching tools are created and deployed by and within the organization itself. However, the next level of knowledge distribution is usually obtained from an external third-party source.

Education

Education is a detailed endeavor in which students and users learn much more than they actually need to know to perform their work tasks. Education is most often associated with users pursuing certification or seeking job promotion. It is typically a requirement for personnel seeking security professional positions. A security professional needs extensive knowledge of security and the local environment for the entire organization and not just for their specific work tasks.



A new development in the security education arena is micro-training (aka micro-learning, micro-education, or fast learning). Micro-training is a training and learning approach that involves delivering short, focused, and bite-sized learning modules or content to learners. These brief learning units are typically designed to be highly specific, addressing a single learning objective or a small set of related objectives in a concise and easily digestible format. Micro-training is characterized by its brevity and effectiveness in conveying information, making it well suited for the fast-paced and attention-challenged digital age. Often, micro-training is delivered via mobile apps.

Improvements

The following are techniques for improving security awareness and training:

- Change the target focus of the training. Sometimes you want to focus on the individual, sometimes on customers and clients, and other times on the organization.
- Change around topic orders or emphasis; maybe focus on social engineering during one training, then next time focus on mobile device security, and then family and travel security after that.
- Use a variety of presentation methods, such as in-person instruction, prerecorded videos, computer software/simulations, virtual reality (VR) experiences, off-site training, interactive websites, or assigned reading of either prepared courseware or off-the-shelf books (such as *Scam Me If You Can: Simple Strategies to Outsmart Today's Rip-off Artists* or *The Art of the Con: How to Think Like a Real Hustler and Avoid Being Scammed*, both by Frank Abagnale).
- Use role-playing by providing attendees with parts in a reenactment both as attacker and defender, but allow various people to offer ideas related to defending or responding to the attacks.

Develop and encourage *security champions*. These are people who take the lead in a project, such as development, leadership, or training, to enable, support, and encourage the adoption of security knowledge and practices through peer leadership, behavior demonstration, and social encouragement. Often a security champion is a member of a group who decides (or is assigned) to take charge of leading the adoption and integration of security concepts into the group's work activities. Security champions are often non-security employees who take up the mantle to encourage others to support and adopt more security practices and behaviors. Security champions are often found in software development, but this concept can be useful in any group of employees in any department.

Security awareness and training can often be improved through gamification. *Gamification* is a means to encourage compliance and engagement by integrating common elements of gameplay into other activities, such as security compliance and behavior change. This can include rewarding compliance behaviors and potentially punishing violating behaviors. Many aspects of gameplay (derived from card games, board games, sports, video games, and so on) can be integrated into security

training and adoption, such as scoring points, earning achievements or badges, competing/cooperating with others, following a set of common/standard rules, having a defined goal, seeking rewards, developing group stories/experiences, and avoiding pitfalls or negative game events. Well-applied game dynamics can result in improved worker engagement with training, an increase in organizational application of lessons, expansion of the comprehension of application of concepts, more efficient workflow, integration of more group activities such as crowdsourcing and brainstorming, increased knowledge retention, and a reduction of worker apathy. In addition to gamification, ways to improve security training include capture-the-flag drills, phishing simulations, computer-based training (CBT), and role-based training, among many others.

Effectiveness Evaluation

It is also important to perform periodic content reviews of all training materials. Reviews help ensure that the training materials and presentation stay in line with business goals, organizational mission, and security objectives. This periodic evaluation of training materials also provides the opportunity to adjust focus, add/remove topics (especially related to emerging technologies and trends), and integrate new training techniques into the courseware.



The Exam Outline objective 1.12.2 gives examples of emerging technologies and trends that should be integrated into training materials. These examples are cryptocurrency and blockchain (which are covered in Chapters 7 and 9) and artificial intelligence (Al) (which is covered in Chapter 17). These are only a few of the new concepts that every organization should be addressing as part of their security management processes, including training personnel on how to use, handle, avoid, or detect.

Additionally, new bold and subtle methods and techniques to present awareness and training should be implemented to keep the content fresh and relevant. Without periodic reviews for content relevancy, materials will

become stale and workers will likely resort to making up their own guidelines and procedures. It is the responsibility of the security governance team to establish security rules as well as provide training and education to further the implementation of those rules.

Troubleshooting personnel issues should include verifying that all personnel have attended awareness training on standard foundational security behaviors and requirements, evaluating the access and activity logs of users, and determining whether violations were intentional, coerced, accidental, or due to ignorance.

A policy violation occurs when a user breaks a rule. Users must be trained on the organization's policies and know their specific responsibilities with regard to abiding by those security rules. If a violation occurs, an internal investigation should evaluate whether it was an accident or an intentional event. If accidental, the worker should be trained on how to avoid the accident in the future, and new countermeasures may need to be implemented. If intentional, the severity of the issue may dictate a range of responses, including retraining, reassignment, and termination.

An example of a policy violation is the distribution of an internal company memo to external entities via a social network posting. Depending on the content of the memo, this could be a minor violation (such as posting a memo due to humorous or pointless content according to the worker) or a major issue (such as posting a memo that discloses a company secret or private information related to customers).

Company policy violations are not always the result of an accident or oversight on the part of the worker, nor are they always an intentional malicious choice. In fact, many internal breaches of company security are the result of intentional manipulation by malicious third parties.

Training and awareness program effectiveness evaluation should take place on an ongoing or continuous basis. Never assume that just because a worker was marked as attending or completing a training event they actually learned anything or will be changing their behavior. Some means of verification should be used to measure whether the training is beneficial or a waste of time and resources. In some circumstances, a quiz or test can be administered to workers immediately after a training session. A follow-up quiz should be performed three to six months later to see if they retained the

information presented in the training. Event and incident logs should be reviewed for the rate of occurrences of security violations due to employee actions and behaviors to see if there is any noticeable difference in the rate of occurrence or trends of incidents before and after a training presentation. Good training (and teachable employees) would be confirmed with a marked difference in user behaviors, especially a reduction of security infractions. High scores on subsequent security quizzes months later demonstrate that security concepts are retained. A combination of these processes of evaluation can help determine if a training or awareness program is being effective and is reducing the security incident rate and related response and management costs. A well-designed, engaging, and successful security training program should result in a measurable reduction in employee-related security incident management costs, hopefully far exceeding the cost of the training program itself. This would, therefore, be a good return on security investment.

Summary

When designing and deploying security solutions, you need to protect your environment from potential human threats. The aspects of secure hiring practices, defining roles, setting policies, following standards, reviewing guidelines, detailing procedures, performing risk management, providing awareness training, and cultivating management planning all contribute to protecting assets.

Secure hiring practices require detailed job descriptions. Job descriptions are used as a guide for selecting candidates and properly evaluating them for a position. Job responsibilities are the specific work tasks an employee is required to perform on a regular basis.

Employment candidate screening, background checks, reference checks, education verification, and security clearance validation are essential elements in proving that a candidate is adequate, qualified, and trustworthy for a secured position.

Onboarding involves integrating a new hire into the organization, which includes organizational socialization and orientation. When a new employee is hired, they should sign an employment agreement/contract and possibly a

nondisclosure agreement (NDA). These documents define the responsibilities and legal liabilities of the relationship between the employee and the organization.

Throughout the employment lifetime of personnel, managers should regularly review or audit the job descriptions, work tasks, privileges, and responsibilities for every staff member. For some industries, mandatory vacations may be needed. Collusion and other privilege abuses can be reduced through strict monitoring of special privileges.

Offboarding is the removal of an employee's identity from the IAM system, or it may be a part of the process of employee transfer to another division of the organization. A termination policy is needed to protect an organization and its remaining employees. The termination procedure should include an exit interview, reminder of NDAs, return of company property, and disabling of network access.

Vendor, consultant, and contractor controls (i.e., an SLA) are used to define the levels of performance, expectation, compensation, and consequences for external entities, persons, or organizations.

Compliance is the act of conforming to or adhering to rules, policies, regulations, standards, or requirements. Compliance is an important concern for security governance.

The primary goal of risk management is to reduce risk to an acceptable level. Determining this level depends on the organization, the value of its assets, and the size of its budget. Risk analysis/assessment is the process by which risk management is achieved and includes inventorying assets, analyzing an environment for threats, and evaluating each risk as to its likelihood of occurring and the cost of the resulting damage. Risk response is the assessing of the cost of various countermeasures for each risk and creating a cost/benefit report for safeguards to present to upper management.

Social engineering is a form of attack that exploits human nature and human behavior. Social engineering attacks take two primary forms: convincing someone to perform an unauthorized operation or convincing someone to reveal confidential information. The most effective defense against social engineering attacks is user education and awareness training.

The common social engineering principles are authority, intimidation, consensus, scarcity, familiarity, trust, and urgency. Eliciting information is the activity of gathering or collecting information from systems or people. Social engineering attacks include phishing, spear phishing, business email compromise (BEC), whaling, smishing, vishing, spam, shoulder surfing, invoice scams, hoaxes, impersonation, masquerading, tailgating, piggybacking, baiting, dumpster diving, identity fraud, typosquatting, and influence campaigns.

For a security solution to be successfully implemented, user behavior must change. Behavior modification involves some level of learning on the part of the user. There are three commonly recognized learning levels: awareness, training, and education.

Security-focused awareness and training programs should be reassessed and revised regularly. Some security awareness and training programs can benefit from security champions or gamification.

Study Essentials

Understand the security implications of hiring new employees. To properly plan for security, you must have standards in place for job descriptions, job classification, work tasks, job responsibilities, prevention of collusion, candidate screening, background checks, security clearances, employment agreements, and nondisclosure agreements. By deploying such mechanisms, you ensure that new hires are aware of the required security standards, thus protecting your organization's assets.

Understand onboarding and offboarding. Onboarding is the process of adding new employees to the organization using socialization and orientation. Offboarding is the removal of an employee's identity from the IAM system once that person has left the organization.

Know the principle of least privilege. The principle of least privilege states that users should be granted the minimum amount of access necessary for them to complete their required work tasks or job responsibilities.

Know about employee oversight. Throughout the employment lifetime of personnel, managers should regularly review or audit the job

descriptions, work tasks, privileges, and responsibilities for every staff member.

Know why mandatory vacations are necessary. Mandatory vacations of one to two weeks are used to audit and verify the work tasks and privileges of employees. This often results in easy detection of abuse, fraud, or negligence.

Know about UBA and UEBA. User behavior analytics (UBA) and user and entity behavior analytics (UEBA) are the concepts of analyzing the behavior of users, subjects, visitors, customers, etc. for some specific goal or purpose.

Understand employee transfers. Personnel transfers may be treated as a termination/rehire rather than a personnel move. This depends on the organization's policies and the means they have determined to best manage this change. Some of the elements that go into making the decision as to which procedure to use include whether the same user account will be retained, if their clearance will be adjusted, if their new work responsibilities are similar to the previous position, and if a “clean slate” account is required for auditing purposes in the new job position.

Be able to explain proper termination policies. A termination policy defines the procedure for terminating employees. It should include items such as always having a witness, disabling the employee's network access, and performing an exit interview. A termination policy should also include escorting the terminated employee off the premises and requiring the return of security tokens and badges and company property.

Be able to define overall risk management. The process of identifying factors that could damage or disclose data, evaluating those factors in light of data value and countermeasure cost, and implementing cost-effective solutions for mitigating or reducing risk is known as risk management. By performing risk management, you lay the foundation for reducing risk overall.

Understand risk analysis and the key elements involved. Risk analysis is the process by which upper management is provided with details to make decisions about which risks are to be mitigated, which should be transferred, and which should be accepted. To fully evaluate risks and

subsequently take the proper precautions, you must analyze the following: assets, asset valuation, threats, vulnerability, exposure, risk, realized risk, safeguards, countermeasures, attacks, and breaches.

Know how to evaluate threats. Threats can originate from numerous sources, including IT, humans, and nature. Threat assessment should be performed as a team effort to provide the widest range of perspectives. By fully evaluating risks from all angles, you reduce your system's vulnerability.

Understand qualitative risk analysis. Qualitative risk analysis is based more on scenarios than calculations. Exact dollar figures are not assigned to possible losses; instead, threats are ranked on a scale to evaluate their risks, costs, and effects. Such an analysis assists those responsible for creating proper risk management policies.

Understand quantitative risk analysis. Quantitative risk analysis focuses on hard values and percentages. A complete quantitative analysis is not possible because of intangible aspects of risk. The process involves valuing assets and identifying threats and then determining a threat's potential frequency and the resulting damage, which leads to the risk response tasks of the cost/benefit analysis of safeguards.

Know what single loss expectancy (SLE) is and how to calculate it. SLE is an element of quantitative risk analysis that represents the cost associated with a single realized risk against a specific asset. The formula is SLE = asset value (AV) * exposure factor (EF).

Know what annualized loss expectancy (ALE) is and how to calculate it. ALE is an element of quantitative risk analysis that represents the possible yearly cost of all instances of a specific realized threat against a specific asset. The formula is ALE = single loss expectancy (SLE) * annualized rate of occurrence (ARO).

Know the formula for safeguard evaluation. In addition to determining the annual cost of a safeguard, you must calculate the ALE for the asset if the safeguard is implemented. Use this formula: ALE before safeguard – ALE after implementing the safeguard – annual cost of safeguard = value of the safeguard to the company, or (ALE1 – ALE2) – ACS.

Know the options for handling risk. Reducing risk, or risk mitigation, is the implementation of safeguards and countermeasures. Assigning risk or transferring a risk places the cost of loss a risk represents onto another entity or organization. Purchasing insurance is one form of assigning or transferring risk. Risk deterrence is the process of implementing deterrents to would-be violators of security and policy. Risk avoidance is the process of selecting alternate options or activities that have less associated risk than the default, common, expedient, or cheap option. Accepting risk means management has evaluated the cost/benefit analysis of possible safeguards and has determined that the cost of the countermeasure greatly outweighs the possible cost of loss due to a risk. It also means that management has agreed to accept the consequences and the loss if the risk is realized.

Understand security control assessment (SCA). An SCA is the formal evaluation of a security infrastructure's individual mechanisms against a baseline or reliability expectation.

Understand security monitoring and measurement. Security controls should provide benefits that can be monitored and measured. If a security control's benefits cannot be quantified, evaluated, or compared, then it does not actually provide any security.

Understand risk reporting. Risk reporting involves the production of a risk report and a presentation of that report to the interested/relevant parties. A risk report should be accurate, timely, comprehensive of the entire organization, clear and precise to support decision-making, and updated on a regular basis.

Understand the Risk Maturity Model (RMM). The Risk Maturity Model (RMM) is a means to assess the key indicators and activities of a mature, sustainable, and repeatable risk management process. The RMM levels are ad hoc, preliminary, defined, integrated, and optimized.

Know about legacy system security risk. Legacy systems are often a threat because they may not be receiving security updates from their vendors. End of life (EOL) is the point at which a manufacturer no longer produces a product. End of service life (EOSL) or end of support (EOS) are those that are no longer receiving updates and support from the vendor.

Understand social engineering. Social engineering is a form of attack that exploits human nature and human behavior. The common social engineering principles are authority, intimidation, consensus, scarcity, familiarity, trust, and urgency. Such attacks may be used to elicit information or gain access through the use of pretexting and/or prepending. Social engineering attacks include phishing, spear phishing, business email compromise (BEC), whaling, smishing, vishing, spam, shoulder surfing, invoice scams, hoaxes, impersonation, masquerading, tailgating, piggybacking, baiting, dumpster diving, identity fraud, typosquatting, and influence campaigns.

Know how to implement security awareness training and education. Before actual training can take place, awareness of security as a recognized entity must be created for users. Once this is accomplished, training, or teaching employees to perform their work tasks and to comply with the security policy, can begin. All new employees require some level of training so that they will be able to comply with all standards, guidelines, and procedures mandated by the security policy. Education is a more detailed endeavor in which students/users learn much more than they actually need to know to perform their work tasks. Education is most often associated with users pursuing certification or seeking job promotion.

Know about the need for periodic content reviews and effectiveness evaluations. It is important to perform periodic content reviews of all training materials. This is to ensure that the training materials and presentation stays in line with business goals, organizational mission, and security objectives. Some means of verification should be used to measure whether the training is beneficial or a waste of time and resources.

Written Lab

1. Name six different administrative controls used to secure personnel.
2. What are the basic formulas or values used in quantitative risk assessment?
3. Describe the process or technique used to reach an anonymous consensus during a qualitative risk assessment.

4. Discuss the need to perform a balanced risk assessment. What are the techniques that can be used and why is this necessary?
5. What are the main types of social engineering principles?
6. Name several types or methods of social engineering.

Review Questions

1. You have been tasked with overseeing the security improvement project for your organization. The goal is to reduce the current risk profile to a lower level without spending considerable amounts of money. You decide to focus on the largest concern mentioned by your CISO. Which of the following is likely the element of the organization that is considered the weakest?

    A. Software products

    B. Internet connections

    C. Security policies

    D. Humans

2. Due to recent organization restructuring, the CEO believes that new workers should be hired to perform necessary work tasks and support the mission and goals of the organization. When seeking to hire new employees, what is the first step?

    A. Create a job description.

    B. Set position classification.

    C. Screen candidates.

    D. Request résumés.

3. _____ is the process of adding new employees to the organization, having them review and sign policies, be introduced to managers and coworkers, and be trained in employee operations and logistics.

    A. Reissue

B. Onboarding

C. Background checks

D. Site survey

4. After repeated events of retraining, a particular worker was caught for the fourth time attempting to access documents that were not relevant to their job position. The CSO decides this was the last chance, and the worker is to be fired. The CSO reminds you that the organization has a formal termination process that should be followed. Which of the following is an important task to perform during the termination procedure to reduce future security issues related to this former employee?

   A. Return the exiting employee's personal belongings.

   B. Review the nondisclosure agreement.

   C. Evaluate the exiting employee's performance.

   D. Cancel the exiting employee's parking permit.

5. Which of the following is a true statement in regard to vendor, consultant, and contractor controls?

   A. Using business email compromise (BEC) is a means to ensure that organizations providing services maintain an appropriate level of service agreed on by the service provider, vendor, or contractor and the customer organization.

   B. Outsourcing can be used as a risk response option known as acceptance or appetite.

   C. Multiparty risk exists when several entities or organizations are involved in a project. The risk or threats are often due to the variations of objectives, expectations, timelines, budgets, and security priorities of those involved.

   D. Risk management strategies implemented by one party do not cause additional risks against or from another party.

6. Match the term to its definition:

   1. Asset

2. Threat
3. Vulnerability
4. Exposure
5. Risk

I. The weakness in an asset, or the absence or the weakness of a safeguard or countermeasure.
II. Anything used in a business process or task.
III. Being susceptible to asset loss because of a threat; there is the possibility that a vulnerability can or will be exploited.
IV. The possibility or likelihood that a threat will exploit a vulnerability to cause harm to an asset and the severity of damage that could result.
V. Any potential occurrence that may cause an undesirable or unwanted outcome for an organization or for a specific asset.

A. 1-II, 2-V, 3-I, 4-III, 5-IV
B. 1-I, 2-II, 3-IV, 4-II, 5-V
C. 1-II, 2-V, 3-I, 4-IV, 5-III
D. 1-IV, 2-V, 3-III, 4-II, 5-I

7. While performing a risk analysis, you identify a threat of fire and a vulnerability of things being flammable because there are no fire extinguishers. Based on this information, which of the following is a possible risk?

A. Virus infection
B. Damage to equipment
C. System malfunction
D. Unauthorized access to confidential information

8. During a meeting of company leadership and the security team, discussion focuses on defining the value of assets in dollars, inventorying threats, predicting the specific amount of harm of a

breach, and determining the number of times a threat could cause harm to the company each year. What is being performed?

A. Qualitative risk assessment

B. Delphi technique

C. Risk avoidance

D. Quantitative risk assessment

9. You have performed a risk assessment and determined the threats that represent the most significant concern to your organization. When evaluating safeguards, what is the rule that should be followed in most cases?

A. The expected annual cost of asset loss should not exceed the annual costs of safeguards.

B. The annual costs of safeguards should equal the value of the asset.

C. The annual costs of safeguards should not exceed the expected annual cost of asset value loss.

D. The annual costs of safeguards should not exceed 10 percent of the security budget.

10. During a risk management project, an evaluation of several controls determines that none are cost-effective in reducing the risk related to a specific important asset. What risk response is being exhibited by this situation?

A. Mitigation

B. Ignoring

C. Acceptance

D. Assignment

11. During the annual review of the company's deployed security infrastructure, you have been reevaluating each security control selection. How is the value of a safeguard to a company calculated?

A. ALE before safeguard – ALE after implementing the safeguard – annual cost of safeguard

B. ALE before safeguard * ARO of safeguard

C. ALE after implementing safeguard + annual cost of safeguard – controls gap

D. Total risk – controls gap

12. Which of the following are valid definitions for risk? (Choose all that apply.)

A. An assessment of probability, possibility, or chance

B. Anything that removes a vulnerability or protects against one or more specific threats

C. Risk = threat * vulnerability

D. The presence of a vulnerability when a related threat exists

13. A new web application was installed onto the company's public web server last week. Over the weekend a malicious attacker was able to exploit the new code and gained access to data files hosted on the system. This is an example of what issue?

A. Inherent risk

B. Risk matrix

C. Qualitative assessment

D. Residual risk

14. Your organization is courting a new business partner. During the negotiations the other party defines several requirements of your organization's security that must be met prior to the signing of the SLA and business partners agreement (BPA). One of the requirements is that your organization demonstrate their level of achievement on the Risk Maturity Model (RMM). The requirement is specifically that a common or standardized risk framework is adopted organization-wide. Which of the five possible levels of RMM is being required of your organization?

A. Preliminary

B. Integrated

C. Defined

D. Optimized

15. The Risk Management Framework (RMF) provides a disciplined, structured, and flexible process for managing security and privacy risk that includes information security categorization; control selection, implementation, and assessment; system and common control authorizations; and continuous monitoring. The RMF has seven steps or phases. Which phase of the RMF focuses on determining whether system or common controls based on a determination that the risk to organizational operations and assets, individuals, other organizations, and the Nation are acceptable?

    A. Categorize

    B. Authorize

    C. Assess

    D. Monitor

16. Company proprietary data is discovered on a public social media posting by the CEO. While investigating, a significant number of similar emails were discovered to have been sent to employees, which included links to malicious sites. Some employees report that they had received similar messages to their personal email accounts as well. What improvements should the company implement to address this issue? (Choose two.)

    A. Deploy a web application firewall.

    B. Block access to personal email from the company network.

    C. Update the company email server.

    D. Implement multifactor authentication (MFA) on the company email server.

    E. Perform an access review of all company files.

    F. Prohibit access to social networks on company equipment.

17. What process or event is typically hosted by an organization and is targeted to groups of employees with similar job functions?

    A. Education

    B. Awareness

    C. Training

    D. Termination

18. Which of the following could be classified as a form of social engineering attack? (Choose all that apply.)

    A. A user logs in to their workstation and then decides to get a soda from the vending machine in the stairwell. As soon as the user walks away from their workstation, another person sits down at their desk and copies all the files from a local folder onto a network share.

    B. You receive an email warning about a dangerous new virus spreading across the Internet. The message tells you to look for a specific file on your hard drive and delete it, since it indicates the presence of the virus.

    C. A website claims to offer free temporary access to their products and services but requires that you alter the configuration of your web browser and/or firewall in order to download the access software.

    D. A secretary receives a phone call from a person claiming to be a client who is running late to meet the CEO. The caller asks for the CEO's private cell phone number so that they can call them.

19. Often a _____ is a member of a group who decides (or is assigned) to take charge of leading the adoption and integration of security concepts into the group's work activities. _____ are often non-security employees who take up the mantle to encourage others to support and adopt more security practices and behaviors.

    A. CISO(s)

    B. Security champion(s)

C. Security auditor(s)

D. Custodian(s)

20. The CSO has expressed concern that after years of security training and awareness programs, the level of minor security violations has actually increased. A new security team member reviews the training materials and notices that it was crafted four years ago. They suggest that the materials be revised to be more engaging and to include elements that allow for the ability to earn recognition, team up with coworkers, and strive toward a common goal. They claim these efforts will improve security compliance and foster security behavior change. What is the approach that is being recommended?

A. Program effectiveness evaluation

B. Onboarding

C. Compliance enforcement

D. Gamification

Chapter 3
Business Continuity Planning

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 1.0: Security and Risk Management**
  - 1.7 Identify, analyze, assess, prioritize, and implement Business Continuity (BC) requirements
    - 1.7.1 Business impact analysis (BIA)
    - 1.7.2 External dependencies
- ✓ **Domain 7.0: Security Operations**
  - 7.13 Participate in Business Continuity (BC) planning and exercises

Despite our best intentions, disasters of one form or another eventually strike every organization. Whether it's a natural disaster such as a hurricane, earthquake, or pandemic, or a person-made calamity such as a building fire, burst water pipe, or economic crisis, every organization will encounter events that threaten their operations or even their very existence.

Resilient organizations have plans and procedures in place to help mitigate the effects a disaster has on their continuing operations and to speed the return to normal operations. Recognizing the importance of planning for business continuity (BC) and disaster recovery (DR), ISC2 has included these two processes in the objectives for the CISSP program. Knowledge of these fundamental topics will help you prepare for the exam and help you prepare your organization for the unexpected.

In this chapter, we'll explore the concepts behind business continuity planning (BCP). Chapter 18, “Disaster Recovery Planning,” will continue the discussion and delve into the specifics of the technical controls that

organizations can put in place to restore operations as quickly as possible after disaster strikes.

Planning for Business Continuity

Business continuity planning (BCP) involves assessing the risks to organizational processes and creating policies, plans, and procedures to minimize the impact those risks might have on the organization if they were to occur. BCP is used to maintain the continuous operation of a business in the event of an emergency. The goal of BCP planners is to implement a combination of policies, procedures, and processes such that a potentially disruptive event has as little impact on the business as possible.

BCP focuses on maintaining business operations with reduced or restricted infrastructure capabilities or resources. As long as the continuity of the organization's ability to perform its mission-critical work tasks is maintained, BCP can be used to manage and restore the environment.

Business Continuity Planning vs. Disaster Recovery Planning

CISSP candidates often become confused about the difference between business continuity planning (BCP) and disaster recovery planning (DRP). They might try to sequence them in a particular order or draw firm lines between the two activities. The reality of the situation is that these lines are blurry in real life and don't lend themselves to neat and clean categorization.

The distinction between the two is one of perspective. Both activities help prepare an organization for a disaster. They intend to keep operations running continuously, when possible, and recover functions as quickly as possible if a disruption occurs. The perspective difference is that business continuity activities are typically strategically focused at a high level and center themselves on business processes and operations. Disaster recovery plans tend to be more tactical and describe technical activities such as recovery sites, backups, and fault tolerance.

In any event, don't get hung up on the difference between the two. It's much more important that you understand the processes and technologies involved in these two related disciplines.

You'll learn more about disaster recovery planning in Chapter 18.

The overall goal of BCP is to provide a quick, calm, and efficient response in the event of an emergency and to enhance a company's ability to recover from a disruptive event promptly. The BCP process has four main elements:

- Project scope and planning
- Business impact analysis
- Continuity planning
- Plan approval and implementation

The next four sections of this chapter cover each of these phases in detail. The last portion of this chapter will introduce some of the critical elements

you should consider when compiling documentation of your organization's business continuity plan.



The top priority of BCP and DRP is always *people*. The primary concern is to get people out of harm's way; then you can address IT recovery and restoration issues.

Project Scope and Planning

As with any formalized business process, the development of a resilient business continuity plan requires the use of a proven methodology. Organizations should approach the planning process with several goals in mind:

- *Organizational review:* Perform a structured review of the business's organization from a crisis planning point of view.
- *BCP team selection:* Create a BCP team with the approval of senior management.
- *Resource requirements:* Assess the resources available to participate in business continuity activities.
- *External dependencies:* Analyze the legal and regulatory landscape that governs an organization's response to a catastrophic event.

The exact process you use will depend on the size and nature of your organization and its business. There isn't a “one-size-fits-all” guide to business continuity project planning. You should consult with project planning professionals in your organization and determine the approach that will work best within your organizational culture.

The purpose of this phase is to ensure that the organization dedicates sufficient time and attention to both developing the project scope and plan and then documenting those activities for future reference.

Organizational Review

One of the first responsibilities of the individuals responsible for business continuity planning is to perform an analysis of the business organization to identify all departments and individuals who have a stake in the BCP process. Here are some areas to consider:

- Operational departments that are responsible for the core services the business provides to its clients
- Critical support services, such as the IT department, facilities and maintenance personnel, and other groups responsible for the upkeep of systems that support the operational departments
- Corporate security teams responsible for physical security, since they are many times the first responders to an incident and are also responsible for the physical safeguarding of the primary facility and alternate processing facility
- Senior executives and other key individuals essential for the ongoing viability of the organization

This identification process is critical for two reasons. First, it provides the groundwork necessary to help identify potential members of the BCP team (see the next section). Second, it builds the foundation for the remainder of the BCP process.

Typically, the individuals spearheading the BCP effort perform the business organization analysis. Some organizations employ a dedicated business continuity manager to lead these efforts, whereas others treat it as a part-time responsibility for another IT leader. Either approach is acceptable because the output of the analysis commonly guides the selection of the remaining BCP team members. However, a thorough review of this analysis should be one of the first tasks assigned to the full BCP team when it convenes. This step is critical because the individuals performing the initial analysis may have overlooked critical business functions known to BCP team members that represent other parts of the organization. If the team were to continue without revising the organizational analysis, the entire BCP process might be negatively affected, resulting in the development of a

plan that does not fully address the emergency-response needs of the organization as a whole.



When developing a business continuity plan, be sure to consider the location of both your headquarters and any branch offices. The plan should account for a disaster that occurs at any location where your organization conducts its business, including your own physical locations and those of your cloud service providers.

BCP Team Selection

In some organizations, the IT and/or security departments bear sole responsibility for business continuity planning, and no other operational or support departments provide input. Those departments may not even know of the plan's existence until a disaster looms on the horizon or actually strikes the organization. This is a critical flaw. The isolated development of a business continuity plan can spell disaster in two ways. First, the plan itself may not take into account knowledge possessed only by the individuals responsible for the day-to-day operation of the business. Second, it keeps operational elements “in the dark” about plan specifics until implementation becomes necessary. These two factors may lead to disengaged units disagreeing with provisions of the plan and failing to implement it properly. They also deny organizations the benefits achieved by a structured training and testing program for the plan.

To prevent these situations from adversely impacting the BCP process, the individuals responsible for the effort should take special care when selecting the BCP team. The team should include, at a minimum, the following individuals:

- Representatives from each of the organization's departments responsible for the core services performed by the business
- Business unit team members from the functional areas identified by the organizational analysis

- IT subject-matter experts with technical expertise in areas covered by the BCP
- Cybersecurity team members with knowledge of the BCP process
- Physical security and facility management teams responsible for the physical plant
- Attorneys familiar with corporate legal, regulatory, and contractual responsibilities
- Human resources team members who can address staffing issues and the impact on individual employees
- Public relations team members who need to conduct similar planning for how they will communicate with stakeholders and the public in the event of a disruption
- Senior management representatives with the ability to set the vision, define priorities, and allocate resources

Tips for Selecting an Effective BCP Team

Select your team carefully. You need to strike a balance between representing different points of view and creating a team with explosive personality differences. Your goal should be to create a group that is as diverse as possible and still operates in harmony.

Take some time to think about the BCP team membership and who would be appropriate for your organization's technical, financial, and political environment. Who would you include?

Each team member brings a unique perspective to the BCP process and will have individual biases. For example, representatives from operational departments will often consider their department the most critical to the organization's continued viability. Although these biases may at first seem divisive, the leader of the BCP effort should embrace them and harness them productively. If used effectively, the biases will help achieve a healthy balance in the final plan as each representative advocates the needs of their department. On the other hand, without effective leadership, these biases

may devolve into destructive turf battles that derail the BCP effort and harm the organization as a whole.

Senior Management and BCP

The role of senior management in the BCP process varies widely from organization to organization. It depends on the culture of the business, management interest in the plan, and the regulatory environment. Critical roles played by senior management usually include setting priorities, providing staff and financial resources, and arbitrating disputes about the criticality (i.e., relative importance) of services.

One of the authors recently completed a BCP consulting engagement with a large nonprofit institution. At the beginning of the engagement, he had a chance to sit down with one of the organization's senior executives to discuss his goals and objectives for their work together. During that meeting, the senior executive asked the consultant, “Is there anything you need from me to complete this engagement?”

The senior executive must have expected a perfunctory response because his eyes widened when the consultant said, “Well, as a matter of fact…” The executive then learned that his active participation in the process was critical to its success.

When working on a business continuity plan, the BCP team leader must seek and obtain as active a role as possible from a senior executive. Visible senior-level support conveys the importance of the BCP process to the entire organization. It also fosters the active participation of individuals who might write BCP off as a waste of time that they might otherwise spend on operational activities. Furthermore, laws and regulations might require the active participation of those senior leaders in the planning process. If you work for a publicly traded company, you may want to remind executives that courts may find the officers and directors of the firm personally liable if a disaster cripples the business after they failed to exercise due diligence in their contingency planning.

You may also have to convince management that BCP and DRP spending are not discretionary expenses. Management's fiduciary responsibilities to the organization's shareholders require them to at least ensure that adequate BCP measures are in place.

In the case of this BCP engagement, the executive acknowledged the importance of his support and agreed to participate. He sent an email to all employees introducing the effort and stating that it had his full backing. He also attended several of the high-level planning sessions and mentioned the effort in an organization-wide “town hall” meeting.

Resource Requirements

After the team validates the organizational review, it should turn to an assessment of the resources required by the BCP effort. This assessment involves the resources needed by three distinct BCP phases:

BCP Development The BCP team will require some resources to perform the four elements of the BCP process (project scope and planning, business impact analysis, continuity planning, and plan approval and implementation). It's more than likely that the major resource consumed by this BCP phase will be effort and time expended by members of the BCP team and the support staff they call on to assist in the development of the plan.

BCP Testing, Training, and Maintenance The testing, training, and maintenance components of this phase of the BCP will require some hardware and software commitments. Still, once again, the major commitment in this phase will be the effort of the employees involved in those activities.

BCP Implementation When a disaster strikes and the BCP team deems it necessary to conduct a full-scale implementation of the business continuity plan, the implementation will require significant resources. Those resources include a large amount of effort (BCP will likely become the focus of a large part, if not all, of the organization) as well as direct financial expenses. For this reason, the team must use its BCP implementation powers judiciously yet decisively.

An effective business continuity plan requires the expenditure of significant resources, ranging from the purchase and deployment of redundant computing facilities to the pencils and paper used by team members scratching out the first drafts of the plan. However, as you saw earlier,

personnel are one of the most significant resources consumed by the BCP process. Many security professionals overlook the importance of accounting for labor, but you can rest assured that senior management will not. Business leaders are keenly aware of the effect that time-consuming side activities have on the operational productivity of their organizations and the real cost of personnel in terms of salary, benefits, and lost opportunities. These concerns become especially paramount when you are requesting the time of senior executives.

You should expect that leaders responsible for resource utilization management will put your BCP proposal under a microscope, and you should prepare to defend the necessity of your plan with coherent, logical arguments that address the business case for BCP.



Real World Scenario

Explaining the Benefits of BCP

At a recent conference, one of the authors discussed business continuity planning with the chief information security officer (CISO) of a health system from a medium-sized U.S. city. The CISO's attitude was shocking. His organization had not conducted a formal BCP process, and he was confident that an informal approach would work fine in the unlikely event of a disaster.

This attitude is one of the most common arguments against committing resources to BCP. In many organizations, the attitude that the business has always survived, and the key leaders will figure something out in the event of a disaster, pervades corporate thinking. If you encounter this objection, you might want to point out to management the costs that will be incurred by the business (both direct costs and the indirect cost of lost opportunities) for each day that the business is down. Then, ask them to consider how long a disorganized recovery might take when compared to an orderly, planned continuity of operations (COOP).

Conducting a formal BCP effort is particularly important in healthcare organizations, where the unavailability of systems could have life-or-death consequences. In October 2020, the U.S. Cybersecurity and Infrastructure Security Agency (CISA) issued an alert notifying healthcare organizations of an outbreak of ransomware activity specifically targeting their work. Strong continuity plans play an essential role in defending against these availability attacks.

External Dependencies

When crafting a robust BCP, you need to understand and mitigate the risks associated with external dependencies. These dependencies range from technology vendors supplying critical hardware, software, and cloud services, to legal and regulatory frameworks that shape your operational

landscape. Each external factor carries potential risks that could, if unaddressed, disrupt your business operations. Consequently, a comprehensive BCP doesn't just look inward at the organization's processes, but also outward, ensuring that external parties' roles and responsibilities are clearly understood and that contingency plans are in place to tackle any disruptions in these areas.

Vendors

As you develop your BCP, it's crucial to consider the role of all technology vendors, not just those offering cloud services. These vendors, encompassing cloud service providers, hardware suppliers, and software developers, are integral to your organization's operational resilience. Their own business continuity arrangements can significantly impact your organization's ability to maintain business operations during disruptive incidents.

Consider, for example, a firm that outsources email and calendaring to a third-party software-as-a-service (SaaS) provider. Does the contract with that provider include details about the provider's service-level agreement (SLA) and commitments for restoring operations in the event of a disaster?

Also, remember that a contract is not normally sufficient due diligence when choosing a vendor. You should also verify that the vendor has the controls in place to deliver on their contractual commitments. Although it may not be possible for you to physically visit the vendor's facilities to verify their control implementation, you can always do the next best thing—send someone else!

Now, before you go off identifying an emissary and booking flights, realize that many of your vendor's customers are probably asking the same question. For this reason, the vendor may have already hired an independent auditing firm to conduct an assessment of its controls. They can make the results of this assessment available to you in the form of a System and Organization Controls (SOC) report. We cover SOC reports in more detail in Chapter 15, “Security Assessment and Testing.”

Keep in mind that there are three different versions of the SOC report. The simplest of these, a SOC 1 report, covers only internal controls over financial reporting. If you want to verify the security, processing integrity,

confidentiality, privacy, or availability controls, you'll want to review either a SOC 2 or a SOC 3 report. The American Institute of Certified Public Accountants (AICPA) sets and maintains the standards surrounding these reports to maintain consistency between auditors from different accounting firms.

For more information on this topic, see the AICPA's document comparing the SOC report types at www.aicpa-cima.com/resources/landing/system-and-organization-controls-soc-suite-of-services.

Legal and Regulatory Requirements

Many industries may find themselves bound by sector-specific, federal, state, and local laws or regulations that require them to implement various degrees of BCP. We've already discussed one example in this chapter—the officers and directors of publicly traded firms have a fiduciary responsibility to exercise due diligence in the execution of their business continuity duties. In other circumstances, the requirements (and consequences of failure) might be even more severe. Emergency services, such as police, fire, and emergency medical operations, have a responsibility to the community to continue operations in the event of a disaster. Indeed, their services become even more critical in an emergency that threatens public safety. Failure to implement an effective BCP could result in the loss of life or property and decreased public confidence in the government.

In many countries, financial institutions, such as banks, brokerages, and the firms that process their data, are subject to strict government and international banking and securities regulations. These regulations are necessarily strict because their purpose is to ensure the continued operation of the institution as a crucial part of the economy. When pharmaceutical manufacturers must produce products in less-than-optimal circumstances following a disaster or in response to a rapidly emerging pandemic, they are required to certify the purity of their products to government regulators. There are countless other examples of industries that are necessary to continue operating in the event of an emergency by various laws and regulations.

Even if you're not bound by any of these considerations, you might have contractual obligations to your clients that require you to implement sound BCP practices. If your contracts include commitments to customers expressed as *service-level agreements* (SLAs), you might find yourself in breach of those contracts if a disaster interrupts your ability to service your clients. Many clients may feel sorry for you and want to continue using your products/services, but their own business requirements might force them to sever the relationship and find new suppliers.

On the flip side of the coin, developing a strong, documented business continuity plan can help your organization win new clients and additional business from existing clients. If you can show your customers the sound procedures you have in place to continue serving them in the event of a disaster, they'll place greater confidence in your firm and might be more likely to choose you as their preferred vendor. That's not a bad position to be in!

All of these concerns point to one conclusion—it's essential to include your organization's legal counsel in the BCP process. They are intimately familiar with the legal, regulatory, and contractual obligations that apply to your organization. They can help your team implement a plan that meets those requirements while ensuring the continued viability of the organization to the benefit of all—employees, shareholders, suppliers, and customers alike.



Laws regarding computing systems, business practices, and disaster management change frequently. They also vary from jurisdiction to jurisdiction. Be sure to keep your attorneys involved throughout the lifetime of your BCP, including the testing and maintenance phases. If you restrict their involvement to a pre-implementation review of the plan, you may not become aware of the impacts that changing laws and regulations have on your corporate responsibilities.

Business Impact Analysis

Once your BCP team completes the four stages of preparing to create a business continuity plan (organizational review, BCP team selection, resource requirements, external dependencies), it's time to dive into the heart of the work—the *business impact analysis* (BIA). We approach the BIA in several stages:

1. Identifying priorities
2. Risk identification
3. Likelihood assessment
4. Impact analysis
5. Resource prioritization

The BIA identifies the business processes and tasks that are critical to an organization's ongoing viability and the threats posed to those resources. It also assesses the likelihood that each threat will occur and the impact those occurrences will have on the business. The results of the BIA provide you with quantitative and qualitative measures that can help you prioritize the commitment of business continuity resources to the various local, regional, and global risk exposures facing your organization.

It's important to realize that there are two different types of analyses that business planners use when facing a decision:

Quantitative Impact Assessment Involves the use of numbers and formulas to reach a decision. This type of data often expresses options in terms of the dollar value to the business.

Qualitative Impact Assessment Takes non-numerical factors, such as reputation, investor/customer confidence, workforce stability, and other concerns, into account. This type of data often results in categories of prioritization (such as high, medium, and low).



Quantitative assessment and qualitative assessment both play an essential role in the BCP process. However, most people tend to favor one type of analysis over the other. When selecting the individual members of the BCP team, try to achieve a balance between people who prefer each strategy. This approach helps develop a well-rounded BCP and will benefit the organization in the long run.

The BIA process described in this chapter approaches the problem from both quantitative and qualitative points of view. However, it's tempting for a BCP team to “go with the numbers” and perform a quantitative assessment while neglecting the somewhat more subjective qualitative assessment. The BCP team should perform a qualitative analysis of the factors affecting your BCP process. For example, if your business is highly dependent on a few important clients, your management team is probably willing to suffer a significant short-term financial loss to retain those clients in the long term. The BCP team must sit down and discuss (preferably with the involvement of senior management) qualitative concerns to develop a comprehensive approach that satisfies all stakeholders.



As you work your way through the BIA process, you will find that it is quite similar to the risk assessment process covered in Chapter 2, “Personnel Security and Risk Management Concepts.” The techniques used are very similar because both use standard risk evaluation techniques. The major difference is that the risk assessment process is focused on individual assets, whereas the BCP focuses on business processes and tasks.

Identifying Priorities

The first BIA task facing the BCP team is identifying business priorities. Depending on your line of business, certain activities are essential to your day-to-day operations when disaster strikes. You should create a

comprehensive list of critical business functions and rank them in order of importance. Although this task may seem somewhat daunting, it's not as hard as it sounds.

These critical business functions will vary from organization to organization, based on each organization's mission. They are the activities that, if disrupted, would jeopardize the organization's ability to achieve its goals. For example, an online retailer would treat the ability to sell products from their website and fulfill those orders promptly as critical business functions.

A great way to divide the workload of this process among the team members is to assign each participant responsibility for drawing up a prioritized list that covers the business functions for which their department is responsible. When the entire BCP team convenes, team members can use those prioritized lists to create a master prioritized list for the organization as a whole. One caution with this approach—if your team is not truly representative of the organization, you may miss critical priorities. Be sure to gather input from all parts of the organization, especially from any areas not represented on the BCP team.

This process helps identify business priorities from a qualitative point of view. Recall that we're describing an attempt to develop both qualitative and quantitative BIAs simultaneously. To begin the quantitative assessment, the BCP team should sit down and draw up a list of organization assets and then assign an *asset value (AV)* in monetary terms to each asset. These values form the basis of risk calculations performed later in the BIA.

The second quantitative measure that the team must develop is the *maximum tolerable downtime (MTD)*, sometimes also known as *maximum tolerable outage (MTO)*. The MTD is the maximum length of time a business function can tolerate a disruption before suffering irreparable harm. The MTD provides valuable information when you're performing both BCP and disaster recovery planning (DRP) planning efforts. The organization's list of critical business functions plays a crucial role in this process. The MTD for critical business functions should be lower than the MTD for activities not identified as critical. Returning to the example of an online retailer, the MTD for the website selling products may be only a few

minutes, whereas the MTD for their internal email system might be measured in hours.

The *recovery time objective (RTO)* for each business function is the amount of time in which you think you can feasibly recover the function in the event of a disruption. This value is closely related to the MTD. Once you have defined your recovery objectives, you can design and plan the procedures necessary to accomplish the recovery tasks.

As you conduct your BCP work, ensure that your RTOs are less than your MTDs, resulting in a situation in which a function should never be unavailable beyond the maximum tolerable downtime.

While the RTO and MTD measure the time to recover operations and the impact of that recovery time on operations, organizations must also pay attention to the potential data loss that might occur during an availability incident. Depending on the way that information is collected, stored, and processed, some data loss may take place.

The *recovery point objective (RPO)* is the data loss equivalent to the time-focused RTO. The RPO defines the point in time before the incident where the organization should be able to recover data from a critical business process. For example, an organization might perform database transaction log backups every 15 minutes. In that case, the RPO would be 15 minutes, meaning that the organization may lose up to 15 minutes' worth of data after an incident. If an incident takes place at 8:30 a.m., the last transaction log backup must have occurred sometime between 8:15 a.m. and 8:30 a.m. Depending on the precise timing of the incident and the backup, the organization may have irretrievably lost between 0 and 15 minutes of data.

Risk Identification

The next phase of the BIA is the identification of risks posed to your organization. Recall from Chapter 1, “Security Governance Through Principles and Policies,” that a risk occurs when an asset has a vulnerability and a threat exists that might exploit that vulnerability. During this phase, you'll have an easy time identifying some common threats, but you might need to exercise some creativity to come up with more obscure (but very real) threats to assets.

Hazards come in two forms: natural and person-made.

The following list includes some events that pose natural threats:

- Violent storms/hurricanes/tornadoes/blizzards
- Lightning strikes
- Natural wildfire
- Earthquakes
- Mudslides/avalanches
- Volcanic eruptions
- Pandemics

Person-made threats may include the following events:

- Terrorist acts/wars/civil unrest
- Workplace violence
- Theft/vandalism
- Fires/arson/explosions
- Prolonged power outages
- Building collapses
- Transportation failures
- Internet disruptions
- Service provider outages
- Economic crises

Remember, these are by no means all-inclusive lists. They merely identify some common threats that many organizations face. You may want to use them as a starting point, but a full listing of risks facing your organization will require input from all members of the BCP team.

The risk identification portion of the process is purely qualitative. At this point in the process, the BCP team should not be concerned about the

likelihood that each type of risk will materialize or the amount of damage such an occurrence would inflict upon the continued operation of the business. The results of this analysis will drive both the qualitative and quantitative portions of the remaining BIA tasks.

Likelihood Assessment

The preceding step consisted of the BCP team's drawing up a comprehensive list of the events that may pose a risk to an organization. You probably recognized that some events are much more likely to happen than others. For example, an earthquake is a much more plausible risk than a tropical storm for a business located in Southern California. A company based in Florida might have the exact opposite likelihood that each threat would occur.

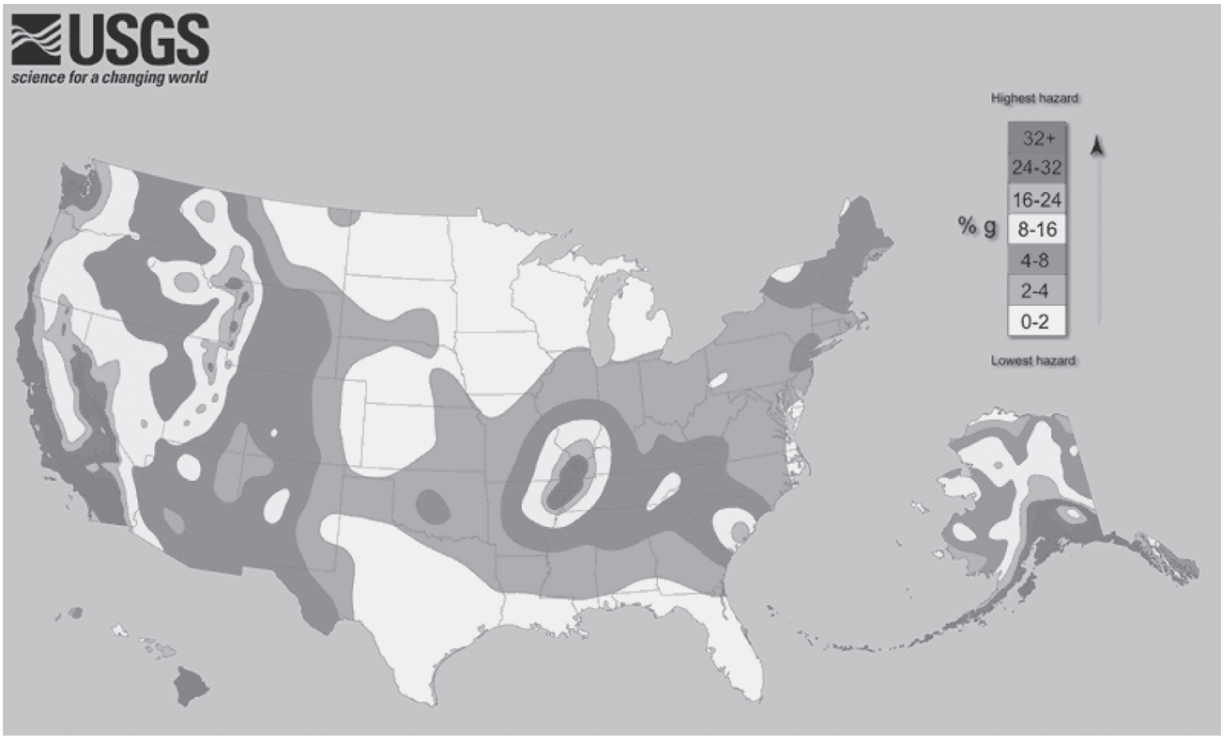
To account for these differences, the next phase of the business impact analysis identifies the likelihood that each threat will occur. We describe this likelihood using the same process used for the risk assessment in Chapter 2. First, we determine the *annualized rate of occurrence (ARO)* that reflects the number of times a business expects to experience a given disaster each year. This annualization process simplifies comparing the magnitude of very different risks.

The BCP team should sit down and determine an ARO for each threat identified in the previous section. Base these numbers on corporate history, professional experience of team members, and advice from experts, such as meteorologists, seismologists, fire prevention professionals, and other consultants, as needed.



In addition to the government resources identified in this chapter, insurance companies develop large repositories of risk information as part of their actuarial processes. You may be able to obtain this information from them to assist in your BCP efforts. After all, you have a mutual interest in preventing damage to your business!

In many cases, you may be able to find likelihood assessments for some hazards prepared by experts at no cost to you. For example, the U.S. Geological Survey (USGS) developed the earthquake hazard map shown in Figure 3.1. This map illustrates the ARO for earthquakes in various regions of the United States. Similarly, the Federal Emergency Management Agency (FEMA) coordinates the development of detailed flood maps of local communities throughout the United States. These resources are available online and offer a wealth of information to organizations performing a business impact analysis.



Source: U.S. Geological survey/www.usgs.gov/programs/earthquake-hazards/hazards

FIGURE 3.1 Earthquake hazard map of the United States

TIP One useful online tool is the nonprofit First Street Foundation's Flood Factor, which helps you quickly identify a property's risk of flooding. See www.floodfactor.com.

Impact Analysis

As you may have surmised based on its name, the impact analysis phase is one of the most critical portions of the business impact analysis. In this phase, you analyze the data gathered during earlier phases to determine the impact that each identified risk would have on the business if it were to occur.

From a quantitative point of view, we will cover three specific metrics: the exposure factor, the single loss expectancy, and the annualized loss expectancy. Each one of these values describes a particular risk/asset combination evaluated during the previous phases.

The *exposure factor* (EF) is the amount of damage that the risk poses to the asset, expressed as a percentage of the asset's value. For example, if the BCP team consults with fire experts and determines that a building fire would destroy 70 percent of the building, the exposure factor of the building to fire is 70 percent.

The *single loss expectancy (SLE)* is the monetary loss expected each time the risk materializes. You can compute the SLE using the following formula:

$$\text{SLE} = \text{Asset Value}\,(\text{AV}) \times \text{Exposure Factor}\,(\text{EF})$$

Continuing with the preceding example, if the building is worth $500,000, the single loss expectancy would be 70 percent of $500,000, or $350,000. You can interpret this figure to mean that you could expect a single fire in the building would cause $350,000 worth of damage.

The *annualized loss expectancy (ALE)* is the monetary loss that the business expects to suffer as a result of the risk harming the asset during a typical year. The SLE is the amount of damage you expect each time a disaster strikes, and the ARO (from the likelihood analysis) is the number of times you expect a disaster to occur each year. You compute the ALE by simply multiplying those two numbers:

$$\text{ALE} = \text{SLE} \times \text{ARO}$$

Returning once again to our building example, fire experts might predict that a fire will occur in the building approximately once every 30 years, specifically determining that there is a 0.03 chance of a fire in any given

year. The ALE is then 3 percent of the $350,000 SLE, or $10,500. You can interpret this figure to mean that the business should expect to lose $10,500 each year due to a fire in the building.

Obviously, a fire will not occur each year—this figure represents the average cost over the approximately 30 years between fires. It's not especially useful for budgeting considerations but proves invaluable when attempting to prioritize the assignment of BCP resources to a given risk. Of course, a business leader may decide that the risk of fire remains unacceptable and take actions that contradict the quantitative assessment. That's where qualitative assessment comes into play.



Be sure you're familiar with the quantitative formulas contained in this chapter, and the concepts of asset value, exposure factor, the annualized rate of occurrence, single loss expectancy, and annualized loss expectancy. Know the formulas and be able to work through a scenario.

From a qualitative point of view, you must consider the nonmonetary impact that interruptions might have on your business. For example, you might want to consider the following:

- Loss of goodwill among your client base
- Loss of employees to other jobs after prolonged downtime
- Social/ethical responsibilities to the community
- Negative publicity

It's difficult to put dollar values on items like these to include them in the quantitative portion of the impact analysis, but they are equally important. After all, if you decimate your client base, you won't have a business to return to when you're ready to resume operations.

Resource Prioritization

The final step of the BIA is to prioritize the allocation of business continuity resources to the various risks that you identified and assessed in earlier phases of the BIA.

From a quantitative point of view, this process is relatively straightforward. You simply create a list of all the risks you analyzed during the BIA process and sort them in descending order according to the ALE computed during the impact analysis phase. This step provides you with a prioritized list of the risks that you should address. Select as many items as you're willing and able to handle simultaneously from the top of the list and work your way down. Eventually, you'll reach a point at which you've exhausted either the list of risks (unlikely) or all your available resources (much more likely).

Recall from the previous section that we also stressed the importance of addressing qualitatively important concerns. In earlier sections about the BIA, we treated quantitative and qualitative analyses as mainly separate functions with some overlap. Now it's time to merge the two prioritized lists, which is more of an art than a science. You must sit down with the BCP team and representatives from the senior management team and combine the two lists into a single prioritized list.

Qualitative concerns may justify elevating or lowering the priority of risks that already exist on the ALE-sorted quantitative list. For example, if you run a fire suppression company, your number-one priority might be the prevention of a fire in your principal place of business even though an earthquake might cause more physical damage. The potential loss of reputation within the business community resulting from the destruction of a fire suppression company by fire might be too challenging to overcome and result in the eventual collapse of the business, justifying the increased priority.

Continuity Planning

The first two phases of the BCP process (project scope and planning and business impact analysis) focus on determining how the BCP process will work and prioritizing the business assets that you must protect against interruption. The next phase of the BCP development, continuity planning,

focuses on developing and implementing a continuity strategy to minimize the impact realized risks might have on protected assets.

Two primary subtasks are involved in continuity planning:

- Strategy development
- Provisions and processes

In this section, you'll learn about both strategy development and the provisions and processes that are essential in continuity planning. The goal of this process is to create a *continuity of operations plan (COOP)*. The continuity of operations plan focuses on how an organization will carry out critical business functions beginning shortly after a disruption occurs and extending for up to one month of sustained operations.

Strategy Development

Strategy development bridges the gap between the business impact analysis and the continuity planning elements of BCP development. The BCP team must now take the prioritized list of concerns raised by the quantitative and qualitative resource prioritization exercises and determine which risks will be addressed by the business continuity plan. Fully addressing all the contingencies would require the implementation of provisions and processes that maintain a zero-downtime posture in the face of every possible risk. For obvious reasons, implementing a policy this comprehensive is impossible.

The BCP team should look back to the MTD estimates created during the early stages of the BIA and determine which risks are deemed acceptable and which must be mitigated by BCP continuity provisions. Some of these decisions are obvious—the risk of a blizzard striking an operations facility in Egypt is negligible and constitutes an acceptable risk. The risk of a monsoon in New Delhi is severe enough that BCP provisions must mitigate it. Each of these risk assessments includes cost considerations. It's normally only appropriate to mitigate a risk if the cost of mitigation is less than the expected cost of the risk itself.

Once the BCP team determines which risks require mitigation and the level of resources that will be committed to each mitigation task, they are ready

to move on to provisions and processes.

Provisions and Processes

The provisions and processes subtask of continuity planning is the meat of the entire business continuity plan. In this subtask, the BCP team designs the specific procedures and mechanisms that will mitigate the risks deemed unacceptable during the strategy development subtask. Three categories of assets must be protected through BCP provisions and processes: people, buildings/facilities, and infrastructure. In the next three sections, we'll explore some of the techniques you can use to safeguard these categories.

People

First, you must ensure that the people within your organization are safe before, during, and after an emergency. Once you've achieved that goal, you must make provisions to allow your employees to conduct both their BCP and operational tasks in as normal a manner as possible, given the circumstances.



Don't lose sight of the fact that people are your most valuable asset. The safety of people must always come before the organization's business goals. Make sure that your business continuity plan makes adequate provisions for the security of your employees, customers, suppliers, and any other individuals who may be affected.

Management should provide team members with all the resources they need to complete their assigned tasks. At the same time, if circumstances dictate that people be present in the workplace for extended periods, arrangements must be made for shelter and food. Any continuity plan that requires these provisions should include detailed instructions for the BCP team in the event of a disaster. The organization should maintain stockpiles of provisions sufficient to feed the operational and support groups for an extended time in an accessible location. Plans should specify the periodic rotation of those stockpiles to prevent spoilage.

Buildings and Facilities

Many businesses require specialized facilities to carry out their critical operations. These might include standard office facilities, manufacturing plants, operations centers, warehouses, distribution/logistics centers, and repair/maintenance depots, among others. When you perform your BIA, you will identify those facilities that play a critical role in your organization's continued viability. Your continuity plan should address two areas for each critical facility:

Hardening Provisions Your BCP should outline mechanisms and procedures that can be put in place to protect your existing facilities against the risks defined in the strategy development phase. Hardening provisions might include steps as simple as patching a leaky roof or as complex as installing reinforced hurricane shutters and fireproof walls.

Alternate Sites If it's not feasible to harden a facility against a risk, your BCP should identify alternate site(s) where business activities can resume immediately (or at least in a time that's shorter than the maximum tolerable downtime for all affected critical business functions). Chapter 18 describes a few of the facility types that might be useful in this stage. Typically, an alternate site is associated with disaster recovery planning (DRP) rather than BCP. The organization might identify the need for an alternate site during BCP development, but it takes an actual interruption to trigger the use of the site, making it fall under the DRP.

Infrastructure

Every business depends on some sort of infrastructure for its critical processes. For many companies, a vital part of this infrastructure is an IT backbone of communications and computer systems that process orders, manage the supply chain, handle customer interaction, and perform other business functions. This backbone consists of servers, workstations, and critical communications links between sites. The BCP must address how the organization will protect these systems against risks identified during the strategy development phase. As with buildings and facilities, there are two main methods of providing this protection:

Physically Hardening Systems You can protect systems against the risks by introducing protective measures such as electronic-friendly fire suppression systems and uninterruptible power supplies.

Alternative Systems You can also protect business functions by introducing redundancy (either redundant components or completely redundant systems/communications links that rely on different facilities).

These same principles apply to whatever infrastructure components serve your critical business processes—transportation systems, electrical power grids, banking and financial systems, water supplies, and so on.

As organizations move many of their technology operations to the cloud, this doesn't reduce their reliance on physical infrastructure. Although the company may no longer operate the infrastructure themselves, they still rely on the physical infrastructure of their cloud service providers and should take measures to ensure they are comfortable with the level of continuity planning conducted by those providers. A disruption at a key cloud provider that affects one of the organization's own critical business functions can be just as damaging as a failure of the organization's own infrastructure.

Plan Approval and Implementation

Once the BCP team completes the design phase of the BCP document, it's time to gain top-level management endorsement of the plan. If you have had senior management involvement throughout the development phases of the plan, this should be a relatively straightforward process. On the other hand, if this is your first time approaching management with the BCP document, you should be prepared to provide a lengthy explanation of the plan's purpose and specific provisions.



Senior management buy-in is essential to the success of the overall BCP effort.

Plan Approval

If possible, you should attempt to have the plan endorsed by the top executive in your business—the chief executive officer, chairperson, president, or similar business leader. This move demonstrates the importance of the plan to the entire organization and showcases the business leader's commitment to business continuity. The signature of such an individual on the plan also gives it much greater weight and credibility in the eyes of other senior managers, who might otherwise brush it off as a necessary but trivial IT initiative.

Plan Implementation

Once you've received approval from senior management, it's time to dive in and start implementing your plan. The BCP team should get together and develop an implementation schedule that utilizes the resources dedicated to the program to achieve the stated process and provision goals in as prompt a manner as possible, given the scope of the modifications and the organization's attitude toward continuity planning.

After fully deploying resources, the BCP team should supervise the design and implementation of a BCP maintenance program. This program ensures that the plan remains responsive to evolving business needs.

Communication, Training and Education

Communication, training and education are essential elements of the BCP implementation. All personnel who will be involved in the plan (either directly or indirectly) should receive some sort of training on the overall plan, as well as their individual responsibilities. These responsibilities should be clearly communicated to everyone involved.

Everyone in the organization should receive at least a plan overview briefing. These briefings provide employees with the confidence that business leaders have considered the possible risks posed to the continued operation of the business and have put a plan in place to mitigate the impact on the organization should a disruption occur.

People with direct BCP responsibilities should be trained and evaluated on their specific BCP tasks to ensure that they can complete them efficiently

when disaster strikes. Furthermore, at least one backup person should be trained for every BCP task to provide redundancy in the event personnel are injured or cannot reach the workplace during an emergency.

BCP Documentation

Documentation is a critical step in the business continuity planning process. Committing your BCP methodology to paper provides several significant benefits:

- It ensures that BCP personnel have a written continuity document to reference in the event of an emergency, even if senior BCP team members are not present to guide the effort.
- It provides a historical record of the BCP process that will be useful to future personnel seeking to both understand the reasoning behind various procedures and implement necessary changes in the plan.
- It forces the team members to commit their thoughts to paper—a process that often facilitates the identification of flaws in the plan. Having the plan on paper also allows draft documents to be distributed to individuals not on the BCP team for a “sanity check.”

In the following sections, we'll explore some of the essential components of the written business continuity plan.

Continuity Planning Goals

First, the plan should describe the goals of continuity planning as set forth by the BCP team and senior management. These goals should be decided on at or before the first BCP team meeting and will most likely remain unchanged throughout the life of the BCP.

The most common goal of the BCP is quite simple: to ensure the continuous operation of the business in the face of an emergency. Other goals may also be inserted in this section of the document to meet organizational needs. For example, you might have an objective that your customer call center experiences no more than 15 consecutive minutes of downtime or that your backup servers be able to handle 75 percent of your processing load within one hour of activation.

Statement of Importance

The statement of importance reflects the criticality of the BCP to the organization's continued viability. This document commonly takes the form of a letter to the organization's employees, stating the reason that the organization devoted significant resources to the BCP development process and requesting the cooperation of all personnel in the BCP implementation phase.

Here's where the importance of senior executive buy-in comes into play. If you can put out this letter under the signature of the chief executive officer (CEO) or an officer at a similar level, the plan will carry tremendous weight as you attempt to implement changes throughout the organization. If you have the signature of a lower-level manager, you may encounter resistance as you try to work with portions of the organization outside of that individual's direct control.

Statement of Priorities

The statement of priorities flows directly from the identifying priorities phase of the business impact analysis. It simply involves listing the functions considered critical to continued business operations in a prioritized order. When listing these priorities, you should also include a statement that they were developed as part of the BCP process and reflect the importance of the functions to continued business operations in the event of an emergency and nothing more. Otherwise, the list of priorities could be used for unintended purposes and result in a political turf battle between competing organizations to the detriment of the business continuity plan.

Statement of Organizational Responsibility

The statement of organizational responsibility also comes from a senior-level executive and can be incorporated into the same letter as the statement of importance. It echoes the sentiment that “business continuity is everyone's responsibility.” The statement of organizational responsibility restates the organization's commitment to business continuity planning. It informs employees, vendors, and affiliates that the organization expects them to do everything they can to assist with the BCP process.

Statement of Urgency and Timing

The statement of urgency and timing expresses the criticality of implementing the BCP and outlines the implementation timetable decided on by the BCP team and agreed to by upper management. The wording of this statement will depend on the actual urgency assigned to the BCP process by your organization's leadership. Consider including a detailed implementation timeline to foster a sense of urgency.

Risk Assessment

The risk assessment portion of the BCP documentation essentially recaps the decision-making process undertaken during the business impact analysis. It should include a discussion of all the critical business functions considered during the BIA as well as the quantitative and qualitative analyses performed to assess the risks to those functions. Include the actual AV, EF, ARO, SLE, and ALE figures in the quantitative analysis. Also, describe the thought process behind the analysis to the reader. Finally, keep in mind that the assessment reflects a point-in-time evaluation, and the team must update it regularly to reflect changing conditions.

Risk Acceptance/Mitigation

The risk acceptance/mitigation section of the BCP documentation contains the outcome of the strategy development portion of the BCP process. It should cover each risk identified in the risk analysis portion of the document and outline one of two thought processes:

- For risks that were deemed acceptable, it should outline the reasons the risk was considered acceptable as well as potential future events that might warrant a reconsideration of this determination.
- For risks that were deemed unacceptable, it should outline the risk management provisions and processes put into place to reduce the risk to the organization's continued viability.



It's far too easy to look at a difficult risk mitigation challenge and say, “We accept this risk” before moving on to less difficult things. Business continuity planners should resist these statements and ask business leaders to document their risk acceptance decisions formally. If auditors later scrutinize your business continuity plan, they will most certainly look for formal artifacts of any risk acceptance decisions made in the BCP process.

Vital Records Program

The BCP documentation should also outline a vital records program for the organization. This document states where critical business records will be stored and the procedures for making and storing backup copies of those records.

One of the biggest challenges in implementing a vital records program is often identifying the essential records in the first place. As many organizations transitioned from paper-based to digital workflows, they often lost the rigor that existed around creating and maintaining formal file structures. Vital records may now be distributed among a wide variety of IT systems and cloud services. Some may be stored on central servers accessible to groups, whereas others may be located in digital repositories assigned to an individual employee.

If that messy state of affairs sounds like your current reality, you may want to begin your vital records program by identifying the records that are truly critical to your business. Sit down with functional leaders and ask, “If we needed to rebuild our organization today in a completely new location without access to any of our computers or files, what records would you need?” Asking the question in this way forces the team to visualize the actual process of re-creating operations and, as they walk through the steps in their minds, will produce an inventory of the organization's vital records. This inventory may evolve as people remember other important information sources, so you should consider using multiple conversations to finalize it.

Once you've identified the records that your organization considers vital, the next task is a formidable one: find them. You should be able to identify the storage locations for each document identified in your vital records inventory. Once you've completed this task, you can then use this vital records inventory to inform the rest of your business continuity planning efforts.

Emergency Response Guidelines

The emergency response guidelines outline the organizational and individual responsibilities for immediate response to an emergency. This document provides the first employees to detect an emergency with the steps they should take to activate provisions of the BCP that do not start automatically. This documentation should include the following:

- Immediate response procedures (security and safety procedures, fire suppression procedures, notification of appropriate emergency-response agencies, etc.)
- A list of the individuals to notify of the incident (executives, BCP team members, etc.)
- Secondary response procedures that first responders should follow while waiting for the BCP team to assemble

Your documentation should be easily accessible to everyone in the organization who may be among the first responders to a crisis incident. Any time a disruption strikes, time is of the essence. Slowdowns in activating your business continuity procedures may result in undesirable downtime for your business operations.

Maintenance

The BCP documentation and the plan itself must be living documents. Every organization encounters nearly constant change, and this dynamic nature ensures that the business's continuity requirements will also evolve. The BCP team should not disband after the plan is developed but should still meet periodically to discuss the plan and review the results of plan tests to ensure that it continues to meet organizational needs.

Minor changes to the plan do not require conducting the full BCP development process from scratch; the BCP team may make them at an informal meeting by unanimous consent. However, keep in mind that drastic changes in an organization's mission or resources may require going back to the BCP drawing board and beginning again.

Any time you make a change to the BCP, you must practice reasonable version control. All older versions of the BCP should be physically destroyed and replaced by the most current version so that no confusion exists as to the correct implementation of the BCP.

It is also a good practice to include BCP components in job descriptions to ensure that the BCP remains fresh and to increase the likelihood that team members carry out their BCP responsibilities correctly. Including BCP responsibilities in an employee's job description also makes them fair game for the performance review process.

Testing and Exercises

The BCP documentation should also outline a formalized exercise program to ensure that the plan remains current. Exercises also verify that team members receive adequate training to perform their duties in the event of a disaster. The testing process is quite similar to that used for the disaster recovery plan, so we'll reserve the discussion of the specific test types for Chapter 18.

Summary

Every organization dependent on technological resources for its survival should have a comprehensive business continuity plan in place to ensure the sustained viability of the organization when emergencies take place. Several important concepts underlie solid business continuity planning practices, including project scope and planning, business impact analysis, continuity planning, and approval and implementation.

Every organization must have plans and procedures in place to help mitigate the effects a disaster can have on continuing operations and to accelerate the return to normal operations. To determine the risks to your critical business functions that require mitigation, you must work with a

cross-functional team to conduct a business impact analysis from both quantitative and qualitative points of view. You must take the appropriate steps in developing a continuity strategy for your organization and know what to do to weather future disasters.

Finally, you must create the documentation required to ensure the effective communication of your plan to current and future BCP team participants. Such documentation should include the continuity of operations plan (COOP). The business continuity plan must also contain statements of importance, priorities, organizational responsibility, and timing. Also, the documentation should include plans for risk assessment, acceptance, and mitigation; a vital records program; emergency-response guidelines; and procedures for maintenance and testing.

Chapter 18 will take this planning to the next step—developing and implementing a disaster recovery plan that includes the technical controls required to keep your business running in the face of a disaster.

Study Essentials

Understand the four steps of the business continuity planning process. Business continuity planning involves four distinct elements: project scope and planning, business impact analysis, continuity planning, and approval and implementation. Each element contributes to the overall goal of ensuring that business operations continue uninterrupted in the face of an emergency.

Describe how to perform the business organization analysis. In the business organization analysis, the individuals responsible for leading the BCP process determine which departments and individuals have a stake in the business continuity plan. This analysis serves as the foundation for BCP team selection and, after validation by the BCP team, is used to guide the next stages of BCP development.

List the necessary members of the business continuity planning team. The BCP team should contain, at a minimum, representatives from each of the operational and support departments; technical experts from the IT department; physical and IT security personnel with BCP skills; legal representatives familiar with corporate legal, regulatory, and contractual

responsibilities; human resources members; public relations members; and representatives from senior management. Additional team members depend on the structure and nature of the organization.

Know the legal and regulatory requirements that face business continuity planners. Business leaders must exercise due diligence to ensure that shareholders' interests are protected in the event disaster strikes. Some industries are also subject to federal, state, and local regulations that mandate specific BCP procedures. Many businesses also have contractual obligations to their clients that they must meet before, during, and after a disaster.

Explain the stages of the business impact analysis process. The five stages of the business impact analysis process are identifying priorities, risk identification, likelihood assessment, impact analysis, and resource prioritization.

Describe the process used to develop a continuity strategy. During the strategy development subtask, the BCP team determines which risks they will mitigate. In the provisions and processes subtask, the team designs mechanisms and procedures that will mitigate identified risks. The plan must then be approved by senior management and implemented. Personnel must also receive training on their roles in the BCP process.

Explain the importance of comprehensively documenting an organization's business continuity plan. Committing the plan to writing provides the organization with a written record of the procedures to follow when disaster strikes. It prevents the “it's in my head” syndrome and ensures the orderly progress of events in an emergency.

Written Lab

1. Why is it essential to include legal representatives on your business continuity planning team?
2. What is wrong with the “seat-of-the-pants” approach to business continuity planning?
3. What is the difference between quantitative and qualitative assessment?

4. What critical components should you include in your business continuity training plan?
5. What are the four main elements of the business continuity planning process?

Review Questions

1. James was recently asked by his organization's CIO to lead a core team of four experts through a business continuity planning process for his organization. What is the first step that this core team should undertake?
    A. BCP team selection
    B. Business organization analysis
    C. Resource requirements analysis
    D. Legal and regulatory assessment
2. Tracy is preparing for her organization's annual business continuity exercise and encounters resistance from some managers who don't see the exercise as important and feel that it is a waste of resources. She has already told the managers that it will only take half a day for their employees to participate. What argument could Tracy make to best address these concerns?
    A. The exercise is required by policy.
    B. The exercise is already scheduled and canceling it would be difficult.
    C. The exercise is crucial to ensuring that the organization is prepared for emergencies.
    D. The exercise will not be very time-consuming.
3. The board of directors of Clashmore Circuits conducts an annual review of the business continuity planning process to ensure that adequate measures are in place to minimize the effect of a disaster on the organization's continued viability. What obligation are they satisfying by this review?

A. Corporate responsibility

B. Disaster requirement

C. Due diligence

D. Going concern responsibility

4. Darcy is leading the BCP effort for her organization and is currently in the project scope and planning phase. What should she expect will be the major resource consumed by the BCP process during this phase?

   A. Hardware

   B. Software

   C. Processing time

   D. Personnel

5. Ryan is assisting with his organization's annual business impact analysis effort. He's been asked to assign quantitative values to assets as part of the priority identification exercise. What unit of measure should he use?

   A. Monetary

   B. Utility

   C. Importance

   D. Time

6. Renee is reporting the results of her organization's BIA to senior leaders. They express frustration at all of the detail, and one of them says, “Look, we just need to know how much we should expect these risks to cost us each year.” What measure could Renee provide to best answer this question?

   A. ARO

   B. SLE

   C. ALE

   D. EF

7. Jake is conducting a business impact analysis for his organization. As part of the process, he asks leaders from different units to provide input on how long the enterprise resource planning (ERP) system could be unavailable without causing irreparable harm to the organization. What measure is he seeking to determine?

    A. SLE

    B. EF

    C. MTD

    D. ARO

8. You are concerned about the risk that an avalanche poses to your $3 million shipping facility. Based on expert opinion, you determine that there is a 5 percent chance that an avalanche will occur each year. Experts advise you that an avalanche would completely destroy your building and require you to rebuild on the same land. Ninety percent of the $3 million value of the facility is attributed to the building, and 10 percent is attributed to the land itself. What is the single loss expectancy (SLE) of your shipping facility to avalanches?

    A. $3 million

    B. $2,700,000

    C. $270,000

    D. $135,000

9. Referring to the scenario in question 8, what is the annualized loss expectancy?

    A. $3 million

    B. $2,700,000

    C. $270,000

    D. $135,000

10. You are concerned about the risk that a hurricane poses to your corporate headquarters in South Florida. The building itself is valued at $15 million. After consulting with the National Weather Service, you determine that there is a 10 percent likelihood that a hurricane will

strike over the course of a year. You hired a team of architects and engineers, who determined that the average hurricane would destroy approximately 50 percent of the building. What is the annualized loss expectancy (ALE)?

A. $750,000

B. $1.5 million

C. $7.5 million

D. $15 million

11. Chris is completing the risk acceptance documentation for his organization's business continuity plan. Which one of the following items is Chris *least* likely to include in this documentation?

    A. Listing of risks deemed acceptable

    B. Listing of future events that might warrant reconsideration of risk acceptance decisions

    C. Risk mitigation controls put in place to address acceptable risks

    D. Rationale for determining that risks were acceptable

12. Brian is developing continuity plan provisions and processes for his organization. What resource should he protect as the highest priority in those plans?

    A. Physical plant

    B. Infrastructure

    C. Financial

    D. People

13. Ricky is conducting the quantitative portion of his organization's business impact analysis. Which one of the following concerns is *least* suitable for quantitative measurement during this assessment?

    A. Loss of a plant

    B. Damage to a vehicle

    C. Negative publicity

D. Power outage

14. Lighter than Air Industries expects that it would lose $10 million if a tornado struck its aircraft operations facility. It expects that a tornado might strike the facility once every 100 years. What is the single loss expectancy for this scenario?

    A. 0.01

    B. $10 million

    C. $100,000

    D. 0.10

15. Referring to the scenario in question 14, what is the annualized loss expectancy?

    A. 0.01

    B. $10 million

    C. $100,000

    D. 0.10

16. In which business continuity planning task would you actually design procedures and mechanisms to mitigate risks deemed unacceptable by the BCP team?

    A. Strategy development

    B. Business impact analysis

    C. Provisions and processes

    D. Resource prioritization

17. Matt is supervising the installation of redundant communications links in response to a finding during his organization's BIA. What type of mitigation provision is Matt overseeing?

    A. Hardening systems

    B. Defining systems

    C. Reducing systems

D. Alternative systems

18. Helen is working on her organization's resilience plans, and her manager asks her whether the organization has sufficient technical controls in place to recover operations after a disruption. What type of plan would address the technical controls associated with alternate processing facilities, backups, and fault tolerance?

    A. Business continuity plan

    B. Business impact analysis

    C. Disaster recovery plan

    D. Vulnerability assessment

19. Darren is concerned about the risk of a serious power outage affecting his organization's data center. He consults the organization's business impact analysis and determines that the ARO of a power outage is 20 percent. He notes that the assessment took place three years ago, and no power outage has occurred. What ARO should he use in this year's assessment, assuming that none of the circumstances underlying the analysis have changed?

    A. 20 percent

    B. 50 percent

    C. 75 percent

    D. 100 percent

20. Of the individuals listed, who would provide the best endorsement for a business continuity plan's statement of importance?

    A. Vice president of business operations

    B. Chief information officer

    C. Chief executive officer

    D. Business continuity manager

Chapter 4
Laws, Regulations, and Compliance

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 1.0: Security and Risk Management**
  - 1.4 Understand legal, regulatory, and compliance issues that pertain to information security in a holistic context
    - 1.4.1 Cybercrimes and data breaches
    - 1.4.2 Licensing and Intellectual Property requirements
    - 1.4.3 Import/export controls
    - 1.4.4 Transborder data flow
    - 1.4.5 Issues related to privacy (e.g., General Data Protection Regulation (GDPR), California Consumer Privacy Act, Personal Information Protection Law, Protection of Personal Information Act)
    - 1.4.6 Contractual, legal, industry standards, and regulatory requirements

The world of compliance is a complex and dynamic landscape of legal and regulatory matters for information technology and cybersecurity professionals. National, state, and local governments have all passed overlapping laws regulating different components of cybersecurity in a patchwork manner. This leads to an incredibly confusing landscape for security professionals, who must reconcile the laws of multiple jurisdictions. Things become even more complicated for multinational companies, which must navigate the variations between international laws as well.

Law enforcement agencies have tackled the issue of cybercrime with gusto in recent years. The legislative branches of governments around the world

have at least attempted to address issues of cybercrime. Many law enforcement agencies have full-time, well-trained computer crime investigators with advanced security training. Agencies without full-time investigators should know where to turn when they require this sort of experience.

In this chapter, we'll cover the various types of laws that deal with computer security issues. We'll examine the legal issues surrounding computer crime, privacy, intellectual property, and a number of other related topics. We'll also cover basic investigative techniques, including the pros and cons of calling in assistance from law enforcement.

Categories of Laws

Three main categories of laws play a role in the U.S. legal system. Each is used to cover a variety of circumstances, and the penalties for violating laws in the different categories vary widely. In the following sections, you'll learn how criminal law, civil law, and administrative law interact to form the complex web of our justice system.

Criminal Law

Criminal law forms the bedrock of the body of laws that preserve the peace and keep our society safe. Many high-profile court cases involve matters of criminal law; these are the laws that the police and other law enforcement agencies concern themselves with. Criminal law contains prohibitions against acts such as murder, assault, robbery, and arson. Penalties for violating criminal statutes fall in a range that includes mandatory hours of community service, monetary penalties in the form of fines (small and large), and deprivation of civil liberties in the form of prison sentences.

Real World Scenario

Don't Underestimate Technology Crime Investigators

A good friend of one of the authors is a technology crime investigator for the local police department. He often receives cases of computer abuse involving threatening emails and website postings.

Recently, he shared a story about a bomb threat that had been emailed to a local high school. The perpetrator sent a threatening note to the school principal declaring that the bomb would explode at 1 p.m. and warning him to evacuate the school. The author's friend received the alert at 11 a.m., leaving him with only two hours to investigate the crime and advise the principal on the best course of action.

He quickly began issuing emergency subpoenas to internet service providers and traced the email to a computer in the school library. At 12:15 p.m., he confronted the suspect with surveillance tapes showing him at the computer in the library as well as audit logs conclusively proving that he had sent the email. The student quickly admitted that the threat was nothing more than a ploy to get out of school a couple of hours early. His explanation? “I didn't think there was anyone around here who could trace stuff like that.”

He was wrong.

A number of criminal laws serve to protect society against computer crime. In later sections of this chapter, you'll learn how some laws, such as the Computer Fraud and Abuse Act (CFAA), the Electronic Communications Privacy Act (ECPA), and the Identity Theft and Assumption Deterrence Act (ITADA) (among others), provide criminal penalties for serious cases of computer crime. Technically savvy prosecutors teamed with concerned law enforcement agencies have dealt serious blows to the “hacking underground” by using the court system to slap lengthy prison terms on offenders guilty of what used to be considered harmless pranks.

In the United States, legislative bodies at all levels of government establish criminal laws through elected representatives. At the federal level, both the House of Representatives and the Senate must pass criminal law bills by a majority vote (in most cases), and then the president must normally sign it in order for the bill to become law. Once passed, these laws then become federal law and apply in all cases where the federal government has jurisdiction (mainly cases that involve interstate commerce, cases that cross state boundaries, or cases that are offenses against the federal government itself). If federal jurisdiction does not apply, state authorities handle the case using laws passed in a similar manner by state legislators.

All federal and state laws must comply with the ultimate authority that dictates how the U.S. system of government works—the U.S. Constitution. All laws are subject to judicial review by regional courts with the right of appeal all the way to the Supreme Court of the United States. If a court finds that a law is unconstitutional, it has the power to strike it down and render it invalid.

Keep in mind that criminal law is a serious matter. If you find yourself involved—as a witness, defendant, or victim—in a matter where criminal authorities become involved, you'd be well advised to seek advice from an attorney familiar with the criminal justice system and specifically with matters of computer crime. It's not wise to “go it alone” in such a complex system.

Civil Law

Civil laws form the bulk of the U.S. body of laws. They are designed to provide for an orderly society and govern matters that are not crimes but that require an impartial arbiter to settle between individuals and organizations. Examples of the types of matters that may be judged under civil law include contract disputes, real estate transactions, employment matters, and estate/probate procedures. Civil laws also are used to create the framework of government that the executive branch uses to carry out its responsibilities. These laws provide budgets for governmental activities and lay out the authority granted to the executive branch to create administrative laws (see the next section).

Civil laws are enacted in the same manner as criminal laws. They must pass through the legislative process before enactment and are subject to the same constitutional parameters and judicial review procedures. At the federal level, both criminal and civil laws are embodied in the United States Code (USC).

The major difference between civil laws and criminal laws is the way in which they are enforced. Usually, law enforcement authorities do not become involved in matters of civil law beyond taking action necessary to restore order. In a criminal prosecution, the government, through law enforcement investigators and prosecutors, brings action against a person accused of a crime. In civil matters, it is incumbent upon the person who thinks they have been wronged to obtain legal counsel and file a civil lawsuit against the person they think is responsible for their grievance. The government (unless it is the plaintiff or defendant) does not take sides in the dispute or argue one position or the other. The only role of the government in civil matters is to provide the judges, juries, and court facilities used to hear civil cases and to play an administrative role in managing the judicial system in accordance with the law.

As with criminal law, it is best to obtain legal assistance if you think you need to file a civil lawsuit or if someone files a civil lawsuit against you. Although civil law does not impose the threat of imprisonment, the losing party may face severe financial penalties. You don't need to look any further than the daily news for examples—multimillion-dollar cases against tobacco companies, major corporations, and wealthy individuals are filed every day.

Administrative Law

The executive branch of the U.S. government charges numerous agencies with wide-ranging responsibilities to ensure that government functions effectively. It is the duty of these agencies to abide by and enforce the criminal and civil laws enacted by the legislative branch. However, as can be easily imagined, criminal and civil law can't possibly lay out rules and procedures that should be followed in every possible situation. Therefore, executive branch agencies have some leeway to enact administrative law, in the form of executive orders, policies, procedures, and regulations that govern the daily operations of the agency. Administrative law covers topics

as mundane as the procedures to be used within a federal agency to obtain a desk telephone to more substantial issues such as the immigration policies that will be used to enforce the laws passed by Congress. Administrative law is published in the Code of Federal Regulations (CFR).

Although administrative law does not require an act of the legislative branch to gain the force of law, it must comply with all existing civil and criminal laws. Government agencies may not implement regulations that directly contradict existing laws passed by the legislature. Furthermore, administrative laws (and the actions of government agencies) must also comply with the U.S. Constitution and are subject to judicial review.

To understand compliance requirements and procedures, you must be fully versed in the complexities of the law. From administrative law to civil law to criminal law (and, in some countries, even religious law), navigating the regulatory environment is a daunting task. You should focus on the generalities of law, regulations, investigations, and compliance as they affect organizational security efforts. Specifically, you will need to:

- Understand legal and regulatory issues that pertain to information security in a holistic concept.
- Determine compliance and other requirements that apply to your organization.

However, it is your responsibility to seek out professional help (i.e., an attorney) to guide and support you in your efforts to maintain legal and legally supportable security.

Laws

In this section, we'll examine a number of laws that relate to information technology. We'll examine several U.S. laws. We'll also look briefly at several high-profile non-U.S. laws, such as the European Union's General Data Protection Regulation (GDPR). Regardless, if you operate in an environment that involves foreign jurisdictions, you should retain local legal counsel to guide you through the system.



Every information security professional should have a basic understanding of the law as it relates to information technology. However, the most important lesson to be learned is knowing when it's necessary to call in an attorney. If you think you're in a legal “gray area,” it's best to seek professional advice.

Computer Crime

The first computer security issues addressed by legislators were those involving computer crime. Early computer crime prosecutions were attempted under traditional criminal law, and many were dismissed because judges thought that applying traditional law to this modern type of crime was too far a stretch. Legislators responded by passing specific statutes that defined computer crime and laid out specific penalties for various crimes. In the following sections, we'll cover several of those statutes.



The U.S. laws discussed in this chapter are federal laws. But keep in mind that almost every state in the union has also enacted some form of legislation regarding computer security issues. Because of the global reach of the Internet, most computer crimes cross state lines and, therefore, fall under federal jurisdiction and are prosecuted in the federal court system. However, in some circumstances, state laws can be more restrictive than federal laws and impose harsher penalties.

Computer Fraud and Abuse Act

The Counterfeit Access Device and Computer Fraud and Abuse Act of 1984 was the first major piece of cybercrime-specific legislation in the United States. This act was then amended in 1986 to create the modern Computer Fraud and Abuse Act (CFAA), which has since been amended periodically. Congress had earlier enacted computer crime law as part of the Comprehensive Crime Control Act (CCCA) of 1984, but the CFAA was

carefully written to exclusively cover computer crimes that crossed state boundaries to avoid infringing on states' rights and treading on thin constitutional ice. The major provisions of the original CCCA made it a crime to perform the following:

- Access classified information or financial information in a federal system without authorization or in excess of authorized privileges.
- Access a computer used exclusively by the federal government without authorization.
- Use a federal computer to perpetrate a fraud (unless the only object of the fraud was to gain use of the computer itself).
- Cause malicious damage to a federal computer system in excess of $1,000.
- Modify medical records in a computer when doing so impairs or may impair the examination, diagnosis, treatment, or medical care of an individual.
- Traffic in computer passwords if the trafficking affects interstate commerce or involves a federal computer system.

When Congress passed the CFAA, it raised the threshold of damage from $1,000 to $5,000 but also dramatically altered the scope of the regulation. Instead of merely covering federal computers that processed sensitive information, the act was changed to cover all “federal interest” computers. This widened the coverage of the act to include the following:

- Any computer used exclusively by the U.S. government
- Any computer used exclusively by a financial institution
- Any computer used by the government or a financial institution when the offense impedes the ability of the government or institution to use that system
- Any combination of computers used to commit an offense when they are not all located in the same state



Be sure you're able to briefly describe the purpose of each law discussed in this chapter.

CFAA Amendments

In 1994, Congress recognized that the face of computer security had drastically changed since the CFAA was last amended in 1986 and made a number of sweeping changes to the act. Collectively, these changes are referred to as the Computer Abuse Amendments Act of 1994 and included the following provisions:

- Outlawed the creation of any type of malicious code that might cause damage to a computer system
- Modified the CFAA to cover any computer used in interstate commerce rather than just “federal interest” computer systems
- Allowed for the imprisonment of offenders, regardless of whether they actually intended to cause damage
- Provided legal authority for the victims of computer crime to pursue civil action to gain injunctive relief and compensation for damages

Since the initial CFAA amendments in 1994, Congress passed additional amendments in 1996, 2001, 2002, and 2008 as part of other cybercrime legislation. We'll discuss those as they come up in this chapter.

Although the CFAA may be used to prosecute a variety of computer crimes, it is also criticized by many in the security and privacy community as an overbroad law. Under some interpretations, the CFAA criminalizes the violation of a website's terms of service. This law was used to prosecute Aaron Swartz for downloading a large number of academic research papers from a database accessible on the MIT network. Swartz died by suicide in 2013 and inspired the drafting of a CFAA amendment that would have excluded the violation of website terms of service from the CFAA. That bill, dubbed Aaron's Law, never reached a vote on the floor of Congress.

Ongoing legislative and judicial actions may affect the broad interpretations of the CFAA in the United States. For example, in the 2020 case *Sandvig v. Barr*, a federal court ruled that the CFAA did not apply to the violations of the terms of use of a website because that would effectively allow website operators to define the boundaries of criminal activity. In 2021, the U.S. Supreme Court ruled in the case, *Van Buren v. United States*, that someone violates CFAA if they first access a computer system that they are authorized to access but then obtain information from files, folders, or databases that they are not authorized to access.

National Information Infrastructure Protection Act of 1996

In 1996, the U.S. Congress passed yet another set of amendments to the Computer Fraud and Abuse Act designed to further extend the protection it provides. The National Information Infrastructure Protection Act (NIIPA) included the following main new areas of coverage:

- Broadens the CFAA to cover computer systems used in international commerce in addition to systems used in interstate commerce
- Extends similar protections to portions of the national infrastructure other than computing systems, such as railroads, gas pipelines, electric power grids, and telecommunications circuits
- Treats any intentional or reckless act that causes damage to critical portions of the national infrastructure as a felony

Federal Information Security Management Act of 2002

The Federal Information Security Management Act (FISMA), passed in 2002, requires that federal agencies implement an information security program that covers the agency's operations. FISMA also requires that government agencies include the activities of contractors in their security management programs. FISMA repealed and replaced two earlier laws: the Computer Security Act of 1987 and the Government Information Security Reform Act of 2000.

The National Institute of Standards and Technology (NIST), responsible for developing the FISMA implementation guidelines, outlines the following elements of an effective information security program:

- Periodic assessments of risk, including the magnitude of harm that could result from the unauthorized access, use, disclosure, disruption, modification, or destruction of information and information systems that support the operations and assets of the organization
- Policies and procedures that are based on risk assessments, cost-effectively reducing information security risks to an acceptable level and ensuring that information security is addressed throughout the life cycle of each organizational information system
- Subordinate plans for providing adequate information security for networks, facilities, information systems, or groups of information systems, as appropriate
- Security awareness training to inform personnel (including contractors and other users of information systems that support the operations and assets of the organization) of the information security risks associated with their activities and their responsibilities in complying with organizational policies and procedures designed to reduce these risks
- Periodic testing and evaluation of the effectiveness of information security policies, procedures, practices, and security controls to be performed with a frequency depending on risk, but no less than annually
- A process for planning, implementing, evaluating, and documenting remedial actions to address any deficiencies in the information security policies, procedures, and practices of the organization
- Procedures for detecting, reporting, and responding to security incidents
- Plans and procedures to ensure continuity of operations for information systems that support the operations and assets of the organization

FISMA places a significant burden on federal agencies and government contractors, who must develop and maintain substantial documentation of their FISMA compliance activities.

Federal Cybersecurity Laws of 2014

In 2014, President Barack Obama signed a series of bills into law that modernized the federal government's approach to cybersecurity issues.

The first of these was the Federal Information Security Modernization Act of 2014, which amended the 2002 version of FISMA. The 2014 FISMA modified the rules of the 2002 FISMA by centralizing federal cybersecurity responsibility with the Department of Homeland Security (DHS). There are two exceptions to this centralization: defense-related cybersecurity issues remain the responsibility of the secretary of defense, and the director of national intelligence bears responsibility for intelligence-related issues.

Second, Congress passed the Cybersecurity Enhancement Act of 2014, which charges NIST with responsibility for coordinating nationwide work on voluntary cybersecurity standards. This act was amended in 2022 to create an ongoing and voluntary public/private partnership to improve cybersecurity, strengthen cybersecurity research and development, develop the cybersecurity workforce, and build public cybersecurity awareness.

NIST produces the 800 series of Special Publications related to computer security in the federal government. These are useful for all security practitioners and are available for free online at `http://csrc.nist.gov/publications/sp800`.

The following are commonly used NIST standards:

- NIST SP 800-53: *Security and Privacy Controls for Information Systems and Organizations*. This standard is required for use in federal computing systems and is also commonly used as an industry cybersecurity benchmark.
- NIST SP 800-171: *Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations*. Compliance with this standard's security controls (which are quite similar to those found in NIST 800-53) is often included as a contractual requirement by government agencies. Federal contractors must often comply with NIST SP 800-171.
- The *NIST Cybersecurity Framework* (CSF) is a set of standards designed to serve as a voluntary risk-based framework for securing information and systems.

The third law from this wave of new requirements was the National Cybersecurity Protection Act of 2014. This law charged the Department of Homeland Security with establishing a national cybersecurity and communications integration center. The role of this center is to serve as the interface between federal agencies and civilian organizations for sharing cybersecurity risks, incidents, analysis, and warnings.

Intellectual Property (IP)

America's role in the global economy is shifting away from a manufacturer of goods and toward a provider of services. This trend also shows itself in many of the world's large industrialized nations. With this shift toward providing services, *intellectual property (IP)* takes on an increasingly important role in many firms. Indeed, it is arguable that the most valuable assets of many large multinational companies are simply the brand names that we've all come to recognize. Company names such as Dell, Procter & Gamble, and Merck bring instant credibility to any product. Publishing companies, movie producers, and artists depend on their creative output to earn their livelihood. Many products depend on secret recipes or production techniques—take the legendary secret formula for Coca-Cola or KFC's secret blend of herbs and spices, for example.

These intangible assets are collectively referred to as intellectual property (IP), and a whole host of laws exist to protect the rights of their owners. After all, it simply wouldn't be fair if a bookstore bought only one copy of each author's book and made copies for all of its customers—that would deprive the author of the benefits of their labor. In the following sections, we'll explore the laws surrounding the four major types of intellectual property—copyrights, trademarks, patents, and trade secrets. We'll also discuss how these concepts specifically concern information security professionals. Many countries protect (or fail to protect) these rights in different ways, but the basic concepts ring true throughout the world.



Some countries are notorious for violating IP rights and are world renowned for their blatant disregard of copyright and patent law. If you're planning to do business in countries where this is a problem, you should definitely consult with an attorney who specializes in this area.

Copyright and the Digital Millennium Copyright Act

Copyright law guarantees the creators of “original works of authorship” protection against the unauthorized duplication of their work. Eight broad categories of works qualify for copyright protection:

- Literary works
- Musical works
- Dramatic works
- Pantomimes and choreographic works
- Pictorial, graphical, and sculptural works
- Motion pictures and other audiovisual works
- Sound recordings
- Architectural works

There is precedent for copyrighting computer software—it's done under the scope of literary works. However, it's important to note that copyright law protects only the expression inherent in computer software—that is, the actual source code. It does not protect the ideas or process behind the software. There has also been some question over whether copyrights can be extended to cover the “look and feel” of a software package's graphical user interface. Court decisions have gone in both directions on this matter; if you will be involved in this type of issue, you should consult a qualified intellectual property attorney to determine the current state of legislation and case law.

There is a formal procedure to obtain a copyright that involves sending copies of the protected work along with an appropriate registration fee to the U.S. Copyright Office. For more information on this process, visit the office's website at www.copyright.gov. However, officially registering a copyright is not a prerequisite for copyright enforcement. Indeed, the law states that the creator of a work has an automatic copyright from the instant the work is created. If you can prove in court that you were the creator of a work (perhaps by publishing it), you will be protected under copyright law. Official registration merely provides the government's acknowledgment that they received your work on a specific date.

Copyright ownership always defaults to the creator of a work. The exceptions to this policy are works for hire. A work is considered “for hire” when it is made for an employer during the normal course of an employee's workday. For example, when an employee in a company's public relations department writes a press release, the press release is considered a work for hire. A work may also be considered a work for hire when it is made as part of a written contract declaring it as such.

Current copyright law provides for a lengthy period of protection. Works by one or more authors are protected until 70 years after the death of the last surviving author. Works for hire and anonymous works are provided protection for 95 years from the date of first publication or 120 years from the date of creation, whichever is shorter.

In 1998, Congress recognized the rapidly changing digital landscape that was stretching the reach of existing copyright law. To help meet this challenge, it enacted the hotly debated Digital Millennium Copyright Act (DMCA). The DMCA also serves to bring U.S. copyright law into compliance with terms of two World Intellectual Property Organization (WIPO) treaties.

The first major provision of the DMCA is the prohibition of attempts to circumvent copyright protection mechanisms placed on a protected work by the copyright holder. This clause was designed to protect copy-prevention mechanisms placed on digital media such as compact discs (CDs) and digital video discs (DVDs). The DMCA provides for penalties of up to $1 million and 10 years in prison for repeat offenders. Nonprofit institutions such as libraries and schools are exempted from this provision.

The DMCA also limits the liability of Internet service providers (ISPs) when their circuits are used by criminals violating the copyright law. The DMCA recognizes that ISPs have a legal status similar to the “common carrier” status of telephone companies and does not hold them liable for the “transitory activities” of their users. To qualify for this exemption, the service provider's activities must meet the following requirements (quoted directly from the Digital Millennium Copyright Act of 1998, U.S. Copyright Office Summary, December 1998):

- The transmission must be initiated by a person other than the provider.
- The transmission, routing, provision of connections, or copying must be carried out by an automatic technical process without selection of material by the service provider.
- The service provider must not determine the recipients of the material.
- Any intermediate copies must not ordinarily be accessible to anyone other than anticipated recipients and must not be retained for longer than reasonably necessary.
- The material must be transmitted with no modification to its content.

The DMCA also exempts activities of service providers related to system caching, search engines, and the storage of information on a network by individual users. However, in those cases, the service provider must take prompt action to remove copyrighted materials upon notification of the infringement.

Congress also included provisions in the DMCA that allow the creation of backup copies of computer software and any maintenance, testing, or routine usage activities that require software duplication. These provisions apply only if the software is licensed for use on a particular computer, the usage is in compliance with the license agreement, and any such copies are immediately deleted when no longer required for a permitted activity.

Finally, the DMCA spells out the application of copyright law principles to the streaming of audio and/or video content over the Internet. The DMCA states that these uses are to be treated as “eligible nonsubscription transmissions.”

Trademarks

Copyright laws are used to protect creative works; there is also protection for *trademarks*, which are words, slogans, and logos used to identify a company and its products or services. For example, a business might obtain a copyright on its sales brochure to ensure that competitors can't duplicate its sales materials. That same business might also seek to obtain trademark protection for its company name and the names of specific products and services that it offers to its clients.

The main objective of trademark protection is to avoid confusion in the marketplace while protecting the intellectual property rights of people and organizations. As with copyright protection, trademarks do not need to be officially registered to gain protection under the law. If you use a trademark in the course of your public activities, you are automatically protected under any relevant trademark law and can use the ™ symbol to show that you intend to protect words or slogans as trademarks. If you want official recognition of your trademark, you can register it with the United States Patent and Trademark Office (USPTO). This process generally requires an attorney to perform a due diligence comprehensive search for existing trademarks that might preclude your registration. The entire registration process can take more than a year from start to finish. Once you've received your registration certificate from the USPTO, you can denote your mark as a registered trademark with the ® symbol.

One major advantage of trademark registration is that you may register a trademark that you intend to use but are not necessarily already using. This type of application is called an *intent to use* application and conveys trademark protection as of the date of filing provided that you actually use the trademark in commerce within a certain time period. If you opt not to register your trademark with the PTO, your protection begins only when you first use the trademark.

The acceptance of a trademark application in the United States depends on these two main requirements:

- The trademark must not be confusingly similar to another trademark—you should determine this during your attorney's due diligence search.

There will be an open opposition period during which other companies may dispute your trademark application.

- The trademark should not be descriptive of the goods and services that you will offer. For example, “Mike's Software Company” would not be a good trademark candidate because it describes the product produced by the company. The USPTO may reject an application if it considers the trademark descriptive.

In the United States, trademarks are granted for an initial period of 10 years and can be renewed for unlimited successive 10-year periods.

Patents

Utility patents protect the intellectual property rights of inventors. They provide a period of 20 years from the time of the invention (from the date of initial application) during which the inventor is granted exclusive rights to use the invention (whether directly or via licensing agreements). At the end of the patent exclusivity period, the invention is in the public domain available for anyone to use.

Patents have three main requirements:

- The invention must be new. Inventions are patentable only if they are original ideas.
- The invention must be useful. It must actually work and accomplish some sort of task.
- The invention must not be obvious. You could not, for example, obtain a patent for your idea to use a drinking cup to collect rainwater. This is an obvious solution. You might, however, be able to patent a specially designed cup that optimizes the amount of rainwater collected while minimizing evaporation.

Protecting Software

There is some ongoing controversy over how the intellectual property contained in software should be protected. Software seems to clearly qualify for copyright protection, but litigants have disputed this notion in court.

Similarly, companies have applied for and received patents covering the way that their software “inventions” function. Cryptographic algorithms, such as RSA and Diffie–Hellman, both enjoyed patent protection at one point. This, too, is a situation that poses some legal controversy.

In the technology field, patents have long been used to protect hardware devices and manufacturing processes. There is plenty of precedent on the side of inventors in those areas. Recent patents have also been issued covering software programs and similar mechanisms, but these patents have become somewhat controversial because many of them are viewed by the technical community as overly broad. The issuance of these broad patents led to the evolution of businesses that exist solely as patent holding companies that derive their revenue by engaging in legal action against companies that they feel infringe upon the patents held in their portfolio. These companies are known by many in the technology community under the derogatory name “patent trolls.”

Design and Plant Patents

The patents described in this section are utility patents, a type of patent that protects the intellectual property around how an invention functions.

Inventors may also take advantage of design patents. These patents cover the appearance of an invention and last for only 15 years. They do not protect the idea of an invention, only the form of the invention, so they are generally seen as a weaker form of intellectual property protection than utility patents, but they are also easier to obtain.

A third type of patent, plant patents, cover new species of plants that are created by people. Those aren't normally very relevant to cybersecurity matters, unless you work in an agricultural industry!

Trade Secrets

Many companies have intellectual property that is absolutely critical to their business, and significant damage would result if it were disclosed to competitors and/or the public—in other words, *trade secrets*. We previously mentioned two examples of this type of information from popular culture—the secret formula for Coca-Cola and KFC's "secret blend of herbs and spices." Other examples are plentiful; a manufacturing company may want to keep secret a certain manufacturing process that only a few key employees fully understand, or a statistical analysis company might want to safeguard an advanced model developed for in-house use.

Two of the previously discussed intellectual property tools—copyrights and patents—could be used to protect this type of information, but with these two major disadvantages:

- Filing a copyright or patent application requires that you publicly disclose the details of your work or invention. This automatically removes the "secret" nature of your property and may harm your firm by removing the mystique surrounding a product or by allowing unscrupulous competitors to copy your property in violation of international intellectual property laws.

- Copyrights and patents both provide protection for a limited period of time. Once your legal protection expires, other firms are free to use your work at will (and they have all the details from the public disclosure you made during the application process!).

There actually is an official process regarding trade secrets. By their nature you don't register them with anyone; you keep them to yourself. To preserve trade secret status, you must implement adequate controls within your organization to ensure that only authorized personnel with a need to know the secrets have access to them. You must also ensure that anyone who does have this type of access is bound by a nondisclosure agreement (NDA) that prohibits them from sharing the information with others and provides penalties for violating the agreement. Consult an attorney to ensure that the agreement lasts for the maximum period permitted by law. In addition, you must take steps to demonstrate that you value and protect your intellectual property. Failure to do so may result in the loss of trade secret protection.

Trade secret protection is one of the best ways to protect computer software. As discussed in the previous section, patent law does not provide adequate protection for computer software products. Copyright law protects only the actual text of the source code and doesn't prohibit others from rewriting your code in a different form and accomplishing the same objective. If you treat your source code as a trade secret, it keeps it out of the hands of your competitors in the first place. This is the technique used by large software development companies such as Microsoft to protect their core base of intellectual property.

Economic Espionage

Trade secrets are often the crown jewels of major corporations, and the U.S. government recognized the importance of protecting this type of intellectual property when Congress enacted the Economic Espionage Act of 1996. This law has these two major provisions:

- Any individual found guilty of stealing trade secrets from a U.S. corporation with the intention of benefiting a foreign government or agent may be fined up to $500,000 and imprisoned for up to 15 years. Organizations found guilty may be fined up to $10,000,000.
- Any individual found guilty of stealing trade secrets under other circumstances may be fined and imprisoned for up to 10 years. Organizations found guilty may be fined up to $5,000,000.

The terms of the Economic Espionage Act give true teeth to the intellectual property rights of trade secret owners. Enforcing this law requires that companies take adequate steps to ensure that their trade secrets are well protected and not accidentally placed into the public domain.

The law was extended by the Defend Trade Secrets Act of 2016. The original Economic Espionage Act required that the government bring criminal enforcement charges. The Defend Trade Secrets Act added a civil right of action, allowing companies to file a civil suit in federal court claiming theft of trade secrets.

Software Licensing

As the software industry has evolved, so too have the complexities surrounding software licensing. Today's security professionals must navigate a landscape of diverse licensing options and agreements. Here are some contemporary software licensing types:

Perpetual Licenses This type of license allows users to pay a onetime fee for the software and use it indefinitely without any time

limitations. Typically, support and updates might require additional fees.

Subscription Licenses Unlike perpetual licenses, subscription licenses are time-bound. Users pay a recurring fee to use the software, often monthly or annually. This often includes updates and support as part of the subscription.

Open-Source Licenses Open-source software is usually free to use, modify, and distribute. However, there are various open source licenses, each with its conditions, like the GNU General Public License (GPL) or the MIT License.

Freeware Software that is available free of charge. It might come with restrictions on usage or lack features available in a paid version.

Enterprise License Agreements (ELAs) These are comprehensive agreements between software vendors and large organizations. ELAs allow for the deployment of software throughout the organization under favorable terms, often at a discounted price.

End-User License Agreements (EULAs) EULAs define the rights and restrictions that apply when using the software. These are typically presented during the installation or initial setup process.

Concurrent Use Licenses This allows a set number of users to access the software at the same time. Once the limit is reached, additional users must wait until a slot becomes available.

Named User Licenses This type of license is tied to specific users, typically identified by their login credentials, ensuring only designated individuals can access the software.

Cloud Services License Agreements These agreements pertain to software-as-a-service (SaaS) provided over the Internet. Users often encounter them when registering for online services. The agreement might present as a link to terms or flash legal information on the screen, requiring user affirmation before accessing the service.



Industry groups provide guidance and enforcement activities regarding software licensing. You can get more information from their websites. One major group is the Software Alliance at http://bsa.org.

Import/Export

The federal government recognizes that the very same computers and encryption technologies that drive the Internet and e-commerce can be extremely powerful tools in the hands of a military force. For this reason, during the Cold War, the government developed a complex set of regulations governing the export of sensitive hardware and software products to other nations. The regulations include the management of *transborder data flow* of new technologies, intellectual property, and personally identifiable information (PII).

Until recently, it was difficult to export high-powered computers outside the United States, except to a select handful of allied nations. The controls on exporting encryption software were even more severe, rendering it virtually impossible to export any encryption technology outside the country. Recent changes in federal policy have relaxed these restrictions and provided for more open commerce.

Two sets of federal regulations governing imports and exports are of particular interest to cybersecurity professionals:

- The International Traffic in Arms Regulations (ITAR) controls the manufacture, export, and import of items that are specifically designated as military and defense items, including technical information related to those items. The items covered under ITAR appear on a list called the United States Munitions List (USML), maintained in 22 CFR 121.
- The Export Administration Regulations (EAR) cover a broader set of items that are designed for commercial use but may have military applications. Items covered by EAR appear on the Commerce Control

List (CCL) maintained by the U.S. Department of Commerce. Notably, EAR includes an entire category covering information security products.

Countries of Concern

Currently, U.S. firms can export high-performance computing systems to virtually any country without receiving prior approval from the government. There are exceptions to this rule for countries designated by the Department of Commerce's Bureau of Industry and Security (BIS) as countries of concern based on the fact that they pose a threat of nuclear proliferation, they are classified as state sponsors of terrorism, or other concerns. These countries include Cuba, Iran, North Korea, and Syria.



You can find a list of countries and their corresponding computer export tiers on the Department of Commerce's website at www.bis.doc.gov.

Encryption Export Controls

The Department of Commerce's Bureau of Industry and Security (BIS) sets forth regulations on the export of encryption products outside the United States. Under previous regulations, it was virtually impossible to export even relatively low-grade encryption technology outside the United States. This placed U.S. software manufacturers at a great competitive disadvantage to foreign firms that faced no similar regulations. After a lengthy lobbying campaign by the software industry, the president directed the Commerce Department to revise its regulations to foster the growth of the American security software industry.



If you're thinking to yourself, “These regulations are confusing and overlapping,” you're not alone! Export controls are a highly specialized area of the law that require expert legal advice if you encounter them in your work.

Current regulations now designate the categories of retail and mass market security software. The rules now permit firms to submit these products for review by the Commerce Department, but the review is supposed to take no longer than 30 days. After successful completion of this review, companies may freely export these products. However, government agencies often exceed legislated deadlines, and companies must either wait until the review is complete or take the matter to court in an attempt to force a decision.

Privacy

The right to privacy has for years been a hotly contested issue in the United States. The main source of this contention is that the Constitution's Bill of Rights does not explicitly provide for a right to privacy. However, this right has been upheld by numerous courts and is vigorously pursued by organizations such as the American Civil Liberties Union (ACLU).

Europeans have also long been concerned with their privacy. Indeed, countries such as Switzerland are world renowned for their ability to keep financial secrets. Later in this chapter, we'll examine how the European Union (EU) data privacy laws impact companies and Internet users.

U.S. Privacy Law

Although there is no explicit constitutional guarantee of privacy, a myriad of federal laws (many enacted in recent years) are designed to protect the private information the government maintains about citizens as well as key portions of the private sector such as financial, educational, and healthcare institutions. In the following sections, we'll examine a number of these federal laws.

Fourth Amendment The basis for privacy rights is in the Fourth Amendment to the U.S. Constitution. It reads as follows:

> The right of the people to be secure in their persons, houses, papers, and effects, against unreasonable searches and seizures, shall not be violated, and no Warrants shall issue, but upon probable cause, supported by Oath or affirmation, and particularly describing the place to be searched, and the persons or things to be seized.

The direct interpretation of this amendment prohibits government agents from searching private property without a warrant and probable cause. The courts have expanded their interpretation of the Fourth Amendment to include protections against wiretapping and other invasions of privacy.

Privacy Act of 1974 The Privacy Act of 1974 is perhaps the most significant piece of privacy legislation restricting the way the federal government may deal with private information about individual citizens. The Privacy Act mandates that U.S. federal agencies maintain only the records that are necessary for conducting their business and that they destroy those records when they are no longer needed for a legitimate function of government. It provides a formal procedure for individuals to gain access to records the government maintains about them and to request that incorrect records be amended. It also severely limits the ability of federal government agencies to disclose private information to other people or agencies without the prior written consent of the affected individuals. It does provide for exceptions involving the census, law enforcement, the National Archives, health and safety, and court orders.



The Privacy Act of 1974 applies *only* to federal government agencies. Many people misunderstand this law and believe that it applies to how companies and other organizations handle sensitive personal information, but that is not the case.

Electronic Communications Privacy Act of 1986 The Electronic Communications Privacy Act (ECPA) makes it a crime to invade the

electronic privacy of an individual. This act broadened the Federal Wiretap Act, which previously covered communications traveling via a physical wire, to apply to any illegal interception of electronic communications or to the intentional, unauthorized access of electronically stored data. It prohibits the interception or disclosure of electronic communication and defines those situations in which disclosure is legal. It protects against the monitoring of email and voicemail communications and prevents providers of those services from making unauthorized disclosures of their content.

One of the most notable provisions of the ECPA is that it makes it illegal to monitor mobile telephone conversations. In fact, such monitoring is punishable by a fine of up to $500 and a prison term of up to five years.

Communications Assistance for Law Enforcement Act (CALEA) of 1994 The Communications Assistance for Law Enforcement Act (CALEA) of 1994 amended the Electronic Communications Privacy Act of 1986. CALEA requires all communications carriers to make wiretaps possible for law enforcement with an appropriate court order, regardless of the technology in use.

Economic Espionage Act of 1996 The Economic Espionage Act of 1996 extends the definition of property to include proprietary economic information so that the theft of this information can be considered industrial or corporate espionage. This changed the legal definition of theft so that it was no longer restricted by physical constraints.

Health Insurance Portability and Accountability Act of 1996 In 1996, Congress passed the Health Insurance Portability and Accountability Act (HIPAA), which made numerous changes to the laws governing health insurance and health maintenance organizations (HMOs). Among the provisions of HIPAA are privacy and security regulations requiring strict security measures for hospitals, physicians, insurance companies, and other organizations that process or store private medical information about individuals.

HIPAA also clearly defines the rights of individuals who are the subject of medical records and requires organizations that maintain

such records to disclose these rights in writing.



The HIPAA privacy and security regulations are quite complex. You should be familiar with the broad intentions of the act, as described here. If you work in the healthcare industry, consider devoting time to an in-depth study of this law's provisions.

Health Information Technology for Economic and Clinical Health Act of 2009 In 2009, Congress amended HIPAA by passing the Health Information Technology for Economic and Clinical Health (HITECH) Act. This law updated many of HIPAA's privacy and security requirements and was implemented through the HIPAA Omnibus Rule in 2013.

One of the changes mandated by the new regulations is a change in the way the law treats business associates, which are organizations that handle protected health information (PHI) on behalf of a HIPAA-covered entity. Any relationship between a covered entity and a business associate must be governed by a written contract known as a business associate agreement (BAA). Under the new regulation, business associates are directly subject to HIPAA and HIPAA enforcement actions in the same manner as a covered entity.

HITECH also introduced new data breach notification requirements. Under the HITECH Breach Notification Rule, HIPAA-covered entities that experience a data breach must notify affected individuals of the breach and must also notify both the Secretary of Health and Human Services and the media when the breach affects more than 500 individuals. In those cases, notification must take place without unreasonable delay and no more than 60 days after discovery of the breach.

Data Breach Notification Laws

HITECH's data breach notification rule is unique in that it is a federal law mandating the notification of affected individuals. Outside of this requirement for healthcare records, data breach notification requirements vary widely from state to state.

In 2002, California passed the Senate Bill SB 1386 and became the first state to immediately disclose to individuals the known or suspected breach of personally identifiable information. This includes unencrypted copies of a person's name in conjunction with any of the following information:

- Social Security number
- Driver's license number
- State identification card number
- Account number, credit or debit card number, in combination with any required security code, access code, or password that would permit access to an individual's financial account.

In the years following SB 1386, other states passed similar laws modeled on the California data breach notification law. In 2018, 16 years after the passage of SB 1386, Alabama and South Dakota became the last two states to pass data breach notification laws.



For a complete listing of state data breach notification laws, see www.ncsl.org/research/telecommunications-and-information-technology/security-breach-notification-laws.aspx.

Children's Online Privacy Protection Act of 1998 In April 2000, provisions of the federal Children's Online Privacy Protection Act (COPPA) became the law of the land in the United States. COPPA

makes a series of demands on websites that cater to children or knowingly collect information from children:

> Websites must have a privacy notice that clearly states the types of information they collect and what it's used for, including whether any information is disclosed to third parties. The privacy notice must also include contact information for the operators of the site.
>
> Parents must be provided with the opportunity to review any information collected from their children and permanently delete it from the site's records.
>
> Parents must give verifiable consent to the collection of information about children younger than the age of 13 prior to any such collection. Exceptions in the law allow websites to collect minimal information solely for the purpose of obtaining such parental consent.

Gramm–Leach–Bliley Act of 1999 Until the Gramm–Leach–Bliley Act (GLBA) became law in 1999, there were strict governmental barriers between financial institutions. Banks, insurance companies, and credit providers were severely limited in the services they could provide and the information they could share with each other. GLBA somewhat relaxed the regulations concerning the services each organization could provide. When Congress passed this law, it realized that this increased latitude could have far-reaching privacy implications. Because of this concern, it included a number of limitations on the types of information that could be exchanged even among subsidiaries of the same corporation and required financial institutions to provide written privacy policies to all their customers.

USA PATRIOT Act of 2001 Congress passed the Uniting and Strengthening America by Providing Appropriate Tools Required to Intercept and Obstruct Terrorism (USA PATRIOT) Act of 2001 in direct response to the September 11, 2001, terrorist attacks in New York City and Washington, DC. The PATRIOT Act greatly broadened the powers of law enforcement organizations and intelligence agencies across a number of areas, including when monitoring electronic communications.

One of the major changes prompted by the PATRIOT Act revolves around the way government agencies obtain wiretapping authorizations. Previously, police could obtain warrants for only one circuit at a time, after proving that the circuit was used by someone subject to monitoring. Provisions of the PATRIOT Act allow authorities to obtain a blanket authorization for a person and then monitor all communications to or from that person under the single warrant.

Another major change is in the way the government deals with Internet service providers (ISPs). Under the terms of the PATRIOT Act, ISPs may voluntarily provide the government with a large range of information. The PATRIOT Act also allows the government to obtain detailed information on user activity through the use of a subpoena (as opposed to a wiretap).

Finally, the USA PATRIOT Act amends the Computer Fraud and Abuse Act (yes, another set of amendments!) to provide more severe penalties for national security related criminal acts. The PATRIOT Act provides for jail terms of up to 20 years and once again expands the coverage of the CFAA.

The PATRIOT Act has a complex legislative history. Many of the key provisions of the PATRIOT Act expired in 2015 when Congress failed to pass a renewal bill. However, Congress later passed the Uniting and Strengthening America by Fulfilling Rights and Ensuring Effective Discipline Over Monitoring (USA FREEDOM) Act in June 2015, which restored key provisions of the PATRIOT Act. The provisions expired again in March 2020 and were once again renewed.

Clarifying Lawful Overseas Use of Data (CLOUD) Act The Clarifying Lawful Overseas Use of Data (CLOUD) Act was enacted in 2018, establishing procedures that govern access to data held by technology companies across national borders. This piece of legislation was introduced as a way to improve law enforcement's ability to gather digital evidence stored on servers regardless of where the servers are located, provided that the company is based within the United States or subject to U.S. jurisdiction.

Key aspects of the CLOUD Act include:

- It authorizes the U.S. government to enter into bilateral agreements with other countries to provide reciprocal rights to data relevant to criminal investigations and proceedings.
- U.S.-based technology companies can receive and must comply with lawful orders for data disclosure issued by foreign governments with which the U.S. has an executive agreement, bypassing the need for U.S. government intervention if certain conditions and human rights standards are met.
- It clarifies that U.S. law enforcement can compel U.S.-based service providers via warrant or subpoena to disclose electronic data in their possession, custody, or control, even if the data is stored on servers located outside the United States.
- The act provides mechanisms for technology companies to challenge or seek a modification to the data requests if they believe the order violates the rights of the customer or the laws of a foreign jurisdiction.

Family Educational Rights and Privacy Act The Family Educational Rights and Privacy Act (FERPA) is another specialized privacy bill that affects any educational institution that accepts any form of funding from the federal government (the vast majority of schools). It grants certain privacy rights to students 18 or older (or younger than 18 and attending a postsecondary institution) and the parents of minor students. Specific FERPA protections include the following:

- Parents/students have the right to inspect any educational records maintained by the institution on the student.
- Parents/students have the right to request correction of records they think are erroneous and the right to include a statement in the records contesting anything that is not corrected.
- Schools may not release personal information from student records without written consent, except under certain circumstances.

Identity Theft and Assumption Deterrence Act In 1998, the president signed the Identity Theft and Assumption Deterrence Act into law. In the past, the only legal victims of identity theft were the creditors who were defrauded. This law was extended by the Identity Theft Penalty Enhancement Act in 2004. Together, these laws make identity theft a crime against the person whose identity was stolen and provide severe criminal penalties (up to a 15-year prison term and/or substantial fines) for anyone found guilty of violating this law.



Real World Scenario

Privacy in the Workplace

One of the authors of this book had an interesting conversation with a relative who works in an office environment. At a family gathering, the author's relative casually mentioned a story he had read online about a local company that had fired several employees for abusing their Internet privileges. He was shocked and couldn't believe that a company would violate their employees' right to privacy.

As you've read in this chapter, the U.S. court system has long upheld the traditional right to privacy as an extension of basic constitutional rights. However, the courts have maintained that a key element of this right is that privacy should be guaranteed only when there is a “reasonable expectation of privacy.” For example, if you mail a letter to someone in a sealed envelope, you may reasonably expect that it will be delivered without being read along the way—you have a reasonable expectation of privacy. On the other hand, if you send your message on a postcard, you do so with the awareness that one or more people might read your note before it arrives at the other end—you do not have a reasonable expectation of privacy.

Recent court rulings have found that employees do not have a reasonable expectation of privacy while using employer-owned communications equipment in the workplace. If you send a message using an employer's computer, Internet connection, telephone, or other communications device, your employer can monitor it as a routine business procedure.

That said, if you're planning to monitor the communications of your employees, you should take reasonable precautions to ensure that there is no implied expectation of privacy. Here are some common measures to consider:

- Clauses in employment contracts that state the employee has no expectation of privacy while using corporate equipment

- Similar written statements in corporate acceptable use and privacy policies
- Logon banners warning that all communications are subject to monitoring
- Warning labels on computers and telephones warning of monitoring

As with many of the issues discussed in this chapter, it's a good idea to consult with your legal counsel before undertaking any communications-monitoring efforts.

European Union Privacy Law

The European Union (EU) has served as a leading force in the world of information privacy, passing a series of regulations designed to protect individual privacy rights. These laws function in a comprehensive manner, applying to almost all individually identifiable information, unlike U.S. privacy laws, which generally apply to specific industries or categories of information.

European Union General Data Protection Regulation

The European Union passed a comprehensive law covering the protection of personal information in 2016. The General Data Protection Regulation (GDPR) went into effect in 2018 and replaced the earlier Data Protection Directive (DPD). The main purpose of this law is to provide a single, harmonized law that covers data throughout the European Union, bolstering the personal privacy protections originally provided by the DPD.

A major difference between the GDPR and the data protection directive is the widened scope of the regulation. The new law applies to all organizations that collect data from EU residents or process that information on behalf of someone who collects it. Importantly, the law even applies to organizations that are *not based in the EU*, if they collect information about EU residents. The ability of the EU to enforce this law globally remains an open question.

The key provisions of the GDPR include the following:

- *Lawfulness, fairness, and transparency* says that you must have a legal basis for processing personal information, you must not process data in a manner that is misleading or detrimental to data subjects, and you must be open and honest about data processing activities.
- *Purpose limitation* says that you must clearly document and disclose the purposes for which you collect data and limit your activity to disclosed purposes.
- *Data minimization* says that you must ensure that the data you process is adequate for your stated purpose and limited to what you actually need for that purpose.
- *Accuracy* says that the data you collect, create, or maintain is correct and not misleading, that you maintain updated records, and that you correct or erase inaccurate data.
- *Storage limitation* says that you keep data only for as long as it is needed to fulfill a legitimate, disclosed purpose and that you comply with the "right to be forgotten" that allows individuals to require companies to delete their information if it is no longer needed.
- *Integrity and confidentiality* says that you must have appropriate security, integrity and confidentiality controls in place to protect data.
- *Accountability* says that you must take responsibility for actions you take with protected data and that the data controller must be able to demonstrate compliance.

Cross-Border Information Sharing

GDPR is of particular concern when transferring information across international borders. Organizations needing to conduct transfers between their subsidiaries have two options available for complying with EU regulations:

- Organizations may adopt a set of standard contractual clauses (SCCs) that have been approved for use in situations where information is being transferred outside of the EU. Those clauses are found on the European Commission website and are available for integration into contracts.

- Organizations may adopt binding corporate rules (BCRs) that regulate data transfers between internal units of the same firm. This is a very time-consuming process—the rules must be approved by every EU member nation where they will be used, so typically this path is only adopted by very large organizations.

In the past, the European Union and the United States operated a safe harbor agreement called Privacy Shield. Organizations were able to certify their compliance with privacy practices through independent assessors and, if awarded the privacy shield, were permitted to transfer information.

However, a 2020 ruling by the Court of Justice of the European Union (CJEU) in a case called Schrems II declared the EU/US Privacy Shield invalid. Currently, companies may not rely on the Privacy Shield and must use either standard contractual clauses or binding corporate rules. At the time this book went to press, efforts were underway to implement a new safe harbor program designed to meet EU requirements.

In some cases, conflicts arise between laws of different nations. For example, electronic discovery rules in the United States might require the production of evidence that is protected under GDPR. In those cases, privacy professionals should consult with attorneys to identify an appropriate course of action.



The Asia-Pacific Economic Cooperation (APEC) publishes a privacy framework that incorporates many standard privacy practices, such as preventing harm, notice, collection limitation, use of personal information, choice, integrity of personal information, security safeguards, access and correction, and accountability. This framework is used to promote the smooth cross-border flow of information between APEC member nations.

Canadian Privacy Law

Canadian law affects the processing of personal information related to Canadian residents. Chief among these, the *Personal Information*

Protection and Electronic Documents Act (PIPEDA) is a national-level law that restricts how commercial businesses may collect, use, and disclose personal information.

Generally speaking, PIPEDA covers information about an individual that is identifiable to that individual. The Canadian government provides the following examples of information covered by PIPEDA:

- Race, national, or ethnic origin
- Religion
- Age
- Marital status
- Medical, education, or employment history
- Financial information
- DNA
- Identifying numbers
- Employee performance records

The law excludes information that does not fit the definition of personal information, including the following examples provided by the Office of the Privacy Commissioner of Canada:

- Information that is not about an individual, because the connection with a person is too weak or far-removed
- Information about an organization such as a business
- Information that has been rendered anonymous, as long as it is not possible to link that data back to an identifiable person
- Certain information about public servants such as their name, position, and title
- A person's business contact information that an organization collects, uses, or discloses for the sole purpose of communicating with that person in relation to their employment, business, or profession

- Government information
- An individual's collection, use, or disclosure of personal information strictly for personal purposes (e.g., personal greeting card list)

PIPEDA may also be superseded by province-specific laws that are deemed substantially similar to PIPEDA. These laws currently exist in Alberta, British Columbia, and Quebec. PIPEDA generally does not apply to nonprofit organizations or political parties and associations. Provincial laws apply to municipalities, universities, schools, and hospitals.

Chinese Privacy Law

In recent years, China has significantly advanced its legal framework related to data protection and privacy, culminating in the *Personal Information Protection Law (PIPL)*, which came into effect in 2021. The PIPL is China's first comprehensive national standard in data privacy law, somewhat analogous to the GDPR in the EU, and it imposes stringent regulations on personal data processing activities.

Key aspects of the PIPL include:

Consent and Legitimate Purpose The PIPL mandates that data processing should be specific, clear, and legitimate. Explicit consent is required for data processing, especially for sensitive data, and individuals have the right to withdraw their consent.

Minimum Necessary Data Collection Similar to the GDPR's data minimization principle, the PIPL requires that organizations only collect personal data that is directly relevant and necessary for the stated purpose.

Data Subject Rights The law empowers individuals with several rights concerning their personal data, including the right to access, correction, deletion, and to be informed of data breaches. It also allows individuals to object to data processing.

Cross-Border Data Transfer The PIPL imposes restrictions on transferring personal data outside of China. Data exporters must conduct a security assessment and ensure that the receiving country's

data protection measures are effectively equivalent to those in China, among other obligations.

Heavy Penalties Noncompliance with the PIPL can result in severe consequences, including financial penalties, suspension of business activities, or revocation of business licenses.

South African Privacy Law

South Africa's primary legislation governing data protection is the *Protection of Personal Information Act (POPIA)*, which went into effect in 2020. POPIA promotes the protection of personal information processed by public and private bodies and introduces specific conditions for the lawful processing of personal information, closely mirroring principles seen in the GDPR.

Important provisions under POPIA include:

Lawful Processing POPIA sets out eight conditions for the lawful processing of personal information, which includes accountability, processing limitation, purpose specification, further processing limitation, information quality, openness, security safeguards, and data subject participation.

Consent Personal information must be collected directly from the data subject, with specific consent required for processing. Consent can be withdrawn, and data subjects also have the right to object to the processing of personal information.

Special Personal Information The act puts strict conditions on the processing of special personal information, such as religious or philosophical beliefs, race or ethnic origin, trade union membership, political persuasion, health or sex life, criminal behavior, or biometric information.

Processing of Personal Information of Children POPIA recognizes the vulnerability of children in the digital age and thus places heightened restrictions on the processing of their personal information. Consent is required from a competent person (e.g., a parent or guardian) where the data subject is a child. Additionally, organizations must ensure that they apply appropriate safeguards when

processing children's data, making sure that it is treated with utmost care and not used for exploitative purposes.

Cross-Border Information Transfers POPIA restricts the transfer of personal information outside South Africa unless the recipient country has similar privacy protections or the data subject consents to the transfer.

Enforcement and Penalties The Information Regulator is the enforcement authority under POPIA, with the power to investigate and fine responsible parties for noncompliance. Penalties for violating POPIA can be severe, including both monetary fines and imprisonment.

State Privacy Laws

In addition to the federal and international laws affecting the privacy and security of information, organizations must be aware of the laws passed by states, provinces, and other jurisdictions where they do business. As with the data breach notification laws discussed earlier in this chapter, states often lead the way in creating privacy regulations that spread across the country and may eventually serve as the model for federal law.

The *California Consumer Privacy Act (CCPA)* is an excellent example of this principle in action. California passed this sweeping privacy law in 2018, modeling it after the European Union's GDPR. Provisions of the law went into effect in 2020, providing consumers with the following:

- The right to know what information businesses are collecting about them and how the organization uses and shares that information
- The right to be forgotten, allowing consumers to request that the organization delete their personal information, in some circumstances
- The right to opt out of the sale of their personal information
- The right to exercise their privacy rights without fear of discrimination or retaliation for their use



California passed other privacy laws that extended CCPA, and other states have passed similar laws in recent years.

Compliance

Over the past decade, the regulatory environment governing information security has grown increasingly complex. Organizations may find themselves subject to a wide variety of laws (many of which were outlined earlier in this chapter) and regulations imposed by regulatory agencies or contractual obligations.



Real World Scenario

Payment Card Industry Data Security Standard

The Payment Card Industry Data Security Standard (PCI DSS) is an excellent example of a compliance requirement that is not dictated by law but by contractual obligation. PCI DSS governs the security of credit card and debit card information and is enforced through the terms of a merchant agreement between a business that accepts credit cards and/or debit cards and the bank that processes the business's transactions.

PCI DSS 4.0 has 12 main requirements:

- Install and maintain network security controls.
- Apply secure configurations to all system components.
- Protect stored account data.
- Protect cardholder data with strong cryptography during transmission over open, public networks.
- Protect all systems and networks from malicious software.
- Develop and maintain secure systems and software.
- Restrict access to system components and cardholder data by business need to know.
- Identify users and authenticate access to system components.
- Restrict physical access to cardholder data.
- Log and monitor all access to systems and networks regularly.
- Test security of systems and networks regularly.
- Support information security with organizational policies and programs.

Each of these requirements is spelled out in detail in the full PCI DSS standard, which can be found at http://pcisecuritystandards.org. Organizations subject to PCI DSS may be required to conduct annual compliance assessments, depending on the number of transactions they process and their history of cybersecurity breaches.

Dealing with the many overlapping, and sometimes contradictory, compliance requirements facing an organization requires careful planning. Many organizations employ full-time IT compliance staff responsible for tracking the regulatory environment, monitoring controls to ensure ongoing compliance, facilitating compliance audits, and meeting the organization's compliance reporting obligations.



Organizations that are not merchants but that store, process, or transmit credit card information on behalf of merchants must also comply with PCI DSS. For example, the requirements apply to shared hosting providers who must protect the cardholder data environment.

Organizations may be subject to compliance audits, either by their standard internal and external auditors or by regulators or their agents. For example, an organization's financial auditors may conduct an IT controls audit designed to ensure that the information security controls for an organization's financial systems are sufficient to ensure compliance with the Sarbanes–Oxley Act (SOX). Some regulations, such as PCI DSS, may require the organization to retain approved independent auditors to verify controls and provide a report directly to regulators.

In addition to formal audits, organizations often must report regulatory compliance to a number of internal and external stakeholders. For example, an organization's board of directors (or, more commonly, that board's audit committee) may require periodic reporting on compliance obligations and status. Similarly, PCI DSS requires organizations that are not compelled to conduct a formal third-party audit to complete and submit a self-assessment report outlining their compliance status.

Contracting and Procurement

The increased use of cloud services and other external vendors to store, process, and transmit sensitive information leads organizations to a new focus on implementing security reviews and controls in their contracting and procurement processes. Security professionals should conduct reviews of the security controls put in place by vendors, both during the initial vendor selection and evaluation process and as part of ongoing vendor governance reviews.

These are some questions to cover during these vendor governance reviews:

- What types of sensitive information are stored, processed, or transmitted by the vendor?
- What controls are in place to protect the organization's information?
- How is your organization's information segregated from that of other clients?
- If encryption is relied on as a security control, what encryption algorithms and key lengths are used? How is key management handled?
- What types of security audits does the vendor perform, and what access does the client have to those audits?
- Does the vendor rely on any other third parties to store, process, or transmit data? How do the provisions of the contract related to security extend to those third parties?
- Where will data storage, processing, and transmission take place? If outside the home country of the client and/or vendor, what implications does that have?
- What is the vendor's incident response process, and when will clients be notified of a potential security breach?
- What provisions are in place to ensure the ongoing integrity and availability of client data?

This is just a brief listing of some of the concerns you may have. Tailor the scope of your security review to the specific concerns of your organization, the type of service provided by the vendor, and the information that will be shared with them.

Summary

Computer security necessarily entails a high degree of involvement from the legal community. In this chapter, you learned about the laws that govern security issues such as computer crime, intellectual property, data privacy, and software licensing.

Three major categories of law impact information security professionals. Criminal law outlines the rules and sanctions for major violations of the public trust. Civil law provides us with a framework for conducting business. Government agencies use administrative law to promulgate the day-to-day regulations that interpret existing law.

The laws governing information security activities are diverse and cover all three categories. Some, such as the Electronic Communications Privacy Act and the Digital Millennium Copyright Act, are criminal laws where violations may result in criminal fines and/or prison time. Others, such as trademark and patent laws, are civil laws that govern business transactions. Finally, many government agencies promulgate administrative law, such as the HIPAA Security Rule, that affects specific industries and data types.

Information security professionals should be aware of the compliance requirements specific to their industry and business activities. Tracking these requirements is a complex task and should be assigned to one or more compliance specialists who monitor changes in the law, changes in the business environment, and the intersection of those two realms.

It's also not sufficient to simply worry about your own security and compliance. With increased adoption of cloud computing, many organizations now share sensitive and personal data with vendors that act as service providers. Security professionals must take steps to ensure that vendors treat data with as much care as the organization itself would and also meet any applicable compliance requirements.

Study Essentials

Understand the differences between criminal law, civil law, and administrative law. Criminal law protects society against acts that violate the basic principles we believe in. Violations of criminal law are prosecuted by federal and state governments. Civil law provides the framework for disputes between people or the transaction of business between people and organizations. Violations of civil law are brought to the court and argued by the two affected parties. Administrative law is used by government agencies to effectively carry out their day-to-day business.

Be able to explain the basic provisions of the major laws designed to protect society against computer crime. The Computer Fraud and Abuse Act (as amended) protects computers used by the government or in interstate commerce from a variety of abuses. The Electronic Communications Privacy Act (ECPA) makes it a crime to invade the electronic privacy of an individual.

Know the differences among copyrights, trademarks, patents, and trade secrets. Copyrights protect original works of authorship, such as books, articles, poems, and songs. Trademarks are names, slogans, and logos that identify a company, product, or service. Patents provide protection to the creators of new inventions. Trade secret law protects the operating secrets of a firm.

Be able to explain the basic provisions of the Digital Millennium Copyright Act of 1998. The Digital Millennium Copyright Act prohibits the circumvention of copy protection mechanisms placed in digital media and limits the liability of Internet service providers for the activities of their users.

Know the basic provisions of the Economic Espionage Act of 1996. The Economic Espionage Act provides penalties for individuals found guilty of the theft of trade secrets. Harsher penalties apply when the individual knows that the information will benefit a foreign government.

Understand the various types of software license agreements. Perpetual licenses allow indefinite use after a onetime fee, while subscription licenses are time-bound with recurring fees. Open source licenses offer usage freedom with conditions, and enterprise agreements

provide licenses for large organizations. EULAs define user rights and restrictions, concurrent licenses set simultaneous user limits, and named user licenses tie to specific users. Click-through agreements require active consent during installation, and cloud service licenses pertain to online services, with terms presented upon registration.

Understand the notification requirements placed on organizations that experience a data breach. California's SB 1386 implemented the first statewide requirement to notify individuals of a breach of their personal information. All other states eventually followed suit with similar laws. Currently, federal law only requires the notification of individuals when a HIPAA-covered entity breaches their protected health information.

Understand the major laws that govern privacy of personal information in the United States, the European Union, Canada, China, and South Africa. The United States has a number of privacy laws that affect the government's use of information as well as the use of information by specific industries, such as financial services companies and healthcare organizations that handle sensitive information. The EU has a more comprehensive General Data Protection Regulation that governs the use and exchange of personal information. In Canada, the Personal Information Protection and Electronic Documents Act (PIPEDA) governs the use of personal information. China includes privacy protections in the Personal Information Protection Law (PIPL), while South Africa's are embedded in the Protection of Personal Information Act (POPIA).

Explain the importance of a well-rounded compliance program. Most organizations are subject to a wide variety of legal and regulatory requirements related to information security. Building a compliance program ensures that you become and remain compliant with these often overlapping requirements.

Know how to incorporate security into the procurement and vendor governance process. The expanded use of cloud services by many organizations requires added attention to conducting reviews of information security controls during the vendor selection process and as part of ongoing vendor governance.

Be able to determine compliance and other requirements for information protection. Cybersecurity professionals must be able to

analyze a situation and determine what jurisdictions and laws apply. They must be able to identify relevant contractual, legal, regulatory, and industry standards and interpret them for their given situation.

Know legal and regulatory issues and how they pertain to information security. Understand the concepts of cybercrime and data breaches and be able to apply them in your environment when incidents arise. Understand what licensing and intellectual property protections apply to your organization's data and your obligations when encountering data belonging to other organizations. Understand the privacy and export control issues associated with transferring information across international borders.

Written Lab

1. What are the two primary mechanisms that an organization may use to share information outside the European Union under the terms of GDPR?
2. What are some common questions that organizations should ask when considering outsourcing information storage, processing, or transmission?
3. What are some common steps that employers can take to notify employees of system monitoring?

Review Questions

1. Brianna is working with a U.S. software firm that uses encryption in its products and plans to export their product outside of the United States. What federal government agency has the authority to regulate the export of encryption software?

   A. NSA

   B. NIST

   C. BIS

   D. FTC

2. Wendy recently accepted a position as a senior cybersecurity administrator at a U.S. government agency and is concerned about the legal requirements affecting her new position. Which law governs information security operations at federal agencies?

    A. FISMA

    B. FERPA

    C. CFAA

    D. ECPA

3. What type of law does not require an act of Congress to implement at the federal level but rather is enacted by the executive branch in the form of regulations, policies, and procedures?

    A. Criminal law

    B. Common law

    C. Civil law

    D. Administrative law

4. What U.S. state was the first to pass a comprehensive privacy law modeled after the requirements of the European Union's General Data Protection Regulation?

    A. California

    B. New York

    C. Vermont

    D. Texas

5. Congress passed CALEA in 1994, requiring that what type of organizations cooperate with law enforcement investigations?

    A. Financial institutions

    B. Communications carriers

    C. Healthcare organizations

    D. Websites

6. What law protects the right of citizens to privacy by placing restrictions on the authority granted to government agencies to search private residences and facilities?

    A. Privacy Act

    B. Fourth Amendment

    C. Second Amendment

    D. Gramm–Leach–Bliley Act

7. Matthew recently authored an innovative algorithm for solving a mathematical problem, and he wants to share it with the world. However, prior to publishing the software code in a technical journal, he wants to obtain some sort of intellectual property (IP) protection. Which type of protection is best suited to his needs?

    A. Copyright

    B. Trademark

    C. Patent

    D. Trade secret

8. Mary is the cofounder of Acme Widgets, a manufacturing firm. Together with her partner, Joe, she has developed a special oil that will dramatically improve the widget manufacturing process. To keep the formula secret, Mary and Joe plan to make large quantities of the oil by themselves in the plant after the other workers have left. They want to protect this formula for as long as possible. What type of intellectual property (IP) protection best suits their needs?

    A. Copyright

    B. Trademark

    C. Patent

    D. Trade secret

9. Richard recently developed a great name for a new product that he plans to begin using immediately. He spoke with his attorney and filed the appropriate application to protect his product name but has not yet received a response from the government regarding his application. He

wants to begin using the name immediately. What symbol should he use next to the name to indicate its protected status?

A. ©

B. ®

C. ™

D. †

10. Tom is an adviser to a federal government agency that collects personal information from constituents. He would like to facilitate a research relationship between that firm that involves the sharing of personal information with several universities. What law prevents government agencies from disclosing personal information that an individual supplies to the government under protected circumstances?

A. Privacy Act

B. Electronic Communications Privacy Act

C. Health Insurance Portability and Accountability Act

D. Gramm–Leach–Bliley Act

11. Renee's organization is establishing a partnership with a firm located in France that will involve the exchange of personal information. Her partners in France want to ensure that the transfer will be compliant with the GDPR. What mechanism would be most appropriate?

A. Binding corporate rules

B. Privacy Shield

C. Privacy Lock

D. Standard contractual clauses

12. The Children's Online Privacy Protection Act (COPPA) was designed to protect the privacy of children using the Internet. What is the minimum age a child must be before companies can collect personal identifying information from them without parental consent?

A. 13

B. 14

C. 15

D. 16

13. Kevin is assessing his organization's obligations under state data breach notification laws. Which one of the following pieces of information would generally not be covered by a data breach notification law when it appears in conjunction with a person's name?

    A. Social Security number

    B. Driver's license number

    C. Credit card number

    D. Student identification number

14. Roger is the CISO at a healthcare organization covered under HIPAA. He would like to enter into a partnership with a vendor who will manage some of the organization's data. As part of the relationship, the vendor will have access to protected health information (PHI). Under what circumstances is this arrangement permissible under HIPAA?

    A. This is permissible if the service provider is certified by the Department of Health and Human Services.

    B. This is permissible if the service provider enters into a business associate agreement.

    C. This is permissible if the service provider is within the same state as Roger's organization.

    D. This is not permissible under any circumstances.

15. Frances learned that a user in her organization recently signed up for a cloud service without the knowledge of her supervisor and is storing corporate information in that service. Which one of the following statements is correct?

    A. If the user did not sign a written contract, the organization has no obligation to the service provider.

    B. The user most likely agreed to a click-through license agreement binding the organization.

    C. The user's actions likely violate federal law.

D. The user's actions likely violate state law.

16. Greg recently accepted a position as the cybersecurity compliance officer with a privately held bank. What law most directly impacts the manner in which his organization handles personal information?

    A. HIPAA

    B. GLBA

    C. SOX

    D. FISMA

17. Ruth recently obtained a utility patent covering a new invention that she created. How long will she retain legal protection for her invention?

    A. 14 years from the application date

    B. 14 years from the date the patent is granted

    C. 20 years from the application date

    D. 20 years from the date the patent is granted

18. Ryan is reviewing the terms of a proposed vendor agreement between the financial institution where he works and a cloud service provider. Which one of the following items should represent the *least* concern to Ryan?

    A. What security audits does the vendor perform?

    B. What provisions are in place to protect the confidentiality, integrity, and availability of data?

    C. Is the vendor compliant with HIPAA?

    D. What encryption algorithms and key lengths are used?

19. Justin is a cybersecurity consultant working with a retailer on the design of their new point-of-sale (POS) system. What compliance obligation relates to the processing of credit card information that might take place through this system?

    A. SOX

B. HIPAA

C. PCI DSS

D. FERPA

20. Leonard and Sheldon recently coauthored a paper describing a new superfluid vacuum theory. How long will the copyright on their paper last?

A. 70 years after publication

B. 70 years after completion of the first draft

C. 70 years after the death of the first author

D. 70 years after the death of the last author

Chapter 5
Protecting Security of Assets

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 2.0: Asset Security**
  - 2.1 Identify and classify information and assets
    - 2.1.1 Data classification
    - 2.1.2 Asset classification
  - 2.2 Establish information and asset handling requirements
  - 2.4 Manage data lifecycle
    - 2.4.1 Data roles (i.e., owners, controllers, custodians, processors, users/subjects)
    - 2.4.2 Data collection
    - 2.4.3 Data location
    - 2.4.4 Data maintenance
    - 2.4.5 Data retention
    - 2.4.6 Data remanence
    - 2.4.7 Data destruction
  - 2.5 Ensure appropriate asset retention (e.g., end of life (EOL), end of support)
  - 2.6 Determine data security controls and compliance requirements
    - 2.6.1 Data states (e.g., in use, in transit, at rest)
    - 2.6.2 Scoping and tailoring
    - 2.6.3 Standards selection
    - 2.6.4 Data protection methods (e.g., Digital Rights Management (DRM), data loss prevention (DLP), cloud access security broker (CASB))

The Asset Security domain focuses on collecting, handling, and protecting information throughout its life cycle. A primary step in this domain is classifying information based on its value to the organization. All follow-on actions vary depending on the classification. For example, highly classified data requires stringent security controls. In contrast, unclassified data uses fewer security controls.

Identifying and Classifying Information and Assets

Managing the data life cycle refers to protecting it from the cradle to the grave. Steps need to be taken to protect the data when it is first created until it is destroyed.

One of the first steps in the life cycle is identifying and classifying information and assets. Organizations often include classification definitions within a security policy. Personnel then label assets appropriately based on the security policy requirements. In this context, assets include sensitive data, the hardware used to process it, and the media used to hold it.

Defining Sensitive Data

Sensitive data is any information that isn't public or unclassified. It can include confidential, proprietary, protected, or any other type of data that an organization needs to protect due to its value to the organization, or to comply with existing laws and regulations.

Personally Identifiable Information

Personally identifiable information (PII) is any information that can identify an individual. National Institute of Standards and Technology (NIST) Special Publication (SP) 800-122 provides a more formal definition:

Any information about an individual maintained by an agency, including

1. any information that can be used to distinguish or trace an individual's identity, such as name, social security number, date and place of birth, mother's maiden name, or biometric records; and
2. any other information that is linked or linkable to an individual, such as medical, educational, financial, and employment information.

The key is that organizations have a responsibility to protect PII. This includes PII related to employees and customers. Many laws require organizations to notify individuals if a data breach results in a compromise of PII.



Protection for personally identifiable information (PII) drives privacy and confidentiality requirements for rules, regulations, and legislation worldwide (especially in North America and the European Union). NIST SP 800-122—Guide to Protecting the Confidentiality of Personally Identifiable Information (PII), provides more information on how to protect PII. It is available from the NIST Special Publications (800 Series) download page: http://csrc.nist.gov/publications/sp800.

Protected Health Information

In the United States, the Health Insurance Portability and Accountability Act (HIPAA) mandates the protection of some health records. To fully understand what information is covered by HIPAA, we need to look at a few definitions. First, the general definition of health information is:

> Health information means any information, whether oral or recorded in any form or medium, that—
>
> (A) is created or received by a health care provider, health plan, public health authority, employer, life insurer, school or university, or health care clearinghouse; and
>
> (B) relates to the past, present, or future physical or mental health or condition of any individual, the provision of health care to an individual, or the past, present, or future payment for the provision of health care to an individual.

Protected health information (PHI) is any health information that is transmitted in electronic form, maintained in electronic media, or transmitted or maintained in any other form or media. Education records, employment records of a covered entity, and records relating to individuals who have been deceased more than 50 years are excluded from the definition of PHI.

Some people think that only medical care providers, such as doctors and hospitals, need to protect PHI. However, HIPAA defines PHI much more broadly. The law applies to healthcare providers, health insurers, and health information clearinghouses, as well as business associates of those organizations that handle PHI. Employers that provide health insurance may handle PHI, so HIPAA applies to a large percentage of organizations in the United States.

Proprietary Data

Proprietary data refers to any data that helps an organization maintain a competitive edge. It could be software code it developed, technical plans for products, internal processes, intellectual property, or trade secrets. If competitors gain access to the proprietary data, it can seriously affect the primary mission of an organization.

Although copyright, patent, and trade secret laws provide a level of protection for proprietary data, this isn't always enough. Many criminals ignore copyrights, patents, and laws. Similarly, foreign entities have stolen a significant amount of proprietary data.

Defining Data Classifications

Organizations typically include data classifications in their security policy or a data policy. A *data classification* identifies the value of the data to the organization and is critical to protect data confidentiality and integrity. The policy identifies classification labels used within the organization. It also identifies how data owners can determine the proper classification and how personnel should protect data based on its classification.

As an example, government data classifications include top secret, secret, confidential, and unclassified. Anything above unclassified is sensitive data, but clearly, these have different values. The U.S. government provides clear definitions for these classifications. As you read them, note that the wording of each definition is close except for a few key words. *Top secret* uses the phrase "exceptionally grave damage," *secret* uses the phrase "serious damage," and *confidential* uses "damage":

Top Secret The top secret label is "applied to information, the unauthorized disclosure of which reasonably could be expected to cause exceptionally grave damage to the national security that the original classification authority is able to identify or describe."

Secret The secret label is "applied to information, the unauthorized disclosure of which reasonably could be expected to cause serious damage to the national security that the original classification authority is able to identify or describe."

Confidential The confidential label is "applied to information, the unauthorized disclosure of which reasonably could be expected to cause damage to the national security that the original classification authority is able to identify or describe."

Unclassified Unclassified refers to any data that doesn't meet one of the descriptions for top secret, secret, or confidential data. Within the United States, unclassified data is available to anyone, though it often requires individuals to request the information using procedures identified in the Freedom of Information Act (FOIA).

There are additional subclassifications of unclassified, such as for official use only (FOUO), sensitive but unclassified (SBU), and

controlled unclassified information (CUI). Documents with these designations have strict controls limiting their distribution. As an example, the U.S. Internal Revenue Service (IRS) uses SBU for individual tax records, restricting access to these records.

A classification authority is the entity that applies the original classification to the sensitive data, and there are strict rules that identify who can do so. For example, the U.S. president, vice president, and agency heads can classify data in the United States. Additionally, individuals in any of these positions can delegate permission for others to classify data.



Although the focus of classifications is often on data, these classifications also apply to hardware assets. This includes any computing system or media that processes or holds this data.

Nongovernmental organizations rarely need to classify their data based on potential damage to national security. However, management is concerned about potential damage to the organization. For example, if attackers accessed the organization's data, what is the potential adverse impact? In other words, an organization doesn't just consider the sensitivity of the data but also the criticality of the data. They could use the same phrases of “exceptionally grave damage,” “serious damage,” and “damage” that the U.S. government uses when describing top secret, secret, and confidential data.

Some nongovernmental organizations use labels such as Class 3, Class 2, Class 1, and Class 0. Other organizations use more meaningful labels such as confidential (or proprietary), private, sensitive, and public. Figure 5.1 shows the relationship between these different classifications, with the government classifications on the left and the nongovernment (or civilian) classifications on the right. Just as the government can define the data based on the potential adverse impact from a data breach, organizations can use similar descriptions.

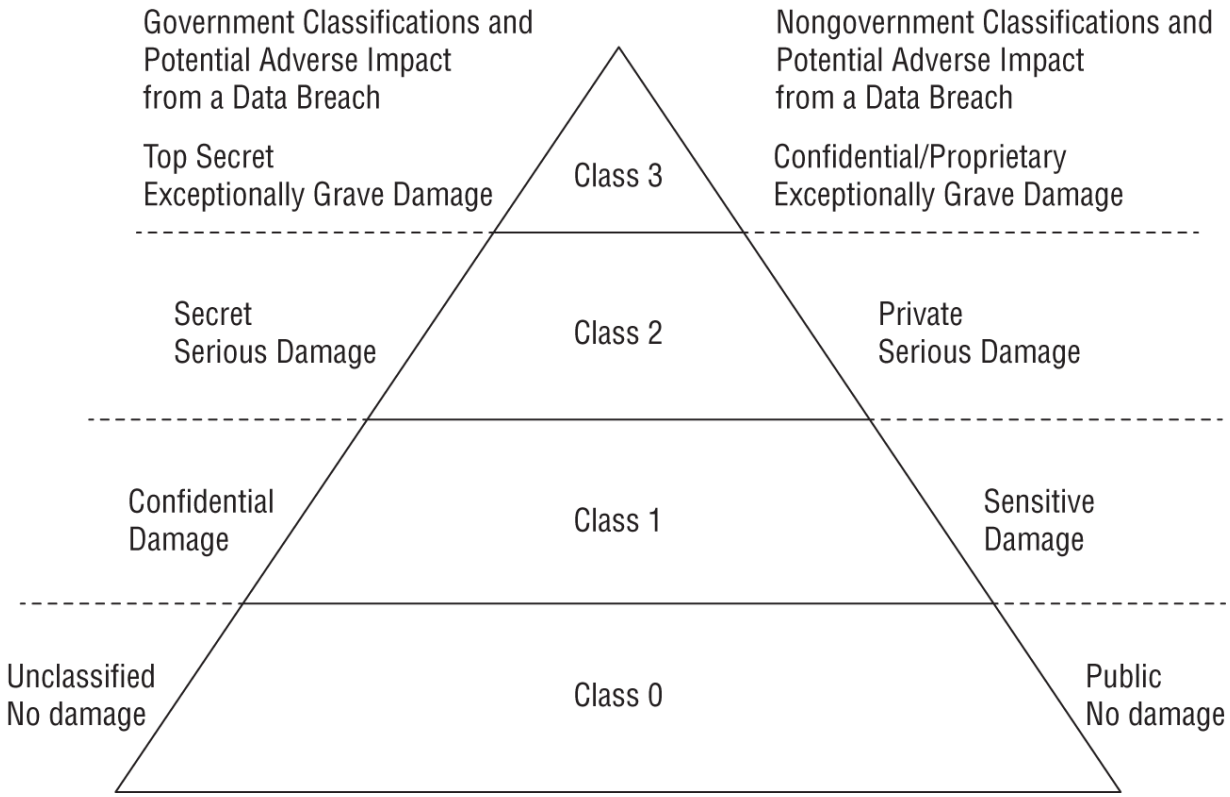


FIGURE 5.1 Data classifications

Both government and civilian classifications identify the relative value of the data to the organization, with top secret representing the highest classification for governments and confidential representing the highest classification for organizations in Figure 5.1. However, it's important to remember that organizations can use any labels they desire. The following sections identify the meaning of some common nongovernment classifications. Remember, even though these are commonly used, there is no standard that all private organizations must use.

Confidential or Proprietary The confidential or *proprietary* label typically refers to the highest level of classified data. In this context, a data breach would cause exceptionally grave damage to the mission of the organization. As an example, attackers have repeatedly attacked Sony, stealing more than 100 terabytes of data, including full-length versions of unreleased movies. These quickly showed up on file-sharing sites, and security experts estimate that people downloaded these movies up to a million times. With pirated versions of the movies available, many people skipped seeing them when Sony ultimately released them. This directly

affected Sony's bottom line. The movies were proprietary, and the organization might have considered it exceptionally grave damage. In retrospect, they may choose to label movies as confidential or proprietary and use the strongest access controls to protect them.

Private The *private* label refers to data that should stay private within the organization but that doesn't meet the definition of confidential or proprietary data. In this context, a data breach would cause serious damage to the mission of the organization. Many organizations label PII and PHI data as private. It's also common to label internal employee data and some financial data as private. As an example, the payroll department of a company would have access to payroll data, but this data is not available to regular employees.

Sensitive *Sensitive data* is similar to confidential data. In this context, a data breach would cause damage to the mission of the organization. As an example, IT personnel within an organization might have extensive data about the internal network, including the layout, devices, operating systems, software, Internet Protocol (IP) addresses, and more. If attackers have easy access to this data, it makes it much easier for them to launch attacks. Management may decide they don't want this information available to the public, so they might label it as sensitive.

Public *Public data* is similar to unclassified data. It includes information posted on websites, in brochures, or any other public source. Although an organization doesn't protect the confidentiality of public data, it does take steps to protect its integrity. For example, anyone can view public data posted on a website. However, an organization doesn't want attackers to modify this data, so it takes steps to protect it.

Civilian organizations aren't required to use any specific classification labels. However, it is important to classify data in some manner and ensure personnel understand the classifications. No matter what labels an organization uses, it still has an obligation to protect sensitive information.

After classifying the data, an organization takes additional steps to manage it based on its classification. Unauthorized access to sensitive information can result in significant losses to an organization. However, basic security practices, such as properly marking, handling, storing, and destroying data and hardware assets based on classifications, helps prevent losses.

Defining Asset Classifications

Asset classifications should match the data classifications. In other words, if a computer is processing top secret data, the computer should also be classified as a top secret asset. Similarly, if media such as internal or external drives hold top secret data, the media should also be classified as top secret.

It is common to use clear marking on the hardware assets so that personnel are reminded of data that can be processed or stored on the asset. For example, if a computer is used to process top secret data, the computer and the monitor will have clear and prominent labels reminding users of the classification of data that can be processed on the computer.

Understanding Data States

It's important to protect data in all *data states*, including while it is at rest, in transit, and in use.

Data at Rest Data at rest (sometimes called data on storage) is any data stored on media such as system hard drives, solid-state drives (SSDs), external USB drives, storage area networks (SANs), and backup tapes. Strong symmetric encryption protects data at rest.

Data in Transit Data in transit (sometimes called data in motion or being communicated) is any data being transmitted over a network. This includes data being transmitted over an internal network using wired or wireless methods and data being transmitted over public networks such as the Internet. A combination of symmetric and asymmetric encryption protects data in transit.

Data in Use Data in use (also known as data being processed) refers to data in memory or temporary storage buffers while an application is using it. Applications often decrypt encrypted data before placing it in memory. This allows the application to work on it, but it's important to flush these buffers when the data is no longer needed. In some cases, it's possible for an application to work on encrypted data using homomorphic encryption. This limits the risk because memory doesn't hold unencrypted data.

The best way to protect the confidentiality of data is to use strong encryption protocols, discussed extensively in Chapter 6, “Cryptography

and Symmetric Key Algorithms.” Additionally, strong authentication and authorization controls help prevent unauthorized access.

As an example, consider a web application that retrieves credit card data for quick access and reuse with the user's permission for an e-commerce transaction. The credit card data is stored in a database server and protected while at rest, while in transit, and while in use.

Database administrators take steps to encrypt sensitive data stored in the database server (data at rest). They would typically encrypt columns holding sensitive data such as credit card data. Additionally, they would implement strong authentication and authorization controls to prevent unauthorized entities from accessing the database.

When the web application sends a request for data from the web server, the database server verifies that the web application is authorized to retrieve the data and, if so, the database server sends it. However, this entails several steps. For example, the database management system first retrieves and decrypts the data and formats it in a way that the web application can read it. The database server then uses a transport encryption algorithm to encrypt the data before transmitting it. This ensures that the data in transit is secure.

The web application server receives the data in an encrypted format. It decrypts the data and sends it to the web application. The web application stores the data in temporary memory buffers while it uses it to authorize the transaction. When the web application no longer needs the data, it takes steps to purge memory buffers, ensuring the complete removal of all residual sensitive data.



The Identity Theft Resource Center (ITRC) routinely tracks data breaches. They post reports through their website (http://idtheftcenter.org) that are free to anyone. In 2023, they tracked 3,205 data breaches, exposing the information of more than 353 million people.

Determining Compliance Requirements

Every organization has a responsibility to learn what legal requirements apply to them and ensure they meet all the compliance requirements. This is especially important if an organization handles PII in different countries. Chapter 4, “Laws, Regulations, and Compliance,” covers a wide assortment of laws and regulations that apply to organizations around the world. For any organization involved in e-commerce, this can get complex very quickly. An important point to remember is that an organization needs to determine what laws apply to it.

Imagine a group of college students work together and create an app that solves a problem for them. On a whim, they start selling the app from the Apple App Store and it goes viral. People around the world are buying the app, bringing cash windfalls to these students. It also brings major headaches. Suddenly these college students need to be knowledgeable about laws around the world that apply to them.

Some organizations have created a formal position called a compliance officer. The person filling this role ensures that the organization is conducting all business activities by following the laws and regulations that apply to the organization. Of course, this starts by first determining everywhere the organization operates and what compliance requirements apply.

Determining Data Security Controls

After defining data and asset classifications, you must define the security requirements and identify security controls to implement those requirements. Imagine that your organization has decided to use the data labels Confidential/Proprietary, Private, Sensitive, and Public, as described earlier. Management then decides on a data security policy dictating the use of specific security controls to protect data in these categories. The policy will likely address data stored in files, in databases, on servers such as email servers, on user systems, sent via email, and stored in the cloud.

For this example, we're limiting the type of data to email only. Your organization has defined how it wants to protect email in each of the data categories. They've decided that any email in the Public category doesn't need to be encrypted. However, email in all other categories

(Confidential/Proprietary, Private, and Sensitive) must be encrypted when being sent (data in transit) and while stored on an email server (data at rest).

Encryption converts cleartext data into scrambled ciphertext and makes it more difficult to read. Using strong encryption methods such as the Advanced Encryption Standard with 256-bit keys (AES 256) makes it almost impossible for unauthorized personnel to read the text.

Table 5.1 shows other security requirements for email that management has defined in their data security policy. Notice that data in the highest level of classification category (Confidential/Proprietary in this example) has the most security requirements defined in the security policy.

TABLE 5.1 Securing email data

Classification	Security requirements for email
Confidential/Proprietary (highest level of protection for any data)	Email and attachments must be encrypted with AES 256. Email and attachments remain encrypted except when viewed. Email can be sent only to recipients within the organization. Email can be opened and viewed only by recipients (forwarded emails cannot be opened). Attachments can be opened and viewed, but not saved. Email content cannot be copied and pasted into other documents. Email cannot be printed.
Private (examples include PII and PHI)	Email and attachments must be encrypted with AES 256. Email and attachments remain encrypted except when viewed. Email can be sent only to recipients within the organization.
Sensitive (lowest level of protection for classified data)	Email and attachments must be encrypted with AES 256.
Public	Email and attachments can be sent in cleartext.



The requirements listed in Table 5.1 are provided as an example only. Any organization could use these requirements or define other requirements that work for them.

Security administrators use the requirements defined in the security policy to identify security controls. For Table 5.1, the primary security control is

strong encryption using AES 256. Administrators should identify methodologies, making it easy for employees to meet the requirements.

Although it's possible to meet all the requirements for securing email shown in Table 5.1, doing so might require implementing other solutions. For example, several software companies sell a range of products that organizations can use to automate these tasks. Users apply relevant labels (such as confidential, private, sensitive, and public) to emails before sending them. These emails pass through a data loss prevention (DLP) server that detects the labels and applies the required protection. The settings for these DLP solutions can be configured for an organization's specific needs.

Table 5.1 shows possible requirements that your organization might want to apply to email. However, you shouldn't stop there. Any type of data that your organization wants to protect needs similar security definitions. For example, you should define requirements for data stored on assets such as servers, data backups stored on-site and off-site, and proprietary data.

Additionally, identity and access management security controls help ensure that only authorized personnel can access resources. Chapter 13, “Managing Identity and Authentication,” and Chapter 14, “Controlling and Monitoring Access,” cover identity and access management security controls in more depth.

Establishing Information and Asset Handling Requirements

A key goal of managing sensitive data is to prevent data breaches. A data breach is an event in which an unauthorized entity can view or access sensitive data. If you pay attention to the news, you probably hear about data breaches quite often. Large data breaches such as the Marriott data breach of 2020 hit the mainstream news. Marriott reported that attackers stole personal data, including names, addresses, email addresses, employer information, and phone numbers, of approximately 5.2 million guests.

The following sections identify basic steps people within an organization should follow to limit the possibility of data breaches.

Data Maintenance

Data maintenance refers to ongoing efforts to organize and care for data throughout its lifetime. In general, if an organization stores all sensitive data on one server, it is relatively easy to apply all the appropriate controls to this one server. In contrast, if sensitive data is stored throughout an organization on multiple servers and end-user computers and mixed with nonsensitive data, it becomes much harder to protect it.

One option would be for one network to process only unclassified data while another network processes classified data. Techniques such as air gaps ensure the two networks never physically touch each other. An *air gap* is a physical security control and means that systems and cables from the classified network never physically touch systems and cables from the unclassified network. Additionally, the classified network can't access the Internet, and Internet attackers can't access it.

Still, there are times when personnel need to add data to the classified network, such as when devices, systems, and applications need updates. One way is manual; personnel copy the data from the unclassified network to a USB device and carry it to the classified network. Another method is to use a unidirectional network bridge; this connects the two networks but allows the data to travel in only one direction, from the unclassified network to the classified network. A third method is to use a technical guard solution, which is a combination of hardware and software placed between the two networks. A guard solution allows properly marked data to travel between the two networks.

Additionally, an organization should routinely review data policies to ensure that they are kept up-to-date and that personnel are following the policies. It's often a good practice to review the causes of recent data breaches and ensure that similar mistakes are not causing needless vulnerabilities.

Data Loss Prevention

Data loss prevention (DLP) solutions attempt to detect and block data exfiltration attempts. These solutions have the capability of scanning unencrypted data looking for keywords and data patterns. For example, imagine that your organization uses data classifications of Confidential,

Proprietary, Private, and Sensitive. A DLP system can scan files for these words and detect them.

Pattern-matching DLP systems look for specific patterns. For example, U.S. Social Security numbers have a pattern of *nnn-nn-nnnn* (three numbers, a dash, two numbers, a dash, and four numbers). The DLP can look for this pattern and detect it. Administrators can set up a DLP system to look for any patterns based on their needs. Cloud DLP solutions can look for the same keywords or patterns.

There are three types of DLP solutions:

Network DLP A network DLP scans all outgoing data in a traditional network looking for specific data. Administrators place it on the edge of the network to scan all data leaving the organization. If a user sends out a file containing restricted data, the DLP system will detect it and prevent it from leaving the organization. The DLP system will send an alert, such as an email to an administrator.

Endpoint DLP An endpoint DLP can scan files stored on a system as well as files sent to external devices, such as printers. For example, an organization's endpoint DLP can prevent users from copying sensitive data to USB flash drives or sending sensitive data to a printer. Administrators configure the DLP to scan the files with the appropriate keywords, and if it detects files with these keywords, it will block the copy or print job. It's also possible to configure an endpoint DLP solution to regularly scan files (such as on a file server) for files containing specific keywords or patterns, or even for unauthorized file types, such as MP3 files.

Cloud DLP Cloud DLP is a subset of network DLP designed and tailored for cloud-native environments.

DLP solutions typically can perform deep-level examinations. For example, if users embed the files in compressed zip files, a DLP solution can still detect the keywords and patterns. However, a DLP solution can't decrypt data or examine encrypted data.

Most DLP solutions also include discovery capabilities. The goal is to discover the location of valuable data within an internal network. When security administrators know where the data is, they can take additional steps to protect it. As an example, a database server may include

unencrypted credit card numbers. When the DLP discovers and reports this, database administrators can ensure the numbers are encrypted. As another example, company policy may dictate that employee laptops do not contain any PII data. A DLP content discovery system can search these and discover any unauthorized data. Additionally, many content discovery systems can search cloud resources used by an organization.

Labeling Sensitive Data and Assets

Labeling (often called security labeling) sensitive information ensures that users can easily identify the classification level of any data. The most important information that a tag or a label provides is the classification of the data. For example, a label of top secret makes it clear to anyone who sees the label that the information or asset is classified top secret. When users know the value of the data or asset, they are more likely to take appropriate steps to control and protect it based on the classification. Security labeling includes both physical and electronic tags and labels.

Physical labels indicate the security classification for the data stored on assets such as media or processed on a system. For example, if a backup tape includes secret data, a physical label attached to the tape makes it clear to users that it holds secret data.

Similarly, if a computer processes sensitive information, the computer would have a label indicating the highest classification of information that it processes. A computer used to process confidential, secret, and top secret data should be marked with a label indicating that it processes top secret data. Physical labels remain on the system or media throughout its lifetime.

Security labeling also includes using digital tags or labels. A simple method is to include the classification as a header or footer in a document or embed it as a watermark. A benefit of these methods is that they also appear on printouts. Even when users include headers and footers on printouts, most organizations require users to place printed sensitive documents within a folder that includes a label or cover page clearly indicating the classification. Headers aren't limited to files. Backup tapes often include header information, and the classification can be included in this header.

Another benefit of headers, footers, and watermarks is that DLP systems can identify documents that include sensitive information and apply the

appropriate security controls. Some DLP systems will also add metadata tags to the document when they detect that the document is classified. These tags provide insight into the document's contents and help the DLP system handle it appropriately.

Similarly, some organizations mandate specific desktop backgrounds on their computers. For example, a system used to process proprietary data might have a black desktop background with the word *Proprietary* in white and a wide orange border. The background could also include statements such as “This computer processes proprietary data” and statements reminding users of their responsibilities to protect the data.

In many secure environments, personnel also use labels for unclassified media and equipment. This prevents an error of omission where sensitive information isn't marked. For example, if a backup tape holding sensitive data isn't marked, a user might assume it only holds unclassified data. However, if the organization marks unclassified data, too, unlabeled media would be easily noticeable, and the user would view an unmarked tape with suspicion.

Organizations often identify procedures to downgrade media. For example, if a backup tape includes confidential information, an administrator might want to downgrade the tape to unclassified. The organization would identify trusted procedures that will purge the tape of all usable data. After administrators purge the tape, they can then downgrade it and replace the labels.

However, many organizations prohibit downgrading media at all. For example, a data policy might prohibit downgrading a backup tape that contains top secret data. Instead, the policy might mandate destroying this tape when it reaches the end of its life cycle. Similarly, it is rare to downgrade a system. In other words, if a system has been processing top secret data, it would be rare to downgrade it and relabel it as an unclassified system. In any event, approved procedures would need to be created to inform personnel what can be downgraded and what should be destroyed.



If media or a computing system needs to be downgraded to a less sensitive classification, it must be sanitized using appropriate procedures, as described in the section “Data Destruction,” later in this chapter. However, it's often safer and easier just to purchase new media or equipment rather than follow through with the sanitization steps for reuse.

Handling Sensitive Information and Assets

Handling refers to the secure transportation of media through its lifetime. Personnel handle data differently based on its value and classification, and as you'd expect, highly classified information needs much greater protection. Even though this is common sense, people still make mistakes. Many times, people get accustomed to handling sensitive information and become lackadaisical about protecting it.

A common occurrence is the loss of control of backup tapes. Backup tapes should be protected with the same level of protection as the data that they contain. In other words, if confidential information is on a backup tape, the backup tape should be protected as a confidential asset.

Similarly, data stored in the cloud needs to be protected with the same level of protection with which it is protected on-site. Amazon Web Services (AWS) Simple Storage Service (S3) is a cloud-based object storage service. Data is stored in S3 *buckets*, which are like folders on Windows systems. Just as you set permissions on any folder, you set permissions on AWS buckets. Unfortunately, this concept eludes many AWS users. As an example, a bucket owned by THSuite, a cannabis retailer, exposed the PII of more than 30,000 individuals in early 2020. Another example from 2020 involved 900,000 before and after cosmetic surgery images and videos stored in an unsecured bucket. Many of these included clear views of the patients' faces, along with all parts of their bodies.

Policies and procedures need to be in place to ensure that people understand how to handle sensitive data. This starts by ensuring that systems and media are labeled appropriately. Additionally, as President Reagan famously said

when discussing relations with the Soviet Union, “Trust, but verify.” Chapter 17, “Preventing and Responding to Incidents,” discusses the importance of logging, monitoring, and auditing. These controls verify that sensitive information is handled appropriately before a significant loss occurs. If a loss does occur, investigators use audit trails to help discover what went wrong. Any incidents that occur because personnel didn't handle data appropriately should be quickly investigated and actions taken to prevent a reoccurrence.

Data Collection Limitation

One of the easiest ways to prevent the loss of data is to simply not collect it. As an example, consider a small e-commerce company that allows customers to make purchases with a credit card. It uses a credit card processor to process credit card payments. If the company just passes the credit card data to the processor for approval and never stores it on a company server, the company cannot lose the credit card data in a later breach.

In contrast, imagine a different e-commerce company sells products online. Every time a customer makes a purchase, the company collects as much information as possible on the customer, such as the name, email address, physical address, phone number, credit card data, and more. It suffers a data breach and all this data is exposed, resulting in significant liabilities for the company.

The guideline is clear. If the data doesn't have a clear purpose for use, don't collect it and store it. This is also why many privacy regulations mention limiting data collection.

Data Location

Data location refers to the location of data backups or data copies. Imagine a small organization's primary business location is in Norfolk, Virginia. The organization stores all the data on-site. However, they regularly perform backups of the data.

A best practice is to keep a backup copy on-site and another backup copy off-site. If a disaster, such as a fire, destroys the primary business location, the organization would still have a backup copy stored off-site.

The decision of how far off-site to store the backup needs to be considered. If it's stored in a business located in the same building, it could be destroyed in the same fire. Even if the backup was stored 5 miles away, it is possible a hurricane or flood could destroy both locations.

Some organizations maintain data in large data centers. It's common to replicate this data to one or more other data centers to maintain the availability of the critical data. These data centers are typically located in separate geographical locations. When using cloud storage for backups, some organizations may need to verify the location of the cloud storage to ensure it is in a separate geographical location.

Storing Sensitive Data

Sensitive data should be stored in such a way that it is protected against any type of loss. Encryption methods prevent unauthorized entities from accessing the data even if they obtain databases or hardware assets.

If sensitive data is stored on physical media such as portable disk drives or backup tapes, personnel should follow basic physical security practices to prevent losses due to theft. This includes storing these devices in locked safes or vaults, or within a secure room that includes several additional physical security controls. For example, a server room includes physical security measures to prevent unauthorized access, so storing portable media within a locked cabinet in a server room would provide strong protection.

Additionally, environmental controls protect the media. This includes temperature and humidity controls such as heating, ventilation, and air conditioning (HVAC) systems.

Here's a point that end users often forget: the value of any sensitive data is much greater than the value of the media holding the sensitive data. In other words, it's cost-effective to purchase high-quality media, especially if the data will be stored for a long time, such as on backup tapes. Similarly, the purchase of high-quality USB flash drives with built-in encryption is worth the cost. Some of these USB flash drives include biometric authentication mechanisms using fingerprints, which provide added protection.



Encryption of sensitive data provides an additional layer of protection and should be considered for any data at rest. If data is encrypted, it becomes much more difficult for an attacker to access it, even if it is stolen.

Data Destruction

When an organization no longer needs sensitive data, personnel should destroy it. Proper destruction ensures that it cannot fall into the wrong hands and result in unauthorized disclosure. Highly classified data requires different steps to destroy it than data classified at a lower level. An organization's security policy or data policy should define the acceptable methods of destroying data based on the data's classification. For example, an organization may require the complete destruction of media holding highly classified data, but allow personnel to use software tools to overwrite data files classified at a lower level.

NIST SP 800-88, Rev. 1—Guides for Media Sanitization provides comprehensive details on different sanitization methods. Sanitization methods (such as clearing, purging, and destroying) help ensure that data cannot be recovered. Proper sanitization steps remove all sensitive data before disposing of a computer. This includes removing or destroying data on nonvolatile memory, internal hard drives, and solid-state drives (SSDs). It also includes removing all CDs/DVDs and Universal Serial Bus (USB) drives.

Sanitization can refer to the destruction of media or using a trusted method to purge classified data from the media without destroying it.

Eliminating Data Remanence

Data remanence is the data that remains on media after the data was supposedly erased. It typically refers to data on a hard drive as residual magnetic flux or slack space. If media includes any type of private and sensitive data, it is important to eliminate data remanence.

Slack space is the unused space within a disk cluster. Operating systems store files on hard disk drives in clusters, which are groups of sectors (the smallest storage unit on a hard disk drive). Sector and cluster sizes vary, but for this example, imagine a cluster size of 4,096 bytes and a file size of 1,024 bytes. After storing the file, the cluster would have 3,072 bytes of unused space or slack space.

Some operating systems fill this slack space with data from memory. If a user was working on a top secret file a moment ago and then creates a small unclassified file, the small file might contain top secret data pulled from memory. This is one of the reasons why personnel should never process classified data on unclassified systems.

Using system tools to delete data generally leaves much of the data remaining on the media, and widely available tools can easily undelete it. Even when you use sophisticated tools to overwrite the media, traces of the original data may remain as less perceptible magnetic fields. This is like a ghost image that can remain on some older TV and computer monitors if the same data is displayed for long periods of time. Forensics experts and attackers have tools they can use to retrieve this data even after it has been supposedly overwritten.

One way to remove data remanence is with a degausser. A degausser generates a heavy magnetic field, which realigns the magnetic fields in magnetic media such as traditional hard drives, magnetic tape, and floppy disk drives. Degaussers using power will reliably rewrite these magnetic fields and remove data remanence. However, they are only effective on magnetic media.

In contrast, SSDs use integrated circuitry instead of magnetic flux on spinning platters. Because of this, degaussing SSDs won't remove data. However, even when using other methods to remove data from SSDs, data remnants often remain.

Some SSDs include built-in erase commands to sanitize the entire disk, but unfortunately, these weren't effective on some SSDs from different manufacturers. Due to these risks, the best method of sanitizing SSDs is destruction. The U.S. National Security Agency (NSA) requires the destruction of SSDs using an approved disintegrator. Approved disintegrators shred the SSDs to a size of 2 millimeters (mm) or smaller.

Many organizations sell multiple information destruction and sanitization solutions used by government agencies and organizations in the private sector that the NSA has approved.

Another method of protecting SSDs is to ensure that all stored data is encrypted. If a sanitization method fails to remove all the data remnants, the remaining data would be unreadable.



Be careful when performing any type of clearing, purging, or sanitization process. The human operator or the tool involved in the activity may not properly perform the task of completely removing data from the media. Software can be flawed, magnets can be faulty, and either can be used improperly. Always verify that the desired result is achieved after performing any sanitization process.

Common Data Destruction Methods

The following list includes some common terms associated with destroying data:

Erasing *Erasing* media is simply performing a delete operation against a file, a selection of files, or the entire media. In most cases, the deletion or removal process removes only the directory or catalog link to the data. The actual data remains on the drive. As new files are written to the media, the system eventually overwrites the erased data, but depending on the size of the drive, how much free space it has, and several other factors, the data may not be overwritten for months. Anyone can typically retrieve the data using widely available undelete tools.

Clearing *Clearing*, or *overwriting*, is a process of preparing media for reuse and ensuring that the cleared data cannot be recovered using traditional recovery tools. When media is cleared, unclassified data is written over all addressable locations on the media. One method writes a single character, or a specific bit pattern, over the entire media. A more thorough method writes a single character over the entire media, writes the character's complement over the entire media, and finishes by writing random bits over the entire media. It repeats this in three separate passes, as

shown in Figure 5.2. Although this sounds like the original data is lost forever, it may be possible to retrieve some of the original data using sophisticated laboratory or forensics techniques. Additionally, not all types of data storage respond well to clearing techniques. For example, spare sectors on hard drives, sectors labeled as “bad,” and areas on many modern SSDs are not necessarily cleared and may still retain data.

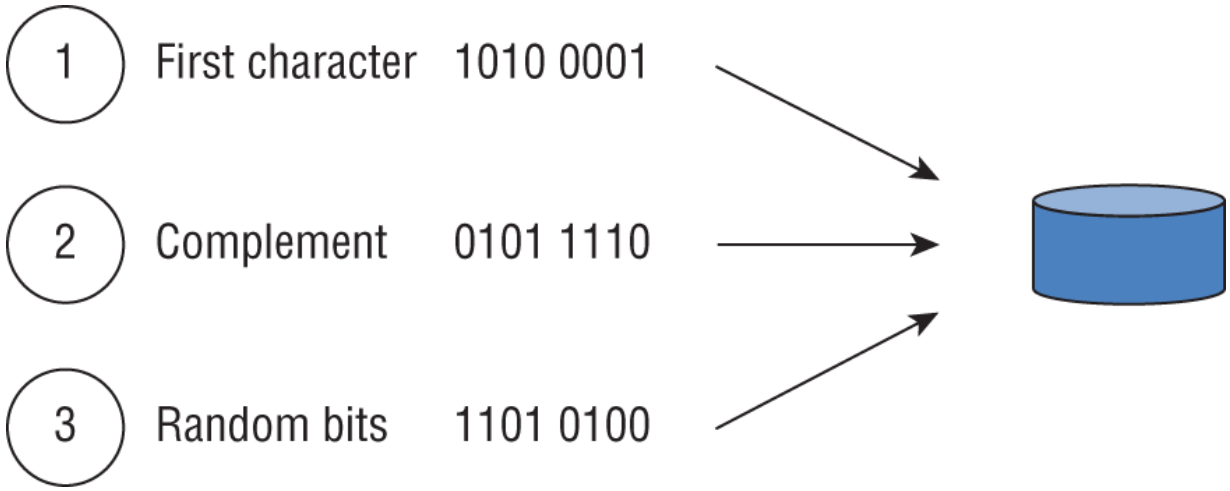


FIGURE 5.2 Clearing a hard drive

Purging *Purging* is a more intense form of clearing that prepares media for reuse in less secure environments. It provides a level of assurance that the original data is not recoverable using any known methods. A purging process will repeat the clearing process multiple times in order to completely remove the data. Even though purging is intended to remove all data remnants, it isn't always trusted. For example, the U.S. government doesn't consider any purging method acceptable to purge top secret data. Media labeled top secret will always remain top secret until it is destroyed.

Degaussing A degausser creates a strong magnetic field that erases data on some types of media in a process called *degaussing*. Technicians commonly use degaussing methods to remove data from magnetic tapes and magnetic hard disk drives (HDDs) with the goal of removing data from that media. Degaussing may render a hard drive unusable so it is not a good option when you intend to reuse the media.

Degaussing does not affect optical discs (CDs, DVDs, Blu-rays) or flash storage media (SD cards, USB flash drives, SSDs).

Destruction Destruction is the final stage in the life cycle of media and is the most secure method of sanitizing media. When destroying media, ensure that the media cannot be reused or repaired and that data cannot be extracted from the destroyed media. Methods of destruction include incineration, shredding, disintegration, pulverizing, and melting. Some organizations remove the platters in highly classified disk drives and destroy them separately.



When organizations donate or sell used computer equipment, they often remove and destroy storage devices that hold sensitive data rather than attempting to purge them. This eliminates the risk that the purging process wasn't complete, which would have resulted in a loss of confidentiality.

Declassification involves any process that purges media or a system in preparation for reuse in an unclassified environment. Sanitization methods can be used to prepare media for declassification, but often the efforts required to securely declassify media are significantly greater than the cost of new media for a less secure environment. Additionally, even though purged data is not recoverable using any known methods, there is a remote possibility that an unknown method is or becomes available. Instead of taking the risk, many organizations choose not to declassify any media and instead destroy it when it is no longer needed.

Cryptographic Erasure

If data is encrypted on a device, it's possible to use cryptographic erasure or cryptoshredding to destroy the data. However, these terms are misleading. They don't erase or shred the data. Instead, they destroy the associated keys. With the cryptographic keys erased, data remains encrypted and can't be accessed.

When using this method, you should use another method to overwrite the data. If the original encryption isn't strong, someone may be able to decrypt it without the key. Additionally, there are often backups of cryptographic keys, and if someone discovers a backup key, they can still access the data.

When using cloud storage, destroying the cryptographic keys may be the only form of secure deletion available to an organization.

Ensuring Appropriate Data and Asset Retention

Retention requirements apply to data or records, media holding sensitive data, systems that process sensitive data, and personnel who have access to sensitive data. Record retention and media retention are the most important elements of asset retention. Chapter 3, “Business Continuity Planning,” covers a vital records program, which can be referenced to identify records to retain.

Record retention involves retaining and maintaining important information as long as it is needed and destroying it when it is no longer needed. An organization's security policy or data policy typically identifies retention time frames. Some laws and regulations dictate the length of time that an organization should retain data, such as three years, seven years, or even indefinitely. Organizations have the responsibility of identifying laws and regulations that apply and complying with them. However, even in the absence of external requirements, an organization should still identify how long to retain data.

As an example, many organizations require the retention of all audit logs for a specific amount of time. The period can be dictated by laws, regulations, requirements related to partnerships with other organizations, or internal management decisions. These audit logs allow the organization to reconstruct the details of past security incidents. When an organization doesn't have a retention policy, administrators may delete valuable data earlier than management expects them to or attempt to keep data indefinitely. The longer an organization retains data, the more it costs in terms of media, locations to store it, and personnel to protect it.

End of life (EOL) and *end of support* can apply to either software or hardware. In the context of asset retention, they apply directly to hardware assets. Most vendors refer to EOL as the time when they stop producing and offering a product for sale. However, they will still support the products they've sold, at least for a while. End of support refers to the time when this support ends. Most hardware is on a refresh cycle based on the EOL and

end-of-support time frames. Organizations sometimes retain legacy hardware to access older data, such as data on tape drives.



Real World Scenario

Retention Policies Can Reduce Liabilities

Saving data longer than necessary also presents unnecessary legal issues. As an example, aircraft manufacturer Boeing was once the target of a class action lawsuit. Attorneys for the claimants learned that Boeing had a warehouse filled with 14,000 email backup tapes and demanded the relevant tapes. Not all the tapes were relevant to the lawsuit, but Boeing had to first restore the 14,000 tapes and examine the content before they could turn them over. Boeing ended up settling the lawsuit for $92.5 million, and analysts speculated that there would have been a different outcome if those 14,000 tapes hadn't existed.

The Boeing lawsuit is an extreme example, but it's not the only one. These events have prompted many companies to implement aggressive email retention policies. It is not uncommon for an email policy to require the deletion of all emails older than six months. These policies are often implemented using automated tools that search for old emails and delete them without any user or administrator intervention.

It is important, however, to understand that companies may *never* delete data when they can reasonably anticipate litigation. In fact, companies who believe that legal action may be forthcoming have a proactive obligation to preserve data and suspend any automated processes that might delete data.

Data Protection Methods

One of the primary methods of protecting the confidentiality of data is encryption, as discussed in the “Understanding Data States” section, earlier in this chapter. DLP methods (discussed in the “Data Loss Prevention” section, earlier in this chapter) help prevent data from leaving the network

or even leaving a computer system. This section covers some additional data protection methods.

Digital Rights Management

Digital rights management (DRM) methods attempt to provide copyright protection for copyrighted works. The purpose is to prevent the unauthorized use, modification, and distribution of copyrighted works such as intellectual property. Here are some methods associated with DRM solutions:

DRM License A license grants access to a product and defines the terms of use. A DRM license is typically a small file that includes the terms of use, along with a decryption key that unlocks access to the product.

Persistent Online Authentication Persistent online authentication (also known as always-on DRM) requires a system to be connected with the Internet to use a product. The system periodically connects with an authentication server, and if the connection or authentication fails, DRM blocks the use of the product.

Continuous Audit Trail A continuous audit trail tracks all use of a copyrighted product. When combined with persistence, it can detect abuse, such as concurrent use of a product simultaneously but in two geographically different locations.

Automatic Expiration Many products are sold on a subscription basis. For example, you can often rent new streaming movies, but these are only available for a limited time, such as 30 days. When the subscription period ends, an automatic expiration function blocks any further access.

As an example, imagine you dreamed up a fantastic idea for a book. When you awoke, you vigorously wrote down everything you remembered. In the following year, you spent every free moment you had developing the idea and eventually published your book. To make it easy for some people to read your book, you included a Portable Document Format (PDF) version of the book. You were grateful to see it skyrocket onto bestseller lists. You're on track for financial freedom to develop another great idea that came to you in another dream.

Unfortunately, someone copied the PDF file and posted it on the dark web. People from around the world found it and then began selling it online for next to nothing, claiming that they had your permission to do so. Of course, you didn't give them permission. Instead, they were collecting money from your year of work, while your revenue sales began to tumble.

This type of copying and distribution, commonly called pirating, has enriched criminals for years. Not only do they sell books they didn't write, but they also copy and sell music, videos, video games, software, and more.

Some DRM methods attempt to prevent the copying, printing, and forwarding of protected materials. Digital watermarks are sometimes placed within audio or video files using steganography. They don't prevent copying but can be used to detect the unauthorized copying of a file. They can also be used for copyright enforcement and prosecution. Similarly, metadata is sometimes placed into files to identify the buyer.

Many organizations and individuals are opposed to DRM. They claim it restricts the fair use of materials they purchase. For example, after paying for some songs, they want to copy them onto both an MP3 player and a smartphone. Additionally, people against DRM claim it isn't effective against people that want to bypass it but instead complicates the usage for legitimate users.

Chapter 4 covers intellectual property, copyrights, trademarks, patents, and trade secrets in more depth. DRM methods are used to protect copyrighted data, but they aren't used to protect trademarks, patents, or trade secrets.

Cloud Access Security Broker

A cloud access security broker (CASB) is software placed logically between users and cloud-based resources. It can be on-premises or within the cloud. Anyone who accesses the cloud goes through the CASB software. It monitors all activity and enforces administrator-defined security policies.

As a simple example, imagine a company has decided to use a cloud provider for data storage but management wants all data stored in the cloud to be encrypted. The CASB can monitor all data going to the cloud and ensure that it arrives and is stored in an encrypted format.

A CASB would typically include authentication and authorization controls and ensure only authorized users can access the cloud resources. The CASB can also log all access, monitor activity, and send alerts on suspicious activity. In general, any security controls that an organization has created internally can be replicated to a CASB. This includes any DLP functions implemented by an organization.

CASB solutions can also be effective at detecting *shadow IT*. Shadow IT is the use of IT resources (such as cloud services) without the approval of, or even the knowledge of, the IT department. If the IT department doesn't know about the usage, it can't manage it. One way a CASB solution can detect shadow IT is by collecting and analyzing logs from network firewalls and web proxies. Chapter 16, “Managing Security Operations,” covers other cloud topics.

Pseudonymization

Pseudonymization refers to the process of using pseudonyms to represent other data. When pseudonymization is performed effectively, it can result in less stringent requirements that would otherwise apply under the European Union (EU) General Data Protection Regulation (GDPR), covered in Chapter 4.



The EU GDPR replaced the European Data Protection Directive (Directive 95/46/EC), and it became enforceable on May 25, 2018. It applies to all EU member states and to all countries transferring data to and from the EU and anyone residing in the EU.

A pseudonym is an alias. As an example, *Harry Potter* author J. K. Rowling published a book titled *The Cuckoo's Calling* under the pseudonym of Robert Galbraith. No one knew it was her, at least for a few months. Someone leaked that Galbraith was a pseudonym, and her agent later confirmed the rumor. Now, if you know the pseudonym, you'll know that any books attributed to Robert Galbraith are written by J. K. Rowling.

Similarly, pseudonymization can prevent data from directly identifying an entity, such as a person. As an example, consider a medical record held by a doctor's office. Instead of including personal information such as the patient's name, address, and phone number, it could just refer to the patient as Patient 23456 in the medical record. The doctor's office still needs this personal information, and it could be held in another database linking it to the patient pseudonym (Patient 23456).

Note that in the example, the pseudonym (Patient 23456) refers to several pieces of information on the person. It's also possible for a pseudonym to refer to a single piece of information. For example, you can use one pseudonym for a first name and another pseudonym for a last name. The key is to have another resource (such as another database) that allows you to identify the original data using the pseudonym.

The doctor's office can release pseudonymized data to medical researchers without compromising patients' privacy information. However, the doctor's office can still reverse the process to discover the original data if necessary.

The GDPR refers to pseudonymization as replacing data with artificial identifiers. These artificial identifiers are pseudonyms.

Tokenization

Tokenization is the use of a token, typically a random string of characters, to replace other data. It is often used with credit card transactions.

As an example, imagine Becky Smith has associated a credit card with her smartphone. Tokenization with a credit card typically works like this:

Registration When she first associated the credit card with her smartphone's digital wallet, the digital wallet's service provider securely sent the actual credit card information to a payment network (Visa, Mastercard, American Express) or the issuing bank. The payment network sent the credit card number to a tokenization vault controlled by the payment network or a third-party service provider. The vault created a token (a string of characters) and recorded the token along with the encrypted credit card number, and associated it with the user's phone. The token was then sent to the digital wallet provider, which saved it to her smartphone.

Usage Later, Becky goes to a Starbucks and buys a cup of coffee with her smartphone. Her smartphone passes the token to the point-of-sale (POS) system. The POS system sends the token to the credit card processor to authorize the charge.

Validation The credit card processor sends the token to the tokenization vault. The vault answers with the unencrypted credit card data, and the credit card processor then processes the charge.

Completing the Sale The credit card processor sends only a reply to the POS system indicating the charge is approved or declined and, if approved, credits the seller for the purchase.

In the past, credit card data has been intercepted and stolen at the POS system. However, when tokenization is used, the credit card number is never used or known to the POS system. The user transfers it once to the payment network, and the payment network stores an encrypted copy of the credit card number along with a token matched to this credit card. Later, the user presents the token, and the payment processor validates the token through the tokenization vault.

E-commerce sites that have recurring charges also use tokenization. Instead of the e-commerce site collecting and storing credit card data, the site obtains a token from the payment gateway or processor. The token is created by a tokenization service, which stores an encrypted copy of the credit card data and sends the token to the e-commerce site. The site processes a charge the same way as it does for a POS system. However, the e-commerce site doesn't hold any sensitive data. Even if an attacker obtained a token and tried to make a charge with it, it would fail because the charges are only accepted from the e-commerce site.



Tokenization is similar to pseudonymization. Pseudonymization uses pseudonyms to represent other data. Tokenization uses tokens to represent other data. Neither the pseudonym nor the token has any meaning or value outside the process that creates them and links them to the other data. Pseudonymization is most useful when releasing a dataset to a third party (such as researchers aggregating data) without releasing any privacy-related data to the third party. Tokenization allows a third party (such as a payment network) to know the token and the original data. However, no one else knows both the token and the original data.

Anonymization

If you don't need personal data, another option is to use anonymization. *Anonymization* is the process of removing all relevant data so that it is theoretically impossible to identify the original subject or person. If done effectively, the GDPR is no longer relevant for the anonymized data. However, it can be difficult to truly anonymize the data. Data inference techniques may be able to identify individuals, even if personal data is removed. This is sometimes referred to as reidentification of anonymized data.

As an example, consider a database that includes a listing of all the actors who have starred or co-starred in movies in the last 75 years, along with the money they earned for each movie. The database has three tables. The Actor table includes the actor names, the Movie table lists the movie names, and the Payment table reports the amount of money each actor earned for each movie. The three tables are linked so that you can query the database and easily identify how much money any actor earned for any movie.

If you removed the names from the Actor table, it no longer includes personal data, but it is not truly anonymized. For example, Gene Hackman has been in more than 70 movies, and no other actor has been in all the same movies. If you identify those movies, you can now query the database and learn exactly how much he earned for each of those movies. Even

though his name was removed from the database, and that was the only obvious personal data in the database, data inference techniques can identify records applying to him.

Randomized masking can be an effective method of anonymizing data. Randomized masking swaps (shuffles) data in individual data columns so that records no longer represent the actual data. However, the data still maintains aggregate values that can be used for other purposes, such as scientific purposes. As an example, Table 5.2 shows four records in a database with the original values. An example of aggregated data is the average age of the four people, which is 29.

TABLE 5.2 Unmodified data within a database

First Name	Last Name	Age
Joe	Smith	25
Sally	Jones	28
Bob	Johnson	37
Maria	Doe	26

Table 5.3 shows the records after data has been swapped around, effectively masking the original data. Notice that this becomes a random set of first names, a random set of last names, and a random set of ages. It looks like real data, but none of the columns relate to each other. However, it is still possible to retrieve aggregated data from the table. The average age is still 29.

TABLE 5.3 Masked data

First Name	Last Name	Age
Sally	Doe	37
Maria	Johnson	25
Bob	Smith	28
Joe	Jones	26

Someone familiar with the dataset may be able to reconstruct some of the data if the table has only three columns and only four records. However,

this is an effective method of anonymizing data if the table has a dozen columns and thousands of records.

Unlike pseudonymization and tokenization, anonymization cannot be reversed. After the data is randomized using an anonymization process, it cannot be returned to the original state.

Understanding Data Roles

Many people within an organization manage, handle, and use data, and they have different requirements based on their roles. Different documentation refers to these roles a little differently. Some of the terms you may see match the terminology used in some NIST documents, and other terms match some of the terminology used in the EU GDPR. When appropriate, we've listed the source so that you can dig into these terms a little deeper if desired.

One of the most important concepts here is ensuring that personnel know who owns information and assets. The owners have a primary responsibility of protecting the data and assets.

Data Owners

The *data owner* is the person who has ultimate organizational responsibility for data. The owner is typically the chief executive officer (CEO), president, or a department head. Data owners identify the classification of data and ensure that it is labeled properly. They also ensure that it has adequate security controls based on the classification and the organization's security policy requirements. Owners may be liable for negligence if they fail to perform due diligence in establishing and enforcing security policies to protect and sustain sensitive data.

NIST SP 800-18, Rev. 1—Guide for Developing Security Plans for Federal Information Systems outlines the following responsibilities for the information owner, which can be interpreted the same as the data owner:

- Establishes the rules for appropriate use and protection of the subject data/information (rules of behavior)

- Provides input to information system owners regarding the security requirements and security controls for the information system(s) where the information resides
- Decides who has access to the information system and with what types of privileges or access rights
- Assists in the identification and assessment of the common security controls where the information resides



NIST SP 800-18 frequently uses the phrase “rules of behavior,” which is effectively the same as an acceptable use policy (AUP). Both outline the responsibilities and expected behavior of individuals and state the consequences of not complying with the rules or AUP. Additionally, individuals are required to periodically acknowledge that they have read, understand, and agree to abide by the rules or AUP. Many organizations post these on a website and allow users to acknowledge that they understand and agree to abide by them using an online electronic digital signature.

Data Controllers and Processors

Data controllers are the persons and organizations responsible for the collection and use of data. In the language of GDPR, “the data controller determines the purposes for which and the means by which personal data is processed.” In other words, the data controller is the entity that determines the “how” and the “why” of personal data collection and use. This is true even if the data controller doesn't handle the data themselves.

In many cases, data controllers outsource some data handling tasks to other organizations. These organizations are known as *data processors*. Under GDPR, a data processor is “a natural or legal person, public authority, agency, or other body, which processes personal data solely on behalf of the data controller.”

As an example, an employer that collects personal information on employees for payroll is a data controller. If they pass this information to a

third-party company to process payroll, the payroll company is the data processor. In this example, the payroll company (the data processor) must not use the data for anything other than processing payroll at the direction of the data controller.

The GDPR restricts data transfers to countries outside the EU. Companies that violate privacy rules in the GDPR may face fines of up to 4 percent of their global revenue or 20 million Euros, whichever is higher. Unfortunately, the GDPR is filled with legalese, presenting many challenges for organizations. As an example, clause 107 includes this single sentence statement:

> Consequently the transfer of personal data to that third country or international organisation should be prohibited, unless the requirements in this Regulation relating to transfers subject to appropriate safeguards, including binding corporate rules, and derogations for specific situations are fulfilled.

As a result, many organizations have created dedicated roles, such as a data privacy officer (DPO), to oversee the control of data and ensure the organization follows all relevant laws and regulations. The GDPR has mandated the role of a data protection officer for any organization that must comply with the GDPR. The person in this role is responsible for ensuring the organization applies the laws to protect individuals' private data.

Data Custodians

Data owners often delegate day-to-day tasks to a *data custodian*. A custodian helps protect the integrity and security of data by ensuring that it is properly stored and protected. For example, custodians would ensure that the data is backed up by following guidelines in a backup policy. If administrators have configured auditing on the data, custodians would also maintain these logs.

In practice, personnel within an IT department or system security administrators would typically be the custodians. They might be the same administrators responsible for assigning permissions to data.

Users and Subjects

A *user* is any person who accesses data via a computing system to accomplish work tasks. Users should have access only to the data they need to perform their work tasks. You can also think of users as employees or end users.

The GDPR defines a *data subject* as a person who can be identified through an identifier, such as a name, an identification number, location data, an online identifier, or other means. As an example, if a file includes PII on Sally Smith, Sally Smith is the data subject.

Using Security Baselines

Once an organization has identified and classified its assets, it will typically want to secure them. That's where security baselines come in. Baselines provide a starting point and ensure a minimum security standard. One common baseline that organizations use is imaging. Chapter 16 covers system imaging in the context of configuration management in more depth. As an introduction, administrators configure a single system with desired settings, capture it as an image, and then deploy the image to other systems. This ensures that systems are deployed in a similar secure state, which helps to protect the privacy of data.

After deploying systems in a secure state, auditing processes periodically check the systems to ensure they remain in a secure state. For example, Microsoft Group Policy can periodically check systems and reapply settings to match the security baseline.

NIST SP 800-53, Rev. 5—Security and Privacy Controls for Information Systems and Organizations mentions *security control baseline* and identifies it as the set of minimum security controls defined for an information system. It stresses that a single set of security controls does not apply to all situations. Still, any organization can select a set of baseline security controls and tailor the baseline to its needs. NIST SP 800-53B—Control Baselines for Information Systems and Organizations includes a comprehensive list of security controls and has identified many of them to include in various baselines. Specifically, they present three security control baselines (determined by the impact level of the system) and a privacy control baseline. These are based on the potential impact to an

organization's mission if there is a loss of confidentiality, integrity, or availability of a system. The baselines are as follows:

Low-Impact System Controls in this baseline are recommended if any loss of confidentiality, integrity, and/or availability will have a low impact on the organization's mission.

Moderate-Impact System Controls in this baseline are recommended if it is possible that a loss of confidentiality, integrity, or availability will have a moderate impact on the organization's mission.

High-Impact System Controls in this baseline are recommended if it is possible that a loss of confidentiality, integrity, or availability will have a high impact on the organization's mission.

Privacy Control Baseline This baseline provides an initial baseline for any systems that process PII. Organizations may combine this baseline with one of the other baselines.

These refer to the worst-case potential impact if a system is compromised and a data breach occurs. For example, imagine a system is compromised. You would try to predict the impact of the compromise on the confidentiality, integrity, or availability of the system and any data it holds:

- If the compromise would cause privacy data to be compromised, you would consider adding the security controls identified as privacy control baseline items to your baseline.
- If the impact is low for all three of the security objectives, you would consider adding the security controls identified as low-impact controls to your baseline.
- If the impact of this compromise is moderate for any one of the security objectives, you would consider adding the security controls identified as moderate-impact, in addition to the low-impact controls.
- If the impact is high for any one of the security objectives, you would consider adding all the controls listed as high-impact in addition to the low-impact and moderate-impact controls.

It's worth noting that many of the items in these lists are basic security practices. Additionally, implementing basic security principles such as the

least privilege principle shouldn't surprise anyone. Of course, just because these are basic security practices, it doesn't mean organizations implement them. Unfortunately, many organizations have yet to discover or enforce the basics.

Comparing Tailoring and Scoping

After selecting a control baseline, organizations fine-tune it with tailoring and scoping processes. A big part of the tailoring process is aligning the controls with an organization's specific security requirements. As a comparison, think of a clothes tailor who alters or repairs clothes. If a person buys a suit at a high-end retailer, a tailor modifies the suit to fit the person perfectly. Similarly, tailoring a baseline ensures it is a good fit for the organization.

Tailoring refers to modifying the list of security controls within a baseline to align with the organization's mission. NIST SP 800-53B formally defines it as “part of an organization-wide risk management process that includes framing, assessing, responding to, and monitoring information security and privacy risks” and indicates it includes the following activities:

- Identifying and designating common controls
- Applying scoping considerations
- Selecting compensating controls
- Assigning values to organization-defined control parameters via explicit assignment and selection operations
- Supplementing baselines with additional controls and control enhancements
- Providing specification information for control implementation

A selected baseline may not include commonly implemented controls. However, just because a security control isn't included in the baseline doesn't mean it should be removed. For example, imagine that a data center includes video cameras covering the external entry, the internal exit, and every row of servers, but the baseline only recommends a video camera cover the external entry. During the tailoring process, personnel will

evaluate these extra cameras and determine if they are needed. They may decide to remove some to save costs or keep them.

An organization might decide that a set of baseline controls applies perfectly to computers in their central location but that some controls aren't appropriate or feasible in a remote office location. In this situation, the organization can select compensating security controls to tailor the baseline to the remote site. For example, imagine the account lockout policy is set to lock out users if they enter an incorrect password five times. In this example, the control value is 5, but the tailoring process may change it to 3.

Scoping is a part of the tailoring process and refers to reviewing a list of baseline security and privacy controls and selecting only those security and privacy controls that apply to the IT systems you're trying to protect. Or, in the simplest terms, scoping processes eliminate controls that are recommended in a baseline. For example, if a system doesn't allow any two people to log on to it simultaneously, there's no need to apply a concurrent session control. During this part of the tailoring process, the organization looks at every control in the baseline and vigorously defends (in writing) any decision to omit a control from the baseline.

Standards Selection

When selecting security controls within a baseline, or otherwise, organizations need to ensure that the controls comply with external security standards. External elements typically define compulsory requirements for an organization. For example, the Payment Card Industry Data Security Standard (PCI DSS) defines requirements that businesses must follow to process major credit cards. Similarly, organizations that collect or process data belonging to EU citizens must abide by the requirements in the GDPR.

Obviously, not all organizations have to comply with these standards. Organizations that don't store, process, or transmit payment card transactions do not need to comply with PCI DSS. Similarly, organizations that do not collect or process EU citizens' data do not need to comply with GDPR requirements. Organizations need to identify the standards that apply and ensure that the security and privacy controls they select fully comply with those standards.

Even if your organization isn't legally required to comply with a specific standard, using a well-designed community standard can be helpful. For example, U.S. government organizations are required to comply with many of the standards published by NIST SP 800 documents. These same documents are used by many organizations in the private sector to help them develop and implement their own security standards.

Summary

Asset security focuses on collecting, handling, and protecting information throughout its life cycle. This includes sensitive information stored or processed on computing systems or transferred over a network and the assets used in these processes. Sensitive information is any information that an organization keeps private and can include multiple levels of classifications. Proper destruction methods ensure that data can't be retrieved after destruction.

Data protection methods include digital rights management (DRM) and using cloud access security brokers (CASBs) when using cloud resources. DRM methods attempt to protect copyrighted materials. A CASB is software placed logically between users and cloud-based resources. It can ensure that cloud resources have the same protections as resources within a network. Entities that must comply with the EU GDPR use additional data protection methods such as pseudonymization, tokenization, and anonymization.

Personnel can fulfill many different roles when handling data. Data owners are ultimately responsible for classifying, labeling, and protecting data. System owners are responsible for the systems that process the data. The GDPR defines data controllers, data processors, and data custodians. Data controllers decide what data to process, the purpose of data collection, and how to process it. A data controller can hire a third party to process data, and in this context, the third party is the data processor. Data processors have a responsibility to protect the privacy of the data and not use it for any purpose other than directed by the data controller. A custodian is delegated day-to-day responsibilities for properly storing and protecting data.

Security baselines provide a set of security controls that an organization can implement as a secure starting point. Some publications (such as NIST SP 800-53B) identify security control baselines. However, these baselines don't apply equally to all organizations. Instead, organizations use scoping and tailoring techniques to identify the security controls to implement after selecting baselines. Additionally, organizations ensure that they implement security controls mandated by external standards that apply to their organization.

Study Essentials

Understand the importance of data and asset classifications. Data owners are responsible for defining data and asset classifications and ensuring that data and systems are properly tagged. Additionally, data owners define requirements to protect data at different classifications, such as encrypting sensitive data at rest, in transit, and in use. Data classifications are typically defined within security policies or data policies.

Define PII and PHI. Personally identifiable information (PII) is any information that can identify an individual. Protected health information (PHI) is any health-related information that can be related to a specific person and is subject to HIPAA. Many laws and regulations mandate the protection of PII and PHI.

Know how to manage sensitive information. Sensitive information is any type of classified information, and proper management helps prevent unauthorized disclosure resulting in a loss of confidentiality. Proper management includes tagging, handling, storing, and destroying sensitive information. The two areas where organizations often miss the mark are adequately protecting backup media holding sensitive information and sanitizing media or equipment when it is at the end of its life cycle.

Describe the three data states. The three data states are at rest, in transit, and in use. Data at rest is any data stored on media such as hard drives or external media. Data in transit is any data transmitted over a network. Encryption methods protect data at rest and in transit. Data in use refers to data in memory and used by an application. Applications should flush memory buffers to remove data after it is no longer needed.

Define DLP. Data loss prevention (DLP) solutions detect and block data exfiltration attempts by scanning unencrypted files and looking for keywords and data patterns. Network DLP solutions (including cloud DLP solutions) scan files before they leave the network. Endpoint DLP solutions prevent users from copying or printing some files.

Compare data destruction methods. Erasing a file doesn't delete it. Clearing media overwrites it with characters or bits. Purging repeats the clearing process multiple times and removes data so that the media can be reused. Degaussing removes data from tapes and magnetic hard disk drives, but it does not affect optical media or SSDs. Destruction methods include incineration, shredding, and disintegration, pulverizing, and melting.

Describe data remanence. Data remanence is the data that remains on media after it should have been removed. Hard disk drives sometimes retain residual magnetic flux that can be read with advanced tools. Advanced tools can read slack space on a disk, which is unused space in clusters. Erasing data on a disk leaves data remanence. For solid-state drives (SSDs), data remanence can persist due to the wear-leveling algorithms they employ, making traditional data erasure methods less effective and potentially allowing remnants of data to remain on unaddressed memory cells.

Understand record retention policies. Record retention policies ensure that data is kept in a usable state while it is needed and destroyed when it is no longer needed. Many laws and regulations mandate keeping data for a specific amount of time, but in the absence of formal regulations, organizations specify the retention period within a policy. Audit trail data needs to be kept long enough to reconstruct past incidents, but the organization must identify how far back they want to investigate. A current trend in many organizations is to reduce legal liabilities by implementing short retention policies with email.

Know the difference between end of life and end of support. End of life (EOL) is the date announced by a vendor when production and sales of a product stop. However, the vendor still supports the product after EOL. End of support identifies the date when a vendor will no longer support a product.

Explain DRM. Digital rights management (DRM) methods provide copyright protection for copyrighted works. The purpose is to prevent the

unauthorized use, modification, and distribution of copyrighted works.

Explain CASB. A cloud access security broker (CASB) is a solution placed logically between users and cloud resources. It can apply internal security controls to cloud resources. The CASB solution can be placed on-premises or in the cloud.

Define pseudonymization. Pseudonymization is the process of replacing some data elements with pseudonyms or aliases. It removes privacy data so that a dataset can be shared. However, the original data remains available in a separate dataset.

Define tokenization. Tokenization replaces data elements with a string of characters or a token. Credit card processors replace credit card data with a token, and a third party holds the mapping to the original data and the token.

Define anonymization. Anonymization replaces privacy data with useful but inaccurate data. The dataset can be shared and used for analysis purposes, but anonymization removes individual identities. Anonymization is permanent.

Know the responsibilities of data roles. The data owner is the person responsible for classifying, labeling, and protecting data. Data controllers decide what data to process, the purpose of data collection, and how to process data. Data processors are third-party entities that process data for an organization at the direction of the data controller. A user accesses data while performing work tasks. The data subject is the person described in the PII. A custodian has day-to-day responsibilities for protecting and storing data.

Know about security control baselines. Security control baselines provide a listing of controls that an organization can apply as a baseline. Not all baselines apply to all organizations. Organizations apply scoping and tailoring techniques to adapt a baseline to their needs.

Written Lab

1. Describe sensitive data.

2. Identify the difference between EOL and EOS.
3. Identify common uses of pseudonymization, tokenization, and anonymization.
4. Describe the difference between scoping and tailoring.

Review Questions

1. Which of the following provides the best protection against the loss of confidentiality for sensitive data?

   A. Data labels

   B. Data classifications

   C. Data handling

   D. Data degaussing methods

2. Administrators regularly back up data on all the servers within your organization. They annotate an archive copy with the server it came from and the date it was created, and transfer it to an unstaffed storage warehouse. Later, they discover that someone leaked sensitive emails sent between executives on the Internet. Security personnel discovered some archive tapes are missing, and these tapes probably included the leaked emails. Of the following choices, what would have prevented this loss without sacrificing security?

   A. Mark the media kept off-site.

   B. Don't store data off-site.

   C. Destroy the backups off-site.

   D. Use a secure off-site storage facility.

3. Administrators have been using tapes to back up servers in your organization. However, the organization is converting to a different backup system, storing backups on disk drives. What is the final stage in the life cycle of tapes used as backup media?

   A. Degaussing

B. Destruction

C. Declassification

D. Retention

4. You are updating your organization's data policy, and you want to identify the responsibilities of various roles. Which one of the following data roles is responsible for classifying data?

   A. Controller

   B. Custodian

   C. Owner

   D. User

5. You are tasked with updating your organization's data policy, and you need to identify the responsibilities of different roles. Which data role is responsible for implementing the protections defined by the security policy?

   A. Data custodian

   B. Data user

   C. Data processor

   D. Data controller

6. A company maintains an e-commerce server used to sell digital products via the Internet. When a customer makes a purchase, the server stores the following information on the buyer: name, physical address, email address, and credit card data. You're hired as an outside consultant and advise them to change their practices. Which of the following can the company implement to avoid an apparent vulnerability?

   A. Anonymization

   B. Pseudonymization

   C. Move the company location

   D. Collection limitation

7. You are performing an annual review of your company's data policy, and you come across some confusing statements related to security labeling. Which of the following could you insert to describe security labeling accurately?

    A. Security labeling is only required on digital media.

    B. Security labeling identifies the classification of data.

    C. Security labeling is only required for hardware assets.

    D. Security labeling is never used for nonsensitive data.

8. A database file includes personally identifiable information (PII) on several individuals, including Karen C. Park. Which of the following is the best identifier for the record on Karen C. Park?

    A. Data controller

    B. Data subject

    C. Data processor

    D. Data owner

9. Administrators regularly back up all the email servers within your company, and they routinely purge on-site emails older than six months to comply with the organization's security policy. They keep a copy of the backups on-site and send a copy to one of the company warehouses for long-term storage. Later, they discover that someone leaked sensitive emails sent between executives over three years ago. Of the following choices, what policy was ignored and allowed this data breach?

    A. Media destruction

    B. Record retention

    C. Configuration management

    D. Versioning

10. An executive is reviewing governance and compliance issues and ensuring the security or data policy addresses them. Which of the following security controls is most likely driven by a legal requirement?

A. Data remanence

B. Data destruction

C. Data user role

D. Record retention

11. Your organization is donating several computers to a local school. Some of these computers include solid-state drives (SSDs). Which of the following choices is the *most* reliable method of destroying data on these SSDs?

    A. Erasing

    B. Degaussing

    C. Deleting

    D. Purging

12. A technician is about to remove magnetic disk drives from several computers. His supervisor told him to ensure that the disk drives do *not* hold any sensitive data. Which of the following methods will meet the supervisor's requirements?

    A. Overwriting the disks multiple times

    B. Formatting the disks

    C. Deleting the files

    D. Defragmenting the disks

13. The IT department is updating the budget for the following year, and they want to include enough money for a hardware refresh for some older systems. Unfortunately, there is a limited budget. Which of the following should be a top priority?

    A. Systems with an end-of-life (EOL) date that occurs in the following year

    B. Systems used for data loss prevention

    C. Systems used to process sensitive data

D. Systems with an end-of-support (EOS) date that occurs in the following year

14. Developers created an application that routinely processes sensitive data. The data is encrypted and stored in a database. When the application processes the data, it retrieves it from the database, decrypts it for use, and stores it in memory. Which of the following methods can protect the data in memory after the application uses it?

    A. Encrypt it with asymmetric encryption.

    B. Encrypt it in the database.

    C. Implement data loss prevention.

    D. Purge memory buffers.

15. Your organization's security policy mandates the use of symmetric encryption for sensitive data stored on servers. Which one of the following guidelines are they implementing?

    A. Protecting data at rest

    B. Protecting data in transit

    C. Protecting data in use

    D. Protecting the data life cycle

16. An administrator is planning to deploy a database server and wants to ensure it is secure. She reviews a list of baseline security controls and identifies the security controls that apply to this database server. What is this called?

    A. Tokenization

    B. Scoping

    C. Standards selection

    D. Imaging

17. An organization is planning to deploy an e-commerce site hosted on a web farm. IT administrators have identified a list of security controls they say will provide the best protection for this project. Management

is now reviewing the list and removing any security controls that do not align with the organization's mission. What is this called?

A. Tailoring

B. Sanitizing

C. Asset classification

D. Minimization

18. An organization is planning to use a cloud provider to store some data. Management wants to ensure that all data-based security policies implemented in the organization's internal network can also be implemented in the cloud. Which of the following will support this goal?

A. CASB

B. DLP

C. DRM

D. EOL

19. Management is concerned that users may be inadvertently transmitting sensitive data outside the organization. They want to implement a method to detect and prevent this from happening. Which of the following can detect outgoing, sensitive data based on specific data patterns and is the best choice to meet these requirements?

A. Antimalware software

B. Data loss prevention systems

C. Security information and event management systems

D. Intrusion prevention systems

20. A software developer created an application and wants to protect it with DRM technologies. Which of the following is she *most* likely to include? (Choose three.)

A. Virtual licensing

B. Persistent online authentication

C. Automatic expiration

D. Continuous audit trail

Chapter 6
Cryptography and Symmetric Key Algorithms

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 3.0: Security Architecture and Engineering**
  - 3.5 Assess and mitigate the vulnerabilities of security architectures, designs, and solution elements
    - 3.5.4 Cryptographic systems
  - 3.6 Select and determine cryptographic solutions
    - 3.6.1 Cryptographic life cycle (e.g., keys, algorithm selection)
    - 3.6.2 Cryptographic methods (e.g., symmetric)

Cryptography provides confidentiality, integrity, authentication, and nonrepudiation for sensitive information while it is stored (at rest), traveling across a network (in transit/in motion), and existing in memory (in use/in processing). Cryptography is an extremely important security technology that is embedded in many of the controls used to protect information from unauthorized visibility and use.

Over the years, mathematicians and computer scientists have developed a series of increasingly complex cryptographic algorithms designed to increase the level of protection provided to data. While cryptographers spent time developing strong encryption algorithms, malicious hackers and governments alike devoted significant resources to undermining them. This led to an “arms race” in cryptography and resulted in the development of the extremely sophisticated algorithms in use today.

This chapter looks at the basics of cryptographic communications and the fundamental principles of symmetric key (secret key) cryptosystems. The

next chapter continues the discussion of cryptography by examining asymmetric key (public key) cryptosystems and the various techniques attackers use to defeat cryptography.

Cryptographic Foundations

The study of any science must begin with a discussion of the fundamental principles on which it is built. The following sections lay this foundation with a review of the goals of cryptography, an overview of the basic concepts of cryptographic technology, and a look at the major mathematical principles used by cryptographic systems.

Goals of Cryptography

Security practitioners use cryptographic systems to meet four fundamental goals: confidentiality, integrity, authentication, and nonrepudiation. Achieving each of these goals requires the satisfaction of a number of design requirements, and not all cryptosystems are intended to achieve all four goals. In the following sections, we'll examine each goal in detail and give a brief description of the technical requirements necessary to achieve it.

Confidentiality

Confidentiality ensures that data remains private in three different situations: when it is at rest, when it is in transit, and when it is in use.

Confidentiality is perhaps the most widely cited goal of cryptosystems—the preservation of secrecy for stored information or for communications between individuals and groups. Two main types of cryptosystems enforce confidentiality:

- *Symmetric cryptosystems* use a shared secret key available to all users of the cryptosystem.
- *Asymmetric cryptosystems* use individual pairs of public and private keys for each user of the system.

Both of these concepts are explored in the section “Modern Cryptography,” later in this chapter.

When developing a cryptographic system for the purpose of providing confidentiality, you must think about the three different types of data that we discussed in Chapter 5, “Protecting Security of Assets”:

- *Data at rest*, or stored data, resides in a fixed location awaiting access. Examples of data at rest include data stored on hard drives, backup tapes, cloud storage services, USB devices, and other storage media.
- *Data in transit*, data in motion, or data on the wire is data being transmitted across a network between two systems. Data in motion might be traveling on a corporate network, a wireless network, or the Internet.
- *Data in use* is data that is stored in the active memory of a computer system, where it may be accessed by a process running on that system.



You should also know that data in transit is also commonly called data *on the wire*, referring to the network cables that carry data communications.

Each of these situations poses different types of confidentiality risks that cryptography can protect against. For example, data in transit may be susceptible to eavesdropping attacks, whereas data at rest is more susceptible to the theft of physical devices. Data in use may be accessed by unauthorized processes if the operating system does not properly implement process isolation.

Integrity

Integrity ensures that data is not altered without authorization. If integrity mechanisms are in place, the recipient of a message can be certain that the message received is identical to the message that was sent. Similarly, integrity checks can ensure that stored data was not altered between the time it was created and the time it was accessed. Integrity controls protect

against all forms of alteration, including intentional alteration by a third party attempting to insert false information, intentional deletion of portions of the data, and unintentional alteration by faults in the transmission process.

Message integrity is enforced through the use of encrypted message digests, known as *digital signatures*, created upon transmission of a message. The recipient of the message simply verifies that the message's digital signature is valid, ensuring that the message was not altered in transit. Integrity can be enforced by both public and secret key cryptosystems. This concept is discussed in detail in Chapter 7, "PKI and Cryptographic Applications." The use of cryptographic hash functions to protect file integrity is discussed in Chapter 21, "Malicious Code and Application Attacks."

Authentication

Authentication verifies the claimed identity of system users and is a major function of cryptosystems. For example, suppose that Bob wants to establish a communications session with Alice and they are both participants in a shared secret communications system. Alice might use a challenge-response authentication technique to ensure that Bob is who he claims to be.

Figure 6.1 shows how this challenge-response protocol would work in action. In this example, the shared-secret code used by Alice and Bob is quite straightforward—the letters of each word are simply reversed. Bob first contacts Alice and identifies himself. Alice then sends a challenge message to Bob, asking him to encrypt a short message using the secret code known only to Alice and Bob. Bob replies with the encrypted message. After Alice verifies that the encrypted message is correct, she trusts that Bob himself is truly on the other end of the connection.

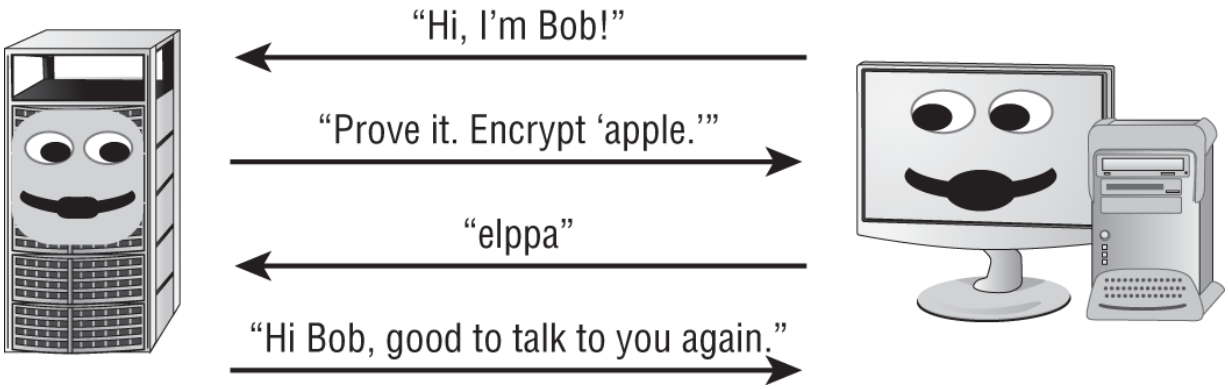


FIGURE 6.1 Challenge-response authentication protocol

Nonrepudiation

Nonrepudiation provides assurance to the recipient that the message was originated by the sender and not someone masquerading as the sender. It also prevents the sender from claiming that they never sent the message in the first place (also known as *repudiating* the message). Secret key, or symmetric key, cryptosystems (such as simple substitution ciphers) do not provide this guarantee of nonrepudiation. If Jim and Bob participate in a secret key communication system, they can both produce the same encrypted message using their shared secret key. Nonrepudiation is offered only by public key, or asymmetric, cryptosystems, a topic discussed in greater detail in Chapter 7.

Cryptography Concepts

As with any science, you must be familiar with certain terminology before studying cryptography. Let's take a look at a few of the key terms used to describe codes and ciphers. Before a message is put into a coded form, it is known as a *plaintext* message and is represented by the letter *P* when encryption functions are described. The sender of a message uses a cryptographic algorithm to *encrypt* the plaintext message and produce a *ciphertext* message, represented by the letter *C*. This message is transmitted by some physical or electronic means to the recipient. The recipient then uses a predetermined algorithm to *decrypt* the ciphertext message and retrieve the plaintext version. (For an illustration of this process, see Figure 6.3 later in this chapter.)

All cryptographic algorithms rely on *keys* to maintain their security. For the most part, a key is nothing more than a number. It's usually a very large binary number, but it's a number nonetheless. Every algorithm has a specific *key space*. The key space is the range of values that are valid for use as a key for a specific algorithm. A key space is defined by its *bit size*. Bit size is nothing more than the number of binary bits (0s and 1s) in the key. The key space is the range between the key that has all 0s and the key that has all 1s. Or to state it another way, the key space is the range of numbers from 0 to 2^n, where n is the bit size of the key. So, a 128-bit key can have a value from 0 to 2^{128} (which is roughly $3.40282367 \times 10^{38}$, a very big number!). It is absolutely critical to protect the security of secret keys and private keys. In fact, all of the security you gain from cryptography rests on your ability to keep the secret and private keys confidential.

Kerckhoffs's Principle

All cryptography relies on algorithms. An *algorithm* is a set of rules, usually mathematical, that dictates how encryption and decryption processes are to take place. Most cryptographers follow Kerckhoffs's principle, a concept that makes algorithms known and public, allowing anyone to examine and test them. Specifically, *Kerckhoffs's principle* (also known as Kerckhoffs's assumption) is that a cryptographic system should be secure even if everything about the system, except the key, is public knowledge. The principle can be summed up as “The enemy knows the system.”

A large number of cryptographers adhere to this principle, but not all agree. In fact, some believe that better overall security can be maintained by keeping both the algorithm and the key private. Kerckhoffs's adherents retort that the opposite approach includes the dubious practice of “security through obscurity” and believe that public exposure produces more activity and exposes more weaknesses more readily, leading to the abandonment of insufficiently strong algorithms and quicker adoption of suitable ones.

As you'll learn in this chapter and the next, different types of algorithms require different types of keys. In symmetric key (or secret key) cryptosystems, all participants use a single shared key. In public key cryptosystems, each participant has their own pair of public and private keys. Cryptographic keys are sometimes referred to as *cryptovariables*, particularly in U.S. government applications.

The art of creating and implementing secret codes and ciphers is known as *cryptography*. This practice is paralleled by the art of *cryptanalysis*—the study of methods to defeat codes and ciphers. Together, cryptography and cryptanalysis are commonly referred to as *cryptology*. Specific implementations of a code or cipher in hardware and software are known as *cryptosystems*.

Federal Information Processing Standard (FIPS) 140–3, “Security Requirements for Cryptographic Modules,” defines the hardware and software requirements for cryptographic modules that the federal government uses.

Cryptographic Mathematics

Cryptography is no different from most computer science disciplines in that it finds its foundations in the science of mathematics. To fully understand cryptography, you must first understand the basics of binary mathematics and the logical operations used to manipulate binary values. The following sections present a brief look at some of the most fundamental concepts with which you should be familiar.

Boolean Mathematics

Boolean mathematics defines the rules used for the bits and bytes that form the nervous system of any computer. You're most likely familiar with the decimal system. It is a base 10 system in which an integer from 0 to 9 is used in each place and each place value is a multiple of 10. It's likely that our reliance on the decimal system has biological origins—human beings have 10 fingers that can be used to count.



Boolean math can be very confusing at first, but it's worth the investment of time to learn how logical functions work. You need to know these concepts to truly understand the inner workings of cryptographic algorithms.

Similarly, the computer's reliance on the Boolean system has electrical origins. In an electrical circuit, there are only two possible states—on (representing the presence of electrical current) and off (representing the absence of electrical current). All computation performed by an electrical device must be expressed in these terms, giving rise to the use of Boolean computation in modern electronics. In general, computer scientists refer to the on condition as a *true* value and the off condition as a *false* value.

Logical Operations

The Boolean mathematics of cryptography uses a variety of logical functions to manipulate data. We'll take a brief look at several of these operations.

AND

The AND operation (represented by the ∧ symbol) checks to see whether two values are both true. Table 6.1 shows a truth table that illustrates all four possible outputs for the AND function. In this truth table, the first two columns, X and Y, show the input values to the AND function. Remember, the AND function takes only two variables as input. In Boolean math, there are only two possible values for each of these variables (0=FALSE and 1=TRUE), leading to four possible inputs to the AND function. The X ∧ Y column shows the output of the AND function for the input values shown in the two adjacent columns. It's this finite number of possibilities that makes it extremely easy for computers to implement logical functions in hardware. Notice in Table 6.1 that only one combination of inputs (where both inputs are TRUE) produces an output value of true.

Logical operations are often performed on entire Boolean words rather than single values. Take a look at the following example:

TABLE 6.1 AND operation truth table

X	Y	X ∧ Y
0	0	0
0	1	0
1	0	0
1	1	1

```
X:      0 1 1 0 1 1 0 0
Y:      1 0 1 0 0 1 1 1
_____________________________
X ∧ Y: 0 0 1 0 0 1 0 0
```

Notice that the AND function is computed by comparing the values of X and Y in each column. The output value is TRUE only in columns where both X and Y are true.

OR

The OR operation (represented by the ∨ symbol) checks to see whether at least one of the input values is true. Refer to the truth table in Table 6.2 for all possible values of the OR function. Notice that the only time the OR function returns a false value is when both of the input values are false.

TABLE 6.2 OR operation truth table

X	Y	X ∨ Y
0	0	0
0	1	1
1	0	1
1	1	1

We'll use the same example we used in the previous section to show you what the output would be if X and Y were fed into the OR function rather than the AND function:

```
X:      0 1 1 0 1 1 0 0
Y:      1 0 1 0 0 1 1 1
```

```
_____________________________
X v Y: 1 1 1 0 1 1 1 1
```

NOT

The NOT operation (represented by the ~ symbol) simply reverses the value of an input variable. This function operates on only one variable at a time. Table 6.3 shows the truth table for the NOT function.

TABLE 6.3 NOT operation truth table

X	~X
0	1
1	0

In this example, you take the value of `X` from the previous examples and run the NOT function against it:

```
X:  0 1 1 0 1 1 0 0
_____________________________
~X: 1 0 0 1 0 0 1 1
```

Exclusive OR

The final logical function you'll examine in this chapter is perhaps the most important and most commonly used in cryptographic applications—the exclusive OR (XOR) function. It's referred to in mathematical literature as the XOR function and is commonly represented by the ⊕ symbol. The XOR function returns a true value when only one of the input values is true. If both values are false or both values are true, the output of the XOR function is false. Table 6.4 provides the truth table for the XOR operation.

TABLE 6.4 Exclusive OR operation truth table

X	Y	X ⊕ Y
0	0	0
0	1	1
1	0	1
1	1	0

The following operation shows the X and Y values when they are used as input to the XOR function:

```
X:      0 1 1 0 1 1 0 0
Y:      1 0 1 0 0 1 1 1
_____________________________
X ⊕ Y: 1 1 0 0 1 0 1 1
```

Modulo Function

The *modulo* (mod) function is extremely important in the field of cryptography. Think back to the early days when you first learned division. At that time, you weren't familiar with decimal numbers and compensated by showing a remainder value each time you performed a division operation. Computers don't naturally understand the decimal system either, and these remainder values play a critical role when computers perform many mathematical functions. The modulo function is, quite simply, the remainder value left over after a division operation is performed.



The modulo function is just as important to cryptography as the logical operations are. Be sure you're familiar with its functionality and can perform simple modular math.

The modulo function is usually represented in equations by the abbreviation *mod*, although it's also sometimes represented by the % symbol. Here are several inputs and outputs for the modulo function:

```
8 mod 6 = 2
6 mod 8 = 6
10 mod 3 = 1
10 mod 2 = 0
32 mod 8 = 0
32 mod 26 = 6
```

We'll revisit this function in Chapter 7 when we explore the RSA public key encryption algorithm (named after Ron Rivest, Adi Shamir, and Leonard Adleman, its inventors).

One-Way Functions

A *one-way function* is a mathematical operation that easily produces output values for each possible combination of inputs but makes it impossible to retrieve the input values. Public key cryptosystems are all based on some sort of one-way function. In practice, however, it's never been proven that any specific known function is truly one-way. Cryptographers rely on functions that they believe are one-way, but it's always possible that they might be broken by future cryptanalysts.

Here's an example. Imagine you have a function that multiplies three numbers together. If you restrict the input values to single-digit numbers, it's a relatively straightforward matter to reverse-engineer this function and determine the possible input values by looking at the numerical output. For example, the output value 15 was created by using the input values 1, 3, and 5. However, suppose you restrict the input values to five-digit prime numbers. It's still quite simple to obtain an output value by using a computer or a good calculator, but reverse-engineering is not quite so simple. Can you figure out which three prime numbers were used to obtain the output value 10,718,488,075,259? Not so simple, eh? (As it turns out, the number is the product of the prime numbers 17,093; 22,441; and 27,943.) There are actually 8,363 five-digit prime numbers, so this problem might be attacked using a computer and a brute-force algorithm, but there's no easy way to figure it out in your head, that's for sure!

Nonce

Cryptography often gains strength by adding randomness to the encryption process. One method by which this is accomplished is through the use of a nonce. A *nonce* (from the phrase “number used once”) is a random number that acts as a placeholder variable in mathematical functions. When the function is executed, the nonce is replaced with a random number generated at the moment of processing for one-time use. The nonce must be a unique number each time it is used. One of the more recognizable examples of a nonce is an initialization vector (IV), a random bit string that is the same length as the block size (the amount of data to be encrypted in each operation) and is XORed with the message. IVs are used to create unique ciphertexts every time the same message is encrypted using the same key.

Zero-Knowledge Proof

One of the benefits of cryptography is found in the mechanism to prove your knowledge of a fact to a third party without revealing the fact itself to that third party. This is often done with passwords and other secret authenticators.

The classic example of a *zero-knowledge proof* involves two individuals: Peggy and Victor. Peggy knows the password to a secret door located inside a circular cave, as shown in Figure 6.2. Victor would like to buy the password from Peggy, but he wants Peggy to prove that she knows the password before paying her for it. Peggy doesn't want to tell Victor the password for fear that he won't pay her later. The zero-knowledge proof can solve their dilemma.

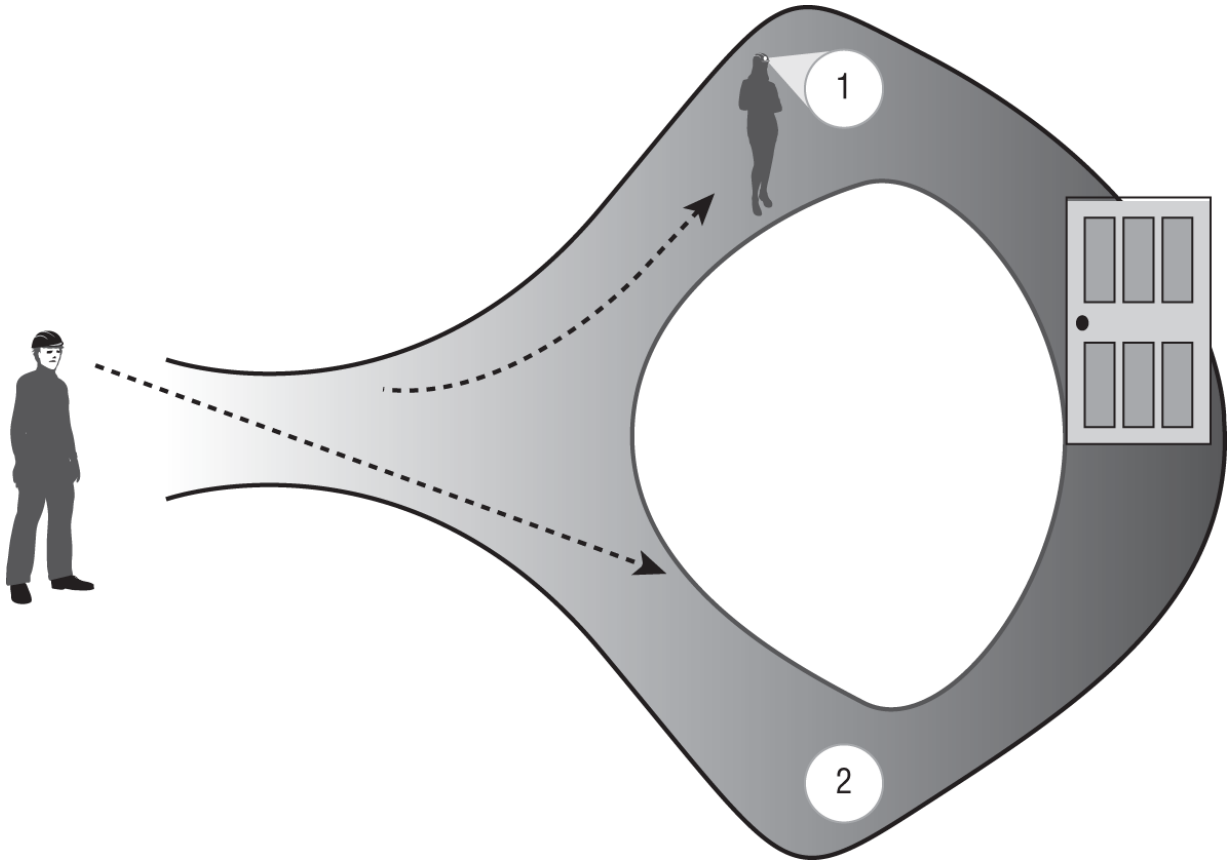


FIGURE 6.2 The magic door

Victor can stand at the entrance to the cave and watch Peggy depart down path 1. Peggy then reaches the door and opens it using the password. She then passes through the door and returns via path 2. Victor saw her leave

down path 1 and return via path 2, proving that she must know the correct password to open the door.

Zero-knowledge proofs appear in cryptography in cases where one individual wants to demonstrate knowledge of a fact (such as a password or key) without actually disclosing that fact to the other individual. This may be done through complex mathematical operations, such as discrete logarithms and graph theory.

Split Knowledge

When the information or privilege required to perform an operation is divided among multiple users, no single person has sufficient privileges to compromise the security of an environment. This separation of duties and two-person control contained in a single solution is called *split knowledge*. The best example of split knowledge is seen in the concept of *key escrow*. In a key escrow arrangement, a cryptographic key is stored with a third party for safekeeping. When certain circumstances are met, the third party may use the escrowed key to either restore an authorized user's access or decrypt the material themselves. This third party is known as the recovery agent. In arrangements that use only a single-key escrow recovery agent, there exists an opportunity for fraud and abuse of this privilege, as the single recovery agent could unilaterally decide to decrypt the information. The *M of N Control* concept requires that a minimum number of agents (`M`) out of the total number of agents (`N`) work together to perform high-security tasks. So, implementing three of eight controls would require three people out of the eight with the assigned work task of key escrow recovery agent to work together to pull a single key out of the key escrow database (thereby also illustrating that `M` is always less than or equal to `N`).

Work Function

You can measure the strength of a cryptography system by measuring the effort in terms of cost and/or time using a *work function* or work factor. Usually the time and effort required to perform a complete brute-force attack against an encryption system is what the work function represents. The security and protection offered by a cryptosystem is directly proportional to the value of the work function/factor. The size of the work function should be matched against the relative value of the protected asset.

The work function need be only slightly greater than the time value of that asset. In other words, all security, including cryptography, should be cost-effective and cost-efficient. Spend no more effort to protect an asset than it warrants, but be sure to provide sufficient protection. Thus, if information loses its value over time, the work function needs to be only large enough to ensure protection until the value of the data is gone.

In addition to understanding the length of time that the data will have value, security professionals selecting cryptographic systems must understand how emerging technologies may impact cipher-cracking efforts. For example, researchers may discover a flaw in a cryptographic algorithm next year that renders information protected with that algorithm insecure. Similarly, technological advancements in cloud-based parallel computing and quantum computing may make brute-force efforts much more feasible down the road.

Ciphers

Cipher systems have long been used by individuals and governments interested in preserving the confidentiality of their communications. In the following sections, we'll cover the definition of a cipher and explore several common cipher types that form the basis of modern ciphers. It's important to remember that these concepts seem somewhat basic, but when used in combination, they can be formidable opponents and cause cryptanalysts many hours of frustration.

Codes vs. Ciphers

People often use the words *code* and *cipher* interchangeably, but technically, they aren't interchangeable. There are important distinctions between the two concepts. *Codes*, which are cryptographic systems of symbols that represent words or phrases, are sometimes secret, but they are not necessarily meant to provide confidentiality. A common example of a code is the “10 system” of communications used by law enforcement agencies. Under this system, the sentence “I received your communication and understand the contents” is represented by the code phrase “10-4.” Semaphores (visual signaling) and Morse code are also examples of codes. These codes are commonly known by the public and provide for ease of communication. Some codes are secret. They may convey confidential

information using a secret codebook where the meaning of the code is known only to the sender and recipient. For example, a spy might transmit the sentence “The eagle has landed” to report the arrival of an enemy aircraft.

Ciphers, on the other hand, are always meant to hide the true meaning of a message. They use a variety of techniques to alter and/or rearrange the characters or bits of a message to achieve confidentiality. Ciphers convert messages from plaintext to ciphertext on a bit basis (that is, a single digit of a binary code), character basis (that is, a single character of an ASCII or Unicode message or another encoding), or block basis (that is, a fixed-length segment of a message, usually expressed in number of bits). The following sections cover several common ciphers in use today.



An easy way to keep the difference between codes and ciphers straight is to remember that codes generally work on words and phrases, whereas ciphers work on individual characters, bits, and blocks.

Transposition Ciphers

Transposition ciphers use an encryption algorithm to rearrange the letters of a plaintext message, forming the ciphertext message. The decryption algorithm simply reverses the encryption transformation to retrieve the original message.

In the challenge-response protocol example in Figure 6.1 earlier in this chapter, a simple transposition cipher was used to reverse the letters of the message so that *apple* became *elppa*. Transposition ciphers can be much more complicated than this. For example, you can use a keyword to perform a *columnar transposition*. In the following example, we're attempting to encrypt the message “The fighters will strike the enemy bases at noon” using the secret key *attacker*. Our first step is to take the letters of the keyword and number them in alphabetical order. The first appearance of the letter *A* receives the value 1; the second appearance is numbered 2. The

next letter in sequence, *C*, is numbered 3, and so on. This results in the following sequence:

```
A T T A C K E R
 1 7 8 2 3 5 4 6
```

Next, the letters of the message are written in order underneath the letters of the keyword:

```
A T T A C K E R
1 7 8 2 3 5 4 6
T H E F I G H T
E R S W I L L S
T R I K E T H E
E N E M Y B A S
E S A T N O O N
```

Finally, the sender enciphers the message by reading down each column; the order in which the columns are read corresponds to the numbers assigned in the first step. This produces the following ciphertext:

```
T E T E E F W K M T I I E Y N H L H A O G L T B O T S E S N H R
R N S E S I E A
```

On the other end, the recipient reconstructs the eight-column matrix using the ciphertext and the same keyword and then simply reads the plaintext message across the rows.

Substitution Ciphers

Substitution ciphers use the encryption algorithm to replace each character or bit of the plaintext message with a different character. One of the earliest known substitution ciphers was used by Julius Caesar to communicate with Cicero in Rome while he was conquering Europe. Caesar knew that there were several risks when sending messages—one of the messengers might be an enemy spy or might be ambushed while en route to the deployed forces. For that reason, Caesar developed a cryptographic system now known as the *Caesar cipher*. The system is extremely simple. To encrypt a message, you simply shift each letter of the alphabet three places to the right. For example, *A* would become *D*, and *B* would become *E*. If you reach the end of the alphabet during this process, you simply wrap around to the beginning so that *X* becomes *A*, *Y* becomes *B*, and *Z* becomes *C*. For

this reason, the Caesar cipher also became known as the ROT3 (or Rotate 3) cipher. The Caesar cipher is a substitution cipher that is mono-alphabetic.



Although the Caesar cipher uses a shift of 3, the more general shift cipher uses the same algorithm to shift any number of characters desired by the user. For example, the ROT12 cipher would turn an *A* into an *M*, a *B* into an *N*, and so on.

Here's an example of the Caesar cipher in action. The first line contains the original sentence, and the second line shows what the sentence looks like when it is encrypted using the Caesar cipher.

```
THE DIE HAS BEEN CAST
WKH GLH KDV EHHQ FDVW
```

To decrypt the message, you simply shift each letter three places to the left.



Although the Caesar cipher is easy to use, it's also easy to crack. It's vulnerable to a type of attack known as *frequency analysis*. The most common letters in the English language are *E, T, A, O, I, N, S, H, R, D, L, and U*. This is also known as "ETAOIN SHRDLU." An attacker seeking to break a Caesar-style cipher encoding message that was written in English merely needs to find the most common letters in the encrypted text and experiment with substitutions of these common letters to help determine the pattern.

You can express the ROT3 cipher in mathematical terms by converting each letter to its decimal equivalent (where *A* is 0 and *Z* is 25). You can then add three to each plaintext letter (P) to determine the ciphertext (C). You account for the wrap-around by using the modulo function discussed in the section "Cryptographic Mathematics," earlier in this chapter. The final encryption function for the Caesar cipher is then this:

```
C = (P + 3) mod 26
```

The corresponding decryption function is as follows:

```
P = (C - 3) mod 26
```

As with transposition ciphers, there are many substitution ciphers that are more sophisticated than the examples provided in this chapter. Polyalphabetic substitution ciphers use multiple alphabets (such as a keyword) in the same message to hinder decryption efforts. One of the most notable examples of a polyalphabetic substitution cipher system is the Vigenère cipher. The Vigenère cipher uses a single encryption/decryption chart, as shown here:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Notice that the chart is simply the alphabet written repeatedly (26 times) under the master heading, shifting by one letter each time. You need a key to use the Vigenère system. For example, the plaintext could be

LAUNCHNOW and the key could be *MILES*. Then, you would perform the following encryption process:

1. Write out the plaintext.
2. Underneath, write out the encryption key, repeating the key as many times as needed to establish a line of text that is the same length as the plaintext.
3. Convert each letter position from plaintext to ciphertext.
    a. Locate the column headed by the first plaintext character (*L*).
    b. Next, locate the row headed by the first character of the key (*M*).
    c. Finally, locate where these two items intersect, and write down the letter that appears there (*X*). This is the ciphertext for that letter position.
4. Repeat steps 1 through 3 for each letter in the plaintext version. The results are shown in Table 6.5.

TABLE 6.5 Using the Vigenère system

Stage of the process	Letters
Plaintext	L A U N C H N O W
Key	M I L E S M I L E
Ciphertext	X I F R U T V Z A

Although polyalphabetic substitution protects against direct frequency analysis, it is vulnerable to a second-order form of frequency analysis called *period analysis*, which is an examination of frequency based on the repeated use of the key.

One-Time Pads

A *one-time pad* is an extremely powerful type of substitution cipher. One-time pads use a different substitution alphabet for each letter of the plaintext message. They can be represented by the following encryption function, where *K* is the encryption key used to encrypt the plaintext letter *P* into the ciphertext letter *C*:

```
C = P ⊕ K
P = C ⊕ K
```

Usually, one-time pads are written as a very long series of numbers to be plugged into the function.



One-time pads are also known as *Vernam ciphers*, after the name of their inventor, Gilbert Sandford Vernam of AT&T Bell Labs.

The great advantage of one-time pads is that, when used properly, they are an unbreakable encryption scheme. There is no repeating pattern of alphabetic substitution, rendering cryptanalytic efforts useless. However, several requirements must be met to ensure the integrity of the algorithm:

- The one-time pad encryption key must be randomly generated. Using a phrase or a passage from a book would introduce the possibility that cryptanalysts could break the code.
- The one-time pad must be physically protected against disclosure. If the enemy has a copy of the pad, they can easily decrypt the enciphered messages.

  You may be thinking at this point that the Caesar cipher, Vigenère cipher, and one-time pad sound very similar. They are! The difference is the key length. The Caesar shift cipher uses a key of length one, the Vigenère cipher uses a longer key (usually a word or sentence), and the one-time pad uses a key that is as long as the message itself.

- Each one-time pad must be used only once. If pads are reused, cryptanalysts can compare similarities in multiple messages encrypted with the same pad and possibly determine the key plaintext values

used. In fact, a common practice when using paper pads is to destroy the page of keying material after it is used to prevent reuse.

- The key must be at least as long as the message to be encrypted. This is because each character of the key is used to encode only one character of the message.



These one-time pad security requirements are essential knowledge for any network security professional. All too often, people attempt to implement a one-time pad cryptosystem but fail to meet one or more of these fundamental requirements. Read on for an example of how an entire Soviet code system was broken because of carelessness in this area.

If any one of these requirements is not met, the impenetrable nature of the one-time pad instantly breaks down. In fact, one of the major intelligence successes of the United States resulted when cryptanalysts broke a top-secret Soviet cryptosystem that relied on the use of one-time pads. In this project, code-named VENONA, a pattern in the way the Soviets generated the key values used in their pads was discovered. The existence of this pattern violated the first requirement of a one-time pad cryptosystem: the keys must be randomly generated without the use of any recurring pattern. The entire VENONA project was recently declassified and is publicly available on the National Security Agency website.

One-time pads have been used throughout history to protect extremely sensitive communications. The major obstacle to their widespread use is the difficulty of generating, distributing, and safeguarding the lengthy keys required. One-time pads can realistically be used only for short messages, because of key lengths.



If you're interested in learning more about one-time pads, there is a great description with photos and examples at www.cryptomuseum.com/crypto/otp/index.htm.

Running Key Ciphers

Many cryptographic vulnerabilities surround the limited length of the cryptographic key. As you learned in the previous section, one-time pads avoid these vulnerabilities by using a key that is at least as long as the message. However, one-time pads are awkward to implement because they require the physical exchange of pads.

One common solution to this dilemma is the use of a *running key cipher* (also known as a *book cipher*). In this cipher, the encryption key is as long as the message itself and is often chosen from a common book, newspaper, or magazine. For example, the sender and recipient might agree in advance to use the text of a chapter from *Moby Dick*, beginning with the third paragraph, as the key. They would both simply use as many consecutive characters as necessary to perform the encryption and decryption operations.

Let's look at an example. Suppose you wanted to encrypt the message “Richard will deliver the secret package to Matthew at the bus station tomorrow” using the key just described. This message is 66 characters in length, so you'd use the first 66 characters of the running key: “With much interest I sat watching him. Savage though he was, and hideously marred.” Any algorithm could then be used to encrypt the plaintext message using this key. Let's look at the example of modulo 26 addition, which converts each letter to a decimal equivalent, adds the plaintext to the key, and then performs a modulo 26 operation to yield the ciphertext. If you assign the letter *A* the value 0 and the letter *Z* the value 25, Table 6.6 shows the encryption operation for the first two words of the ciphertext.

```
C = (P + K) mod 26
P = (C - K) mod 26
```

TABLE 6.6 The encryption operation

Operation component	**x**	**x**	**x**	**x**	**x**	**x**	**x**	**x**	**x**	**x**	**x**
Plaintext	R	I	C	H	A	R	D	W	I	L	L
Key	W	I	T	H	M	U	C	H	I	N	T
Numeric plaintext	17	8	2	7	0	17	3	22	8	11	11
Numeric key	22	8	19	7	12	20	2	7	8	13	19
Numeric ciphertext	13	16	21	14	12	11	5	3	16	24	4
Ciphertext	N	Q	V	O	M	L	F	D	Q	Y	E

When the recipient receives the ciphertext, they use the same key and then subtract the key from the ciphertext, perform a modulo 26 operation, and then convert the resulting plaintext back to alphabetic characters.

Block Ciphers

Block ciphers operate on "chunks," or blocks, of a message and apply the encryption algorithm to an entire message block at the same time. The transposition ciphers are examples of block ciphers. The simple algorithm used in the challenge-response algorithm takes an entire word and reverses its letters. The more complicated columnar transposition cipher works on an entire message (or a piece of a message) and encrypts it using the transposition algorithm and a secret keyword. Most modern encryption algorithms implement some type of block cipher.

Stream Ciphers

Stream ciphers operate on one character or bit of a message (or data stream) at a time. The Caesar cipher is an example of a stream cipher. The one-time pad is also a stream cipher because the algorithm operates on each letter of the plaintext message independently. Stream ciphers can also function as a type of block cipher. In such operations there is a buffer that fills up to real-time data that is then encrypted as a block and transmitted to the recipient.

Confusion and Diffusion

Cryptographic algorithms rely on two basic operations to obscure plaintext messages—confusion and diffusion. *Confusion* occurs when the

relationship between the plaintext and the key is so complicated that an attacker can't merely continue altering the plaintext and analyzing the resulting ciphertext to determine the key. *Diffusion* occurs when a change in the plaintext results in multiple changes spread throughout the ciphertext. Consider, for example, a cryptographic algorithm that first performs a complex substitution and then uses transposition to rearrange the characters of the substituted ciphertext. In this example, the substitution introduces confusion, and the transposition introduces diffusion.

Modern Cryptography

Modern cryptosystems use computationally complex algorithms and long cryptographic keys to meet the cryptographic goals of confidentiality, integrity, authentication, and nonrepudiation. This section covers the roles cryptographic keys play in the world of data security and examine three types of algorithms commonly used today: symmetric encryption algorithms, asymmetric encryption algorithms, and hashing algorithms.

Cryptographic Keys

In the early days of cryptography, one of the predominant principles was “security through obscurity.” Some cryptographers thought the best way to keep an encryption algorithm secure was to hide the details of the algorithm from outsiders. Old cryptosystems required communicating parties to keep the algorithm used to encrypt and decrypt messages secret from third parties. Any disclosure of the algorithm could lead to compromise of the entire system by an adversary.

Modern cryptosystems do not rely on the secrecy of their algorithms. In fact, the algorithms for most cryptographic systems are widely available for public review on the Internet. Opening algorithms to public scrutiny actually improves their security. Widespread analysis of algorithms by the computer security community allows practitioners to discover and correct potential security vulnerabilities and ensure that the algorithms they use to protect their communications are as secure as possible.

Instead of relying on secret algorithms, modern cryptosystems rely on the secrecy of one or more cryptographic keys used to personalize the

algorithm for specific users or groups of users. Recall from the discussion of transposition ciphers that a keyword is used with the columnar transposition to guide the encryption and decryption efforts. The algorithm used to perform columnar transposition is well known—you just read the details of it in this book! However, columnar transposition can be used to securely communicate between parties as long as a keyword is chosen that would not be guessed by an outsider. As long as the security of this keyword is maintained, it doesn't matter that third parties know the details of the algorithm.



Although the public nature of the algorithm does not compromise the security of columnar transposition, the method does possess several inherent weaknesses that make it vulnerable to cryptanalysis. It is therefore an inadequate technology for use in modern secure communication.

In the discussion of one-time pads earlier in this chapter, you learned that the main strength of the one-time pad algorithm is derived from the fact that it uses an extremely long random key. In fact, for that algorithm, the key is at least as long as the message itself. Most modern cryptosystems do not use keys quite that long, but the length of the key is still an extremely important factor in determining the strength of the cryptosystem and the likelihood that the encryption will not be compromised through cryptanalytic techniques. Longer keys provide higher levels of security by increasing the size of the key space, rendering brute-force attacks more difficult.

The rapid increase in computing power allows you to use increasingly long keys in your cryptographic efforts. However, this same computing power is also in the hands of cryptanalysts attempting to defeat the algorithms you use. Therefore, it's essential that you outpace adversaries by using sufficiently long keys that will defeat contemporary cryptanalysis efforts. Additionally, if you want to improve the chance that your data will remain safe from cryptanalysis some time into the future, you must strive to use keys that will outpace the projected increase in cryptanalytic capability

during the entire time period the data must be kept safe. For example, the advent of quantum computing may transform cryptography, rendering current cryptosystems insecure, as discussed earlier in this chapter.

When the Data Encryption Standard (DES) was created in 1975, a 56-bit key was considered sufficient to maintain the security of any data. However, there is now widespread agreement that the 56-bit DES algorithm is no longer secure because of advances in cryptanalysis techniques and supercomputing power. Modern cryptographic systems use at least a 128-bit key to protect data against prying eyes. Remember, the length of the key directly relates to the work function of the cryptosystem: the longer the key, the harder it is to break the cryptosystem.

In addition to choosing keys that are long and will remain secure for the expected length of time that the information will remain confidential, you should also implement some other key management practices:

- Always store secret keys securely and, if you must transmit them over a network, do so in a manner that protects them from unauthorized disclosure.
- Select keys using an approach that has as much randomness as possible, taking advantage of the entire key space.
- Destroy keys securely when they are no longer needed.

Symmetric Key Algorithms

Symmetric key algorithms rely on a “shared secret” encryption key that is distributed to all members who participate in the communications. This key is used by all parties to both encrypt and decrypt messages, so the sender and the receiver both possess a copy of the shared key. The sender encrypts with the shared secret key, and the receiver decrypts with it. When large-sized keys are used, symmetric encryption is very difficult to break. It is primarily employed to perform bulk encryption and provides only for the security service of confidentiality. Symmetric key cryptography can also be called *secret key cryptography* and *private key cryptography*. Figure 6.3 illustrates the symmetric key encryption and decryption processes (with “C” representing a ciphertext message and “P” representing a plaintext message).

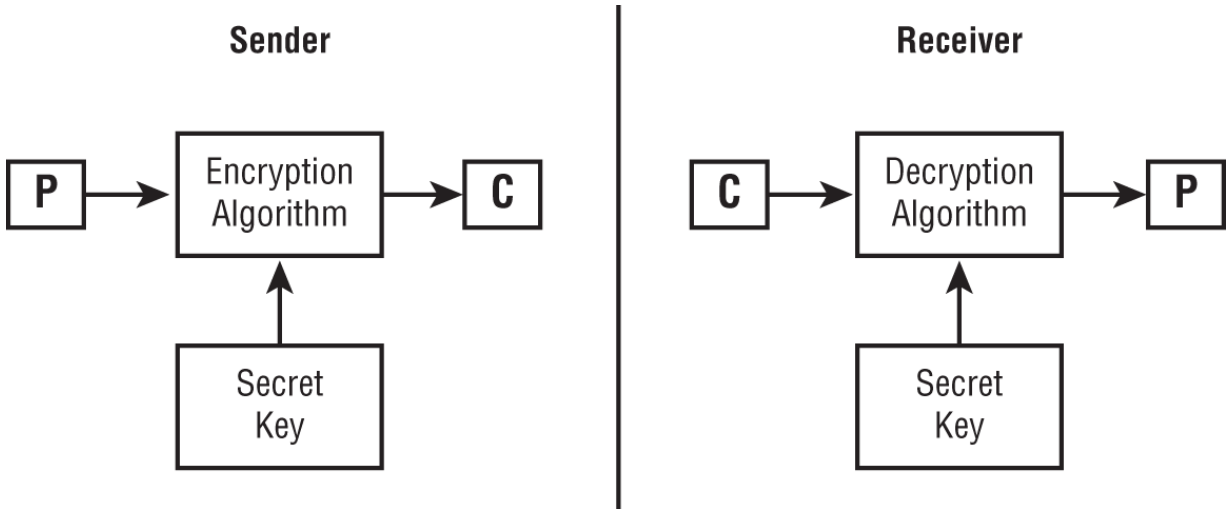


FIGURE 6.3 Symmetric key cryptography



If you find yourself getting confused about the difference between symmetric and asymmetric cryptography, it may be helpful to remember that "same" is a synonym for "symmetric" and "different" is a synonym for asymmetric. In symmetric cryptography, the message is encrypted and decrypted with the same key, whereas in asymmetric cryptography, encryption and decryption use different (but related) keys.

In some cases, symmetric cryptography may be used with temporary keys that exist only for a single session. In those cases, the secret key is known as an *ephemeral key*. The most common example of this is the Transport Layer Security (TLS) protocol, which uses asymmetric cryptography to set up an encrypted channel and then switches to symmetric cryptography using an ephemeral key. You'll learn more about this topic in Chapter 7.



The use of the term *private key* can be tricky because it is part of three different terms that have two different meanings. The term *private key* by itself always means the private key from the key pair of public key cryptography (aka asymmetric). However, both *private key cryptography* and *shared private key* refer to symmetric cryptography. The meaning of the word *private* is stretched to refer to two people sharing a secret that they keep confidential. (The true meaning of *private* is that only a single person has a secret that's kept confidential.) Be sure to keep these confusing terms straight in your studies.

Symmetric key cryptography has several weaknesses:

Key distribution is a major problem. Parties must have a secure method of exchanging the secret key before establishing communications with a symmetric key protocol. If a secure electronic channel is not available, an offline key distribution method must often be used (that is, out-of-band exchange).

Symmetric key cryptography does not implement nonrepudiation. Because any communicating party can encrypt and decrypt messages with the shared secret key, there is no way to prove where a given message originated.

The algorithm is not scalable. It is extremely difficult for large groups to communicate using symmetric key cryptography. Secure private communication between individuals in the group could be achieved only if each possible combination of users shared a secret key.

Keys must be regenerated often. Each time a participant leaves the group, all secret keys known by that participant must be discarded. In automated encryption systems, keys may be regenerated based on the length of time that has passed, the amount of data exchanged, or the fact that a session goes idle or is terminated.

The major strength of symmetric key cryptography is the great speed at which it can operate. Symmetric key encryption is very fast, often 1,000 to 10,000 times faster than asymmetric algorithms. By nature of the

mathematics involved, symmetric key cryptography also naturally lends itself to hardware implementations, creating the opportunity for even higher-speed operations and bulk encryption tasks.

The section “Symmetric Cryptography,” later in this chapter, provides a detailed look at the major secret key algorithms in use today.

Asymmetric Key Algorithms

Asymmetric key algorithms provide a solution to the weaknesses of symmetric key encryption. *Public key algorithms* are the most common example of asymmetric algorithms. In these systems, each user has two keys: a public key, which is shared with all users, and a private key, which is kept secret and known only to the user. But here's a twist: opposite and related keys must be used in tandem to encrypt and decrypt. In other words, if the public key encrypts a message, then only the corresponding private key from the key pair can decrypt it, and vice versa.

Figure 6.4 shows the algorithm used to encrypt and decrypt messages in a public key cryptosystem (with “C” representing a ciphertext message and “P” representing a plaintext message). Consider this example. If Alice wants to send a message to Bob using public key cryptography, she creates the message and then encrypts it using Bob's public key. The only possible way to decrypt this ciphertext is to use Bob's private key, and the only user with access to that key is Bob. Therefore, Alice can't even decrypt the message herself after she encrypts it. If Bob wants to send a reply to Alice, he simply encrypts the message using Alice's public key, and then Alice reads the message by decrypting it with her own private key.

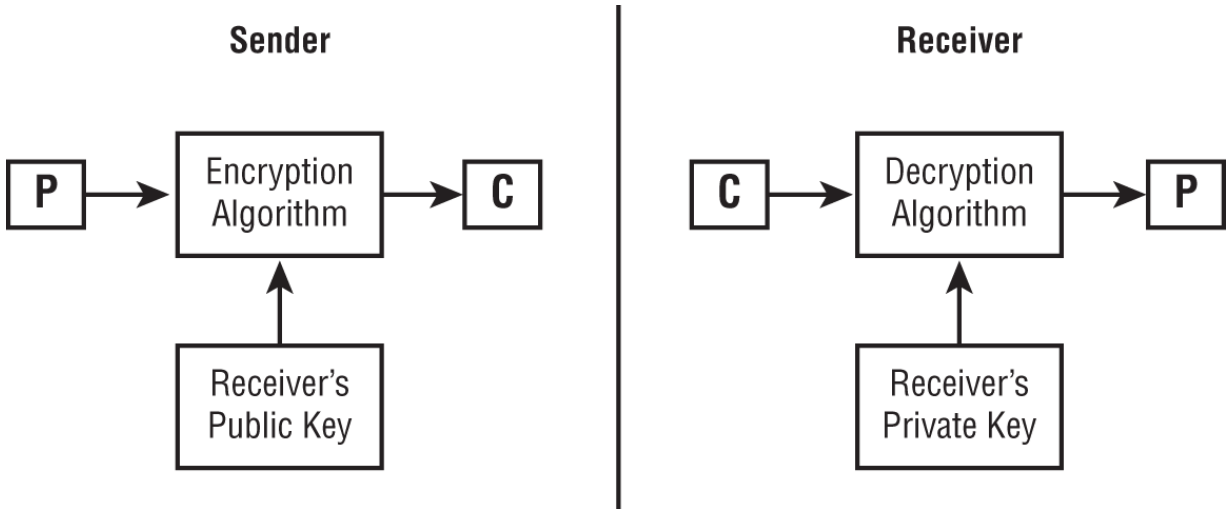


FIGURE 6.4 Asymmetric key cryptography



Real World Scenario

Key Requirements

In a class one of the authors of this book taught recently, a student wanted to see an illustration of the scalability issue associated with symmetric encryption algorithms. The fact that symmetric cryptosystems require each pair of potential communicators to have a shared private key makes the algorithm nonscalable. The total number of keys required to completely connect *n* parties using symmetric cryptography is given by the following formula:

$$\text{Number of Keys} = \frac{n(n-1)}{2}$$

Now, this might not sound so bad (and it's not for small systems), but consider the figures shown in Table 6.7. Obviously, the larger the population, the less likely a symmetric cryptosystem will be suitable to meet its needs.

TABLE 6.7 Symmetric and asymmetric key comparison

Number of participants	Number of symmetric keys required	Number of asymmetric keys required
2	1	4
3	3	6
4	6	8
5	10	10
10	45	20
100	4,950	200
1,000	499,500	2,000
10,000	49,995,000	20,000

Asymmetric key algorithms also provide support for digital signature technology. Basically, if Bob wants to assure other users that a message with his name on it was actually sent by him, he first creates a message digest by using a hashing algorithm (you'll find more on hashing algorithms in the next section). Bob then encrypts that digest using his private key. Any user who wants to verify the signature simply decrypts the message digest using Bob's public key and then verifies that the decrypted message digest is accurate. Chapter 7 explains this process in greater detail.

The following is a list of the major strengths of asymmetric key cryptography:

The addition of new users requires the generation of only one public-private key pair. This same key pair is used to communicate with all users of the asymmetric cryptosystem. This makes the algorithm extremely scalable.

Users can be removed far more easily from asymmetric systems. Asymmetric cryptosystems provide a key revocation mechanism that allows a key to be canceled, effectively removing a user from the system.

Key regeneration is required only when a user's private key is compromised. If a user leaves the community, the system administrator simply needs to invalidate that user's keys. No other keys are compromised and therefore key regeneration is not required for any other user.

Asymmetric key encryption can provide confidentiality, integrity, authentication, and nonrepudiation. If a user does not share their private key with other individuals, a message signed by that user can be shown to be accurate and from a specific source and cannot be later repudiated. Asymmetric cryptography may be used to create digital signatures that provide nonrepudiation, as discussed in Chapter 7.

Key distribution is a simple process. Users who want to participate in the system simply make their public key available to anyone with whom they want to communicate. There is no method by which the private key can be derived from the public key.

No preexisting communication link needs to exist. Two individuals can begin communicating securely from the moment they start communicating. Asymmetric cryptography does not require a preexisting relationship to provide a secure mechanism for data exchange.

The major weakness of public key cryptography is its slow speed of operation. For this reason, many applications that require the secure transmission of large amounts of data use public key cryptography to establish a connection and then exchange a symmetric secret key. The remainder of the session then uses symmetric cryptography. This approach of combining symmetric and asymmetric cryptography is known as *hybrid cryptography*.

Table 6.8 compares the symmetric and asymmetric cryptography systems. Close examination of this table reveals that a weakness in one system is matched by a strength in the other.

TABLE 6.8 Comparison of symmetric and asymmetric cryptography systems

Symmetric	Asymmetric
Single shared key	Key pair sets
Out-of-band exchange	In-band exchange
Not scalable	Scalable
Fast	Slow
Bulk encryption	Small blocks of data, digital signatures, digital certificates
Confidentiality	Confidentiality, integrity (via hashing), authenticity, nonrepudiation (via digital signatures)



Chapter 7 provides technical details on modern public key encryption algorithms and some of their applications.

Hashing Algorithms

In the previous section, you learned that public key cryptosystems can provide digital signature capability when used in conjunction with a message digest. Message digests (also known as hash values or fingerprints) are fixed-length summaries of a message's content (not unlike a file checksum) produced by a hashing algorithm. It's extremely difficult, if not impossible, to derive a message from an ideal hash function, and it's very unlikely that two messages will produce the same hash value. Cases where a hash function produces the same message digest value for two different messages are known as *collisions*, and the existence of collisions typically leads to the deprecation of a hashing algorithm.

Chapter 7 provides details on contemporary hashing algorithms and explains how they are used to provide digital signature capability, which helps meet the cryptographic goals of integrity and nonrepudiation.

Symmetric Cryptography

You've learned the basic concepts underlying symmetric key cryptography, asymmetric key cryptography, and hashing functions. In this section, we'll take an in-depth look at several common symmetric cryptosystems.

Block Cipher Modes of Operation

The symmetric key cryptography block cipher modes of operation describe the different ways that cryptographic algorithms may transform data to achieve sufficient complexity that offers protection against attack. The major modes of operation are:

- Electronic Codebook (ECB) mode
- Cipher Block Chaining (CBC) mode
- Cipher Feedback (CFB) mode
- Output Feedback (OFB) mode
- Counter (CTR) mode
- Galois/Counter mode (GCM)
- Counter with Cipher Block Chaining Message Authentication Code (CCM) mode

Electronic Codebook Mode

Electronic Codebook (ECB) mode is the simplest mode to understand and the least secure. Each time the algorithm processes a fixed block of data, it simply encrypts the block using the chosen secret key. This means that if the algorithm encounters the same block multiple times, it will produce the same encrypted block. If an enemy were eavesdropping on the communications, they could simply build a “codebook” of all the possible encrypted values. After a sufficient number of blocks were gathered, cryptanalytic techniques could be used to decipher some of the blocks and break the encryption scheme.

This vulnerability makes it impractical to use ECB mode on any but the shortest transmissions. In everyday use, ECB is used only for exchanging

small amounts of data, such as keys and parameters used to initiate other cryptographic modes as well as the cells in a database.

Cipher Block Chaining Mode

In *Cipher Block Chaining (CBC)* mode, each block of unencrypted text is XORed with the block of ciphertext immediately preceding it before it is encrypted. The decryption process simply decrypts the ciphertext and reverses the XOR operation. CBC implements an initialization vector (IV) and XORs it with the first block of the message, producing a unique output every time the operation is performed. The IV must be sent to the recipient, perhaps by tacking the IV onto the front of the completed ciphertext in plain form or by protecting it with ECB mode encryption using the same key used for the message. One important consideration when using CBC mode is that errors propagate—if one block is corrupted during transmission, it becomes impossible to decrypt that block and the next block as well.

Cipher Feedback Mode

Cipher Feedback (CFB) mode is the streaming cipher version of CBC. In other words, CFB operates against data produced in real time. However, instead of breaking a message into blocks, it uses memory buffers of the same block size. As the buffer becomes full, it is encrypted and then sent to the recipients. Then the system waits for the next buffer to be filled as the new data is generated before it is in turn encrypted and then transmitted. Other than the change from preexisting data to real-time data, CFB operates in the same fashion as CBC. It uses an IV, and it uses chaining.

Output Feedback Mode

In *Output Feedback (OFB) mode,* ciphers operate in almost the same fashion as they do in CFB mode. However, instead of XORing an encrypted version of the previous block of ciphertext, OFB XORs the plaintext with a seed value. For the first encrypted block, an IV is used to create the seed value. Future seed values are derived by running the algorithm on the previous seed value. The major advantages of OFB mode are that there is no chaining function and transmission errors do not propagate to affect the decryption of future blocks.

Counter Mode

Counter (CTR) mode uses a stream cipher similar to that used in CFB and OFB modes. However, instead of creating the seed value for each encryption/decryption operation from the results of the previous seed values, it uses a simple counter that increments for each operation. As with OFB mode, errors do not propagate in CTR mode.



CTR mode allows you to break an encryption or decryption operation into multiple independent steps. This makes CTR mode well suited for use in parallel computing.

Galois/Counter Mode

Galois/Counter mode (GCM) takes the standard CTR mode of encryption and adds data authenticity controls to the mix, providing the recipient assurances of the integrity of the data received. This is done by adding *authentication tags* to the encryption process.

Counter with Cipher Block Chaining Message Authentication Code Mode

Similar to GCM, *Counter with Cipher Block Chaining Message Authentication Code mode (CCM)* combines a confidentiality mode with a data authenticity process. In this case, CCM ciphers combine the Counter (CTR) mode for confidentiality with the Cipher Block Chaining Message Authentication Code (CBC-MAC) algorithm for data authenticity.

Currently, CCM is used only with block ciphers that have a 128-bit block length (such as the AES algorithm) and require the use of a nonce that must be changed for each transmission.



GCM and CCM modes both include data authenticity in addition to confidentiality. They are, therefore, known as authenticated modes of encryption. ECB, CBC, CFB, OFB, and CTR modes only provide confidentiality and are, therefore, known as unauthenticated modes.

Data Encryption Standard

The U.S. government published the Data Encryption Standard (DES) in 1977 as a proposed standard cryptosystem for all government communications. Because of flaws in the algorithm, cryptographers and the federal government no longer consider DES secure. It is widely believed that intelligence agencies routinely decrypt DES-encrypted information. DES was superseded by the Advanced Encryption Standard in December 2001. It is still important to understand DES because it is the building block of Triple DES (3DES), a stronger encryption algorithm discussed in the next section.

DES is a 64-bit block cipher that has five modes of operation: Electronic Codebook (ECB) mode, Cipher Block Chaining (CBC) mode, Cipher Feedback (CFB) mode, Output Feedback (OFB) mode, and Counter (CTR) mode. All of the DES modes operate on 64 bits of plaintext at a time to generate 64-bit blocks of ciphertext. The key used by DES is 56 bits long.

DES uses a long series of exclusive OR (XOR) operations to generate the ciphertext. This process is repeated 16 times for each encryption/decryption operation. Each repetition is commonly referred to as a *round* of encryption, explaining the statement that DES performs 16 rounds of encryption. Each round generates a new key that is then used as the input to subsequent rounds.



As mentioned, DES uses a 56-bit key to drive the encryption and decryption process. However, you may read in some literature that DES uses a 64-bit key. This is not an inconsistency—there's a perfectly logical explanation. The DES specification calls for a 64-bit key. However, of those 64 bits, only 56 actually contain keying information. The remaining 8 bits are supposed to contain parity information to ensure that the other 56 bits are accurate. In practice, however, those parity bits are rarely used. You should commit the 56-bit figure to memory.

Triple DES

As mentioned in previous sections, the Data Encryption Standard's (DES) 56-bit key is no longer considered adequate in the face of modern cryptanalytic techniques and supercomputing power. However, an adapted version of DES, Triple DES (3DES), uses the same algorithm to produce encryption that is stronger but that is no longer considered adequate to meet modern requirements. For this reason, 3DES encryption should be avoided, although it may still be supported by some products.



As of January 1, 2024, the U.S. federal government formally disallowed the use of 3DES for government data. The algorithm is still used in private industry.

There are several different variants of 3DES that each use different numbers of independent keys. The first two, DES-EDE3 and DES-EEE3, use three independent keys: K_1, K_2, and K_3. The difference between the two are the operations used, which are represented by the letter *E* for encryption and *D* for decryption. DES-EDE3 encrypts the data with K1, decrypts the resulting ciphertext with K2, and then encrypts that text with K3. DES-EDE3 can be expressed using the following notation, where `E(K,P)` represents the

encryption of plaintext *P* with key *K*, and `D(K,C)` represents the decryption of ciphertext *C* with key *K*:

```
E(K1,D(K2,E(K3,P)))
```

DES-EEE3, on the other hand, encrypts the data with all three keys in sequential order, and may be represented as follows:

```
E(K1,E(K2,E(K3,P)))
```



If you find yourself wondering why there is a decryption operation in the middle of EDE3 mode, that's an arcane artifact of the process used to create the algorithm and provide backward compatibility with DES. Encryption and decryption are reversible operations, so even though the decryption function is used, it can still be thought of as a round of encryption.

Mathematically, DES-EEE3 and DES-EDE3 should have an effective key length of 168 bits. However, known attacks against this algorithm reduce the effective strength to 112 bits.



This discussion raises an obvious question—what happened to Double DES (2DES)? You'll read in Chapter 7 that Double DES was tried but quickly abandoned when it was proven that an attack known as the meet-in-the-middle attack rendered it no more secure than standard DES.

International Data Encryption Algorithm

The International Data Encryption Algorithm (IDEA) block cipher was developed in response to complaints about the insufficient key length of the DES algorithm. Like DES, IDEA operates on 64-bit blocks of plaintext/ciphertext. However, it begins its operation with a 128-bit key. This key is broken up in a series of operations into 52 16-bit subkeys. The

subkeys then act on the input text using a combination of XOR and modulus operations to produce the encrypted/decrypted version of the input message. IDEA is capable of operating in the same five modes used by DES: ECB, CBC, CFB, OFB, and CTR.



All of this material on key length block size and the number of rounds of encryption may seem dreadfully boring; however, it's important material, so be sure to brush up on it while preparing for the exam.

The IDEA algorithm was patented by its Swiss developers. However, the patent expired in 2012, and it is now available for unrestricted use. One popular implementation of IDEA is found in Phil Zimmermann's popular Pretty Good Privacy (PGP) secure email package. Chapter 7 covers PGP in further detail.

Blowfish

Bruce Schneier's Blowfish block cipher is another alternative to DES and IDEA. Like its predecessors, Blowfish operates on 64-bit blocks of text. However, it extends IDEA's key strength even further by allowing the use of variable-length keys ranging from a relatively insecure 32 bits to an extremely strong 448 bits. Obviously, the longer keys will result in corresponding increases in encryption/decryption time. However, time trials have established Blowfish as a much faster algorithm than both IDEA and DES. Also, Schneier released Blowfish for public use with no license required. Blowfish encryption is built into a number of commercial software products and operating systems. A number of Blowfish libraries are also available for software developers, and it is often used for Secure Shell (SSH) connections.

SKIPJACK

Developed by the U.S. National Security Agency (NSA), the algorithm code named SKIPJACK was approved for use by the U.S. government in Federal Information Processing Standard (FIPS) 185, the Escrowed

Encryption Standard (EES) in 1994. SKIPJACK uses an 80-bit key and operates on 64-bit blocks of text. SKIPJACK was quickly embraced by the U.S. government and provided the cryptographic routines supporting the Clipper and Capstone encryption chips.

However, SKIPJACK had an added twist—it supports the escrow of encryption keys. At the time, two government agencies, the National Institute of Standards and Technology (NIST) and the U.S. Department of the Treasury, each held a portion of the information required to reconstruct a SKIPJACK key. When law enforcement authorities obtained legal authorization, they would contact the two agencies to obtain the pieces of the key, enabling them to decrypt communications between the affected parties.

SKIPJACK and the Clipper chip were not embraced by the cryptographic community at large because of its mistrust of the escrow procedures in place within the U.S. government.

Rivest Ciphers

Ron Rivest, of Rivest-Shamir-Adleman (RSA) Data Security, created a series of symmetric ciphers over the years known as the Rivest Ciphers (RC) family of algorithms. Several of these, RC4, RC5, and RC6, have particular importance today.

Rivest Cipher 4 (RC4)

RC4 is a stream cipher developed by Rivest in 1987 and very widely used during the decades that followed. It uses a single round of encryption and allows the use of variable-length keys ranging from 40 bits to 2,048 bits. RC4's adoption was widespread because it was integrated into the Wired Equivalent Privacy (WEP), Wi-Fi Protected Access (WPA), Secure Sockets Layer (SSL), and Transport Layer Security (TLS) protocols.

A series of attacks against this algorithm render it insecure for use today. WEP, WPA, and SSL no longer meet modern security standards for both this and other reasons. TLS no longer allows the use of RC4 as a stream cipher.

Rivest Cipher 5 (RC5)

RC5 is a block cipher of variable block sizes (32, 64, or 128 bits) that uses key sizes between 0 (zero) length and 2,040 bits. It is important to note that RC5 is not simply the next version of RC4. In fact, it is completely unrelated to the RC4 cipher. Instead, RC5 is an improvement on an older algorithm called RC2 that is no longer considered secure.

RC5 is the subject of brute-force cracking attempts. A large-scale effort leveraging massive community computing resources cracked a message encrypted using RC5 with a 64-bit key, but this effort took more than four years to crack a single message.

Rivest Cipher 6 (RC6)

RC6 is a block cipher that was developed as the next version of RC5. It uses a 128-bit block size and allows the use of 128-, 192-, or 256-bit symmetric keys. This algorithm was one of the candidates for selection as the Advanced Encryption Standard (AES) discussed in the next section, but it was not selected and is not widely used today.

Advanced Encryption Standard

In October 2000, NIST announced that the Rijndael (pronounced “rhine-doll”) block cipher had been chosen as the replacement for DES. In November 2001, NIST released FIPS 197, which mandated the use of AES/Rijndael for the encryption of all sensitive but unclassified data by the U.S. government.

The Advanced Encryption Standard (AES) cipher allows the use of three key strengths: 128 bits, 192 bits, and 256 bits. AES only allows the processing of 128-bit blocks, but Rijndael exceeded this specification, allowing cryptographers to use a block size equal to the key length. The number of encryption rounds depends on the key length chosen:

- 128-bit keys require 10 rounds of encryption.
- 192-bit keys require 12 rounds of encryption.
- 256-bit keys require 14 rounds of encryption.

CAST

The CAST algorithms, named after their creators, Carlisle Adams and Stafford Tavares, are another family of symmetric key block ciphers that are integrated into some security solutions. The CAST algorithms use a Feistel network and come in two forms:

- CAST-128 uses either 12 or 16 rounds of Feistel network encryption with a key size between 40 and 128 bits on 64-bit blocks of plaintext.
- CAST-256 uses 48 rounds of encryption with a key size of 128, 160, 192, 224, or 256 bits on 128-bit blocks of plaintext.

The CAST-256 algorithm was a candidate for the Advanced Encryption Standard but was not selected for that purpose.

Comparison of Symmetric Encryption Algorithms

There are many symmetric encryption algorithms you need to be familiar with. Table 6.9 lists several common and well-known symmetric encryption algorithms along with their block size and key size.

TABLE 6.9 Symmetric encryption memorization chart

Algorithm Name	Block size (Bits)	Key size (Bits)
Advanced Encryption Standard (AES)	128	128, 192, 256
Rijndael	Variable	128, 192, 256
Blowfish	64	32–448
Data Encryption Standard (DES)	64	56
International Data Encryption Algorithm (IDEA)	64	128
Rivest Cipher 4 (RC4)	N/A (Stream cipher)	40–2,048
Rivest Cipher 5 (RC5)	32, 64, 128	0–2,040
Rivest Cipher 6 (RC6)	128	128, 192, 256
SKIPJACK	64	80
Triple DES (3DES)	64	112 or 168
CAST-128	64	40–128
CAST-256	128	128, 160, 192, 224, 256

Symmetric Key Management

Because cryptographic keys contain information essential to the security of the cryptosystem, it is incumbent upon cryptosystem users and administrators to take extraordinary measures to protect the security of the keying material. These security measures are collectively known as *key management practices*. They include safeguards surrounding the creation, distribution, storage, destruction, recovery, and escrow of symmetric (secret) keys.

Creation and Distribution of Symmetric Keys

As previously mentioned, one of the major problems underlying symmetric encryption algorithms is the secure distribution of the secret keys required to operate the algorithms. The three main methods used to exchange secret

keys securely are offline distribution, public key encryption, and the Diffie–Hellman key exchange algorithm.

Offline Distribution The most technically simple (but physically inconvenient) method involves the physical exchange of key material. One party provides the other party with a sheet of paper or piece of storage media containing the secret key. In many hardware encryption devices, this key material comes in the form of an electronic device that resembles an actual key that is inserted into the encryption device. However, every offline key distribution method has its own inherent flaws. If keying material is sent through the mail, it might be intercepted. Telephones can be wiretapped. Papers containing keys might be inadvertently thrown in the trash or lost. The use of offline distribution is cumbersome for end users, particularly when they are located in geographically distant locations.

Public Key Encryption Many communicators want to obtain the speed benefits of secret key encryption without the hassles of key distribution. For this reason, many people use public key encryption to set up an initial communications link. Once the link is successfully established and the parties are satisfied as to each other's identity, they exchange a secret key over the secure public key link. They then switch communications from the public key algorithm to the secret key algorithm and enjoy the increased processing speed. In general, secret key encryption is thousands of times faster than public key encryption.

Diffie–Hellman In some cases, neither public key encryption nor offline distribution is sufficient. Two parties might need to communicate with each other, but they have no physical means to exchange key material, and there is no public key infrastructure in place to facilitate the exchange of a secret key. In situations like this, key exchange algorithms like the Diffie–Hellman algorithm prove to be extremely useful mechanisms. You'll find a complete discussion of Diffie–Hellman in Chapter 7.

Storage and Destruction of Symmetric Keys

Another major challenge with the use of symmetric key cryptography is that all of the keys used in the cryptosystem must be kept secure. This includes following best practices surrounding the storage of encryption keys:

- Never store an encryption key on the same system where encrypted data resides. This just makes it easier for the attacker!
- For sensitive keys, consider providing two different individuals with half of the key. They then must collaborate to re-create the entire key. This is known as the principle of *split knowledge* (discussed earlier in this chapter).

When a user with knowledge of a secret key leaves the organization or is no longer permitted access to material protected with that key, the keys must be changed, and all encrypted materials must be reencrypted with the new keys.

When choosing a key storage mechanism, you have three major options available to you:

- *Software-based storage mechanisms* store keys as digital objects on the system where they are used. For example, this might involve storing the key on the local file system. More advanced software-based mechanisms may use specialized applications to protect those keys, including the use of secondary encryption to prevent unauthorized access to the keys. Software-based approaches are generally simple to implement but introduce the risk of the software mechanism being compromised.
- *Hardware-based storage mechanisms* are dedicated hardware devices used to manage cryptographic keys. These may be personal devices, such as flash drives or smartcards that store a key used by an individual, or they may be enterprise devices, called *hardware security modules (HSMs)*, that manage keys for an organization. Hardware approaches are more complex and expensive to implement than software approaches, but they offer added security.
- *Cloud-based storage mechanisms* take the HSM approach and implement it in the data centers of cloud service providers. Major cloud service providers, including AWS and Microsoft, offer HSM solutions for secure key management.

Key Escrow and Recovery

Cryptography is a powerful tool. Like most tools, it can be used for a number of beneficent purposes, but it can also be used with malicious intent. To gain a handle on the explosive growth of cryptographic technologies, governments around the world have floated ideas to implement key escrow systems. These systems allow the government, under limited circumstances such as a court order, to obtain the cryptographic key used for a particular communication from a central storage facility.

Two major approaches to key escrow have been proposed over the past decade:

Fair Cryptosystems In this escrow approach, the secret keys used in a communication are divided into two or more pieces, each of which is given to an independent third party. Each of these pieces is useless on its own, but they may be recombined to obtain the secret key. When the government obtains legal authority to access a particular key, it provides evidence of the court order to each of the third parties and then reassembles the secret key.

Escrowed Encryption Standard This escrow approach provides the government or another authorized agent with a technological means to decrypt ciphertext. It was the approach proposed for the Clipper chip.

It's highly unlikely that government regulators will ever overcome the legal and privacy hurdles necessary to implement key escrow on a widespread basis. The technology is certainly available, but the general public will likely never accept the potential government intrusiveness it facilitates.

There are, however, legitimate uses for key escrow within an organization. Key escrow and recovery mechanisms prove useful when an individual leaves the organization and other employees require access to their encrypted data, or when a key is simply lost. In these approaches, key *recovery agents (RAs)* have the ability to recover the encryption keys assigned to individual users. This is, of course, an extremely powerful privilege, as an RA could gain access to any user's encryption key. For this reason, many organizations choose to adopt a mechanism known as *M of N control* for key recovery. In this approach, there is a group of individuals of size N in an organization who are granted RA privileges. If they wish to recover an encryption key, a subset of at least M of them must agree to do so. For example, in an M-of-N control system where M=3 and N=12, there

are 12 authorized recovery agents, of whom 3 must collaborate to retrieve an encryption key.

Cryptographic Life Cycle

With the exception of the one-time pad, all cryptographic systems have a limited life span. Moore's law, a commonly cited trend in the advancement of computing power, states that the processing capabilities of a state-of-the-art microprocessor will double approximately every two years. This means that, eventually, processors will reach the amount of strength required to simply guess the encryption keys used for a communication.

Security professionals must keep this cryptographic life cycle in mind when selecting an encryption algorithm and have appropriate governance controls in place to ensure that the algorithms, protocols, and key lengths selected are sufficient to preserve the integrity of a cryptosystem for however long it is necessary to keep the information it is protecting secret. Security professionals can use the following algorithm and protocol governance controls:

- Specifying the cryptographic algorithms (such as AES and RSA) acceptable for use in an organization
- Identifying the acceptable key lengths for use with each algorithm based on the sensitivity of information transmitted
- Enumerating the secure transaction protocols (such as TLS) that may be used

For example, if you're designing a cryptographic system to protect the security of business plans that you expect to execute next week, you don't need to worry about the theoretical risk that a processor capable of decrypting them might be developed a decade from now. On the other hand, if you're protecting the confidentiality of information that could be used to construct a nuclear bomb, it's virtually certain that you'll still want that information to remain secret 10 years in the future!

Summary

Cryptographers and cryptanalysts are in a never-ending race to develop more secure cryptosystems and advanced cryptanalytic techniques designed to circumvent those systems.

Cryptography dates back as early as Caesar and has been an ongoing topic of study for many years. In this chapter, you learned some of the fundamental concepts underlying the field of cryptography and gained a basic understanding of the terminology used by cryptographers.

This chapter also examined the similarities and differences between symmetric key cryptography (where communicating parties use the same key) and asymmetric key cryptography (where each communicator has a pair of public and private keys). You learned how hashing may be used to guarantee integrity and how hashes play a role in the digital signature process that guarantees nonrepudiation.

We then analyzed some of the symmetric algorithms currently available and their strengths and weaknesses. We wrapped up the chapter by taking a look at the cryptographic life cycle and the role of algorithm/protocol governance in enterprise security.

The next chapter expands this discussion to cover contemporary public key cryptographic algorithms. Additionally, some of the common cryptanalytic techniques used to defeat both types of cryptosystems will be explored.

Study Essentials

Understand the role that confidentiality, integrity, and nonrepudiation play in cryptosystems. Confidentiality is one of the major goals of cryptography. It protects the secrecy of data while it is both at rest and in transit. Integrity provides the recipient of a message with the assurance that data was not altered (intentionally or unintentionally) between the time it was created and the time it was accessed. Nonrepudiation provides undeniable proof that the sender of a message actually authored it. It prevents the sender from subsequently denying that they sent the original message.

Know how cryptosystems can be used to achieve authentication goals. Authentication provides assurances as to the identity of a user. One possible scheme that uses authentication is the challenge-response protocol,

in which the remote user is asked to encrypt a message using a key known only to the communicating parties. Authentication can be achieved with both symmetric and asymmetric cryptosystems.

Be familiar with the basic terminology of cryptography. When a sender wants to transmit a private message to a recipient, the sender takes the plaintext (unencrypted) message and encrypts it using an algorithm and a key. This produces a ciphertext message that is transmitted to the recipient. The recipient then uses a similar algorithm and key to decrypt the ciphertext and re-create the original plaintext message for viewing.

Understand the difference between a code and a cipher and explain the basic types of ciphers. Codes are cryptographic systems of symbols that operate on words or phrases and are sometimes secret but don't always provide confidentiality. Ciphers, however, are always meant to hide the true meaning of a message. Know how the following types of ciphers work: transposition ciphers, substitution ciphers (including one-time pads), stream ciphers, and block ciphers.

Know the requirements for successful use of a one-time pad. For a one-time pad to be successful, the key must be generated randomly without any known pattern. The key must be at least as long as the message to be encrypted. The pads must be protected against physical disclosure, and each pad must be used only one time and then discarded.

Understand split knowledge. Split knowledge means that the information or privilege required to perform an operation is divided among multiple users. This ensures that no single person has sufficient privileges to compromise the security of the environment. M of N Control is an example of split knowledge used in key recovery and other sensitive tasks.

Understand work function (work factor). Work function, or work factor, is a way to measure the strength of a cryptography system by measuring the effort in terms of cost and/or time to decrypt messages. Usually, the time and effort required to perform a complete brute-force attack against an encryption system is what a work function rating represents. The security and protection offered by a cryptosystem is directly proportional to the value of its work function/factor.

Understand the importance of key security. Cryptographic keys provide the necessary element of secrecy to a cryptosystem. Modern cryptosystems utilize keys that are at least 128 bits long to provide adequate security.

Know the differences between symmetric and asymmetric cryptosystems. Symmetric key cryptosystems (or secret key cryptosystems) rely on the use of a shared secret key. They are much faster than asymmetric algorithms, but they lack support for the following: scalability, easy key distribution, and nonrepudiation. Asymmetric cryptosystems use public-private key pairs for communication between parties but operate much more slowly than symmetric algorithms.

Be able to explain the basic operational modes of symmetric cryptosystems. Symmetric cryptosystems operate in several discrete modes, including Electronic Codebook (ECB) mode, Cipher Block Chaining (CBC) mode, Cipher Feedback (CFB) mode, Output Feedback (OFB) mode, Counter (CTR) mode, Galois/Counter mode (GCM), and Counter with Cipher Block Chaining Message Authentication Code mode (CCM). ECB mode is considered the least secure and is used only for short messages. 3DES uses three iterations of DES with two or three different keys to increase the effective key strength to 112 or 168 bits, respectively.

Know the Advanced Encryption Standard (AES). The Advanced Encryption Standard (AES) uses the Rijndael algorithm and is the U.S. government standard for the secure exchange of sensitive but unclassified data. AES uses key lengths of 128, 192, and 256 bits and a fixed block size of 128 bits to achieve a much higher level of security than that provided by the older DES algorithm.

Written Lab

1. What is the major obstacle preventing the widespread adoption of one-time pad cryptosystems to ensure data confidentiality?
2. Encrypt the message “I will pass the CISSP exam and become certified next month” using columnar transposition with the keyword SECURE.

3. Decrypt the message “F R Q J U D W X O D W L R Q V B R X J R W L W” using the Caesar ROT3 substitution cipher.

Review Questions

1. Ryan is responsible for managing the cryptographic keys used by his organization. Which of the following statements are correct about how he should select and manage those keys? (Choose all that apply.)

   A. Keys should be sufficiently long to protect against future attacks if the data is expected to remain sensitive.

   B. Keys should be chosen using an approach that generates them from a predictable pattern.

   C. Keys should be maintained indefinitely.

   D. Longer keys provide greater levels of security.

2. John recently received an email message from Bill. What cryptographic goal would need to be met to convince John that Bill was actually the sender of the message?

   A. Nonrepudiation

   B. Confidentiality

   C. Availability

   D. Integrity

3. You are implementing AES encryption for files that your organization plans to store in a cloud storage service and wish to have the strongest encryption possible. What key length should you choose?

   A. 192 bits

   B. 256 bits

   C. 512 bits

   D. 1,024 bits

4. You are creating a security product that must facilitate the exchange of symmetric encryption keys between two parties that have no way to

securely exchange keys in person. What algorithm might you use to facilitate the exchange?

A. Rijndael

B. Blowfish

C. Vernam

D. Diffie–Hellman

5. What occurs when the relationship between the plaintext and the key is complicated enough that an attacker can't merely continue altering the plaintext and analyzing the resulting ciphertext to determine the key?

A. Confusion

B. Transposition

C. Polymorphism

D. Diffusion

6. Randy is implementing an AES-based cryptosystem for use within his organization. He is planning to use it to ensure that only authorized individuals are able to read sensitive information. What goal of cryptogrpahy is Randy attempting to achieve?

A. Nonrepudiation

B. Confidentiality

C. Authentication

D. Integrity

7. Brian encountered encrypted data left on one of his systems by attackers who were communicating with one another. He has tried many cryptanalytic techniques and was unable to decrypt the data. He believes that the data may be protected with an unbreakable system. When correctly implemented, what is the only cryptosystem known to be unbreakable?

A. Transposition cipher

B. Substitution cipher

C. Advanced Encryption Standard

D. One-time pad

8. Helen is planning to use a one-time pad to meet a unique cryptographic requirement in her organization. She is trying to identify the requirements for using this cryptosystem. Which of the following are requirements for the use of a one-time pad? (Choose all that apply.)

   A. The encryption key must be at least one-half the length of the message to be encrypted.

   B. The encryption key must be randomly generated.

   C. Each one-time pad must be used only once.

   D. The one-time pad must be physically protected against disclosure.

9. Brian administers a symmetric cryptosystem used by 20 users, each of whom has the ability to communicate privately with any other user. One of those users lost control of their account and Brian believes that user's keys were compromised. How many keys must he change?

   A. 1

   B. 2

   C. 19

   D. 190

10. Which one of the following cipher types operates on large pieces of a message rather than individual characters or bits of a message?

    A. Stream cipher

    B. Caesar cipher

    C. Block cipher

    D. ROT3 cipher

11. James is the administrator for his organization's symmetric key cryptographic system. He issues keys to users when the need arises. Mary and Beth recently approached him and presented a need to be able to exchange encrypted files securely with each other. How many keys must James generate?

A. One

B. Two

C. Three

D. Four

12. Dave is developing a key escrow system that requires multiple people to retrieve a key but does not depend on every participant being present. What type of technique is he using?

    A. Split knowledge

    B. M of N Control

    C. Work function

    D. Zero-knowledge proof

13. What is used to increase the strength of cryptography by creating a unique ciphertext every time the same message is encrypted with the same key?

    A. Initialization vector (IV)

    B. Vigenère cipher

    C. Steganography

    D. Stream cipher

14. Tammy is choosing a mode of operation for a symmetric cryptosystem that she will be using in her organization. She wants to choose a mode that is capable of providing both confidentiality and data authenticity. What mode would best meet her needs?

    A. ECB

    B. GCM

    C. OFB

    D. CTR

15. Julie is designing a highly secure system and is concerned about the storage of unencrypted data in RAM. What use case is she considering?

A. Data in transit

B. Data at rest

C. Data in destruction

D. Data in use

16. Renee conducted an inventory of encryption algorithms used in her organization and found that they are using all of the algorithms below. Which of these algorithms should be discontinued? (Choose all that apply.)

    A. AES

    B. DES

    C. 3DES

    D. RC5

17. Which one of the following encryption algorithm modes suffers from the undesirable characteristic of errors propagating between blocks?

    A. Electronic Codebook mode

    B. Cipher Block Chaining mode

    C. Output Feedback mode

    D. Galois/Counter mode

18. Which one of the following secret key distribution methods is most cumbersome when users are located in different geographic locations?

    A. Diffie–Hellman

    B. Public key encryption

    C. Offline key distribution

    D. Key escrow

19. Victoria is choosing an encryption algorithm for use within her organization and would like to choose the most secure symmetric algorithm from a list of those supported by the software package she intends to use. If the package supports the following algorithms, which would be the best option?

A. CAST-256

B. 3DES

C. RC4

D. SKIPJACK

20. The Jones Institute has six employees and uses a symmetric key encryption system to ensure confidentiality of communications. If each employee needs to communicate privately with every other employee, how many keys are necessary?

A. 1

B. 6

C. 15

D. 30

Chapter 7
PKI and Cryptographic Applications

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 3:0 Security Architecture and Engineering**
  - 3.5 Assess and mitigate the vulnerabilities of security architectures, designs, and solution elements
    - 3.5.4 Cryptographic systems
  - 3.6 Select and determine cryptographic solutions
    - 3.6.1 Cryptographic life cycle (e.g., keys, algorithm selection)
    - 3.6.2 Cryptographic methods (e.g., asymmetric)
    - 3.6.3 Public key infrastructure (PKI) (e.g., quantum key distribution)
    - 3.6.4 Key management practices (e.g., rotation)
    - 3.6.5 Digital signatures and digital certificates (e.g., non-repudiation, integrity)
  - 3.7 Understand methods of cryptanalytic attacks
    - 3.7.1 Brute force
    - 3.7.2 Ciphertext only
    - 3.7.3 Known plaintext
    - 3.7.4 Frequency analysis
    - 3.7.5 Chosen ciphertext
    - 3.7.6 Implementation attacks
    - 3.7.7 Side-channel
    - 3.7.8 Fault injection
    - 3.7.9 Timing
    - 3.7.10 Man-in-the-middle (MITM)

In Chapter 6, “Cryptography and Symmetric Key Algorithms,” we introduced basic cryptography concepts and explored a variety of secret key cryptosystems. The symmetric cryptosystems discussed in that chapter offer fast, secure communication but introduce the substantial challenge of key exchange between previously unrelated parties.

This chapter explores the world of asymmetric (or public key) cryptography and the public key infrastructure (PKI) that supports secure communication between individuals who don't necessarily know each other prior to the communication. Asymmetric algorithms provide convenient key exchange mechanisms and are scalable to very large numbers of users, addressing the two most significant challenges for users of symmetric cryptosystems.

This chapter also explores several practical applications of asymmetric cryptography: securing portable devices, email, web communications, and networking. The chapter concludes with an examination of a variety of attacks malicious individuals might use to compromise weak cryptosystems.

Asymmetric Cryptography

The section “Modern Cryptography” in Chapter 6 introduced the basic principles behind both secret (symmetric) and public (asymmetric) key cryptosystems. You learned that symmetric key cryptosystems require that both communicating parties possess the same shared secret key, creating the problem of secure key distribution. You also learned that asymmetric cryptosystems avoid this hurdle by using pairs of public and private keys to facilitate secure communication without the overhead of complex key distribution systems.

In the following sections, we'll explore the concepts of public key cryptography in greater detail and look at four of the more common asymmetric cryptosystems in use today: Rivest–Shamir–Adleman (RSA), Diffie–Hellman, ElGamal, and elliptic curve cryptography (ECC). We'll also explore the emerging world of quantum cryptography.

Public and Private Keys

Recall from Chapter 6 that *public key cryptosystems* assign each user a pair of keys: a public key and a corresponding private key. As the names imply, public key cryptosystem users make their public keys freely available to anyone with whom they want to communicate. The mere possession of the public key by third parties does not introduce any weaknesses into the cryptosystem. The private key, on the other hand, is reserved for the sole use of the individual or entity who owns the key. Users should not normally share their private keys with any other cryptosystem user, outside of key escrow and recovery arrangements. Normal communication between public key cryptosystem users follows the process shown in Figure 7.1.

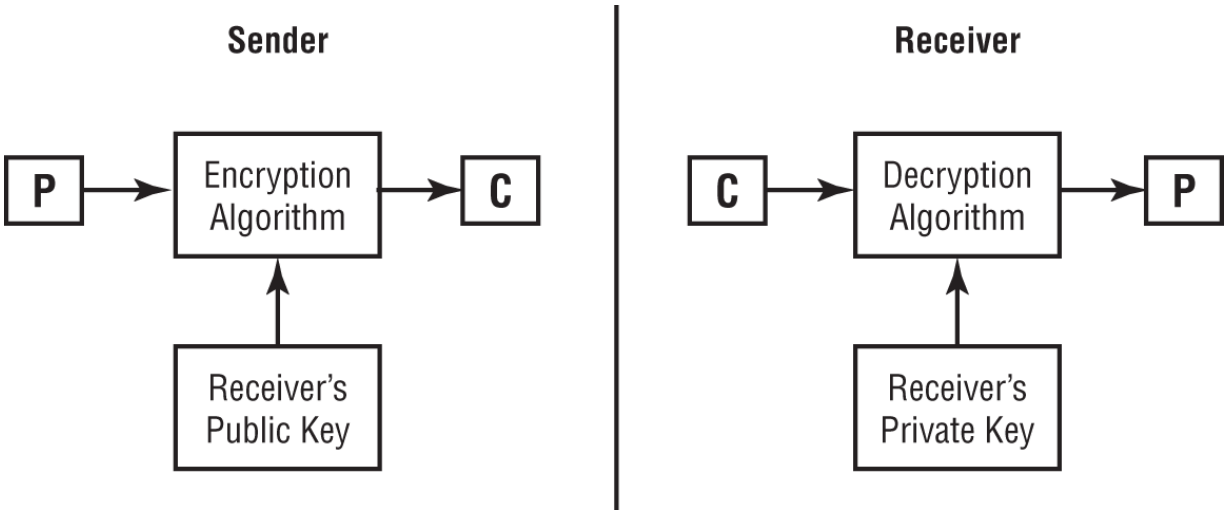


FIGURE 7.1 Asymmetric key cryptography

Notice that the process does not require the sharing of private keys. The sender encrypts the plaintext message (*P*) with the recipient's public key to create the ciphertext message (*C*). When the recipient opens the ciphertext message, they decrypt it using their private key to view the original plaintext message.

Once the sender encrypts the message with the recipient's public key, no user (including the sender) can decrypt that message without knowing the recipient's private key (the second half of the public-private key pair). This is the beauty of public key cryptography—public keys can be freely shared using insecure communications and then used to create secure communications' channels between users previously unknown to each other.

You also learned in the previous chapter that public key cryptography entails a higher degree of computational complexity. Keys used within

public key systems must be longer than those used in secret key systems to produce cryptosystems of equivalent strengths.



Because of the high computational requirements associated with public key cryptography, architects often prefer to use symmetric cryptography on anything other than short messages. Later in this chapter, you'll learn how hybrid cryptography combines the benefits of symmetric and asymmetric cryptography.

RSA

The most famous public key cryptosystem is named after its creators. In 1977, Ronald Rivest, Adi Shamir, and Leonard Adleman proposed the *RSA (Rivest-Shamir-Adleman) public key algorithm*, which remains a worldwide standard today. They patented their algorithm and formed a commercial venture known as RSA Security to develop mainstream implementations of their security technology. Today, the RSA algorithm has been released into the public domain and is widely used for secure communication.

The RSA algorithm depends on the computational difficulty inherent in factoring the product of large prime numbers. Each user of the cryptosystem generates a pair of public and private keys that are mathematically related using the algorithm described in the following steps:

1. Choose two large prime numbers (approximately 200 digits each), labeled p and q.
2. Compute the product of those two numbers: $n = p * q$.
3. Select a number, e, that satisfies the following two requirements:
    a. e is less than n.
    b. e and $(p - 1)(q - 1)$ are relatively prime—that is, the two numbers have no common factors other than 1.
4. Find a number, d, such that $ed = 1 \bmod ((p - 1)(q - 1))$.

5. Distribute *e* and *n* as the public key to all cryptosystem users. Keep *d* secret as the private key.

If Alice wants to send an encrypted message to Bob, she generates the ciphertext (*C*) from the plaintext (*P*) using the following formula (where *e* is Bob's public key and *n* is the product of *p* and *q* created during the key generation process):

$$C = P^e \bmod n$$

When Bob receives the message, he performs the following calculation to retrieve the plaintext message:

$$P = C^d \bmod n$$

Merkle–Hellman Knapsack

Another early asymmetric algorithm, the Merkle–Hellman Knapsack cryptosystem, was developed the year after RSA was publicized. Like RSA, it's based on the difficulty of performing factoring operations, but it relies on a component of set theory known as *super-increasing sequence* rather than on large prime numbers. Merkle–Hellman was proven ineffective when it was broken in 1984.

Importance of Key Length

The length of the cryptographic key is perhaps the most important security parameter that can be set at the discretion of the security administrator. It's important to understand the capabilities of your encryption algorithm and choose a key length that provides an appropriate level of protection. This judgment can be made by weighing the difficulty of defeating a given key length (measured in the amount of processing time required to defeat the cryptosystem) against the importance of the data.

Generally speaking, the more critical your data, the stronger the key you should use to protect that data. Timeliness of the data is also an important consideration. You must take into account the rapid growth of computing power—Moore's law suggests that computing power doubles approximately every two years. If it takes current computers one year of processing time to break your code, it will take only three months if the attempt is made with contemporary technology about four years down the road. If you expect that your data will still be sensitive at that time, you should choose a much longer cryptographic key that will remain secure well into the future.

Also, as attackers are now able to leverage cloud computing resources, they are able to more efficiently attack encrypted data. The cloud allows attackers to rent scalable computing power, including powerful graphic processing units (GPUs) on a per-hour basis, and offers significant discounts when using excess capacity during nonpeak hours. This brings powerful computing well within the reach of many attackers.

The strengths of various key lengths also vary greatly according to the cryptosystem you're using. The key lengths shown in the following table for three cryptosystems all provide equal protection because of differences in the way that the algorithms use the keying material:

Cryptosystem	**Key length**
Symmetric	128 bits
RSA	3,072 bits

Cryptosystem	Key length
Elliptic curve cryptography	256 bits

ElGamal

In Chapter 6, you learned how the Diffie–Hellman algorithm uses large integers and modular arithmetic to facilitate the secure exchange of secret keys over insecure communications channels. In 1985, Dr. Taher Elgamal published an article describing how the mathematical principles behind the Diffie–Hellman key exchange algorithm could be extended to support an entire public key cryptosystem used for encrypting and decrypting messages.

At the time of its release, one of the major advantages of ElGamal over the RSA algorithm was that it was released into the public domain. Elgamal did not obtain a patent on his extension of Diffie–Hellman, and it is freely available for use, unlike the then-patented RSA technology. (RSA released its algorithm into the public domain in 2000.)

However, ElGamal also has a major disadvantage—the algorithm doubles the size of any message that it encrypts. This presents a major hardship when encrypting large amounts of data that must be sent over a network.

Elliptic Curve Cryptography

The same year that Elgamal published his algorithm, two other mathematicians, Neal Koblitz from the University of Washington and Victor Miller from IBM, independently proposed the application of *elliptic curve cryptography (ECC)*.



The mathematical concepts behind elliptic curve cryptography are quite complex and well beyond the scope of this book. However, you should be generally familiar with the elliptic curve algorithm and its potential applications. If you are interested in learning the detailed mathematics behind elliptic curve cryptosystems, an excellent tutorial exists at www.certicom.com/content/certicom/en/ecc-tutorial.html.

Any elliptic curve can be defined by the following equation:

$$y^2 = x^3 + ax + b$$

In this equation, *x*, *y*, *a*, and *b* are all real numbers. Each elliptic curve has a corresponding *elliptic curve group* made up of the points on the elliptic curve along with the point *O*, located at infinity. Two points within the same elliptic curve group (*P* and *Q*) can be added together with an elliptic curve addition algorithm. This operation is expressed, quite simply, as follows:

$$P + Q$$

This problem can be extended to involve multiplication by assuming that *Q* is a multiple of *P*, meaning the following:

$$Q = k.P$$

Computer scientists and mathematicians believe that it is extremely hard to find the integer *k*, even if *P* and *Q* are already known. This difficult problem, known as the elliptic curve discrete logarithm problem (ECDLP), forms the basis of elliptic curve cryptography. It is widely believed that this problem is harder to solve than both the prime factorization problem that the RSA cryptosystem is based on and the standard discrete logarithm problem (DLP) utilized by Diffie–Hellman and ElGamal. This is illustrated by the data shown in the table in the sidebar "Importance of Key Length," which noted that a 3,072-bit RSA key is cryptographically equivalent to a 256-bit elliptic curve cryptosystem key.

Diffie–Hellman Key Exchange

In Chapter 6, you learned how the Diffie–Hellman algorithm is an approach to key exchange that allows two individuals to generate a shared secret key over an insecure communications channel. With knowledge of asymmetric cryptography under your belt, we can now dive a little more into the details of how this algorithm actually works, as Diffie–Hellman key exchange is an example of public key cryptography.

The beauty of this algorithm lies in the ability of two users to generate a shared secret that they both know without ever actually transmitting that secret. Hence, they may use public key cryptography to generate a shared secret key that they then use to communicate with a symmetric encryption algorithm. This is one example of an approach known as *hybrid cryptography*, which we discuss in more detail later in this chapter.

The Diffie–Hellman algorithm works by using the mathematics of prime numbers, similar to the RSA algorithm. Imagine that Richard and Sue would like to communicate over a secure, encrypted connection but they are in different places and have no shared secret key. Richard or Sue could simply create such a key, but then they would have no way to share it with each other without exposing it to eavesdropping. So, instead, they use the Diffie–Hellman algorithm, following this process:

1. Richard and Sue agree on two large numbers: p (which is a prime number) and g (which is an integer), such that $1 < g < p$.
2. Richard chooses a large random integer r and performs the following calculation:

   $$R = g^r \bmod p$$

3. Sue chooses a large random integer s and performs the following calculation:

   $$S = g^s \bmod p$$

4. Richard sends R to Sue and Sue sends S to Richard.
5. Richard then performs this calculation:

   $$K = S^r \bmod p$$

6. Sue then performs this calculation:

$$K = R^s \bmod p$$

At this point, Richard and Sue both have the same value, K, and can use this for secret key communication between the two parties.

It is important to note that Diffie–Hellman is not an encryption protocol in and of itself. It is technically a key exchange protocol. However, it is commonly used to create a shared secret key for use in Transport Layer Security (TLS), where it is referred to as either DHE (Diffie-Hellman Ephemeral) or EDH (Ephemeral Diffie-Hellman. We discuss this use of Diffie–Hellman later in this chapter.



The Diffie–Hellman key exchange algorithm relies on the use of large prime numbers. The ECDHE (Elliptic Curve Diffie-Hellman Ephemeral) key exchange algorithm is a variant of this approach that uses the elliptic curve problem to perform a similar shared secret key agreement process.

Quantum Cryptography

Quantum computing is an area of advanced theoretical research in computer science and physics. The theory behind them is that we can use principles of quantum mechanics to replace the binary 1 and 0 bits of digital computing with multidimensional quantum bits known as *qubits*.

Quantum computing remains an emerging field, and currently, quantum computers are confined to theoretical research. Nobody has yet developed a practical implementation of a useful quantum computer. That said, if quantum computers do come on the scene, they have the potential to revolutionize the world of computer science by providing the technological foundation for the most powerful computers ever developed. Those computers would quickly upend many of the principles of modern cybersecurity.

The most significant impact of quantum computing on the world of cryptography resides in the potential that quantum computers may be able

to solve problems that are not possible to solve on contemporary computers. This concept is known as *quantum supremacy* and, if achieved, may be able to easily solve the factorization problems upon which many classical asymmetric encryption algorithms rely. If this occurs, it could render popular algorithms such as RSA and Diffie–Hellman insecure.

However, quantum computers may also be used to create newer, more complex cryptographic algorithms. These *quantum cryptography* systems may be more resistant to quantum attacks and could usher in a new era of cryptography. Researchers have already developed lab implementations of *quantum key distribution (QKD)*, an approach to use quantum computing to create a shared secret key between two users, similar to the goal of the Diffie–Hellman algorithm. Like quantum cryptography in general, QKD has not yet reached the stage of practical use.

Post-Quantum Cryptography

The most practical implication of quantum computing today is that cybersecurity professionals should be aware of the length of time that their information will remain sensitive. It is possible that an attacker could retain stolen copies of encrypted data for an extended period of time and then use future developments in quantum computing to decrypt that data. If the data remains sensitive at that point, the organization may suffer injury. The most important point here for security professionals is that they must be thinking today about the security of their current data in a post-quantum world.

Also, it is quite possible that the first major practical applications of quantum computing to cryptanalytic attacks may occur in secret. An intelligence agency or other group discovering a practical means to break modern cryptography would benefit most if they kept that discovery secret and used it to their own advantage. It is even possible that such discoveries have already occurred in secret.

Hash Functions

Later in this chapter, you'll learn how cryptosystems implement digital signatures to provide proof that a message originated from a particular user of the cryptosystem and to ensure that the message was not modified while in transit between the two parties. Before you can completely understand that concept, we must first explain the concept of *hash functions*. We will explore the basics of hash functions and look at several common hash functions used in modern digital signature algorithms.

Hash functions have a very simple purpose—they take a potentially long message and generate a unique output value derived from the content of the message. This value is commonly referred to as the *message digest*. Message digests can be generated by the sender of a message and transmitted to the recipient along with the full message for two reasons.

First, the recipient can use the same hash function to recompute the message digest from the full message. They can then compare the computed message digest to the transmitted one to ensure that the message sent by the originator is the same one received by the recipient. If the message digests do not match, that means the message was somehow modified while in transit. It is important to note that the messages must be *exactly* identical for the digests to match. If the messages have even a slight difference in spacing, punctuation, or content, the message digest values will be completely different. It is not possible to tell the degree of difference between two messages by comparing the digests. Even a slight difference will generate a totally different digest value.

Second, the message digest can be used to implement a digital signature algorithm. This concept is covered in the section “Digital Signatures,” later in this chapter.

In most cases, a message digest is 128 bits or larger. However, a single-digit value can be used to perform the function of parity, a low-level or single-digit checksum value used to provide a single individual point of verification. In most cases, the longer the message digest, the more reliable its verification of integrity.

According to RSA Security, there are five basic requirements for a cryptographic hash function:

- The input can be of any length.

- The output has a fixed length.
- The hash function is relatively easy to compute for any input.
- The hash function is one-way (meaning that it is extremely hard to determine the input when provided with the output). One-way functions and their usefulness in cryptography are described in Chapter 6.
- The hash function is collision resistant (meaning that it is extremely hard to find two messages that produce the same hash value).

The bottom line is that hash functions create a value that uniquely represents the data in the original message but cannot be reversed, or "de-hashed." Access to the hashed value does not allow someone to determine what the original message contained. Access to the recipient's message, the original message's hashed value (message digest), and the hash function's identifier allows one to verify whether the message has been changed from the original. This is done by comparing the hashed value of the original message with the hashed value of the recipient's message. If the hashes match, the hash function was run on the same input data, so the input data has not changed.

In the following sections, we'll look at some common hashing algorithms:

- Secure Hash Algorithm (SHA)
- Message Digest 5 (MD5)
- RACE Integrity Primitives Evaluation Message Digest (RIPEMD).

Hash-Based Message Authentication Code (HMAC) is also discussed later in this chapter.



In addition to SHA, MD5, RIPEMD, and HMAC, you should recognize HAVAL. Hashing Algorithm with Variable Length (HAVAL) is a modification of MD5. HAVAL uses 1,024-bit blocks and produces hash values of 128, 160, 192, 224, and 256 bits.

SHA Family

The Secure Hash Algorithm (SHA) and its successors, SHA-1, SHA-2, and SHA-3, are government standard hash functions promoted by the National Institute of Standards and Technology (NIST) and are specified in an official government publication—the Secure Hash Standard (SHS), also known as Federal Information Processing Standards (FIPS) 180-4.

SHA-1 takes an input of virtually any length (in reality, there is an upper bound of approximately 2,097,152 terabytes on the algorithm) and produces a 160-bit message digest. The SHA-1 algorithm processes a message in 512-bit blocks. Therefore, if the message length is not a multiple of 512, the SHA algorithm pads the message with additional data until the length reaches the next highest multiple of 512.

Cryptanalytic attacks demonstrated that there are weaknesses in the SHA-1 algorithm, and therefore, NIST deprecated SHA-1 and no longer recommends its use for any purpose, including digital signatures and digital certificates. Web browsers dropped support for SHA-1 in 2017.

As a replacement, NIST announced the SHA-2 standard, which has four major variants:

- SHA-256 produces a 256-bit message digest using a 512-bit block size.
- SHA-224 uses a truncated version of the SHA-256 hash that drops 32 bits to produce a 224-bit message digest using a 512-bit block size.
- SHA-512 produces a 512-bit message digest using a 1,024-bit block size.
- SHA-384 uses a truncated version of the SHA-512 hash that drops 128 bits to produce a 384-bit digest using a 1,024-bit block size.



Although it might seem trivial, you should take the time to memorize the size of the message digests produced by each one of the hash algorithms described in this chapter.

The cryptographic community generally considers the SHA-2 algorithms secure, but they theoretically suffer from the same weakness as the SHA-1 algorithm. In 2015, the federal government announced the release of the Keccak algorithm as the SHA-3 standard. The SHA-3 family was developed to serve as a drop-in replacement for the SHA-2 family of hash functions, offering the same variants and hash lengths using a different computational algorithm. SHA-3 provides the same level of security as SHA-2, but it is slower than SHA-2, so SHA-3 is not commonly used outside of some specialized cases where the algorithm is efficiently implemented in hardware.

MD5

The Message Digest 2 (MD2) hash algorithm was developed by Ronald Rivest (the same Rivest of Rivest, Shamir, and Adleman fame) in 1989 to provide a secure hash function for 8-bit processors. In 1990, Rivest enhanced his message digest algorithm to support 32-bit processors and increase the level of security with a version called MD4.

In 1992, Rivest released the next version of his message digest algorithm, which he called MD5. It also processes 512-bit blocks of the message, but it uses four distinct rounds of computation to produce a digest of the same length as the MD2 and MD4 algorithms (128 bits). MD5 has the same padding requirements as MD4—the message length must be 64 bits less than a multiple of 512 bits.

MD5 implements additional security features that reduce the speed of message digest production significantly. Unfortunately, cryptanalytic attacks demonstrated that the MD5 protocol is subject to collisions, preventing its use for ensuring message integrity. Specifically, Arjen Lenstra and others demonstrated in 2005 that it is possible to create two digital certificates from different public keys that have the same MD5 hash.

Some tools and systems still rely on MD5, so you may see it in use today, but it is now far better to rely on more secure hashing algorithms, such as SHA-2.

RIPEMD

The RACE Integrity Primitives Evaluation Message Digest (RIPEMD) series of hash functions is an alternative to the SHA family that is used in some applications, such as Bitcoin cryptocurrency implementations. The family contains a series of increasingly sophisticated functions:

- RIPEMD produced a 128-bit digest and contained some structural flaws that rendered it insecure.
- RIPEMD-128 replaced RIPEMD, also producing a 128-bit digest, but it is also no longer considered secure.
- RIPEMD-160 is the replacement for RIPEMD-128 that remains secure today and is the most commonly used of the RIPEMD variants. It produces a 160-bit hash value.



You may also see references to RIPEMD-256 and RIPEMD-320. These functions are actually based on RIPEMD-128 and RIPEMD-160, respectively. They do not add any security; they simply create longer hash values for cases where a longer value is needed. RIPEMD-256 has the same level of insecurity as RIPEMD-128, while RIPEMD-320 has the same level of security as RIPEMD-160. This leads to the unusual-sounding situation where RIPEMD-160 is secure, but RIPEMD-256 is not.

Comparison of Hash Function Value Lengths

Table 7.1 lists well-known hashing functions and their resultant hash value lengths in bits. Earmark this page for memorization.

TABLE 7.1 Hash algorithm memorization chart

Hash function name	Hash value length (bits)
HAVAL	128, 160, 192, 224, and 256 bits
HMAC	Variable
MD5	128
SHA-1	160
SHA2-224/SHA3-224	224
SHA2-256/SHA3-256	256
SHA2-384/SHA3-384	384
SHA2-512/SHA3-512	512
RIPEMD-128	128
RIPEMD-160	160
RIPEMD-256	256 (but with equivalent security to 128)
RIPEMD-320	320 (but with equivalent security to 160)

Digital Signatures

Once you have chosen a cryptographically sound hash function and a public key cryptographic algorithm, you can use them to implement a *digital signature* system. Digital signature infrastructures have two distinct goals:

- Digitally signed messages assure the recipient that the message truly came from the claimed sender. They enforce nonrepudiation (that is, they preclude the sender from later claiming that the message is a forgery).
- Digitally signed messages assure the recipient that the message was not altered while in transit between the sender and recipient. This protects against both malicious modification (a third party altering the meaning of the message) and unintentional modification (because of faults in the communications process, such as electrical, optical, or radio frequency [RF] interference).

Digital signature algorithms rely on a combination of the two major concepts already covered in this chapter—public key cryptography and hashing functions.

If Alice wants to digitally sign a message she's sending to Bob, she performs the following actions:

1. Alice generates a message digest (i.e., hash) of the original plaintext message using one of the cryptographically sound hashing algorithms, such as SHA2-512.
2. Alice then encrypts only the message digest using her private key. This encrypted message digest is the digital signature.
3. Alice appends the signed message digest to the plaintext message.
4. Alice transmits the appended message to Bob.

When Bob receives the digitally signed message, he reverses the procedure, as follows:

1. Bob decrypts the digital signature using Alice's public key.
2. Bob uses the same hashing function to create a message digest of the full plaintext message received from Alice.
3. Bob then compares the decrypted message digest he received from Alice with the message digest he computed himself. If the two digests match, he can be assured that the message he received was sent by Alice. If they do not match, either the message was not sent by Alice or the message was modified while in transit.



Digital signatures are used for more than just messages. Software vendors often use digital signature technology to authenticate code distributions that you download from the Internet, such as applets and software patches.

Note that the digital signature process does not provide confidentiality in and of itself. It only ensures that the cryptographic goals of integrity, authentication, and nonrepudiation are met. Let's break that down. If the hash generated by the sender and the hash generated by the recipient match, then we know that the two hashed messages are identical and we have integrity. If the digital signature was verified with the public key of the sender, then we know that it was created using that sender's private key. That private key should be known only to the holder of the key (the sender, in this case), so the verification of the private key proves to the recipient that the digital signature came from the sender, providing origin authentication. The recipient (or anyone else) can then demonstrate that process to a third party, providing nonrepudiation.

If Alice also wanted to ensure the confidentiality of her message to Bob, she could add an additional step to the message creation process. After appending the digitally signed message digest to the plaintext message, Alice could encrypt the entire message with Bob's public key. Then, when Bob would receive the message, he would first decrypt it with his own private key before following the steps just outlined.

HMAC

The Hash-Based Message Authentication Code (HMAC), also known as Keyed-Hash Message Authentication Code, symmetric (secret) key algorithm implements a *partial* digital signature—it guarantees the integrity of a message during transmission, but it does not provide for nonrepudiation.

Which Key Should I Use?

If you're new to public key cryptography, selecting the correct key for various applications can be quite confusing. Encryption, decryption, message signing, and signature verification all use the same algorithm with different key inputs. Here are a few simple rules to help keep these concepts straight in your mind:

- If you want to encrypt a confidential message, use the recipient's public key.
- If you want to decrypt a confidential message sent to you, use your private key.
- If you want to digitally sign a message you are sending to someone else, use your private key.
- If you want to verify the digital signature on a message sent by someone else, use the sender's public key.

These four rules are the core principles of public key cryptography and digital signatures. If you understand each of them, you're off to a great start.

HMAC can be combined with any standard hash function, such as MD5, SHA-2, or SHA-3, by using a shared secret key. Therefore, only communicating parties who know the secret key can generate or verify the *partial* digital signature. If the recipient decrypts the message digest but cannot successfully compare it to a message digest generated from the original plaintext message, that means the message was altered in transit.

Because HMAC relies on a shared secret key, it does not provide any nonrepudiation functionality (as previously mentioned). However, it operates in a more efficient manner than the digital signature standard (DSS) described in the following section and may be suitable for applications in which symmetric key cryptography is appropriate. In short, HMAC represents a halfway point between the unencrypted use of a hash

function and the computationally expensive digital signature algorithms (DSA) based on public key cryptography.

Digital Signature Standard

NIST specifies the digital signature algorithms acceptable for federal government use in FIPS 186-5, also known as the Digital Signature Standard (DSS). This document specifies that all federally approved digital signature algorithms must use the SHA-3 hashing functions.

DSS also specifies the encryption algorithms that can be used to support a digital signature infrastructure. There are three currently approved standard encryption algorithms in FIPS 186-5:

- The Rivest–Shamir–Adleman (RSA) digital signature algorithm, as specified in IETF RFC 8017.
- The Elliptic Curve Digital Signature Algorithm (ECDSA), as specified in FIPS 186-5. This algorithm provides inherent support for nonrepudiation.
- The Edwards Curve Digital Signature Algorithm (EdDSA), as specified in IETF RFC 8032.

Public Key Infrastructure

The major strength of public key encryption is its ability to facilitate communication between parties previously unknown to each other. This is made possible by the *public key infrastructure (PKI)* hierarchy of trust relationships. These trusts permit combining asymmetric cryptography with symmetric cryptography along with hashing and digital certificates, giving us hybrid cryptography.

In the following sections, you'll learn the basic components of the public key infrastructure and the cryptographic concepts that make secure global communications possible. You'll learn the composition of a digital certificate, the role of certificate authorities, and the process used to generate and destroy digital certificates.

Certificates

Digital *certificates* provide communicating parties with the assurance that the people they are communicating with truly are who they claim to be. Digital certificates are essentially endorsed copies of an individual's public key. When users verify that a certificate was signed by a trusted certificate authority (CA), they know that the public key is legitimate.

Digital certificates contain specific identifying information, and their construction is governed by the International Telecommunications Union (ITU), an international standard—X.509. Certificates that conform to X.509 contain the following data:

- Version of X.509 to which the certificate conforms
- Serial number (from the certificate creator)
- Signature algorithm identifier (specifies the technique used by the certificate authority to digitally sign the contents of the certificate)
- Issuer name (identification of the certificate authority [CA] that issued the certificate)
- Validity period (specifies the dates and times—a starting date and time and an expiration date and time—during which the certificate is valid)
- Subject's name (contains the common name [CN] of the certificate as well as the distinguished name [DN] of the entity that owns the public key contained in the certificate)
- Subject's public key (the meat of the certificate—the actual public key the certificate owner used to set up secure communications)

Certificates may be issued for a variety of purposes. These include providing assurance for the public keys of:

- Computers/machines
- Individual users
- Email addresses
- Developers (code-signing certificates)

The subject of a certificate may include a wildcard in the certificate name, indicating that the certificate is good for subdomains as well. The wildcard is designated by an asterisk character. For example, a wildcard certificate issued to `*.example.org` would be valid for all of the following domains:

- `example.org`
- `www.example.org`
- `mail.example.org`
- `secure.example.org`



Wildcard certificates are only good for one level of subdomain. Therefore, the `*.example.org` certificate would not be valid for the `www.cissp.example.org` subdomain.

Certificate Authorities

Certificate authorities (CAs) are the glue that binds the public key infrastructure together. These neutral organizations offer notarization services for digital certificates. To obtain a digital certificate from a reputable CA, you must prove your identity to the satisfaction of the CA. The following list includes some of the major CAs who provide widely accepted digital certificates:

- IdenTrust
- AWS Certificate Manager (ACM)
- GlobalSign
- ComodoCA
- Certum
- GoDaddy
- DigiCert

- SECOM Trust Systems
- Entrust
- Actalis
- Trustwave

Nothing is preventing any organization from simply setting up shop as a certificate authority. However, the certificates issued by a CA are only as good as the trust placed in the CA that issued them. This is an important item to consider when receiving a digital certificate from a third party. If you don't recognize and trust the name of the CA that issued the certificate, you shouldn't place any trust in the certificate at all. PKI relies on a hierarchy of trust relationships. If you configure your browser to trust a CA, it will automatically trust all of the digital certificates issued by that CA. Browser developers preconfigure browsers to trust the major CAs to avoid placing this burden on users.



Let's Encrypt is a well-known CA because they offer free certificates in an effort to encourage the use of encryption. You can learn more about this free service at `http://letsencrypt.org`.

Registration authorities (RAs) assist CAs with the burden of verifying users' identities prior to the issuance of digital certificates. They do not directly issue certificates themselves, but they play an important role in the certification process, allowing CAs to remotely validate user identities.

Certificate authorities must carefully protect their own private keys to preserve their trust relationships. To do this, they often use an *offline CA* to protect their *root certificate*, the top-level certificate for their entire PKI. This offline CA is disconnected from networks and powered down until it is needed. The offline CA uses the root certificate to create subordinate *intermediate CAs* that serve as the *online CAs* used to issue certificates on a routine basis.

In the CA trust model, the use of a series of intermediate CAs is known as *certificate chaining*. To validate a certificate, the browser verifies the

identity of the intermediate CA(s) first and then traces the path of trust back to a known root CA, verifying the identity of each link in the chain of trust.

Certificate authorities do not need to be third-party service providers. Many organizations operate internal CAs that provide *self-signed certificates* for use exclusively inside an organization. These certificates won't be trusted by the browsers of external users, but internal systems may be configured to trust the internal CA, saving the expense of purchasing certificates from a third-party CA.

Certificate Life Cycle

The technical concepts behind the public key infrastructure (PKI) are relatively simple. In the following sections, we'll cover the processes used by certificate authorities (CAs) to create, validate, and revoke client certificates.

Enrollment

When you want to obtain a digital certificate, you must first prove your identity to the CA in some manner; this process is called *enrollment*. As mentioned in the previous section, this sometimes involves physically appearing before an agent of the certificate authority with the appropriate identification documents. Some certificate authorities provide other means of verification, including the use of credit report data and identity verification by trusted community leaders.

Once you've satisfied the certificate authority regarding your identity, you provide them with your public key in the form of a *certificate signing request (CSR)*. The CA next creates an X.509 digital certificate containing your identifying information and a copy of your public key. The CA then digitally signs the certificate using the CA's private key and provides you with a copy of your signed digital certificate. You may then safely distribute this certificate to anyone with whom you want to communicate securely.

Certificate authorities issue different types of certificates depending on the level of identity verification that they perform. The simplest, and most common, certificates are *Domain Validation (DV) certificates*, where the CA simply verifies that the certificate subject has control of the domain name. *Extended Validation (EV) certificates* provide a higher level of

assurance, and the CA takes steps to verify that the certificate owner is a legitimate business before issuing the certificate.

Verification

When you receive a digital certificate from someone with whom you want to communicate, you *verify* the certificate by checking the CA's digital signature using the CA's public key. You then must check the validity period of the certificate to ensure that the current date is after the starting date of the certificate and that the certificate has not yet expired. Finally, you must check and ensure that the certificate was not revoked using a *certificate revocation list (CRL)* or the *Online Certificate Status Protocol (OCSP)*. At this point, you may assume that the public key listed on the certificate is authentic, provided that it satisfies the following requirements:

- The digital signature of the CA is authentic.
- You trust the CA.
- The certificate is not listed on a CRL.
- The certificate actually contains the data you are trusting.

The last point is a subtle but extremely important item. Before you trust an identifying piece of information about someone, be sure that it is actually contained within the certificate. If a certificate contains the email address (`billjones@foo.com`) but not the individual's name, you can be certain only that the public key contained therein is associated with that email address. The CA is not making any assertions about the actual identity of the `billjones@foo.com` email account. However, if the certificate contains the name Bill Jones along with an address and telephone number, the CA is vouching for that information as well.

Digital certificate verification algorithms are built into a number of popular web browsing and email clients, so you won't often need to get involved in the particulars of the process. However, it's important to have a solid understanding of the technical details taking place behind the scenes to make appropriate security judgments for your organization. It's also the reason that, when purchasing a certificate, you choose a CA that is widely trusted. If a CA is not included in, or is later pulled from, the list of CAs

trusted by a major browser, it will greatly limit the usefulness of your certificate.

In 2017, a significant security failure occurred in the digital certificate industry. Symantec, through a series of affiliated companies, issued several digital certificates that did not meet industry security standards. In response, Google announced that the Chrome browser would no longer trust Symantec certificates. As a result, Symantec wound up selling off its certificate-issuing business to DigiCert, which agreed to properly validate certificates prior to issuance. This demonstrates the importance of properly validating certificate requests. A series of seemingly small lapses in procedure can decimate a CA's business.

Certificate pinning approaches instruct browsers to attach (pin) a certificate to a host for an extended period of time. When sites use certificate pinning, the browser associates that site (domain or subdomain) with their public key. This allows users or administrators to notice and intervene if a certificate unexpectedly changes.

Revocation

Occasionally, a certificate authority needs to *revoke* a certificate. This might occur for one of the following reasons:

- The certificate was compromised (for example, the certificate owner accidentally gave away the private key).
- The certificate was erroneously issued (for example, the CA mistakenly issued a certificate without proper verification).
- The details of the certificate changed (for example, the subject's name changed).
- The security association changed (for example, the subject is no longer employed by the organization sponsoring the certificate).



The revocation request grace period is the maximum response time within which a CA will perform any requested revocation. This is defined in the *Certificate Practice Statement (CPS)*. The CPS states the practices a CA employs when issuing or managing certificates.

You can use three techniques to verify the authenticity of certificates and identify revoked certificates:

Certificate Revocation Lists Certificate revocation lists (CRLs) are maintained by the various certificate authorities and contain the serial numbers of certificates that have been issued by a CA and that have been revoked, along with the date and time the revocation went into effect. The major disadvantage to certificate revocation lists is that they must be downloaded and cross-referenced periodically, introducing a period of latency between the time a certificate is revoked and the time end users are notified of the revocation.

Online Certificate Status Protocol (OCSP) This protocol eliminates the latency inherent in the use of certificate revocation lists by providing a means for real-time certificate verification. When a client receives a certificate, it sends an OCSP request to the CA's OCSP server. The server then responds with a status of good, revoked, or unknown. The browser uses this information to determine whether the certificate is valid.

Certificate Stapling The primary issue with OCSP is that it places a significant burden on the OCSP servers operated by certificate authorities. These servers must process requests from every single visitor to a website or other user of a digital certificate, verifying that the certificate is valid and not revoked.

Certificate stapling is an extension to the Online Certificate Status Protocol that relieves some of the burden placed on certificate authorities by the original protocol. In the absence of OCSP, when a user visits a website and initiates a secure connection, the web server sends its digital certificate to the user's browser. The user's browser would normally then

be responsible for contacting an OCSP server to verify the certificate's validity.

In certificate stapling, the web server contacts the OCSP server itself and receives a signed and timestamped response from the OCSP server, which it then attaches, or staples, to the digital certificate. Then, when a user requests a secure web connection, the web server sends the digital certificate with the stapled OCSP response to the user's browser. The user's browser then verifies whether the certificate is authentic and also validates that the stapled OCSP response is genuine and recent. Because the CA signed the OCSP response, the user's browser knows that it is from the CA, and the timestamp provides the user's browser with assurance that the CA recently validated the certificate. From there, communication may continue as normal.

The time savings come when the next user visits the website. The web server can simply reuse the stapled certificate without recontacting the OCSP server. As long as the timestamp is recent enough, the user's browser will accept the stapled certificate without needing to contact the CA's OCSP server again. It's common to have stapled certificates with a validity period of 24 hours. That reduces the burden on an OCSP server from handling one request per user over the course of a day, which could be millions of requests, to handling one request per certificate per day. That's a tremendous reduction.

Certificate Formats

Digital certificates are stored in files, and those files come in a variety of formats, both binary and text-based:

- The most common binary format is the Distinguished Encoding Rules (DER) format. DER certificates are normally stored in files with the `.der`, `.crt`, or `.cer` extension.
- The Privacy-Enhanced Mail (PEM) certificate format is an ASCII text version of the DER format. PEM certificates are normally stored in files with the `.pem` or `.crt` extension.



You may have picked up on the fact that the `.crt` file extension is used for both binary DER files and text PEM files. That's very confusing. You should remember that you can't tell whether a CRT certificate is binary or text without actually looking at the contents of the file.

- The Personal Information Exchange (PFX) format is commonly used by Windows systems. PFX certificates may be stored in binary form, using either `.pfx` or `.p12` file extensions.
- Windows systems also use P7B certificates, which are stored in ASCII text format using the `.p7b` file extension.

Table 7.2 provides a summary of certificate formats.

TABLE 7.2 Digital certificate formats

File format names	Format	File extension(s)
Distinguished Encoding Rules (DER)	Binary	`.der`, `.crt`, `.cer`
Privacy-Enhanced Mail (PEM)	Text	`.pem`, `.crt`
Personal Information Exchange (PFX)	Binary	`.pfx`, `.p12`
P7B	Text	`.p7b`

Asymmetric Key Management

When working within the public key infrastructure, you must comply with several best practice requirements to maintain the security of your communications.

First, choose your encryption system wisely. As you learned earlier, “security through obscurity” is not an appropriate approach. Choose an encryption system with an algorithm in the public domain that has been thoroughly vetted by industry experts. Be wary of systems that use a “black-box” approach and maintain that the secrecy of their algorithm is critical to the integrity of the cryptosystem.

You must also select your keys in an appropriate manner. Use a key length that balances your security requirements with performance considerations. Also, ensure that your key is truly random. Any patterns within the key increase the likelihood that an attacker will be able to break your encryption and degrade the security of your cryptosystem.

When using public key encryption, keep your private key secret. Do not, under any circumstances, allow anyone else to gain access to your private key. Remember, allowing someone access even once permanently compromises all communications that take place (past, present, or future) using that key and allows the third party to successfully impersonate you.

Retire keys when they've served a useful life. Many organizations have mandatory key rotation requirements to protect against undetected key compromise. If you don't have a formal policy that you must follow, select an appropriate interval based on the frequency with which you use your key. You might want to change your key pair every few months, if practical.

Back up your key. If you lose the file containing your private key because of data corruption, disaster, or other circumstances, you'll certainly want to have a backup available. You may want to either create your own backup or use a key escrow service that maintains the backup for you. In either case, ensure that the backup is handled in a secure manner. After all, it's just as important as your primary key file.

Hardware security modules (HSMs) also provide an effective way to manage encryption keys. These hardware devices store and manage encryption keys in a secure manner that prevents humans from ever needing to work directly with the keys. Many of them are also capable of improving the efficiency of cryptographic operations, in a process known as hardware acceleration. HSMs range in scope and complexity from very simple devices, such as the YubiKey, that store encrypted keys on a USB drive for personal use, to more complex enterprise products that reside in a data center. HSMs include tamper-resistance mechanisms to prevent someone who gains physical access to the device from accessing the cryptographic material it maintains. Cloud service providers, such as Amazon and Microsoft, also offer cloud-based HSMs that provide secure key management for infrastructure-as-a-service (IaaS) cloud computing resources.

Hybrid Cryptography

You've now learned about the two major categories of cryptographic systems: symmetric and asymmetric algorithms. You've also learned about the major advantages and disadvantages of each. Chief among these are the facts that symmetric algorithms are fast but introduce key distribution challenges, and though asymmetric algorithms solve the key distribution problem, they are also computationally intensive and slow. If you're choosing between these approaches, you're forced to make a decision between convenience and speed.

Hybrid cryptography combines symmetric and asymmetric cryptography to achieve the shared secret key distribution benefits of asymmetric cryptosystems with the speed of symmetric algorithms. These approaches work by setting up an initial connection between two communicating entities using asymmetric cryptography. That connection is used for only one purpose: the exchange of a randomly generated shared secret key, known as an *ephemeral key*. The two parties then exchange whatever data they wish using the shared secret key with a symmetric algorithm. When the communication session ends, they discard the ephemeral key and then repeat the same process if they wish to communicate again later.

The beauty behind this approach is that it uses asymmetric cryptography for the shared secret key distribution, a task that requires the encryption of only a small amount of data. Then it switches to the faster symmetric algorithm for the vast majority of data exchanged.

Transport Layer Security (TLS) is the most well-known example of hybrid cryptography, and we discuss that approach later in this chapter.

Applied Cryptography

Up to this point, you've learned a great deal about the foundations of cryptography, the inner workings of various cryptographic algorithms, and the use of the public key infrastructure (PKI) to distribute identity credentials using digital certificates. You should now feel comfortable with the basics of cryptography and be prepared to move on to higher-level

applications of this technology to solve everyday communications problems.

In the following sections, we'll examine the use of cryptography to secure data at rest, such as that stored on portable devices, as well as data in transit, using techniques that include secure email, encrypted web communications, and networking.

Portable Devices

The now ubiquitous nature of laptop computers, smartphones, and tablets brings new risks to the world of computing. Those devices often contain highly sensitive information that, if lost or stolen, could cause serious harm to an organization and its customers, employees, and affiliates. For this reason, many organizations turn to encryption to protect the data on these devices in the event they are misplaced.

Current versions of popular operating systems now include disk encryption capabilities that make it easy to apply and manage encryption on portable devices. For example, Microsoft Windows includes the BitLocker and Encrypting File System (EFS) technologies, macOS includes FileVault encryption, and the VeraCrypt open source package allows the encryption of disks on Linux, Windows, and Mac systems.

Trusted Platform Module

Modern computers often include a specialized cryptographic component known as a Trusted Platform Module (TPM). The TPM is a chip that resides on the motherboard of the device. The TPM serves a number of purposes, including the storage and management of cryptographic keys used for full-disk encryption (FDE) solutions. The TPM provides the operating system with access to the keys only if the user successfully authenticates. This prevents someone from removing the drive from one device and inserting it into another device to access the drive's data.

A wide variety of commercial tools are available that provide added features and management capability. The major differentiators between these tools are how they protect keys stored in memory, whether they

provide full-disk or volume-only encryption, and whether they integrate with hardware-based Trusted Platform Modules (TPMs) to provide added security. Any effort to select encryption software should include an analysis of how well the alternatives compete on these characteristics.



Don't forget about smartphones when developing your portable device encryption policy. Most major smartphone and tablet platforms include enterprise-level functionality that supports encryption of data stored on the phone.

Email

We have mentioned several times that security should be cost-effective. When it comes to email, simplicity is the most cost-effective option, but sometimes cryptography functions provide specific security services that you can't avoid using. Since ensuring security is also cost-effective, here are some simple rules about encrypting email:

- If you need confidentiality when sending an email message, encrypt the message.
- If your message must maintain integrity, you must hash the message.
- If your message needs authentication, integrity, and/or nonrepudiation, you should digitally sign the message.
- If your message requires confidentiality, integrity, origin authentication, and nonrepudiation, you should encrypt and digitally sign the message.

It is always the responsibility of the sender to put proper mechanisms in place to ensure that the security (that is, confidentiality, integrity, authenticity, and nonrepudiation) of a message or transmission is maintained.



The coverage of email in this chapter focuses on the use of cryptography to provide secure communications between two parties. You'll find more coverage of email security topics in Chapter 12, “Secure Communications and Network Attacks.”

One of the most in-demand applications of cryptography is encrypting and signing email messages. Until recently, encrypted email required the use of complex, awkward software that in turn required manual intervention and complicated key exchange procedures. An increased emphasis on security in recent years resulted in the implementation of strong encryption technology in mainstream email packages. Next, we'll look at some of the secure email standards in widespread use today.

Pretty Good Privacy

Phil Zimmermann's Pretty Good Privacy (PGP) secure email system appeared on the computer security scene in 1991. It combines the CA hierarchy described earlier in this chapter with the “web of trust” concept—that is, you must become trusted by one or more PGP users to begin using the system. You then accept their judgment regarding the validity of additional users and, by extension, trust a multilevel “web” of users descending from your initial trust judgments.

PGP initially encountered a number of hurdles to widespread use. The most difficult obstruction was the U.S. government export regulations, which treated encryption technology as munitions and prohibited the distribution of strong encryption technology outside the United States. Fortunately, this restriction has since been repealed, and PGP may be freely distributed to most countries.

PGP is available in two versions: a commercial product called PGP Encryption Solutions, which is available through Broadcom's Symantec Enterprise Division, and an open source variant called OpenPGP. These products allow for the use of modern encryption algorithms, hash functions, and signature standards within the PGP framework.

PGP messages are often sent in text-encoded format to facilitate compatibility with other email systems. Here is an example of how an encrypted message appears when sent using PGP:

```
-----BEGIN PGP MESSAGE-----

hQGMAyHB9q9kWbl7AQwAmgyZoaXC2Xvo3jrVIWains3/UvUImp3YEbcEmlLK+26
o
TNGBSNi5jLi2A62e8TLGbPkJv5vN3JZH4F27ZvYIhqANwk2nTI1sE0bA2Rzlw6P
c
XCUooGhNY/rmmWTLvWNVRdSXZj2i28fk2gi2QJlrEwYLkKJdUxzKldSLht+Bc+V
2
NbvQrTzJ0LmRq9FKvZ4lz5v7Qj/f1GdKF/5HCTthUWxJMxxuSzCp46rFR6sKAQX
G
tHdi2IzrroyQLR23HO6KuleisGf1X2wzfWENlXMUNGNLxPi2YNvo3MaFMMw3o1d
F
Zj28ptpCH8eGOVIAa05ZNnCk2a6alqTf9aKH8932uCS/AcYG3xqVcRCz7qyaLqD
5
NFg4GXq10KD8Jo1VP/HncOx7/39MGRDuzJqFieQzsVo0uCwVB2zJYC0SeJyMHky
D
TaAxz4HMQxzm8FubreTfisXKuUfPbYAuT855kc2iBKTGo9Cz1WjhQo6mveI6hvu
0
qYUaX5sGgfbD4bzCMFJj0nUBUdMni0jqHJ2XuZerEd8m0DioUOBRJybLlohtRki
k
Gzra/+WGE1ckQmzch5LDPdIEZphvV+5/DbhHdhxN7QMWe6ZkaADAZRgu77tkQK6
c
QvrBPZdk22uS0vzdwzJzzvybspzq1HkjD+aWR9CpSZ9mukZPXew=
=7NWG
-----END PGP MESSAGE-----
```

Similarly, digitally signed messages contain the text of the message followed by a PGP signature. Here is an example:

```
-----BEGIN PGP SIGNED MESSAGE-----
Hash: SHA256

I am enjoying studying cybersecurity.
-----BEGIN PGP SIGNATURE-----

iQGzBAEBCAAdFiEE75kumjjPhsn37slI+Cb2Pddh6OYFAmAF4FMACgkQ+Cb2Pdd
h
6Oba4gv8D4ybEtYidHdlfDYfbF+wYAz8JZ0Mw//f41iwkBG6BO6RtKtNPV202Ng
b
3Uxqjody48ndmDM4q60x3EMy+97ZXNoZL7fY5vv2viDa1so4BqevtRKYe6sfjxM
g
XImhPVxUknWhJUlUopQvsetBe51nqiqhpVONx/GRDXR9gdmGO89gD7XSCy0vHhE
```

```
W
AuoBVNBjbXqmxWdBPdrGcA9zFhdvxzmc6iI4zYe2mQxk1Nt1K6PRXNGjJLIxqch
L
sD7rLVYG1I7+CLGYreJH0siW0Xltbr96qT++1u4tMo1ng1UraoB21zTPVcHA0pJ
u
DLrlXB0GFxVbDHpttOhYDPFZPk4NpzztDuAeNCA5/Oi3JJMjzBRrRuoIH7abmeP
X
qc0Bl1/DAbbiYd5uX01i8ejIveLoeb4OZfLZH/j+bJZT5762Wx0DwkVtm8smk6n
l
+whpAZb5MV6SaS1xEcsRpU+w/O61OPteZ6eIHkU9pDu0yXM6IdtfRpqEw3LKVN/
M
zblGsAq4
=GXp+
-----END PGP SIGNATURE-----
```

The preceding example sends the message in plaintext with a PGP signature appended to the bottom. If you add encryption to protect the confidentiality of the message, the encryption is applied after the message is digitally signed, producing output that appears similar to any other encrypted message. For example, here is that same digitally signed message with encryption added:

```
-----BEGIN PGP MESSAGE-----

owEBBwL4/ZANAwAIAfgm9j3XYejmAaxAYgh0ZXN0LnR4dGAF4N1JIGFtIGVuam9
5
aW5nIG15IHByZXBhcmF0aW9uIGZvciB0aGUgQ0lTU1AgZXhhbS4KCokBswQAAQg
A
HRYhBO+ZLpo4z4bJ9+7JSPgm9j3XYejmBQJgBeDdAAoJEPgm9j3XYejmLfoL/RR
W
oDUl+AeZGffqwnYiJH2gB+Tn+pLjnXAhdf/YV4OsWEsjqKBvItctgcQuSOFJzuO
+
jNgoCAFryi6RrwJ6dTh3F50QJYyJYlgIXCbkyVlaV6hXCZWPT40Bk/pI+HX9A6l
4
J272xabjFf63/HiIEUJDHg/9u8FXKVvBImV3NuMMjJEqx9RcivwvpPn6YLJJ1MW
y
zlUhu3sUIGDWNlArJ4SdskfY32hWAvHkgOAY8JSYmG6L6SVhvbRgv3d+rOOlutq
K
4bVIO+fKMvxycnluPuwmVH99I1Ge8p1ciOMYCVg0dBEP/DeoFlQ4tvKMCPJG0w0
E
ZgLgKyKQpjmNU9BheGvIfzRt1dKYeMx7lGZPlu7rr1Fk0oX/yMiaePWy5NYE2O5
I
D6op9EcJImcMn8wmPM9YTZbmcfcumSpaG1i0EzzAT5eMXn3BoDij12JJrkCCbhY
y
34u2CFR4WycGIIoFHV4RgKqu5TTuV+SCc//vgBaN20Qh9p7gRaNfOxHspto6fA=
=
```

```
=oTCB
-----END PGP MESSAGE-----
```

As you can see, it is not possible to tell that this message is digitally signed until after it is decrypted.

Many commercial providers also offer PGP-based email services as web-based cloud email offerings, mobile device applications, or webmail plug-ins. These services appeal to administrators and end users because they remove the complexity of configuring and maintaining encryption certificates and provide users with a managed secure email service. Some products in this category include Proton Mail, StartMail, Mailvelope, SafeGmail Chrome extension, and Hushmail.

S/MIME

The Secure/Multipurpose Internet Mail Extensions (S/MIME) protocol has emerged as a de facto standard for encrypted email. S/MIME uses the RSA encryption algorithm and has received the backing of major industry players, including RSA Security. S/MIME has already been incorporated in a large number of commercial products, including these:

- Microsoft Outlook and Microsoft 365
- Apple's iCloud Mail
- Google Workspace Enterprise Plus edition

S/MIME relies on the use of X.509 certificates for exchanging cryptographic keys. The public keys contained in these certificates are used for digital signatures and for the exchange of symmetric keys used for longer communications sessions. Users who receive a message signed with S/MIME will be able to verify that message by using the sender's digital certificate. Users who wish to use S/MIME for confidentiality or want to create their own digitally signed messages must obtain their own certificates.

Despite strong industry support for the S/MIME standard, technical limitations have prevented its widespread adoption. Although major desktop mail applications support S/MIME email, mainstream web-based

email systems do not support it out of the box (the use of browser extensions is required).

Web Applications

Encryption is widely used to protect web transactions. This is mainly because of the strong movement toward ecommerce and the desire of both ecommerce vendors and consumers to securely exchange financial information (such as credit card information) over the web. We'll look at the two technologies that are responsible for the small lock icon within web browsers—Secure Sockets Layer (SSL) and Transport Layer Security (TLS).

Secure Sockets Layer (SSL)

SSL was originally developed by Netscape to provide client/server encryption for web traffic sent using the Hypertext Transfer Protocol Secure (HTTPS) over port 443. Over the years, security researchers discovered a number of critical flaws in the SSL protocol that render it insecure for use today. However, SSL serves as the technical foundation for its successor, Transport Layer Security (TLS), which remains widely used today.



Even though TLS has been in existence for more than a decade, many people still mistakenly call it SSL. When you hear people use the term *SSL*, that's a red flag that you should further investigate to ensure that they're really using the modern, secure TLS and not the outdated SSL.

Transport Layer Security (TLS)

TLS relies on the exchange of server digital certificates to negotiate encryption/decryption parameters between the browser and the web server. TLS's goal is to create a secure communications channel that remains open for an entire web browsing session. It depends on a combination of symmetric and asymmetric cryptography. The following steps are involved in a TLS 1.3 connection:

1. When a user accesses a website, their browser and the web server negotiate a cipher suite that is supported by both.
2. The browser retrieves the web server's digital certificate and extracts the server's public key from it.
3. The browser creates a random symmetric key (known as the ephemeral key), uses the web server's public key to encrypt the ephemeral key, and sends the encrypted ephemeral key to the web server.
4. The web server decrypts the ephemeral key using its own private key, and the two systems exchange all future messages using the ephemeral key.

This approach allows TLS to leverage the advanced functionality of asymmetric cryptography while encrypting and decrypting the vast majority of the data exchanged using the faster symmetric algorithm.

When TLS was first proposed as a replacement for SSL, not all browsers supported the more modern approach. To ease the transition, early versions of TLS supported downgrading communications to SSL v3.0 when both parties did not support TLS. However, in 2011, TLS v1.2 dropped this backward compatibility.

In 2014, an attack known as the Padding Oracle On Downgraded Legacy Encryption (POODLE) demonstrated a significant flaw in the SSL 3.0 fallback mechanism of TLS. In an effort to remediate this vulnerability, many organizations completely dropped SSL support and now rely solely on TLS security.

The original version of TLS, TLS 1.0, was simply an enhancement to the SSL 3.0 standard. TLS 1.1, developed in 2006 as an upgrade to TLS 1.0, also contains known security vulnerabilities. TLS 1.2, released in 2008, is now considered the minimum secure option. TLS 1.3, released in 2018, is secure and adds performance improvements. As of 2024, NIST requires that U.S. federal agencies support TLS 1.3 and recommends the same for all other organizations.

It's important to understand that TLS is not an encryption algorithm itself. It is a protocol within which encryption algorithms may function. Therefore, it isn't sufficient to verify that a system is using a secure version of TLS.

Security professionals must also ensure that the algorithms being used with TLS are secure as well.

Each system supporting TLS provides a listing of the cipher suites that it supports. These are combinations of encryption algorithms that it is willing to use together, and these lists are used by two systems to identify a secure option that both systems support. In TLS 1.3, a cipher suite consists of two components:

- The bulk encryption algorithm that will be used for symmetric encryption. For example, a server might support multiple versions of AES and 3DES.
- The hash algorithm that will be used to create message digests. For example, a server might support different versions of the SHA algorithm.

TLS 1.3 cipher suites are usually expressed in long strings that combine both of these elements. For example, the cipher suite:

```
TLS AES_256_CBC_SHA384
```

means that the server supports TLS using AES CBC mode with a 256-bit key for bulk encryption. Hashing will take place using the SHA-384 algorithm.



TLS 1.3 uses variants of the Diffie-Hellman key exchange algorithm. The systems participating in TLS 1.3 communication automatically determine the version of Diffie-Hellman to use and, therefore, the key exchange algorithm is not included in the cipher suite negotiation, as it was in earlier versions of TLS.

Tor and the Dark Web

Tor, formerly known as The Onion Router, provides a mechanism for anonymously routing traffic across the Internet using encryption and a set of relay nodes. It relies on a technology known as *perfect forward secrecy*

(PFS), where layers of encryption prevent nodes in the relay chain from reading anything other than the specific information they need to accept and forward the traffic. By using perfect forward secrecy in combination with a set of three or more relay nodes, Tor allows for both anonymous browsing of the standard Internet, as well as the hosting of completely anonymous sites on the dark web.

Steganography and Watermarking

Steganography is the art of using cryptographic techniques to embed secret messages within another message. Steganographic algorithms work by making alterations to the least significant bits of the many bits that make up image files. The changes are so minor that there is no appreciable effect on the viewed image. This technique allows communicating parties to conceal messages in plain sight—for example, they might embed a secret message within an illustration on an otherwise innocent web page.



It is also possible to embed messages inside larger excerpts of text. This approach is known as a concealment cipher.

Steganographers often embed their secret messages within images or WAV files because these files are often so large that the secret message would easily be missed by even the most observant inspector. Steganography techniques are often used for illegal activities, such as espionage and child pornography.

Steganography can also be used for legitimate purposes, however. Adding digital watermarks to documents to protect intellectual property is accomplished by means of steganography. The hidden information is known only to the file's creator. If someone later creates an unauthorized copy of the content, the watermark can be used to detect the copy and (if uniquely watermarked files are provided to each original recipient) trace the offending copy back to the source.



Steganography commonly works by modifying the least significant bit (LSB) of a pixel value in its binary representation. For example, in the RGB color model, each pixel is described by using three decimal numbers, each ranging from 0 to 255. The first number represents the degree of red color in a pixel, the second represents green, and the third represents blue. If a pixel has the blue value of 64 (binary value of 1000000), changing the LSB to 1 would result in the binary value of 1000001 or the decimal equivalent of 65. This is an imperceptible change but does allow the encoding of a bit of steganographic data.

Steganography is an extremely simple technology to use, with free tools openly available on the Internet. Figure 7.2 shows the entire interface of one such tool, iSteg. It simply requires that you specify a text file containing your secret message and an image file that you wish to use to hide the message. Figure 7.3 shows an example of a picture with an embedded secret message; the message is impossible to detect with the human eye because the text file was added into the message by modifying only the least significant bits of the file. Those do not survive the printing process, and in fact, even if you examined the original full-color, high-resolution digital images, you would not be able to detect the difference.

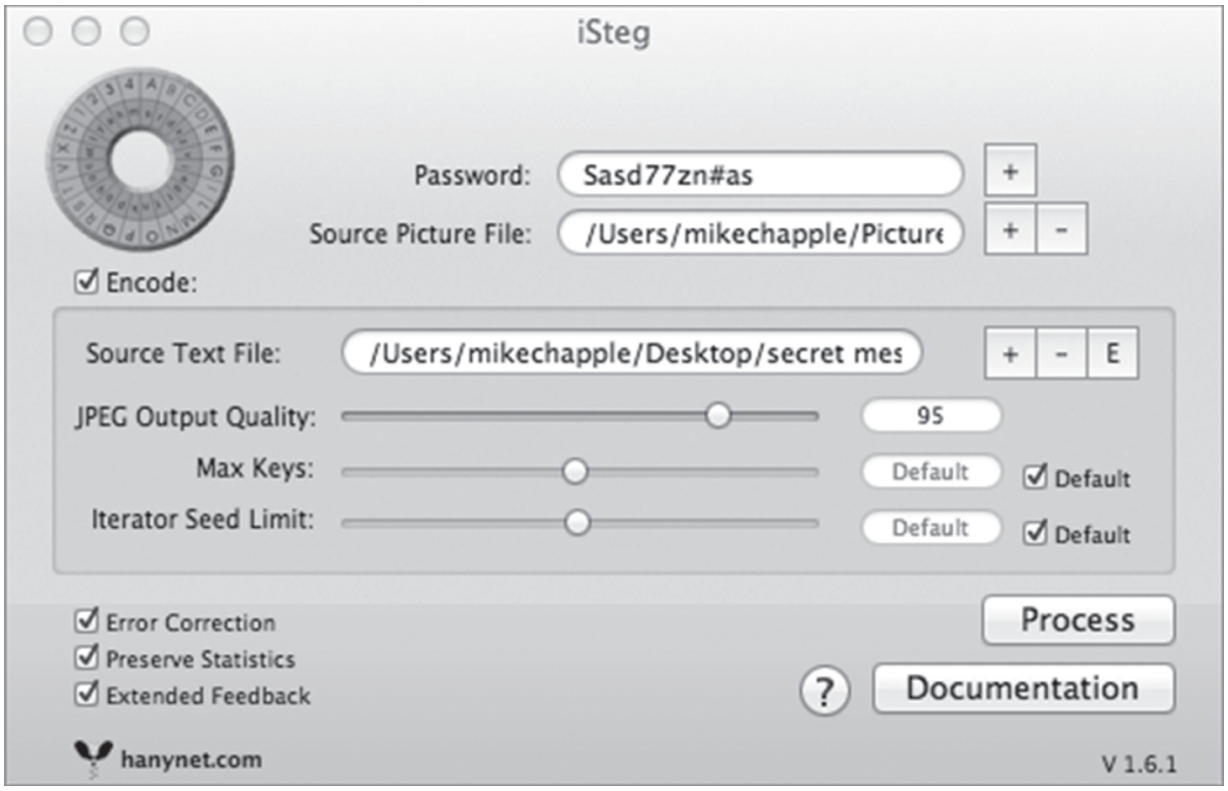


FIGURE 7.2 Steganography tool



FIGURE 7.3 Image with embedded message

Networking

The final application of cryptography we'll explore in this chapter is the use of cryptographic algorithms to provide secure networking services. In the following sections, we'll take a brief look at methods used to secure communications circuits.

Circuit Encryption

Security administrators use two types of encryption techniques to protect data traveling over networks:

- *Link encryption* protects entire communications circuits by creating a secure tunnel between two points using either a hardware solution or a software solution that encrypts all traffic entering one end of the tunnel and decrypts all traffic leaving the other end of the tunnel. For

example, a company with two offices connected via a data circuit might use link encryption to protect against attackers monitoring at a point in between the two offices.

- *End-to-end encryption* protects communications between two parties (for example, a client and a server) and is performed independently of link encryption. An example of end-to-end encryption would be the use of TLS to protect communications between a user and a web server.

The critical difference between link and end-to-end encryption is that in link encryption, all the data, including the header, trailer, address, and routing data, is also encrypted. Therefore, each packet has to be decrypted at each hop so that it can be properly routed to the next hop and then reencrypted before it can be sent along its way, which slows the routing. End-to-end encryption does not encrypt the header, trailer, address, and routing data, so it moves faster from point to point but is more susceptible to sniffers and eavesdroppers.

When encryption happens at the higher OSI layers, it is usually end-to-end encryption, and if encryption is done at the lower layers of the OSI model, it is usually link encryption.

Secure Shell (SSH) is a good example of an end-to-end encryption technique. This suite of programs provides encrypted alternatives to common Internet applications such as the File Transfer Protocol (FTP), Telnet, and rlogin. There are two versions of SSH. SSH-1 is now considered insecure. SSH-2 drops support for some insecure algorithms and adds several security enhancements, including support for the Diffie–Hellman key exchange protocol and the ability to run multiple sessions over a single SSH connection (channel multiplexing). SSH-2 adds support for secure file transfer (SFTP). SSH-2 provides added protection against on-path attacks, eavesdropping, and IP/DNS spoofing.

IPSec

Various security architectures are in use today, each one designed to address security issues in different environments. One such architecture that supports secure communications is the Internet Protocol Security (IPSec) standard. IPSec is a standard architecture set forth by the Internet

Engineering Task Force (IETF) for setting up a secure channel to exchange information between two entities.

The IP Security (IPSec) protocol provides a complete infrastructure for secured network communications. IPSec has gained widespread acceptance and is now offered in several commercial operating systems out of the box. IPSec relies on security associations, and there are two main components:

- The Authentication Header (AH) provides assurances of message integrity. AH also provides authentication and access control and prevents replay attacks.
- The Encapsulating Security Payload (ESP) provides confidentiality and integrity of packet contents. It provides encryption and limited authentication and prevents replay attacks.



ESP also provides some limited authentication, but not to the degree of the AH. Though ESP is sometimes used without AH, it's rare to see AH used without ESP.

IPSec provides for two discrete modes of operation. When IPSec is used in *transport mode* for end-to-end encryption, only the packet payload is encrypted. This mode is designed for peer-to-peer communication. When it's used in *tunnel mode*, the entire packet, including the header, is encrypted. This mode is designed for link encryption.

At runtime, you set up an IPSec session by creating a *security association (SA)*. The SA represents the communication session and records any configuration and status information about the connection. The SA represents a simplex connection. If you want a two-way channel, you need two SAs, one for each direction. Also, if you want to support a bidirectional channel using both AH and ESP, you will need to set up four SAs.

Some of IPSec's greatest strengths come from being able to filter or manage communications on a per-SA basis so that clients or gateways between which security associations exist can be rigorously managed in terms of what kinds of protocols or services can use an IPSec connection. Also,

without a valid security association defined, pairs of users or gateways cannot establish IPSec links.

Further details of the IPSec algorithm are provided in Chapter 11, “Secure Network Architecture and Components.”

Emerging Applications

Cryptography plays a central role in many emerging areas of cybersecurity and technology. Let's look at a few of these concepts: the blockchain, lightweight cryptography, and homomorphic encryption.

Blockchain

The *blockchain* is, in its simplest description, a distributed and immutable public ledger. This means that it can store records in a way that distributes those records among many different systems located around the world and do so in a manner that prevents anyone from tampering with those records. The blockchain creates a data store that nobody can tamper with or destroy.

The first major application of the blockchain is *cryptocurrency*. The blockchain was originally invented as a foundational technology for Bitcoin, allowing the tracking of Bitcoin transactions without the use of a centralized authority. In this manner, the blockchain allows the existence of a currency that has no central regulator. Authority for Bitcoin transactions is distributed among all participants in the Bitcoin blockchain.

Although cryptocurrency is the blockchain application that has received the most attention, there are many other uses for a distributed immutable ledger—so much so that new applications of blockchain technology seem to be appearing every day. For example, property ownership records could benefit tremendously from a blockchain application. This approach would place those records in a transparent, public repository that is protected against intentional or accidental damage. Blockchain technology might also be used to track supply chains, providing grocery consumers, for example, with confidence that their produce came from reputable sources and allowing regulators to easily track down the origin of recalled produce.

Lightweight Cryptography

There are many specialized use cases for cryptography that you may encounter during your career where computing power and energy might be limited.

Some devices operate at extremely low power levels and put a premium on conserving energy. For example, imagine sending a satellite into space with a limited power source. Thousands of hours of engineering go into getting as much life as possible out of that power source. Similar cases happen here on Earth, where remote sensors must transmit information using solar power, a small battery, or other equipment.

Smartcards are another example of a low-power environment. They must be able to securely communicate with smartcard readers but only using the energy either stored on the card or transferred to it by a magnetic field.

In these cases, cryptographers often design specialized hardware that is purpose-built to implement lightweight cryptographic algorithms with as little power expenditure as possible. You won't need to know the details of how these algorithms work, but you should be familiar with the concept that specialized hardware can minimize power consumption.

Another specialized use for cryptography is in cases where you need very low latency. That simply means that the encryption and decryption should not take a long time. Encrypting network links is a common example where low-latency cryptography is desirable. The data is moving quickly across a network, and the encryption should be done as quickly as possible to avoid becoming a bottleneck.

Specialized encryption hardware also fulfills many low-latency requirements. For example, a dedicated VPN hardware device may contain cryptographic hardware that implements encryption and decryption operations in highly efficient form to maximize speed.

High resiliency requirements exist when it is extremely important that data be preserved and not accidentally destroyed during an encryption operation. In cases where resiliency is extremely important, the easiest way to address the issue is for the sender of data to retain a copy until the recipient confirms the successful receipt and decryption of the data.

Homomorphic Encryption

Privacy concerns also introduce some specialized use cases for encryption. In particular, we sometimes have applications where we want to protect the privacy of individuals but still want to perform calculations on their data. *Homomorphic encryption* technology allows this, encrypting data in a way that preserves the ability to perform computation on that data. When you encrypt data with a homomorphic algorithm and then perform computation on that data, you get a result that, when decrypted, matches the result you would have received if you had performed the computation on the plaintext data in the first place.

Cryptographic Attacks

As with any security mechanism, malicious individuals have found a number of attacks to defeat cryptosystems. It's important that you understand the threats posed by various cryptographic attacks to minimize the risks posed to your systems:

Brute-Force Attack Brute-force attacks are quite straightforward. Such an attack attempts every possible valid combination for a key or password. They involve using massive amounts of processing power to methodically guess the key used to secure cryptographic communications.

Analytic Attack This is an algebraic manipulation that attempts to reduce the complexity of the algorithm. Analytic attacks focus on the logic of the algorithm itself.

Implementation Attack This is a type of attack that exploits weaknesses in the implementation of a cryptography system. It focuses on exploiting the hardware or the software code, not just errors and flaws but the methodology employed to program the encryption system.

Statistical Attack A statistical attack exploits statistical weaknesses in a cryptosystem, such as the inability to produce truly random numbers. Statistical attacks may be attempted against a database. Also, a vulnerability in the hardware or operating system hosting the cryptography application may be exploited.

Fault Injection Attack In these attacks, the attacker attempts to compromise the integrity of a cryptographic device by causing some type of external fault. For example, they might use high-voltage electricity, high or

low temperature, or other factors to cause a malfunction that undermines the security of the device.

Side-Channel Attack Computer systems generate characteristic footprints of activity, such as changes in processor utilization, electricity consumption, or electromagnetic radiation. Side-channel attacks seek to use this information to monitor system activity and retrieve information that is actively being encrypted.

Timing Attack Timing attacks are an example of a side-channel attack where the attacker measures precisely how long cryptographic operations take to complete, gaining information about the cryptographic process that may be used to undermine its security.

For a nonflawed protocol, the average amount of time required to discover the key through a brute-force attack is directly proportional to the length of the key. A brute-force attack will always be successful given enough time. Every additional bit of key length doubles the time to perform a brute-force attack because the number of potential keys doubles.

There are two modifications that attackers can make to enhance the effectiveness of a brute-force attack:

- Rainbow tables provide precomputed values for cryptographic hashes. These are commonly used for cracking passwords stored on a system in hashed form.
- Specialized, scalable computing hardware designed specifically for the conduct of brute-force attacks may greatly increase the efficiency of this approach.

Salting Saves Passwords

Salt might be hazardous to your health, but it can save your password. To help combat the use of brute-force attacks, including those aided by dictionaries and rainbow tables, cryptographers make use of a technology known as *cryptographic salt*.

The cryptographic salt is a random value that is added to the end of the password before the operating system hashes the password. The salt is then stored in the password file along with the hash. When the operating system wishes to compare a user's proffered password to the password file, it first retrieves the salt and appends it to the password. It feeds the concatenated value to the hash function and compares the resulting hash with the one stored in the password file.

Specialized password hashing functions, such as PBKDF2 (Password-Based Key Derivation Function 2), bcrypt, and scrypt, allow for the creation of hashes using salts and also incorporate a technique known as *key stretching* that makes it more computationally difficult to perform a single password guess.

The use of salting, especially when combined with key stretching, dramatically increases the difficulty of brute-force attacks. Anyone attempting to build a rainbow table must build a separate table for each possible value of the cryptographic salt.

Frequency Analysis and the Ciphertext-Only Attack In many cases, the only information you have at your disposal is the encrypted ciphertext message, a scenario known as the *ciphertext-only attack*. In this case, one technique that proves helpful against simple ciphers is frequency analysis—counting the number of times each letter appears in the ciphertext. Using your knowledge that the letters *E*, *T*, *A*, *O*, *I*, *N* are the most common in the English language, you can then test several hypotheses:

- If these letters are also the most common in the ciphertext, the cipher was likely a transposition cipher, which rearranged the characters of the plaintext without altering them.

- If other letters are the most common in the ciphertext, the cipher is probably some form of substitution cipher that replaced the plaintext characters.

This is a simple overview of frequency analysis, and many sophisticated variations on this technique can be used against polyalphabetic ciphers and other sophisticated cryptosystems.

Known Plaintext Attack In the known plaintext attack, the attacker has a copy of the encrypted message along with the plaintext message used to generate the ciphertext (the copy). This knowledge greatly assists the attacker in breaking weaker codes. For example, imagine the ease with which you could break the Caesar cipher described in Chapter 6 if you had both a plaintext copy and a ciphertext copy of the same message.

Chosen Plaintext Attack In this attack, the attacker obtains the ciphertexts corresponding to a set of plaintexts of their own choosing. This allows the attacker to attempt to derive the key used and thus decrypt other messages encrypted with that key. This can be difficult, but it is not impossible. Advanced methods such as differential cryptanalysis are types of chosen plaintext attacks.

Chosen Ciphertext Attack In a chosen ciphertext attack, the attacker has access to the algorithm. They have the ability to decrypt chosen portions of the ciphertext message and use the decrypted portion of the message to discover the key.

Meet-in-the-Middle Attack Attackers might use a meet-in-the-middle attack to defeat encryption algorithms that use two rounds of encryption. This attack is the reason that Double DES (2DES) was quickly discarded as a viable enhancement to the DES encryption (it was replaced by Triple DES, or 3DES).

In the meet-in-the-middle attack, the attacker uses a known plaintext message. The plaintext is then encrypted using every possible key (k1), and the equivalent ciphertext is decrypted using all possible keys (k2). When a match is found, the corresponding pair (k1, k2) represents both portions of the double encryption. This type of attack generally takes only double the time necessary to break a single round

of encryption (or 2^n rather than the anticipated $2^n * 2^n$), offering minimal added protection.

Man-in-the-Middle Attack In the man-in-the-middle (MITM) attack, a malicious individual sits between two communicating parties and intercepts all communications (including the setup of the cryptographic session). The attacker responds to the originator's initialization requests and sets up a secure session with the originator. The attacker then establishes a second secure session with the intended recipient using a different key and posing as the originator. The attacker can then "sit in the middle" of the communication and read all traffic as it passes between the two parties.



Be careful not to confuse the meet-in-the-middle attack with the man-in-the-middle attack. They may have similar names, but they are quite different.

Birthday Attack The birthday attack, also known as a *collision attack* or *reverse hash matching* (see the discussion of brute-force and dictionary attacks in Chapter 14, "Controlling and Monitoring Access"), seeks to find flaws in the one-to-one nature of hashing functions. In this attack, the malicious individual seeks to replace the content of a digitally signed communication with a different message that produces the same message digest, thereby maintaining the validity of the original digital signature.



Don't forget that social engineering techniques can also be used in cryptanalysis. If you're able to obtain a decryption key by simply asking the sender for it, that's much easier than attempting to crack the cryptosystem.

Replay Attack The replay attack is used against cryptographic algorithms that don't incorporate temporal protections. In this attack, the malicious individual intercepts an encrypted message between two

parties (often a request for authentication) and then later “replays” the captured message to open a new session. This attack can be defeated by incorporating a timestamp and expiration period into each message, using a challenge-response mechanism, and encrypting authentication sessions with ephemeral session keys.



Many other attacks make use of cryptographic techniques as well. For example, Chapter 14 describes the use of cryptographic techniques in pass-the-hash and Kerberos exploitation, and Chapter 21, “Malicious Code and Application Attacks,” describes the use of cryptography in ransomware attacks.

Summary

Asymmetric key cryptography, or public key encryption, provides an extremely flexible infrastructure, facilitating simple, secure communication between parties that do not necessarily know each other prior to initiating the communication. It also provides the framework for the digital signing of messages to ensure nonrepudiation and message integrity.

This chapter explored public key encryption, which provides a scalable cryptographic architecture for use by large numbers of users. We also described some popular cryptographic algorithms, and the use of link encryption and end-to-end encryption. We introduced you to the public key infrastructure, which uses certificate authorities (CAs) to generate digital certificates containing the public keys of system users and digital signatures, which rely on a combination of public key cryptography and hashing functions. You also learned how to use the PKI to obtain integrity and nonrepudiation through the use of digital signatures. You learned how to ensure consistent security throughout the cryptographic life cycle by adopting key management practices and other mechanisms.

We also looked at some of the common applications of cryptographic technology in solving everyday problems. You learned how cryptography can be used to secure email (using PGP and S/MIME), web

communications (using TLS), and both peer-to-peer and gateway-to-gateway networking (using IPSec).

Finally, we covered some of the more common attacks used by malicious individuals attempting to interfere with or intercept encrypted communications between two parties. Such attacks include cryptanalytic, replay, brute-force, known plaintext, chosen plaintext, chosen ciphertext, on-path, man-in-the-middle, and birthday attacks. It's important for you to understand these attacks in order to provide adequate security against them.

Study Essentials

Understand the key types used in asymmetric cryptography. Public keys are freely shared among communicating parties, whereas private keys are kept secret. To encrypt a message, use the recipient's public key. To decrypt a message, use your own private key. To sign a message, use your own private key. To validate a signature, use the sender's public key.

Be familiar with the three major public key cryptosystems. RSA is the most famous public key cryptosystem; it was developed by Rivest, Shamir, and Adleman in 1977. It depends on the difficulty of factoring the product of prime numbers. ElGamal is an extension of the Diffie–Hellman key exchange algorithm that depends on modular arithmetic. Elliptic curve cryptography depends on the elliptic curve discrete logarithm problem and provides more security than other algorithms when both are used with keys of the same length.

Know the fundamental requirements of a hash function. Good hash functions have five requirements. They must allow input of any length, provide fixed-length output, make it relatively easy to compute the hash function for any input, provide one-way functionality, and be collision-resistant.

Be familiar with the major hashing algorithms. The Secure Hash Algorithm SHA-3 is the government standard message digest function. SHA-2 supports variable-length message digests, ranging up to 512 bits. SHA-3 improves upon the security of SHA-2 and supports the same hash lengths.

Know how cryptographic salts improve the security of password hashing. When straightforward hashing is used to store passwords in a password file, attackers may use rainbow tables of precomputed values to identify commonly used passwords. Adding salts to the passwords before hashing them reduces the effectiveness of rainbow table attacks. Common password hashing algorithms that use key stretching to further increase the difficulty of attack include PBKDF2, bcrypt, and scrypt.

Understand how digital signatures are generated and verified. To digitally sign a message, first use a hashing function to generate a message digest; then encrypt the digest with your private key. To verify the digital signature on a message, decrypt the signature with the sender's public key and then compare the original message digest to one you generate yourself. If they match, the message is authentic.

Understand the public key infrastructure (PKI). In the public key infrastructure, certificate authorities (CAs) generate digital certificates containing the public keys of system users. Users then distribute these certificates to people with whom they want to communicate. Certificate recipients verify a certificate using the CA's public key.

Know the common applications of cryptography to secure email. The emerging standard for encrypted messages is the S/MIME protocol. Another popular email security tool is Phil Zimmermann's Pretty Good Privacy (PGP). Most users of email encryption rely on having this technology built into their email client or their web-based email service.

Know the common applications of cryptography to secure web activity. The de facto standard for secure web traffic is the use of HTTP over Transport Layer Security (TLS). This approach relies on hybrid cryptography using asymmetric cryptography to exchange an ephemeral session key, which is then used to carry on symmetric cryptography for the remainder of the session.

Know the common applications of cryptography to secure networking. The IPSec protocol standard provides a common framework for encrypting network traffic and is built into a number of common operating systems. In IPSec transport mode, packet contents are encrypted for peer-to-peer communication. In tunnel mode, the entire packet,

including header information, is encrypted for gateway-to-gateway communications.

Be able to describe IPSec. IPSec is a security architecture framework that supports secure communication over IP. IPSec establishes a secure channel in either transport mode or tunnel mode. It can be used to establish direct communication between computers or to set up a VPN between networks. IPSec uses two protocols: Authentication Header (AH) and Encapsulating Security Payload (ESP).

Be able to explain common cryptographic attacks. Ciphertext-only attacks require access only to the ciphertext of a message. One example of a ciphertext-only attack is the brute-force attack, which attempts to randomly find the correct cryptographic key. Frequency analysis, another ciphertext-only attack, counts characters in the ciphertext to reverse substitution ciphers. Known plaintext, chosen ciphertext, and chosen plaintext attacks require the attacker to have some extra information in addition to the ciphertext. The on-path attack fools both parties into communicating with the attacker instead of directly with each other. The birthday attack is an attempt to find collisions in hash functions. The replay attack is an attempt to reuse authentication requests.

Written Lab

1. Explain the process Bob should use if he wants to send a confidential message to Alice using asymmetric cryptography.
2. Explain the process Alice would use to decrypt the message Bob sent in question 1.
3. Explain the process Bob should use to digitally sign a message to Alice.
4. Explain the process Alice should use to verify the digital signature on the message from Bob in question 3.

Review Questions

1. Brian computes the digest of a single sentence of text using a SHA-2 hash function. He then changes a single character of the sentence and computes the hash value again. Which one of the following statements is true about the new hash value?

    A. The new hash value will be one character different from the old hash value.

    B. The new hash value will share at least 50 percent of the characters of the old hash value.

    C. The new hash value will be unchanged.

    D. The new hash value will be completely different from the old hash value.

2. Alan believes that an attacker is collecting information about the electricity consumption of a sensitive cryptographic device and using that information to compromise encrypted data. What type of attack does he suspect is taking place?

    A. Brute-force

    B. Side-channel

    C. Known plaintext

    D. Frequency analysis

3. If Richard wants to send an encrypted message to Sue using a public key cryptosystem, which key does he use to encrypt the message?

    A. Richard's public key

    B. Richard's private key

    C. Sue's public key

    D. Sue's private key

4. If a 2,048-bit plaintext message were encrypted with the ElGamal public key cryptosystem, how long would the resulting ciphertext message be?

    A. 1,024 bits

    B. 2,048 bits

C. 4,096 bits

D. 8,192 bits

5. Acme Widgets currently uses a 3,072-bit RSA encryption standard companywide. The company plans to convert from RSA to an elliptic curve cryptosystem. If the company wants to maintain the same cryptographic strength, what ECC key length should it use?

   A. 256 bits

   B. 512 bits

   C. 1,024 bits

   D. 2,048 bits

6. John wants to produce a message digest of a 2,048-byte message he plans to send to Mary. If he uses the SHA-2 hashing algorithm, what is a possible size for the message digest generated?

   A. 160 bits

   B. 512 bits

   C. 1,024 bits

   D. 2,048 bits

7. After conducting a survey of encryption technologies used in her organization, Melissa suspects that some may be out of date and pose security risks. Which one of the following technologies is considered flawed and should no longer be used?

   A. SHA-3

   B. TLS 1.3

   C. IPSec

   D. SSL 3.0

8. You are developing an application that compares passwords to those stored in a Unix password file. The hash values you compute are not correctly matching those in the file. What might have been added to the stored password hashes?

A. Salt

B. Double hash

C. Added encryption

D. One-time pad

9. Richard received an encrypted message sent to him from Sue. Sue encrypted the message using the RSA encryption algorithm. Which key should Richard use to decrypt the message?

    A. Richard's public key

    B. Richard's private key

    C. Sue's public key

    D. Sue's private key

10. Richard wants to digitally sign a message he's sending to Sue so that Sue can be sure the message came from him without modification while in transit. Which key should he use to encrypt the message digest?

    A. Richard's public key

    B. Richard's private key

    C. Sue's public key

    D. Sue's private key

11. Which one of the following algorithms is not supported by the Digital Signature Standard under FIPS 186-5?

    A. Edwards-Curve DSA

    B. RSA

    C. ElGamal DSA

    D. Elliptic Curve DSA

12. Which International Telecommunications Union (ITU) standard governs the creation and endorsement of digital certificates for secure electronic communication?

A. X.500

B. X.509

C. X.900

D. X.905

13. Ron believes that an attacker accessed a highly secure system in his data center and applied high-voltage electricity to it in an effort to compromise the cryptographic keys that it uses. What type of attack does he suspect?

    A. Implementation attack

    B. Fault injection

    C. Timing

    D. Chosen ciphertext

14. Brandon is analyzing network traffic and is searching for user attempts to access websites over secure TLS connections. What TCP port should Brandon add to his search filter because it would normally be used by this traffic?

    A. 22

    B. 80

    C. 443

    D. 1433

15. Beth is assessing the vulnerability of a cryptographic system to attack. She believes that the cryptographic keys are properly secured and that the system is using a modern, secure algorithm. Which one of the following attacks would most likely still be possible against the system by an external attacker who did not participate in the system and did not have physical access to the facility?

    A. Ciphertext-only

    B. Known plaintext

    C. Chosen plaintext

D. Fault injection

16. Which of the following tools can be used to improve the effectiveness of a brute-force password cracking attack?

    A. Rainbow tables

    B. Hierarchical screening

    C. TKIP

    D. Random enhancement

17. Chris is searching a Windows system for binary key files and wishes to narrow his search using file extensions. Which one of the following certificate formats is closely associated with Windows binary certificate files?

    A. CCM

    B. PEM

    C. PFX

    D. P7B

18. What is the major disadvantage of using certificate revocation lists?

    A. Key management

    B. Latency

    C. Record keeping

    D. Vulnerability to brute-force attacks

19. Which one of the following encryption algorithms is now considered insecure?

    A. Advanced Encryption Standard

    B. RSA

    C. Elliptic Curve Cryptography

    D. Merkle–Hellman Knapsack

20. Brian is upgrading a system to support SSH-2 rather than SSH-1. Which one of the following advantages will he achieve?

A. Support for multifactor authentication

B. Support for simultaneous sessions

C. Support for 3DES encryption

D. Support for IDEA encryption

Chapter 8
Principles of Security Models, Design, and Capabilities

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 3.0: Security Architecture and Engineering**
  - 3.1 Research, implement and manage engineering processes using secure design principles
    - 3.1.4 Secure defaults
    - 3.1.5 Fail securely
    - 3.1.7 Keep it simple and small
    - 3.1.8 Zero trust or trust but verify
    - 3.1.9 Privacy by design
    - 3.1.11 Secure access service edge
  - 3.2 Understand the fundamental concepts of security models (e.g. Biba, Star Model, Bell-LaPadula)
  - 3.3 Select controls based upon systems security requirements
  - 3.4 Understand security capabilities of Information Systems (IS) (e.g., memory protection, Trusted Platform Module (TPM), encryption/decryption)
  - 3.10 Manage the information system lifecycle
    - 3.10.1 Stakeholders needs and requirements
    - 3.10.2 Requirements analysis
    - 3.10.3 Architectural design
    - 3.10.4 Development/implementation
    - 3.10.5 Integration
    - 3.10.6 Verification and validation
    - 3.10.7 Transition/deployment
    - 3.10.8 Operations and maintenance/sustainment
    - 3.10.9 Retirement/disposal

Understanding the philosophy behind security solutions helps limit your search for the best controls for your specific security needs. In this chapter, we discuss secure system design principles, security models, the Common Criteria, and security capabilities of information systems.

Domain 3 includes a variety of topics that are discussed in other chapters, including the following:

- Chapter 1, “Security Governance Through Principles and Policies”
- Chapter 6, “Cryptography and Symmetric Key Algorithms”
- Chapter 7, “PKI and Cryptographic Applications”
- Chapter 8, “Principles of Security Models, Design, and Capabilities”
- Chapter 9, “Security Vulnerabilities, Threats, and Countermeasures”
- Chapter 10, “Physical Security Requirements”
- Chapter 14, “Controlling and Monitoring Access”
- Chapter 16, “Managing Security Operations”
- Chapter 20, “Software Development Security”
- Chapter 21, “Malicious Code and Application Attacks”

Secure Design Principles

Security should be a consideration at every stage of a system's development. Programmers, developers, engineers, and so on should strive to build security into every application or system they develop, with greater levels of security provided to critical applications and those that process sensitive information. It's imperative to consider the security implications of a development project in the early stages because it's much easier to build security into a system during development than adding security to an existing system. Developers should research, implement, and manage engineering processes using secure design principles.

Objects and Subjects

Controlling access to any resource in a secure system involves two entities. The *subject* is the active entity that requests access to a resource. A subject is commonly a user, but it can also be a process, program, computer, or organization. The *object* is the passive entity that the subject wants to access. An object is commonly a resource, such as a file or printer, but it can also be a user, process, program, computer, or organization. You want to keep a broad understanding of the terms subject and object, rather than only considering users and files. Access is the relationship between a subject and object, including reading, writing, modifying, deleting, printing, moving, backing up, and many other operations or activities. Authorization or access control is the management of the relationship between subjects and objects.

Remember that the actual entities referenced by the terms *subject* and *object* are specific to an individual access request. The entity serving as the object in one access event could serve as the subject in another. For example, process A may ask for data from process B. To satisfy process A's request, process B must ask for data from process C. In this example (Table 8.1), process B is the object of the first request and the subject of the second request.

TABLE 8.1 Subjects and objects

Request	Subject	Object
First request	Process A	Process B
Second request	Process B	Process C

This also serves as an example of transitive trust. *Transitive trust* is the concept that if A trusts B and B trusts C, then A inherits the trust of C through the transitive property (Figure 8.1)—which works as it would in a mathematical equation: if a = b and b = c, then a = c. In the previous example, when A requests data from B and then B requests data from C, the data that A receives is essentially from C. Transitive trust is a serious security concern because it may enable the bypassing of restrictions or limitations between A and C, especially if A and C both support interaction with B. An example would be when an organization blocks access to Facebook or YouTube to increase worker productivity. Thus, workers (A) do not have access to certain Internet sites (C). However, if workers are able

to have access to a web proxy, virtual private network (VPN), or anonymization service, then this can serve as a means to bypass the local network restriction. In other words, if workers (A) are accessing VPN service (B), and the VPN service (B) can access the blocked internet service (C), then A can access C through B via a transitive trust exploitation.

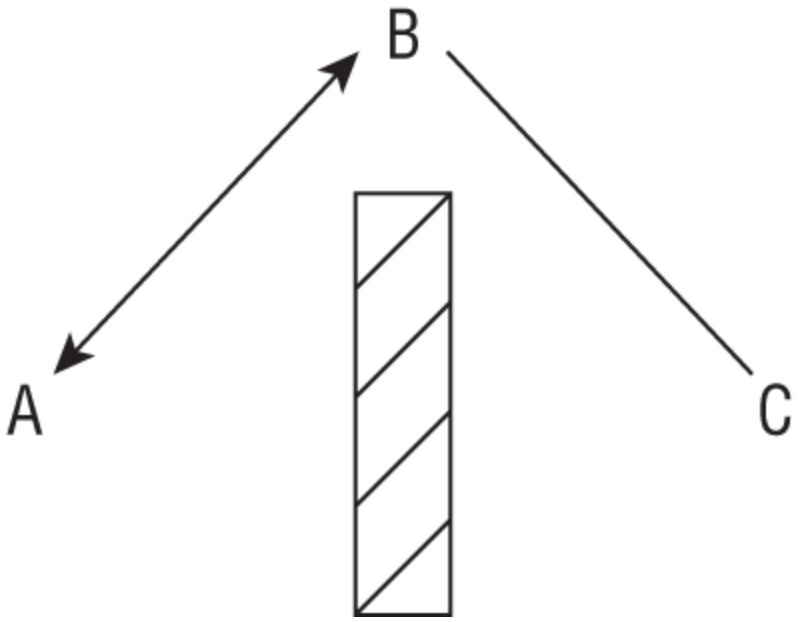


FIGURE 8.1 Transitive trust

Closed and Open Systems

Systems are designed and built according to one of two differing philosophies. A *closed system* is designed to work well with a narrow range of other systems, generally all from the same manufacturer. The standards for closed systems are often proprietary and not normally disclosed. *Open systems*, on the other hand, are designed using agreed-on industry standards. Open systems are much easier to integrate with systems from different manufacturers that support the same standards or that use compatible application programming interfaces (APIs).

An API is a defined set of interactions allowed between computing elements, such as applications, services, networking, firmware, and hardware. An API defines the types of requests that can be made, the exact means to make the requests, the data forms of the exchange, and other related requirements (such as authentication and/or session encryption). APIs make interoperability of computing elements possible. Without APIs,

computing components would be unable to interact directly, and information sharing would not be easy. APIs are what make modern computing and the Internet possible. The app on your smartphone talks to the phone's operating system via an API; the phone's operating system talks over the telco or Wi-Fi network via an API to reach the cloud service's API to submit a request and receive a response.

Closed systems are harder to integrate with unlike systems, but this “feature” could make them more secure. A closed system is often composed of proprietary hardware and software that does not incorporate industry standards or offer an open API. This lack of integration ease means that attacks that typically focus on generic system components either will not work or must be customized to be successful. In many cases, attacking a closed system is harder than launching an attack on an open system, since a unique exploit of a unique vulnerability would be required. In addition to the lack of known vulnerable components on a closed system, it is often necessary to possess more in-depth knowledge of the specific target system to launch a successful attack.

Open systems are generally far easier to integrate with other open systems. It is easy, for example, to create a local area network (LAN) with a Microsoft Windows Server machine, a Linux machine, and a Macintosh machine. Although all three computers use different operating systems and could represent up to three different hardware architectures, each supports industry standards and open APIs, which makes it easy for network (or other) communications to occur. This ease of interoperability comes at a price, however. Because standard communications components are incorporated into each of these three open systems, there are far more predictable entry points and methods for launching attacks. In general, their openness makes them more vulnerable to attack, and their widespread availability makes it possible for attackers to find plenty of potential targets. Also, open systems are more popular and widely deployed than closed systems and thus attract more attention from attackers. An attacker who develops basic attacking skills will find more targets that are open systems than closed ones. Inarguably, there's a greater body of shared experience and knowledge on how to attack open systems than there is for closed systems. The security of an open system is therefore more dependent on the

use of secure and defensive coding practices and a thoughtful defense-in-depth deployment strategy (see Chapter 1).

Open Source vs. Closed Source

It's also helpful to keep in mind the distinction between open-source and closed-source systems. An *open-source* solution is one where the source code, and other internal logic, is exposed to the public. A closed-source solution is one where the source code and other internal logic is hidden from the public. Open-source solutions often depend on public inspection and review to improve the product over time. *Closed-source* solutions are more dependent on the vendor/programmer to revise the product over time. Both open-source and closed-source solutions can be available for sale or at no charge, but the term *commercial* typically implies closed-source. However, closed-source code is sometimes revealed through either vendor compromise or through decompiling or disassembly. The former is always a breach of ethics and often the law, whereas the latter is a standard element in ethical reverse engineering or systems analysis.

It is also the case that a closed-source program can be either an open system or a closed system, and an open-source program can be either an open system or a closed system. Since these terms are so similar, it is essential to read questions carefully. Additional coverage of open-source and other software issues is included in Chapter 20, “Software Development Security.”



CISSP Objective 3.1 lists 11 secure design principles. Six of them are covered in this chapter (i.e., secure defaults, fail securely, keep it simple and small, zero trust or trust but verify, privacy by design, and secure access service edge); the other five are covered in other chapters where they integrate best with broader coverage of similar topics. For threat modeling and defense in depth, see Chapter 1; for least privilege and segregation of duties, see Chapter 16; and for shared responsibility, see Chapter 9.

Secure Defaults

You have probably heard the phrase “the tyranny of the default.” But do you know what this means? Tyranny has several definitions, but the one that applies here is “a rigorous condition imposed by some outside agency or force” (attributed to American historian Dixon Wecter). Many assume that the settings present in a software or hardware product when it is first installed are optimal. This is based on the assumption that the designers and developers of a product know the most about that product, so the settings they made are likely the best ones. However, this assumption overlooks the fact that often, the default settings of a product are selected to minimize installation problems to avoid increased load on the technical support services. For example, consider the fact that most devices have a default password, which minimizes the costs of support when installing or using the product for the first time. Unfortunately, default settings often make the discovery and exploitation of equipment trivial for attackers.

Never assume that the default settings of any product are secure. They typically are not because secure settings would likely get in the way of existing business tasks or system operations. It is always up to the system's administrator and/or company security staff to alter a product's settings to comply with the organization's security policies. Unless your organization hired the developer, that developer did not craft the code or choose settings specifically for your organization's use of their product.

A much better assumption is that the default settings of a product are the worst possible options for your organization. Therefore, you need to review every setting to determine what it does and what you need configured to do to optimize security while supporting business operations.

Fortunately, there is some movement toward more *secure defaults*. Some products, especially security products, may now be designed with their most secure settings enabled by default. However, such a locked-down product will have fewer enabled capabilities and will likely be less user-friendly. Thus, while being more secure, secure defaults may be an obstacle for those who only want their systems to “just work.”

If you are a developer, you must create detailed explanations of each of your product's configuration options. You can't assume that customers know everything about your product, especially the configuration settings and what each option does to alter its features, operations, communications, etc. You may be required to have default settings to make the product as easy to install as possible. Still, you may be able to provide one or more configurations in either written instructional form or in a file that can be imported or applied. This will go a long way to assist customers with gaining the most advantage from your product while minimizing the security risks.



Restrictive defaults refer to a practice or policy where the default settings or options are intentionally configured to be more limiting or restrictive to enhance security, privacy, or compliance. This concept is often applied in software, systems, or services where administrators or users are provided with a preconfigured set of options and the default settings prioritize security, safety, or regulatory compliance.

Fail Securely

System failures can occur due to a wide range of causes. Once the failure event occurs, how the system or environment handles the failure is important. The most desired result is for an application to *fail securely*. The

first type of failure management is programmatic error handling (aka *exception handling*). This is the process where a programmer codes in mechanisms to anticipate and defend against errors to avoid the termination of execution. Error handling includes code that will attempt to handle errors when they arise before they can cause harm or interrupt execution.

One such mechanism, supported by many languages, is a `try..catch` statement. This logical block statement is used to place code that could result in an error on the `try` branch and then code that will be executed if there is an error on the `catch` branch. This is similar to `if..then..else` statements, but it is designed to handle errors deftly.

Other mechanisms are to avoid or prevent errors, especially regarding user input. Input sanitization, input filtering, and input validation are terms used to refer to this concept. This often includes checking the input for length, filtering against a block list of unwanted input, and escaping metacharacters. See more about secure coding practices in Chapter 9; Chapter 15, "Security Assessment and Testing"; and Chapter 20.

There are several similar terms that can be confusing and thus require a bit of focus to comprehend. These terms are fail-soft, fail-secure, fail-safe, fail-open, and fail-closed. Typically, confusion occurs when not understanding the context where these terms are used. The two primary contexts are the physical world and the digital environment. In the physical world, entities primarily prioritize the protection of people. However, there are some circumstances where assets are protected in priority over people. In the digital world, entities focus on protecting assets, but the type of protection may vary among the CIA Triad.

When a program fails securely, it is able to do so only because it was designed and programmed to. When secure failure is integrated into a system, the designer must make a few difficult choices about what the results of a failure event will be. The first question to be resolved is whether the system can operate in a fail-soft mode. To *fail-soft* is to allow a system to continue to operate after a component fails. This is an alternative to having a failure cause a complete system failure. An example is a typical multitasking operating system that can support numerous simultaneous applications. If one application fails, the others can typically continue to operate.

If fail-soft isn't a viable option, then the designer needs to consider the type of product, its deployment scenarios, and the priorities related to failure response. In other words, when the product fails without a fail-soft design, it will fail completely. The designer/developer must decide what type of complete failure to perform and what to protect or sacrifice to achieve the planned failure result. There are numerous scenarios to consider. The initial distinction is whether the product is something that affects the physical world, such as a door-locking mechanism, or primarily a digital asset–focused product, such as a firewall. If a product can affect the physical world, then the life and safety of humans must be considered and likely prioritized. This human protection prioritization is called *fail-safe*. The idea is that when a failure occurs, the system, device, or product will revert to a state that protects the health and safety of people. For example, a fail-safe door will open easily in an emergency to allow people to escape a building. However, this implies that the protection of assets may be sacrificed in favor of personnel safety. However, in some physical world situations, a product could be designed and intended to protect assets in priority above people, such as a bank vault, medical lab, or even a data center. A fail-secure system prioritizes the physical security of assets over any other consideration. For example, a vault door may automatically close and lock when the building enters a state of emergency. This prioritization of asset protection may occur at the potential cost of harming personnel who could be trapped inside. Obviously, the prioritization of physical world products should be considered carefully. In the context of the physical world, the term *fail-open* is a synonym for fail-safe, and fail-closed is a synonym for fail-secure.

If the product is primarily digital, then the focus of security is completely on digital assets. That means the designer must then prioritize the security aspect—namely, availability or confidentiality and integrity. If the priority is for maintaining availability, then when the product fails, the connection or communication is allowed to continue. This is known as fail-open. If the priority is for maintaining confidentiality and integrity, then when the product fails, the connection or communication is cut off. This is known as fail-secure, fail-closed, and/or fail-safe (again, in the context of a digital environment).



The Internet Engineering Task Force (IETF) recommends avoiding using the term fail-safe when discussing digital-only issues as it introduces the concept of human safety, which is not a concern in a digital context and thus causes unnecessary confusion.

However, when the context switches from the physical world to the digital world, the definition of fail-safe changes. An example could be a firewall, which, if designed to fail-open, would allow communications without filtering. In contrast, implementing a fail-secure, fail-closed, or fail-safe solution would cut off communications. The fail-open state protects availability by sacrificing confidentiality and integrity, whereas the fail-closed state sacrifices availability to preserve confidentiality and integrity. Another example of a digital environment event following a fail-secure, fail-closed, and/or fail-safe procedure is when an operating system encounters a processing or memory isolation violation, it terminates all executions then initiates a reboot. This mechanism is known as a stop error or the Blue Screen of Death (BSoD) in Windows.

A condensed summary of the context and protection priority of these terms is presented in Table 8.2.

TABLE 8.2 Fail terms' definitions related to physical and digital products

Physical	State	Digital
Protect People	Fail-Open	Protect Availability
Protect People	Fail-Safe	Protect Confidentiality and Integrity
Protect Assets	Fail-Closed	Protect Confidentiality and Integrity
Protect Assets	Fail-Secure	Protect Confidentiality and Integrity

Keep It Simple and Small

Keep it simple is a shortened form of the classic statement “keep it simple, stupid” or “keep it stupid simple.” This is sometimes called the KISS principle. In security, this concept encourages avoiding overcomplicating the environment, organization, or product design. The more complex a

system, the more difficult it is to secure. The more lines of code, the more challenging it is to test it thoroughly. The more parts there are, the more places there are for things to go wrong. The more features and capabilities, the larger the attack surface. Thus, keeping a system's design or software coding simple and small will directly relate to the difficulty of establishing, testing, and verifying its security.

There are many other concepts that have a similar or related emphasis, such as the following:

“Don't Repeat Yourself” (DRY) The idea is to eliminate redundancy in software by not repeating the same code in multiple places. Otherwise, duplicating code would increase the difficulty if changes are needed.

Computing Minimalism Crafting code to use the least necessary hardware and software resources possible is computing minimalism. This is also the goal of the program evaluation and review technique (PERT), which is discussed in Chapter 20.

Rule of Least Power Use the least powerful programming language that is suitable for the needed solution.

“Worse Is Better” (aka New Jersey Style) The quality of software does not necessarily increase with increased capabilities and functions; there is often a worse software state (i.e., fewer functions), which is the better (i.e., preferred, maybe more secure) option.

“You Aren't Gonna Need It” (YAGNI) Programmers should not write capabilities and functions until they are necessary, so rather than create them when you think of them, create them only when you need them.

It is easy to get caught up in adding complexity to a system, whether that system is a software program or an organizational IT security structure. The KISS principle encourages us all to avoid the overly complex in favor of the streamlined, optimized, and reduced solution. Simpler solutions are easier to secure, easier to troubleshoot, and easier to verify.

Zero-Trust

Zero trust is a security concept where nothing and no person inside the organization is automatically trusted. There has long been an assumption that everything on the inside is trusted and everything on the outside is untrusted. This has led to a significant security focus on endpoint devices, the locations where users interact with company resources. An endpoint device could be a user's workstation, a tablet, a smartphone, an Internet of Things (IoT) device, an industrial control system (ICS), an edge computing sensor, or any public-facing servers in a screened subnet or extranet. The idea that a security perimeter exists between the safe inside and the harmful outside is problematic. There have been too many occurrences of security breaches caused by insiders as well as external attacker breaches that gained the freedom to perform lateral movement internally once they breached the security barrier.

The concept of a security perimeter is further complicated by the use of mobile devices, the cloud, and the proliferation of endpoint devices. If a device can operate inside a private network, then be used externally with direct internet access, and then returned to the private network, there is no actual security perimeter. For most organizations, there is no longer a clearly defined line between inside and outside.

Zero trust is an alternate approach to security where nothing and no person is automatically trusted. Instead, each request for activity or access is assumed to be from an unknown and untrusted location until otherwise verified. The concept is “never trust, always verify.” Since anyone and anything could be malicious, every transaction should be verified before it is allowed to occur. The zero-trust model is based on “assume breach,” meaning that you should always assume a security breach has occurred and that whoever or whatever is making a request could be malicious. The goal is to have every access request be authenticated, authorized, and encrypted prior to the access being granted to a resource or asset. The implementation of a zero-trust architecture does involve a significant shift from historical security management concepts. This shift typically requires internal microsegmentation and strong adherence to the principle of least privilege. This approach prevents lateral movement so that if there is a breach or even a malicious insider, their ability to move about the environment is severely restricted.



Microsegmentation is dividing up an internal network into numerous subzones. Each zone is separated from the others by internal segmentation firewalls (ISFWs), subnets, or virtual local area network (VLANs). Zones could be as small as a single device, such as a high-value server or even a client or endpoint device. Any and all communications between zones are filtered, may be required to authenticate, often require session encryption, and may be subjected to allow list and block list control.

Zero trust is implemented using a wide range of security solutions, including internal segmentation firewalls (ISFWs), multifactor authentication (MFA), identity and access management (IAM), and next-generation endpoint security (see Chapter 11). A zero-trust approach to security management can only be successful if a means to validate and monitor user activities continuously is implemented. If a one-time validation mechanism is used, then the opportunity to abuse the system remains since threats, users, and connection characteristics are always subject to change. Thus, zero-trust networking can only work if real-time vetting and visibility into user activities are maintained.

A summary of zero trust is that all devices are segmented and isolated from each other to prevent any and all communications. Then, inter-device transactions must be authorized, authenticated, encrypted, monitored/analyzed, and logged.



In some situations, complete isolation may be needed instead of controlled and filtered interaction. This type of isolation is achieved using an air gap. An *air gap* is a network security measure employed to ensure that a secure system is physically isolated from other systems. Air gap implies that neither cabled nor wireless network links are available. Due to the proliferation of wireless connectivity options, a Faraday cage may be necessary to enforce air gap isolation (see Chapter 8).

To implement a zero-trust system, an organization must be capable of and willing to abandon some long-held assumptions about security. First and foremost, it must be understood that there is no such thing as a trusted source. No entity, asset, or subject—internal or external—is to be trusted by default. Instead, always assume attackers are already on the inside, on every system. From this new "no assumed trust" position, it is obvious that traditional default access controls are insufficient. Each and every subject, each and every time, needs to be authenticated, authorized, and encrypted. From there, a continuous real-time monitoring system should be established to look for violations and suspicious events. But even with zero trust integrated into the IT architecture, it is only an element of a holistic security strategy that is integrated into the entire organization's management processes.

Zero trust has been formalized in NIST SP 800-207, "Zero Trust Architecture." Please consult this document to learn more about this revolution in security design.

Trust but Verify

The phrase "trust but verify" (a quote from a Russian proverb) was made famous by former president Ronald Reagan when discussing U.S. relations with the Soviet Union. However, our focus on this phrase is on its use in the security realm. A more traditional security approach of trusting subjects and devices within the company's security perimeter (i.e., internal entities) automatically can be called "*trust but verify*." This type of security

approach leaves an organization vulnerable to insider attacks and grants intruders the ability to easily perform lateral movement among internal systems. Often, the trust but verify approach depends on an initial authentication process to gain access to the internal “secured” environment and then relies on generic access control methods. Due to the rapid growth and changes in the modern threatscape, the trust but verify model of security is no longer sufficient. Most security experts now recommend designing organizational security around the zero-trust model. So, in regard to the question of “zero trust or trust but verify?”, today's answer should only be zero trust.

Privacy by Design

Privacy by design (PbD) is a guideline to integrate privacy protections into products during the early design phase rather than attempting to tack it on at the end of development. It is effectively the same overall concept as “security by design” or “integrated security,” where security is to be an element of the design and architecture of a product starting at initiation and being maintained throughout the software development life cycle (SDLC).

As described in Ann Cavoukian's paper “Privacy by Design – The 7 Foundational Principles: Implementation and Mapping of Fair Information Practices,” the PbD framework is based on seven foundational principles:

- Proactive, not reactive; preventive, not remedial
- Privacy as the default
- Privacy embedded into design
- Full functionality – positive-sum, not zero-sum
- End-to-end life cycle protection
- Visibility and transparency
- Respect for user privacy

The goal of PbD is to have developers integrate privacy protections into their solutions to avoid privacy violations in the first place. The overall concept focuses on prevention rather than remedies for violations.

PbD is also the driving factor behind an initiative to have privacy protections integrated throughout an organization, not just by developers. Business operations and systems design can also integrate privacy protections into their core functions. This in turn, has led to the *Global Privacy Standard (GPS)*, which was crafted to create a single set of universal and harmonized privacy principles. GPS is to be adopted by countries to use as a guide in developing privacy legislation, used by organizations to integrate privacy protection into their operations, and used by developers to integrate privacy into the products they produce. There is some integration of a few of the principles of PbD in the EU's GDPR (see Chapter 4, “Laws, Regulations, and Compliance”).

For more on PbD and GPS, please visit `gpsbydesign.org`, review the Cavoukian paper mentioned earlier, and read an additional paper, “Privacy by Design in Law, Policy and Practice.” Learn more about privacy in Chapter 4 and about software development security in Chapter 20.

Secure Access Service Edge (SASE)

Secure Access Service Edge (SASE) is a framework that combines network security functions with wide area network (WAN) capabilities, catering to the dynamic, secure access needs of modern organizations. SASE is designed to respond to the evolving IT landscape marked by trends such as cloud adoption, a mobile workforce, and an increased emphasis on network security.

SASE features a cloud-native architecture, unifying traditionally separate network and security services. This cloud-native approach allows organizations to deploy and manage their network and security services from the cloud, eliminating the need for on-premises hardware and offering scalability and flexibility.

A core principle of SASE is identity-centric security, prioritizing the identity of users and devices over the traditional perimeter-based security model. This is particularly relevant in the current environment where users access resources from diverse locations and devices. This core principle is implemented through zero trust network access (ZTNA), which is founded on the concept that no entity, whether inside or outside the organization's network, should be trusted by default. Every user and device must undergo

authentication (with MFA for users when possible) and authorization processes for access.

SASE also leverages edge computing, bringing security and networking closer to users and devices, reducing latency, and improving performance, especially for cloud-based applications. The framework emphasizes a globally distributed network, ensuring consistent security and performance for users regardless of their geographical location. SASE is often delivered as a service, allowing organizations to subscribe to the specific capabilities they require, simplifying management, reducing capital expenditures, and enabling scalability.

Continuous monitoring of user behavior and network conditions is a key aspect of SASE, enabling adaptive security measures that respond to changes in real time, contributing to an enhanced overall security posture.

Implementing SASE addresses the challenges posed by the modern IT landscape, providing a more agile and scalable approach to network and security services. It aligns well with the needs of a distributed and mobile workforce and accommodates the increasing reliance on cloud-based applications and resources.

Techniques for Ensuring CIA

To ensure the confidentiality, integrity, and availability (CIA) of data, you must ensure that all components that have access to data are secure and well behaved. Software designers use different techniques to ensure that programs do only what is required and nothing more. Although the concepts we discuss in the following sections all relate to software programs, they are also commonly used in all areas of security. For example, physical confinement guarantees that all physical access to hardware is controlled.

Confinement

Software designers use process confinement to restrict the actions of a program. Simply put, process *confinement* allows a process to read from and write to only certain memory locations and resources. This is also known as *sandboxing*. It is the application of the principle of least privilege

to processes. The goal of confinement is to prevent data leakage to unauthorized programs, users, or systems.

The operating system, or some other security component, disallows illegal read/write requests. If a process attempts to initiate an action beyond its granted authority, that action will be denied. In addition, further actions, such as logging the violation attempt, may be taken. Generally, the offending process is terminated. Confinement can be implemented in the operating system itself (such as through process isolation and memory protection), through the use of a confinement application or service (for example, Sandboxie at `sandboxie.com`), or through a virtualization or hypervisor solution (such as VMware or Oracle's VirtualBox).

Bounds

Each process that runs on a system is assigned an authority level. The authority level tells the operating system what the process can do. In simple systems, there may be only two authority levels: user and kernel. The authority level tells the operating system how to set the bounds for a process. The *bounds* of a process consist of limits set on the memory addresses and resources it can access. The bounds state the area within which a process is confined or contained. In most systems, these bounds segment logical areas of memory for each process to use. It is the responsibility of the operating system to enforce these logical bounds and to disallow access to other processes. More secure systems may require physically bounded processes. Physical bounds require each bounded process to run in an area of memory that is physically separated from other bounded processes, not just logically bounded in the same memory space. Physically bounded memory can be very expensive, but it's also more secure than logical bounds. Bounds can be a means to enforce confinement.

Isolation

When a process is confined through enforcing access bounds, that process runs in isolation. Process isolation ensures that any behavior will affect only the memory and resources associated with the isolated process. *Isolation* is used to protect the operating environment, the kernel of the operating system, and other independent applications. Isolation is an essential component of a stable operating system. Isolation is what prevents an

application from accessing the memory or resources of another application, whether for good or ill. Isolation allows for a fail-soft environment so that separate processes can operate normally or fail/crash without interfering or affecting other processes. Isolation is achieved through the enforcement of containment using bounds. Hardware and software isolation implementations are discussed throughout Chapter 9.

These three concepts (confinement, bounds, and isolation) make designing secure programs and operating systems more difficult, but they also make it possible to implement more secure systems. Confinement is making sure that an active process can only access specific resources (such as memory). Bounds is the limitation of authorization assigned to a process to limit the resources the process can interact with and the types of interactions allowed. Isolation is the means by which confinement is implemented through the use of bounds. The goals of these concepts are to ensure that the predetermined scope of resource access is not violated and that any failure or compromise of a process has minimal to no effect on any other process.

Access Controls

To ensure the security of a system, you need to allow subjects to access only authorized objects. Access controls limit the access of a subject to an object. Access rules state which objects are valid for each subject. Further, an object might be valid for one type of access and be invalid for another type of access. There are a wide range of options for access controls, such as discretionary, role-based, and mandatory. Please see Chapter 14 for an in-depth discussion of access controls.

Trust and Assurance

A *trusted system* is one in which all protection mechanisms work together to process sensitive data for many types of users while maintaining a stable and secure computing environment. In other words, trust is the presence of a security mechanism, function, or capability. *Assurance* is the degree of confidence in the satisfaction of security needs. In other words, assurance is how reliable the security mechanisms are at providing security. Assurance must be continually maintained, updated, and reverified. This is true if the secured system experiences a known change (good or bad—i.e., a vendor patch or a malicious exploit) or if a significant amount of time has passed.

In either case, change has occurred at some level. Change is often the antithesis of security; it often diminishes security. This is why change management, patch management, and configuration management are so important to security management.

Assurance varies from one system to another and often must be established on individual systems. However, there are grades or levels of assurance that can be placed across numerous systems of the same type, systems that support the same services, or systems that are deployed in the same geographic location. Thus, trust can be built into a system by implementing specific security features, whereas assurance is an assessment of the reliability and usability of those security features in a real-world situation.

Understand the Fundamental Concepts of Security Models

In information security, models provide a way to formalize security policies. Such models can be abstract or intuitive, but all are intended to provide an explicit set of rules that a computer can follow to implement the fundamental security concepts, processes, and procedures of a security policy. A *security model* provides a way for designers to map abstract statements into a security policy that prescribes the algorithms and data structures necessary to build hardware and software. Thus, a security model gives software designers something against which to measure their design and implementation.

Tokens, Capabilities, and Labels

Several different methods are used to describe the necessary security attributes for an object. A security *token* is a separate object that is associated with a resource and describes its security attributes. This token can communicate security information about an object prior to requesting access to the actual object. In other implementations, various lists are used to store security information about multiple objects. A *capabilities list* maintains a row of security attributes for each controlled object. Although not as flexible as the token approach, a capabilities list generally offers quicker lookups when a subject requests access to an object. A third common type of attribute storage is called a *security label*, which is generally a permanent part of the object to which it's attached. Once a security label is set, it usually cannot be altered. This permanence provides another safeguard against tampering that neither tokens nor capabilities lists provide.

You'll explore several security models in the following sections; all of them can shed light on how security enters into computer architectures and operating system design:

- Trusted computing base
- State machine model
- Information flow model
- Noninterference model
- Take-grant model
- Access control matrix
- Bell–LaPadula model
- Biba model
- Clark–Wilson model
- Brewer and Nash model

Other Security Models

There are several more security models you can learn about if you formally study computer security, systems design, or application development. Some of those include the object-capability model, Lipner's Model, the Boebert and Kain Integrity model, the two-compartment exchange (Kärger) model, Gong's JDK Security Model, the Lee–Shockley model, the Jueneman model, and more.

Trusted Computing Base

The *trusted computing base (TCB)* design principle is the combination of hardware, software, and controls that work together to form a trusted base to enforce your security policy. The TCB is a subset of a complete information system. It should be as small as possible so that a detailed analysis can reasonably ensure that the system meets design specifications and requirements. The TCB is the only portion of that system that can be trusted to adhere to and enforce the security policy. It is the responsibility of TCB components to ensure that a system behaves properly in all cases and that it adheres to the security policy under all circumstances.

Security Perimeter

The *security perimeter* of your system is an imaginary boundary that separates the TCB from the rest of the system (Figure 8.2). This boundary ensures that no insecure communications or interactions occur between the TCB and the remaining elements of the computer system. For the TCB to communicate with the rest of the system, it must create secure channels, also called *trusted paths*. A trusted path is a channel established with strict standards to allow necessary communication to occur without exposing the TCB to security exploitations.

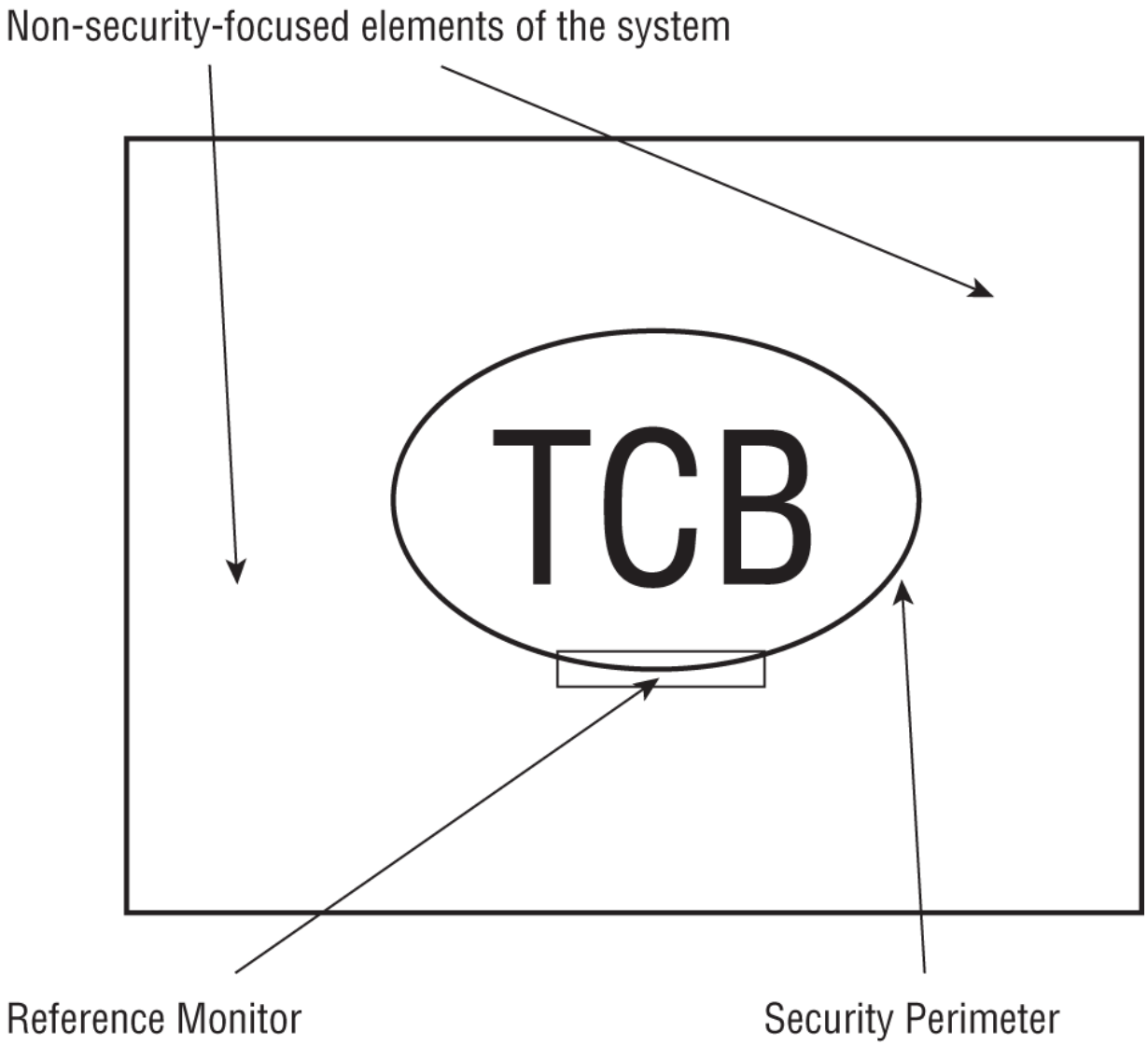


FIGURE 8.2 The TCB, security perimeter, and reference monitor

Reference Monitors and Kernels

The part of the TCB that validates access to every resource prior to granting access requests is called the *reference monitor* (Figure 8.2). The reference monitor stands between every subject and object, verifying that a requesting subject's credentials meet the object's access requirements before any requests are allowed to proceed. Effectively, the reference monitor is the access control enforcer for the TCB. The reference monitor enforces access control or authorization based on the desired security model, whether discretionary, mandatory, role-based, or some other form of access control.

The collection of components in the TCB that work together to implement reference monitor functions is called the *security kernel*. The reference monitor is a concept or theory that is put into practice via the implementation of a security kernel in software and hardware. The purpose of the security kernel is to launch appropriate components to enforce reference monitor functionality and resist all known attacks. The security kernel mediates all resource access requests, granting only those requests that match the appropriate access rules in use for a system.

State Machine Model

The *state machine model* describes a system that is always secure no matter what state it is in. It's based on the computer science definition of a *finite state machine (FSM)*. An FSM combines an external input with an internal machine state to model all kinds of complex systems, including parsers, decoders, and interpreters. Given an input and a state, an FSM transitions to another state and may create an output. Mathematically, the next state is a function of the current state and the input next state—that is, the next state = F(input, current state). Likewise, the output is also a function of the input and the current state output—that is, the output = F(input, current state).

According to the state machine model, a *state* is a snapshot of a system at a specific moment in time. If all aspects of a state meet the requirements of the security policy, that state is considered secure. A transition occurs when accepting input or producing output. A transition always results in a new state (also called a *state transition*). All state transitions must be evaluated. If each possible state transition results in another secure state, the system can be called a *secure state machine*. A secure state machine model system always boots into a secure state, maintains a secure state across all transitions, and allows subjects to access resources only in a secure manner compliant with the security policy. The secure state machine model is the basis for many other security models.

Information Flow Model

The *information flow model* focuses on controlling the flow of information. Information flow models are based on the state machine model. Information flow models don't necessarily deal with only the direction of information flow; they can also address the type of flow.

Information flow models are designed to prevent unauthorized, insecure, or restricted information flow, often between different levels of security (known as multilevel models). Information flow can be between subjects and objects at the same or different classification levels. An information flow model allows all authorized information flows, and prevents all unauthorized information flows.

Another interesting perspective on the information flow model is that it is used to establish a relationship between two versions or states of the same object when those two versions or states exist at different points in time. Thus, information flow dictates the transformation of an object from one state at one point in time to another state at another point in time. The information flow model also addresses covert channels by specifically excluding all undefined flow pathways.

Noninterference Model

The *noninterference model* is loosely based on the information flow model. However, instead of being concerned about the flow of information, the noninterference model is concerned with how the actions of a subject at a higher security level affect the system state or the actions of a subject at a lower security level. Basically, the actions of subject A (high) should not affect or interfere with the actions of subject B (low) or even be noticed by subject B. If such violations occur, subject B may be placed into an insecure state or be able to deduce or infer information about a higher level of classification. This is a type of information leakage and implicitly creates a covert channel. Thus, the noninterference model can be imposed to provide a form of protection against damage caused by malicious programs, such as Trojan horses, backdoors, and rootkits.

Composition Theories

Some other models that fall into the information flow category build on the notion of inputs and outputs between multiple systems. These are called composition theories because they explain how outputs from one system relate to inputs to another system. There are three composition theories:

- Cascading: Input for one system comes from the output of another system.

- Feedback: One system provides input to another system, which reciprocates by reversing those roles (so that system A first provides input to system B and then system B provides input to system A).
- Hookup: One system sends input to another system but also sends input to external entities.

Take-Grant Model

The *take-grant model* employs a *directed graph* (Figure 8.3) to dictate how rights can be passed from one subject to another or from a subject to an object. Simply put, a subject (X) with the grant right can grant another subject (Y) or another object (Z) any right that subject (X) possesses. Likewise, a subject (X) with the take right can take a right from another subject (Y). In addition to these two primary rules, the take-grant model has a create rule and a remove rule to generate or delete rights. The key to this model is that using these rules allows you to figure out when rights in the system can change and where leakage (that is, unintentional distribution of permissions) can occur.

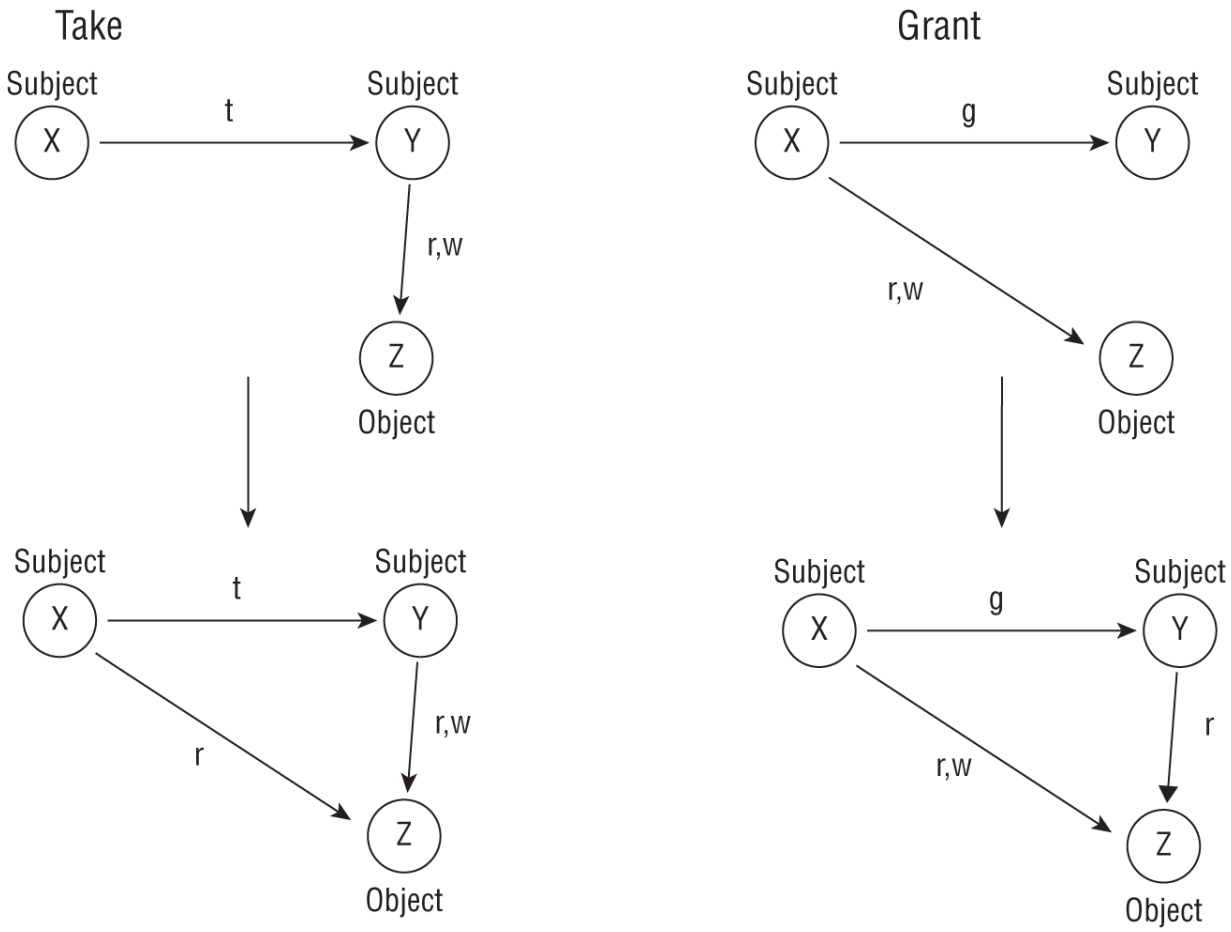


FIGURE 8.3 The take-grant model's directed graph

In essence, here are the four rules of the take-grant model:

- *Take rule:* Allows a subject to take rights over an object
- *Grant rule:* Allows a subject to grant rights to an object
- *Create rule:* Allows a subject to create new rights
- *Remove rule:* Allows a subject to remove rights it has

It is interesting to ponder that the take and grant rules are effectively a copy function. This can be recognized in modern operating systems in the process of inheritance, such as subjects inheriting a permission from a group or a file inheriting access control list (ACL) values from a parent folder. The two additional rules (create and remove), not defined by a directed graph, are also commonly present in modern operating systems. For example, to obtain permission on an object, that permission does not

have to be copied from a user account that already has that permission; instead, it is simply created by an account with the privilege capability of create or assign permissions (which can be the owner of an object or a subject with full control or administrative privileges over the object).

Access Control Matrix

An *access control matrix* is a table of subjects and objects that indicates the actions or functions that each subject can perform on each object. Each column of the matrix is an ACL pulled from objects. Once sorted, each row of the matrix is a capabilities list for each listed subject. An ACL is tied to an object; it lists the valid actions each subject can perform. A capability list is tied to the subject; it lists valid actions that can be taken on each object included in the matrix.

From an administration perspective, using only capability lists for access control is a management nightmare. A capability list method of access control can be accomplished by storing on each subject a list of rights the subject has for every object. This effectively gives each user a key ring of access and rights to objects within the security domain. To remove access to a particular object, every user (subject) that has access to it must be individually manipulated. Thus, managing access on each user account is much more difficult than managing access on each object (in other words, via ACLs). A capabilities table can be created by pivoting an access control matrix; this results in the columns being subjects and the rows being ACLs from objects.

The access control matrix shown in Table 8.3 is for a discretionary access control system. A mandatory or role-based matrix can be constructed simply by replacing the subject names with classifications or roles. Access control matrices are used by systems to quickly determine whether the requested action by a subject for an object is authorized.

TABLE 8.3 An access control matrix

Subjects	Document file	Printer	Network folder share
Bob	Read	No Access	No Access
Mary	No Access	No Access	Read
Amanda	Read, Write	Print	No Access
Mark	Read, Write	Print	Read, Write
Kathryn	Read, Write	Print, Manage Print Queue	Read, Write, Execute
Colin	Read, Write, Change Permissions	Print, Manage Print Queue, Change Permissions	Read, Write, Execute, Change Permissions

Bell–LaPadula Model

The *Bell–LaPadula model* was developed for the U.S. Department of Defense (DoD) in the 1970s based on the DoD's multilevel security policies. The multilevel security policy states that a subject with any level of clearance can access resources at or below its clearance level. However, within clearance levels, access to compartmentalized objects is granted only on a need-to-know basis.

By design, the Bell–LaPadula model prevents the leaking or transfer of classified information to less secure clearance levels. This is accomplished by blocking lower-classified subjects from accessing higher-classified objects. With these restrictions, the Bell–LaPadula model is focused on maintaining confidentiality and does not address any other aspects of object security.

Lattice-Based Access Control

This general category for nondiscretionary access controls is covered in Chapter 13, “Managing Identity and Authentication.” Here's a quick preview on that more detailed coverage of this subject (which drives the underpinnings for most access control security models): Subjects under *lattice-based access controls* are assigned positions in a lattice (i.e., a multilayered security structure or multileveled security domains). Subjects can access only those objects that fall into the range between the least upper bound (LUB) (the nearest security label or classification higher than their lattice position) and the greatest (i.e., highest) lower bound (GLB) (the nearest security label or classification lower than their lattice position) of the labels or classifications for their lattice position.

This model is built on a state machine concept and the information flow model. It also employs mandatory access controls and is a lattice-based access control concept. The *lattice* tiers are the *classification levels* defined by the organization's security policy.

There are three basic properties of this state machine:

- The *Simple Security Property* (i.e., the ss-Property) states that a subject may not read information at a higher sensitivity level (no read-up).
- The **-Property (star-property)* states that a subject may not write information to an object at a lower sensitivity level (no write-down).
- The *Discretionary Security Property* states that the system uses an access matrix to enforce discretionary access control.

These first two properties define the states into which the system can transition. No other transitions are allowed. All states accessible through these two rules are secure states. Thus, Bell–LaPadula–modeled systems offer state machine model security (see Figure 8.4).

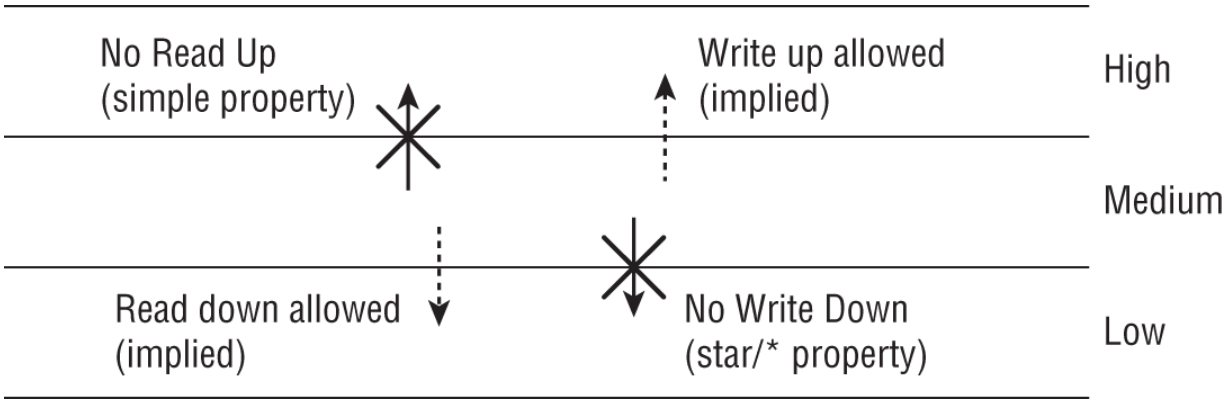


FIGURE 8.4 The Bell–LaPadula model

The Bell–LaPadula properties are in place to protect data confidentiality. A subject cannot read an object that is classified at a higher level than the subject is cleared for. Because objects at one level have data that is more sensitive or secret than data in objects at a lower level, a subject (who is not a trusted subject) cannot write data from one level to an object at a lower level. That action would be similar to pasting a top-secret memo into an unclassified document file. The third property enforces a subject's job/role-based need to know to access an object.



An exception in the Bell–LaPadula model states that a "trusted subject" is not constrained by the *-Property. A trusted subject is defined as "a subject that is guaranteed not to consummate a security-breaching information transfer even if it is possible." This means that a trusted subject is allowed to violate the *-Property and perform a write-down, which is necessary when performing valid object declassification or reclassification.

The Bell–LaPadula model was designed in the 1970s, so it does not support many operations that are common today, such as file sharing and networking. It also assumes secure transitions between security layers and does not address covert channels (see Chapter 9).

Biba Model

The *Biba model* was designed after the Bell–LaPadula model, but it focuses on integrity. The Biba model is also built on a state machine concept, is based on information flow, and is a multilevel model. In fact, the Biba model is the inverted Bell–LaPadula model. The properties of the Biba model are as follows:

- The *Simple Integrity Property* states that a subject cannot read an object at a lower integrity level (no read-down).
- The * *(star) Integrity Property* states that a subject cannot modify an object at a higher integrity level (no write-up).
- The *Invocation Property* states that a process from below cannot request higher access (neither read nor write); only with subjects at an equal or lower level.



In both the Biba and Bell–LaPadula models, there are two properties that are inverses of each other: simple and * (star). However, they may also be labeled as axioms, principles, or rules. What you should focus on is the *simple* and *star* designations. Take note that *simple* is always about reading, and *star* is always about writing. In both cases, the rules define what cannot or should not be done. Usually, what is not prevented or blocked is allowed. Thus, even though a rule is stated as a No declaration, its opposite direction is implied as allowed.

Figure 8.5 illustrates these Biba model properties.

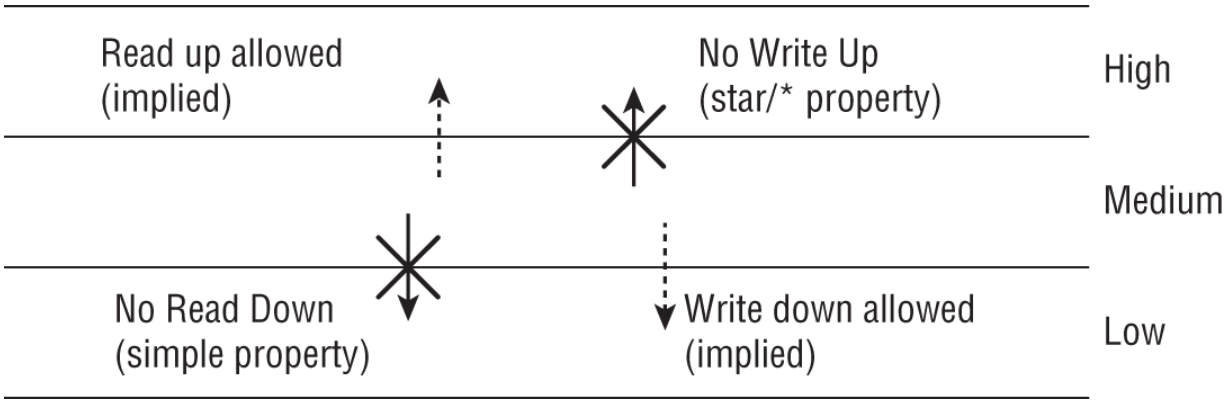
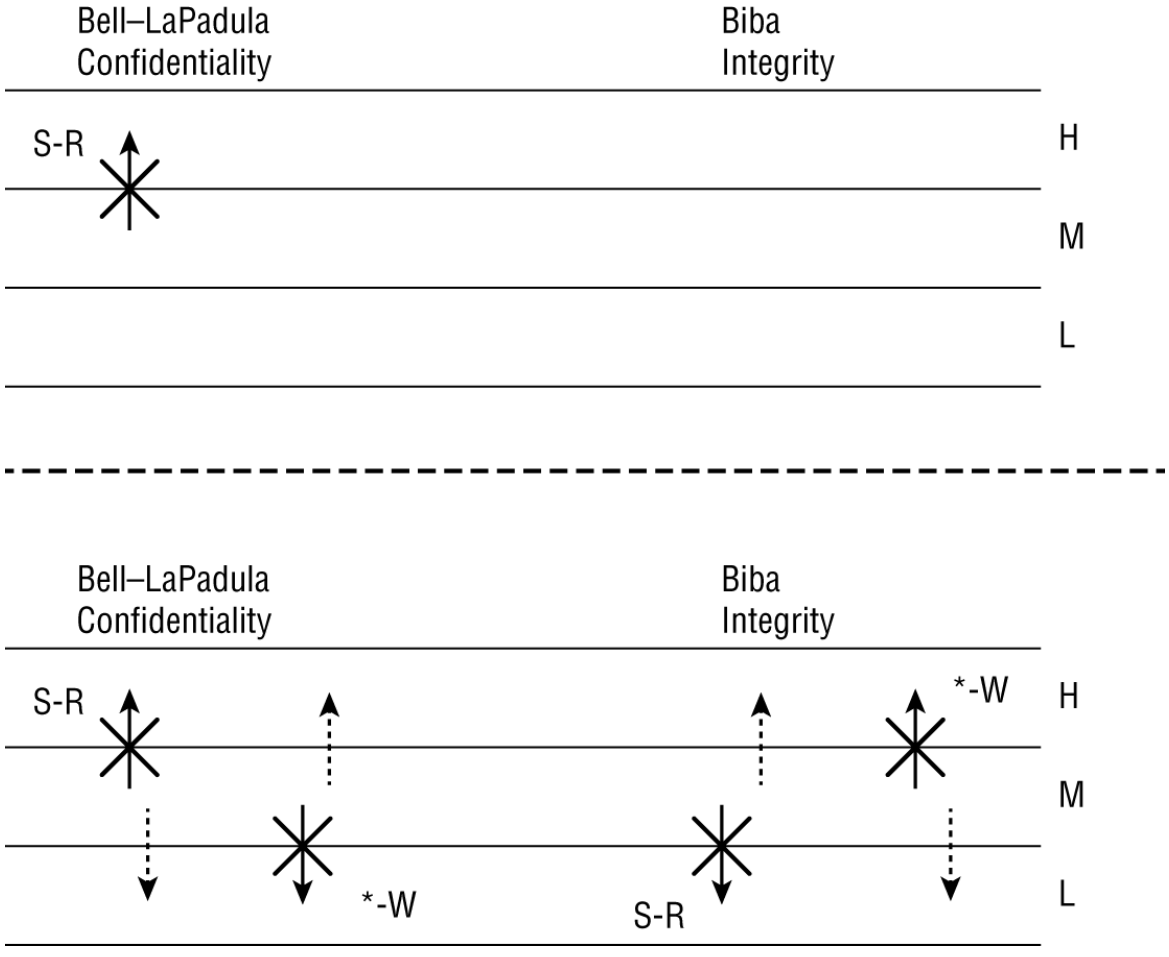


FIGURE 8.5 The Biba model

Consider the Biba properties. The second property of the Biba model is pretty straightforward. A subject cannot write to an object at a higher integrity level. That makes sense. What about the first property? Why can't a subject read an object at a lower integrity level? The answer takes a little thought. Think of integrity levels as being like the purity level of air. You would not want to pump air from the smoking section into the clean room environment. The same applies to data. When integrity is important, you do not want unvalidated data read into validated documents. The potential for data contamination is too great to permit such access.



Memorizing the properties of Bell–LaPadula and Biba can be challenging, but there is a shortcut. If you can memorize the graphical layout in the following figure above the dotted line, then you can figure out the rest. Notice that Bell–LaPadula is placed on the left and Biba is on the right, and the security benefit of each is listed below the model name. Then, only the Bell–LaPadula model's simple property is listed. That property is “No Read Up,” which is represented by an arrow pointing upward that is crossed out and labeled by an “S” for simple and an “R” for read. From there, all of the other rules are the opposing element of the pair or inverted.



Biba requires that all subjects and objects have a classification label (it is still a DoD-derived security model). Thus, data integrity protection is

dependent on data classification.

Critiques of the Biba model reveal a few drawbacks:

- It addresses only integrity, not confidentiality or availability.
- It focuses on protecting objects from external threats; it assumes that internal threats are handled programmatically.
- It does not address access control management, and it doesn't provide a way to assign or change an object's or subject's classification level.
- It does not prevent covert channels.

Clark–Wilson Model

The *Clark–Wilson model* uses a multifaceted approach to enforcing data integrity. Instead of defining a formal state machine, the Clark–Wilson model defines each data item and allows modifications through only a limited or controlled intermediary program or interface.

The Clark–Wilson model does not require the use of a lattice structure; rather, it uses a three-part relationship of subject/program/object (or subject/transaction/object) known as a triple or an *access control triplet*. Subjects do not have direct access to objects. Objects can be accessed only through programs. Through the use of two principles—well-formed transactions and separation of duties—the Clark–Wilson model provides an effective means to protect integrity.

Well-formed transactions take the form of programs. A subject is able to access objects only by using a program, interface, or access portal (Figure 8.6). Each program has specific limitations on what it can and cannot do to an object (such as a database or other resource). This effectively limits the subject's capabilities. This is known as a constrained, limiting, or restrictive interface. If the programs are properly designed, then the triple relationship provides a means to protect the integrity of the object.

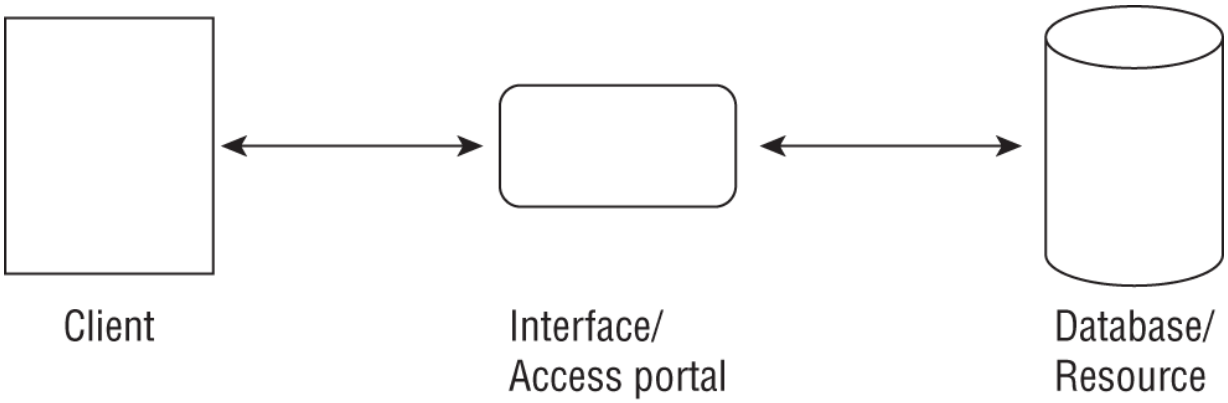


FIGURE 8.6 The Clark–Wilson model

Clark–Wilson defines the following items and procedures:

- A *constrained data item (CDI)* is any data item whose integrity is protected by the security model.
- An *unconstrained data item (UDI)* is any data item that is not controlled by the security model. Any data that is to be input and hasn't been validated, or any output, would be considered an unconstrained data item.
- An *integrity verification procedure (IVP)* is a procedure that scans data items and confirms their integrity.
- *Transformation procedures (TPs)* are the only procedures that are allowed to modify a CDI. The limited access to CDIs through TPs forms the backbone of the Clark–Wilson integrity model.

The Clark–Wilson model uses security labels to grant access to objects, but only through transformation procedures and a *restricted interface model*. A restricted interface model uses classification-based restrictions to offer only subject-specific authorized information and functions. One subject at one classification level will see one set of data and have access to one set of functions, whereas another subject at a different classification level will see a different set of data and have access to a different set of functions. The different functions made available to different levels or classes of users may be implemented by either showing all functions to all users but deactivating those that are not authorized for a specific user or by showing only those functions granted to a specific user. Through these mechanisms, the Clark–

Wilson model ensures that data is protected from unauthorized changes from any user. In effect, the Clark–Wilson model enforces separation of duties. The Clark–Wilson design makes it a common model for commercial applications.



The Clark–Wilson model was designed to protect integrity using the access control triplet. However, though the intermediary interface can be programmed to limit what can be done to an object by a subject, it can just as easily be programmed to limit or restrict what objects are shown to a subject. Thus, this concept can lend itself readily to protect confidentiality. In many situations, there is an intermediary program between a subject and an object. If the focus of that intermediary is to protect integrity, then it is an implementation of the Clark–Wilson model. If it is intended to protect confidentiality, then they are benefiting from an alternate use of the intermediary program.

Brewer and Nash Model

The *Brewer and Nash model* was created to permit access controls to change dynamically based on a user's previous activity (making it a kind of state machine model as well). This model applies to a single integrated database; it seeks to create security domains that are sensitive to the notion of conflict of interest (for example, someone who works at company C who has access to proprietary data for company A should not also be allowed access to similar data for company B if those two companies compete with each other). This model creates a class of data that defines which security domains are potentially in conflict and prevents any subject with access to one domain that belongs to a specific conflict class from accessing any other domain that belongs to the same conflict class. Metaphorically, this puts a wall around all other information in any conflict class. Thus, this model also uses the principle of data isolation within each conflict class to keep users out of potential conflict-of-interest situations (for example, management of company datasets). Because company relationships change all the time, dynamic updates to members of and definitions for conflict classes are important.

Another way of looking at or thinking of the Brewer and Nash model is of an administrator having full control access to a wide range of data in a system based on their assigned job responsibilities and work tasks. However, at the moment an action is taken against any data item, the administrator's access to any conflicting data items is temporarily blocked. Only data items that relate to the initial data item can be accessed during the operation. Once the task is completed, the administrator's access returns to full control.

The Brewer and Nash model was sometimes known as the Chinese Wall model, but this term is deprecated. Instead, other terms of “ethical wall” and “cone of silence” have been used to describe Brewer and Nash.

Disambiguating the Word “Star” in Models

The term *star* presents a few challenges when it comes to security models. For one thing, there is no formal security model named “Star Model.” However, both the Bell–LaPadula and the Biba models have a *star property*, which is discussed in their respective sections in this chapter.

Although not a model, the Cloud Security Alliance (CSA) also has a STAR program. CSA's Security Trust Assurance and Risk (STAR) program focuses on improving cloud service provider (CSP) security through auditing, transparency, and integration of standards.

Although not related to security, there is also Galbraith's Star Model, which helps businesses organize divisions and departments to achieve business missions and goals and adjust over time for long-term viability. This model is based on five main areas of business administration that need to be managed, balanced, and harnessed toward the mission and goals of the organization. The five areas of Galbraith's Star Model are Strategy, Structure, Processes, Rewards, and People.

Understanding how “star” is used in the context of the Bell–LaPadula and Biba models, CSA's STAR program, and Galbraith's Star Model will help you distinguish what is meant when you see the word used in different contexts.

Select Controls Based on Systems Security Requirements

Those who purchase information systems for certain kinds of applications —for example, national security agencies whose sensitive information may be extremely valuable (or dangerous in the wrong hands) or central banks or securities traders that have certain data that may be worth billions of dollars—often want to understand the security strengths and weaknesses of systems prior to acquisition. Such buyers are often willing to consider only systems that have been subjected to formal evaluation processes in advance and have received some kind of security rating.

Often, trusted third parties are used to perform security evaluations; the most important result from such testing is their “seal of approval” that the system meets all essential criteria.

Common Criteria

The *Common Criteria (CC)* defines various levels of testing and confirmation of systems' security capabilities. Nevertheless, it's wise to observe that even the highest CC ratings do not equate to a guarantee that such systems are completely secure or that they are entirely devoid of vulnerabilities or susceptibilities to exploit. The Common Criteria was designed as a dynamic subjective product evaluation model and replaced previous static systems, such as the U.S. Department of Defense's Trusted Computer System Evaluation Criteria (TCSEC) and the EU's Information Technology Security Evaluation Criteria (ITSEC).

A document titled “Arrangement on the Recognition of Common Criteria Certificates in the Field of Information Technology Security” was signed by representatives from government organizations in Canada, France, Germany, the United Kingdom, and the United States in 1998, making the document an international standard. Since then, 23 additional countries have signed the arrangement. The arrangement documentation was formally adopted as a standard and published as ISO/IEC 15408:2022 and labeled as “Information security, cybersecurity and privacy protection: Evaluation criteria for IT security.”



The latest versions of ISO/IEC standards are available at standards.iso.org/ittf/PubliclyAvailableStandards/index.html. The current (as of this writing) version of the Common Criteria is to be replaced with ISO/IEC WD 15408-1 (www.iso.org/standard/88134.html). This could occur in 2024.

The objectives of the CC guidelines are as follows:

- To add to buyers' confidence in the security of evaluated, rated IT products
- To eliminate duplicate evaluations (among other things, this means that if one country, agency, or validation organization follows the CC in rating specific systems and configurations, others elsewhere need not repeat this work)
- To keep making security evaluations more cost-effective and efficient
- To make sure evaluations of IT products adhere to high and consistent standards
- To promote evaluation and increase the availability of evaluated, rated IT products
- To evaluate the functionality (what the system does) and assurance (how much can you trust the system) of the target of evaluation (TOE)

The Common Criteria process is based on two key elements: protection profiles and security targets. *Protection profiles (PPs)* specify the security requirements and protections for a product that is to be evaluated (the TOE), which are considered the security desires, or the “I want” of a customer. *Security targets (STs)* specify the claims of security from the vendor that are built into a TOE. STs are considered the implemented security measures, or the “I will provide” from the vendor. In addition to offering security targets, vendors may offer packages of additional security features. A package is an intermediate grouping of security requirement components that can be added to or removed from a TOE (like the option

packages when purchasing a new vehicle). This system of the PP and ST allows for flexibility, subjectivity, and customization of an organization's specific security functional and assurance requirements over time.

An organization's PP is compared to various STs from the selected vendor's TOEs. The closest or best match is what the client purchases. The client initially selects a vendor based on published or marketed *evaluation assurance levels (EALs)* for currently available systems. Using Common Criteria to choose a vendor allows clients to request exactly what they need for security rather than having to use static fixed security levels. It also allows vendors more flexibility on what they design and create. A well-defined set of Common Criteria supports subjectivity and versatility, and it automatically adapts to changing technology and threat conditions. Furthermore, the EALs provide a method for comparing vendor systems that is more standardized (like the old TCSEC).

Table 8.4 summarizes EALs 1 through 7. For a complete description of EALs, consult the CC standard documents.

TABLE 8.4 Common Criteria evaluation assurance levels

Level	Assurance level	Description
EAL1	Functionally tested	Applies when some confidence in correct operation is required but where threats to security are not serious. This is of value when independent assurance that due care has been exercised in protecting personal information is necessary.
EAL2	Structurally tested	Applies when delivery of design information and test results are in keeping with good commercial practices. This is of value when developers or users require low to moderate levels of independently assured security. It is especially relevant when evaluating legacy systems.
EAL3	Methodically tested and checked	Applies when security engineering begins at the design stage and is carried through without substantial subsequent alteration. This is of value when developers or users require a moderate level of independently assured security, including thorough investigation of TOE and its development.
EAL4	Methodically designed, tested and reviewed	Applies when rigorous, positive security engineering and good commercial development practices are used. This does not require substantial specialist knowledge, skills, or resources. It involves independent testing of all TOE security functions.
EAL5	Semi-formally verified designed and tested	Uses rigorous security engineering and commercial development practices, including specialist security engineering techniques. This applies when developers or users require a high level of independently assured security in a planned development approach, followed by rigorous development.

Level	Assurance level	Description
EAL6	Semi-formally verified design and tested	Uses the application of high assurance security engineering techniques to a rigorous development environment in order to produce a premium TOE for protecting high-value assets against significant risks. It is therefore applicable to the development of security TOEs for application in high risk situations where the value of the protected assets justifies the additional costs.
EAL7	Formally verified design and tested	Used only for highest-risk situations or where high-value assets are involved. This is limited to TOEs where tightly focused security functionality is subject to extensive formal analysis and testing.

Though the CC guidelines are flexible and accommodating enough to capture most security needs and requirements, they are by no means perfect. As with other evaluation criteria, the CC guidelines do nothing to make sure that how users act on data is also secure. The CC guidelines also do not address administrative issues outside the specific purview of security. As with other evaluation criteria, the CC guidelines do not include evaluation of security *in situ*—that is, they do not address controls related to personnel, organizational practices and procedures, or physical security. Likewise, controls over electromagnetic emissions are not addressed, nor are the criteria for rating the strength of cryptographic algorithms explicitly laid out. Nevertheless, the CC guidelines represent some of the best techniques whereby systems may be rated for security.



Additional Common Criteria documentation is available at commoncriteriaportal.org. Visit this site to get information on the current version of the CC guidelines and guidance on using the CC along with lots of other useful, relevant information.

Authorization to Operate

For many environments, it is necessary to obtain an official approval to use secured equipment for operational objectives. This is often referred to as an *authorization to operate (ATO)*. ATO is the current term for this concept as defined by the Risk Management Framework (RMF) (see Chapter 2, "Personnel Security and Risk Management Concepts"), which replaces the previous term of accreditation. An ATO is an official authorization to use a specific collection of secured IT/IS systems to perform business tasks and accept the identified risk. The assessment and assignment of an ATO is performed by an *authorizing official (AO)*. An AO is an authorized entity who can evaluate an IT/IS system, its operations, and its risks, and potentially issue an ATO.



NIST maintains an excellent glossary with references at csrc.nist.gov/glossary.

A typical ATO is issued for 3 years (although assigned time frames vary and the AO can adjust the time frame even after issuing an ATO) and must be reobtained whenever one of the following conditions occurs:

- The ATO time frame has expired.
- The system experiences a significant security breach.
- The system experiences a significant security change.

The AO has the discretion to determine which breaches or security changes result in a loss of ATO. Either a modest intrusion event or the application of a substantial security patch could cause the negation of an ATO.

An AO can issue four types of authorization decisions:

Authorization to Operate This decision is issued when risk is managed to an acceptable level.

Common Control Authorization This decision is issued when a security control is inherited from another provider and when the risk

associated with the common control is at an acceptable level and already has a ATO from the same AO.

Authorization to Use This decision is issued when a third-party provider (such as a cloud service) provides IT/IS servers that are deemed to have risk at an acceptable level; it is also used to allow for reciprocity in accepting another AO's ATO.

Denial of Authorization This decision is issued when risk is unacceptable.

Please see NIST SP 800-37r2 for more on the Risk Management Framework and authorization.

Understand Security Capabilities of Information Systems

The security capabilities of information systems include memory protection, virtualization, Trusted Platform Module (TPM), encryption/decryption, interfaces, and fault tolerance. It is important to carefully assess each aspect of the infrastructure to ensure that it sufficiently supports security. Without an understanding of the security capabilities of information systems, it is impossible to evaluate them or implement them properly.

Memory Protection

Memory protection is a core security component that must be designed and implemented into an operating system. It must be enforced regardless of the programs executing in the system. Otherwise, instability, violation of integrity, denial of service, and disclosure are likely results. Memory protection is used to prevent an active process from interacting with an area of memory that was not specifically assigned or allocated to it.

Memory protection is discussed throughout Chapter 9 in relation to the topics of isolation, virtual memory, segmentation, memory management, and protection rings, as well as protections against buffer (i.e., memory) overflows.

Virtualization

Virtualization technology is used to host one or more operating systems within the memory of a single host computer or to run applications that are not compatible with the host OS. Virtualization can be a tool to isolate operating systems, test suspicious software, or implement other security protections. See Chapter 9 for more information about virtualization.

Trusted Platform Module (TPM)

The Trusted Platform Module (TPM) is both a specification for a cryptoprocessor chip on a mainboard and the general name for the implementation of the specification. A TPM can be used to implement a broad range of cryptography-based security protection mechanisms. A TPM chip is often used to store and process cryptographic keys for a hardware-supported or OS-implemented local storage device encryption system. A TPM is an example of a hardware security module (HSM). An HSM is a cryptoprocessor used to manage and store digital encryption keys, accelerate crypto operations, support faster digital signatures, and improve authentication. An HSM can be a chip on a motherboard, an external peripheral, a network-attached device, or an extension card (which is inserted into a device, such as a router, firewall, or rack-mounted server blade). HSMs include tamper protection to prevent their misuse even if an attacker gains physical access.

Interfaces

A *constrained* or *restricted interface* is implemented within an application to restrict what users can do or see based on their privileges. Users with full privileges have access to all the capabilities of the application. Users with restricted privileges have limited access.

Applications constrain the interface using different methods. A common method is to hide the capability if the user doesn't have permission to use it. Commands might be available to administrators via a menu or by right-clicking an item, but if a regular user doesn't have permissions, the command does not appear. Other times, the command is shown but is dimmed or deactivated. The regular user can see it but will not be able to use it.

The purpose of a constrained interface is to limit or restrict the actions of both authorized and unauthorized users. The use of such an interface is a practical implementation of the Clark–Wilson model of security.

Fault Tolerance

Fault tolerance is the ability of a system to suffer a fault but continue to operate. Fault tolerance is achieved by adding redundant components such as additional disks within a redundant array of independent disks (RAID) (aka redundant array of inexpensive disks [RAID]) array, or additional servers within a failover clustered configuration. Fault tolerance is an essential element of security design. It is also considered part of avoiding single points of failure and the implementation of redundancy. For more details on fault tolerance, redundant servers, RAID, and failover solutions, see Chapter 18, “Disaster Recovery Planning.”

Encryption/Decryption

Encryption is the process of converting plaintext to ciphertext, whereas decryption reverses that process. Symmetric and asymmetric methods of encryption and decryption can be used to support a wide range of security solutions to protect confidentiality and integrity. Please see the full coverage of cryptography in Chapters 6 and 7.

Manage the Information System Life Cycle

Managing the information system life cycle is a comprehensive process that involves various stages, each with specific activities and considerations. Managing the information system life cycle involves a structured and organized approach to developing, deploying, and maintaining an information system.

The typical stages of the information system life cycle are:

Stakeholders' Needs and Requirements This initial phase focuses on identifying and understanding the needs, expectations, and requirements of stakeholders who will interact with the information system. It encompasses a thorough analysis of the diverse set of

requirements presented by end users, managers, regulatory bodies, and other relevant parties.

Requirements Analysis This phase involves a detailed examination of these requirements, including determining both functional and nonfunctional requirements, considering constraints, and ensuring alignment with the overall goals of the organization.

Architectural Design In this phase, a blueprint for the information system is created, defining the overall structure, components, modules, data flow, and interfaces of the system.

Development/Implementation This phase is where the actual coding and development of the information system take place. Developers work on creating the software, configuring hardware, and integrating various components to bring the system to life.

Integration This phase is the process of combining different modules or components of the system to ensure they work together seamlessly. The goal here is to ensure that the individual elements of the system function as a unified whole.

Verification and Validation This phase is focused on confirming that the developed system meets the specified requirements. Verification ensures that each component of the system is correctly implemented, whereas validation ensures that the system as a whole fulfills its intended purpose.

Transition/Deployment This phase is when the system is deployed for actual use. Transition involves the migration of the system from the development environment to the operational environment, making it available to end users.

Operations and Maintenance/Sustainment In this phase, the system is actively used in the operational environment. Operations involve day-to-day management, monitoring, and support, while maintenance ensures that the system continues to function correctly by addressing issues, applying updates, and making improvements. Sustainment refers to the ongoing focus on maintaining and supporting the operational functionality of an information system over an extended period.

Retirement/Disposal Eventually, the information system reaches the end of its life cycle. The retirement or disposal phase involves decommissioning the system in an organized manner, considering data disposal, ensuring compliance with regulations, and making decisions about the future of the system or its replacement.

Effectively managing the information system life cycle requires collaboration among different stakeholders, adherence to best practices, and continuous improvement to meet evolving needs and technological advancements.

Summary

Secure systems are not just assembled; they are designed to support security. Systems that must be secure are judged for their ability to support and enforce the security policy. Programmers should strive to build security into every application they develop, with greater levels of security provided to critical applications and those that process sensitive information.

There are numerous issues related to the establishment and integration of security into a product, including managing subjects and objects and their relationships, using open or closed systems, managing secure defaults, designing a system to fail securely, abiding by the “keep it simple” postulate, implementing zero trust (instead of trust but verify), incorporating privacy by design, and using Secure Access Service Edge (SASE). CIA can be protected using confinement, bounds, and isolation. Access controls are used to implement security protections.

Proper security concepts, controls, and mechanisms must be integrated before and during the design and architectural period to produce a reliably secure product. A trusted system is one in which all protection mechanisms work together to process sensitive data for many types of users while maintaining a stable and secure computing environment. In other words, trust is the presence of a security mechanism or capability. Assurance is the degree of confidence in the satisfaction of security needs. In other words, assurance is how reliable the security mechanisms are at providing security.

When security systems are designed, it is often helpful to derive security mechanisms from standard security models. Some of the security models

that should be recognized include the trusted computing base, state machine model, information flow model, noninterference model, take-grant model, access control matrix, Bell–LaPadula model, Biba model, Clark–Wilson model, and the Brewer and Nash model.

Several security criteria exist for evaluating computer security systems. The Common Criteria uses a subjective system to meet security needs and a standard evaluation assurance level (EAL) to evaluate reliability.

The NIST Risk Management Framework (RMF) establishes an authorization to operate (ATO) issued by an authorizing official (AO) to ensure that only systems with acceptable risk levels are used to perform IT operations.

It is important to carefully assess each aspect of the infrastructure to ensure that it sufficiently supports security. Without an understanding of the security capabilities of information systems, it is impossible to evaluate them, nor is it possible to implement them properly. The security capabilities of information systems include memory protection, virtualization, Trusted Platform Module (TPM), encryption/decryption, interfaces, and fault tolerance.

Managing the information system life cycle is a comprehensive process that involves various stages of a structured and organized approach to developing, deploying, and maintaining an information system.

Study Essentials

Be able to describe open and closed systems. Open systems are designed using industry standards and are usually easy to integrate with other open systems. Closed systems are generally proprietary hardware and/or software. Their specifications are not normally published, and they are usually harder to integrate with other systems.

Know about secure defaults. Never assume the default settings of any product are secure. It is always up to the system administrator and/or company security staff to alter a product's settings to comply with the organization's security policies.

Understand the concept of fail securely. Failure management includes programmatic error handling (aka exception handling) and input sanitization; secure failure is integrated into the system (fail-safe versus fail-secure).

Know about the principle of "keep it simple." "Keep it simple" is the encouragement to avoid overcomplicating the environment, organization, or product design. The more complex a system, the more difficult it is to secure.

Understand zero trust. Zero trust is a security concept where nothing inside the organization is automatically trusted. Each request for activity or access is assumed to be from an unknown and untrusted location until otherwise verified. The concept is "never trust, always verify." The zero-trust model is based around "assume breach" and microsegmentation.

Know about privacy by design. Privacy by design (PbD) is a guideline to integrate privacy protections into products during the early design phase rather than attempting to tack them on at the end of development. The PbD framework is based on seven foundational principles.

Understand trust and assurance. A trusted system is one in which all protection mechanisms work together to process sensitive data for many types of users while maintaining a stable and secure computing environment. In other words, trust is the presence of a security mechanism or capability. Assurance is the degree of confidence in the satisfaction of security needs. In other words, assurance is how reliable the security mechanisms are at providing security.

Define a trusted computing base (TCB). A TCB is the combination of hardware, software, and controls that form a trusted base that enforces the security policy.

Know details about each of the security models. The state machine model ensures that all instances of subjects accessing objects are secure. The information flow model is designed to prevent unauthorized, insecure, or restricted information flow. The noninterference model prevents the actions of one subject from affecting the system state or actions of another subject. The take-grant model dictates how rights can be passed from one subject to another or from a subject to an object. An access control matrix is

a table of subjects and objects that indicates the actions or functions that each subject can perform on each object. Bell–LaPadula subjects have a clearance level that allows them to access only those objects with the corresponding classification levels, which protects confidentiality. Biba prevents subjects with lower security levels from writing to objects at higher security levels. Clark–Wilson is an integrity model that relies on the access control triplet (subject/program/object).

Know the controls used for evaluating computer security systems. The Common Criteria (ISO/IEC 15408) is a subjective security function evaluation tool that uses protection profiles (PPs) and security targets (STs) and assigns an evaluation assurance level (EAL). Authorization to operate (ATO) (from the RMF) is a formal approval to operate IT/IS based on an acceptable risk level based on the implementation of an agreed-on set of security and privacy controls.

Understand the security capabilities of information systems. Common security capabilities include memory protection, virtualization, Trusted Platform Module (TPM), encryption/decryption, interfaces, and fault tolerance.

Know about the information system life cycle. Managing the information system life cycle is a comprehensive process that involves various stages, each with specific activities and considerations. Managing the information system life cycle involves a structured and organized approach to developing, deploying, and maintaining an information system. Know the nine stages.

Written Lab

1. Name at least seven security models and the primary security benefit of using each.
2. Describe the primary components of TCB.
3. What are the two primary rules or principles of the Bell–LaPadula security model? Also, what are the two rules of Biba?
4. What is the difference between open and closed systems and open and closed source?

5. Name at least four design principles and describe them.

Review Questions

1. You have been working on crafting a new expansion service to link to the existing computing hardware of a core business function. However, after weeks of research and experimentation, you are unable to get the systems to communicate. The CTO informs you that the computing hardware you are focusing on is a closed system. What is a closed system?

   A. A system designed around final, or closed, standards

   B. A system that includes industry standards

   C. A proprietary system that uses unpublished protocols

   D. Any machine that does not run Windows

2. A compromise of a newly installed Wi-Fi-connected baby monitor enabled an attacker to virtually invade a home and play scary sounds to a startled toddler. How was the attacker able to gain access to the baby monitor in this situation?

   A. Outdated malware scanners

   B. A WAP supporting 5 GHz channels

   C. Performing a social engineering attack against the parents

   D. Exploiting default configuration

3. While working against a deadline, you are frantically trying to finish a report on the current state of security of the organization. You are pulling records and data items from over a dozen sources, including a locally hosted database, several documents, a few spreadsheets, and numerous web pages from an internal server. However, as you start to open another file from your hard drive, the system crashes and displays the Windows Blue Screen of Death. This event is formally known as a stop error and is an example of a(n) _____ approach to software failure.

   A. Fail-open

B. Fail-secure

C. Limit check

D. Object-oriented

4. As a software designer, you want to limit the actions of the program you are developing. You have considered using bounds and isolation but are not sure they perform the functions you need. Then, you realize that the limitation you want can be achieved using confinement. Which best describes a confined or constrained process?

   A. A process that can run only for a limited time

   B. A process that can run only during certain times of the day

   C. A process that can access only certain memory locations

   D. A process that controls access to an object

5. When a trusted subject violates the star property of Bell–LaPadula to write an object into a lower level, what valid operation could be taking place?

   A. Perturbation

   B. Noninterference

   C. Aggregation

   D. Declassification

6. What security method, mechanism, or model reveals a capabilities list of a subject across multiple objects?

   A. Separation of duties

   B. Access control matrix

   C. Biba

   D. Clark–Wilson

7. What security model has a feature that in theory has one name or label but, when implemented into a solution, takes on the name or label of the security kernel?

   A. Information flow model

B. Biba model

C. Trusted computing base

D. Brewer and Nash model

8. The Clark–Wilson model uses a multifaceted approach to enforcing data integrity. Instead of defining a formal state machine, the Clark–Wilson model defines each data item and allowable data transformations. Which of the following is not part of the access control relationship of the Clark–Wilson model?

   A. Object

   B. Interface

   C. Input sanitization

   D. Subject

9. While researching security models to base your new computer design around, you discover the concept of the TCB. What is a trusted computing base (TCB)?

   A. Hosts on your network that support secure transmissions

   B. The operating system kernel, other OS components, and device drivers

   C. The combination of hardware, software, and controls that work together to enforce a security policy

   D. The predetermined set or domain (i.e., a list) of objects that a subject can access

10. What is a security perimeter? (Choose all that apply.)

    A. The boundary of the physically secure area surrounding your system

    B. The imaginary boundary that separates the TCB from the rest of the system

    C. The network where your firewall resides

    D. Any connections to your computer system

11. The trusted computing base (TCB) is a combination of hardware, software, and controls that work together to form a trusted base to enforce your security policy. What part of the TCB concept validates access to every resource prior to granting the requested access?

    A. TCB partition

    B. Trusted library

    C. Reference monitor

    D. Security kernel

12. A security model provides a way for designers to map abstract statements into a solution that prescribes the algorithms and data structures necessary to build hardware and software. Thus, a security model gives software designers something against which to measure their design and implementation. Which of the following is the best definition of a security model?

    A. A security model states policies an organization must follow.

    B. A security model provides a framework to implement a security policy.

    C. A security model is a technical evaluation of each part of a computer system to assess its concordance with security standards.

    D. A security model is used to host one or more operating systems within the memory of a single host computer or to run applications that are not compatible with the host OS.

13. The state machine model describes a system that is always secure no matter what state it is in. A secure state machine model system always boots into a secure state, maintains a secure state across all transitions, and allows subjects to access resources only in a secure manner compliant with the security policy. Which security models are built on a state machine model?

    A. Bell–LaPadula and take-grant

    B. Biba and Clark–Wilson

C. Clark–Wilson and Bell–LaPadula

D. Bell–LaPadula and Biba

14. You are tasked with designing the core security concept for a new government computing system. The details of its use are classified, but it will need to protect confidentiality across multiple classification levels. Which security model addresses data confidentiality in this context?

    A. Bell–LaPadula

    B. Biba

    C. Clark–Wilson

    D. Brewer and Nash

15. The Bell–LaPadula multilevel security model was derived from the DoD's multilevel security policies. The multilevel security policy states that a subject with any level of clearance can access resources at or below its clearance level. Which Bell–LaPadula property keeps lower-level subjects from accessing objects with a higher security level?

    A. (Star) security property

    B. No write-up property

    C. No read-up property

    D. No read-down property

16. The Biba model was designed after the Bell–LaPadula model. Whereas the Bell–LaPadula model addresses confidentiality, the Biba model addresses integrity. The Biba model is also built on a state machine concept, is based on information flow, and is a multilevel model. What is the implied meaning of the simple property of Biba?

    A. Write-down

    B. Read-up

    C. No write-up

    D. No read-down

17. The Common Criteria defines various levels of testing and confirmation of systems' security capabilities. What part of the Common Criteria specifies the claims of security from the vendor that are built into a target of evaluation?

    A. Protection profiles

    B. Evaluation assurance levels

    C. Authorizing official

    D. Security target

18. The authorizing official (AO) has the discretion to determine which breaches or security changes result in a loss of the authorization to operate (ATO). The AO can also issue four types of authorization decisions. Which of the following are examples of these ATOs? (Choose all that apply.)

    A. Common control authorization

    B. Mutual authorization

    C. Denial of authorization

    D. Authorization to transfer

    E. Authorization to use

    F. Verified authorization

19. A new operating system update has made significant changes to the prior system. While testing, you discover that the system is highly unstable, allows for integrity violations between applications, can be affected easily by local denial-of-service attacks, and allows for information disclosure between processes. You suspect that a key security mechanism has been deactivated or broken by the update. What is a likely cause of these problems?

    A. Use of virtualization

    B. Lack of memory protections

    C. Not following the Clark–Wilson model

    D. Support for storage and transmission encryption

20. As an application designer, you need to implement various security mechanisms to protect the data that will be accessed and processed by your software. What would be the purpose of implementing a constrained or restricted interface?

    A. To limit the actions of authorized and unauthorized users

    B. To enforce identity verification

    C. To track user events and check for violations

    D. To swap datasets between primary and secondary memory

Chapter 9
Security Vulnerabilities, Threats, and Countermeasures

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 3.0: Security Architecture and Engineering**
  - 3.1 Research, implement and manage engineering processes using secure design principles
    - 3.1.10 Shared responsibility
  - 3.5 Assess and mitigate the vulnerabilities of security architectures, designs, and solution elements
    - 3.5.1 Client-based systems
    - 3.5.2 Server-based systems
    - 3.5.5 Industrial control systems (ICS)
    - 3.5.7 Distributed systems
    - 3.5.8 Internet of Things (IoT)
    - 3.5.9 Microservices (e.g., application programming interface (API))
    - 3.5.10 Containerization
    - 3.5.12 Embedded systems
    - 3.5.13 High-Performance Computing systems
    - 3.5.14 Edge computing systems
    - 3.5.15 Virtualized systems

Security professionals must pay careful attention to the IT system and ensure that their higher-level protective controls are not built on a shaky

foundation. After all, the most secure firewall configuration in the world won't do much good if the underlying system has a fundamental security flaw that allows malicious individuals to bypass the firewall altogether.

In this chapter, we'll cover those underlying security concerns by surveying a field known as *computer architecture*: the physical design of computers from various components.

The Security Architecture and Engineering domain addresses various concerns and issues, including secure design elements, security architecture, vulnerabilities, threats, and associated countermeasures. Additional elements of this domain are discussed in various chapters:

- Chapter 1, “Security Governance Through Principles and Policies”
- Chapter 6, “Cryptography and Symmetric Key Algorithms”
- Chapter 7, “PKI and Cryptographic Applications”
- Chapter 8, “Principles of Security Models, Design, and Capabilities”
- Chapter 9, “Security Vulnerabilities, Threats, and Countermeasures”
- Chapter 10, “Physical Security Requirements”
- Chapter 14, “Controlling and Monitoring Access”
- Chapter 16, “Managing Security Operations”
- Chapter 20, "Software Development Security"
- Chapter 21, “Malicious Code and Application Attacks”

Please be sure to review all of these chapters to have a complete perspective on the topics of this domain.

Shared Responsibility

Shared responsibility is the security design principle that indicates that organizations do not operate in isolation. Instead, they are intertwined with the world in numerous ways. We all use the same basic technology, we follow the same communication protocol specifications, we use the same Internet, we use common foundations of operating systems and

programming languages, and most of our IT/IS is implemented using *off-the-shelf solutions* (whether commercial or open source). Thus, we are automatically integrated with the rest of the world and share the responsibility of establishing and maintaining security.

We must realize this shared responsibility and take our role in this situation seriously. Here are several aspects of this concept to ponder:

- Everyone in an organization has some level of security responsibility. It is the job of the CISO and security team to establish security and maintain it. It is the job of the regular employees to perform their tasks within the confines of security. It is the job of the auditor to monitor the environment for violations.
- Organizations are responsible to their stakeholders to make good security decisions to sustain the organization. Otherwise, the needs of the stakeholders may be violated.
- When working with third parties, especially with cloud providers, each entity needs to understand their portion of the shared responsibility of performing work operations and maintaining security. This is often referenced as the cloud-shared responsibility model, which is discussed further in Chapter 16.
- As we become aware of new vulnerabilities and threats, we should consider it our responsibility (if not our duty) to responsibly disclose that information to the proper vendor or to an information sharing center (also known as a threat intelligence source or service).



Automated indicator sharing (AIS) is an initiative by the Department of Homeland Security (DHS) to facilitate the open and free exchange of indicators of compromise (IoCs) and other cyberthreat information between the U.S. federal government and the private sector in an automated and timely manner (described as “machine speed”). An *indicator* is an observable along with a hypothesis about a threat. An *observable* is an identified fact of occurrence, such as the presence of a malicious file, usually accompanied by a hash.

AIS uses *Structured Threat Information eXpression (STIX)* and *Trusted Automated eXchange of Intelligence Information (TAXII)* to share threat indicators. STIX is a standardized language expressing structured information about cyberthreats and a common framework for organizations to share and analyze threat intelligence. TAXII defines protocols and services for automated sharing of structured threat information.

AIS is managed by the National Cybersecurity and Communications Integration Center (NCCIC). For more information on the AIS program, please visit `us-cert.gov/ais`.

Because we participate in shared responsibility, we must research, implement, and manage engineering processes using secure design principles.

Data Localization and Data Sovereignty

Data localization refers to storing and processing data within a specific country or region's physical borders or geographical boundaries. This concept is often driven by regulatory requirements or government policies that mandate certain data, especially sensitive or personal information, to be kept within the jurisdiction's borders where it was generated or where the data subject resides. Data localization aims to exert greater control over data privacy, security, and compliance with local laws. It can involve restrictions on cross-border data transfer and may influence how businesses

structure their data storage and processing infrastructure to adhere to these regulations.

Data sovereignty refers to the concept that digital data is subject to the laws and regulations of the country or region in which it is located or originates. It emphasizes that governments have authority over the data collected within their jurisdiction, and organizations must comply with local data protection and privacy laws. Data sovereignty is closely related to concerns about privacy, security, and compliance, and it often influences decisions regarding where data is stored, processed, and managed to align with legal and regulatory requirements in a given geographical area.

While data localization and data sovereignty are related concepts, they have distinct focuses and implications:

- *Focus:*
  - *Data localization:* Primarily centers on the physical location of data storage and processing. It mandates that certain types of data, especially sensitive or personal information, must be kept within specific geographical borders.
  - *Data sovereignty:* Encompasses a broader set of principles related to the authority and control that a country or region asserts over the data generated within its jurisdiction. This includes where the data is stored and concerns about legal jurisdiction and compliance with local data protection laws.
- *Scope:*
  - *Data localization:* Primarily addresses the geographical aspect of data storage and processing. It often involves regulations specifying that data should reside on servers physically located within the borders of a specific country or region.
  - *Data sovereignty:* Encompasses a broader range of concerns, including legal jurisdiction, regulatory compliance, and the overarching authority that a government has over data collected within its boundaries. It extends beyond mere storage location to include control and governance aspects.
- *Drivers:*

    - *Data localization:* Often driven by specific regulations or laws that mandate the physical presence of data within a particular jurisdiction. This can be motivated by data privacy, security, or economic concerns.
    - *Data sovereignty:* Driven by legal, political, and cultural considerations. Governments may enact data sovereignty regulations to protect the privacy of their citizens, ensure compliance with local laws, and assert control over data handling within their borders.
- *Implications:*
    - *Data localization:* The primary implication is that organizations must establish data centers or use cloud services within the specified jurisdiction to comply with local regulations. This may impact the efficiency of cross-border data flow and increase operational costs.
    - *Data sovereignty:* Besides impacting data storage location, data sovereignty considerations may influence how organizations handle data governance, legal compliance, and interactions with third-party service providers. It can require a more comprehensive approach to data management and legal compliance.

While data localization is a subset of data sovereignty, focusing specifically on where data is stored and processed, data sovereignty encompasses a broader set of principles related to data governance, jurisdiction, and legal compliance within a specific geopolitical boundary. Both concepts, however, underscore the importance of aligning data practices with local regulations and legal frameworks.



Data portability refers to the ability of individuals to easily and securely move their personal data from one system, service, or application to another. It allows users to transfer their data between different platforms, promoting user control and facilitating competition among service providers. Data portability empowers individuals by allowing them to choose services based on their preferences while maintaining access and control over their personal information. This concept is often associated with data protection and privacy regulations emphasizing user rights and data control.

Assess and Mitigate the Vulnerabilities of Security Architectures, Designs, and Solution Elements

Computer architecture is an engineering discipline concerned with designing and constructing computing systems at a logical level. Technical mechanisms that can be implemented via computer architecture are the controls that system designers can build into their systems. These include layering (see Chapter 1, "Security Governance Through Principles and Policies"), abstraction (see Chapter 1), data hiding (see Chapter 1), trusted recovery (see Chapter 18, "Disaster Recovery Planning"), process isolation (later in this chapter), and hardware segmentation (later in this chapter).



The more complex a system, the less assurance it provides. More complexity means more areas for vulnerabilities exist, and more areas must be secured against threats. More vulnerabilities and more threats mean that the subsequent security the system provides is less trustworthy. See Chapter 8 for more on "keep it simple."

Hardware

The term *hardware* encompasses any tangible part of a computer that you can reach out and touch, from the keyboard and monitor to its CPU(s), storage media, and memory chips. Remember that although the physical portion of a storage device (such as a hard disk or flash memory) may be considered hardware, the contents of those devices—the collections of 0s and 1s that make up the software and data stored within them—may not.

Processor

The *central processing unit (CPU)*, generally called the *processor* or the *microprocessor*, is the computer's nerve center—it is the chip (or chips in a multiprocessor system) that governs all major operations and either directly performs or coordinates the complex symphony of calculations that allows a computer to perform its intended tasks. Surprisingly, the CPU can perform only a limited set of computational and logical operations, despite the complexity of the tasks it allows the overall computer system to perform. The operating system and compilers or interpreters are responsible for translating high-level programming languages into simple instructions that a CPU understands. This limited range of functionality is intentional—it allows a CPU to perform computational and logical operations at blazing speeds.

Execution Types

As computer processing power increased, users demanded more advanced features to enable these systems to process information at greater rates and to manage multiple functions simultaneously:



At first blush, the terms *multitasking*, *multicore*, *multiprocessing*, *multiprogramming*, and *multithreading* may seem nearly identical. However, they describe very different ways of approaching the “doing two things at once” problem. We strongly advise you to review the distinctions between these terms until you feel comfortable with them.

Multitasking In computing, *multitasking* means handling two or more tasks simultaneously. In the past, most systems did not truly multitask because they relied on the OS to simulate multitasking by carefully structuring the sequence of commands sent to the CPU for execution (see multiprogramming). A single-core multitasking system is able to juggle more than one task or process at any given time. However, with that single-core CPU, it only executes a single process at any given moment. This is similar to juggling three balls, where your hands usually touch only one ball at any given instant, but the coordination of movements keeps all three balls moving.

Multicore Today, most CPUs are *multicore*. This means that the CPU is now a chip containing two, four, eight, dozens, or more independent execution cores that can operate simultaneously and/or independently. There are even some specialty chips with over 10,000 cores.

Multiprocessing In a *multiprocessing* environment, a multiprocessor system harnesses the power of more than one processor to complete the execution of a multithreaded application. See the section “Large-Scale Parallel Data Systems,” later in this chapter.



Some multiprocessor systems may assign or dedicate a process or execution thread to a specific CPU (or core). This is called *affinity*.

Multiprogramming *Multiprogramming* is similar to multitasking. It involves the pseudo-simultaneous execution of two tasks on a single processor coordinated by the OS as a way to increase operational efficiency. For the most part, multiprogramming is a way to batch or serialize multiple processes so that when one process stops to wait on a peripheral, its state is saved, and the next process in line begins to process. The first program does not return to processing until all other processes in the batch have had their chance to execute, and they, in turn, stop for a peripheral. This methodology causes significant delays

in completing a task for any single program. However, across all processes in the batch, the total time to complete all tasks is reduced.

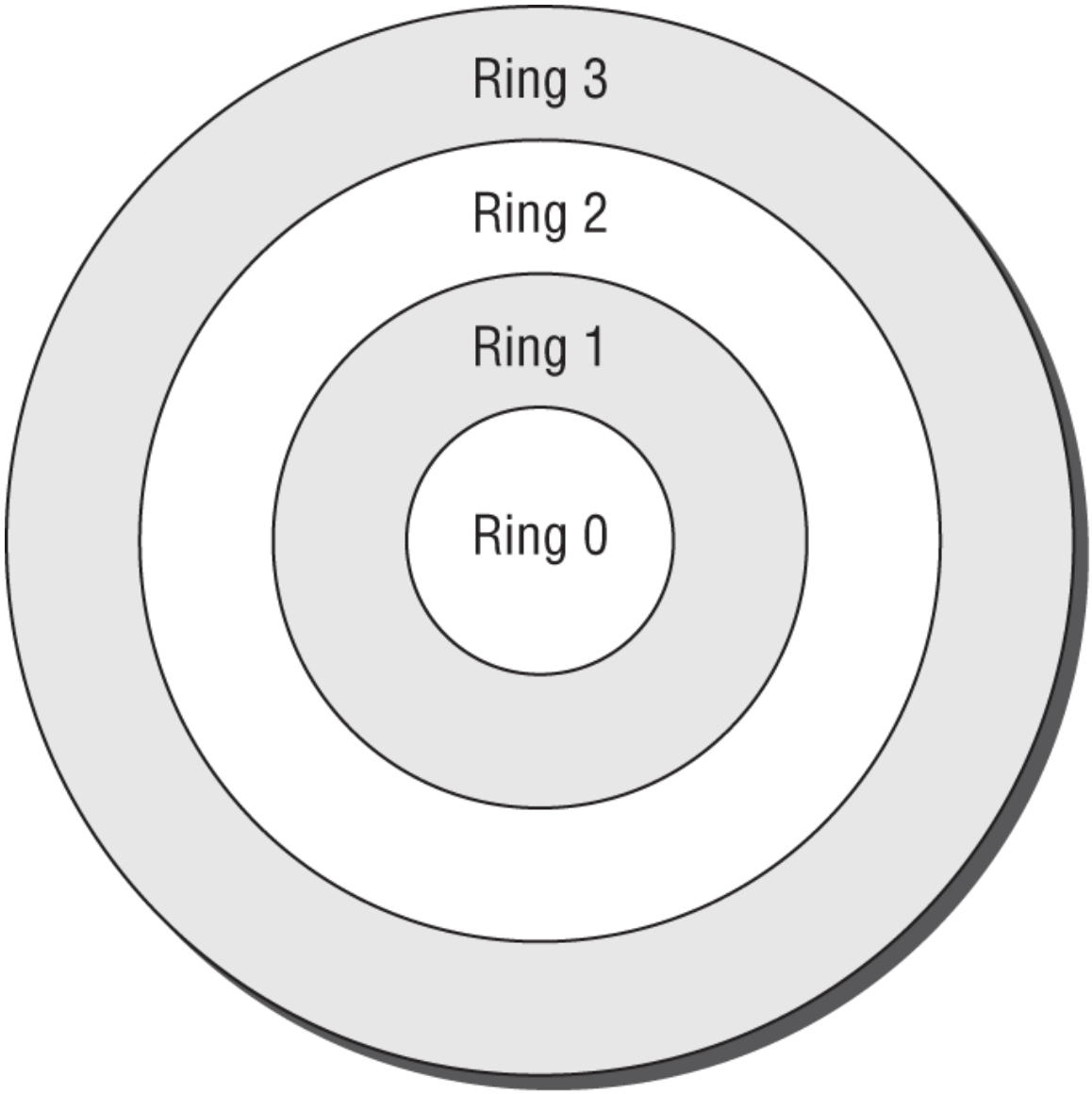
Multithreading *Multithreading* permits multiple concurrent tasks to be performed within a single process. Unlike multitasking, where multiple tasks consist of multiple processes, multithreading enables multiple tasks to operate within a single process. A thread is a self-contained sequence of instructions that can execute in parallel with other threads that are part of the same parent process. Multithreading is often used in applications where frequent context switching between multiple active processes causes excessive overhead and reduces efficiency; switching between threads incurs far less overhead and is, therefore, more efficient.

Protection Mechanisms

When a computer is running, it operates a runtime environment that represents the combination of the OS and whatever applications may be active. Within that runtime environment, it's necessary to integrate security controls to protect the OS's integrity, manage which users are allowed to access specific data items, authorize or deny operations requested against such data, and so forth. How running computers implement and handle security at runtime may be broadly described as a collection of protection mechanisms, such as protection rings and operational states.

PROTECTION RINGS

From a security standpoint, *protection rings* organize code and components in an OS (as well as applications, utilities, or other code that runs under the OS's control) into concentric rings, as shown in Figure 9.1. The deeper inside the circle, the higher the privilege level associated with the code that occupies a specific ring. Though the original Multics implementation allowed up to seven rings (numbered 0 through 6), most modern OSs use a four-ring model (numbered 0 through 3).



Ring 0: OS Kernel/Memory (Resident Components)
Ring 1: Other OS Components
Ring 2: Drivers, Protocols, etc.
Ring 3: User-Level Programs and Applications

Rings 0–2 run in supervisory or privileged mode.
Ring 3 runs in user mode.

FIGURE 9.1 The four-layer protection ring model

As the innermost ring, 0 has the highest level of privilege and can basically access any resource, file, or memory location. The part of an OS that always remains resident in memory (so that it can run on demand at any time) is called the *kernel*. It occupies ring 0 and can preempt code running at any other ring. The remaining parts of the OS—those that come and go as various tasks are requested, operations are performed, processes are switched, and so forth—occupy ring 1. Ring 2 is also somewhat privileged in that it's where I/O drivers and system utilities reside; these are able to access peripheral devices, special files, and so forth that applications and other programs cannot themselves access directly. Those applications and programs occupy the outermost ring, ring 3.

The essence of the ring model lies in priority, privilege, and memory segmentation. Any process that wants to execute must get in line (a pending process queue). The process associated with the lowest ring number always runs before processes associated with higher-numbered rings. Processes in lower-numbered rings can access more resources and interact with the OS more directly than those in higher-numbered rings. Those processes that run in higher-numbered rings must generally ask a handler or driver in a lower-numbered ring for services they need (aka system call), sometimes called a *mediated-access model*. In practice, many modern OSs use only two rings or divisions: one for system-level access (rings 0 through 2), often called *kernel mode* or *privileged mode*, and one for user-level programs and applications (ring 3), often called *user mode*.

From a security standpoint, the ring model enables an OS to protect and insulate itself from users and applications. It also permits the enforcement of strict boundaries between highly privileged OS components (such as the kernel) and less privileged parts of the OS (such as other parts of the OS, plus drivers and utilities).

The ring that a process occupies determines its access level to system resources. Processes may access objects directly only if they reside within their own ring or within some outside ring. Before any such request can be honored, the called ring must check to make sure that the calling process has the proper credentials and authorization to access the data and to perform the operation(s) involved in satisfying the request.

Rings Compared to Levels

Many of the protecting ring concept's features also apply to a multilayer or multilevel system. The top of a layered or multilevel system is the same as the center ring (i.e., ring 0) of a protection ring scheme. Likewise, the bottom of a layered or multilevel system is the same as the outer ring of a protection ring scheme. Levels, layers, domains, and rings are similar in terms of protection and access concepts.

PROCESS STATES

Process states or *operating states* are various forms of execution in which a process may run. Where the OS is concerned, it can be in one of two modes at any given moment: operating in a privileged, all-access mode known as *supervisor state* (aka Kernel mode) or operating in what's called the *problem state* associated with user mode, where privileges are low and all access requests must be checked against credentials for authorization before they are granted or denied. The latter is called the problem state not because problems are guaranteed to occur but because the unprivileged nature of user access means that problems can occur and the system must take appropriate measures to protect security, integrity, and confidentiality.

Processes line up for execution in an OS in a processing queue, where they will be scheduled to run as a processor becomes available. Most OSs allow processes to consume processor time only in fixed increments or chunks; if a process consumes its entire processing time (called a *time slice*) without completing, it returns to the processing queue for another time slice the next time its turn comes around. Also, the process scheduler usually selects the highest-priority process for execution, so reaching the front of the line doesn't always guarantee access to the CPU (because a process may be preempted at the last instant by another process with higher priority).

According to whether a process is running, it can operate in one of several states:

Ready In the *ready state*, a process is ready to resume or begin processing as soon as it is scheduled for execution. If the CPU is

available when the process reaches this state, it will transition directly into the running state; otherwise, it sits in the ready state until its turn comes up.

Running The *running state* or *problem state* is when a process executes on the CPU and keeps going until it finishes, its time slice expires, or it is blocked for some reason (usually because it has generated an interrupt for I/O). If the time slice ends and the process isn't completed, it returns to the ready state; if the process is paused while waiting for I/O, it goes into the waiting state.

Waiting The *waiting state* is when a process is ready for continued execution but is waiting for I/O to be serviced before it can continue processing. Once I/O is complete, then the process typically returns to the ready state, where it waits in the process queue to be assigned time again on the CPU for further processing.

Supervisory The *supervisory state* is used when the process must perform an action that requires privileges that are greater than the problem state's set of privileges, including modifying system configuration, installing device drivers, or modifying security settings. Basically, any function not occurring in the user mode (ring 3) or problem state takes place in the supervisory mode. This state is not shown in Figure 9.2, but it effectively replaces the running state when a process is run with higher-level privileges.

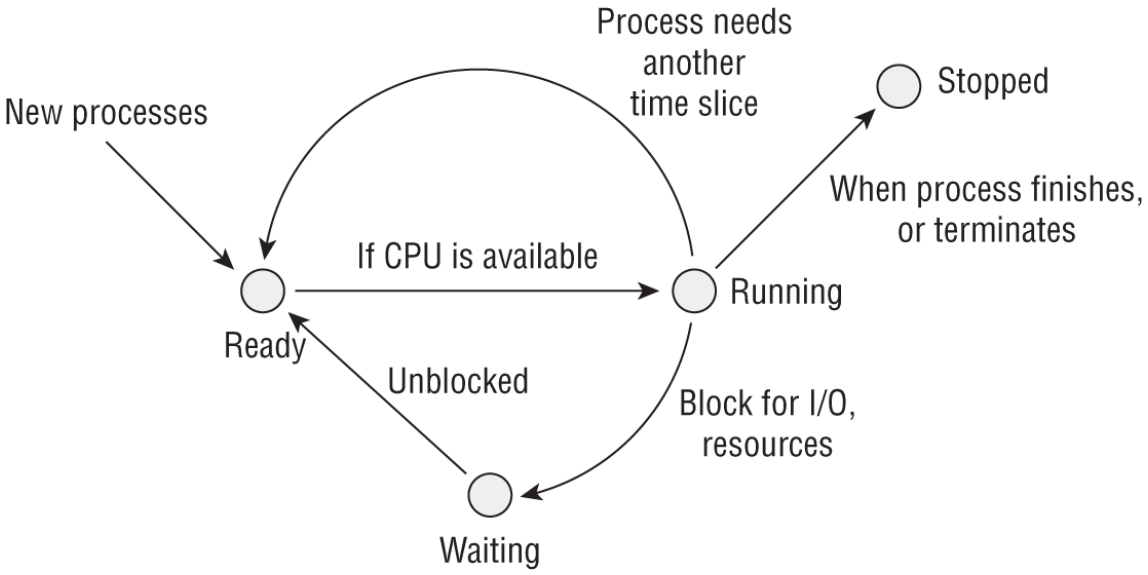


FIGURE 9.2 The life cycle of an executed process

Stopped When a process finishes or must be terminated (because an error occurs, a required resource is not available, or a resource request can't be met), it goes into a *stopped state*. At this point, the OS can recover all memory and other resources allocated to the process and reuse them for other processes as needed.

Figure 9.2 shows a diagram of how these various states relate to one another. New processes always transition into the ready state. When the OS decides which process to run next, it checks the ready queue and takes the highest-priority job that's ready to run.

Memory

The second major hardware component of a system is *memory*, the storage bank for information that the computer needs to keep readily available. There are many different kinds of memory, each suitable for different purposes, and we'll take a look at each in the sections that follow.

Read-Only Memory

Read-only memory (ROM) works like the name implies—it's memory the system can read but can't change (no writing allowed). The contents of a standard ROM chip are burned in at the factory, and the end user simply cannot alter it. ROM chips often contain “bootstrap” information that

computers use to start up prior to loading an OS from disk. This includes the *power-on self-test (POST)* series of diagnostics that run each time you boot a PC.

ROM's primary advantage is that it can't be modified. This attribute makes ROM extremely desirable for orchestrating a computer's innermost workings.

There is a type of ROM that may be altered to some extent. It is known as programmable read-only memory (PROM), and its several subtypes:

Programmable Read-Only Memory (PROM) A basic *programmable read-only memory (PROM)* chip is similar to a ROM chip in functionality, but with one exception. During manufacturing, a PROM chip's contents aren't “burned in” at the factory as with standard ROM chips. Instead, a PROM incorporates special functionality that allows an end user to burn in the chip's contents later. Once data is written to a PROM chip, no further changes are possible.

Erasable Programmable Read-Only Memory (EPROM) EPROM combines the relatively high cost of PROM chips and software developers' inevitable desire to tinker with their code once it's written. You then have the rationale for the development of *erasable PROM (EPROM)*. There are two main subcategories of EPROM: UVEPROM and EEPROM (see the next item). *Ultraviolet EPROMs (UVEPROMs)* can be erased with UV light. These chips have a small window that, when illuminated with a special ultraviolet light, causes the contents of the chip to be erased. After this process is complete, end users can burn new information into the UVEPROM as if it had never been programmed before.

Electronically Erasable Programmable Read-Only Memory (EEPROM) A more flexible, friendly alternative to UVEPROM is *electronically erasable PROM (EEPROM)*, which uses electric voltages delivered to the pins of the chip to force erasure.

Flash Memory *Flash memory* is a derivative concept from EEPROM. It is a nonvolatile form of storage media that can be electronically erased and rewritten. The primary difference between EEPROM and flash memory is that EEPROM can be erased and

written at the byte level, whereas flash memory can be erased and written in blocks or pages. The most common type of flash memory is NAND flash. It is widely used in memory cards, thumb drives, mobile devices, and SSDs (solid-state drives).

Random Access Memory

Random access memory (RAM) is readable and writable memory that contains information a computer uses during processing. RAM retains its contents only when power is continuously supplied to it. Unlike with ROM, when a computer is powered off, all data stored in RAM disappears. For this reason, RAM is useful only for temporary storage. Critical data should never be stored solely in RAM; a backup copy should always be kept on another storage device to prevent its disappearance in the event of a sudden loss of electrical power. The following are types of RAM:

Real Memory *Real memory* (also known as *main memory* or *primary memory*) is typically the largest RAM storage resource available to a computer. It is normally composed of a number of dynamic RAM chips and, therefore, must be refreshed by the CPU periodically (see the sidebar "Dynamic vs. Static RAM" for more information on this subject).

Cache RAM Computer systems contain many caches that improve performance by taking data from slower devices and temporarily storing it in faster devices when repeated use is likely; this is *cache RAM*. The processor normally contains an onboard cache of extremely fast memory used to hold data on which it will operate. This can be referred to as L1, L2, L3, and even L4 cache (with the L being short for level). Many modern CPUs include up to three levels of on-chip cache, with some caches (usually L1 and/or L2) dedicated to a single processor core, whereas L3 may be a shared cache between cores. Some CPUs can involve L4 cache, which may be located on the mainboard/motherboard or on the GPU (graphics processing unit). Likewise, real memory often contains a cache of information pulled or read from a storage device.

Many peripherals also include onboard caches to reduce the storage burden they place on the CPU and OS. Many storage devices, such as hard disk

drives (HDDs), solid-state drives (SSDs), and some thumb drives, contain caches to assist with improving read and write speed. However, these caches must be flushed to the permanent or secondary storage area before disconnection or power loss to avoid data loss of cache resident data.

Dynamic vs. Static RAM

There are two main types of RAM: dynamic RAM and static RAM. Most computers contain a combination of both types and use them for different purposes.

To store data, dynamic RAM uses a series of capacitors, tiny electrical devices that hold a charge. These capacitors either hold a charge (representing a 1 bit in memory) or do not hold a charge (representing a 0 bit). However, because capacitors naturally lose their charges over time, the CPU must spend time refreshing the contents of dynamic RAM to ensure that 1 bits don't unintentionally change to 0 bits, thereby altering memory contents.

Static RAM uses more sophisticated technology—a logical device known as a *flip-flop*, which to all intents and purposes, is simply an on/off switch that must be moved from one position to another to change a 0 to 1, or vice versa. More importantly, static memory maintains its contents unaltered as long as power is supplied and imposes no CPU overhead for periodic refresh operations.

Dynamic RAM is cheaper than static RAM because capacitors are cheaper than flip-flops. However, static RAM runs much faster than dynamic RAM. This creates a trade-off for system designers, who combine static and dynamic RAM modules to strike the right balance of cost versus performance.

Registers

The CPU also includes a limited amount of onboard memory, known as *registers*, that provide it with directly accessible memory locations that the brain of the CPU, the *arithmetic-logical unit (ALU)*, uses when performing calculations or processing instructions. The size and number of registers

vary, but typical CPUs have 8 to 32 registers and are often either 32 or 64 bits in size. In fact, any data that the ALU is to manipulate must be loaded into a register unless it is directly supplied as part of the instruction. The main advantage of this type of memory is that it is part of the ALU itself and, therefore, operates in lockstep with the CPU at typical CPU speeds.

Memory Addressing

When using memory resources, the processor must have some means of referring to various locations in memory. The solution to this problem is known as *memory addressing*, and several different addressing schemes are used in various circumstances. The following are five of the most common addressing schemes:

Register Addressing As you learned in the previous section, registers are small memory locations directly in the CPU. When the CPU needs information from one of its registers to complete an operation, it uses a *register address* (for example, “register 1”) to access its contents.

Immediate Addressing *Immediate addressing* is not a memory addressing scheme per se but rather a way of referring to data that is supplied to the CPU as part of an instruction. For example, the CPU might process the command “Add 2 to the value in register 1.” This command uses two addressing schemes. The first is immediate addressing—the CPU is being told to add the value 2 and does not need to retrieve that value from a memory location—it's supplied as part of the command. The second is register addressing; it's instructed to retrieve the value from register 1.

Direct Addressing In *direct addressing*, the CPU is provided with an actual address of the memory location to access. The address must be located on the same memory page as the instruction being executed. Direct addressing is more flexible than immediate addressing since the contents of the memory location can be changed more readily than reprogramming the immediate addressing's hard-coded data.

Indirect Addressing *Indirect addressing* uses a scheme similar to direct addressing. However, the memory address supplied to the CPU as part of the instruction doesn't contain the actual value that the CPU

is to use as an operand. Instead, the memory address contains another memory address. The CPU reads the indirect address to learn the address where the desired data resides and then retrieves the actual operand from that address.

Base+Offset Addressing *Base+offset addressing* uses a value stored in one of the CPU's registers or pointers as the base location to begin counting. The CPU then adds the offset supplied with the instruction to that base address and retrieves the operand from that computed memory location.



A *pointer* is a basic element or object in many programming languages that is used to store a memory address. Basically, a pointer holds the address of something stored in memory so that when the program reads the pointer, it points to the location of the data actually needed by the application. Effectively, a pointer references a memory location. The act of accessing a pointer to read that memory location is known as *dereferencing*. Pointers can store the memory address used in direct, indirect, or base addressing. Another potential issue is a *race condition*, which occurs when a system or device tries to perform two or more operations at the same time. This can cause null pointer errors in which an application dereferences a pointer that it expects to be valid but is really null (or corrupted), resulting in a system crash.

Secondary Memory

Secondary memory is a term commonly used to refer to magnetic, optical, or flash-based media or other storage devices that contain data not immediately available to the CPU. For the CPU to access secondary memory data, the OS must first read the data and store it in real memory.

Virtual memory is a special type of secondary memory that is used to expand the addressable space of real memory. The most common type of virtual memory is the *pagefile* or *swapfile* that most OSs manage as part of their memory management functions. This specially formatted file contains previously stored RAM data but not recently used in real memory. When

the OS needs to access addresses stored in the pagefile, it checks to see whether the page is memory-resident (in which case it can access it immediately) or whether it has been swapped to disk, in which case it reads the data from disk back into real memory (this process is called *paging*).

Virtual memory's primary drawback is that the paging operations that occur when data is exchanged between primary and secondary memory are relatively slow. The need for virtual memory is reduced with larger banks of actual physical RAM, and the performance hit of virtual memory can be reduced by using flash media (NVMe [nonvolatile memory express] SSDs, SSDs, and USB flash drives) to host the virtual memory paging file.

Data Storage Devices

Data storage devices are used to store information that may be used by a computer at any time after it's written.

Primary vs. Secondary

Primary memory, also known as *primary storage*, is the RAM that a computer uses to keep necessary information readily available to the CPU while the computer is running. Secondary memory (or *secondary storage*) includes all the familiar long-term storage devices you use often. Secondary storage consists of magnetic, optical, and flash media such as HDDs, SSDs, flash drives, magnetic tapes, CDs, DVDs, and flash memory cards.

Volatile vs. Nonvolatile

The volatility of a storage device is simply a measure of how likely it is to lose its data when power is turned off or cycled. Devices designed to retain their data (such as magnetic media, flash media, ROMs, and optical media) are classified as *nonvolatile*, whereas devices such as static or dynamic RAM modules, which lose their data when power is removed, are classified as *volatile*.

Random vs. Sequential

Storage devices may be accessed in one of two fashions. *Random access storage* devices allow an OS to read (and sometimes write) immediately from any point within the device by using some type of addressing system.

Almost all primary storage devices are random access devices. You can use a memory address to access information stored at any point within a RAM chip without reading the data that is physically stored before it. Most secondary storage devices are also random access.

Sequential storage devices, on the other hand, do not provide this flexibility. They require you to read (or speed past) all the data physically stored before the desired location. A common example of a sequential storage device is a magnetic tape drive.

Memory Security Issues

Memory stores and processes your data—some of which may be extremely sensitive. Any memory devices that may retain sensitive data should be purged before they are allowed to leave your organization for any reason. This is especially true for secondary memory and ROM/PROM/EPROM/EEPROM devices designed to retain data even after the power is turned off.

However, memory data retention issues are not limited to secondary memory (i.e., storage devices). It is technically possible that the electrical components used in volatile primary memory could retain some of their charge for a limited period of time after power is turned off. A technically sophisticated individual could theoretically retrieve portions of the data stored on such devices.

A memory compromise called the cold boot attack freezes memory chips to delay the decay of resident data when the system is turned off or the RAM is pulled out of the motherboard (see en.wikipedia.org/wiki/Cold_boot_attack). There are even attacks and tools that focus on memory image dumps or system crash dumps to extract encryption keys (see www.passware.com/kit-forensic).

Storage Media Security

There are several concerns when it comes to the security of secondary storage devices:

- Data may remain on secondary storage devices even after it has been erased. This condition is known as *data remanence*. Utilities are

available that can retrieve files from a disk even after they have been deleted or reformatted. If you truly want to remove data from a secondary storage device, you must use a specialized utility designed to overwrite all traces of data on the device (commonly called *sanitizing*) or damage or destroy it beyond possible repair. (See Chapter 5, "Protecting Security of Assets.")



SSDs are large-capacity flash memory secondary storage devices. Many SSDs include additional reserved memory blocks, which can be used in place of bad blocks. As blocks are written to or erased, they deteriorate at a predictable failure rate. Many SSD manufacturers counter this failure rate with two main techniques: reserved blocks and wear leveling. When a block stops working reliably, it is marked as bad, and a reserve block is then used in its place. This is similar to an HDD's bad sectors. Wear leveling attempts to perform write and erase events evenly across the entire drive's capacity of blocks to maximize use lifetime.

- A traditional zeroization wipe is less effective for SSDs because bad blocks/cells are likely not overwritten.
- Secondary storage devices are also prone to theft. Economic loss is not the major factor (after all, how much does a backup tape or a hard drive cost?), but the loss of confidential information poses the great risks. For this reason, it is important to use full-disk encryption to reduce the risk of an unauthorized entity gaining access to your data. Many HDDs, SSDs, and flash devices offer on-device native encryption.
- Removable media pose a significant information disclosure risk, so securing them often requires encryption technologies.

Emanation Security

Many electrical devices emanate electrical signals or radiation that can be intercepted and may contain confidential, sensitive, or private data. Obvious examples of emanation devices are wireless networking equipment and

mobile phones, but many other devices are vulnerable to emanation interception that you might not expect, including monitors, network cables, modems, and internal or external media drives (hard drives, USB thumb drives, CDs, and so on). With the right equipment, adversaries can intercept electromagnetic or radio frequency signals (collectively known as *emanations*) from these devices and interpret them to extract confidential data.



There are many valid uses of emanations, such as Wi-Fi, Bluetooth, GPS, and mobile phone signals.

The types of countermeasures and safeguards used to protect against emanation attacks are derived from TEMPEST. *TEMPEST* (Telecommunications Electronics Material Protected from Emanating Spurious Transmissions) was originally a government research study aimed at protecting electronic equipment. It refers to a set of standards and guidelines used to minimize the unintentional electromagnetic signals that could be emitted by electronic equipment and potentially intercepted, leading to unauthorized access to sensitive information. The term TEMPEST has become less common, and the corresponding security discipline is now commonly known as EMSEC. Emission Security (EMSEC) involves implementing various measures to prevent unauthorized individuals from obtaining valuable information that could be derived through intercepting and analyzing compromising emanations from cryptographic equipment, automated information systems (AISs), and telecommunications systems. It has since expanded to a general study of monitoring emanations and preventing their interception.

Simply because of the kinds of electronic components from which they're built, many computer hardware devices emit electromagnetic (EM) radiation during normal operation. The process of communicating with other machines or peripheral equipment creates emanations that can be intercepted. These emanation leaks can cause serious security issues but are generally easy to address.

TEMPEST/EMSEC attacks read the electronic emanations that devices produce (known as *Van Eck radiation*) from a distance (this process is known as *Van Eck phreaking*). TEMPEST/EMSEC eavesdropping or Van Eck phreaking countermeasures include the following:

Faraday Cage A *Faraday cage* is a box, mobile room, or entire building designed with an external metal skin, often a wire mesh that fully surrounds an area on all sides. This metal skin acts as an EM-absorbing capacitor that prevents electromagnetic signals (emanations) from exiting or entering the area that the cage encloses.

White Noise

White noise simply means broadcasting false traffic to mask and hide the presence of real emanations. White noise can consist of a real signal from another source that is not confidential, a constant signal at a specific frequency, a randomly variable signal, or even a jam signal that causes interception equipment to fail. Although this is similar to jamming devices, the purpose is to convolute the signal only for the eavesdropper, not the authorized user, rather than stopping even valid uses of emanations.



White noise describes any random sound, signal, or process that can drown out meaningful information. This can vary from audible frequencies to inaudible electronic transmissions, and it may even involve the deliberate act of creating line or traffic noise to disguise origins or disrupt listening devices.

Control Zone A third type of TEMPEST/EMSEC countermeasure, a *control zone*, is simply implementing a Faraday cage and white noise generation to protect a specific area in an environment; the rest of the environment is unaffected. A control zone can be a room, a floor, or an entire building.

In addition to the TEMPEST/EMSEC derived countermeasure technologies concepts, shielding, access control, and antenna management can be helpful

against emanation eavesdropping. *Shielding* of cables (networking and otherwise) may be sufficient to reduce or block emanation access. This may be an element included in the manufacture of equipment, such as shielded twisted pair (STP) cabling, or may be accomplished by using shielding conduits or just replacing copper network cables with fiber-optic cables.

Input and Output Devices

Input and output devices can present security risks to a system. Security professionals should be aware of these risks and ensure that appropriate controls are in place to mitigate them.

Monitors

TEMPEST/EMSEC eavesdropping technology can compromise the security of data displayed on a monitor. This may be accomplished by eavesdropping on the video cable or the monitor itself. It is arguable that the biggest risk with any monitor is still shoulder surfing or cameras. Don't forget shoulder surfing is a concern for desktop displays, laptop displays, tablets, and mobile phones.

Printers

Printers also represent a security risk that is easy to overlook. Depending on the physical security controls used at your organization, it may be much easier to walk out with sensitive information in printed form than to walk out with a flash drive or magnetic media. If printers are shared, users may forget to retrieve their sensitive printouts, leaving them vulnerable to prying eyes. Many modern printers also store data locally, often on a hard drive, and some retain copies of printouts indefinitely. Printers are usually exposed on the network for convenient access and are often not designed to be secure systems.

Concerns should also apply to *multifunction printers (MFPs)*, especially those that include fax capabilities and that are network-attached (whether wired or wireless). In 2018, researchers discovered that it is still possible to take control of a computer system over a *public switched telephone network (PSTN)* line using ancient *AT commands* supported by telephone modems and fax modems/machines. See the researcher's DEFCON 26 presentation PDF at

media.defcon.org/DEF%20CON%2026/DEF%20CON%2026%20presentations/DEFCON-26-Yaniv-Balmas-What-The-FAX.pdf. If you don't always need fax capabilities, don't leave the telephone line plugged in. If you do need always-available fax capabilities, use a stand-alone fax machine.

Keyboards/Mice

Keyboards, mice, and similar input devices are not immune to security vulnerabilities. All of these devices are vulnerable to TEMPEST/EMSEC monitoring, whether wired or wireless. Also, keyboards are vulnerable to less sophisticated bugging. A simple device can be placed inside a keyboard or along its connection cable to intercept all the keystrokes that take place and transmit them to a remote receiver using a radio signal. This has the same effect as TEMPEST/EMSEC monitoring but can be done with much less expensive gear. Additionally, if your keyboard and mouse are wireless, including Bluetooth, their radio signals can be intercepted.

POTS Telephone Modems

With the advent of ubiquitous broadband and wireless connectivity, telephone modems are becoming a scarce legacy computer component. If your organization is still using older equipment, there is a chance that a modem is part of the hardware configuration. Modems allow users to create uncontrolled access points into your network. If improperly configured, they can create extremely serious security vulnerabilities that allow an outsider to bypass all your perimeter protection mechanisms and directly access your network resources. Modems create an alternate egress channel that insiders can use to funnel data outside your organization. But keep in mind that these vulnerabilities can be exploited only if the modem is connected to an operational telephone landline.



The same risk of creating a security perimeter bypass exists when systems have both wired and wireless NICs. Systems should typically be restricted to using only one method/means of connection at a time. For example, if a cable is connected to the system's RJ45 jack, then the wireless interface should be deactivated. It may also be worth considering that for devices that exit and enter the premises a geofencing type system be used, where wireless connection devices are deactivated as the equipment enters the facility. See more on this in Chapter 11, “Secure Network Architecture and Components.”

You should seriously consider an outright ban on telephone modems in your organization's security policy unless you truly need them for business reasons. In those cases, security officials should know the physical and logical locations of all modems on the network, ensure that they are correctly configured, and make certain that appropriate protective measures are in place to prevent their illegitimate use.

Firmware

Firmware (also known as *microcode*) is a term used to describe software that is stored in a ROM or an EEPROM chip. This type of software is changed infrequently (actually, never, if it's stored on a true ROM chip as opposed to an EEPROM or flash chip) and often drives the basic operation of a computing device.

Many hardware devices, such as printers and modems, need some limited set of instructions and processing power to complete their tasks while minimizing the burden placed on the OS itself. In many cases, these “mini” OSs are entirely contained in firmware chips onboard the devices they serve. Many devices commonly use firmware as their primary OS, including mobile devices, Internet of Things (IoT) equipment, edge computing devices, fog computing devices, and industrial control systems.

Basic input/output system (BIOS) is the legacy basic low-end firmware or software embedded in a motherboard's EEPROM or flash chip. The BIOS contains the OS-independent primitive instructions that a computer needs to

start up and load the OS from disk. The BIOS identifies and initiates the basic system hardware components, such as the hard drive, optical drive, and video card, so that the bootstrapping process of loading an OS can begin. In most modern systems, the BIOS has been replaced by UEFI.

Unified Extensible Firmware Interface (UEFI) provides support for all of the same functions as BIOS with many improvements, such as support for larger hard drives (especially for booting), faster boot times, enhanced security features, and even the ability to use a mouse when making system changes (BIOS was limited to keyboard control only). UEFI also includes a CPU-independent architecture, a flexible pre-OS environment with networking support, measured boot, boot attestation (aka secure boot), and backward and forward compatibility. It also runs CPU-independent drivers (for system components, drive controllers, and hard drives).

The process of updating the UEFI, BIOS, or firmware is known as *flashing*. If bad actors or malware can alter the UEFI, BIOS, or firmware of a system, they may be able to bypass security features or initiate otherwise prohibited activities. There have been a few examples of malicious code embedding itself into UEFI, BIOS, or firmware. There is also an attack known as *phlashing*, in which a malicious variation of official BIOS or firmware is installed that introduces remote control or other malicious features into a device.

Boot attestation or *secure boot* is a feature of UEFI that aims to protect the local OS by preventing the loading or installing of device drivers or an OS that is not signed by a preapproved digital certificate. Secure boot thus protects systems against a range of low-level or boot-level malware, such as certain rootkits and backdoors. Secure boot intends that only drivers and OSs that pass attestation (the verification and approval process accomplished through the validation of a digital signature) are allowed to be installed and loaded on the local system. Unfortunately, there are now forms of malware which can bypass or abuse UEFI security features on specific systems.

Measured boot is an optional feature of UEFI that takes a hash calculation of every element involved in the booting process. The hashes are performed by and stored in the Trusted Platform Module (TPM). If foul play is detected in regard to booting, the hashes of the most recent boot can be

accessed and compared against known-good values to determine which (if any) of the boot components have been compromised. Measured boot does not interrupt or stop the process of booting; it just records the hash IDs of the elements used in the boot. Thus, it is like a security camera. It does not prevent a malicious action; it just records whatever occurs in its area of view.

Client-Based Systems

Client-based vulnerabilities place the user, their data, and their system at risk of compromise and destruction. A client-side attack is any attack that is able to harm a client. Generally, when attacks are discussed, it's assumed that the primary target is a server or a server-side component. A client-side or client-focused attack is one where the client itself, or a process on the client, is the target. A common example of a client-side attack is a malicious website that transfers malicious mobile code (such as an applet) to a vulnerable browser running on the client. Client-side attacks can occur over any communications protocol, not just Hypertext Transfer Protocol (HTTP). Another potential vulnerability that is client-based is the risk of poisoning of local caches.

Mobile Code

Applets are code objects sent from a server to a client to perform some action. In fact, applets are actually self-contained miniature programs that execute independently of the server that sent them—that is, mobile code. The arena of the web is undergoing constant flux. The use of applets is not as common today as it was in the early 2010s. However, applets are not absent from the web, and many browsers still support them (or still have add-ons present that support them). Thus, even when your organization does not intentionally use applets in your internal or public web design, your web browsers could encounter them while surfing the public web.

Imagine a web server that offers a variety of financial tools to web users. One of these tools might be a mortgage calculator that processes a user's financial information and provides a monthly mortgage payment based on the loan's principal and term and the borrower's credit information. Instead of processing this data and returning the results to the client system, the

remote web server might send an applet to the local system that enables it to perform those calculations itself. This provides a number of benefits to both the remote server and the end user:

- The processing burden is shifted to the client, freeing up resources on the web server to process requests from more users.
- The client is able to produce data using local resources rather than waiting for a response from the remote server. In many cases, this results in a quicker response to changes in the input data.
- In a properly programmed applet, the web server does not receive any data provided to the applet as input, therefore maintaining the security and privacy of the user's financial data.

However, applets introduce security concerns. Any time you execute someone else's code, you are at risk of the code being malicious or having exploitable flaws. Security administrators must take steps to ensure that code sent to systems on their network is safe and properly screened for malicious activity. Also, unless the code is analyzed line by line, the end user can never be certain that the applet doesn't contain a Trojan horse, backdoor, rootkit, ransomware, or some other malware component. For example, the mortgage calculator might indeed transmit sensitive financial information to the web server without the end user's knowledge or consent.

Two historical examples of applet types are Java applets and ActiveX controls. *Java* is a platform-independent programming language developed by Sun Microsystems (now owned by Oracle). *ActiveX* controls were Microsoft's response to Sun's Java. Java is still in use for internal development and business software, but its use on the Internet is rare. ActiveX is now a legacy technology and is both EOL and EOS. It was only supported by internet Explorer. Most modern internet-capable systems no longer support these applet forms, but it is important to ensure that before assuming a system is secure.

JavaScript is the most widely used web scripting language in the world and is embedded into (included inside of) HTML documents using `<script>` `</script>` enclosure tags. JavaScript is usually dependent on its HTML host document, but it can operate as a stand-alone script file (both client-side and server-side). However, it is generally not considered an applet—it

is usually embedded code. It is automatically downloaded along with the primary web documents from most web servers you access. JavaScript enables dynamic web pages and supports web applications as well as a plethora of client-side activities and page behaviors.

Most browsers support JavaScript via a dedicated JavaScript engine. Most of the implementations use sandbox isolation to restrict JavaScript to web-related activities while minimizing its ability to perform general-purpose programming tasks. Also, most browsers default to enforcing the same-origin policy. The same-origin policy prohibits JavaScript code from accessing content from another origin. The origin is typically defined by a combination of protocol (i.e., HTTP versus HTTPS), domain/IP address, and port number. If other content has any one of these origin elements different from the origin of the JavaScript code, the code will be blocked from accessing that content.

However, there are ways of abusing JavaScript. Threat actors can create believable fake websites that look and act like a valid site, including duplicating the JavaScript dynamic elements. But since the JavaScript code is in the HTML document sent to the browser, a malicious actor could alter that code to perform harmful actions, such as copying or cloning credentials and distributing them to the attacker. Threat actors have also found means to breach the sandbox isolation and even violate same-original policies from time to time, so JavaScript should be considered a threat. Whenever you allow code from an unknown and thus untrusted source to execute on your system, you are putting your system at risk of compromise. XSS and XSRF/CSRF exploit methods can be used to compromise JavaScript support in browsers.

Here are some responses to these risks:

- Keep browsers updated (client side).
- Implement secure JavaScript subsets/libraries (server side).
- Use a content security policy (CSP) that attempts to rigidly enforce same-origin restrictions for most browser-side active technologies (integrated into browsers and referenced by HTML header values).

As with most web applications, insertion attacks are common, so watch out for the injection of odd or abusive JavaScript code in the input being received by a web server.

As a client, you may gain some benefit by being behind a web application firewall (WAF) or next-generation firewall (NGFW). We do not recommend deactivating JavaScript outright—that would cause most of the web to stop functioning in your browser. Instead, the use of security-focused add-ons, browser helper objects (BHOs), and extensions may reduce the risk of JavaScript. However, the use of add-ons and extensions can also put your system at greater risks to other threats.

For more on web-related vulnerabilities, attacks, and countermeasures, see Chapter 21, “Malicious Code and Application Attacks.”

Local Caches

There are many types of local caches, including DNS cache, ARP cache, and temporary internet files. See Chapter 11 for details about DNS cache and ARP cache abuses.

Temporary internet files or the *internet files cache* is the temporary storage of files downloaded from Internet sites that are being held by the client's utility (typically a browser) for current and possibly future use. This cache mostly contains website content, but other Internet services can also use a file cache. A variety of exploitations, such as the split-response attack, can cause the client to download content that was not an intended element of a requested web page and store it in the cache. DOM XSS may be able to access and use locally cached files to execute malicious code or exfiltrate data (see Chapter 21). Mobile code scripting attacks could also be used to plant false content in the cache. Once files have been poisoned in the cache, then even when a legitimate web document calls on a cached item, the malicious item will be activated.

Client utilities should be managing the local files cache, but those utilities might not always be doing the best job. Often, the defaults are for efficiency and performance, not for security. Consider reconfiguring the cache to only retain files for a short period of time, minimize the cache size, and deactivate the preloading of content. Keep in mind that these changes can reduce browsing performance when on slower or high-latency connections.

You may want to configure the browser to delete all cookies and cache upon exit. Although you can typically perform a manual cache wipe, you would have to remember to do that. Another option is to use an automated tool that can be configured to wipe temporary Internet files on a schedule or upon a targeted program close.

Additional coverage of client-based endpoint security concerns is in Chapter 11.

Server-Based Systems

An important area of server-based concern, which may include clients as well, is the issue of *data flow control*. Data flow is the movement of data between processes, between devices, across a network, or over communication channels. Management of data flow ensures not only efficient transmission with minimal delays or latency, but also reliable throughput using hashing and confidentiality protection with encryption. Data flow control also ensures that receiving systems are not overloaded with traffic, especially to the point of dropping connections or being subject to a malicious or even self-inflicted denial of service. When data overflow occurs, data may be lost or corrupted or may trigger a need for retransmission. These results are undesirable, and data flow control is often implemented to prevent these issues from occurring. Data flow control may be provided by networking devices, including routers and switches, as well as network applications and services.

A *load balancer* spreads or distributes network traffic load across several network links or devices. A load balancer may be able to provide more control over data flow. The purpose of load balancing is to obtain more optimal infrastructure utilization, minimize response time, maximize throughput, reduce overloading, and eliminate bottlenecks. Although load balancing can be used in a variety of situations, a common implementation is spreading a load across multiple members of a server farm or cluster. A load balancer might use a variety of techniques to perform load distribution, including random choice, round robin, load/utilization monitoring, and preferencing. See Chapter 12, “Secure Communications and Network Attacks,” for more on load balancing.

A denial-of-service (DoS) attack can severely deter data flow control. It is important to monitor for DoS attacks and implement mitigations. See Chapter 17, “Preventing and Responding to Incidents,” for a discussion of these attacks and potential defenses.

For more on server protections, see Chapter 18.

Large-Scale Parallel Data Systems

Parallel data systems or *parallel computing* is a computation system designed to perform numerous calculations simultaneously. However, parallel data systems often go far beyond basic multiprocessing capabilities. They often include the concept of dividing up a large task into smaller elements, and then distributing each subelement to a different processing subsystem for parallel computation. This implementation is based on the idea that some problems can be solved efficiently if broken into smaller tasks that can be worked on concurrently. Parallel data processing can be accomplished by using distinct CPUs or multicore CPUs, virtual systems, or any combination of these. *Large-scale parallel data systems* must also be concerned with performance, power consumption, and reliability/stability issues.

Within the arena of multiprocessing or parallel processing there are several divisions. The first division is between symmetric multiprocessing (SMP) and asymmetric multiprocessing (AMP).

The scenario where a single computer contains multiple processors that are treated equally and controlled by a single OS is called *symmetric multiprocessing (SMP)*. In SMP, processors share not only a common OS but also a common data bus and memory resources. In this type of arrangement, systems may use a large number of processors. The collection of processors works collectively on a single or primary task, code, or project.

In *asymmetric multiprocessing (AMP)*, the processors are often operating independently of one another. Usually, each processor has its own OS and/or task instruction set, as well as a dedicated data bus and memory resources. Under AMP, processors can be configured to execute only specific code or operate on specific tasks (or specific code or tasks are

allowed to run only on specific processors; this might be called affinity in some circumstances).

A variation of AMP is *massive parallel processing (MPP)*, where numerous AMP systems are linked together to work on a single primary task across multiple processes in multiple linked systems. Some computationally intensive operations, such as those that support the research of scientists and mathematicians, require more processing power than a single OS can deliver. MPP may best serve such operations. MPP systems house hundreds or even thousands of processors, each of which has its own OS and memory/bus resources. Some MPPs have over 10 million execution cores. When the software that coordinates the entire system's activities and schedules them for processing encounters a computationally intensive task, it assigns responsibility for the task to a single processor (not so different from the Master Control Program [MCP] in the popular movie “Tron”). This processor breaks the task into manageable parts and distributes them to other processors for execution. Those processors return their results to the coordinating processor, where they are assembled and returned to the requesting application. MPP systems are extremely powerful (not to mention extremely expensive!) and are used in a great deal of computing or computational-based research.

Both types of multiprocessing provide unique advantages and are suitable for different types of situations. SMP systems are adept at processing simple operations at extremely high rates, whereas MPP systems are uniquely suited for processing very large, complex, computationally intensive tasks that lend themselves to decomposition and distribution into a number of subordinate parts.

The arena of large-scale parallel data systems is still evolving. It is likely that many management issues are yet to be discovered and solutions to known issues are still being sought. Large-scale parallel data management is likely a key tool in managing big data and will often involve cloud computing (see Chapter 16), grid computing, or peer-to-peer computing solutions.

Grid Computing

Grid computing is a form of parallel distributed processing that loosely groups a significant number of processing nodes to work toward a specific processing goal. Members of the grid can enter and leave the grid at random intervals. Often, members join the grid only when their processing capacities are not being taxed for local workloads. When a system is otherwise in an idle state, it could join a grid group, download a small portion of work, and begin calculations. When a system leaves the grid, it saves its work and may upload completed or partial work elements back to the grid. Many interesting uses of grid computing have developed, including projects seeking out intelligent aliens, performing protein folding, predicting the weather, modeling earthquakes, planning financial decisions, and solving for primes.

The biggest security concern with grid computing is that the content of each work packet is potentially exposed to the world. Many grid computing projects are open to the world, so there is no restriction on who can run the local processing application and participate in the grid's project. This also means that grid members could keep copies of each work packet and examine the contents. Thus, grid projects will not likely be able to maintain secrecy and are not appropriate for private, confidential, or proprietary data.

Grid computing can also vary greatly in computational capacity from moment to moment. Work packets are sometimes not returned, returned late, or returned corrupted. This requires significant reworking and causes instability in the speed, progress, responsiveness, and latency of the project as a whole and with individual grid members. Time-sensitive projects might not be given sufficient computational time to finish by a specific chronological deadline.

Grid computing often uses a central primary core of servers to manage the project, track work packets, and integrate returned work segments. If the central servers are overloaded or go offline, complete failure or crashing of the grid can occur. However, when central grid systems are inaccessible, grid members usually complete their current local tasks and then regularly poll to discover when the central servers come back online. There is also a potential risk that a compromise of the central grid servers could be leveraged to attack grid members or trick grid members into performing malicious actions instead of the intended purpose of the grid community.

Peer to Peer

Peer-to-peer (P2P) technologies are networking and distributed application solutions that share tasks and workloads among peers. This is similar to grid computing; the primary differences are that there is no central management system, and the services are usually provided in real time rather than as a collection of computational power. Common examples of P2P include many VoIP services, BitTorrent (for data/file distribution), and tools for streaming audio/music distribution.

Security concerns with P2P solutions include a perceived inducement to pirate copyrighted materials, the ability to eavesdrop on distributed content, a lack of central control/oversight/management/filtering, and the potential for services to consume all available bandwidth.

Industrial Control Systems

An *industrial control system (ICS)* is a form of computer-management device that controls industrial processes and machines, also known as *operational technology (OT)* (and also industrial IoT [IIOT]; see the section "Internet of Things"). ICSs are used across a wide range of industries, including manufacturing, fabrication, electricity generation and distribution, water distribution, sewage processing, and oil refining. There are several forms of ICS, such as *programmable logic controllers (PLCs)*, *distributed control systems (DCSs)*, and *supervisory control and data acquisition (SCADA)*.

PLC units are effectively single-purpose or focused-purpose digital computers. They are typically deployed for managing and automating various industrial electromechanical operations, such as controlling systems on an assembly line or a large-scale digital light display (such as a giant display system in a stadium or on a Las Vegas Strip marquee).

DCS units are typically found in industrial process plants where gathering data and implementing control over a large-scale environment from a single location is essential. An important aspect of DCS is that the controlling elements are distributed across the monitored environment, such as a manufacturing floor or a production line, and the centralized monitoring location sends commands out of those localized controllers while gathering

status and performance data. A DCS might be analog or digital, depending on the task or the device being controlled. For example, a liquid flow value DCS would be an analog system, whereas an electric voltage regulator DCS would likely be a digital system.

A SCADA system can operate as a stand-alone device, be networked together with other SCADA systems, or be networked with traditional IT systems. SCADA typically includes a human-machine interface (HMI) to enable personnel to oversee, manage, and control the complex industrial machine and technology systems. SCADA is used to monitor and control a wide range of industrial processes. SCADA can communicate with PLCs and DCS solutions.

A DCS focuses on processes and is state driven, whereas SCADA focuses on data-gathering and is event driven. DCS is more suited to operating on a limited scale, whereas SCADA is suitable for managing systems over large geographic areas.

Legacy SCADA systems were designed with minimal human interfaces. Often, they used mechanical buttons and knobs or simple LCD screen interfaces (similar to what you might have on a business printer or a GPS navigation device). However, modern networked SCADA devices may have more complex remote-control software interfaces.



A PLC is used to control a single device in a stand-alone manner. DCS was used to interconnect several PLCs within a limited physical range to gain centralized control, management, and oversight through networking. SCADA expanded this to large-scale physical areas to interconnect multiple DCSs and individual PLCs. For example, a PLC can control a single transformer, a DCS can manage a power station, and SCADA can oversee a power grid.

Modbus is a widely used communication protocol in industrial automation and control systems. Developed by Modicon (now part of Schneider Electric) in 1979, Modbus has become a de facto standard for connecting and managing devices within industrial environments. It provides a

common language for different devices to exchange data and commands, facilitating communication in SCADA systems, PLCs, and other industrial applications. The open standard nature of Modbus fosters interoperability, a crucial requirement in ICS and SCADA environments where diverse devices from different manufacturers must seamlessly interact.

In theory, the static design of PLC, DCS, and SCADA and their minimal human interfaces should make the system fairly resistant to compromise or modification. Thus, little security was built into these industrial control devices, especially in the past. But there have been several well-known compromises of industrial control systems in recent years; for example, Stuxnet delivered the first-ever-known rootkit to a SCADA system located in a nuclear facility. Many SCADA vendors have started implementing security improvements into their solutions to prevent or at least reduce future compromises. However, in practice, SCADA and ICS systems are still often poorly secured, vulnerable, and infrequently updated, and older versions not designed for security are still in widespread use.

Generally, typical security management and hardening processes can be applied to ICS, DCS, PLC, and SCADA systems to improve on whatever security is or isn't present in the device from the manufacturer. Common important security controls include isolating networks, limiting access physically and logically, restricting code to only essential applications, changing default credentials, and logging all activity. While confidentiality and integrity are often important considerations, the primary concern for OT/ICS is ensuring the availability of real-time control signals. Immediate and continuous availability of control signals is critical for the proper functioning of industrial processes. Otherwise, catastrophic consequences could result.

The ISA99 standards development committee has established and is maintaining guidelines for securing ICS, DCS, PLC, and SCADA systems. Much of their work is integrated into the International Electrotechnical Commission's (IEC) 62443 series of standards. To learn more about these standards, visit www.isa.org and iecee.org. NIST maintains ICS security standards in SP 800-82rs (nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-82r3.pdf). North American Electric Reliability Corporation (NERC) maintains its own security guides for ICS (www.nerc.com), which are

similar to those of the European Reference Network for Critical Infrastructure Protection (ERNCIP) (erncip-project.jrc.ec.europa.eu).

Distributed Systems

A *distributed system* or a *distributed computing environment (DCE)* is a collection of individual systems that work together to support a resource or provide a service. Often, a DCE is perceived by users as a single entity rather than numerous individual servers or components. DCEs are designed to support communication and coordination among their members to achieve a common function, goal, or operation. Some DCE systems are composed of homogenous members; others are composed of heterogeneous systems. Distributed systems can be implemented to provide resiliency, reliability, performance, and scalability benefits. Most DCEs exhibit numerous duplicate or concurrent components, are asynchronous, and allow for fail-soft or independent failure of components. A DCE is also known as (or at least described as) concurrent computing, parallel computing, and distributed computing. DCE solutions are implemented as client-server architectures (see the previous client and server sections as well as endpoint coverage in Chapter 11), as a three-tier architecture (such as basic web applications), as multitiered architectures (such as advanced web applications), and as peer-to-peer architectures (such as BitTorrent and most cryptocurrency blockchain ledgers as discussed in Chapter 7). DCE solutions are often employed for scientific and medical research projects, education projects, and industrial applications requiring extensive computational resources.

DCE forms the backbone of a wide range of modern Internet, business, and communication technologies that you might regularly use, including DNS, single-sign-on, directory services, massively multiplayer online role-playing games (MMORPGs), mobile networks, and most websites. DCE also makes possible a plethora of advanced technologies such as service-oriented architecture (SOA), software-defined networking (SDN), microservices, and so on.

A DCE typically includes an *interface definition language (IDL)*. An IDL is a language used to define the interface between client and server processes or objects in a distributed system. IDL enables the creation of interfaces

between objects when those objects are in varying locations or are using different programming languages; thus, IDL interfaces are language-independent and location-independent. There are numerous examples of DCE IDLs or frameworks, such as remote procedure calls (RPCs), the Common Object Request Broker Architecture (CORBA), and the Distributed Component Object Model (DCOM).

There are some security issues inherent with DCE. The primary security concern is the interconnectedness of the components. This configuration could allow for error or malware propagation, and if an adversary compromises one component, it may grant them the ability to compromise other components in the collective through pivoting and lateral movement. Other common issues to consider and address include the following:

- Access by unauthorized users
- Masquerading, impersonation, and spoofing attacks of users and/or devices
- Security control bypass or deactivating
- Communication eavesdropping and manipulation
- Insufficient authentication and authorization
- A lack of monitoring, auditing, and logging
- Failing to enforce accountability

The issues in this list are not unique to DCE, but they are especially problematic in a distributed system.

Since distributed systems include members that may be distributed geographically, they have a larger potential attack surface than that of a single system. Thus, it is important to consider the collective threats and risks of the individual member components of a DCE as well as the communications interconnections between them. To secure DCE, encryption is needed for storage, transmission, and processing (such as homomorphic encryption). Also, strong multifactor authentication should be implemented. If a strict homogeneous component set is not maintained, heterogenous systems introduce their own risks, whether different OSs are in use or just different versions or patch levels of the same OS. The more

varied the DCE components, the more challenging it is to maintain consistent security configuration, enforcement, monitoring, and oversight. If the DCE is so large or broadly distributed as to cross international boundaries, then data sovereignty issues need to be addressed.

High-Performance Computing (HPC) Systems

High-performance computing (HPC) systems are computing platforms designed to perform complex calculations or data manipulations at extremely high speeds. Super computers and MPP solutions are common examples of HPC systems. HPC systems are used when real-time or near-real-time processing of massive data is necessary for a particular task or application. These applications can include scientific studies, industrial research, medical analysis, societal solutions, and commercial endeavors.

Many of the products and services we use today, including mobile devices and their apps, IoT devices, ICS solutions, streaming media, voice assistants, 3D modeling and rendering, and AI/ML calculations, all depend on HPC to exist. As the population of Internet and computing devices increases, as the datasets being collected continue to increase exponentially, and as new uses of that data and those devices are conceived, HPC will be in even greater demand in the future.

An HPC solution is composed of three main elements: compute resources, network capabilities, and storage capacity. Each element must be able to provide equivalent capabilities to optimize overall performance. If storage is too slow, then data cannot be fed to the application processing on the compute resources. If networking capacity is not sufficient, then users of a resource will experience latency or even a benign denial of service (DoS).



A *benign DoS* occurs when a service is running on insufficient resources, when there has been an unforeseen popularity or traffic spike, or when something about the supporting system fails, such as drive loss, network link drop, or a corrupted configuration. This type of DoS occurs through no direct or intentional malign action on the part of an adversary. It is due to innocent events, unexpected conditions, or mistakes on the part of the owners/operators. For more on DoS, see Chapter 17.

Real-Time Operating Systems

A concept related to HPC is that of the real-time OS (RTOS). Often HPCs implement RTOS compute capability or otherwise attempt to achieve real-time processing and operations.

A *real-time operating system (RTOS)* is designed to process or handle data as it arrives on the system with minimal latency or delay. An RTOS is usually stored on read-only memory (ROM) and is designed to operate in a hard real-time or soft real-time condition. A hard real-time solution is for mission-critical operations where delay must be eliminated or minimized for safety, such as autonomous cars. A soft real-time solution is used when some level of modest delay is acceptable under typical or normal conditions, as it is for most consumer electronics, such as the delay between a digitizing pen and a graphics program on a computer.

RTOSs can be event driven or time-sharing. An event-driven RTOS will switch between operations or tasks based on preassigned priorities. A time-sharing RTOS will switch between operations or tasks based on clock interrupts or specific time intervals. An RTOS is often implemented when scheduling or timing is the most critical part of the task to be performed.

A security concern using RTOSs is that these systems are often focused and single-purpose, leaving little room for security. They often use custom or proprietary code, which may include unknown bugs or flaws that attackers could discover. An RTOS might be overloaded or distracted with bogus

datasets or process requests by malware. When deploying or using RTOSs, use isolation and communication monitoring to minimize abuses.

Internet of Things

Smart devices offer the user a plethora of customization options, typically through installing apps, and may take advantage of on-device or in-the-cloud machine learning (ML) processing. The products that can be labeled “smart devices” are constantly expanding and already include smartphones, tablets, music players, home assistants, extreme sport cameras, virtual reality/augmented reality (VR/AR) systems, and fitness trackers.

The *Internet of Things (IoT)* is a class of Internet-connected smart devices that provide automation, remote control, or AI processing to appliances or devices. IoT may often perform functions similar to an embedded system, but they are different. An IoT device is almost always a separate and distinct hardware device that is used on its own or in conjunction with an existing system (such as a smart ioT thermostat for a heating, ventilation, and air-conditioning [HVAC] system). An embedded system is one where the computer control component has been integrated into the larger mechanism's structure, design, and operation, often even built into the same chassis or case.

The security issues related to IoT are often about access and encryption. An IoT device was often not designed with security as a core concept or even an afterthought. This has resulted in numerous home and office network security breaches. Additionally, once an attacker has remote access to or through an IoT device, they may be able to access other devices on the compromised network. When electing to install IoT equipment, evaluate the security of the device as well as the security reputation of the vendor. If the device does not have the ability to meet or accept your existing security baseline, then don't compromise your security just for a flashy gadget.

One possible secure implementation is to deploy a distinct network segment for the IoT equipment, which is kept separate and isolated from the primary network. This configuration is often known as *three dumb routers* (see www.grc.com/sn/sn-545.pdf or www.pcper.com/reviews/General-Tech/Steve-Gibsons-Three-Router-Solution-IOT-Insecurity). Other

standard security practices are beneficial to IoT, including keeping systems patched, limiting physical and logical access, monitoring all activity, and implementing firewalls and filtering.



Wearable technology or *wearables* are offshoots of smart devices and IoT devices that are specifically designed to be worn by an individual. The most common examples of wearable technology are smart watches and fitness trackers. There are an astounding number of available options, with a wide range of features and security capabilities. When selecting a wearable device, consider the security implications. Is the data being collected in a cloud service that is secured for private use or is it made publicly available? What alternative uses is the collected data going to be used for? Is the communication between the device and the collection service encrypted? And can you delete your data and profile from the service completely if you stop using the device?

Although we often associate smart devices and IoT with home or personal use, they concern every organization. This is partly because of the use of mobile devices by employees within the company's facilities and even on the organizational network.

Another concern is that many IoT or networked automation devices are often used in the business environment. This includes environmental controls, such as HVAC management, air quality control, debris and smoke detection, lighting controls, door automation, personnel and asset tracking, and consumable inventory management and auto-reordering (such as coffee, snacks, printer toner, paper, and other office supplies). Thus, both smart devices and IoT devices are potential elements of a modern business network that need appropriate security management and oversight. For some additional reading on the importance of proper security management of smart devices and IoT equipment, see “NIST Cybersecurity for IOT Program” at www.nist.gov/itl/applied-cybersecurity/nist-cybersecurity-iot-program.

A common IoT device deployed in a business environment is a sensor. Sensors can measure just about anything, including temperature, humidity, light levels, dust particles, movement, acceleration, and air/liquid flow. Sensors can be linked with cyber-physical systems to automatically adjust or alter operations based on the sensor's measurements, such as turning on the air conditioning when the temperature rises above a threshold. Sensors can also be linked to ICS, DCS, and SCADA solutions.

IoT devices—in fact, almost all hardware and software—often have insecure or weak defaults. Never assume defaults are good enough. Always evaluate acquired IoT products' setting and configuration options and make changes that optimize security and support business functions. This is especially relevant to default passwords, which must always be changed and verified.

Industrial Internet of Things (IIoT) is a derivative of IoT that focuses more on industrial, engineering, manufacturing, or infrastructure level oversight, automation, management, and sensing. IIoT is an evolution of ICS and DCS that integrates cloud services to perform data collection, analysis, optimization, and automation. Examples of IIOT include edge computing and fog computing, discussed next.

Edge and Fog Computing

Edge computing is a philosophy of network design where data and the compute resources are located as close as possible to optimize bandwidth use while minimizing latency. In edge computing, the intelligence and processing are contained within each device. Thus, rather than sending data to a master processing entity, each device can process its own data locally. The architecture of edge computing performs computations closer to the data source, which is at or near the edge of the network. This is distinct from performing processing in the cloud on data transmitted from remote locations. Edge computing is often implemented as an element of IIOT solutions, but edge computing is not limited to this type of implementation.

Edge computing can be viewed as the next evolution of computing concepts. Originally, computing was accomplished on core mainframe computers where applications were executed on the central system but

where controlled or manipulated via thin clients. Then the distributed concept of client/server moved computing out to endpoint devices. This allowed for the execution of decentralized applications (i.e., not centrally controlled) that ran locally on the endpoint system. From there, virtualization led to cloud computing. Cloud computing is a type of centralized application execution on remote data center systems, controlled remotely by endpoints. Finally, edge computing is the use of devices that are close to or at the endpoint where applications are centrally controlled, but the actual execution is as close to the user or network edge as possible.

One potential use for edge devices is the deployment of mini-web servers by ISPs to host static or simple pages for popular sites that are located nearer to the bulk of common visitors than the main web servers. This speeds up the initial access to the front page of a popular organization's web presence, but then subsequent page visits are directed to and served by core or primary web servers that may be located elsewhere. Other examples of edge computing solutions include security systems, motion-detecting cameras, image recognition systems, IoT and IIOT devices, self-driving cars, optimized content delivery network (CDN) caching, medical monitoring devices, and videoconferencing solutions.

Fog computing is another example of advanced computation architecture, which is also often used as an element in an IIOT deployment. *Fog computing* relies on sensors, IoT devices, or even edge computing devices to collect data, and then transfers it back to a central location for processing. The fog computing processing location is positioned in the LAN. Thus, with fog computing, intelligence and processing are centralized in the LAN. The centralized computing power processes information gathered from the fog of disparate devices and sensors.

In short, edge computing performs processing on the distributed edge systems, whereas fog computing performs centralized processing of the data collected by the distributed sensors. Both edge and fog computing can often take advantage of or integrate the use of microcontrollers, embedded devices, static devices, cyber-physical systems, and IoT equipment.

Embedded Devices and Cyber-Physical Systems

An *embedded system* is any form of computing component added to an existing mechanical or electrical system for the purpose of providing automation, remote control, and/or monitoring. The embedded system is typically designed around a limited set of specific functions in relation to the larger product to which it is attached. It may consist of the same components found in a typical computer system or a microcontroller (an integrated chip with onboard memory and peripheral ports).

Microcontrollers

A *microcontroller* is similar to, but less complex, than a system on a chip, or SoC (see Chapter 11). A microcontroller may be a component of an SoC. A microcontroller is a small computer consisting of a CPU, various input/output capabilities, RAM, and often nonvolatile storage in the form of flash or ROM/PROM/EEPROM. Examples include Raspberry Pi, Arduino, and a field-programmable gate array (FPGA).

Embedded systems can be a security risk because they are generally static systems, meaning that even the administrators who deploy them have no real means to alter the device's operations to address security vulnerabilities. Some embedded systems can be updated with vendor patches, but patches are often released months after a known exploit is found in the wild. It is essential that embedded systems be isolated from the Internet and from a private production network to minimize exposure to remote exploitation, remote control, or malware compromise.

Security concerns for embedded systems include the fact that most are designed to minimize cost and extraneous features. This often leads to a lack of security and difficulty with upgrades or patches. Because an embedded system may be in control of a mechanism in the physical world, a security breach could cause harm to people and property.

Static Systems

Another concept similar to that of embedded systems is *static systems* (aka *static environments*). A static environment is a set of conditions, events, and surroundings that don't change. In theory, a static environment doesn't offer new or surprising elements once understood. A static IT environment is any

system that is intended to remain unchanged by users and administrators. The goal is to prevent, or at least reduce, the possibility of a user implementing change that could result in reduced security or functional operation. This is also known as a nonpersistent environment or a stateless system, as opposed to a persistent environment or stateful system, which allows changes and retains them between access events and reboots.

Examples of static systems include the check-in kiosk at the airport, an ATM, and often the complimentary guest computer at a hotel or library. Those guest computers are configured to provide the user with a temporary desktop environment to perform a restricted range of tasks. However, when the user terminates their session due to timeout or logging out, the system discards all the previous session's information and changes and restores a pristine version of the environment for the next user. Static systems can be implemented in a variety of ways, including using local VMs or remotely accessed VDI (virtual desktop infrastructure).

In technology, static environments are applications, OSs, hardware sets, or networks configured for a specific need, capability, or function, then set to remain unaltered. However, although the term *static* is used, no truly static systems exist. There is always the chance that a hardware failure, a hardware configuration change, a software bug, a software-setting change, or an exploit may alter the environment, resulting in undesired operating parameters or actual security intrusions.

Sometimes, the phrase *static OS* refers to the concept of a static system/environment or indicates a slight variation. That variation is that the OS itself is beyond the ability of the user to change, but the user can install or use applications. Often, those applications may be limited, restricted, or controlled to avoid allowing an application to alter the otherwise static OS. Some potential examples of static OSs would be smart TVs, gaming systems/consoles, or mobile devices where only applications from a vendor-controlled app store can be installed.

Cyber-Physical Systems

Cyber-physical systems refer to devices that offer a computational means to control something in the physical world. In the past, these might have been referred to as embedded systems, but the category of cyber-physical seems

to focus more on the physical world results rather than the computational aspects. Cyber-physical devices and systems are essentially key elements in robotics and sensor networks. Basically, any computational device that can cause a movement to occur in the real world is considered a robotic element, whereas any such device that can detect physical conditions (such as temperature, light, movement, and humidity) is a sensor. Examples of cyber-physical systems include prosthetics to provide human augmentation or assistance, collision avoidance in vehicles, air traffic control coordination, precision in robot surgery, remote operation in hazardous conditions, and energy conservation in vehicles, equipment, mobile devices, and buildings.

Security Concerns of Embedded and Static Systems

Embedded, static, network-enabled, and cyber-physical are usually more limited or constrained based on their design or hardware capabilities compared to typical endpoint, server, and networking hardware. These constraints can have security implications.

Some embedded systems run on replaceable or rechargeable batteries. Others only receive a small amount of power from a USB plug or special power adapter/converter. These power limitations can restrict the speed of operations, which in turn can limit the execution of security components. If additional power is consumed, the device might overheat. This could result in slower performance, crashing, or destruction.

Most embedded systems use less-capable CPUs. This is due to cost and power savings or limitations. Fewer computing capabilities means fewer functions, which means fewer security operations.

Many embedded systems have limited network capabilities. These network capabilities could be limited to wired only or wireless only. Within wireless, the device could be limited to a specific Wi-Fi version, frequency, speed, and/or encryption. Some devices using wireless are limited to special communication protocols, such as Zigbee or Bluetooth Low Energy (BLE).

Many embedded systems are unable to process high-end encryption. The crypto on these special devices is often limited and may use older algorithms or poor keys, or just lack good key management. Some devices are known to have preshared and/or hard-coded encryption keys.

Some embedded systems are difficult to patch, whereas others might not even offer patching or upgrading. Without update and patch management, vulnerable code will remain at risk.

Some embedded systems do not use authentication to control subjects or restrict updates. Some devices use hard-coded credentials. These should be avoided. Only use equipment that allows for customized credentials, and choose devices that support mutual-certificate authentication.

Some embedded systems have a limited transmission range due to low-power antennae. This can restrict the device's usefulness or require signal boosting to compensate.

Due to the low cost of some embedded systems, they might not include necessary security features. Other devices that do include needed security components may be too costly to be considered.

Similar to supply chain issues, when an embedded system is used, the organization automatically trusts the device vendor and the cloud service behind it. This implied trust may be misguided. Always thoroughly investigate vendors before relying on their product, and even then, segregate embedded systems in their own constrained network segments. See the discussion of zero trust in Chapter 8.

Based on these constraints and other concerns, security management of embedded and static systems must accommodate the fact that most are designed to minimize costs and extraneous features. This often leads to a lack of security mechanisms and difficulty with upgrades or patches.

Static environments, embedded systems, cyber-physical systems, high-performance computing (HPC) systems, edge computing devices, fog computing devices, mobile devices, and other limited or single-purpose computing environments need security management. Although they may not have as broad an attack surface and aren't exposed to as many risks as a general-purpose computer, they still require proper security government. Many of the same general security management principles used over servers and endpoints can be applied to embedded, static, and cyber-physical systems.

Network segmentation involves controlling traffic among networked devices. Complete or physical network segmentation occurs when a

network is isolated from all outside communications, which means transactions can occur only between devices within the segmented network. You can impose logical network segmentation with switches using virtual local area networks (VLANs), or through other traffic-control means, including MAC addresses, IP addresses, physical ports, TCP or UDP ports, protocols, or application filtering, routing, and access control management. Network segmentation can isolate embedded devices and static environments to prevent changes and/or exploits from reaching them. See Chapter 11 for more on segmentation.

An application firewall is a device, server add-on, virtual service, or system filter that defines a strict set of communication rules for a service and all users. It's intended to be an application-specific server-side firewall to prevent application-specific protocol and payload attacks. A network firewall is a hardware device designed for general network filtering, typically called an appliance. A network firewall is designed to provide broad protection for an entire network. An internal segmentation firewall (ISFW) is used to create a network division or segment. Every network needs a network firewall. Many application servers need an application firewall. However, the use of an application firewall generally doesn't negate the need for a network firewall. You should use firewalls in a series to complement each other, rather than seeing them as competitive solutions. See Chapters 11 and 17 for more on firewalls.

Security layers exist where devices with different levels of classification or sensitivity are grouped together and isolated from other groups with different levels. This isolation can be absolute or one-directional. For example, a lower level may not be able to initiate communication with a higher level, but a higher level may initiate with a lower level. Isolation can also be logical or physical. Logical isolation requires the use of classification labels on data and packets, which must be respected and enforced by network management, OSs, and applications. Physical isolation requires implementing network segmentation or air gaps between networks of different security levels. See Chapter 5, “Protecting Security of Assets,” to learn more about managing data and asset classification.

Manual updates should be used in static environments to implement only tested and authorized changes. An automated update system would allow untested updates to introduce unknown security reductions. As with manual

software updates, strict control over firmware in a static environment is important. Firmware updates should be implemented on a manual basis, only after thorough testing and review. Firmware version control or oversight of firmware release should focus on maintaining a stable operating platform while minimizing exposure to downtime or compromise.

Even embedded and static systems should be monitored for performance, violations, compliance, and operational status. Some of these types of devices can perform on-device monitoring, auditing, and logging, whereas others may require external systems to collect activity data. Any and all devices, equipment, and computers within an organization should be monitored to ensure high performance and minimal downtime, and to detect and stop violations and abuse.

As with any security solution, relying on a single security mechanism is unwise. Defense in depth uses multiple types of access controls in literal or theoretical concentric circles or layers. This form of layered security helps an organization avoid a monolithic security stance. A monolithic mentality is the belief that a single security mechanism is all that is required to provide sufficient security. With security control redundancy and diversity, a static environment can avoid the pitfalls of a single security feature failing; the environment has several opportunities to deflect, deny, detect, and deter any threat. Unfortunately, no security mechanism is perfect. Each individual security mechanism has a flaw or a work-around just waiting to be discovered and abused by a malicious actor.

Microservices

It is important to evaluate and understand the vulnerabilities in system architectures, especially in regard to technology and process integration. As multiple technologies and complex processes are intertwined in the act of crafting new and unique business functions, new issues and security problems often surface. As systems are integrated, attention should be paid to potential single points of failure as well as to emergent weaknesses in *service-oriented architecture (SOA)*. An SOA constructs new applications or functions out of existing but separate and distinct software services. The resulting application is often new; thus, its security issues are unknown, untested, and unprotected. All new deployments, especially new

applications or functions, must be thoroughly vetted before they can go live into a production network or the public Internet.

Microservices, or microservices architecture, is an architectural style in software development where an application is structured as a collection of small, independently deployable, and loosely coupled services. Each service, often referred to as a microservice, represents a specific business capability and operates as a self-contained unit with its own database and communication mechanisms. Microservices are designed to be scalable, maintainable, and independently deployable, allowing developers to build and enhance different parts of an application without affecting the entire system. This architectural approach promotes flexibility, agility, and the ability to scale applications more efficiently. It is the conversion or transformation of a capability of one application into a microservice that can be called upon by numerous other applications.

Microservices are often created as a means to provide purpose-specific business capabilities through services that are independently deployed. Often, microservices are small, focused on a singular operation, designed with few dependencies, and based on fast short-term development cycles (similar to Agile). It is also common to deploy microservices based on immutable architecture or infrastructure as code.

Microservices are a popular development strategy because they allow large, complex solutions to be broken into smaller, self-contained functions. This design also enables multiple programming groups to work on crafting separate elements or microservices simultaneously.

In the context of microservices, APIs play a crucial role. Each microservice exposes an API that allows communication and interaction with other services within the architecture. APIs form the communication channels between these microservices, enabling them to interact seamlessly. Microservices expose well-defined APIs that specify how external components or services can request or manipulate data. These APIs facilitate the integration and coordination of various microservices, allowing them to work together cohesively within the larger application.

The microservices architecture, with its emphasis on modularization, scalability, and flexibility, is well suited for dynamic and evolving applications. The use of APIs ensures that each microservice can be

developed, deployed, and updated independently, promoting agility and enabling teams to work on different components of the application concurrently. This approach has gained popularity in modern software development due to its ability to enhance maintainability, scalability, and the overall efficiency of complex applications.

Security is a paramount concern in a microservices architecture due to its architecture's distributed and interconnected nature. Each microservice needs to implement robust security measures to protect sensitive data and ensure the integrity of communication between services. Common security considerations include:

- *Authentication and authorization:* Implementing secure authentication mechanisms to verify the identity of users and services, and defining proper authorization policies to control access to microservices.
- *Data encryption:* Employing encryption techniques to secure data both in transit and at rest. This is particularly important when microservices communicate over networks or store sensitive information.
- *API security:* Ensuring that APIs are secure by implementing proper authentication, authorization, and encryption for data exchanged between microservices. API gateways are often used to manage and secure API communication.
- *Monitoring and logging:* Implementing comprehensive monitoring and logging to detect and respond to security incidents. This includes tracking access patterns, detecting anomalies, and logging security-relevant events.
- *Container security:* If microservices are deployed using containerization technologies (e.g., Docker), ensuring the security of containerized environments is crucial. This includes employing secure images and configurations, implementing access controls, regularly updating software components to address vulnerabilities, monitoring container activity for suspicious behavior, and utilizing tools for vulnerability scanning and compliance enforcement.

The microservices architecture, emphasizing modularization and scalability, introduces challenges and opportunities for security. Properly addressing

security concerns at the level of individual microservices and the interactions between them is essential to building a resilient and secure application.

Infrastructure as Code

Infrastructure as code (IaC) is a change in how hardware management is perceived and handled. Instead of seeing hardware configuration as a manual, direct hands-on, one-on-one administration hassle, it is viewed as just another collection of elements to be managed like software and code are managed under DevSecOps (development, security, and operations). With IaC, the hardware infrastructure is managed in much the same way that software code is managed, including version control, pre-deployment testing, custom-crafted test code, reasonableness checks, regression testing, and consistency in a distributed environment.

This alteration in hardware management approach has allowed many organizations to streamline infrastructure changes to occur more easily, rapidly, securely, and reliably than before. IaC often uses definition files and rule sets that are machine readable to quickly deploy new settings and manage hardware consistently and efficiently. These files can be treated as software code in terms of development, testing, deployment, updates, and management. IaC is not just limited to hardware; it can also be used to oversee and manage virtual machines (VMs), storage area networks (SANs), and software-defined networking (SDN).

Immutable Architecture

Immutable architecture is the concept that a server never changes once it is deployed. If there is a need to update, modify, fix, or otherwise alter, a new server is built or cloned from the current one, the necessary changes are applied, and then the new server is deployed to replace the previous one. Once the new server is validated, the older server is decommissioned. VMs are destroyed and the physical hardware/system is reused for future deployments.

The benefits of immutable architecture are reliability, consistency, and a predictable deployment process. It eliminates issues common in mutable

infrastructures where midstream updates and changes can cause downtime, data loss, or incompatibility.

The mindset of immutable architecture is often described with the analogy of pets versus cattle or snowflakes versus phoenixes. If a server is treated like a pet, when something goes wrong, everyone marshals to the rescue. However, if a server is treated like cattle, it is removed from the herd when something goes wrong, and another is brought in to replace it. If a server is managed uniquely, then it is a snowflake and requires specific focus and attention, causing an increase in administrative time and attention, not to mention complexity for the environment. If a server is always built from scratch, then when changes are needed, a new system can be created with integrated improvements through automated processes, thus rising from the ashes (of previously decommissioned servers) like a phoenix. This minimizes administrative overhead, reduces deployment time, and maintains consistency in the environment.

Software-Defined Networking (SDN)

A derivative of IaC and DCE is software-defined networking (SDN). SDN is the management of networking as a virtual or software resource even though it technically still occurs over hardware (similar to infrastructure as code [IaC]). Just as DCE is a collection of individual systems that work together to support a resource or provide a service, so too is SDN a collection of hardware and software elements designed to virtualize networking management and control. Please see Chapter 11 for details on SDN.

Virtualized Systems

Virtualization technology is used to host one or more OSs within the memory of a single host computer or to run applications that are not compatible with the host OS. This mechanism allows virtually any OS to operate on any hardware. It also allows multiple OSs to work simultaneously on the same hardware.

Organizations may implement virtualization technologies due to the huge cost savings available. For example, an organization may be able to reduce 100 physical servers to just 10 physical servers, with each physical server hosting 10 virtual servers. This reduces HVAC costs, power costs, and overall operating costs.

The *hypervisor*, also known as the *virtual machine monitor/manager (VMM)*, is the component of virtualization that creates, manages, and operates the virtual machines. The computer running the hypervisor is known as the host OS, and the OSs running within a hypervisor-supported virtual machine are known as guest OSs or virtualized systems.

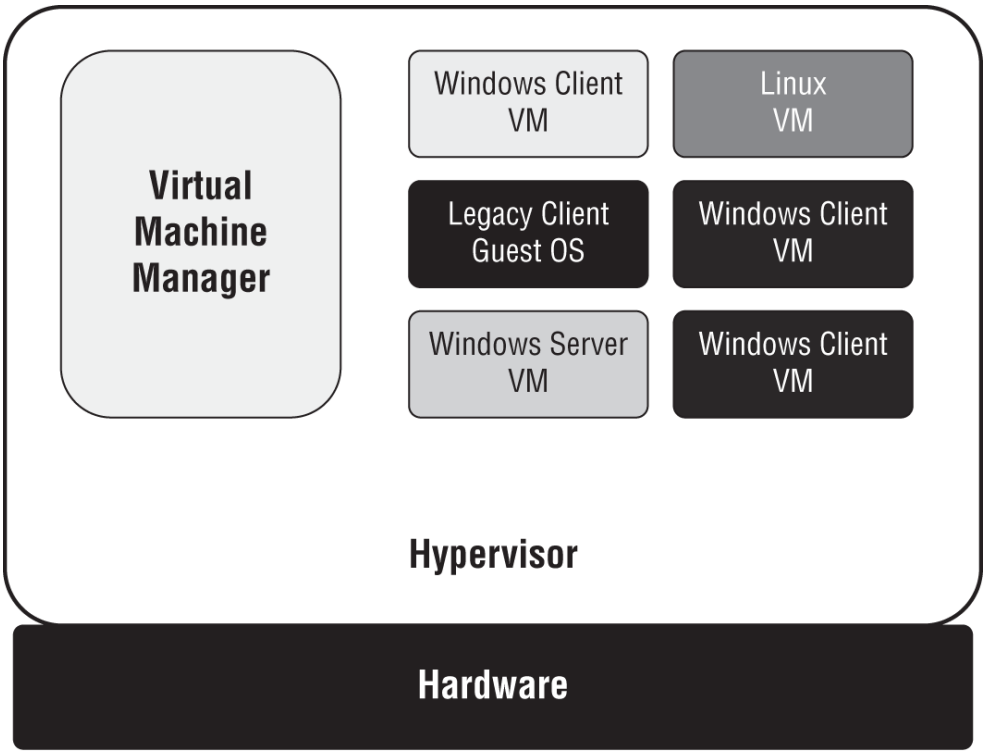
A *type I hypervisor* is a native or bare-metal hypervisor (Figure 9.3, top). In this configuration, there is no host OS; instead, the hypervisor installs directly onto the hardware where the host OS would normally reside. Type 1 hypervisors are often used to support server virtualization. This allows for maximization of the hardware resources while eliminating any risks or resource reduction caused by a host OS.

A *type II hypervisor* is a hosted hypervisor (Figure 9.3, bottom). In this configuration, a standard regular OS is present on the hardware, and then the hypervisor is installed as another software application. Type II hypervisors are often used in relation to desktop deployments, where the guest OSs offer safe sandbox areas to test new code, allow the execution of legacy applications, support apps from alternate OSs, and provide the user with access to the capabilities of a host OS.

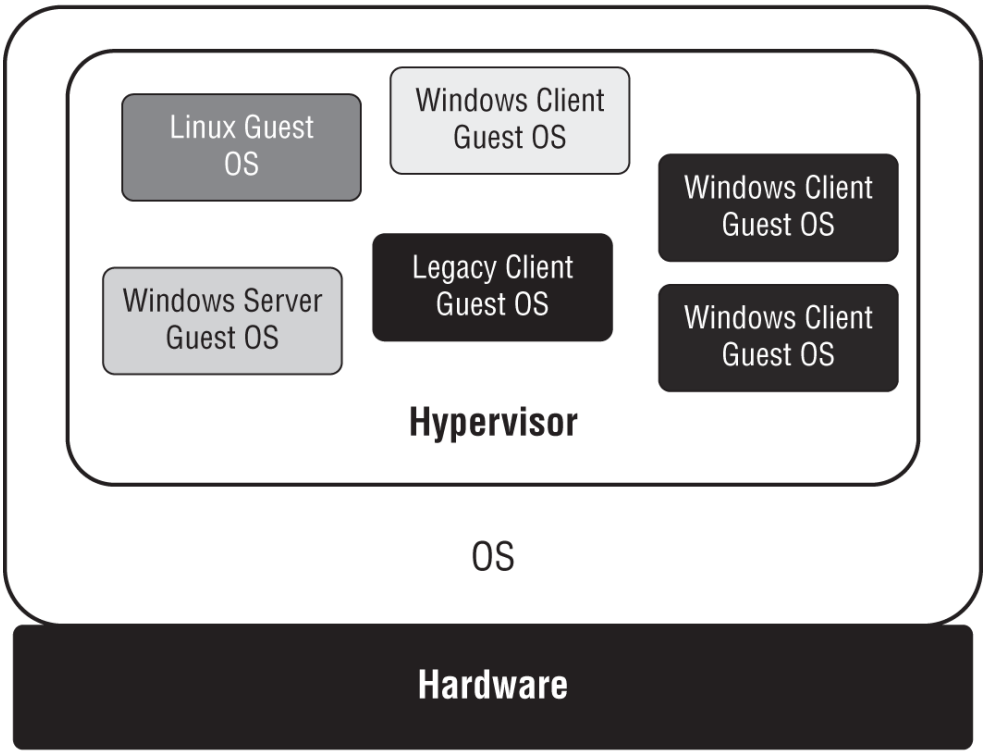
Cloud computing is a natural extension and evolution of virtualization, the Internet, distributed architecture, and the need for ubiquitous access to data and resources. However, it does have some potential security issues, including privacy concerns, regulation compliance difficulties, use of open versus closed source solutions, adoption of open standards, and whether or not cloud-based data is actually secured (or even securable). See Chapter 16 for details on cloud computing.

Virtualization has several benefits, such as launching individual instances of virtual servers or services as needed, real-time scalability, and running the exact OS version needed for a specific application. Virtualized servers and services are indistinguishable from traditional servers and services from a user's perspective. Recovery from damaged, crashed, or corrupted virtual

systems is often quick, simply consisting of replacing the virtual system's main hard drive file with a clean backup version and then relaunching it. Virtualization can also provide a reasonably secure means to continue to operate *end of life (EOL)* and *end of service life (EOSL)/end of support (EOS)* OSs to support legacy business applications.



Type I hypervisor



Type II hypervisor

FIGURE 9.3 Types of hypervisors

Elasticity refers to the flexibility of virtualization and cloud solutions (see Chapter 16) to expand or contract resource utilization based on need. In relation to virtualization, host elasticity means additional hardware hosts can be booted when needed and then used to distribute the workload of the virtualized services over the newly available capacity. As the workload becomes smaller, you can pull virtualized services off unneeded hardware so that it can be shut down to conserve electricity and reduce heat. Elasticity can also refer to the ability of a VM/guest OS to take advantage of any unused hardware resources on the fly as needed, but then release those resources when they are not needed. For example, a hardware host supporting five VM-based guest OSs may have over 30 percent CPU computational capacity unused. If a process-intensive application is launched within one of the VMs, the additional hardware host CPU capacity could be consumed; then once the application completes its intensive work task, the resource would be released. Elasticity has been a common capability of classic stand-alone systems for decades, but now with virtualization, the use of resources is shared among more than just a few processes—it can span across multiple VMs on the same hardware host as well as potentially across numerous hardware hosts.

It is also important to understand scalability in relation to elasticity. These terms are similar, but they are describing different concepts. Elasticity is the expansion or contraction of resources to meet current processing needs (especially in cloud computing), whereas scalability is the ability to take on more work or tasks. Usually, scalability is both a software and hardware characteristic that can handle more tasks or workloads, whereas elasticity is a hardware (physical or virtual) or platform characteristic where resources are optimized to meet the demands of current tasks. Usually, scalability is considered a long-term characteristic, while elasticity is more short-term. An elastic system can adjust resource consumption based on current processing needs, whereas a scalable system can be expanded to handle more processing over time. Thus, a scalable system must also be elastic, but an elastic system does not need to be scalable.

In relation to security, virtualization offers several benefits. It is often easier and faster to make backups of entire virtual systems than the equivalent native hardware-installed system. Snapshots (aka checkpoints) are backups

of virtual machines. Plus, when there is an error or problem, the virtual system can be replaced by a snapshot backup in minutes. Malicious code compromise or infection of virtual systems rarely affects the host OS due to the hypervisor's VM-to-VM and VM-to-host isolation. This allows for safe testing and experimentation.

Virtualization is used for a wide variety of new architectures and system design solutions. Locally (or at least within an organization's private infrastructure), virtualization can be used to host servers, client OSs, limited user interfaces (i.e., virtual desktops), applications, and more.

Virtual Software

A *virtual application* or *virtual software* is a software product deployed in such a way that it is fooled into believing it is interacting with a full host OS. A virtual (or virtualized) application has been packaged or encapsulated so that it can execute but operate without full access to the host OS. A virtual application is isolated from the host OS so that it cannot make any direct or permanent changes to the host OS. Any changes, such as file writes, configuration file or registry modifications, or system setting alterations, are intercepted by the isolation manager and recorded (typically into a single file). This allows the contained software to perceive it has interaction with the OS, without that interaction actually taking place. Thus, the virtualized application executes just like any regularly installed application, but it is only interacting and changing with a virtual representation of the OS, not the actual OS (aka sandboxing).

In many cases, operating an application in a software virtualization tool can effectively transform an installed application into a portable application. This means the application's encapsulation and file can be moved to another OS (with the same software virtualization product), where it can execute. It may also be possible to place the application's encapsulation onto removable media and be able to execute the software from a portable storage device plugged into another computer system.

Some software virtualization solutions enable applications from one OS to be operated on another. For example, Wine allows some Windows software products to be executed on Linux.

The concept of software virtualization has evolved into its own virtualization derivative concept known as *containerization*, which is covered in a later section, “Containerization.”

Virtualized Networking

The concept of OS virtualization has given rise to other virtualization topics, such as virtualized networks. A *virtualized network* or *network virtualization* is the combination of hardware and software networking components into a single integrated entity. The resulting solution allows for software control over all network functions: management, traffic shaping, address assignment, and so on. A single management console or interface can be used to oversee every aspect of the *virtual network*, a task that required physical presence at each hardware component in the past. Virtualized networks have become a popular means of infrastructure deployment and management by corporations worldwide. They allow organizations to implement or adapt other interesting network solutions, including SDNs, virtual SANs, guest OSs, and port isolation.

Custom *virtual network segmentation* can be used in relation to virtual machines to make guest OSs members of the same network division as that of the host, or guest OSs can be placed into alternate network divisions. A virtual machine can be made a member of a different network segment from that of the host or placed into a network that only exists virtually and does not relate to the physical network media (effectively an SDN; see Chapter 11).

Software-Defined Everything

Virtualization extends beyond just servers and networking. *Software-defined everything (SDx)* refers to a trend of replacing hardware with software using virtualization. SDx includes virtualization, virtualized software, virtual networking, containerization, serverless architecture (Chapter 16), XaaS (Chapter 16), infrastructure as code (IaC), SDN (Chapter 11), virtual storage area network (VSAN) (Chapter 11), software-defined storage (SDS) (Chapter 11), VDI, VMI, SDV, and software-defined data center (SDDC).

The SDx examples that are not defined elsewhere (either in this chapter or in Chapter 11) are discussed here.

Virtual desktop infrastructure (VDI) is a means to reduce the security risk and performance requirements of end devices by hosting desktop/workstation OS virtual machines on central servers that are remotely accessed by users. Thus, VDI is also known as a virtual desktop environment (VDE). Users can connect to the server to access their desktop from almost any system, including from mobile devices. Persistent virtual desktops retain a customizable desktop for the user. Nonpersistent virtual desktops are identical and static for all users. If a user makes changes, the desktop reverts to a known state after the user logs off. (See the discussion of static systems earlier in this chapter under “Static Systems.”)

VDI has been adopted into mobile devices and has already been widely used in relation to tablets and laptop computers. It is a means to retain storage control on central servers, gain access to higher levels of system processing and other resources, and allow lower-end devices access to software and services beyond their hardware's capacity. This has led to *virtual mobile infrastructure (VMI)*, where the OS of a mobile device is virtualized on a central server. Thus, most of the actions and activities of the traditional mobile device no longer occur on the mobile device itself. This remote virtualization allows an organization greater control and security than when using a standard mobile device platform. It can also enable personally owned devices to interact with the VDI without increasing the risk profile.

A *thin client* is a computer or mobile device with low to modest capability or a virtual interface that is used to remotely access and control a mainframe, virtual machine, VDI, or VMI. Thin clients were common in the 1980s when most computation took place on a central mainframe computer. Today, thin clients are being reintroduced as a means to reduce the expenses of high-end endpoint devices where local computation and storage are not required or are a significant security risk. A thin client can be used to access a centralized resource hosted on-premises or in the cloud. All processing/storage is performed on the server or central system, so the thin client provides the user with display, keyboard, and mouse/touchscreen functionality.

Software-defined visibility (SDV) is a framework to automate the processes of network monitoring and response. The goal is to enable the analysis of every packet and make deep intelligence-based decisions on forwarding, dropping, or otherwise responding to threats. SDV is intended to benefit companies, security entities, and managed service providers (MSPs). The goal of SDV is to automate detection, reaction, and response. SDV provides security and IT management with oversight into all aspects of the company network, both on-premises and in the cloud, with an emphasis on defense and efficiency. SDV is another derivative of IaC.

Software-defined data center (SDDC) or *virtual data center (VDC)* is the concept of replacing physical IT elements with solutions provided virtually, and often by an external third party, such as a cloud service provider (CSP). SDDC is effectively another XaaS (i.e., anything as a service, i.e., a cloud technology) concept, namely *IT as a service (ITaaS)*. It is similar to infrastructure as a service (IaaS); thus, some claim it is nothing more than a marketing or advertising term of misdirection.

To explore SDx further, you will find a wealth of articles at: `sdxcentral.com`, and you might want to start with “What Is Software Defined Everything – Part 1: Definition of SDx” at `www.sdxcentral.com/cloud/definitions/software-defined-everything-sdx-part-1-definition`.

Virtualization Security Management

The primary software component in virtualization is a hypervisor. The hypervisor manages the VMs, virtual data storage, and virtual network components. As an additional layer of software on the physical server, it represents an additional attack surface. If an attacker can compromise a physical host, the attacker can potentially access all of the virtual systems hosted on the physical server. Administrators often take extra care to ensure that virtual hosts are hardened.

Although virtualization can simplify many IT concepts, it's important to remember that many of the same basic security requirements still apply. Virtualization doesn't lessen the security management requirements of an OS. Thus, patch management is still essential. For example, each VM's guest OS still needs to be updated individually. Updating the host system

doesn't update the guest OSs. Also, don't forget that you need to keep the hypervisor updated as well.

When using virtualized systems, it's important to protect the stability of the host. This usually means avoiding using the host for any purpose other than hosting the virtualized elements, especially in a server-focused deployment. If host availability is compromised, the availability and stability of the virtual systems are also compromised.

Additionally, organizations should maintain backups of their virtual assets. Many virtualization tools include built-in tools to create full backups of virtual systems and create periodic snapshots, allowing relatively easy point-in-time restores.

Virtualized systems should be security tested. The virtualized OSs can be tested in the same manner as hardware installed OSs, such as with vulnerability assessment and penetration testing.

VM sprawl occurs when an organization deploys numerous virtual machines without an overarching IT management or security plan in place. Although VMs are easy to create and clone, they have the same licensing and security management requirements as a metal-installed OS. Uncontrolled VM creation can quickly lead to a situation where manual oversight cannot keep up with system demand. To prevent or avoid VM sprawl, a policy for developing and deploying VMs must be established and enforced. This should include establishing a library of initial or foundation VM images that are to be used to develop and deploy new services. In some instances, VM sprawl relates to the use of lower-powered equipment that results in poorly performing VMs. VM sprawl is a virtual variation of server sprawl and could allow for virtual shadow IT.

Server Sprawl and Shadow IT

Server sprawl or system sprawl is the situation where numerous underutilized servers are operating in your organization's server room. These servers are taking up space, consuming electricity, and placing demands on other resources, but their provided workload or productivity does not justify their presence. This can occur if an organization purchases cheap lower-end hardware in bulk instead of selecting optimal equipment for specific use cases.

Somewhat related to server sprawl is shadow IT. *Shadow IT* is a term used to describe the IT components (physical or virtual) deployed by a department without the knowledge or permission of senior management or the IT group. The existence of shadow IT is often due to complex bureaucracy that makes the acquisition of needed equipment overly difficult and time-consuming. Other terms that might be used to refer to shadow IT include embedded IT, feral IT, stealth IT, hidden IT, secret IT, and client IT.

Shadow IT usually does not follow company security policy, and it might not be kept current and updated with patches. Shadow IT often lacks proper documentation, is not under consistent oversight and control, and may not be reliable or fault-tolerant. Shadow IT greatly increases the risk of disclosure of sensitive, confidential, proprietary, and personal information to unauthorized insiders and outsiders. Shadow IT can be composed of physical devices, virtual machines, or cloud services.

VM escaping occurs when software within a guest OS is able to breach the isolation-protection provided by the hypervisor to violate the container of other guest OSs or to infiltrate a host OS. Several VM escape vulnerabilities have been discovered in a variety of hypervisors. Fortunately, the vendors have been fast to release patches. For example, Virtualized Environment Neglected Operations Manipulation (VENOM) (CVE-2015-3456) was able to breach numerous VM products that employed a compromised open source virtual floppy disk driver to allow malicious code to jump between VMs and even access the host.

VM escaping can be a serious problem, but steps can be implemented to minimize the risk. First, keep highly sensitive systems and data on separate physical machines. An organization should already be concerned about over-consolidation resulting in a single point of failure; running numerous hardware servers so that each supports a handful of guest OSs helps with this risk. Keeping enough physical servers on hand to maintain physical isolation between highly sensitive guest OSs will further protect against a VM escape exploit. Second, keep all hypervisor software current with vendor-released patches. Third, monitor attack, exposure, and abuse indexes for new threats to your environment.



To search for, locate, or research vulnerabilities, exploits, and attacks (whether related to virtualization or not), use exploit-db.com, cve.org, and nvd.nist.gov.

Containerization

Containerization is the next stage in the evolution of the virtualization trend for both internally hosted systems and cloud providers and services. A virtual machine–based system uses a hypervisor installed onto the bare metal of the host server and then operates a full guest OS within each virtual machine, and each virtual machine often supports only a single primary application. This is a resource-wasteful design and reveals its origins as separate physical machines.

Containerization or *OS-virtualization* is based on the concept of eliminating the duplication of OS elements in a virtual machine. Instead, each application is placed into a container that includes only the actual resources needed to support the enclosed application, and the common or shared OS elements are then part of the container engine. Some deployments (such as Docker and Kubernetes) eliminate the hypervisor altogether and use a collection of common binaries and libraries for the containers to call upon when needed. Containerization is able to provide 10 to 100 times more application density per physical server than that provided by traditional hypervisor virtualization solutions.

Application cells or *application containers* (Figure 9.4) are used to virtualize software so that they can be ported to almost any OS.

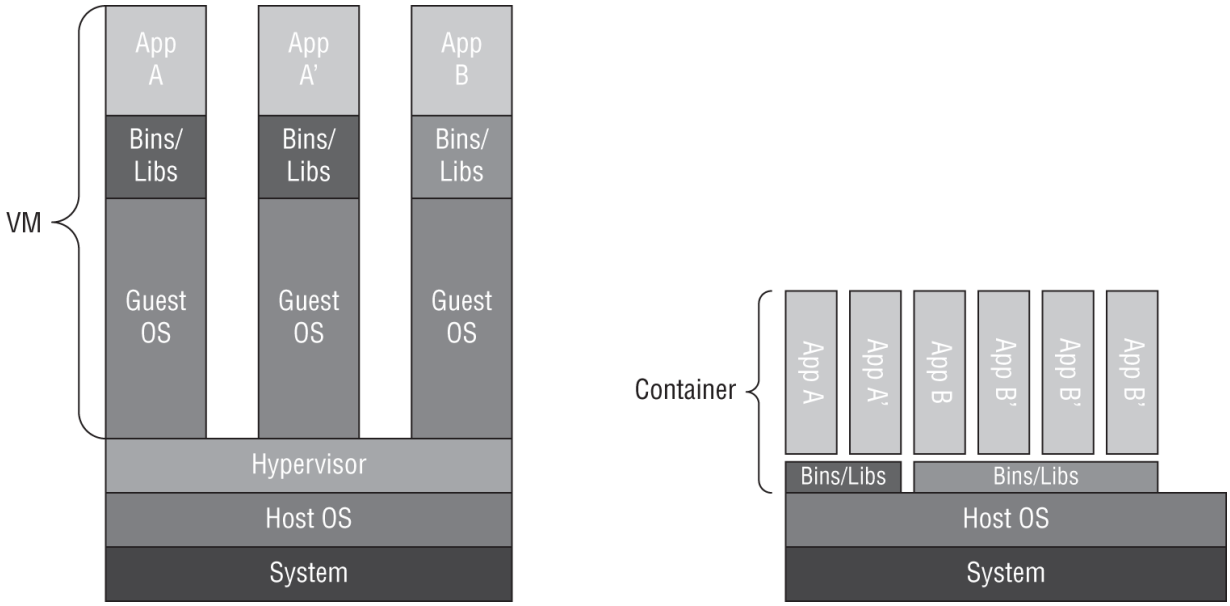


FIGURE 9.4 Application containers versus a hypervisor

There are many different technological solutions that are grouped into the concept of containerization. Some containerization solutions allow for multiple concurrent applications within a single container, whereas others are limited to one per container. Many containerization solutions allow for customization of how much interaction applications in separate containers are allowed.

Mobile Devices

A *mobile device* is anything with a battery (unless you also want to include things that are field powered, solar powered, etc., so generally anything that does not need a power cord to operate). However, we mostly discuss issues related to smartphones, tablets, or portable computers (i.e., notebooks and laptops). It may be tempting to only consider smartphones in relation to mobile device questions, but you should also consider the question in regard to a laptop computer, a tablet, and maybe even a smart watch or fitness tracker. These other perspectives may assist you in answering the question correctly.

Some mobile devices have less than typical default or even available security features because they often run stripped-down OSs or custom mobile OSs without the long history of security improvements found in popular PC OSs. Many of the “features” of a mobile device can be the focus of attacks, compromises, and intrusions. Extra care and attention need to be paid to any mobile device's security for both personal and business/work use.

Smartphones and other mobile devices present an ever-increasing security risk as they become more and more capable of interacting with the Internet as well as corporate networks. These devices have internal memory and may support removable memory cards that can hold a significant amount of data. Additionally, many devices include applications that allow users to read and manipulate different types of files and documents. When personally owned devices are allowed to enter and leave a secured facility without limitation, oversight, or control, the potential for harm is significant.

Malicious insiders can bring in malicious code from outside on various storage devices, including mobile phones, audio players, digital cameras, memory cards, optical discs, and Universal Serial Bus (USB) drives. These same storage devices can be used to leak or steal internal confidential and private data to disclose it to the outside world. Malicious insiders can execute malicious code, visit dangerous websites, or intentionally perform harmful activities.

Mobile devices often contain sensitive data such as contacts, text messages, email, scheduling information, and possibly notes and documents. Any mobile device with a camera feature can take photographs of sensitive information or locations. The loss or theft of a mobile device could mean the compromise of personal and/or corporate secrets.

Additionally, mobile devices aren't immune to eavesdropping. With the right type of sophisticated equipment, most mobile phone conversations can be tapped into—not to mention the fact that anyone nearby can hear you talking. Employees should be coached to be discreet about what they discuss over mobile phones in public spaces.

Android and iOS

Two of the most widely used device OSs are Android and iOS.

Android

Android is a mobile device OS based on Linux, which was acquired by Google in 2005. The Android source code is made open source through the Apache license, but most Android devices also include proprietary software. Although it's mostly intended for use on phones and tablets, Android is being used on a wide range of devices, including televisions, game consoles, digital cameras, microwaves, watches, e-readers, cordless phones, and ski goggles.

The use of Android in phones and tablets allows for a wide range of user customization: you can install Google Play Store apps as well as apps from unknown external sources (such as the Amazon Appstore), and many devices support the replacement of the default version of Android with a customized or alternate version. However, when Android is used on other devices, it can be implemented as something closer to a static system.

Whether static or not, Android has numerous security vulnerabilities. These include exposure to malicious apps, running scripts from malicious websites, and allowing insecure data transmissions.

Improvements are made to Android security as new updates are released. Users can adjust numerous configuration settings to reduce vulnerabilities and risks. Also, users may be able to install apps that add additional security features to the platform.

Security-Enhanced Android (SEAndroid) is a security improvement for Android. SEAndroid is a framework to integrate elements of Security-Enhanced Linux into Android devices. These improvements include adding support for mandatory access control (MAC) and middleware mandatory access control (MMAC), reducing privilege daemon vulnerabilities, sandboxing and isolating apps, blocking app privilege escalation, enabling app privilege adjustments both during installation

and at runtime, and defining a centralized security policy that can be scrutinized.

iOS

iOS is the mobile device OS from Apple that is standard on the *iPhone*. iOS isn't licensed for use on any non-Apple hardware. Thus, Apple is in full control of the features and capabilities of iOS. However, iOS is not really an example of a static environment, because users can install apps from the Apple App Store (although it can be argued that iOS is a static OS).

Mobile Device Security Features

A wide range of security features may be available on mobile devices, such as portable computers, tablets, and smartphones. Not all mobile devices have good security features. Be sure to consider the security options of a new device before you make a purchase decision. But even if security features are available, they're of no value unless they're enabled and properly configured. A security benefit is gained only when the security function is in force. Be sure to check that all desired security features are operating as expected on any device allowed to connect to the organization's network or enter the organization's facility.

The following sections discuss various examples of on-device security features that are often present on or available for mobile devices.

Mobile Device Management

Administrators register employee devices with a mobile device management system. *Mobile device management (MDM)* is a software solution to the challenging task of managing the myriad mobile devices that employees use to access company resources. The MDM system monitors and manages mobile devices and ensures that they are kept up-to-date. The goals of MDM are to improve security, provide monitoring, enable remote management, and support troubleshooting. Many MDM solutions support a wide range of devices and can operate across many service providers. You can use MDM to push or remove apps, manage data, and enforce

configuration settings both over the air (across a carrier network) and over Wi-Fi connections. MDM can be used to manage company-owned devices as well as personally owned devices.

Unified endpoint management (UEM) is a type of software tool that provides a single management platform to control mobile, PC, IoT, wearables, ICS, and other devices. UEM is intended to replace MDM and enterprise mobility management (EMM) products by combining the features of numerous products into one solution.

Device Authentication

Authentication on or to a mobile device is often fairly simple, especially for mobile phones and tablets. This is known as *device authentication*. However, a swipe or pattern access shouldn't be considered true authentication. Whenever possible, use a password, provide a personal identification number (PIN), offer your eyeball or face for recognition, scan your fingerprint, provide a USB key, or use a proximity device such as a near-field communication (NFC) or radio-frequency identification (RFID) ring or tile. These means of device authentication are much more difficult for a thief to bypass if properly implemented. It's also prudent to combine device authentication with device encryption to block access to stored information via a connection cable.



Retina-, iris-, face-, and fingerprint-based authentication are all examples of biometrics. See Chapter 13, “Managing Identity and Authentication,” for the full discussion of biometrics or the “something you are” authentication factor.

A strong password would be a great idea on a phone or other mobile device if locking the phone provided true security. But most mobile devices aren't that secure, so even with a strong password, the device may still be accessible over Bluetooth, wireless, or a USB cable. If a specific mobile device blocked access to the device when the system lock was enabled, this would be a worthwhile feature to set to trigger automatically after a period of inactivity or manual initialization (often related to screen lock). This

benefit is usually obtained when you enable both a device password and storage encryption.



When accessing an online website, service, or cloud offering from a mobile device, a form of multifactor authentication (MFA) may be implemented by combining your user credentials with context-aware authentication. Context-aware authentication evaluates the origin and context of a user's attempt to access a system. If the user originates from a known trusted system, such as a system inside the company facility or the same personal mobile device, then a low-risk context is present and a modest level of authentication is mandated for gaining access. If the context and origin of the user is from an unknown device and/or external/unknown location, the context is high risk. The authentication system will then demand that the user traverse a more complex multifactor authentication gauntlet to gain access. Context-aware authentication is thus an adaptive authentication that may be able to reduce the burden of authentication during low-risk scenarios but thwart impersonation attempts during high-risk scenarios.

Full-Device Encryption

Some mobile devices, including portable computers, tablets, and mobile phones, may offer *full-device encryption (FDE)*. Many mobile devices are either pre-encrypted or can be encrypted by the user/owner. Once a mobile device is encrypted, the user's data is protected whenever the screen is locked, which causes the physical data port on the device to be deactivated. This prevents unauthorized access to data on the device through a physical cable connection as long as the screen remains locked.

If most or all of the storage media of a device can be encrypted, this is usually a worthwhile feature to enable. However, encryption isn't a guarantee of protection for data, especially if the device is stolen while unlocked or if the system itself has a known backdoor attack vulnerability.

Communication Protection

Voice encryption may be possible on mobile devices when Voice over Internet Protocol (VoIP) services are used. VoIP service between computer-like devices is more likely to offer an encryption option than VoIP connections to a traditional landline phone or typical mobile phone. When a voice conversation is encrypted, attempting to listen in on the communications becomes almost worthless because the contents of the conversation are undecipherable.

This concept of communication protection should be applied to any type of transmission, whether video, text, or data. There are numerous apps that provide encrypted communications, many using standard and well-respected cryptography solutions, such as the Signal protocol (see Chapters 6 and 7 for more on encryption).

Remote Wiping

Remote wipe or *remote sanitization* is to be performed if a device is lost or stolen. A remote wipe lets you delete all data and possibly even configuration settings from a device remotely. The wipe process can be triggered over mobile phone service or sometimes over any Internet connection (such as Wi-Fi). However, a remote wipe isn't a guarantee of data security. The wiping trigger signal might not be received by the device. Thieves may be smart enough to prevent connections that would trigger the wipe function while they dump out the data. This could be accomplished by removing the subscriber identity module (SIM) card, deactivating Wi-Fi, and/or placing the device in a Faraday cage.

Additionally, a remote wipe is mostly a deletion operation and resetting the device back to factory conditions. The use of an undelete or data recovery utility can often recover data on a wiped device. To ensure that a remote wipe destroys data beyond recovery, the device should be encrypted (aka full-device encryption [FDE]). Thus, the undelete operation would only be recovering encrypted data, which the attacker should be unable to decipher.

Screen Locks

A *screen lock* is designed to prevent someone from casually picking up and being able to use your phone or mobile device. However, most screen locks can be unlocked by swiping across the screen or drawing a pattern. Neither of these is truly a secure operation. These easy-bypass options may be the

default on the device but should be changed to something more secure and resistive of unauthorized access, such as a PIN, password, or biometric.

Screen locks may have workarounds on some devices, such as accessing the phone application through the emergency calling feature. And a screen lock doesn't necessarily protect the device if a malicious actor connects to it over Bluetooth, wireless, or a USB cable.

Screen locks are often triggered after a timeout period of nonuse. Most devices can be configured to auto-trigger a password-protected screen lock if the system is left idle for a few minutes. Similarly, many tablets and mobile phones can be set to trigger a screen lock and dim or turn off the display after a set time period, such as 30 seconds. The lockout feature ensures that if you leave your device unattended or it's lost or stolen, it will be difficult for anyone else to be able to access your data or applications. To unlock the device, you must enter valid credentials.

Device Lockout

Lockout on a mobile device is similar to account lockout on a company workstation. When a user fails to provide their credentials after repeated attempts, the account or device is deactivated (i.e., locked out) for a period of time or until an administrator clears the lockout flag.

Mobile devices may offer a *device lockout* feature, but it's only effective if a screen lock has been configured. Otherwise, a simple screen swipe to access the device doesn't provide sufficient security, because an authentication process doesn't occur. Some devices trigger ever longer delays between access attempts as a greater number of authentication failures occur. Some devices allow for a set number of attempts (such as three) before triggering a lockout that lasts minutes or hours. Other devices trigger a persistent lockout and require the use of a different account or master password/code to regain access to the device. Some devices may even have a maximum number of logon attempts (such as 10), before securely wiping all data on the device and resetting back to factory settings. Be sure to know the exact nature of a device's lockout mechanism before attempting to guess credentials; otherwise you might inadvertently trigger a security wipe.

GPS and Location Services

The *Global Positioning System (GPS)* is a satellite-based geographical location service. Many mobile devices include a GPS chip to support and benefit from localized services, such as navigation, so it's possible to track those devices. The GPS chip itself is usually just a receiver of signals from orbiting GPS satellites. However, applications on the mobile device can record the GPS location of the device and then report it to an online service. You can use GPS tracking to monitor your own movements, track the movements of others (such as minors or delivery personnel), or track down a stolen device. But for GPS tracking to work, the mobile device must have Internet or wireless phone service over which to communicate its location information. Apps are able to provide location-based services as well as reveal the location of the device (and thus its user/owner) to third parties (sometimes without consent). This risk needs to be evaluated in regard to the organizational security policy and relative location-based risks.

Geolocation data is commonly used in navigation tools, authentication services, and many location-based services, such as offering discounts or coupons to nearby retail stores.

Location-based authorization policies for controlling access can be used to grant or deny resource access based on where the subject is located. This might be based on whether the network connection is local wired, local wireless, or remote. Location-based policies can also grant or deny access based on MAC address, IP address, OS version, patch level, and/or subnet in addition to logical or geographical location, which is a feature of both network access control (NAC) (see Chapter 11) and context-aware authentication. Location-based policies should be used only in addition to standard authentication processes, not as a replacement for them.

Geotagging is the ability of a mobile device to include details about its location in any media created by the device, such as photos, videos, and social media posts. Mobile devices with location services enable embedding geographical location in the form of latitude and longitude as well as date/time information into media created on the device. This allows an adversary (or angry ex) to view photos from social networking or similar sites and determine exactly when and where a photo was taken.

Geotagging can be used for nefarious purposes, such as determining when a person normally performs routine activities. Once a geotagged photo has

been uploaded to the Internet, a potential cyber stalker may have access to more information than the uploader intended. This is prime material for security-awareness briefs for end users.

Other Location Services

The most commonly discussed location service of a mobile device is that of GPS. However, it is important to recognize that there are at numerous location determination services or capabilities in many mobile devices. These include *wireless positioning system (WiPS) or Wi-Fi positioning system (WFPS)*, cellular/mobile service tower triangulation, Bluetooth location services, and environmental sensors.

WiPS uses the known location of wireless access points/base stations to determine a mobile device's location. WiPS is often used as a supplement to GPS when sufficient satellite signals are unavailable, such as when underground, inside buildings, or near tall structures.

Due to U.S. 911 regulations (which established E911), mobile devices can be located using mobile service tower triangulation. However, E911 location tracking is not as accurate as GPS.

iBeacon is a technology developed by Apple to track devices based on their Bluetooth device address and signal properties. Though originally designed to track people inside Apple stores, it is now used in many other contexts by a wide range of organizations to track devices via Bluetooth and their related user/owner.

Environmental sensors on many mobile devices include accelerometers, compasses, thermometers, altimeters (altitude sensors), and barometric pressure sensors. With this vast range of sensing data, if the initial location of a device is known or can be approximated, then its location at any future point in time can be determined if continuous sensor data is recorded.

It is also possible for a device to be located through its camera and microphones, but so far, this method is not as reliable as the others. This final concept measures light levels, intensity, and color to potentially determine if a device is outside or inside and if it is located near a window, which, based on the time of day, may be able to determine the general area (such as a city) based on light levels caused by the sun's position in the sky. This can then be combined with monitoring of background noise via the microphone to further refine the location. But

this requires extensive knowledge of regional sounds, a massive dataset of noise from across the globe, or access to real-time microphone networks or sensors.

Geofencing is the designation of a specific geographical area that is then used to implement features or trigger settings automatically on mobile devices. A geofence can be defined by GPS coordinates, WiPS, or the presence of or lack of a specific wireless signal. A device can be configured to enable or deactivate features based on a geofenced area, such as an onboard camera or the Wi-Fi capability.

Content Management

Content management is the control over mobile devices and their access to content hosted on company systems as well as the control of access to company data stored on mobile devices. Typically, an *MCM (mobile content management) system* is used to control company resources and the means by which they are accessed or used on mobile devices. An MCM can take into account a device's capabilities, storage availability, screen size, bandwidth limitations, memory (RAM), and processor capabilities when rendering or sending data to mobile devices.

The goal of a *content management system (CMS)* for mobile devices is to maximize performance and work benefits while reducing complexity, confusion, and inconvenience. An MCM may also be tied to an MDM to ensure secure use of company data.

A content filter, which may block access to resources, data, or services based on IP address, domain name, protocol, or keyword, is often implemented as a firewall service rather than an on-device mechanism. Therefore, content filtering is usually enforced by the network through which communication occurs.

Application Control

Application control or *application management* is a device-management solution that limits which applications can be installed onto a device. It can also be used to force specific applications to be installed or to enforce the settings of certain applications to support a security baseline or maintain

other forms of compliance. Using application control can often reduce exposure to malicious applications by limiting the user's ability to install apps that come from unknown sources or that offer non-work-related features. This mechanism is often implemented by an MDM. Without application control, users could theoretically install malicious code, run data stealing software, operate apps that reveal location data, or not install business-necessary applications.

Application allow listing (also known as whitelisting) is a security option that prohibits unauthorized software from being able to execute. Allow listing is also known as *deny by default* or *implicit deny*. In application security, allow listing prevents any and all software, including malware, from executing unless it's on the preapproved exception list: the allow list. This is a significant departure from the typical device-security stance, which is to allow by default and deny by exception (also known as deny listing or block listing, also known as blacklisting). Deny listing allows anything and everything, both benign and malicious, to execute by default, unless it is added to the deny list, which prevents execution from that point forward.

Due to the growth of malware, an application allow listing approach is one of the few options remaining that shows real promise in protecting devices and data. However, no security solution is perfect, including allow listing. All known allow listing solutions can be circumvented with kernel-level vulnerabilities and application configuration issues.

Mobile application management (MAM) is similar to an MDM but focuses only on app management rather than managing the entire mobile device.

Push Notifications

Push notification services are able to send information to your device rather than having the device (or its apps) pull information from an online resource. Push notifications are useful in being notified about a concern immediately, but they can also be a nuisance if they are advertising or spam. Many apps and services can be configured to use push and/or pull notifications. Push notifications are mostly a distraction, but it is possible to perform social engineering attacks via these messages as well as distribute malicious code or links to abusive sites and services.

Push notifications are also a concern in browsers for both mobile devices and PCs. Another issue is that malicious or pernicious notifications may capture a user in a push locker. If the user denies agreement to a push prompt, it may redirect them to a subdomain where another push notification is displayed. If they deny again, then they are redirected again to yet another subdomain, to then see another push notification. This can be repeated indefinitely. Until your browser and/or host-based intrusion detection system (HIDS) can detect and respond to push lockers, the only response is to close/terminate the browser and not return to the same URL.

Third-Party Application Stores

The primary, default, and authorized application (aka app) stores of Apple App Store and Google Play Store are the official sources for apps for use on the typical or standard Apple and Android smartphone or device, respectively. *Third-party app stores* often have less rigorous security rules regarding hosting an app. On Android devices, simply enabling a single feature to install apps from unknown sources allows the use of third-party app stores (as well as sideloading; see the section “Sideloading,” later in this chapter). For Android devices, the Amazon Appstore is an example of a third-party app store. For Apple iOS devices, currently you are limited to the official Apple App Store unless you jailbreak or root the device (which is not usually a security recommendation).

When a mobile device is being managed by an organization, especially when using an MDM/UEM/MAM, most third-party sources of apps will be blocked. Such third-party app sources represent a significant increase in risk of data leakage or malware intrusion to an organizational network.

Storage Segmentation

Storage segmentation is used to artificially compartmentalize various types or values of data on a storage medium. On a mobile device, storage segmentation may be used to isolate the device's OS and preinstalled apps from user-installed apps and user data. Some MDMs/UEMs further impose storage segmentation to separate company data and apps from user data and apps. This allows for ownership and rights over user data to be retained by the user, while granting ownership and rights over business data (such as

remote wiping) to the organization, even on devices owned by the employee.

With or without storage segmentation, risk can be reduced by minimizing the storage of nonessential data, sensitive data, and personal data (i.e., PII and PHI) on a device. So, even if a device is lost or stolen, the loss potential is kept to a minimum if there is little to no valuable data on the system for an adversary to gain access to.

Asset Tracking

Asset tracking is the management process used to maintain oversight over an inventory, such as deployed mobile devices. An asset-tracking system can be passive or active. Passive systems rely on the asset itself to check in with the management service on a regular basis, or the device is detected as being present in the office each time the employee arrives at work. An active system uses polling or pushing technology to send out queries to devices to elicit a response.

You can use asset tracking to verify that a device is still in the possession of the assigned authorized user. Some asset-tracking solutions can locate missing or stolen devices.

Some asset-tracking solutions expand beyond hardware inventory management and can oversee the installed apps, app usage, stored data, and data access on a device. You can use this type of monitoring to verify compliance with security guidelines or to check for exposure of confidential information to unauthorized entities.

Removable Storage

Many mobile devices support removable storage. Some devices support microSD cards, which can be used to expand available storage on a mobile device. However, most mobile phones require the removal of a back plate and sometimes removal of the battery to add or remove a storage card. Larger mobile phones, tablets, and laptop computers may support an easily accessible card slot on the side of the device.

Many mobile devices also support external USB storage devices, such as flash drives and external hard drives. Some may require a special on-the-go (OTG) cable. *USB On-The-Go (OTG)* is a specification that allows a mobile

device with a USB port to act as a host and use other standard peripheral USB equipment, such as storage devices, mice, keyboards, and digital cameras. USB OTG is a feature that can be deactivated via MDM/UEM if it is perceived as a risk vector for mobile devices used within an organization.

In addition, mobile storage devices may provide Bluetooth- or Wi-Fi-based access to stored data through an onboard wireless interface.

Organizations need to consider whether the use of removable storage on portable and mobile devices is a convenient benefit or a significant risk vector. If the former, proper access limitations and use training are necessary. If the latter, then a prohibition of removable storage can be implemented via MDM/UEM.

Deactivating Unused Features

Although enabling security features is essential for them to have any beneficial effect, it's just as important to remove apps and deactivate features that aren't essential to business tasks or common personal use. The wider the range of enabled features and installed apps, the greater the chance that an exploitation or software flaw will cause harm to the device and/or the data it contains. Following common security practices, such as hardening, reduces the attack surface of mobile devices.

Rooting or Jailbreaking

Rooting or *jailbreaking* (the special term for rooting Apple devices) is the action of breaking the digital rights management (DRM) security on the bootloader of a mobile device to be able to operate the device with root or full system privileges. Most mobile devices are locked in such a way as to restrict end-user activity to that of a limited user. But a root user can manipulate the OS, enable or deactivate hardware features, and install software applications that are not available to the limited user. Rooting may enable a user to change the core OS or operate apps that are unavailable in the standard app stores. However, this is not without its risks. Operating in rooted status also reduces security, since any executable also launches with full root privileges. Many forms of malicious code cannot gain footing on normal mode devices but can easily take root (pun intended) when the user has rooted or jailbroken their device.

Generally, an organization should prohibit the use of rooted devices on the company network or even access to company resources whenever possible.

It is legal to root a device if you fully own the device, if you are in a one- or two-year contract with a hardware fee, or if you are in a lease-to-own contract and you do not fully own the device until that contract is fulfilled. Legal root does not require a manufacturer, vendor, or telco to honor any warranty. In most cases, any form of system tampering, including rooting, voids your warranty. Rooting may also void your support contract or replacement contract. Rooting is actively suppressed by the telcos, many carriers, and some product vendors, Apple being the main example. A rooted device might be prohibited to operate over a telco network, access resources, download apps, or receive future updates. Thus, though it is often legal to root a device, there are numerous consequences to consider prior to altering a mobile device in that manner.

“Bricking” refers to rendering a device completely nonfunctional or as useless as a brick, typically through a malfunction or intentional action. When a device is “bricked,” it loses its normal functionality and becomes inoperable, often requiring significant efforts to restore its original state. This term is commonly used in the context of electronic devices such as smartphones, tablets, routers, or other hardware that may become unusable due to software errors, firmware corruption, or unauthorized modifications. “Bricking” has become less of a concern with modern devices, which often include a nonreplaceable recovery ROM in addition to its standard flashable firmware (i.e., updatable firmware).

Sideloading

Sideloading is the activity of installing an app on a device by bringing the installer file to the device through some form of file transfer or USB storage method. Most organizations should prohibit user sideloading, because it may be a means to bypass security restrictions imposed by an app store, application allow listing, or the MDM/UEM/MAM. An MDM/UEM/MAM-enforced configuration can require that all apps be digitally signed; this would eliminate sideloading and likely jailbreaking as well.

Custom Firmware

Mobile devices come preinstalled with a vendor- or telco-provided firmware or core OS. If a device is rooted or jailbroken, it can allow the user to install alternate custom firmware in place of the default firmware. Custom firmware may remove bloatware, add or remove features, and streamline the OS to optimize performance. You can find online discussion forums and communities, such as xda-developers.com and howardforums.com, that specialize in custom firmware for Apple and Android devices.

An organization should not allow users to operate mobile devices that have custom firmware unless that firmware is preapproved by the organization.

Carrier Unlocking

Most mobile devices purchased directly from a telco are carrier locked. This means you are unable to use the device on any other telco network until the carrier lock is removed or carrier unlocked. Once you fully own a device, the telco should freely *carrier unlock* the phone, but you will have to ask for it specifically because they don't do so automatically. If you have an account in good standing and are traveling to another country with compatible telco service, you may be able to get a telco to carrier unlock your phone for your trip so that you can use another SIM card (or eSIM) for local telco services.

Having a device carrier unlocked is not the same as rooting. Carrier unlocked status only allows the switching of telco services (which is technically possible only if your device uses the same radio frequencies as the telco). A carrier unlocked device should not represent any additional risk to an organization; thus, there is likely no need for a prohibition of carrier unlocked devices on company networks.

Firmware Over-the-Air (OTA) Updates

Firmware over-the-air (OTA) updates are firmware updates that are downloaded from the telco or vendor over-the-air (via a data connection either provided by the carrier or via Wi-Fi). Generally, as a mobile device owner, you should install new firmware OTA updates onto a device once they become available. However, some updates may alter the device configuration or interfere with MDM/UEM restrictions. You should attempt to test new updates before allowing managed devices to receive them. You

may have to establish a waiting period so that the MDM/UEM vendor can update their management product to properly oversee the deployment and configuration of the new firmware update. An organization's standard patch management, configuration management, and change management policies should be applied to mobile devices.

Credential Management

The storage of credentials in a central location is referred to as credential management. Given the wide range of Internet sites and services, each with its own particular logon requirements, using unique names and passwords can be a burden. *Credential management* solutions offer a means to securely store a plethora of credential sets. Often, these tools employ a master credential set (multifactor being preferred) to unlock the dataset when needed. Some credential-management options can even provide auto-login options for apps and websites.

A *password vault* is another term for a credential manager. These are often software solutions, sometimes hardware-based, sometimes local only, and sometimes using cloud storage. They are used to generate and store credentials for sites, services, devices, and whatever other secrets you want to keep private. The vault itself is encrypted and must be unlocked to regain access to the stored items. Most password vaults use Password-Based Key Derivation Function 2 (PBKDF2) or Bcrypt (see Chapter 7) to convert the vault's master password into a reasonably strong encryption key.

Text Messaging

Short Message Service (SMS), Multimedia Messaging Service (MMS), and Rich Communication Services (RCS) are all useful communication systems, but they also serve as an attack vector (such as smishing and SPIM, discussed in Chapter 2, “Personnel Security and Risk Management Concepts”). These testing and messaging services are primarily operated and supported by the telco providers. Texting can be used as an authentication factor known as SMS-based 2FA. SMS-based 2FA is better than single-factor password-only authentication, but it is not recommended if any other second-factor option is available. See Chapter 13 for more on SMS-based 2FA.

Many non-telco/non-carrier texting and messaging services are supported via apps on mobile devices. It is important to keep any messaging service app updated and restrict its use to nonsensitive content.

Mobile Device Deployment Policies

A number of deployment models are available for allowing and/or providing mobile devices for employees to use while at work and to perform work tasks when away from the office. A *mobile device deployment policy* must address the wide range of security concerns regarding the use of a PED in relation to the organization's IT infrastructure and business tasks.

Users need to understand the benefits, restrictions, and consequences of using mobile devices at work and for work. Reading and signing off on the BYOD, CYOD, COPE, COMS/COBO, etc., policy along with attending an overview or training program may be sufficient to accomplish reasonable awareness. These topics are covered in the next sections.



An alternative to allowing personal or business-provided mobile devices to interact with company resources directly would be to implement a VDI or VMI solution (see section “Software-Defined Everything” earlier this chapter).

Bring Your Own Device (BYOD)

Bring your own device (BYOD) is a policy that allows employees to bring their own personal mobile devices to work and may allow them to use those devices to connect to business resources and/or the Internet through the company network. Although BYOD may improve employee morale and job satisfaction, it increases security risk to the organization. If the BYOD policy is open-ended, any device may be allowed to connect to the company network. Not all mobile devices have sufficient security features, and thus such a policy allows noncompliant devices onto the production network. Employees likely retain full control over their devices, allowing them to disable or bypass security features imposed by the organization.

This is likely the least secure option for the organization since company data and applications will be on the personal mobile device, it exposes the organization's network to malicious code from the PEDs, and the devices will have the widest range of variation and security capabilities (or more likely the lack of security capabilities). Additionally, this option potentially exposes the worker's PII on the device to the organization.

Choose Your Own Device (CYOD)

The concept of *choose your own device (CYOD)* provides users with a list of approved devices from which to select the device to implement. A CYOD policy can be implemented so that employees purchase their own devices from the approved list (a BYOD variant).

This option attempts to keep the expense of devices the responsibility of workers rather than the organization, but it often results in much more complex and challenging situations. For example, how will it handle a situation wherein a worker has already spent considerable money on a device that is not on the preapproved list? Will they be given money to purchase an approved device? What about the person who paid for an approved device—will the company reimburse them because they already paid for someone else's device? What about the person who decides they don't want to use a mobile device for work activities—will they be paid the funds anyway, allowing them to treat it as a paycheck bonus?

Also, this option has the same security issues as COPE: the potential for malware transfer and the comingling of business and personal data on the same device.

Corporate-Owned, Personally Enabled (COPE)

Corporate-owned, personally enabled (COPE) is a mobile policy where the organization purchases devices and provides them to employees. Each user is then able to customize the device and use it for both work activities and personal activities. COPE allows the organization to select exactly which devices are to be allowed on the organizational network—specifically only those devices that can be configured into compliance with the security policy.

This option reduces the mobile devices to those preselected by the organization and that have the minimum security capabilities mandated by company security policy. However, this option still has the risk of exposing company data through user error, exposes the organization to malware via the device, and puts worker PII at risk of being accessed by the organization.

Corporate-Owned Mobile Strategy (COMS)

A *corporate-owned mobile strategy (COMS)* or *corporate-owned, business-only (COBO)* strategy is when the company purchases the mobile devices that can support security compliance with the security policy. These devices are to be used exclusively for company purposes, and users should not perform any personal tasks on the devices. This often requires workers to carry a second device for personal use.

This is the best option for both the organization as well as the individual worker. The option maintains clear separation between work activities and personal activities, since the device is for work use exclusively. This option protects company resources from personal activity risks, and it protects personal data from unauthorized or unethical organizational access. Yes, it is a hassle to carry a second device for personal activities, but that inconvenience is well worth the security benefits for both parties.

Mobile Device Deployment Policy Details

No matter which mobile device deployment policy you select and implement, your policy needs to address the many device security features listed earlier in this section. You can ensure this by defining required features and how they are to be configured for company security policy compliance. The mobile device deployment policy must also address several other concerns that are operational, legal, and logistic-based as well. These are discussed in the following sections.

Data Ownership

When a personal device is used for business tasks, commingling of personal data and business data is likely to occur. Some devices can support storage segmentation, but not all devices can provide data-type isolation. Establishing data ownership can be complicated. For example, if a device is

lost or stolen, the company may wish to trigger a remote wipe, clearing the device of all valuable information. However, the employee will often be resistant to this, especially if there is any hope that the device will be found or returned. A wipe may remove all business and personal data, which may be a significant loss to the individual—especially if the device is recovered, because then the wipe would seem to have been an overreaction. Clear policies about data ownership should be established. Some MDM/UEM solutions can provide data isolation/segmentation and support business data sanitization without affecting personal data.

The mobile device deployment policy regarding data ownership should address backups for mobile devices. Business data and personal data should be protected by a backup solution—either a single solution for all data on the device or separate solutions for each type or class of data. Backups reduce the risk of data loss in the event of a remote-wipe event as well as device failure or damage.

Support Ownership

When an employee's mobile device experiences a failure, a fault, or damage, who is responsible for the device's repair, replacement, or technical support? The mobile device deployment policy should define what support will be provided by the company and what support is left to the individual and, if relevant, their service provider.

Patch and Update Management

The mobile device deployment policy should define the means and mechanisms of secure patch management and update management for a personally owned mobile device. Is the user responsible for installing updates? Should the user install all available updates? Should the organization test updates prior to on-device installation? Are updates to be handled over the air (via service provider) or over Wi-Fi? Are there versions of the mobile OS that cannot be used? What patch or update level is required? These issues should be addressed both for the primary OS of the device and for all apps installed on the device.

Security Product Management

The mobile device deployment policy should dictate whether antivirus, antimalware, antispyware scanners, firewalls, HIDS, or other security tools are to be installed on mobile devices. The policy should indicate which products/apps are recommended for use, as well as the settings for those solutions.

Forensics

The mobile device deployment policy should address forensics and investigations related to mobile devices. Users need to be aware that in the event of a security violation or a criminal activity, their devices might be involved. An investigation would mandate gathering evidence from those devices. Some processes of evidence gathering can be destructive, and some legal investigations require the confiscation of devices. An owner of a personal device may refuse access to the contents of their device, even when that content is, in theory, the property of the organization. A company-owned device could have a secondary account, a master password, or a remote management tool preinstalled that would grant the organization the ability to access the device's contents without the user's consent.



In all legal matters, including mobile device forensics and privacy, consult your own attorney(s) for the best course of action and policy contents.

Privacy

The mobile device deployment policy should address privacy and monitoring. When a personal device is used for business tasks, the user often loses some or all of the privacy they enjoyed prior to using their mobile device at work. Workers may need to agree to be tracked and monitored on their mobile device, even when not on company property and outside work hours. A personal device in use under BYOD or CYOD should be considered by the individual to be quasi-company property.

A primary way for a worker to protect their privacy in regard to a mobile device is to not use a single device for both work and personal activities.

Architecture/Infrastructure Considerations

When implementing mobile device deployment policies, organizations should evaluate their network and security design, architecture, and infrastructure. If every worker brings in a personal device, the number of endpoint devices on the network may double. This requires planning to handle IP assignments, communications isolation, data-priority management, and increased intrusion detection system (IDS)/intrusion prevention system (IPS) monitoring load, as well as increased bandwidth consumption, both internally and across any Internet link. Most mobile devices are wireless enabled, so this will likely require a more robust wireless network and dealing with Wi-Fi congestion and interference. Your mobile device deployment policy must be considered in light of the additional infrastructure costs it will trigger.

Legal Concerns

Company attorneys should evaluate the legal concerns of mobile devices. Using personal devices in the execution of business tasks probably means an increased burden of liability and risk of data leakage. Mobile devices may make employees happy, but they might not be a worthwhile or cost-effective endeavor for your organization if they significantly increase risk and legal liability.

Acceptable Use Policy

The mobile device deployment policy should either reference the company's acceptable use policy (AUP) or include a mobile device–specific version focusing on unique issues. The use of personal mobile devices at work is accompanied by an increased risk of information disclosure, distraction, and access to inappropriate content. Workers should remain mindful that the primary goal when at work is to accomplish productivity tasks.

Onboard Camera/Video

The mobile device deployment policy needs to address mobile devices with onboard cameras. Some environments disallow cameras of any type. This

would require that mobile devices be without a camera. If cameras are allowed, a description of when they may and may not be used should be clearly documented and explained to workers. A mobile device can act as a storage device, provide an alternate wireless connection pathway to an outside provider or service, and may be used to collect images and videos that disclose confidential information or equipment.

If geofencing is available, it may be possible to use MDM/UEM to implement a location-specific hardware-deactivate profile to turn off the camera (or other components) while the device is on company premises but return the feature to operational status once the device leaves the geofenced area.

Recording Microphone

Most mobile devices with a speaker also have a microphone. The microphone can be used to record audio, noise, and voices nearby. Many mobile devices also support external microphones connected by a USB adapter, Bluetooth, or a 1/8″ stereo jack. If microphone recording is deemed a security risk, this feature should be deactivated using an MDM/UEM or deny presence of mobile devices in sensitive areas or meetings.

Tethering and Hotspots

Tethering is the activity of sharing the cellular network data connection of a mobile device with other devices. This is also known as a *hotspot*. This effectively allows the mobile device to act as a portable wireless access point (WAP). The sharing of data connections can take place over Wi-Fi, Bluetooth, or USB cable. Some service providers include tethering in their service plans, whereas others charge an additional fee and some block tethering completely.

Tethering may represent a risk to the organization. It is a means for a user to grant Internet access to devices that are otherwise network-isolated, and it can be used as a means to bypass the company's filtering, blocking, and monitoring of Internet use. Thus, tethering should be blocked while a mobile device is within a company facility.

Hotspot devices that operate as portable WAPs are available and can be used to create a Wi-Fi network linked to a telco's or carrier's data network.

Hotspot devices should be barred from use in most organizations because they provide a direct link to the Internet without a company's security restrictions being enforced.

Contactless Payment Methods

A number of mobile device–based payment systems, called *contactless payment methods*, do not require direct physical contact between the mobile device and the point-of-sale (PoS) device. Some are based on NFC, others on RFID, some on SMS, and still others on optical camera–based solutions, such as scanning *Quick Response (QR) codes*. Mobile payments are convenient for the shopper but might not always be a secure mechanism. Users should only employ mobile payment solutions that require a per-transaction confirmation or that require the device to be unlocked and an app launched to perform a transaction. Without these precautions, it may be possible to clone your device's contactless payment signals and perform transaction abuse.

Your organization is unlikely to see any additional risk based on mobile payment solutions. However, use caution when implementing them on company-owned equipment or when they are linked to your company's financial accounts.

SIM Cloning

Subscriber identity module (SIM) cards are used to associate a device with a subscriber's identity and service at a mobile or wireless telco. SIMs can be easily swapped between devices and cloned to abuse a victim's telco services. If a SIM card is cloned, then the cloned SIMs may be able to connect other devices to the telecommunications services and link the use back to the account of the original owner. Physical control must be maintained on mobile devices and an account or service lock established on mobile services with the telco carrier.

Essential Security Protection Mechanisms

The need for security mechanisms within an OS comes down to one simple fact: Software should not be trusted. Third-party software is inherently untrustworthy, no matter who or where it comes from. This is not to say that

all software is evil. Instead, this is a protection stance—because all third-party software is written by someone other than the OS creator, that software might cause problems. Thus, treating all non-OS software as potentially damaging allows the OS to prevent many disastrous occurrences through the use of software management protection mechanisms. The OS must employ protection mechanisms to keep the computing environment stable and to keep processes isolated from one another. Without these efforts, the security of data could never be reliable or even possible. This is effectively applying the principle of zero trust (see Chapter 8).

Computer system designers should adhere to a number of common protection mechanisms when designing secure systems. These principles are specific instances of the more general security rules that govern safe computing practices. Designing security into a system during the earliest stages of development will help ensure that the overall security architecture has the best chance for success and reliability.

Process Isolation

Process isolation requires that the OS provide separate memory spaces for each process's instructions and data. It also requires that the OS enforce those boundaries, preventing one process from reading or writing data that belongs to another process. There are two major advantages to using this technique:

- It prevents unauthorized data access.
- It protects the integrity of processes.

Without such controls, a poorly designed process could go haywire and write data to memory spaces allocated to other processes, causing the entire system to become unstable rather than affecting only the execution of the errant process. In a more malicious vein, processes could attempt (and perhaps even succeed at) reading or writing to memory spaces outside their scope, intruding on or attacking other processes.

Many modern OSs address the need for process isolation by implementing virtual machines on a per-user or per-process basis. A virtual machine presents a user or process with a processing environment—including memory, address space, and other key system resources and services—that

allows that user or process to behave as though they have sole, exclusive access to the entire computer. This allows each user or process to operate independently without requiring it to take cognizance of other users or processes that might be active simultaneously on the same machine. As part of the mediated access to the system that the OS provides, it maps virtual resources and access in lower-privileged layers/rings, and system calls are used to request access to the corresponding real resources. This not only makes things easier for programmers, but also protects individual users and processes from one another.

Hardware Segmentation

Hardware segmentation is similar to process isolation in purpose—it prevents access to information that belongs to a different process/security level. The main difference is that hardware segmentation enforces these requirements through the use of physical hardware controls rather than the logical process isolation controls imposed by an OS. Such implementations are rare, and they are generally restricted to national security implementations, where the extra cost and complexity is offset by the sensitivity of the information involved and the risks inherent in unauthorized access or disclosure.

Root of Trust

The *root of trust (RoT)* is a foundational concept in cybersecurity and cryptographic systems. It represents the starting point or anchor of a security chain, providing a secure and trustworthy foundation for various security functions. The root of trust is critical for establishing and verifying the integrity, authenticity, and confidentiality of digital information within a system.

A *trust anchor* is a specific entity or component within a system that is inherently trusted. It serves as a reference point for establishing trust in other entities or components within the system. The trust anchor is typically a well-protected and tamper-resistant element, and trust in the overall system is derived from the trustworthiness of the trust anchor.

A hardware-based RoT refers to the implementation of the root of trust using dedicated hardware components. This often involves the integration

of secure hardware modules, such as Trusted Platform Modules (TPMs) or hardware security modules (HSMs), which are designed to provide a secure and isolated environment for cryptographic operations and key management. Hardware-based RoTs enhance security by isolating critical security functions from the general-purpose computing environment, making them more resistant to various forms of attacks. These concepts are crucial for building secure systems and ensuring the integrity and trustworthiness of digital interactions.

System Security Policy

Just as security policy guides the day-to-day security operations, processes, and procedures in organizations, they have an important role to play when designing and implementing systems. This is equally true whether a system is entirely hardware-based, entirely software-based, or a combination of both. In this case, the role of a *system security policy* is to inform and guide the design, development, implementation, testing, and maintenance of a particular system. Thus, this kind of security policy tightly targets a single implementation effort. (Although it may be adapted from other, similar efforts, it should reflect the target as accurately and completely as possible.)

For system developers, a system security policy is best encountered in the form of a document that defines a set of rules, practices, and procedures that describe how the system should manage, protect, and distribute sensitive information. Security policies that prevent information flow from higher security levels to lower security levels are called multilevel security policies. As a system is developed, the security policy should be designed, built, implemented, and tested as it relates to all applicable system components or elements, including any or all of the following: physical hardware components, firmware, software, and how the organization interacts with and uses the system. The overall point is that security must be considered for the entire life of the project. When security is applied only at the end, it typically fails.

Common Security Architecture Flaws and Issues

No security architecture is totally secure. Every computer system has weaknesses and vulnerabilities. The goal of security models and architectures is to address as many known weaknesses as possible. Due to this fact, corrective actions must be taken to resolve security issues. The following sections present some common security issues that affect computer systems in relation to vulnerabilities of security architectures. You should understand each of the issues and how they can degrade the overall security of your system. Some issues and flaws overlap one another and are used in creative ways to attack systems. Although the following discussion covers the most common flaws, the list is not exhaustive. Attackers are very clever.

Many attacks and exploits are covered elsewhere that are also relevant to this chapter's content, such as denial of service (DoS) (Chapter 17), buffer overflow (Chapter 21), malware (Chapter 21), escalation of privilege (Chapter 21), and maintenance hooks/backdoors (Chapter 21). We covered numerous malicious issues earlier in this chapter, such as emanation eavesdropping, the cold boot attack against memory, phlashing, mobile-code-based client-side attacks, exploitation of local Internet caches, and VM escaping. Several additional adversarial threats are included here, such covert channels, design/coding flaws, rootkits, and incremental attacks.

Covert Channels

A *covert channel* is a method that is used to pass information over a path that is not normally used for communication. Because the path is not normally used for communication, it may not be protected by the system's normal security controls. Using a covert channel provides a means to violate, bypass, or circumvent a security policy undetected. Covert channels are one of the important examples of vulnerabilities of security architectures.

As you might imagine, a covert channel is the opposite of an overt channel. An *overt channel* is a known, expected, authorized, designed, monitored, and controlled method of communication. Therefore, a covert channel is an unknown, unexpected, unauthorized, not designed (at least not by the original system designers), unmonitored, and uncontrolled method of data transfer.

There are two basic types of covert channels:

Covert Timing Channel A *covert timing channel* conveys information by altering the performance of a system component or modifying a resource's timing in a predictable manner. Using a covert timing channel is generally a method to secretly transfer data and is very difficult to detect.

Covert Storage Channel A *covert storage channel* conveys information by writing data to a common storage area where another process can read it. When assessing the security of software, be diligent for any process that writes to any area of memory that another process can read.

Examples of covert timing channels include the following:

- Blinking a light visible outside the building so that if a reading is taken every two seconds when the light is on count it as a 1 and when the light is off count it as a 0. With an external camera linked to a recording system, a slow transmission of binary data can occur.
- Using a microphone to listen to the noise occurring in an area or related to a computer system. Then modify a case fan to spin faster (for a 1) or slower (for a 0) to force a change in the noise generated every 10 seconds.
- Monitoring utilization levels of an Internet connection when an insider is artificially padding or restricting traffic every 30 seconds. When traffic is above 80 percent utilization, record a 1; when below 40 percent utilization, record a 0.

Here are examples of covert storage channels; notice that they all involve placing data in a location that is either unseen by the OS or ignored by the OS:

- Writing data into unallocated or unpartitioned space, which may be accomplished using a hex editor
- Writing data directly into a bad sector of an HDD or a bad block on an SSD

- Writing data into the unused space at the end of a cluster, an area known as slack space
- Writing data directly into sectors or clusters without proper registration with the directory system, file container, or header

Both types of covert channels rely on the use of communication techniques to exchange information with otherwise unauthorized subjects. Detecting such abuse can be difficult because the covert channel is outside the normal data transfer environment and security oversight. The best defense is to implement detailed and thorough auditing of all user and application activities and analyze log files for any covert channel activity, which may be anomalous behavior or may elicit known malicious activities via heuristics or pattern matching.

Attacks Based on Design or Coding Flaws

Certain attacks may result from poor design techniques, questionable implementation practices and procedures, or poor or inadequate testing. Some attacks may result from deliberate design decisions when special points of entry, built into code to circumvent access controls, login, or other security checks often added to code while under development, are not removed when that code is put into production. For what we hope are obvious reasons, such points of egress are properly called maintenance hooks or backdoors because they avoid security measures by design. Extensive testing and code review are required to uncover such covert means of access, which are easy to remove during the final phases of development but can be incredibly difficult to detect during the testing and maintenance phases.

Poor coding practices and lack of security consideration are common sources or causes of vulnerabilities in system architectures that can be attributed to failures in design, implementation, prerelease code cleanup, or out-and-out coding mistakes. Although such flaws are avoidable, finding and fixing them requires rigorous security-conscious design from the beginning of a development project and extra time and effort spent in testing and analysis. This helps to explain the often lamentable state of software security, but it does not excuse it! Although functionality testing is commonplace for commercial code and applications, separate testing for

security issues has been gaining attention and credibility only in the past few years, courtesy of widely publicized virus and worm attacks, SQL injection attacks, cross-site scripting attacks, and occasional defacements of or disruptions to widely used public sites online. Check out the OWASP Top 10 Web Application Security Risks report at owasp.org/www-project-top-ten. Most of these coding concerns are addressed in Chapter 20, "Software Development Security," and Chapter 21.

Humans will never write completely secure (flawless) code. Any program that does not gracefully handle any exception is in danger of exiting in an unstable state. It is possible to cleverly crash a program after it has increased its security level to carry out a normal task. If an attacker is successful in crashing the program at the right time, they can attain a higher security level and cause damage to the confidentiality, integrity, and availability of your system. These are just a few of the myriad ways that code can be compromised.

Perfect security might be impossible, but you can definitely take many strong measures to secure your code better. Source code analysis tools implemented throughout the development cycle will minimize the number of flaws in the production release, and the flaws identified prior to production release will cost much less to mitigate. All programs that are executed directly or indirectly must be fully tested to comply with your security model. Make sure you have the latest version of any software installed, and be aware of any known security vulnerabilities. Because each security model, and each security policy, is different, you must ensure that the software you execute does not exceed the authority you allow. Writing secure code is difficult, but it's certainly possible. Make sure all programs you use are designed to address security concerns. The concepts of code review and testing are covered in Chapter 15, "Security Assessment and Testing."

Rootkits

A *rootkit* is malware that embeds itself deep within an OS. The term is a derivative of the concept of rooting and a utility kit of hacking tools. Rooting is gaining total or full control over a system.

A rootkit can manipulate information seen by the OS and displayed to users. A rootkit may replace the OS kernel, shim itself under the kernel, replace device drivers, or infiltrate application libraries so that whatever information it feeds to or hides from the OS, the OS thinks it is normal and acceptable. This allows a rootkit to hide itself from detection, prevent its files from being viewed by file management tools, and prevent its active processes from being viewed by task management or process management tools. Thus, a rootkit is a type of invisibility shield used to hide itself and other malicious tools.

Several rootkit-detection tools are available, some of which are able to remove known rootkits. However, once you suspect a rootkit is on a system, the only truly secure response is to reconstitute or replace the entire computer. Reconstitution involves performing a thorough storage sanitization operation on all storage devices on that system, reinstalling the OS and all applications from trusted original sources, and then restoring files from trusted rootkit-free backups. Obviously, the best protection against rootkits is defense (i.e., don't get infected in the first place) rather than response.

There are often no noticeable symptoms or indicators of compromise related to a rootkit infection. In the moments after the initial rootkit installation, there might be some system sluggishness and unresponsiveness as the rootkit installs itself, but otherwise, it will actively mask any symptoms. In some rootkit infections, the malware's initial infector, dropper, or installer will perform privilege escalation.

A means to potentially detect the presence of a rootkit is to notice when system files, such as device drivers and dynamic-link libraries (DLLs), have a file size and/or hash value change. File hash tracking can be performed manually by an administrator or automatically by HIDSs and system monitoring security tools.

Incremental Attacks

Some forms of attack occur in slow, gradual increments rather than through obvious or recognizable attempts to compromise system security or integrity. Two such forms of *incremental attack* are data diddling and the salami attack.

Data diddling occurs when an attacker gains access to a system and makes small or random changes to data during processing, input, or transaction rather than obviously altering file contents or damaging or deleting entire files. Such changes can be difficult to detect unless files and data are protected by encryption or unless some kind of integrity check (such as a checksum or message digest) is routinely performed and applied each time a file is read or written. Encrypted filesystems, file-level encryption techniques, or some form of file monitoring (which includes integrity checks performed by file integrity monitoring [FIM] tools) usually offer adequate guarantees that no data diddling is underway. Data diddling is often considered an attack performed more often by insiders rather than outsiders (external intruders).

The *salami attack* is more mythical by all published reports. The name of the attack refers to a systematic whittling at assets in accounts or other records with financial value, where very small amounts are deducted from balances regularly and routinely. Metaphorically, the attack may be explained as stealing a very thin slice from a salami each time it's put on the slicing machine when a paying customer is accessing it. Most security experts concede that salami attacks are possible, especially when organizational insiders could be involved. Organizations can prevent or eliminate such an attack only by proper separation of duties and proper control over code. Setting financial transaction monitors to track very small transfers of funds or other items of value should help to detect such activity; regular employee notification of the practice should help to discourage attempts at such attacks.



If you want an entertaining method of learning about the salami attack or the salami technique, view the movies *Office Space* and *Superman III*. You can also read the article from *Wired* about an attack of this nature from 2008: www.wired.com/2008/05/man-allegedly-b.

Summary

Shared responsibility is the security design principle indicating that organizations do not operate in isolation. It is because we participate in shared responsibility that we must research, implement, and manage engineering processes using secure design principles.

Designing secure computing systems begins with an investigation of hardware, software, and firmware and how those pieces fit into the security puzzle. It's important to understand the principles of common computer and network organizations, architectures, and designs; the difference between address space and memory space; and machine types or system variations.

Additionally, a security professional must have a good grasp of operating modes (user, supervisor, privileged), storage types (primary, secondary, real, virtual, volatile, nonvolatile, random, sequential), and common protection mechanisms (such as process isolation and hardware segmentation).

System function, purpose, and design work *toward* establishing and supporting security or *against* it. Client-based systems should be concerned about running code from unknown sources as well as protecting local caches. Server-based systems need to manage data flow and optimize operations using large-scale parallel data systems, grid computing, or peer-to-peer solutions when appropriate. Additional concerns relate to industrial control systems, the Internet of Things, microservices, and infrastructure as code.

Virtualization technology is used to host one or more OSs within the memory of a single host computer. Virtual software, virtual networking, software-defined everything, containerization, serverless architecture, and other related advancements often dictate the need for virtualization security management.

Static environments, embedded systems, cyber-physical systems, HPC systems, edge computing devices, and fog computing devices, mobile devices, and other limited or single-purpose computing environments need security management.

No matter how sophisticated a security architecture is, flaws exist that attackers can exploit. Some flaws are introduced by programmers, whereas others are architectural design issues.

Study Essentials

Understand shared responsibility. The security design principle indicates that organizations do not operate in isolation. It is because we participate in shared responsibility that we must research, implement, and manage engineering processes using secure design principles.

Understand the concept of protection rings. From a security standpoint, protection rings organize code and components in an OS into concentric rings. The deeper inside the circle you go, the higher the privilege level associated with the code that occupies a specific ring.

Describe the different types of memory used by a computer. ROM is nonvolatile and can't be written to by the end user. Data can be written to PROM chips only once. EPROM/UVEPROM chips may be erased with ultraviolet light. EEPROM chips may be erased with electrical current. RAM chips are volatile and lose their contents when the computer is powered off.

Know the security issues surrounding memory components. Some security issues surround memory components: the fact that data may remain on the chip after power is removed and the control of access to memory in a multiuser system.

Know the concepts of memory addressing. Means of memory addressing include register addressing, immediate addressing, direct addressing, indirect addressing, and base+offset addressing.

Describe the different characteristics of storage devices used by computers. Primary storage is the same as memory. Secondary storage consists of magnetic, flash, and optical media that must be first read into primary memory before the CPU can use the data. Random access storage devices can be read at any point, whereas sequential access devices require scanning through all the data physically stored before the desired location.

Know the security issues surrounding secondary storage devices. Three main security issues surround secondary storage devices: removable media can be used to steal data, access controls and encryption must be applied to protect data, and data can remain on the media even after file deletion or media formatting.

Know about emanation security. Many electrical devices emanate electrical signals or radiation that can be intercepted by unauthorized individuals. These signals may contain confidential, sensitive, or private data. TEMPEST/EMSEC countermeasures to Van Eck phreaking (i.e., eavesdropping) include Faraday cages, white noise, control zones, and shielding.

Understand security risks that input and output devices can pose. Input/output devices can be subject to eavesdropping and tapping, are subject to shoulder surfing, are used to smuggle data out of an organization, or are used to create unauthorized, insecure points of entry into an organization's systems and networks. Be prepared to recognize and mitigate such vulnerabilities.

Be aware of JavaScript concerns. JavaScript is the most widely used scripting language in the world and is embedded into HTML documents. Whenever you allow code from an unknown and thus untrusted source to execute on your system, you are putting your system at risk of compromise.

Know about large-scale parallel data systems. Systems designed to perform numerous calculations simultaneously include SMP, AMP, and MPP. Grid computing is a form of parallel distributed processing that loosely groups a significant number of processing nodes to work toward a specific processing goal. Peer-to-peer (P2P) technologies are networking and distributed application solutions that share tasks and workloads among peers.

Be able to define OT/ICS. An industrial control system (ICS) or an operational technology (OT) is a form of computer-management device that controls industrial processes and machines. ICS examples include distributed control systems (DCSs), programmable logic controllers (PLCs), and supervisory control and data acquisition (SCADA).

Be aware of distributed systems. A distributed system or a distributed computing environment (DCE) is a collection of individual systems that work together to support a resource or provide a service. The primary security concern is the interconnectedness of the components.

Understand data sovereignty. Data sovereignty is the concept that, once information has been converted into a binary form and stored as digital

files, it is subject to the laws of the country within which the storage device resides.

Be able to define IoT. The Internet of Things (IoT) is a class of devices that are Internet-connected to provide automation, remote control, or AI processing to appliances or devices. The security issues related to IoT often relate to access and encryption.

Understand microservices. A microservice is simply one element, feature, capability, business logic, or function of a web application that can be called upon or used by other web applications. It is the conversion or transformation of a capability of one web application into a microservice that can be called upon by numerous other web applications. It allows large complex solutions to be broken into smaller self-contained functions.

Be able to define IaC. Infrastructure as code (IaC) is a change in how hardware management is perceived and handled. Instead of seeing hardware configuration as a manual, direct hands-on, one-on-one administration hassle, it is viewed as just another collection of elements to be managed in the same way that software and code are managed under DevSecOps (development, security, and operations).

Understand hypervisors. The hypervisor, also known as the virtual machine monitor/manager (VMM), is the component of virtualization that creates, manages, and operates virtual machines.

Understand virtual software. A virtual application or virtual software is a software product deployed in such a way that it is fooled into believing it is interacting with a full host OS. A virtual (or virtualized) application has been packaged or encapsulated so that it can execute but operate without full access to the host OS. A virtual application is isolated from the host OS so that it cannot make any direct or permanent changes to the host OS.

Know virtual networking. A virtualized network or network virtualization is the combination of hardware and software networking components into a single integrated entity. The resulting solution allows for software control over all network functions: management, traffic shaping, address assignment, and so on.

Know about SDx. Software-defined everything (SDx) refers to a trend of replacing hardware with software using virtualization. SDx includes

virtualization, virtualized software, virtual networking, containerization, serverless architecture, infrastructure as code, SDN, VSAN, software-defined storage (SDS), VDI, VMI, SDV, and software-defined data center (SDDC).

Know about VDI and VMI. Virtual desktop infrastructure (VDI) is a means to reduce the security risk and performance requirements of end devices by hosting desktop/workstation OS virtual machines on central servers that are remotely accessed by users. Virtual mobile infrastructure (VMI) is where the OS of a mobile device is virtualized on a central server.

Be aware of SDV. Software-defined visibility (SDV) is a framework to automate the processes of network monitoring and response. The goal is to enable the analysis of every packet and make deep intelligence-based decisions on forwarding, dropping, or otherwise responding to threats.

Know some of the security issues of virtualization. Virtualization doesn't lessen the security management requirements of an OS. Thus, patch management is still essential. It's important to protect the stability of the host. Organizations should maintain backups of their virtual assets. Virtualized systems should be security tested. VM sprawl occurs when an organization deploys numerous virtual machines without an overarching IT management or security plan in place.

Understand containerization. Containerization or OS virtualization is based on the concept of eliminating the duplication of OS elements in a virtual machine. Each application is placed into a container that includes only the actual resources needed to support the enclosed application, and the common or shared OS elements are then part of the hypervisor.

Understand embedded systems. An embedded system is typically designed around a limited set of specific functions in relation to the larger product to which it is attached.

Be aware of microcontrollers. A microcontroller is similar to but less complex than a system on a chip (SoC). A microcontroller may be a component of an SoC. A microcontroller is a small computer consisting of a CPU (with one or more cores), memory, various input/output capabilities, RAM, and often nonvolatile storage in the form of flash or

ROM/PROM/EEPROM. Examples include Raspberry Pi, Arduino, and FPGA.

Understand embedded systems and static environment security concerns. Static environments, embedded systems, cyber-physical systems, HPC systems, edge computing devices, fog computing devices, mobile devices, and other limited or single-purpose computing environments need security management. These techniques may include network segmentation, security layers, application firewalls, manual updates, firmware version control, and control redundancy and diversity.

Know about HPC systems. High-performance computing (HPC) systems are computing platforms designed to perform complex calculations or data manipulations at extremely high speeds. Supercomputers and MPP solutions are common examples of HPC systems.

Be aware of RTOS. A real-time operating system (RTOS) is designed to process or handle data as it arrives on the system with minimal latency or delay. An RTOS is usually stored on read-only memory (ROM) and is designed to operate in a hard real-time or soft real-time condition.

Understand edge computing. Edge computing is a philosophy of network design where data and the compute resources are located as close as possible to optimize bandwidth use while minimizing latency. In edge computing, the intelligence and processing are contained within each device. Thus, rather than having to send data off to a master processing entity, each device can process its own data locally.

Know about fog computing. Fog computing is another example of advanced computation architectures, which is also often used as an element in an IIoT deployment. Fog computing relies on sensors, IoT devices, or even edge computing devices to collect data, and then transfer it back to a central location for processing. Thus, intelligence and processing is centralized.

Understand mobile device security. Personal electronic device (PED) security features can often be managed using a mobile device management (MDM) or unified endpoint management (UEM) solution. These include device authentication, full-device encryption, communication protection, remote wiping, screen locks, device lockout, GPS and location services

management, content management, application control, push notification management, third-party application store control, storage segmentation, asset tracking, removable storage, deactivating unused features, rooting/jailbreaking, sideloading, custom firmware, carrier unlocking, firmware OTA updates, credential management, and text messaging security.

Understand mobile device deployment policies. A number of deployment models are available for allowing and/or providing mobile devices for employees to use while at work and to perform work tasks when away from the office. Examples include BYOD, CYOD, COPE, and COMS/COBO. You should also consider VDI and VMI options.

Understand process isolation. Process isolation requires that the OS provide separate memory spaces for each process's instructions and data. It also requires that the OS enforce those boundaries, preventing one process from reading or writing data that belongs to another process.

Be aware of hardware segmentation. Hardware segmentation is similar to process isolation in purpose—it prevents the access of information that belongs to a different process/security level. The main difference is that hardware segmentation enforces these requirements through the use of physical hardware controls rather than the logical process isolation controls imposed by an OS.

Understand the need for system security policy. The role of a system security policy is to inform and guide the design, development, implementation, testing, and maintenance of a particular system. Thus, this kind of security policy tightly targets a single implementation effort.

Be able to explain what covert channels are. A covert channel is a method that is used to pass information over a path that is not normally used for communication. Using a covert channel provides a means to violate, bypass, or circumvent a security policy undetected. Basic types are timing and storage.

Know about vulnerabilities due to design and coding flaws. Certain attacks may result from poor design techniques, questionable implementation practices and procedures, or poor or inadequate testing. Some attacks may result from deliberate design decisions when special

points of entry, built into code to circumvent access controls, login, or other security checks often added to code while under development, are not removed when that code is put into production. Poor coding practices and lack of security consideration are common sources or causes of vulnerabilities of system architectures that can be attributed to failures in design, implementation, prerelease code cleanup, or out-and-out coding mistakes.

Written Lab

1. Name three types of ICSs and describe what they do or how they are used.
2. Name the three pairs of aspects or features used to describe storage.
3. Name some vulnerabilities found in distributed architectures.
4. There are numerous server-based technologies that both increase computation and resource access capabilities and also introduce new risks to be managed. Name at least 10 examples of these technologies (over 20 were included in this chapter).
5. In relation to mobile devices, list seven of the potential on-device security features, list the four main deployment models, and list seven of the issues that should be addressed on a mobile device deployment policy.

Review Questions

1. While designing the security for the organization, you realize the importance of not only balancing the objectives of the organization against security goals but also focusing on the shared responsibility of security. Which of the following is considered an element of shared responsibility? (Choose all that apply.)
   A. Everyone in an organization has some level of security responsibility.
   B. Always consider the threat to both tangible and intangible assets.

C. Organizations are responsible to their stakeholders for making good security decisions to sustain the organization.

D. When working with third parties, especially with cloud providers, each entity needs to understand their portion of the shared responsibility of performing work operations and maintaining security.

E. Multiple layers of security are required to protect against adversary attempts to gain access to internal sensitive resources.

F. As we become aware of new vulnerabilities and threats, we should consider it our responsibility (if not our duty) to responsibly disclose that information to the proper vendor or to an information sharing center.

2. Many PC OSs provide functionality that enables them to support the simultaneous execution of multiple applications on single-processor systems. What term is used to describe this capability?

   A. Multistate

   B. Multithreading

   C. Multitasking

   D. Multiprocessing

3. Based on recent articles about the risk of mobile code and web apps, you want to adjust the security configurations of organizational endpoint devices to minimize the exposure. On a modern Windows system with the latest version of Microsoft's browser and all others deactivated or blocked, which of the following is of the highest concern?

   A. Java

   B. Flash

   C. JavaScript

   D. ActiveX

4. Your organization is considering deploying a publicly available screen saver to use spare system resources to process sensitive company data.

What is a common security risk when using grid computing solutions that consume available resources from computers over the Internet?

A. Loss of data privacy

B. Latency of communication

C. Duplicate work

D. Capacity fluctuation

5. Your company is evaluating several cloud providers to determine which is the best fit to host your custom services as a custom application solution. There are many aspects of security controls you need to evaluate, but the primary issues include being able to process significant amounts of data in short periods of time, controlling which applications can access which assets, and being able to prohibit VM sprawl or repetition of operations. Which of the following is not relevant to this selection process?

   A. Collections of entities, typically users, but can also be applications and devices, which can be granted or denied access to perform specific tasks or access certain resources or assets

   B. A VDI or VMI instance that serves as a virtual endpoint for accessing cloud assets and services

   C. The ability of a cloud process to use or consume more resources (such as compute, memory, storage, or networking) when needed

   D. A management or security mechanism able to monitor and differentiate between numerous instances of the same VM, service, app, or resource

6. A large city's central utility company has seen a dramatic increase in the number of distribution nodes failing or going offline. An APT group was attempting to take over control of the utility company and was responsible for the system failures. Which of the following systems has the attacker compromised?

   A. MFP

   B. RTOS

C. SoC

D. SCADA

7. Your organization is concerned about information leaks due to workers taking home retired equipment. Which one of the following types of memory might retain information after being removed from a computer and therefore represents a security risk?

   A. Static RAM

   B. Dynamic RAM

   C. Secondary memory

   D. Real memory

8. Your organization is considering the deployment of a DCE to support a massively multiplayer online role-playing game (MMORPG) based on the characters of a popular movie franchise. What is the primary concern of a DCE that could allow for propagation of malware or making adversarial pivoting and lateral movement easy?

   A. Unauthorized user access

   B. Identity spoofing

   C. Interconnectedness of the components

   D. Poor authentication

9. Your boss wants to automate the control of the building's HVAC system and lighting to reduce costs. He instructs you to keep costs low and use off-the-shelf IoT equipment. When you are using IoT equipment in a private environment, what is the best way to reduce risk?

   A. Use public IP addresses.

   B. Power off devices when not in use.

   C. Keep devices current on updates.

   D. Block access from the IoT devices to the Internet.

10. Service-oriented architecture (SOA) constructs new applications or functions out of existing but separate and distinct software services.

The resulting application is often new; thus, its security issues are unknown, untested, and unprotected. Which of the following is a direct extension of SOA that creates single-use functions that can be employed via an API by other software?

A. Cyber-physical systems

B. Fog computing

C. DCS

D. Microservices

11. A new local VDI has been deployed in the organization. There have been numerous breaches of security due to issues on typical desktop workstations and laptop computers used as endpoints. Many of these issues stemmed from users installing unapproved software or altering the configuration of essential security tools. In an effort to avoid security compromises originating from endpoints in the future, all endpoint devices are now used exclusively as dumb terminals. Thus, no local data storage or application execution is performed on endpoints. Within the VDI, each worker has been assigned a VM containing all of their business necessary software and datasets. These VMs are configured to block the installation and execution of new software code, data files cannot be exported to the actual endpoints, and each time a worker logs out, the used VM is discarded and a clean version copied from a static snapshot replaces it. What type of system has now been deployed for the workers to use?

    A. Cloud services

    B. Nonpersistent

    C. Thin clients

    D. Fog computing

12. A review of your company's virtualization of operations determines that the hardware resources supporting the VMs are nearly fully consumed. The auditor asks for the plan and layout of VM systems but is told that no such plan exists. This reveals that the company is suffering from what issue?

A. Use of EOSL systems

B. VM sprawl

C. Poor cryptography

D. VM escaping

13. A company server is currently operating at near maximum resource capacity, hosting just seven virtual machines. Management has instructed you to deploy six new applications onto additional VMs without purchasing new hardware since the IT/IS budget is exhausted. How can this be accomplished?

A. Data sovereignty

B. Infrastructure as code

C. Containerization

D. Process isolation

14. Which of the following is a primary concern when implementing security measures for industrial control systems (ICSs)?

A. Confidentiality of sensor data

B. Availability of real-time control signals

C. Integrity of historical system logs

D. Nonrepudiation of operator commands

15. You have been tasked with designing and implementing a new security policy to address the new threats introduced by the recently installed embedded systems. What is a security risk of an embedded system that is not commonly found in a standard PC?

A. Software flaws

B. Access to the Internet

C. Control of a mechanism in the physical world

D. Power loss

16. A company is developing a new product to perform simple automated tasks related to indoor gardening. The device will be able to turn lights

on and off and control a pump to transfer water. The technology to perform these automated tasks needs to be small and inexpensive. It only needs minimal computational capabilities and does not need networking. The organization thinks that using an embedded system or a microcontroller may be able to provide the functionality necessary for the product. Which of the following is the best choice to use for this new product?

A. Arduino

B. RTOS

C. MPP

D. FPGA

17. You are developing a new product that is intended to process data to trigger real-world adjustments with minimal latency or delay. The current plan is to embed the code into a ROM chip to optimize for mission-critical operations. What type of solution is most appropriate for this scenario?

A. Containerized application

B. An Arduino

C. DCS

D. RTOS

18. A major online data service wants to provide better response and access times for its users and visitors. They plan on deploying thousands of mini-web servers to ISPs across the nation. These mini-servers will host the few dozen main pages of their website so that users will be routed to the logically and geographically closest server for optimal performance and minimal latency. Only if a user requests data not on these mini-servers will they be connecting to the centralized main web cluster hosted at the company's headquarters. What is this type of deployment commonly known as?

A. Edge computing

B. Fog computing

C. Thin clients

D. Infrastructure as code

19. You are working on improving your organization's policy on mobile equipment. Because of several recent and embarrassing breaches, the company wants to increase security through technology as well as user behavior and activities. What is the most effective means of reducing the risk of losing the data on a mobile device, such as a laptop computer?

    A. Defining a strong logon password

    B. Minimizing sensitive data stored on the mobile device

    C. Using a cable lock

    D. Encrypting the hard drive

20. The CISO has asked you to propose an update to the company's mobile device security policy. The main concerns are the intermingling of personal information with business data and complexities of assigning responsibility over device security, management, updates, and repairs. Which of the following would be the best option to address these issues?

    A. Bring your own device (BYOD)

    B. Corporate-owned personally enabled (COPE)

    C. Choose your own device (CYOD)

    D. Corporate-owned mobile strategy (COMS)

Chapter 10
Physical Security Requirements

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 3.0: Security Architecture and Engineering**
  - 3.8 Apply security principles to site and facility design
  - 3.9 Design site and facility security controls
    - 3.9.1 Wiring closets/intermediate distribution frame
    - 3.9.2 Server rooms/data centers
    - 3.9.3 Media storage facilities
    - 3.9.4 Evidence storage
    - 3.9.5 Restricted and work area security
    - 3.9.6 Utilities and Heating, Ventilation, and Air Conditioning (HVAC)
    - 3.9.7 Environmental issues (e.g., natural disasters, man-made)
    - 3.9.8 Fire prevention, detection, and suppression
    - 3.9.9 Power (e.g., redundant, backup)
- ✓ **Domain 7: Security Operations**
  - 7.14 Implement and manage physical security
    - 7.14.1 Perimeter security controls
    - 7.14.2 Internal security controls

The topic of physical and environmental security is referenced in several domains. The primary occurrences are in Domain 3.0, “Security Architecture and Engineering,” and Domain 7.0, “Security Operations.”

This chapter explores these issues and discusses safeguards and countermeasures to protect against them. You'll often need a disaster recovery plan or a business continuity plan should a severe physical event (such as an explosion, sabotage, or natural disaster) occur. Chapter 3, “Business Continuity Planning,” and Chapter 18, “Disaster Recovery Planning,” cover those topics in detail.

Apply Security Principles to Site and Facility Design

Without control over the physical environment, no collection of administrative, technical, or logical security controls can provide adequate protection. If a malicious person can gain physical access to your facility or equipment, they can do anything, including destruction, disclosure, and alteration.

There are many aspects of implementing and maintaining physical security. A core element is selecting or designing the facility to house your IT infrastructure and your organization's operations. The process of selecting or designing facility security always starts with a plan.

Secure Facility Plan

A *secure facility plan* outlines your organization's security needs and emphasizes methods or mechanisms to employ to provide security. Such a plan is developed through risk assessment and critical path analysis. *Critical path analysis* is a systematic effort to identify relationships between mission-critical applications, processes, and operations and all the necessary supporting elements, both physical and technological. For example, an online store relies on internet access, computer hardware, electricity, temperature control, storage facilities, etc.

When critical path analysis is performed properly, a complete picture of the interdependencies and interactions necessary to sustain the organization is produced. The first step in designing a secure IT infrastructure is providing security for the organization's and its computers' basic requirements. These basic requirements include electricity, environmental controls (in other

words, a building, air conditioning, heating, humidity control, and so on), and water/sewage.

While examining critical paths, it is also important to evaluate completed or potential technology convergence. *Technology convergence* is the tendency for various technologies, solutions, utilities, and systems to evolve and merge over time. Often, this results in multiple systems performing similar or redundant tasks or one system taking over the features and abilities of another. Although, in some instances, this can result in improved efficiency and cost savings, it can also represent a single point of failure and become a more valuable target for malicious actors and intruders. For example, if voice, video, building control, storage (i.e., network-attached storage [NAS]), and productivity traffic all share a single connection path rather than individual paths, a single act of sabotage to the main connection is all that is required for intruders or thieves to sever external communications.

Security staff should participate in site and facility design considerations. Otherwise, many important aspects of physical security essential for the existence of logical security may be overlooked. With security staff involved in the physical facility design, you can be assured that your long-term security goals as an organization will be supported not just by your policies, personnel, and electronic equipment, but also by the building itself.

A secure facility plan is based on a layered defense model. Only with overlapping layers of physical security can a reasonable defense be established against would-be intruders. Physical security should be thought of as establishing an obstacle course or gauntlet that attackers have to attempt to work their way through. Thus, security mechanisms are positioned to operate in series rather than in parallel to optimize the difficulty of breaching the protective infrastructure.

Site Selection

Site selection should be based on the security needs of the organization. Cost, location, and size are important, but addressing security requirements should always take precedence.

Securing assets depends largely on-site security, which involves numerous considerations and situational elements. Site location and construction are

crucial in the overall site selection process.

Proximity to other buildings and businesses is a crucial consideration. What attention do they draw, and how does that affect your operation or facility? If a nearby business attracts too many visitors, generates noise, causes vibrations, or handles dangerous materials, they could harm your employees or buildings. Proximity to emergency-response personnel is another issue to consider.

At a minimum, ensure that the building is designed to withstand local extreme weather conditions and that it can deter or fend off most overt break-in attempts. Vulnerable entry points such as windows and doors tend to dominate such analysis. Still, you should also evaluate objects (trees, shrubs, planters, columns, storage buildings, or other human-made items) that can obscure break-in attempts.

Does your organization need to be easily accessed and thus clearly visible? Or would it be a better design not to stand out? *Industrial camouflage* is the attempt to mask or hide a facility's actual function, purpose, or operations by providing a façade presenting a believable or convincing alternative. For example, a data center may present itself as a food-packing facility.

Facility Design

The top priority of security should always be the protection of the life and safety of personnel. To that end, be sure that all facility designs and physical security controls are in compliance with all applicable laws and regulations. These may include health and safety requirements, building codes, labor restrictions, and more. In the United States, some common regulations to follow in regard to facility security are guidelines and requirements from the Occupational Safety and Health Administration (OSHA) and the Environmental Protection Agency (EPA). For most organizations, having a facility security officer to assist with the design, implementation, management, and oversight of facility security may be worthwhile.

Important issues to consider include combustibility, fire rating, construction materials, load rating, placement, and control of items such as walls, doors, ceilings, flooring, HVAC, power, water, sewage, gas, and so on. Forced intrusion, emergency access, resistance to entry, direction of entries and

exits, use of alarms, and conductivity are other important aspects to evaluate. Every element within a facility should be evaluated in terms of how it could be used for and against the protection of the IT infrastructure and personnel (for example, positive flows of air and water from inside a facility to outside its boundaries).

There's also a well-established school of thought on “secure architecture” that's often called *Crime Prevention Through Environmental Design (CPTED)*. First-generation CPTED addresses facility design, landscaping, entrance concepts, campus layouts, lighting, road placement, and traffic management of vehicles and those on foot, while Second-generation CPTED addresses social cohesion. community culture, connectivity, and threshold capacity.

The core principle of CPTED is that the design of the physical environment can be managed, manipulated, and crafted with the intention to create behavioral effects or changes in people present in those areas that result in a reduction of crime as well as a reduction of the fear of crime. Just think of a dark back alley with sunken doorways and several overflowing trash dumpsters; then compare that to a well-lit street with a broad sidewalk with attractive storefronts. Notice the feelings you have about those locations just by thinking about them. CPTED design-guided locations have an amazing but subtle effect on people's behaviors as well as their perceptions of a location.

CPTED has numerous recommendations and suggestions for improving facility design for security purposes, such as the following:

- Keep planters under 2.5 feet tall—this prevents them from being used to hide behind or as a step to reach a window.
- Keep decorative elements small or far away from the building.
- Locate the data center at the core of the building.
- Provide benches and tables to encourage people to sit and look around; they provide automatic surveillance.
- Mount cameras in full view to act as a deterrent.
- Keep entrances open and clear (i.e., without obstacles like trees or columns) to maintain visibility.

- Keep the number of entrances to a minimum and close off doorways during evenings or weekends when fewer workers are present.
- Provide parking for visitors near the entrance.
- Make delivery access driveways and entrances less visible or noticeable to the public—for example, by positioning them on the back of the building and requiring an alternate road.

First-generation CPTED has four principles: access control, natural surveillance, image and milieu, and territorial control.

Access control is the subtle guidance of those entering and leaving a building through the placement of entranceways, the use of fences and bollards, and the placement of lights. The idea here is to make the entrance point to a building look like an entrance point without having to resort to giant signs saying, “Enter Here!” This can also extend internally by creating security zones to distinguish the general access areas from those of higher security that require certain classifications or job responsibilities to enter. Those areas of the same access level should be open, inviting, and easy to move around in, but those areas that are restricted or closed off should seem more difficult to access and require more effort and intention of the individual to access.

Natural surveillance is any means to make criminals feel uneasy through the increasing opportunities for them to be observed. This can be accomplished by an open and obstacle-free outside area, especially around entrances, with clear lines of sight. This can be further increased by encouraging workers and even the public to loiter around the area by providing a pleasing landscape (not directly against the buildings) with plenty of seating. Walkways and stairways should be open so that others nearby can easily see if someone is present. All areas should be very well lit, especially at night.

Image refers to the visual elements and aesthetics of an environment. A well-maintained, aesthetically pleasing space tends to project a positive image. This positive image can influence people's behavior and perceptions, making them more likely to engage positively with the environment. Conversely, poorly maintained or neglected spaces may project a negative image, potentially attracting criminal activity. *Milieu* encompasses the

broader environment or setting, including the overall ambiance and character of a place. It considers factors such as lighting, landscaping, signage, and the general “feel” of the surroundings. A positive milieu can contribute to a sense of safety and community, whereas a negative or hostile milieu may contribute to feelings of insecurity and vulnerability.

Territorial control is the attempt to make the area feel like an inclusive, caring community. The area should be designed so that it looks cared for and respected, and that it is actively being defended. This can be accomplished with decorations, flags, lighting, landscaping, presentations of company logos, clearly visible building numbers, decorative sidewalks, and other architectural features. This approach may cause intruders to feel like they don't belong and that their activities would be at a higher risk of being detected.

Second-generation CPTED has four principles: social cohesion, community culture, connectivity, and threshold capacity.

Social cohesion refers to the level of connectedness and solidarity within a community. It involves fostering positive relationships among community members. A cohesive community is more likely to be vigilant, look out for one another, and collectively address safety concerns. Second-generation CPTED recognizes the importance of social cohesion in creating a supportive environment that deters crime.

Understanding and respecting the unique *community culture* is essential in second-generation CPTED. This includes considering the values, traditions, and norms that shape the community's identity. Design interventions should align with the community's culture to ensure they are well received and effectively integrated. Respecting cultural diversity contributes to a sense of ownership and pride among community members, fostering a safer and more inclusive environment.

Connectivity involves creating physical and social links within a community. This includes designing spaces that facilitate interaction and communication among residents. Well-connected neighborhoods with clear pathways, parks, and communal spaces promote a sense of belonging and discourage criminal activities by increasing visibility and natural surveillance.

Threshold capacity refers to the ability of a community or neighborhood to absorb and respond to external influences while maintaining its stability and security. Considering the threshold capacity involves assessing how various changes, such as new developments or social programs, might impact the community. Understanding and respecting the threshold capacity help prevent unintended negative consequences that could undermine the safety and well-being of the community.

The International CPTED Association is an excellent source for information on this subject, as is Oscar Newman's book *Creating Defensible Space,* published by the U.S. Department of Housing and Urban Development's Office of Policy Development and Research.

The use of CPTED does not replace the use of actual facility hardening, such as locked doors, security guards, fences, and bollards. However, combining traditional physical barriers and CPTED strategies can provide preventive, detection, and deterrent security.

Implement Site and Facility Security Controls

The grouping of controls named “physical” should probably be called “facility” instead since the controls for protecting a facility include policies, personnel management, computer technology, and physical barriers. So, just calling this grouping physical is not as accurate as it could be, but physical is the accepted terminology.

Administrative physical security controls include facility construction and selection, site management, building design, personnel controls, awareness training, and emergency response and procedures. *Technical physical security controls* include building access controls; intrusion detection; alarms; security cameras; monitoring; heating, ventilation, and air-conditioning (HVAC) power supplies; and fire detection and suppression. *Physical controls for physical security* include fencing, lighting, locks, construction materials, person traps, guard dogs, and security guards.

When designing physical security for an environment, focus on the functional order in which controls should be used. A common order of operations is as follows:

1. Deter
2. Deny
3. Detect
4. Delay
5. Determine
6. Decide

Security controls should be deployed so that initial attempts to access physical assets are *deterred* (boundary restrictions accomplish this). If deterrence fails, then direct access to physical assets should be *denied* (for example, locked vault doors). If denial fails, your system needs to *detect* intrusion (for example, using motion sensors). If the breach is successful, then the intruder should be *delayed* sufficiently in their access attempts to enable authorities to respond (for example, a cable lock on the asset). Security staff or legal authorities should *determine* the cause of the incident or assess the situation to understand what is occurring. Then, based on that assessment, they should *decide* on the response to implement, such as apprehending the intruder or collecting evidence for further investigation.



A cable lock is used to protect smaller devices and equipment by making them more difficult to steal. A cable lock usually isn't an impenetrable security device, since most portable systems are constructed with thin metal and plastic. However, a thief will be reluctant to swipe a cable-locked device, because the damage caused by forcing the cable lock out of the security/lock slot will be obvious when they attempt to pawn or sell the device.

Equipment Failure

Preparing for equipment failure can take many forms. In some non-mission-critical situations, knowing where to purchase replacement parts for a 48-hour replacement timeline is sufficient. In other situations, maintaining on-site replacement parts is mandatory. Remember that the response time in

returning a system to a fully functioning state is directly proportional to the cost involved in maintaining such a solution. Costs include storage, transportation, pre-purchasing, and maintaining on-site installation and restoration expertise. In some cases, keeping replacements on-site is not feasible. Establishing a service-level agreement (SLA) with the hardware vendor is essential for those situations. An SLA clearly defines the response time a vendor will provide during an equipment failure emergency.

Equipment failure is a common cause of a loss of availability. When deciding on strategies to maintain availability, it is often important to understand the criticality of each asset and business process as well as the associated *allowable interruption window (AIW)*, *service delivery objective (SDO)*, and *maximum tolerable downtime/outage (MTD/MTO)* (see Chapters 3 and 18 for more on these concepts). These ranges, boundaries, and objectives help focus on the necessary strategies to maintain availability or at least minimize downtime while optimizing cost efficiency.

Aging hardware should be scheduled for replacement and/or repair. The schedule for such operations should be based on the *mean time to failure (MTTF)* and *mean time to repair (MTTR)* estimates established for each device or on prevailing best organizational practices for managing the hardware life cycle. MTTF is the expected typical functional lifetime of the device given a specific operating environment. Be sure to schedule all devices to be replaced before their MTTF expires. MTTR is the average length of time required to perform a repair on the device. A device can often undergo numerous repairs before a catastrophic failure is expected. An additional measurement is that of the *mean time between failures (MTBF)*. This estimates the time between the first and any subsequent failures. If the MTTF and MTBF values are the same or fairly similar, manufacturers often only list the MTTF to represent both values.

When a device is sent out for repairs, you need to have an alternate solution or a backup device to fill in for the duration of the repair. Often, waiting until a minor failure occurs before a repair is performed is satisfactory, but waiting until a complete failure occurs before replacement is an unacceptable security practice.

Wiring Closets

A *cable plant management policy* defines a facility's physical structure and deployment of network cabling and related devices. A cable plant is the collection of interconnected cables and intermediary devices (such as cross-connects, patch panels, and switches) that establish the physical network. Elements of a cable plant include the following:

- *Entrance facility:* The demarcation point or main distribution frame (MDF) is the entrance point to the building where the cable from the provider connects the internal cable plant.
- *Equipment room:* This is the main wiring closet for the building, often connected to or adjacent to the entrance facility.
- *Backbone distribution system:* This provides wired connections between the equipment and telecommunications rooms, including cross-floor connections.
- *Wiring closet:* This serves the connection needs of a large building's floor or section by providing space for networking equipment and cabling systems. It also serves as the interconnection point between the backbone and horizontal distribution systems. The wiring closet is also known as the *premises wire distribution room, main distribution frame (MDF), intermediate distribution frame (IDF),* and *telecommunications room*.
- *Horizontal distribution system:* This connects the telecommunications room and work areas, often including cabling, cross-connection blocks, patch panels, and supporting hardware infrastructure (such as cable trays, cable hangers, and conduits).

Protected cable distribution or *protective distribution systems (PDSs)* are how cables are protected against unauthorized access or harm. The goals of PDSs are to deter violations, detect access attempts, and otherwise prevent compromise of cables. Elements of PDS implementation can include protective conduits, sealed connections, and regular human inspections. Some PDS implementations require intrusion or compromise detection within the conduits.

Wiring closets or equipment rooms are commonly used to house and manage the wiring for many other important elements of a building,

including alarm systems, circuit breaker panels, telephone punch-down blocks, wireless access points, telephone services, and video systems, including security cameras.

Cable plant security is fundamental. Most of the security for a facility focuses on preventing physical unauthorized access. If an unauthorized intruder gains access to the area, they may be able to steal equipment, pull or cut cables, or even plant a listening device. Thus, the security policy for the building should include a few ground rules, such as the following:

- Never use a wiring closet or equipment room as a general storage area.
- Have adequate locks, which might include biometric elements.
- Keep the area tidy.
- Do not store flammable items in the area.
- Set up video surveillance to monitor activity inside the wiring closet.
- Use a door-open sensor to log entries.
- Do not give keys to anyone except the authorized administrator.
- Perform regular physical inspections of the wiring closet's security and contents.
- Include the entire cable plant in the organization's environmental management and monitoring processes to ensure appropriate environmental control and monitoring, as well as to detect damaging conditions such as flooding or fire.

It is also essential to notify your building management of your cable plant security policy and access restrictions. Doing so will further reduce unauthorized access attempts.

Server Rooms/Data Centers

Server rooms, *data centers*, communications rooms, *server vaults*, and *IT closets* are enclosed, restricted, and protected rooms where your mission-critical servers and network devices are housed. A server room is often configured as a lights-out area, which is generally designed to improve efficiency. A server room is often not optimized for workers but for housing

equipment. Data centers can include gas-based halon-substitute oxygen-displacement fire detection and extinguishing systems, walls with a one-hour minimum fire rating, low temperatures, and little or no lighting (i.e., a lights-out area). Server rooms should be designed to support the optimal operation of the IT infrastructure and to block unauthorized human access or intervention.

Server rooms should be located at the core of the building. Avoid locating the data center on the ground floor, top floor, and basement whenever possible. The server room should also be located away from water, gas, and sewage lines. These pose too large a risk of leakage or flooding, which can cause serious damage and downtime.

For many organizations, their data center and their server room are one and the same. For some organizations, a data center is an external location used to house the bulk of their backend computer servers, data storage equipment, and network management equipment. This could be a separate building near the primary offices, or it could be a remote location. A data center might be owned and managed exclusively by your organization, or it could be a leased service from a data center provider (such as a cloud service provider (CSP) or colocation center). A data center could be a single-tenant configuration or a multitenant configuration.

In many data centers and server rooms, a variety of technical controls are employed as access control mechanisms to manage physical access. These include but are not limited to smart/dumb cards, proximity devices and readers biometrics, intrusion detection systems (IDSs) (focusing on physical intruders), and a design based on in-depth defense.

Smartcards and Badges

Badges, *identification cards*, and *security IDs* are forms of physical identification and/or electronic access control devices. A badge can be as simple as a name tag indicating whether you are a valid employee or a visitor (sometimes called a “dumb card”). Or it can be as complex as a smartcard or token device that employs multifactor authentication (MFA) to verify and prove your identity and provide authentication and authorization to access a facility, specific rooms, or secured workstations. Badges may be color-coded by facility or classification level, and they often include

pictures, magnetic stripes, QR codes or bar codes for optical decoding, smartcard chips, RFID, NFC, and personal details to help a security guard verify identity.

Smartcards are credit card–sized IDs, badges, or security passes with an embedded magnetic stripe, bar code, or integrated circuit chip. They contain information about the authorized bearer that can be used for identification and/or authentication purposes. Some smartcards can even process information or store reasonable amounts of data in a memory chip. Several phrases or terms may be used when referring to a smartcard:

- An identity token containing integrated circuits (ICs)
- A processor IC card
- An IC card with an *ISO 7816 interface* (Figure 10.1)



FIGURE 10.1 A smartcard's ISO 7816 interface

Smartcards are often viewed as a reliable security solution, but they should not be considered complete by themselves. Smartcards represent a "something you have" authentication factor. Like any single security mechanism, smartcards are subject to weaknesses and vulnerabilities. Smartcards can fall prey to physical, logical, Trojan horse, or social engineering attacks. In most cases, a smartcard is used in a multifactor configuration. Thus, theft or loss of a smartcard does not result in easy impersonation. Smartcards can serve dual (or multiple) purposes, such as gaining access to a facility just by waving the card near a wall-mounted reader or gaining access to a computer system by inserting the card into a reader (which is usually followed by a prompt for a personal identification number [PIN] or other authentication factor—i.e., MFA).

Magnetic stripe cards are machine-readable ID cards with a magnetic stripe. Like a credit card, debit card, or ATM card, magnetic stripe cards can retain a small amount of data but are unable to process data like a smartcard. Magnetic stripe cards often function as a type of two-factor control: the card is "something you have" and its PIN is "something you know." However, magnetic stripe cards are easy to copy or duplicate and are insufficient for authentication purposes in a secure environment.

A badge can be used either for identification or for authentication. When a badge is used for identification, it is swiped by a device, and then the badge owner must provide one or more authentication factors, such as a password, passphrase, or biological trait (if a biometric device is used). When a badge is used for authentication, the badge owner provides an ID, username, and so on and then swipes the badge to authenticate.

When an employee is terminated or otherwise departs the organization, badges should be retrieved and destroyed as part of the offboarding process. Facility security may require that each authorized person wear badges in plain view. Badges should be designed with security features to minimize the ability of intruders to replicate or duplicate. Day passes and/or visitor badges should be clearly marked as such with bright colors for easy recognition from a distance, especially for escort-required visitors.

Proximity Devices

In addition to smartcards, proximity devices can be used to control physical access. A *proximity device* can be a passive device, a field-powered device, or a transponder. The proximity device is worn or held by the authorized bearer. When it passes near a proximity reader, the reader device is able to determine who the bearer is and whether they have authorized access.

The *passive proximity device* has no active electronics; it is just a small magnet with specific properties (like antitheft devices commonly found in or on retail product packaging). A passive device reflects or otherwise alters the electromagnetic (EM) field generated by the reader device. This alteration is detected by the reader device, which triggers the alarm, records a log event, or sends a notification.

A *field-powered proximity device* has electronics that activate when the device enters the EM field that the reader generates. Such devices generate

electricity from an EM field to power themselves (such as card readers that only require the access card to be waved within inches of the reader to unlock doors). This is effectively radio-frequency identification (RFID); see Chapter 11, “Secure Network Architecture and Components,” for more.

A *transponder proximity device* is self-powered and transmits a signal received by the reader. This can occur consistently or only at the press of a button (like a garage door opener or car alarm key fob). Such devices may have batteries or capacitors, or may even be solar-powered.



Automatic Request to Exit (AREX) is a security system feature commonly employed in access control systems to automatically signal to unlock a secured door or gate when someone wishes to exit a protected area. This feature enhances security and convenience by automating the exit process. An AREX system typically involves proximity sensors or devices installed near exit points. These devices can include motion detectors, infrared sensors, pressure-sensitive mats, or other technologies that can detect when someone is approaching the exit.

Intrusion Detection Systems

Intrusion detection systems (IDSs) are automated or manual systems designed to detect an attempted physical intrusion, breach, or attack, the use of an unauthorized entry, or the occurrence of some specific event at an unauthorized or abnormal time. Intrusion detection systems used to monitor physical activity may include security guards, automated access controls, motion detectors, and other specialty monitoring techniques. See Chapter 17, “Preventing and Responding to Incidents,” for a discussion of the different type of IDS that is a logical/technical control related to network or host breaches.

Physical intrusion detection systems, also called *burglar alarms*, detect unauthorized activities and notify the authorities (internal security or external law enforcement). The most common type of system uses a simple circuit dry contact switch at entrance points to detect when a door or

window has been opened. Some windows may include an internal wire grid or a surface-mounted foil strip that detects when the glass has been broken. Some systems may even use a light beam–based tripwire mechanism to detect entry into a controlled area. This is similar to the safety mechanism located at the bottom of most automatic garage doors. All of these are examples of perimeter breach detection methods. Most IDSs or burglar alarm systems will include both perimeter breach and internal motion-detection methods (see the later sections “Motion Detectors” and “Perimeter Breach Detection”), which in turn may trigger an authority response or an audible alarm (see the later section “Intrusion Alarms”).

Two aspects of any intrusion detection and alarm system can cause it to fail: how it gets its power and how it communicates. The detection and alarm mechanisms will not function if the system loses power. Thus, a reliable detection and alarm system has a battery backup with enough stored power for at least 24 hours of operation.

If communication lines are cut, an alarm may not function, and security personnel and emergency services will not be notified. Thus, a reliable detection and alarm system incorporates a *heartbeat sensor* for line supervision. A heartbeat sensor is a mechanism by which the communication pathway is either constantly or periodically checked with a test signal. If the receiving station detects a failed heartbeat signal, such as the loss of the constant signal or missing one or two interval checks, the alarm triggers automatically. Both measures are designed to prevent intruders from circumventing the detection and alarm system by cutting power, cutting communication cables, or jamming radio signals.

Motion Detectors

A *motion detector*, or *motion sensor*, is a device that senses movement or sound in a specific area, and it is a common element of intruder detection systems. Many types of motion detectors exist, including the following:

- A *digital motion detector* monitors for significant or meaningful changes in the digital pattern of a monitored area. This is effectively a smart security camera.
- *A passive infrared (PIR) or heat-based motion detector* monitors for significant or meaningful changes in a monitored area's heat levels.

- A *wave pattern motion detector* or *microwave motion detector* transmits a consistent low ultrasonic or high microwave frequency signal into a monitored area and monitors for significant or meaningful changes or disturbances in the reflected pattern.
- A *capacitance motion detector* senses changes in the electrical or magnetic field surrounding a monitored object.
- A *photoelectric motion detector* senses changes in visible light levels for the monitored area. Photoelectric motion detectors are usually deployed in internal rooms with no windows and are kept dark.
- A *passive audio motion detector* listens for abnormal sounds in the monitored area.



"Dual-technology sensors" refer to a type of sensor that combines two different technologies (typically IR motion and microwave motion detection) to enhance the accuracy and reliability of detection. These sensors are designed to minimize false alarms and improve overall performance by leveraging the strengths of multiple technologies.

Perimeter Breach Detection

While motion detection mechanisms can be used to monitor for internal movement, they can also be useful to detect when a perimeter is crossed or breached. Numerous perimeter breach detection technologies may be implemented for this purpose, including contact devices and infrared linear beam sensors.

Contact devices detect the opening of a window or door. Often, these are using a balanced magnetic switch (BMS). These are usually small boxes connected to the frame and a door or window. When the door or window is closed, the two BMS items are close enough to each other to keep an electric circuit open based on a magnet pulling on a metal lever. When the door or window is opened, the switch completes once the magnet pulls far enough away to release the lever. A contact device can be directly

connected to a contact alarm, so that the instant a door or window is opened, an alarm is triggered.

Infrared linear beam sensors can be used to detect when someone or something crosses through a threshold, opening, or a specific area of a room. These are similar to the safety devices located at the bottom of a garage door. If the beam between the transmitter and receiver is blocked by someone walking through the beam, then the sensor notices the beam break. This could result in sounding an alarm, notifying security guards, or triggering a safety device (such as opening or closing a door).

Intrusion Alarms

Whenever an intrusion detector registers a significant or meaningful change in the environment, it triggers an alarm. An *alarm* is a separate mechanism that triggers a deterrent, a repellent, and/or a notification.

- *Deterrent alarms:* Alarms that trigger deterrents may engage additional locks, shut doors, and so on. Such an alarm aims to make further intrusion or attack more difficult.
- *Repellent alarms:* Alarms that trigger repellents usually sound an audio siren or bell and turn on lights. These kinds of alarms are used to discourage intruders or attackers from continuing their malicious or trespassing activities and force them off the premises.
- *Notification alarms:* Alarms that trigger notification are often silent from the intruder/attacker perspective but record data about the incident and notify administrators, security guards, and law enforcement. A recording of an incident can take the form of log files and/or security camera recordings. A silent alarm aims to bring authorized security personnel to the location of the intrusion or attack in hopes of catching the person(s) committing the unwanted or unauthorized acts.

  Alarms are also categorized by location: local, centralized, or auxiliary.

- *Local alarm system:* Local alarm systems must broadcast an audible alarm signal that can be easily heard from a distance. Additionally, they must be protected from tampering. For a local alarm system to be

effective, a security team or guards must be positioned nearby who can respond when the alarm is triggered.

- *Central station system:* The alarm is usually silent locally, but off-site monitoring agents are notified to respond to the security breach. Most residential security systems are of this type. Most central station systems are well-known or national security companies, such as Brinks and ADT. A *proprietary system* is similar to a central station system, but the host organization has its own on-site security staff waiting to respond to security breaches.
- *Auxiliary alarm system:* Auxiliary alarm systems can be added to either local or centralized alarm systems. Emergency services are notified to respond to the incident and arrive at the location when the security perimeter is breached. This can include fire, police, and medical services.

Two or more of these types of intrusion and alarm systems can be incorporated into a single solution.

Secondary Verification Mechanisms

When intrusion detectors, sensors, and alarms are used, *secondary verification mechanisms* should be in place. As the sensitivity of intrusion detection devices increases, false triggers occur more often. Innocuous events such as the presence of animals, birds, bugs, vegetation, trash, or authorized personnel can trigger false alarms. Deploying two or more detection and sensor systems and requiring two or more triggers in quick succession before an alarm is issued may significantly reduce false alarms and increase the likelihood that alarms indicate actual intrusions or attacks.

Security cameras are security mechanisms related to motion detectors, sensors, and alarms. However, a security camera is not an automated detection-and-response system. A security camera usually requires personnel to watch the captured or live video to detect suspicious and malicious activities and to trigger alarms. Security cameras can expand a security guard's effective visible range, increasing the scope of the oversight. A security camera with AI detection capabilities may serve as a primary detection tool, but often, cameras are used as a secondary or

follow-up mechanism that is reviewed after a trigger from a primary detection system occurs.

The same logic used for auditing and audit trails is used for a security camera and recorded events. A visible security camera is a deterrent measure, whereas reviewing recorded events is a detection measure.

Cameras

Video surveillance, video monitoring, closed-circuit television (CCTV), and *security cameras* are all means to deter unwanted activity and create a digital record of the occurrence of events. Cameras should be positioned at exit and entry points. Cameras should also be used to monitor activities around valuable assets and resources as well as to provide additional protection in public areas such as parking structures and walkways.



Closed-circuit television (CCTV) is a security camera system that resides inside an organization's facility and is usually connected to a recording device and monitors for the security guards to view. Most traditional CCTV systems have been replaced by remote-controlled IP cameras (aka security cameras).

Be sure the locations and capabilities of the security cameras are coordinated with the interior and exterior design of the facility. Cameras should be positioned to have clear sight lines of all exterior walls, entrance and exit points, and interior hallways. Security cameras can be overt and obvious to provide a deterrent benefit, or hidden and concealed to provide a detection benefit primarily.

Most security cameras record to local or cloud-based storage. Cameras vary in type, including visible light, infrared, and motion-triggered recording. Some cameras are fixed, whereas others support remote control of automated *pan, tilt, and zoom (PTZ)*.

Some camera systems include a system on a chip (SoC) or embedded components and may be able to perform various specialty functions, such as time-lapse recording, tracking, facial recognition, object detection, or

infrared or color-filtered recording. Such devices may be targeted by attackers, infected by malware, or remotely controlled by malicious actors.

Dummy or decoy cameras can provide deterrence with minimal expense. Many security cameras are network-connectable (i.e., IP cameras), which allows them to be accessed and controlled over a network.

Some cameras or enhanced video surveillance (EVS) systems are capable of object detection, including faces, devices, and weapons. Detection of an object or person could trigger retention of video, notification of security personnel, closing/locking doors, and/or sounding an alarm.

Some cameras are activated through motion recognition. Motion recognition can trigger a retention of video and/or notify security personnel of the event. Some EVSs can even automatically identify individuals and track their motion across the monitored area. This may include gait analysis. *Gait analysis* is the evaluation of the way someone walks as a form of biometric authentication or identification. Each person has a unique walking pattern, which can be used to recognize them. Gait analysis can be used for walking approach authentication as well as intrusion detection. Gait analysis is effectively a biological characteristic that can be used to differentiate between authorized individuals and unauthorized intruders.

Animals, birds, insects, weather, or foliage may fool simple motion recognition or motion-triggered cameras. A secondary verification mechanism should be used to distinguish between a false alarm and an intrusion. Many camera solutions and EVSs can be enhanced using machine learning to improve video monitoring through automation, improved image recognition, and pattern/activity interpretation.

Access Abuses

No matter what form of physical access control is used, a security guard or other monitoring system must be deployed to prevent abuse, such as gaining unauthorized entry. Examples of access abuses of physical access controls include propping open secured doors or fail-safe exits and bypassing locks or access controls. Impersonation and masquerading are using someone else's security ID to gain entry into a facility. Tailgating and piggybacking are means to gain unauthorized entry by exploiting an authorized person. See Chapter 2, “Personnel Security and Risk

Management Concepts," for a discussion of impersonation, masquerading, tailgating, and piggybacking. Detecting abuses like these can be done by creating audit trails, retaining access logs, using security cameras (see the previous "Cameras" section), and using security guards (see the section "Security Guards and Guard Dogs," later in this chapter).

Audit trails and access logs are useful tools even for physical access control. They may need to be created manually by security guards. Or they can be generated automatically if sufficient automated access control mechanisms (such as smartcards and certain proximity devices) are used. The time a subject requests entry, the result of the authentication process, and the length of time the secured gate remains open are important elements to include in audit trails and access logs. In addition to using the electronic or paper trail, consider monitoring entry points with security cameras that enable the comparison of the audit trails and access logs with a visual recording of the events. Such information is critical to reconstruct the events for an intrusion, breach, or attack.

Media Storage Facilities

Media storage facilities should be designed to store blank media, reusable media, and even installation media securely. Whether hard drives, flash memory devices, optical disks, or tapes, media should be protected against theft and corruption. A locked storage cabinet or closet should be sufficient for this purpose, but a safe can be installed if deemed necessary. New blank media should be secured to prevent someone from stealing it or planting malware on it.

Media that is reused, such as thumb drives, flash memory cards, or portable hard drives, should be protected against theft and data remnant recovery. *Data remnants* are the remaining data elements left on a storage device after an insufficient sanitization process is used (see Chapter 5, "Protecting Security of Assets"). Standard deletion or formatting processes clear out the directory structure and mark clusters as available for use but leave the original data in the clusters. A simple un-deletion utility or data recovery scanner can often recover access to these files. Restricting access to media and using secure wiping solutions can reduce this risk.

Installation media must be protected against theft and malware planting. This will ensure that when a new installation needs to be performed, the media is available and safe for use.

Here are some means of implementing secure media storage facilities:

- Store media in a locked cabinet or safe rather than an office supply shelf.
- Have a media librarian or custodian who manages access to the locked media cabinet.
- Use a check-in/checkout process to track who retrieves, uses, and returns media from storage.
- For reusable media, when the device is returned, run a secure drive sanitization or *zeroization* (a procedure that erases data by replacing it with meaningless data such as zeroes) process to remove all data remnants.
- Media can also be verified using a hash-based integrity check mechanism to ensure either that valid files remain valid or that a medium has been properly and fully sanitized to retain no remnants of previous use.



A safe is a movable secured container that is not integrated into a building's construction. A vault is a permanent safe or strongroom that is integrated into a building's construction.

For more security-intensive organizations, placing a security notification label on media may be necessary to indicate its use classification or employ RFID/NFC asset tracking tags on media (see Chapter 11). Higher levels of protection could also include fire, flood, electromagnetic field, and temperature monitoring and protection.

Evidence Storage

Evidence storage is quickly becoming a necessity for all businesses, not just law enforcement–related organizations. A key part of incident response is gathering evidence for root cause analysis (see Chapter 17). As cybercrime events continue to increase, it is important to retain logs, audit trails, and other records of digital events. It may also be necessary to retain image copies of drives or snapshots of virtual machines for future comparison. This may be related to internal corporate investigations or law enforcement–based forensic analysis. In either case, preserving datasets that might be used as evidence is essential to the favorable conclusion to a corporate internal investigation or a law enforcement investigation of cybercrime.

Secure evidence storage is likely to involve the following:

- Using a dedicated storage system distinct from the production network
- Potentially keeping the storage system offline when not actively having new datasets transferred to it
- Blocking internet connectivity to and from the storage system
- Tracking all activities on the evidence storage system
- Calculating hashes for all datasets stored on the system
- Limiting access to the security administrator and legal counsel
- Encrypting all datasets stored on the system

There may be additional security requirements for an evidence storage solution based on your local regulations, industry, or contractual obligations. See Chapter 19, “Investigations and Ethics,” for more.

Work Area Security

The design and configuration of internal security, including work areas and visitor areas, should be considered carefully. There should not be equal access to all locations within a facility. Areas that contain assets of higher value or importance should have more restricted access. For example, anyone who enters the facility should be able to access the restrooms and the public telephone without going into sensitive areas, and only network administrators and security staff should have access to the server room and

wiring closets. Valuable and confidential assets should be located in a facility's center of protection. In effect, you should focus on deploying concentric circles of physical protection. This type of configuration requires increased levels of authorization to gain access to more sensitive areas inside the facility.

Walls or partitions can be used to separate similar but distinct work areas. Such divisions deter casual shoulder surfing or eavesdropping (*shoulder surfing* is the act of gathering information from a system by observing the monitor or the use of the keyboard by the operator). Floor-to-ceiling walls should be used to separate areas with differing sensitivity and confidentiality (where false or suspended ceilings are present, walls should cut these off to provide an unbroken physical barrier between more and less secure areas).

A *clean-desk policy* (or clean-desk-space policy) instructs workers how and why to clean off their desks at the end of each work period. In relation to security, such a policy primarily aims to reduce the disclosure of sensitive information. This can include passwords, financial records, medical information, sensitive plans or schedules, and other confidential materials. If, at the end of each day/shift, a worker places all work materials into a lockable desk drawer or file cabinet, this prevents exposure, loss, and/or theft of these materials.

Each work area should be evaluated and assigned a classification just as IT assets are classified. Only people with clearance or classifications corresponding to the classification of the work area should be allowed access. Areas with different purposes or uses should be assigned different levels of access or restrictions. The more access to assets the equipment within an area offers, the more critical the restrictions that are used to control who enters those areas and what activities they are allowed to perform.

Your facility security design process should support the implementation and operation of internal security. In addition to managing workers in proper workspaces, you must address visitors and visitor control. Should there be an escort requirement for visitors, and what other forms of visitor control should be implemented? In addition to basic physical security tools such as

door locks, person traps, video cameras, written logs, security guards, and RFID ID tags should be implemented.

An example of a secure or restricted work area is the *sensitive compartmented information facility (SCIF)*. An SCIF is often used by government and military agencies, divisions, and contractors to provide a secure environment for highly sensitive data storage and computation. The purpose of an SCIF is to store, view, and update sensitive compartmented information (SCI), which is a type of classified information. An SCIF has restricted access to limit entrance to those individuals with a specific business need and authorization to access the data contained within. This is usually determined by the individual's clearance and SCI approval levels. In most cases, a SCIF is restricted against using or possessing photography, video, or other recording devices in the secured area. An SCIF can be established in a ground-based facility, an aircraft, or a floating platform. It can be a permanent installation or a temporary establishment, and it is typically located within a structure, although an entire structure can be implemented as an SCIF.

Utility Considerations

Reliable operations of IT and continued ability to perform business tasks often depend on consistency in the mundane utilities. The following sections discuss security concerns of power, noise, temperature, and humidity.

Power Considerations

Power supplied by electric companies is not always consistent and clean. Most electronic equipment demands clean power to function properly. Equipment damage from power fluctuations is a common occurrence. Many organizations opt to manage their own power through various means. The first stage or level of power management is using *surge protectors*. However, these only offer protection against power overloads. In the event a spike of power occurs, the surge protector's fuse will trip or blow (i.e., burn out), and all power will be cut off. Surge protectors should be used only when instant termination of electricity will not cause damage to the equipment.

The next level is to use a *power conditioner* or *power-line conditioner*. It is a form of advanced surge protector that is also able to remove or filter line noise.

The third level of power protection is to use an *uninterruptible power supply (UPS)*. A UPS is a type of self-charging battery that can be used to supply consistent, clean power to sensitive equipment. Most UPS devices provide surge protection, power conditioning, and battery-supplied supplemental power. There are two main types of UPSs: double conversion and line interactive. A UPS can also be called a backup UPS or a standby UPS.

A *double conversion UPS* functions by taking power in from the wall outlet, storing it in a battery, pulling power out of it, and then feeding that power to whatever devices are connected. By directing current through its battery, it is able to maintain a consistent, clean power supply to whatever devices are connected to it.

A *line-interactive UPS* has a surge protector, battery charger/inverter, and voltage regulator positioned between the grid power source and the equipment. The battery is not in line under normal conditions. If the grid fails, there is a type of three-position switch that will automatically switch so that power is pulled from the battery through the inverter and voltage regulator to provide power to the equipment. Lower-quality versions of this type of UPS may have a very short moment when power is interrupted. Although most systems should be able to continue operating with this fault, it can be damaging to sensitive devices or cause other equipment to shut down, freeze, or reboot.

The primary purpose of an UPS is the battery-supplied power that can continue to support the operation of electrical devices in the event of power loss or a disconnect from the power grid. A UPS can continue to supply power for minutes or hours, depending on its battery capacity and how much power the equipment attached to it needs (i.e., the load placed on it).

When designing a UPS-based power management solution, consider what systems are critical and thus need continued power versus those that can be allowed to be powered off during any loss of power. This approach can assist with the optimization and distribution of critical power reserves.

Another power option is a large-scale battery backup or a failover battery. This system collects power into a battery but can switch to pulling power from the battery when the power grid fails. Generally, this system is implemented to supply power to an entire building rather than just one or a few devices. Many traditional versions of battery backups were not implemented as a form of UPS, and thus, there was usually a period of time (even if just a moment) of complete power loss to the equipment as the grid source of power failed and a switching event occurred to retrieve power from a battery. Some modern battery backups are implemented more like a UPS so that power is not interrupted. Such battery backups are often associated with solar power or other green or renewable energy solutions. However, they can be used with a grid-only source of power.

The highest level of power protection is the use of *generators*. If maintaining operations for a considerable time despite a brownout or blackout is necessary, on-site electric generators are required. Such generators turn on automatically when a power failure is detected. Most generators operate using a fuel tank of liquid or gaseous propellant that must be maintained to ensure reliability. Electric generators are considered alternate or backup power sources. With sufficient fuel supply, especially if resupply is possible, then a power generator can serve as an alternative power source for a long time.

UPSs should still be used even when a generator is installed to provide continuous alternative power. In this situation, the purpose of the UPS is to provide power long enough to complete a logical shutdown of a system, or until a generator is powered on and provides stable power. It may take a generator several minutes before it is triggered, starts (i.e., turns on), and is warmed up to provide consistent power.

Ideally, power is consistently clean without any fluctuations, but in reality, commercial power suffers from many problems. Here is a list of terms associated with power issues you should know:

- *Fault:* Momentary complete loss of power
- *Blackout:* Prolonged complete loss of power
- *Sag:* Momentary low voltage
- *Brownout*: Prolonged low voltage

- *Spike:* Momentary high voltage
- *Surge:* Prolonged high voltage
- *Inrush:* An initial spike of power usually associated with connecting to a power source
- *Ground:* The wire in an electrical circuit that provides an alternate pathway for electricity to flow safely to the earth (i.e., the ground)

All of these issues can cause problems for electrical equipment. When you're experiencing a power issue, you have to determine where the fault is occurring. If the issue takes place outside your meter, then it is to be repaired by the power company, whereas any internal issues are your responsibility.

Noise

Noise is power interference through disturbance, interruption, or fluctuation. Noise that is not consistent is labeled as *transient noise*. Noise can cause more than just problems with how equipment functions related to its power source; it can also interfere with the quality of communications, transmissions, and playback. Noise generated by electric current, that is, *electromagnetic interference (EMI)*, can affect data transmission that relies on electromagnetic transport mechanisms, such as telephone, cellular, television, audio, radio, and network connections.

Radio-frequency interference (RFI) is another source of noise and interference that can affect many of the same systems as EMI. A wide range of common electrical appliances generate RFI, including fluorescent lights, electrical cables, electric space heaters, computers, elevators, motors, and electric magnets, so it's important to locate all such equipment when deploying IT systems and infrastructure elements.

Protecting your power supply and equipment from noise is essential to maintaining a productive and functioning environment for your IT infrastructure. Steps to take for this kind of protection include providing sufficient power conditioning, establishing proper grounding, using shielded cables, running cables through shielding conduits, switching to fiber-optic cables for networking, and limiting copper cable exposure to EMI and RFI sources.

Temperature, Humidity, and Static

In addition to power considerations, maintaining the environment involves control over the HVAC mechanisms. Rooms intended primarily to house computers should generally be kept between 59 and 89.6 degrees Fahrenheit (15 and 32 degrees Celsius). However, some extreme environments run their equipment 20 degrees Fahrenheit lower or higher than this range. The actual temperature is not as important as keeping devices from reaching a temperature that would cause damage and optimizing temperature related to device performance and humidity management. Some devices may operate more efficiently at higher or lower temperatures. Generally, temperature management is optimized using fans, either directly connected to heat sinks on devices, like CPUs, memory banks, or video cards, or indirectly by being part of their chassis or host storage cabinet (such as a rack-mount cabinet). Fans are used to pull warm/hot air off equipment and out of devices and allow it to be replaced by cooler air.

Hot and cold aisles are a means of maintaining optimum operating temperature in large server rooms. The overall technique is to arrange server racks in lines separated by aisles (Figure 10.2). Then, the airflow system is designed so hot, rising air is captured by air-intake vents on the ceiling, whereas cold air is returned in opposing aisles from either the ceiling or the floor. Thus, every other aisle is hot, then cold.

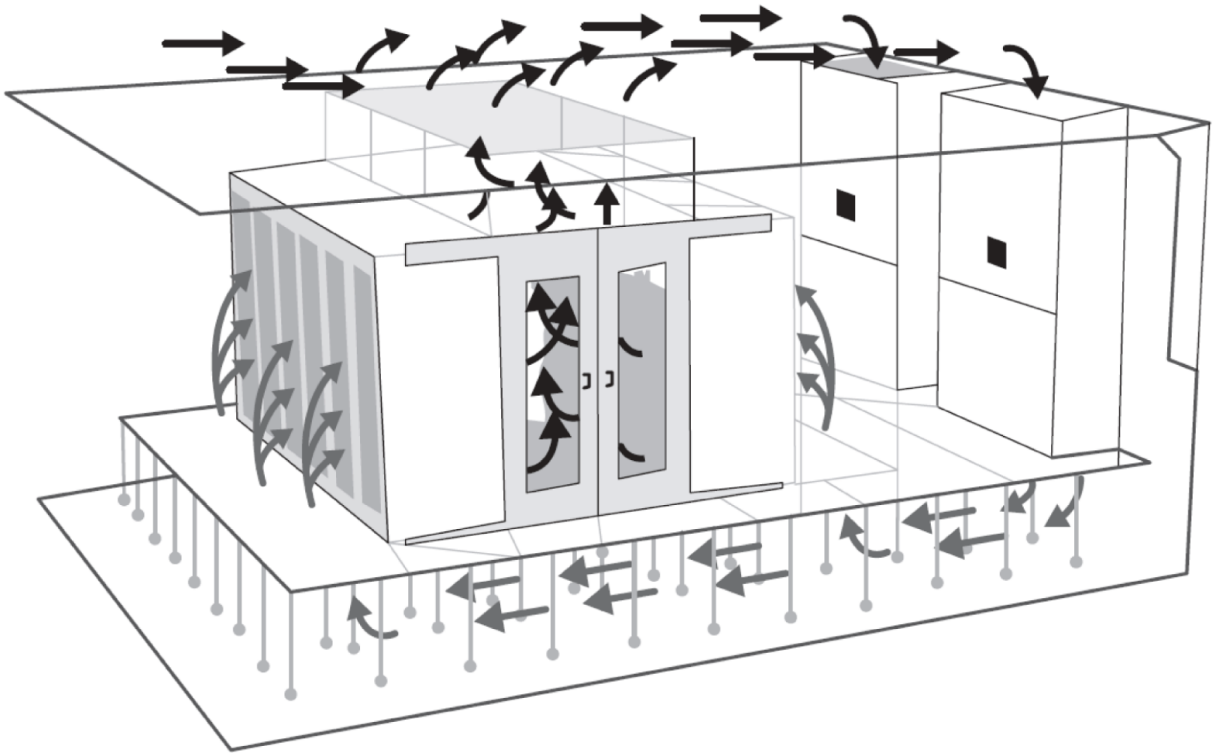


FIGURE 10.2 Hot and cold aisles



A common HVAC-related term is plenum. The plenum consists of boxes and tubes that distribute conditioned air throughout a building. Plenum spaces are the areas of a building designed to contain the HVAC plenum components. Plenum spaces are typically distinct and separate from human-inhabitable spaces within a building. Due to building codes in most countries, anything that is placed into the plenum space must be plenum-rated. This type of fire rating requires that those products produce minimal levels of smoke and/or toxic gases, especially if the building has enclosed spaces that could trap gases. Electrical cables and networking cables are common plenum-rated products.

An important aspect of temperature management is attempting to maintain a stable temperature rather than allowing the temperature to fluctuate up and down. Such heat oscillations can cause the expansion and contraction of

materials. This could cause chip creep (where friction-fit connections work their way out of their sockets) or cracks in soldered connections.

We also recommend that you maintain *positive air pressure* in the data center as well as superior levels of air filtration. These efforts will help reduce dust, debris, microfine particulate matter infiltration, and other contaminants (such as cleaning chemicals or vehicle exhaust). Without such efforts, these unwanted particles can build up over time; dust bunnies can attach to surfaces due to static charges or may cause corrosion.

Additionally, humidity (i.e., relative humidity [RH]) in a computer room should be maintained between 20 and 80 percent. However, some environments allow for RH to be as low as 8 percent and as high as 90 percent. Too much humidity can result in condensation, which causes corrosion. Too little humidity allows for static electricity buildup, which can result in *electrostatic discharge (ESD)*. Even with antistatic carpeting, if the environment has low humidity it is still possible to generate 20,000-volt static discharges from your human body via ESD. Table 10.1 shows that even minimal levels of static discharge can destroy electronic equipment.

TABLE 10.1 Static voltage and damage

Static voltage	**Possible damage**
40	Destruction of sensitive circuits and other electronic components
1,000	Scrambling of monitor displays
1,500	Destruction of data stored on hard drives
2,000	Abrupt system shutdown
4,000	Printer jam or component damage
17,000	Permanent circuit damage



Environmental monitoring is measuring and evaluating the quality of the environment within a given structure. This can focus on general or basic concerns, such as temperature, humidity, dust, smoke, and other debris. However, more advanced systems can include chemical, biological, radiological, and microbiological detectors.

Condition monitoring is monitoring and assessing the operational parameters, performance, and health of machinery, equipment, or systems in real-time or periodically. The primary goal of condition monitoring is to identify any deviations from normal operating conditions that could indicate potential faults, defects, or deterioration. This proactive approach helps predict and prevent equipment failures, minimize downtime, and optimize maintenance strategies.

Water Issues

Your environmental safety policy and procedures should address water issues, such as leakage and flooding. Plumbing leaks are not an everyday occurrence, but when they do happen, they can cause significant damage.

Water and electricity don't mix. If your computer systems come into contact with water, especially while they are operating, damage is sure to occur. Plus, water and electricity create a serious risk of electrocution for nearby personnel. Whenever possible, locate server rooms, data centers, and critical computer equipment away from any water source or transport pipes located in the building. You may also want to install water-detection circuits on the floor (or under the floor with raised flooring data centers) around mission-critical systems. Water-detection circuits will sound an alarm and alert you if water is encroaching upon the equipment.

To minimize emergencies, be familiar with shutoff valves and drainage locations. In addition to monitoring for plumbing leaks, you should evaluate your facility's ability to handle severe rain or flooding in its vicinity. Is the facility located on a hill or in a valley? Is there sufficient drainage? Is there a history of flooding or accumulation of standing water? Is a server room in the basement or on the first floor? Are there water features or landscaping

around the building that might cause flooding or direct heavy rainfall toward and into the building?

Fire Prevention, Detection, and Suppression

Fire prevention, detection, and suppression must not be overlooked. Protecting personnel from harm should always be the most important goal of any security or protection system. In addition to protecting people, fire detection and suppression is designed to keep asset damage caused by fire, smoke, heat, and suppression materials to a minimum.

Standard fire prevention and resolution training involves knowledge of the *fire triangle* (see Figure 10.3). The three corners of the triangle represent fuel, heat, and oxygen. The center of the triangle represents the chemical reaction among these three elements. The purpose of the fire triangle is to illustrate that if you can remove any one of the four items from the fire triangle, the fire can be extinguished. Different suppression mediums address different aspects of the fire:

- Water suppresses the temperature.
- Soda acid and other dry powders suppress the fuel supply.
- Carbon dioxide (CO_2) suppresses the oxygen supply.
- Halon substitutes and other nonflammable gases interfere with the chemistry of combustion and/or suppress the oxygen supply.
- Aqueous film forming foam (AFFF) suppresses temperature and fuel supply.

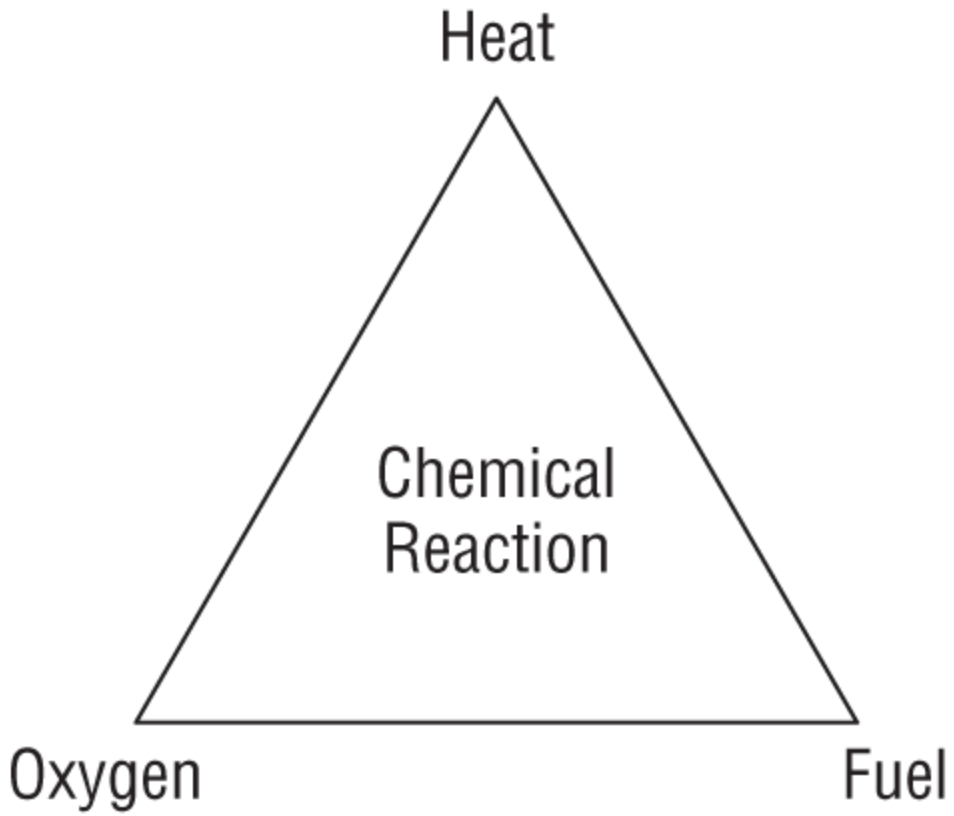


FIGURE 10.3 The fire triangle



Aqueous film forming foam (AFFF) is a type of firefighting foam used to suppress flammable liquid fires. It is a water-based solution containing foaming agents, surfactants, and typically some fluorochemicals. AFFF is designed to quickly spread across the surface of flammable liquids, forming a thin film or barrier that suppresses the release of flammable vapors and prevents the fire from spreading.

When selecting a suppression medium, consider what aspect of the fire triangle it addresses, what this really represents, how effective the suppression medium usually is, and what impact the suppression medium will exert on your environment.

In addition to understanding the fire triangle, you should understand the stages of fire. Fires go through numerous stages, and Figure 10.4 addresses the four most vital stages.

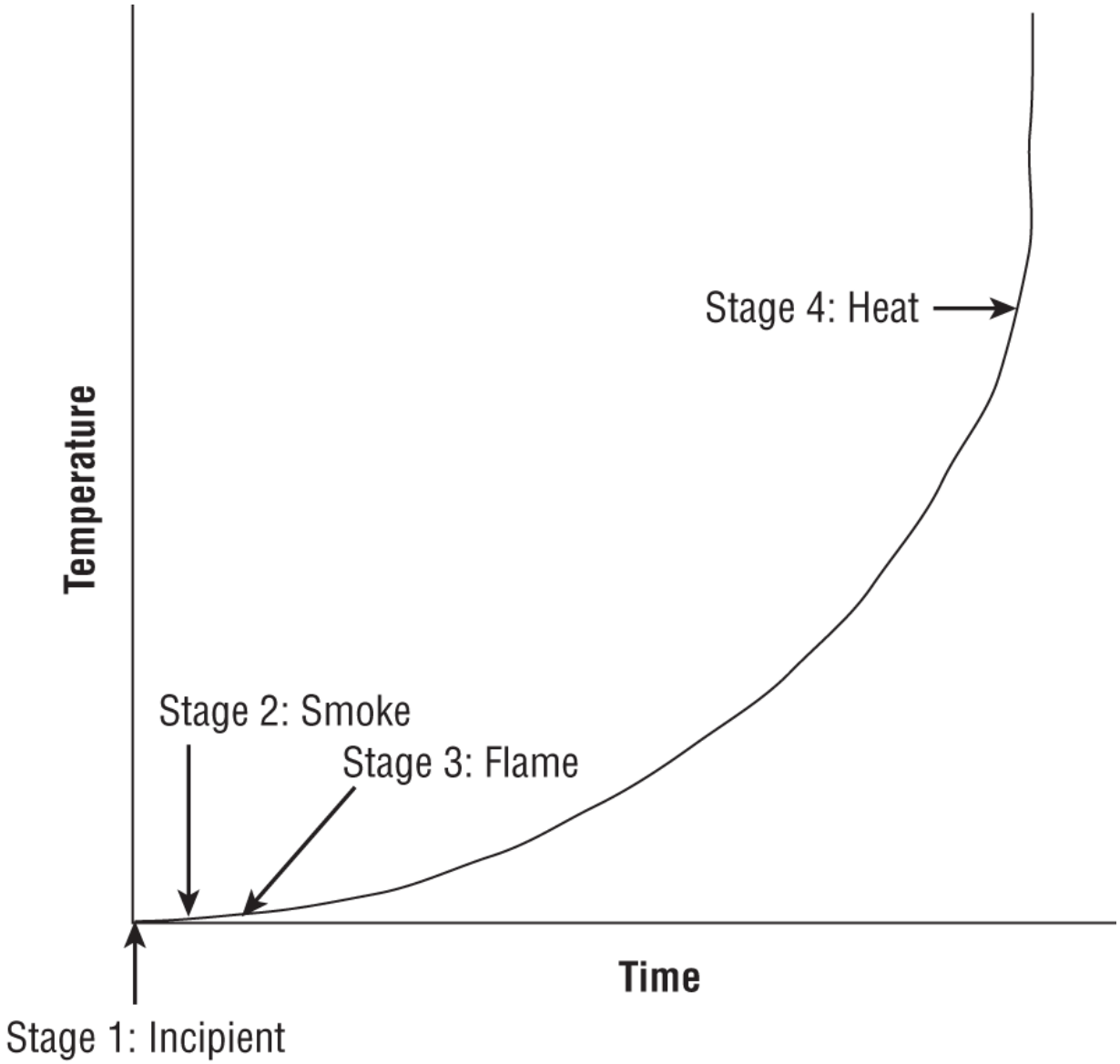


FIGURE 10.4 The four primary stages of fire

Stage 1: The Incipient Stage At this stage, there is only air ionization and no smoke.

Stage 2: The Smoke Stage In Stage 2, smoke is visible from the point of ignition.

Stage 3: The Flame Stage This is when a flame can be seen with the naked eye.

Stage 4: The Heat Stage At Stage 4, the fire is considerably further down the timescale to the point where there is an intense heat buildup

and everything in the area burns.

The earlier a fire is detected, the easier it is to extinguish and the less damage it and its suppression medium(s) can cause.

One of the basics of fire management is proper personnel awareness training. Employees need to be trained in safety and escape procedures. Everyone should be thoroughly familiar with the fire suppression mechanisms in their facility. Everyone should also be familiar with at least two evacuation routes from their primary work area and know how to locate evacuation routes elsewhere in the facility. Typically, evacuation routes are indicated by emergency exit signs, illustrated by maps posted on walls, located in common or central areas (such as near elevators), and defined in personnel training and reference manuals. Personnel should be trained in the location and use of fire extinguishers. Organizations should also preestablish a rendezvous location or safety verification mechanism (such as voicemail) to confirm that all employees escaped a building successfully.

Other items to include in fire or general emergency-response training include cardiopulmonary resuscitation (CPR), emergency shutdown procedures, general first aid, and automated external defibrillator (AED) devices.

Once employees are trained, their training should be tested using drills and simulations. All elements of physical security, especially those related to human life and safety, should be tested on a regular basis. It is mandated by law (in the United States) that fire extinguishers, fire detectors/alarms, and elevators be inspected regularly.



Most fires in a data center are caused by overloaded electrical distribution outlets. A second common cause is improper use of heating devices (such as coffeepots, hot plates, and space heaters) when located near combustible materials (such as paper, cloth, and cardboard).

Fire Extinguishers

If a worker notices a fire before the building detects it, then they may be able to use a handheld fire extinguisher to put out the fire. There are several types of fire extinguishers. Understanding what type to use on various forms of fire is essential to effective fire suppression. If a fire extinguisher is used improperly or the wrong form of fire extinguisher is used, the fire could spread and intensify instead of being quenched. A fire extinguisher may be effective through the first three stages of fire, but is unlikely to be of any use at Stage 4, the heat stage.

Fortunately, local fire regulations and building codes typically dictate the type of fire extinguisher to be present. For most standard office environments, a multiclass extinguisher (likely an ABC) is deployed because it is suitable for the widest range of common fire types in that type of location. Table 10.2 lists common types of fire extinguishers.

TABLE 10.2 Fire extinguisher classes

Class	Type	Suppression material
A	Common combustibles	Water, soda acid (a dry powder or liquid chemical)
B	Liquids	AFFF, CO_2, halon or alternate gas options, soda acid
C	Electrical	CO_2, halon or alternate gas options
D	Metals	Dry powder
K	Cooking media (fats, oil)	Alkaline mixtures (e.g., potassium acetate, potassium citrate, or potassium carbonate) (to cause saponification)



Water and other liquids cannot be used on Class B/K fires because they would vaporize, causing an explosion and spreading the burning liquids all over the area. Water cannot be used on Class C fires because of the potential for electrocution. Oxygen suppression cannot be used on metal fires because burning metal produces its own oxygen.

Fire Detection Systems

Properly protecting a facility from fire requires installing an automated detection and suppression system. There are many types of fire detection systems. *Fixed-temperature detection* systems trigger suppression when a specific temperature is reached. This is the most common type of detector and is present in most office buildings. The potentially visible sprinkler head serves as both the detection and release mechanism. The trigger is usually a metal or plastic component that is in the sprinkler head and melts at a specific temperature. There is also a version with a small glass vial containing chemicals that vaporize to over-pressurize and shatter the container at a specific temperature. This system is inexpensive and reliable, even over long time periods.

Rate-of-rise detection systems trigger suppression when the speed at which the temperature changes reaches a specific level. These are often digital temperature measuring devices, which can be fooled by HVAC heating during winter months and thus are not widely deployed.

Flame-actuated systems trigger suppression based on the infrared energy of flames. This mechanism is fast and reliable but often fairly expensive. Thus, it is often only used in high-risk environments.

Smoke-actuated systems use photoelectric or radioactive ionization sensors as triggers. Either method monitors for light or radiation obstruction or reduction across an air gap caused by particles in the air. It is intended to be triggered by smoke, but dust and steam can sometimes trigger the alarm. The radioactive ionization-based smoke detectors use americium as a source of alpha particles and a Geiger counter to detect the rate of these particles' transmission across the air gap. This element produces such low levels of radiation that a layer of dead skin cells is sufficient to block its transmission.

Incipient smoke detection systems, also known as aspirating sensors, are able to detect the chemicals typically associated with the very early stages of combustion before a fire is otherwise detectable via other means. These devices are even more costly than flame-actuated sensors and are also only used in high-risk or critical environments.

To be effective, fire detectors need to be placed strategically. Don't forget to place them inside dropped ceilings and raised floors, in server rooms, in

private offices and public areas, in HVAC vents, in elevator shafts, in the basement, and so on.

Once a fire-detection device notices the presence of a fire, it typically will trigger the fire alarm. Most fire alarms are loud, piercing beeps or sirens paired with brightly flashing lights. A fire alarm is intended to be obvious, startling, and attention-grabbing. There is usually no mistaking a fire alarm or "not noticing" that it went off. Once a fire alarm occurs, all personnel should follow their safety training and begin to exit the building.

Most fire-detection systems can be linked to fire response service notification mechanisms. When suppression is triggered, such linked systems will contact the local fire response team and request aid using an automated message or alarm.

As for fire suppression mechanisms, they can be based on a water or gas system. Water is common in human-friendly environments, whereas gaseous systems are more appropriate where personnel typically do not reside and generally in non-human-compatible areas, such as engine compartments or equipment panels.

Water Suppression Systems

There are four main types of water suppression systems:

- A *wet pipe system* (also known as a *closed head system*) is always full of water. Water discharges immediately when suppression is triggered.
- A *dry pipe system* contains a compressed inert gas. Once suppression is triggered, the inert gas is released, opening a water valve that causes the pipes to fill and discharge water into the environment moments later.
- A *preaction system* is a variation of the dry pipe system that uses a two-stage detection and release mechanism. The system exists as a dry pipe until the initial stages of a fire (smoke, heat, and so on) are detected, and then the pipes are allowed to fill with water (Stage 1). The water is released only after the sprinkler head activation triggers are triggered by sufficient heat (Stage 2). If the fire is quenched before sprinklers are triggered, pipes can be manually emptied and reset. This

also allows manual intervention (typically via a button mounted on a wall) to stop the release of water before sprinkler release occurs.

- A *deluge system* is a system that uses larger pipes and delivers a significantly larger volume of water compared to a wet pipe system. Also, when one sprinkler head opens, they all open to deluge the area fully with suppressant. Deluge systems are inappropriate for environments that contain electronics and computers.

Preaction systems are the most appropriate water-based system for environments that house both computers and humans together because they provide the opportunity to prevent the release of water in the event of a false alarm or false initial trigger.



The most common cause of failure for a water-based system is human error, such as turning off a water source when a fire occurs or triggering water release when there is no fire.

Gas Discharge Systems

Gas discharge systems use compressed gas to extinguish fire effectively. However, gas discharge systems should not be used in environments in which people are located. Gas discharge systems usually remove the oxygen from the air, thus making them hazardous to personnel. They employ a pressurized gaseous suppression medium, such as carbon dioxide (CO_2), halon, or *FM-200* (a halon replacement, although it too is already slated to be phased out). Benefits of gas-based fire suppression include causing the least damage to computer systems, extinguishing the fire quickly by removing oxygen, and being more effective and faster than a water-based system.

CO_2 is an effective fire suppressant, but it poses a risk to people. If CO_2 leaks into an enclosed space, it can cause asphyxiation at only a 7.5 percent concentration. Fire suppressant use of CO_2 is often at 34 percent or higher concentration. CO_2 is naturally colorless, odorless, and tasteless, so extreme

care must be used when deploying a CO_2 system. There are some additives available to induce an odor. Due to its risks, CO_2 should be implemented only in special circumstances where personnel will not be present and a water-based system is inappropriate, such as engine compartments, generator rooms, around flammable liquids, and large industrial equipment. CO_2 is able to reduce temperatures as well as keep oxygen away from combustion locations.

Halon is an effective fire suppression compound (it starves a fire of oxygen by disrupting the chemical reaction of combustion), but it degrades into toxic gases at 900 degrees Fahrenheit. Also, it is not environmentally friendly (it is an ozone-depleting substance). The 1989 Montreal Protocol (an international agreement) initiated the termination of manufacturing of ozone-depleting substances, including halon. In 1994, the EPA banned the manufacture of halon in the United States and banned importing halon into the country. However, according to the Montreal Protocol, you can obtain halon by contacting a halon recycling facility. The EPA seeks to exhaust existing stocks of halon to take this substance out of circulation, although there are still significant domestic stockpiles of halon.

Due to halon's issues, it is often replaced by a more ecologically friendly and less toxic medium. There are dozens of EPA-approved substitutes for halon. You can also replace halon substitutes with low-pressure water mists, but such systems are usually not employed in computer rooms or electrical equipment storage facilities. A low-pressure water mist is a vapor cloud used to reduce the temperature in an area quickly.

Damage

Addressing fire detection and suppression includes dealing with possible contamination and damage caused by a fire. The destructive elements of a fire include smoke and heat, but they also include the suppression media, such as water or soda acid. Smoke and soot are damaging to storage devices and many computer components. Heat can damage any electronic or computer component. For example, temperatures of 100 degrees Fahrenheit can damage storage tapes, 175 degrees can damage computer hardware (CPU and RAM), and 350 degrees can damage paper products (through warping and discoloration).

Suppression media can cause short circuits, initiate corrosion, or otherwise render equipment useless. All these issues must be addressed when designing a fire response system. Even a small fire might trigger the Incident Response Plan (IRP), Business Continuity Plan (BCP), or Disaster Recovery Plan (DRP).



Don't forget that in the event of a fire, in addition to damage caused by the flames and your chosen suppression medium, fire department members may inflict damage using water hoses and axes while searching for people to rescue and hot spots to extinguish.

Implement and Manage Physical Security

Many types of physical access control mechanisms can be deployed in an environment to control, monitor, and manage access to a facility. These range from deterrents to detection mechanisms. The various sections, divisions, or areas within a site or facility should be clearly designated as public, private, or restricted. Each of these areas requires unique and focused physical access controls, monitoring, and prevention mechanisms. The following sections discuss many such mechanisms that may be used to separate, isolate, and control access to various areas of a site, including perimeter and internal security.

Signage or signs can be used to declare areas off-limits to those who are not authorized, indicate that security cameras are in use, indicate entrances and exits, and disclose safety warnings. Signs are useful in deterring minor criminal activity, establishing a basis for recording events, and guiding people into compliance or adherence with rules or safety precautions. Signs are usually physical displays with words or images, but digital signs and warning banners should also be implemented on both local and remote connections.

If not mandated by regulations, a self-imposed schedule of control testing should be implemented for door locks, fences, gates, person traps, turnstiles, video cameras, and all other physical security controls.

Perimeter Security Controls

The accessibility to the building or campus location is also important. Single entrances are great for providing security, but multiple entrances are better for evacuation during emergencies. What types of roads are nearby, such as residential streets or highways? What means of transportation are easily accessible (trains, highways, airports, shipping)? What about traffic levels throughout the day?

Keep in mind that the need for perimeter security also constrains accessibility. Access and use needs should meld and support the implementation and operation of perimeter security. The use of physical access controls and monitoring personnel and equipment entering and leaving, as well as auditing/logging all physical events, are key elements in maintaining overall organizational security.

Fences, Gates, Turnstiles, and Person Traps

A *fence* is a perimeter-defining device. Fences are used to differentiate between areas under a specific level of security protection and those that aren't. Fencing can include a wide range of components, materials, and construction methods. It can consist of stripes painted on the ground, chain link fences, barbed wire, concrete walls, and even invisible perimeters using laser, motion, or heat detectors. Various types of fences are effective against different types of intruders:

- Fences 3 to 4 feet high deter casual trespassers.
- Fences 6 to 7 feet high are too hard to climb easily and deter most intruders, except determined ones.
- Fences 8 or more feet high with barbed or razor wire strands deter most intruders.

An advanced form of fencing is known as a *perimeter intrusion detection and assessment system (PIDAS)*. A PIDAS is a fence system that has two or three fences used in concert to optimize security. PIDAS fencing is often present around military locations and prisons. Typically, a PIDAS fence has one main tall fence that may be 8 to 20 feet tall. The main fence may be electrified, may have barbed wire/razor wire elements, and/or can include

touch detection technologies. This main fence is then surrounded by an outside fence, which may only be 4 to 6 feet tall. The purpose of this outer fence is to keep animals and casual trespassers from accessing the main fence. This reduces the *nuisance alarm rate (NAR)* or false positives from animals or foliage on interior fences. Additional fences can be located between the main fence and the exterior fence. These additional fences may be electrified or use barbed/razor wire. The space between the fences can serve as a corridor for guard patrols or wandering guard dogs. These corridors are kept free of vegetation.

A *gate* is a controlled exit and entry point in a fence or wall. The deterrent level of a gate must be equivalent to the deterrent level of the fence to sustain the effectiveness of the fence as a whole. Hinges and locking/closing mechanisms should be hardened against tampering, destruction, or removal. When a gate is closed, it should not offer any additional access vulnerabilities. Keep the number of gates to a minimum. They can be monitored by guards. When they're not protected by guards, use of dogs or security cameras is recommended.

A *turnstile* (see Figure 10.5) is a form of gate that prevents more than one person at a time from gaining entry and often restricts movement in one direction. It is used to gain entry but not to exit, or vice versa. A turnstile is basically the fencing equivalent of a secured revolving door. A turnstile can be designed to turn freely to allow easy egress. An ingress turnstile can be implemented with a locking mechanism that requires personnel to provide a code, combination, or credential before it will allow a single person to enter the secured area. A turnstile can be used as a personnel flow control device to limit the direction of travel and the speed of access (i.e., only one person can pass at a time after valid authentication).

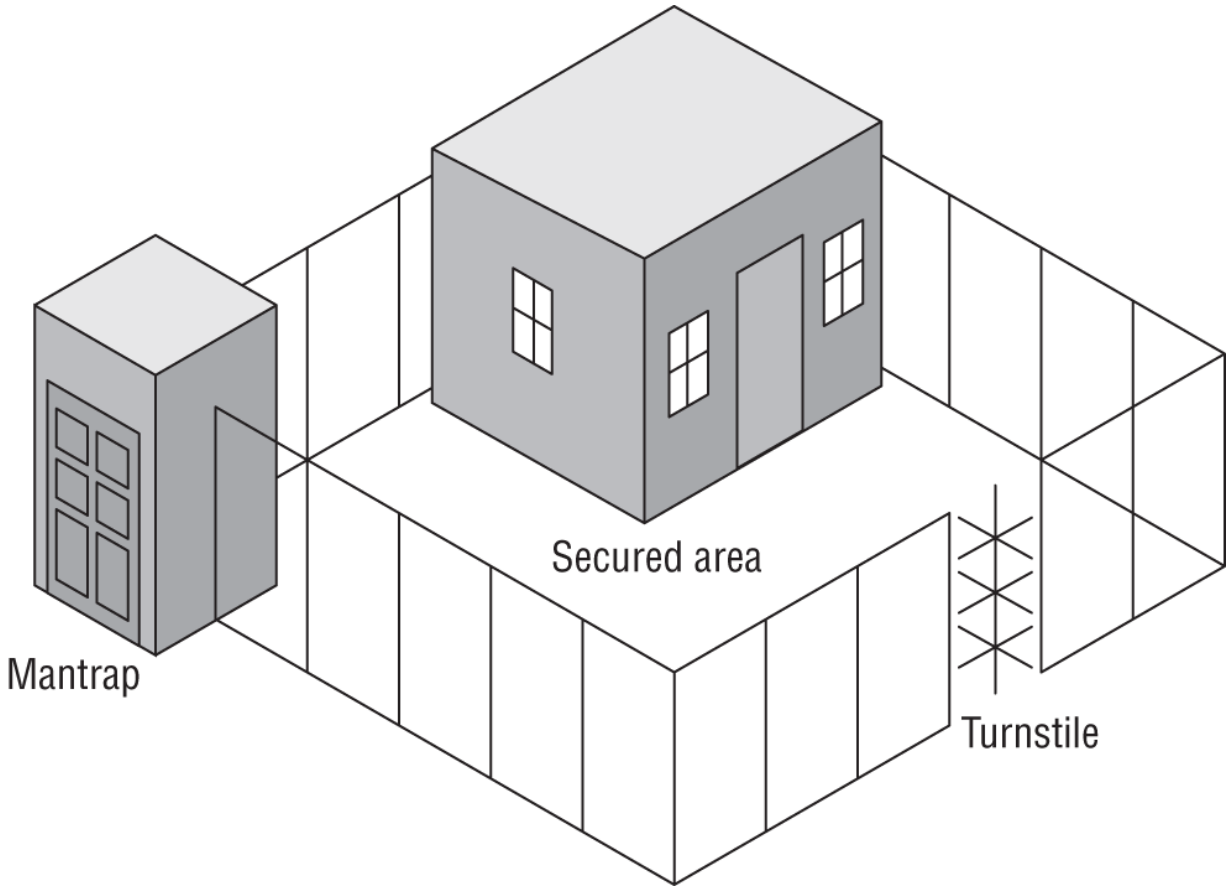


FIGURE 10.5 A secure physical boundary with a person trap and a turnstile

A person trap (also known as a man trap or an *access control vestibule*) is a double set of doors (also shown in Figure 10.5) that is often protected by a guard or some other physical layout that prevents piggybacking and can trap individuals at the discretion of security personnel. The purpose of a person trap is to immobilize a subject until their identity and authentication authority are verified. If a subject is authorized for entry, the inner door opens, allowing entry into the facility or onto the premises. If a subject is not authorized, both doors remain closed and locked until an escort (typically a guard or a police officer) arrives to escort the subject off the property or arrest the subject for trespassing (this is known as a delay feature). Often, a person trap includes a scale to prevent piggybacking or tailgating. Person traps can be used to control entrance into a facility or entrance within a facility to a higher secured area, such as a data center or an SCIF.

Another key element of physical security, especially for data centers, government facilities, and highly secure organizations, is *security bollards*, which prevent vehicles from ramming access points and entrances. These can be permanently fixed in place or automatically rise from their installed base at a fixed time or an alert. They are often disguised as planters or other architectural elements. See the previous discussion of CPTED in the "Facility Design" section.

Barricades, in addition to fencing, are used to control both foot traffic and vehicles. K-rails (often seen during road construction), large planters, zigzag queues, bollards, and tire shredders are all examples of barricades. When used properly, they can control crowds and prevent vehicles from being used to cause damage to your building. Long, straight, and unobstructed vehicle paths should be avoided to prevent the buildup of excessive speed. If generators and fuel storage are present, additional layers of barricade protection may be necessary to prevent tampering or destruction.

Lighting

Lighting is the most commonly used form of perimeter security control, providing the security benefit of deterrence. The primary purpose of lighting is to discourage casual intruders, trespassers, prowlers, or would-be thieves who would rather perform their misdeeds, such as vandalism, theft, and loitering, in the dark. Both interior and exterior lighting should be implemented for security, especially related to parking areas, walkways, and entrances. Exterior lighting should generally be on from dusk until dawn. Interior lighting may be always on, switched manually, or triggered on demand, possibly via motion. Emergency lighting should be implemented in key areas (such as exits and escape routes) and triggered with the loss of power or along with a fire alarm. Lighting is often claimed to be the most commonly deployed physical security mechanism. However, lighting is only a deterrent and not a strong deterrent. It should not be used as the primary or sole protection mechanism except in areas with a low threat level. Your entire site, inside and out, should be well lit. This provides for easy identification of personnel and makes it easier to notice intrusions.

Lighting should not necessarily be used to illuminate the positions of guards, dogs, patrol posts, or other similar security elements. However,

these can be illuminated if knowledge of their presence is to be used as a deterrent. Lighting should be combined with security guards, guard dogs, security cameras, or some other form of intrusion detection or surveillance mechanism. Lighting must not cause a nuisance or problem for nearby residents, roads, railways, airports, and so on. It should also never cause glare or reflective distraction to guards, dogs, and monitoring equipment, which could otherwise aid attackers during break-in attempts. Strong lights used to illuminate a building located on a fence line pointing inward can function as a means to hide intruders. Just think of standing in the dark with someone pointing a flashlight at you—you will be unable to see the other person because the light pointing toward you overpowers your vision.

It is generally accepted as a de facto standard that lighting used for perimeter protection should illuminate critical areas with at least 2 foot-candles of power (which is approximately 2 lumens, or 20 lux). Another common issue for the use of lighting is the placement of the lights. Standards seem to indicate that light poles should be placed the same distance apart as the diameter of the illuminated area created by illumination elements. Thus, if a lighted area is 40 feet in diameter, poles should be 40 feet apart (although it seems to us that placing the poles about 10–20 percent closer is a better option to ensure overlapping of the illuminated areas). This light pole positioning allows for the intersection of lighted areas on the ground, thus preventing an intruder from gaining access under the cover of darkness.

Security Guards and Guard Dogs

All physical security controls, whether static deterrents or active detection and surveillance mechanisms, ultimately rely on personnel to intervene and prevent actual intrusions and attacks. *Security guards* exist to fulfill this need. Guards can be posted around a perimeter or inside to monitor access points or watch detection and surveillance monitors. The real benefit of guards is that they are able to adapt and react to various conditions or situations. Guards can learn and recognize attack and intrusion activities and patterns, can adjust to a changing environment, and can make decisions and judgment calls. Security guards are often an appropriate security control when immediate situation handling and decision-making on-site is necessary.

Guards should perform patrols both internally and externally to look for security violations, unauthorized entities, or other abnormalities throughout the facility and campus grounds. Patrols should be frequent, but at random intervals. This prevents an intruder from observing a pattern of patrols and then timing their break-in accordingly.

Unfortunately, using security guards is not a perfect solution. There are numerous disadvantages to deploying, maintaining, and relying on security guards. Not all environments and facilities support security guards. This may be because of actual human incompatibility or the facility's layout, design, location, and construction. Not all security guards are reliable. Prescreening, bonding, and training do not guarantee that you won't end up with an ineffective or unreliable security guard.

Even if a guard is initially reliable, guards are subject to physical injury and illness, take vacations, can become distracted, are vulnerable to social engineering, and may become unemployable because of substance abuse. In addition, security guards usually offer protection only up to the point at which their life is endangered. Additionally, security guards are usually unaware of the scope of the operations within a facility and are, therefore, not thoroughly equipped to know how to respond to every situation. Though this is considered a disadvantage, the lack of knowledge of the scope of the operations within a facility can also be considered an advantage, because this supports the confidentiality of those operations and thus helps reduce the possibility that a security guard will be involved in the disclosure of confidential information. Finally, security guards are expensive, whether they are employees or are provided by a third-party contractor.

Guard dogs can be an alternative to security guards. They can often be deployed as a perimeter security control. As a detection and deterrent, dogs are extremely effective. However, dogs are costly, require a high level of maintenance (i.e., housing, feeding, health care, training, etc.), and impose serious insurance and liability requirements.

Robot sentries can be used to patrol an area automatically to look for anything out of place. Robot sentries often use facial recognition to identify authorized individuals and potentially identify intruders. Robot sentries can be on wheels or a flying drone (aka uncrewed aerial vehicle [UAV]).

Internal Security Controls

A mechanism to handle visitors is required if a facility is designed with restricted areas to control physical security. Often, an escort is assigned to visitors, and their access and activities are monitored closely. Failing to track the actions of outsiders when they are allowed into a protected area can result in malicious activity against the most protected assets. Visitor control can also benefit from the use of keys, combination locks, badges, motion detectors, intrusion alarms, and more.

Reception can be used as a choke point to block access to unauthorized visitors. The reception area should be segregated from the security areas with locked doors and monitored by security cameras. If a visitor is authorized, then an escort can be assigned to accompany them around the facility. If a valid worker arrives, the receptionist may be able to "buzz" the door open for them. Any unauthorized visitors can be asked to leave, security guards can be brought to bear, or police can be called.

Visitor logs are a manual or automated list of nonemployee entries or access to a facility or location. Employee logs may also be useful for access tracking and verification. Logs of physical access should be maintained. These can be created automatically through the use of smartcards or manually by a security guard. The physical access logs establish context for the interpretation of logical logs. Logs are helpful in an emergency to determine whether everyone has escaped a building safely.

Keys and Combination Locks

Locks keep closed doors closed. They are designed and deployed to prevent access to everyone without proper authorization. A *lock* is a crude form of an identification and authorization mechanism. If you possess the correct key or combination, you are considered authorized and permitted entry. Key-based locks are the most common and inexpensive forms of physical access control devices. These are often known as *preset*, *deadbolt*, or conventional locks. These types of locks are subject to *lock picking*, which is often categorized under a class of lock mechanism attacks called *shimming*. Many conventional locks are also vulnerable to an attack known as bumping. *Bumping* is accomplished using a special bump key that when

properly tapped or bumped causes the lock pins to jump and allows the cylinder to turn.

Programmable or combination locks offer a broader range of control than preset locks. Some programmable locks can be configured with multiple valid access combinations or may include digital or electronic controls employing keypads, smartcards, or cipher devices. For instance, an *electronic access control (EAC) lock* incorporates three elements: an electromagnet to keep the door closed, a credential reader to authenticate subjects and deactivate the electromagnet, and a sensor to reengage the electromagnet when the door is closed. An EAC can monitor the amount of time that a door stays open to trigger a warning buzzer if a door stays open for longer than 5 seconds and trigger an intrusion alarm if the door stays open for longer than 10 seconds (times are examples, not prescriptions).

Locks serve as an alternative to security guards as a perimeter entrance access control device. A gate or door can be opened and closed to allow access by a security guard who verifies your identity before granting access, or the lock itself can serve as the verification device that also grants or restricts entry.

Environmental Issues and Life Safety

An important aspect of physical access control and maintaining the security of a facility is protecting the basic elements of the environment and protecting human life. In all circumstances and under all conditions, the most important aspect of security is protecting people. Thus, preventing harm to people is the most important goal for all security solutions.

Part of maintaining personnel safety is maintaining a facility's basic environment. People can survive for short periods without water, food, power, and air conditioning. But in some cases, the loss of these elements can have disastrous results or be symptoms of more immediate and dangerous problems. Flooding, fires, release of toxic materials, natural disasters, and human-made disasters all threaten human life as well as the stability of a facility. Physical security procedures should focus on protecting human life, restoring the environment's safety, and restoring the utilities necessary for the IT infrastructure to function.

People should always be your top priority. Only after personnel are safe can you consider addressing business continuity. Many organizations adopt *occupant emergency plans (OEPs)* to guide and assist with sustaining personnel safety after a disaster. The OEP guides how to minimize threats to life, prevent injury, manage duress, handle travel, provide safety monitoring, and protect property from damage due to a destructive physical event. The OEP does not address IT issues or business continuity, just personnel and general property. The business continuity plan (BCP) and disaster recovery plan (DRP) address IT and business continuity and recovery issues.

Regulatory Requirements

Every organization operates within a certain industry and jurisdiction. Both of these entities (and possibly additional ones) impose legal requirements, restrictions, and regulations on the practices of organizations that fall within their realm. These legal requirements can apply to the licensed use of software, hiring restrictions, handling of sensitive materials, and compliance with safety regulations.

Complying with all applicable legal requirements is a key part of sustaining security. The legal requirements for an industry and a country (and often also a state and city) must be considered a baseline or foundation on which the remainder of the security infrastructure is built.

Key Performance Indicators of Physical Security

Key performance indicators (KPIs) of physical security should be determined, monitored, recorded, and evaluated. KPIs are metrics or measurements of the operation or failure of various aspects of physical security. The goal of using KPIs is to assess the effectiveness of security efforts. Only with such information can management make informed decisions on altering existing security operations to achieve a higher level of effective security protection. Keep in mind that the overall goal of security is to reduce risk so that the organization's objectives can be achieved cost-effectively.

Here are common and potential examples of physical security KPIs:

- Number of successful intrusions

- Number of successful crimes
- Number of successful incidents
- Number of successful disruptions
- Number of unsuccessful intrusions
- Number of unsuccessful crimes
- Number of unsuccessful incidents
- Number of unsuccessful disruptions
- Time to detect incidents
- Time to assess incidents
- Time to respond to incidents
- Time to recover from incidents
- Time to restore normal conditions after an incident
- Level of organizational impact of incidents
- Number of false positives (i.e., false detection alerts/alarms)

A baseline should be established for each KPI, and a record of each measurement should be maintained. This historical record and baseline are necessary to perform trend analysis and gain an understanding of the performance of the physical security mechanisms. Automatically collected KPIs are often preferred, since they will be recorded reliably. Manual KPI measurements are often more important, but they require attention and focus to collect. Each incident response operation (even if a BCP and DRP level issue), should conclude with a lessons learned phase where/when any additional KPI-related information is gathered or determined and recorded. With reliable KPI assessment, organizations can identify deficiencies, assess improvements, evaluate response measures, and perform return on security investment (ROSI) and cost/benefit analysis for physical security controls.

Summary

In all circumstances and under all conditions, the most important goal of security is protecting people.

Several elements are involved in implementing and maintaining physical security. One core element is selecting or designing the facility to house your IT infrastructure and the operations of your organization. You must start with a plan that outlines the security needs for your organization and develops through a process known as critical path analysis. Additional elements of a secure facility plan are evaluating site selection and visibility requirements and considering facility design elements such as Crime Prevention Through Environmental Design (CPTED).

The security controls implemented to manage physical security can be divided into three groups: administrative (management, managerial, or procedural), technical (logical), and physical. Administrative physical security controls include facility construction and selection, site management, building design, personnel controls, awareness training, and emergency response and procedures. Technical physical security controls include building access controls; intrusion detection; alarms; security cameras; monitoring; heating, ventilation, and air-conditioning (HVAC) power supplies; and fire detection and suppression. Physical controls for physical security include fencing, lighting, locks, construction materials, a person trap, guard dogs, and security guards.

Wiring closets and server rooms are important infrastructure elements that require protection. They often house core networking devices and other sensitive equipment. Protections include adequate locks, smartcards for authentication, proximity devices and readers intrusion detection systems, cameras, surveillance, access control, and regular physical inspections.

An important aspect of physical access control and maintaining a facility's security is protecting the environment's basic elements; this may include the use of media storage facilities, evidence storage, and work area restrictions. Providing clean power sources, minimizing interference, and managing the environment are also important.

Fire detection and suppression must not be overlooked. In addition to protecting people, fire detection and suppression are designed to keep damage caused by fire, smoke, heat, and suppression materials to a minimum, especially regarding the IT infrastructure.

Additional physical security mechanisms to implement and manage include perimeter breach detection, fences, gates, turnstiles, person traps, lighting, security guards, guard dogs, locks, badges, protected cable distribution, motion detectors, intrusion alarms, and secondary verification mechanisms. It is also essential to evaluate regulatory compliance and track KPIs.

Study Essentials

Understand why there is no security without physical security. Without control over the physical environment, no amount of administrative or technical/logical access controls can provide adequate security. If a malicious person can gain physical access to your facility or equipment, they can do just about anything they want, from destruction to disclosure and alteration.

Understand a security facility plan. A secure facility plan outlines your organization's security needs and emphasizes methods or mechanisms to provide security. Such a plan is developed through risk assessment and critical path analysis.

Know about technology convergence. Technology convergence is the tendency for various technologies, solutions, utilities, and systems to evolve and merge over time. Though this can result in improved efficiency and cost savings in some instances, it can also represent a single point of failure and become a more valuable target for malicious actors and intruders.

Understand site selection. Site selection should be based on the security needs of the organization. Cost, location, and size are important, but addressing the requirements of security should always take precedence. The key elements in selecting a site are visibility, composition of the surrounding area, and accessibility.

Know the key elements in designing a facility for construction. A key element in designing a facility for construction is understanding the level of security needed by your organization and planning for it before construction begins.

Know the functional order of controls. These are deter, deny, detect, delay, determine, and decide.

Understand equipment failure. No matter the quality of the equipment your organization chooses to purchase and install, eventually, it will fail. Preparing for equipment failure may include purchasing replacement parts, storing equipment, or having an SLA with a vendor.

Know how to design and configure secure work areas. There should not be equal access to all locations within a facility. Areas that contain assets of higher value or importance should have restricted access. Valuable and confidential assets should be located in the heart or center of protection provided by a facility.

Understand the security concerns of a wiring closet. A wiring closet is where the networking cables for a whole building or just a floor are connected to other essential equipment, such as patch panels, switches, routers, LAN extenders, and backbone channels. Most of the wiring closet security focuses on preventing unauthorized access. If an unauthorized intruder gains access to the area, they may be able to steal equipment, pull or cut cables, or even plant a listening device.

Know about proximity devices and readers. A proximity device can be a passive device, a field-powered device, or a transponder. When it passes near a proximity reader, the reader device is able to determine who the bearer is and whether they have authorized access.

Understand intrusion detection systems. Intrusion detection systems (IDSs) or burglar alarms are automated or manual systems designed to detect an attempted intrusion, breach, or attack; the use of an unauthorized entry point; or the occurrence of some specific event at an unauthorized or abnormal time.

Know about cameras. Video surveillance, video monitoring, closed-circuit television (CCTV), and security cameras are all means to deter unwanted activity and create a digital record of the occurrence of events. Cameras can be overt or hidden; can record locally or to a cloud storage service; may offer pan, tilt, and zoom; may operate in visible or infrared light; may be triggered by movement; and may support time-lapse recording, tracking, facial recognition, gait analysis, object detection, or infrared or color-filtered recording.

Understand security needs for media storage. Media storage facilities should be designed to store blank media, reusable media, and installation media securely. The concerns include theft, corruption, and data remnant recovery. Media storage facility protections include using locked cabinets or safes, using a media librarian/custodian, implementing a check-in/checkout process, and using media sanitization.

Understand the concerns of evidence storage. Evidence storage is used to retain logs, drive images, virtual machine snapshots, and other datasets for recovery, internal investigations, and forensic investigations. Protections include dedicated/isolated storage facilities, offline storage, activity tracking, hash management, access restrictions, and encryption.

Know the common threats to physical access controls. No matter what form of physical access control is used, a security guard or other monitoring system must be deployed to prevent abuse, impersonation, masquerading, tailgating, and piggybacking.

Understand how to control your environment. In addition to power considerations, maintaining the environment involves control over the HVAC mechanisms. Rooms containing primarily computers should be kept at 59 to 89.6 degrees Fahrenheit (15 to 32 degrees Celsius). Humidity in a computer room should be maintained between 20 and 80 percent. Too much humidity can cause corrosion. Too little humidity causes static electricity.

Understand the need to manage water leakage and flooding. Your environmental safety policy and procedures should address water leakage and flooding. Water and electricity don't mix. Locate server rooms and critical computer equipment away from any water source or transport pipes whenever possible.

Understand the importance of fire detection and suppression. Protecting personnel from harm should always be the most important goal of any security or protection system. In addition to protecting people, fire detection and suppression are designed to keep damage caused by fire, smoke, heat, and suppression materials to a minimum, especially in regard to the IT infrastructure.

Know about physical perimeter security controls. Controlled access to a facility can be accomplished using fences, gates, turnstiles, person traps,

bollards, and barricades.

Know about security guards and guard dogs. Guards can be posted around a perimeter or inside to monitor access points or watch detection and surveillance monitors. Guards are able to adapt and react to various conditions or situations and can learn and recognize attack and intrusion activities and patterns, adjust to a changing environment, and make decisions and judgment calls. An alternative to security guards, guard dogs can often be deployed as a perimeter security control and are an extremely effective detection and deterrent.

Understand how to handle visitors in a secure facility. If a facility employs restricted areas to control physical security, then a mechanism to handle visitors is required. Often an escort is assigned to visitors, and their access and activities are monitored closely. Failing to track outsiders' actions when granted access to a protected area can result in malicious activity against the most protected assets.

Understand internal security controls. There are many physical security mechanisms for internal control, including locks, badges, protective distribution systems (PDSs), motion detectors, intrusion alarms, and secondary verification mechanisms.

Know about KPIs of physical security. Physical security's key performance indicators (KPIs) should be determined, monitored, recorded, and evaluated. KPIs are metrics or measurements of the operation or failure of various aspects of physical security.

Written Lab

1. What kind of device helps to define an organization's perimeter and also serves to deter casual trespassing?
2. What is the problem with halon-based fire suppression technology?
3. What kinds of potential issues can an emergency visit from the fire department leave in its wake?
4. What are the three main types of proximity devices and how do they work?

Review Questions

1. Your organization plans on building a new facility to house most on-site workers. The current facility has had numerous security issues, such as loitering, theft, graffiti, and even a few physical altercations between employees and nonemployees. The CEO has asked you to assist in developing the facility plan to reduce these security concerns. While researching options, you discover the concepts of first generation CPTED. Which of the following is not one of its core strategies?

    A. Territorial control

    B. Access control

    C. Natural training and enrichment

    D. Natural surveillance

2. What method is a systematic effort to identify relationships between mission-critical applications, processes, and operations and all the necessary supporting elements when evaluating the security of a facility or designing a new facility?

    A. Log file audit

    B. Critical path analysis

    C. Risk analysis

    D. Taking inventory

3. Which of the following is a true statement in regard to security cameras? (Choose all that apply.)

    A. Cameras should be positioned to watch exit and entry points.

    B. Cameras are not needed around valuable assets and resources or to provide additional protection in public areas such as parking structures and walkways.

    C. Cameras should be positioned to have clear sight lines of all exterior walls, entrance and exit points, and interior hallways.

D. Security cameras should only be overt and obvious to provide a deterrent benefit.

E. Security cameras have a fixed area of view for recording.

F. Some camera systems include a system on a chip (SoC) or embedded components and may be able to perform various specialty functions, such as time-lapse recording, tracking, facial recognition, object detection, or infrared or color-filtered recording.

G. Motion detection or sensing cameras can always distinguish between humans and animals.

4. Your organization is planning on building a new primary headquarters in a new town. You have been asked to contribute to the design process and given copies of the proposed blueprints to review. Which of the following is *not* a security-focused facility or site design element?

   A. Separation of work and visitor areas

   B. Restricted access to areas with higher value or importance

   C. Confidential assets located in the heart or center of a facility

   D. Equal access to all locations within a facility

5. A recent security audit of your organization's facilities has revealed a few items that need to be addressed. A few of them are related to your main data center. But you think at least one of the findings is a false positive. Which of the following does *not* need to be true to maintain the most efficient and secure server room?

   A. It must be optimized for workers.

   B. It must include the use of nonwater fire suppressants.

   C. The humidity must be kept between 20 and 80 percent.

   D. The temperature must be kept between 59 and 89.6 degrees Fahrenheit.

6. A recent security policy update has restricted the use of portable storage devices when they are brought in from outside. As compensation, a media storage management process has been

implemented. Which of the following is *not* a typical security measure implemented in relation to a media storage facility containing reusable, removable media?

A. Employing a media librarian or custodian

B. Using a check-in/checkout process

C. Hashing

D. Using sanitization tools on returned media

7. The company's server room has been updated with raised floors and MFA door locks. You want to ensure that updated facility is able to maintain optimal operational efficiency. What is the ideal humidity range for a server room?

A. 20–40 percent

B. 20–80 percent

C. 80–89.6 percent

D. 70–95 percent

8. You are mapping out the critical paths of network cables throughout the building. Which of the following items do you need to make sure to include and label on your master cabling map as part of crafting the cable plant management policy? (Choose all that apply.)

A. Person trap

B. Entrance facility

C. Equipment room

D. Fire escapes

E. Backbone distribution system

F. Telecommunications room

G. UPSs

H. Horizontal distribution system

I. Loading dock

9. What is the best type of water-based fire suppression system for a computer facility?

    A. Wet pipe system

    B. Dry pipe system

    C. Preaction system

    D. Deluge system

10. Your company has a yearly fire detection and suppression system inspection performed by the local authorities. You start up a conversation with the lead inspector and they ask you, “What is the most common cause of a false positive for a water-based fire suppression system?” What do you answer?

    A. Water shortage

    B. People

    C. Ionization detectors

    D. Placement of detectors in drop ceilings

11. A data center has had repeated hardware failures. An auditor notices that systems are stacked together in dense groupings with no clear organization. What should be implemented to address this issue?

    A. Visitor logs

    B. Industrial camouflage

    C. Gas-based fire suppression

    D. Hot aisles and cold aisles

12. Which of the following are a benefit of a gas-based fire suppression system? (Choose all that apply.)

    A. Can be deployed throughout a company facility

    B. Will cause the least damage to computer systems

    C. Extinguishes the fire by removing oxygen

    D. May be able to extinguish the fire faster than a water discharge system

13. When designing physical security for an environment, it is important to focus on the functional order in which controls should be used. Which of the following is the correct order of the six common physical security control mechanisms?

    A. Decide, Delay, Deny, Detect, Deter, Determine

    B. Deter, Deny, Detect, Delay, Determine, Decide

    C. Deny, Deter, Delay, Detect, Decide, Determine

    D. Decide, Detect, Deny, Determine, Deter, Delay

14. Equipment failure is a common cause of a loss of availability. When deciding on strategies to maintain availability, it is often important to understand the criticality of each asset and business process as well as the organization's capacity to weather adverse conditions. Match the term to the definition.

    I. MTTF

    II. MTTR

    III. MTBF

    IV. SLA

    1. Clearly defines the response time a vendor will provide in the event of an equipment failure emergency
    2. An estimation of the time between the first and any subsequent failures
    3. The expected typical functional lifetime of the device given a specific operating environment
    4. The average length of time required to perform a repair on the device
    5. Select the option below that includes only correct matchings between the terms (using Roman numerals) and the definitions (using Arabic numerals).

        A. I - 1, II - 2, III - 4, IV - 3

        B. I - 4, II - 3, III - 1, IV - 2

C. I - 3, II - 4, III - 2, IV - 1

D. I - 2, II - 1, III - 3, IV - 4

15. You have been placed on the facility security planning team. You've been tasked to create a priority list of issues to address during the initial design phase. What is the most important goal of all security solutions?

    A. Prevention of disclosure

    B. Maintaining integrity

    C. Human safety

    D. Sustaining availability

16. While reviewing the facility design blueprints, you notice several indications of a physical security mechanism being deployed directly into the building's construction. Which of the following is a double set of doors that is often protected by a guard and is used to contain a subject until their identity and authentication are verified?

    A. Gate

    B. Turnstile

    C. Person trap

    D. Proximity detector

17. Due to a recent building intrusion, facility security has become a top priority. You are on the proposal committee that will be making recommendations on how to improve the organization's physical security stance. What is the most common form of perimeter security devices or mechanisms?

    A. Security guards

    B. Fences

    C. CCTV

    D. Lighting

18. Your organization has just landed a new contract for a major customer. This will involve increasing production operations at the primary

facility, which will entail housing valuable digital and physical assets. You need to ensure that these new assets receive proper protections. Which of the following is *not* a disadvantage of using security guards?

A. Security guards are usually unaware of the scope of the operations within a facility.

B. Not all environments and facilities support security guards.

C. Not all security guards are themselves reliable.

D. Prescreening, bonding, and training do not guarantee effective and reliable security guards.

19. While designing the security plan for a proposed facility, you are informed that the budget was just reduced by 30 percent. However, they did not adjust or reduce the security requirements. What is the most common and inexpensive form of physical access control device for both interior and exterior use?

A. Lighting

B. Security guard

C. Key locks

D. Fences

20. While implementing a motion detection system to monitor unauthorized access into a secured area of the building, you realize that the current infrared detectors are causing numerous false positives. You need to replace them with another option. What type of motion detector senses changes in the electrical or magnetic field surrounding a monitored object?

A. Wave

B. Photoelectric

C. Heat

D. Capacitance

Chapter 11
Secure Network Architecture and Components

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

✓ **Domain 4.0: Communication and Network Security**

- 4.1 Apply secure design principles in network architectures
  - 4.1.1 Open System Interconnection (OSI) and Transmission Control Protocol/Internet Protocol (TCP/IP) models
  - 4.1.2 Internet Protocol (IP) version 4 and 6 (IPv6) (e.g., unicast, broadcast, multicast, anycast)
  - 4.1.3 Secure protocols (e.g., Internet Protocol Security (IPSec), Secure Shell (SSH), Secure Sockets Layer (SSL)/Transport Layer Security (TLS))
  - 4.1.4 Implications of multilayer protocols
  - 4.1.5 Converged protocols (e.g., Internet Small Computer Systems Interface (iSCSI), Voice over Internet Protocol (VoIP), InfiniBand over Ethernet, Compute Express Link)
  - 4.1.6 Transport architecture (e.g., topology, data/control/management plane, cut-through/store-and-forward)
  - 4.1.8 Traffic flows (e.g., north-south, east-west)
  - 4.1.9 Physical segmentation (e.g., in-band, out-of-band, air-gapped)
  - 4.1.10 Logical segmentation (e.g., virtual local area networks (VLANs), virtual private networks (VPNs), virtual routing and forwarding, virtual domain)
  - 4.1.11 Micro-segmentation (e.g., network overlays/encapsulation; distributed firewalls, routers, intrusion detection system (IDS)/intrusion prevention system (IPS), zero trust)
  - 4.1.12 Edge networks (e.g., ingress/egress, peering)
  - 4.1.13 Wireless networks (e.g., Bluetooth, Wi-Fi, Zigbee, satellite)

- 4.1.14 Cellular/mobile networks (e.g., 4G, 5G)
- 4.1.15 Content distribution networks (CDN)
- 4.1.16 Software defined networks (SDN) (e.g., application programming interface (API), Software-Defined Wide-Area Network, network functions virtualization)
- 4.1.17 Virtual Private Cloud (VPC)

- 4.2 Secure network components
  - 4.2.1 Operation of infrastructure (e.g., redundant power, warranty, support)
  - 4.2.2 Transmission media (e.g., physical security of media, signal propagation quality)
  - 4.2.3 Network Access Control (NAC) systems (e.g., physical, and virtual solutions)
  - 4.2.4 Endpoint security (e.g., host-based)

✓ **Domain 7.0 Security Operations**

- 7.7 Operate and maintain detection and preventative measures
  - 7.7.1 Firewalls (e.g., next generation, web application, network)

This chapter discusses the Open Systems Interconnection (OSI) model as a guiding principle in networking, cabling, wireless connectivity, Transmission Control Protocol/Internet Protocol (TCP/IP) and related protocols, networking devices, and firewalls. To properly implement secure design principles in network architectures, you must fully understand computer communications technologies.

The Communication and Network Security domain deals with topics related to network components (i.e., network devices and protocols)—specifically, how they function and how they are relevant to security. This domain is discussed in this chapter and in Chapter 12, “Secure Communications and Network Attacks.” Be sure to read and study the materials in both chapters to ensure complete coverage of the essential material.

OSI Model

Communications between computers over networks are made possible by protocols. A *protocol* is a set of rules and restrictions that define how data is transmitted over a network medium (e.g., twisted-pair cable, fiber optics, and wireless transmission). The International Organization for Standardization (ISO) developed the Open Systems Interconnection (OSI) Reference Model for protocols in the late 1970s.

History of the OSI Model

The *OSI Reference Model* (more commonly called the *OSI model*) wasn't the first or only attempt to establish a common communications standard. In fact, the most widely used protocol today, TCP/IP (which is based on the Defense Advanced Research Projects Agency [DARPA] model, also known now as the TCP/IP model) was developed in the early 1970s. The OSI model was not developed until the late 1970s (and not formally published as standard ISO 7498 until 1984).

The *OSI model* was developed to establish a common communication structure or standard for all computer systems. The OSI model serves as a conceptual framework, or theoretical model, for how protocols should function in an ideal world on ideal hardware. The OSI model was developed by ISO to facilitate interoperability between different vendors' systems. The OSI model has become a common reference point.

OSI Functionality

The OSI model divides networking tasks into seven layers. Each layer is responsible for performing specific tasks or operations with the ultimate goal of supporting data exchange (i.e., network communication) between two computers. They are referred to by either their name or their layer number (Figure 11.1). The layers are ordered specifically to indicate how information flows through the various levels of communication. Each layer communicates directly with the layer above it as well as the layer below it.

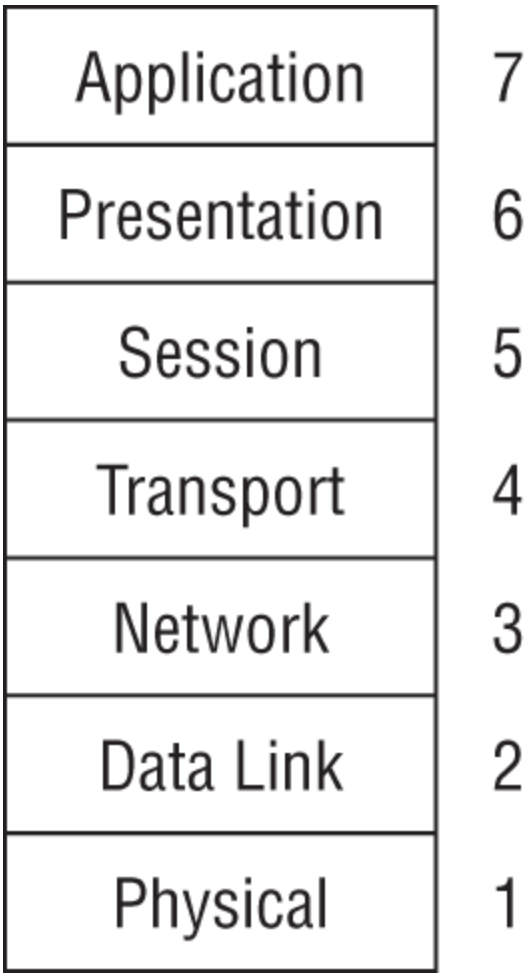


FIGURE 11.1 The OSI model

Encapsulation/Deencapsulation

The OSI model represents a protocol stack, which is a layered collection of multiple protocols (i.e., a multilayered protocol). Communication between protocol layers occurs through encapsulation and deencapsulation. *Encapsulation* is the addition of a header, and possibly a footer, to the data received by each layer from the layer above before it's handed off to the layer below. As the message is encapsulated at each layer, the previous layer's header and payload become the payload of the current layer. The inverse action occurring as data moves up through the OSI model layers from Physical to Application is known as *deencapsulation*. The encapsulation/deencapsulation process is as follows:



The term decapsulation is sometimes used, but the term used by the Internet Engineering Task Force (IETF) is deencapsulation.

1. The Application Layer receives data from software. The Application Layer encapsulates the message by adding information to it. Information is usually added only at the beginning of the message (called a header); however, some layers also add material at the end of the message (called a footer), as shown in Figure 11.2. The Application Layer passes the encapsulated message to the Presentation Layer.

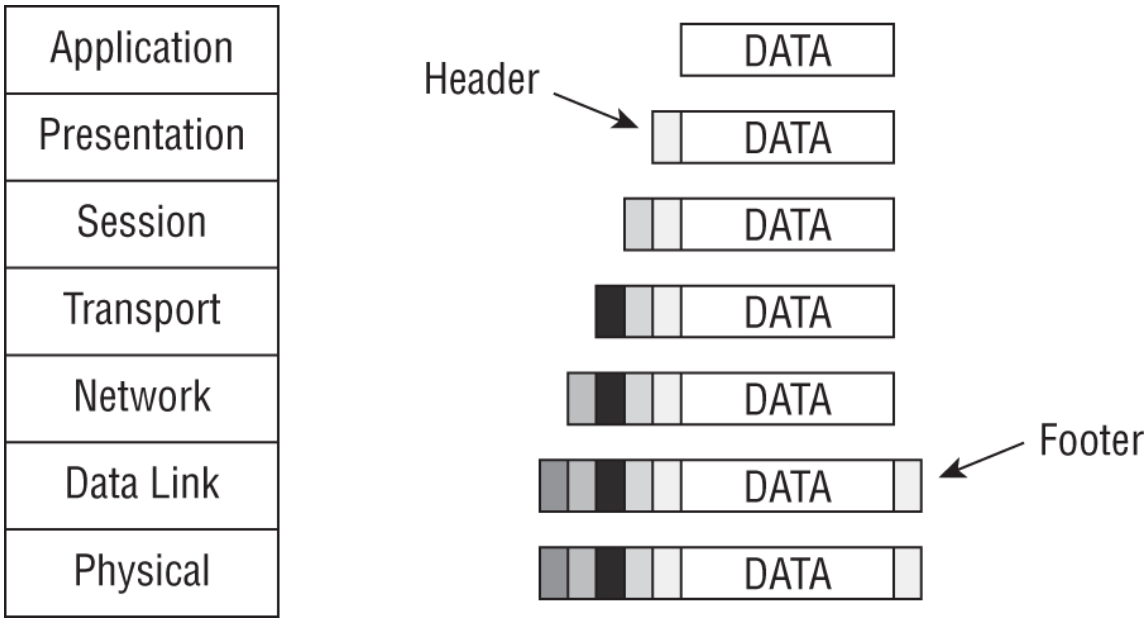


FIGURE 11.2 OSI model encapsulation

2. The process of passing the message down and adding layer-specific information continues until the message reaches the Physical Layer.
3. At the Physical Layer, the message is converted into signals that represent bits and is transmitted over the physical connection.
4. The receiving computer captures the bits from the physical connection, re-creates the message in the Physical Layer, and sends the message up to the Data Link Layer.

5. The Data Link Layer strips its information and sends the message up to the Network Layer.
6. This process of deencapsulation is performed until the message reaches the Application Layer.
7. When the message reaches the Application Layer, the data in the message is sent to the intended software recipient.

The information removed by each layer contains instructions, checksums, and so on that can be understood only by the peer layer that originally added or created the information (see Figure 11.3). This is known as *peer-layer communication*.

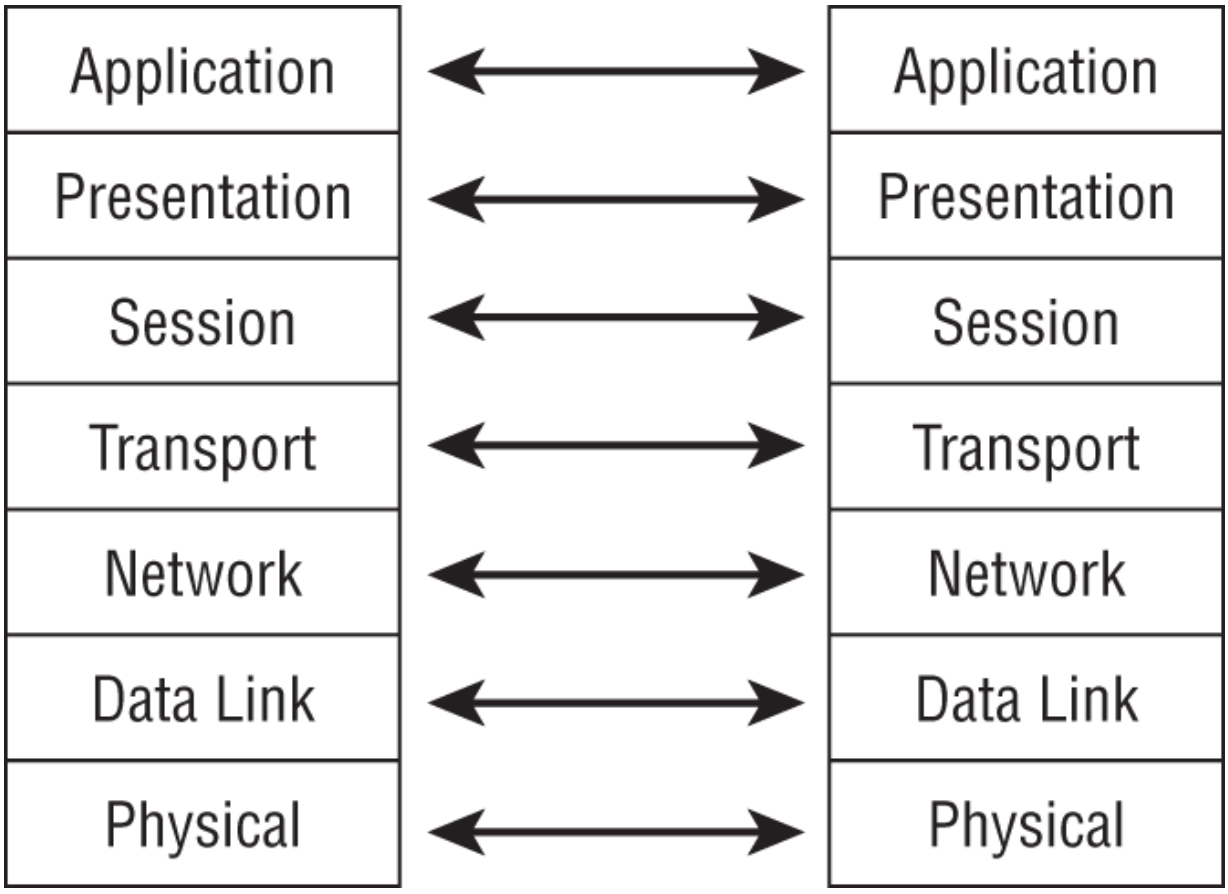


FIGURE 11.3 The OSI model peer layer logical channels

The data sent into the protocol stack at the Application Layer (Layer 7) is encapsulated into a network container. The *protocol data unit (PDU)* is then passed down to the Presentation Layer (Layer 6), which in turn passes it

down to the Session Layer (Layer 5). This network container is known as the PDU at Layers 7, 6, and 5. Once the network container reaches the Transport Layer (Layer 4) it is then called a *segment* (TCP) or a *datagram* (User Datagram Protocol [UDP]). In the Network Layer (Layer 3), it is called a *packet*. In the Data Link Layer (Layer 2), it is called a *frame*. In the Physical Layer (Layer 1), the network container is converted into *bits* for transmission over the physical connection medium. Figure 11.4 shows the label applied to the network container at each layer.

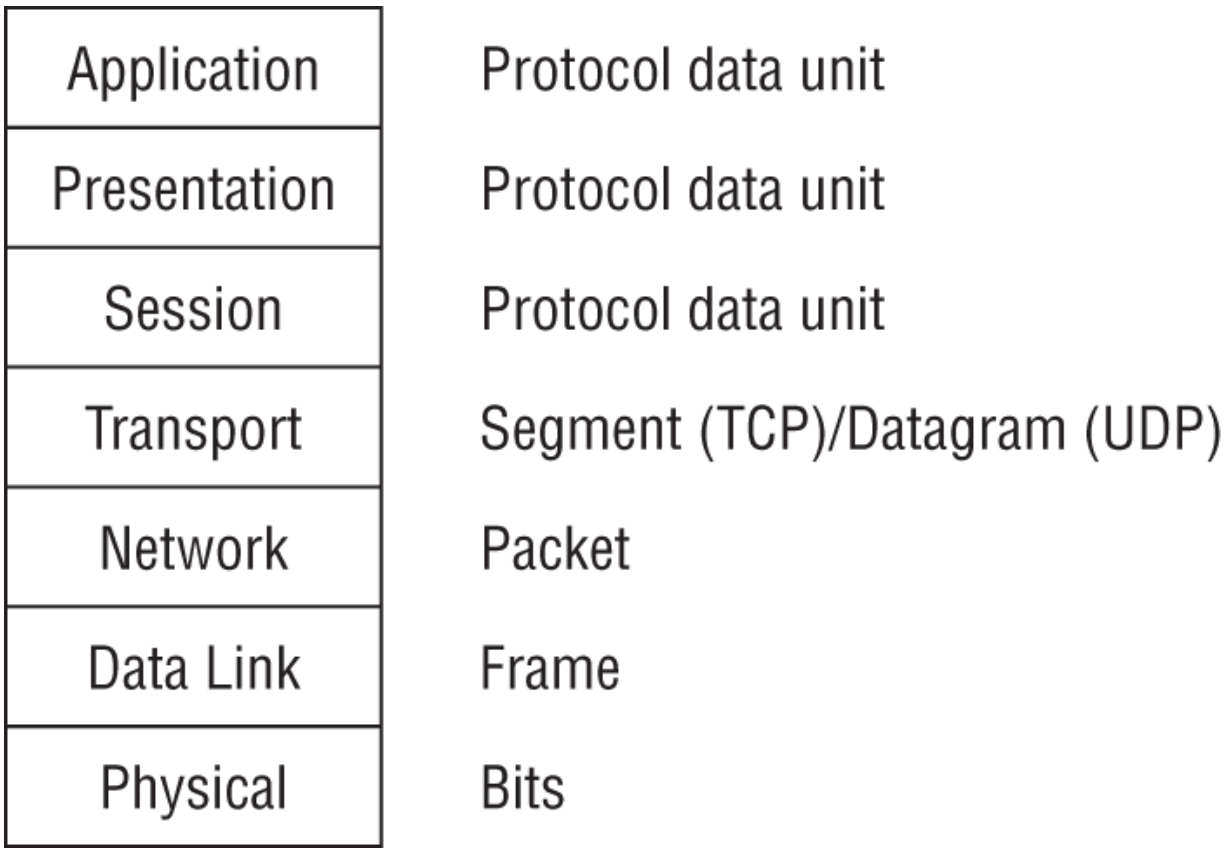


FIGURE 11.4 OSI model layer-based network container names

OSI Layers

Understanding the functions and responsibilities of each layer of the OSI model will help you understand how network communications function, how attacks can be perpetrated, and how security can be implemented to protect network communications.

Remember the OSI

Mnemonics can help you remember the layers of the OSI model in order: Application, Presentation, Session, Transport, Network, Data Link, and Physical (top to bottom). Examples include: “Please Do Not Teach Surly People Acronyms” (Physical Layer up to the Application Layer) and “All Presidents Since Truman Never Did Pot” (Application Layer down to Physical Layer).

Application Layer

The *Application Layer (Layer 7)* is responsible for interfacing user applications, network services, or the operating system with the protocol stack. The software application is not located within this layer; rather, the protocols and services required to transmit files, exchange messages, connect to remote terminals, and so on are found here.

Presentation Layer

The *Presentation Layer (Layer 6)* is responsible for transforming data into a format that any system following the OSI model can understand. It imposes common or standardized structure and formatting rules onto the data. The Presentation Layer is also responsible for encryption and compression.

On TCP/IP networks, there is no actual Presentation Layer. There is no current need to reformat data for network transport, and protocol-stack compression only occurs in concert with some encryption operations. Encryption in relation to network communication can occur in at least five locations:

- Pre-network encryption, where the software encrypts prior to sending the data into the Application Layer
- Transport Layer encryption typically performed by TLS
- VPN encryption, which can occur at Layer 2, 3, or 4 depending on the VPN technology in use (such as L2TP, IPSec, OpenVPN [i.e., TLS VPN], respectively)

- Wireless encryption at the Data Link Layer
- Bulk encryption at the Physical Layer (provided by a device external to the network interface card [NIC])



Many technologies provide encrypted connectivity to individual services or entire systems, such as remote desktop solutions and HTML5. Some entities label these capabilities as VPNs, but they are not VPNs. HTML5 is the encoding language for HTML documents, which is then interpreted by a browser. While it does support capabilities that are similar to a VPN, it is not a true VPN.

Session Layer

The *Session Layer (Layer 5)* is responsible for establishing, maintaining, and terminating communication sessions between two computers. It manages dialog discipline or dialog control (simplex, half-duplex, full-duplex), establishes checkpoints for grouping and recovery, and retransmits PDUs that have failed or been lost since the last verified checkpoint.

On TCP/IP networks, there is no actual Session Layer. Session Layer functions are handled by TCP at the Transport Layer or not at all when UDP is in use.



Communication sessions can operate in one of three different discipline or control modes:

- *Simplex:* One-way communication (as a sender or receiver, but not both)
- *Half-Duplex:* Two-way communication, but only one direction can send or receive data at a time
- *Full-Duplex:* Two-way communication, in which data can be sent and received in both directions simultaneously

Transport Layer

The *Transport Layer (Layer 4)* is responsible for managing the integrity of a connection and controlling the session. The Transport Layer establishes communications between nodes (also known as devices) and defines the rules of a session. Session rules specify how much data each segment can contain, how to verify message integrity, and how to determine whether data has been lost. Session rules are established through a handshaking process. (Please see the section "Transport Layer Protocols," later in this chapter, for the discussion of the SYN/ACK three-way handshake of TCP.)

The Transport Layer establishes a logical connection between two devices and provides end-to-end transport services to support data delivery. This layer includes mechanisms for segmentation, sequencing, error checking, controlling the flow of data, error correction, multiplexing, and network service optimization. The following protocols operate within the Transport Layer:

- Transmission Control Protocol (TCP)
- User Datagram Protocol (UDP)
- Transport Layer Security (TLS)



Since the actual TCP/IP protocol stack does not functionally have a presentation or session layer, as defined by the OSI model, some of the features of those “missing” layers are handled in the Transport Layer by TCP.

Network Layer

The *Network Layer (Layer 3)* is responsible for logical addressing and performing routing. Logical addressing occurs when an address is assigned and used by software or a protocol rather than being provided and controlled by hardware. The Network Layer's packet header includes the source and destination IP addresses.

The Network Layer is responsible for providing routing or delivery guidance, but it is not responsible for verifying guaranteed delivery. The Network Layer also manages error detection and node data traffic (i.e., traffic control).

Non-IP, or Legacy, Protocols

Non-IP protocols are protocols that serve as an alternative to IP at the OSI Network Layer (3). With the dominance and success of TCP/IP, non-IP protocols (i.e., *legacy protocols*) have become the purview of special-purpose networks, such as IPX/SPX, AppleTalk, and NetBEUI. Because non-IP protocols are rare, most firewalls are unable to perform packet header, address, or payload content filtering on those protocols. Also, non-IP protocols can be encapsulated in IP to be communicated across the Internet. Thus, legacy protocols need to be blocked.

A router is the primary network hardware device that functions at Layer 3. Routers determine the best logical path for the transmission of packets based on speed, hops, preference, and so on. Routers use the destination IP address to guide the transmission of packets. A *routed protocol* is a Network Layer protocol whose communications are controlled by routers

and their routing tables. Routers maintain a routing table that includes information about known subnets and the pathway to reach those subnets. The routing table information is used to direct the traffic of a routed protocol to its destination.

Routing Protocols

There are two broad categories of *interior routing protocols*: distance vector and link state. *Distance vector routing protocols* maintain a list of destination networks along with metrics of direction and distance as measured in hops (in other words, the number of routers to cross to reach the destination). *Link state routing protocols* gather router characteristics, such as speed, latency, error rates, and actual monetary cost for use. This information is tabulated to make a next hop routing decision. Common examples of distance vector routing protocols are *Routing Information Protocol (RIP)* and *Interior Gateway Routing Protocol (IGRP)*. Common examples of link state routing protocols are *Open Shortest Path First (OSPF)* and *Intermediate System to Intermediate System (IS-IS)*. There is also a commonly used advanced distance vector routing protocol that replaces IGRP: *Enhanced Interior Gateway Routing Protocol (EIGRP)*.

There is one main category of *exterior routing protocols* that is called path vector. *Path vector routing protocols* make next hop decisions based on the entire remaining path (i.e., vector) to the destination. This is distinct from interior routing protocols, which make next hop decisions based solely on information related to that next immediate hop. Interior routing protocols are myopic, whereas exterior routing protocols are far-sighted. The primary example of a path vector protocol is *Border Gateway Protocol (BGP)*. BGP maintains a routing table of the autonomous systems (AS) across the Internet. An autonomous system (AS) is a collection of IP networks and routers under the control of a single organization that presents a common routing policy to the Internet.

Route security can be enforced by configuring routers to accept route updates only from other authenticated routers. Administrative access to a router should be limited physically and logically to only specific authorized entities. It is also important to keep router firmware updated.

Data Link Layer

The *Data Link Layer (Layer 2)* is responsible for formatting the packet for transmission. The proper format is determined by the hardware, topology, and technology of the network, such as Ethernet (IEEE 802.3).

Part of the processing performed on the network container within the Data Link Layer includes adding the source and destination hardware addresses to the frame. The *hardware address* is the *Media Access Control (MAC) address*, which is a 6-byte (48-bit) binary address written in hexadecimal notation (for example, 00-13-02-1F-58-F5). This address is also known as the *physical address*, the *NIC address*, and the *Ethernet address*. The first 3 bytes (24 bits) of the address is the *organizationally unique identifier (OUI)*, which denotes the vendor or manufacturer of the physical network interface. OUIs are registered with the Institute of Electrical and Electronics Engineers (IEEE), which controls their issuance. The OUI can be used to discover the manufacturer of a NIC through the IEEE website at `http://standards.ieee.org/products-services/regauth/index.html`. The last 3 bytes (24 bits) of the MAC address represent a unique number assigned to that interface by the manufacturer. Some manufacturers will encode information into these final 24 bits, which may represent the make, model, and production run along with a unique value. Thus, some devices (such as mobile devices, IoT equipment, and embedded systems) that use a unique NIC can be identified by their MAC addresses.

Among the protocols at the Data Link Layer (Layer 2) of the OSI model, you should be familiar with Address Resolution Protocol (ARP). See the section “ARP Concerns” later in this chapter.

Network hardware devices that function at Layer 2, the Data Link Layer, are switches and bridges. These devices support MAC-based traffic routing. Switches receive a frame on one port and send it out another port based on the destination's MAC address. MAC address destinations are used to determine whether a frame is transferred over the bridge from one network segment to another.

Physical Layer

The *Physical Layer (Layer 1)* converts a frame into bits for transmission over the physical connection medium, and vice versa for receiving communications.

Network hardware devices that function at Layer 1, the Physical Layer, are NICs, hubs, repeaters, concentrators, and amplifiers. These devices perform hardware-based signal operations, such as sending a signal from one connection port out on all other ports (a hub) or amplifying the signal to support greater transmission distances (a repeater).

TCP/IP Model

The *TCP/IP model* (also called the *DARPA model* or the *DOD model*) consists of only four layers, as opposed to the OSI reference model's seven. The four layers of the TCP/IP model are *Application* (also known as *Process*), *Transport* (also known as *Host-to-Host*), *Internet* (sometimes *Internetworking*), and *Link* (although *Network Interface* and sometimes *Network Access* are also used). Figure 11.5 shows how they compare to the seven layers of the OSI model. The TCP/IP protocol suite was developed before the OSI Reference Model was created.

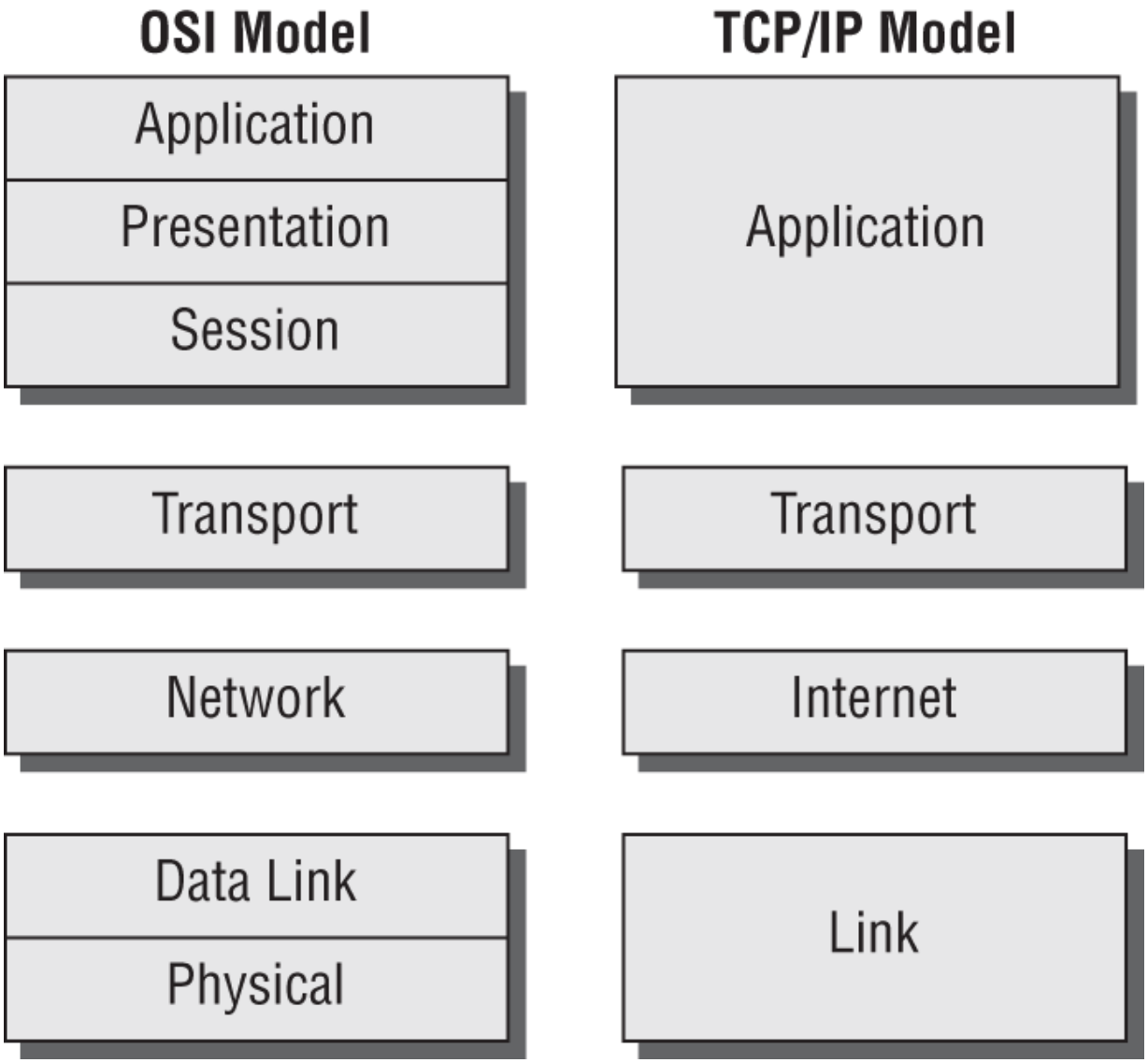


FIGURE 11.5 Comparing the OSI model with the TCP/IP model



Since the TCP/IP model layer names and the OSI model layer names can be used interchangeably, it is important to know which model is being addressed in various contexts. Unless informed otherwise, always assume that the OSI model provides the basis for discussion because it's the most widely used network reference model.

The TCP/IP model was derived directly from the TCP/IP protocol suite or stack comprising hundreds of individual protocols. TCP/IP is a platform-

independent protocol based on open standards. TCP/IP can be found in just about every available operating system, but it consumes a significant amount of resources and is relatively easy to hack, because it was originally designed for ease of use and interoperability rather than for security.

TCP/IP's vulnerabilities are numerous. Improperly implemented TCP/IP stacks in various operating systems are vulnerable to buffer overflows, SYN flood attacks, various denial-of-service (DoS) attacks, fragment attacks, oversized packet attacks, spoofing attacks, adversary-in-the-middle attacks (AitM), hijack attacks, and coding error attacks.

TCP/IP (as well as most protocols) is also subject to passive attacks via monitoring or sniffing. Eavesdropping and other attacks are discussed in more detail at the end of Chapter 12.

Analyzing Network Traffic

Network communications analysis is often an essential function in managing a network. It can be useful in tracking down malicious communications, detecting errors, or resolving transmission problems. However, network eavesdropping may also be used to violate communication confidentiality and/or serve as the information-gathering phase of a subsequent attack.

A *protocol analyzer* is a tool used to examine the contents of network traffic. A protocol analyzer can be a dedicated hardware device or software installed on a typical host system. A protocol analyzer is a frame/packet-capturing tool that can collect network traffic and store it in memory or on a storage device. Once a frame or packet is captured, it can be analyzed either with complex automated tools and scripts or manually. A protocol analyzer may also be called a *sniffer*, *network evaluator*, *network analyzer*, *traffic monitor*, or *packet-capturing utility*. A sniffer is generally a packet- or frame-capturing tool, whereas a protocol analyzer is able to decode and interpret packet/frame contents.

A protocol analyzer usually places the NIC into *promiscuous mode* to see and capture all Ethernet frames on the local network segment. In promiscuous mode, the NIC ignores the destination MAC addresses of Ethernet frames and collects each frame that reaches the interface.

The protocol analyzer can examine individual frames down to the binary level. Most analyzers or sniffers automatically parse out the contents of the header into an expandable outline form. Any configuration or setting can be easily seen in the header details. The payload of packets is often displayed in both hexadecimal and ASCII.

Protocol analyzers typically offer both *capture filters* and *display filters*. A capture filter is a set of rules to govern which frames are saved into the capture file or buffer and which are discarded. A display filter is used to show only those frames from the packet file or buffer that match your requirements.

Protocol analyzers vary from simple raw frame/packet-capturing tools to fully automated analysis engines. There are both open source (such as Wireshark) and commercial (such as Omnipeek, NetWitness, and NetScout) options.

Common Application Layer Protocols

In the Application Layer of the OSI model reside numerous application- or service-specific protocols:

***Telnet*, TCP Port 23** This is a terminal emulation network application that supports remote connectivity for executing commands and running applications but does not support the transfer of files. Telnet should not be used; replace it with SSH.

***File Transfer Protocol (FTP)*, TCP Ports 20 (Active Mode Data Connection)/Ephemeral (Passive Mode Data Connection) and 21 (Control Connection)** This is a network application that supports an exchange of files that requires anonymous or specific authentication. FTP should not be used; replace it with SFTP or FTPS.

Trivial File Transfer Protocol (TFTP), UDP Port 69 This is a network application that supports an exchange of files that does not require authentication. Used to host network device configuration files and can support multicasting. TFTP should not be used.

***Simple Mail Transfer Protocol (SMTP)*, TCP Port 25** This is a protocol used to transmit email messages from a client to an email server and from

one email server to another. Only use if encrypted with TLS to create SMTPS (i.e., STARTTLS, explicit TLS, or opportunistic TLS) over TCP port 587 or implicit SMTPS over TCP port 465.

***Post Office Protocol (POP3),* TCP Port 110** This is a protocol used to pull email messages from an inbox on an email server down to an email client (aka client archiving). Only use if encrypted with TLS to create POPS on TCP port 995.

***Internet Message Access Protocol (IMAP4),* TCP Port 143** This is a protocol used to pull email messages from an inbox on an email server down to an email client. IMAP offers the ability to retrieve only headers from an email server as well as to delete messages directly off the email server (i.e., server archiving). Only use if encrypted with TLS to create IMAPS on TCP port 993.

***Dynamic Host Configuration Protocol (DHCP),* UDP Ports 67 (server) and 68 (client)** DHCP provides for centralized control of TCP/IP configuration settings assigned to systems upon bootup.

***Hypertext Transfer Protocol (HTTP),* TCP Port 80** This is the protocol used to transmit web page elements from a web server to web browsers in cleartext.

***Hypertext Transfer Protocol Secure (HTTPS)* TCP Port 443** This is the TLS-encrypted version of HTTP. (HTTPS with TLS does support use of TCP port 80—but only for server-to-server communications.)

***Line Printer Daemon (LPD),* TCP Port 515** This is a network service that is used to spool print jobs and send print jobs to printers. Consider enclosing in a VPN for use.

***X Window,* TCP Ports 6000–6063** This is a GUI API for command-line operating systems. Consider enclosing it in a VPN for use.

***Network File System (NFS),* TCP Port 2049** This is a network service used to support file sharing between dissimilar systems. Consider enclosing it in a VPN for use.

***Simple Network Management Protocol (SNMP),* UDP Port 161 (UDP Port 162 for Trap Messages)** This is a network service used to collect

network health and status information from a central monitoring station. Use the secure SNMPv3 only.

SNMPv3

Simple Network Management Protocol (SNMP) is a standard network-management protocol supported by most network devices and TCP/IP-compliant hosts. These include routers, switches, WAPs, firewalls, VPNs, printers, and so on. From a management console, you can use SNMP to interact with various network devices to obtain status information, performance data, statistics, and configuration details. Some devices support the modification of configuration settings through SNMP.

Early versions of SNMP relied on plaintext transmission of community strings as authentication. Communities are named collections of network devices. The original default community names were public and private. The latest version of SNMP allows for encrypted communications, as well as robust authentication protection.

UDP port 161 is used by the SNMP agent (that is, the network device) to receive requests, and UDP port 162 is used by the management console to receive responses and notifications (also known as *trap messages*). Trap messages inform the management console when an event or threshold violation occurs on a monitored system.

Transport Layer Protocols

When a connection is established via the Transport Layer, it is done using ports. Since port numbers are 16-digit binary numbers, the total number of ports is 2^16, or 65,536, numbered from 0 through 65,535. Ports allow a single IP address to support multiple simultaneous communications, each using a different port number (i.e., multiplexing over IP). The combination of an IP address and a port number is known as a *socket*.

The first 1,024 of these ports (0–1,023) are called the *well-known ports* or the *service ports*. These ports are reserved for use exclusively by servers.

Ports 1,024 to 49,151 are known as the *registered software ports*. These are ports that have one or more networking software products specifically registered with the Internet Assigned Numbers Authority (IANA) at `http://iana.org`.

Ports 49,152 to 65,535 are known as *random*, *dynamic*, or *ephemeral ports* because they are often used randomly and temporarily by clients as source ports. However, most operating systems allow for any port from 1,024 to be used as a dynamic client source port as long as it is not already in use on that local system.

The two primary Transport Layer protocols of TCP/IP are TCP and UDP. *Transmission Control Protocol (TCP)* is a full-duplex connection-oriented protocol, whereas *User Datagram Protocol (UDP)* is a simplex connectionless protocol.

Transmission Control Protocol (TCP) supports full-duplex communications, is connection-oriented, and employs reliable sessions. TCP is *connection-oriented* because it employs a handshake process between two systems to establish a communication session. The *three-way handshake* process (Figure 11.6) is as follows:

1. The client sends a SYN (synchronize) flagged packet to the server.
2. The server responds with a SYN/ACK (synchronize and acknowledge) flagged packet back to the client.
3. The client responds with an ACK (acknowledge) flagged packet back to the server.

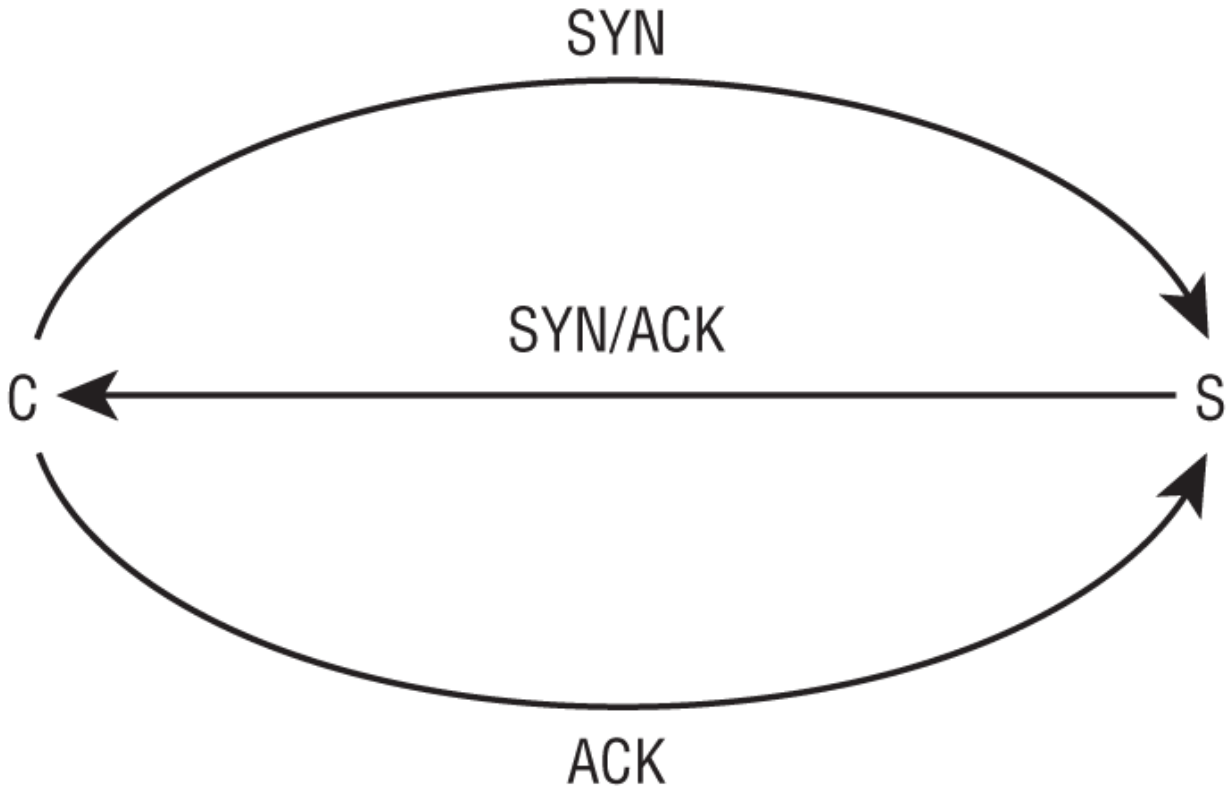


FIGURE 11.6 The TCP three-way handshake

When a communication session is complete, there are two methods to disconnect the TCP session. First, and most common, is the use of FIN (finish) flagged packets to gracefully initiate session shutdown. Second is the use of an RST (reset) flagged packet, which causes an immediate and abrupt session termination.

TCP should be employed when the delivery of data is required. In the event that all packets of a transmission window were not received, no acknowledgment is sent. After a timeout period, the sender will resend the entire transmission window set of packets again. TCP guarantees delivery because it will continue to resend any unacknowledged window of segments until it receives an acknowledgment, it receives an RST, the local application terminates the network communication attempts, or power is removed from the system.

User Datagram Protocol (UDP) also operates at Layer 4 (the Transport Layer) of the OSI model. It is a *connectionless* "best-effort" communications protocol. It offers no standard error detection (other than an optional packet checksum) or correction, does not use sequencing, does

not use flow control mechanisms, does not use a preestablished session, and is considered unreliable. UDP has very low overhead and thus can transmit data quickly. However, UDP should be used only when the delivery of data is not essential. UDP is often employed by real-time or streaming communications for audio and/or video.

Domain Name System

There are three numbering and addressing concepts you should be familiar with:

Domain Name The domain name or computer name is a “temporary” human-friendly convention assigned to an IP address.

IP Address The IP address is a “temporary” logical address assigned over or onto the MAC address.

MAC Address The MAC address, or hardware address, is a “permanent” physical address.

“Permanent” and “Temporary” Addresses

The reason these two adjectives are within quotation marks is that they are not completely accurate. MAC addresses are designed to be permanent physical addresses but often can be changed. When the NIC supports the change, the change occurs on the hardware. When the OS supports the change, the change is only in memory, but it looks like a hardware change to all other network entities (this is known as *MAC spoofing* or *MAC cloning* if duplicating another device's MAC address).

An IP address is temporary because it is a logical address and can be changed at any time, either by DHCP or by an administrator. However, there are instances where systems are statically assigned an IP address. Likewise, computer names or DNS names might appear permanent, but they are logical and thus able to be modified by an administrator.

Domain Name System (DNS) resolves a human-friendly domain name into its IP address equivalent. Then, Address Resolution Protocol (ARP) (see the

later section “ARP Concerns”) resolves the IP address into its MAC address equivalent. It is also possible to resolve an IP address into a domain name via a DNS reverse lookup if a PTR (i.e., pointer) resource record is defined in the domain's zone file. IP addresses are assigned either statically, or dynamically via DHCP.

DNS is the hierarchical naming scheme used in both public and private networks. DNS links IP addresses and human-friendly *fully qualified domain names (FQDNs)* together. An FQDN consists of three main parts:

- *Top-level domain (TLD)*—The `com` in `www.google.com`
- *Registered domain name*—The `google` in `www.google.com`
- *Subdomain(s) or hostname*—The `www` in `www.google.com`

The TLD can be any number of official options, including six of the original seven TLDs—`com`, `org`, `edu`, `mil`, `gov`, and `net`—as well as many newer ones, such as `info`, `museum`, `telephone`, `mobi`, `biz`, and so on. There are also two-letter country variations known as *country codes*. (See `www.iana.org/domains/root/db` for details on current TLDs and country codes.) The seventh original TLD was `int`, for international, which was replaced by the two-letter country codes.

The registered domain name must be officially registered with one of any number of approved domain registrars, such as Network Solutions (`http://networksolutions.com`), `Domains.com`, or IONOS (`http://ionos.com`).

The far-left section of an FQDN can be either a single hostname, such as `www.`, `ftp.`, `blog.`, `images.`, and so on, or a multi-sectioned subdomain designation, such as `server1.group3.bldg5.myexamplecompany.com`.

The total length of an FQDN can't exceed 253 characters (including the dots). Any single section can't exceed 63 characters. FQDNs can only contain letters, numbers, hyphens, and periods. Though not typically shown, there is a dot to the right of the TLD, which represents the root of the entire DNS namespace.

Every registered domain name has an assigned authoritative name server. The *primary authoritative name server* hosts the original editable zone file

for the domain. *Secondary authoritative name servers* can be used to host read-only copies of the zone file. A *zone file* is the collection of *resource records* or details about the specific domain. There are dozens of possible resource records (see `www.iana.org/assignments/dns-parameters/dns-parameters.xhtml`), such as A records linking an FQDN to an IPv4 address and AAAA records linking an FQDN to an IPv6 address. The use of AAAA is sometimes referred to as DNSv6.

Some of the more commonly used resource records are:

- A: Address record; links a FQDN to an IPv4 address
- AAAA: Address record; links a FQDN to an IPv6 address
- PTR: Pointer record; links an IPv4 or IPv6 address to a FQDN
- CNAME: Canonical name or alias record; links a FQDN to another FQDN
- MX: Mail exchange record; identifies SMTP email servers for a domain
- NS: Name server record; identifies the DNS servers for a domain
- SOA: Start of authority record; identifies the primary authoritative DNS server, the responsible email address, serial number (of the zone file), and time intervals of refresh, retry, expire, and default TTL

Originally, DNS was handled by a static local file known as the `hosts` file. The `hosts` file contains hard-coded references for domain names and their associated IP addresses. This file still exists on most TCP/IP capable computers, but a dynamic DNS query system has mostly replaced it. Administrators or threat actors can add content to the `hosts` file.

When client software points to an FQDN, the resolution process first checks the local DNS cache to see whether the answer is already known. The DNS cache consists of the preloaded local `hosts` file plus any DNS query results (that haven't timed out). If the needed answer isn't in the cache, a DNS query is sent to the DNS server indicated in the local IP configuration.

DNS operates over TCP and UDP port 53. TCP port 53 is used for zone transfers. These are zone file exchanges between DNS servers, for special

manual queries, or when a response exceeds 512 bytes. UDP port 53 is used for most typical DNS queries.

Domain Name System Security Extensions (DNSSEC) (http://dnssec.net) is a security improvement to the existing DNS infrastructure. The primary function of DNSSEC is to provide mutual certificate authentication and encrypted sessions between devices during DNS operations. DNSSEC has been implemented across a significant portion of the DNS system. Once fully implemented, DNSSEC will significantly reduce server-focused DNS abuses, such as zone file poisoning and DNS cache poisoning. However, DNSSEC only applies to DNS servers, not to systems performing queries against DNS servers (such as clients).

Non-DNS servers (i.e., mostly client devices), especially when using the Internet, should consider using *DNS over HTTPS (DoH)*. This system creates an encrypted session with a DNS server of TLS-protected HTTP and then uses that session as a form of VPN to protect the DNS query and response. A late 2020 enhancement to DoH is *Oblivious DoH (OdoH)*. OdoH adds a DNS proxy between the client and the DNS resolver so that the identity of the requesting client is isolated from the DNS resolver. Thus, ODoH provides anonymity and privacy to DNS queries. However, you are now trusting the ODoH provider to protect your privacy.



For an excellent primer and advanced discussion on DNS, its operation, and known issues, please visit “An Illustrated Guide to the Kaminsky DNS Vulnerability”: http://unixwiz.net/techtips/iguide-kaminsky-dns-vuln.html.

DNS Poisoning

DNS poisoning is the act of falsifying the DNS information used by a client to reach a desired system. It can take place in many ways. Whenever a client needs to resolve a DNS name into an IP address, it may go through the following process:

1. Check the local cache (which includes content from the `hosts` file).
2. Send a DNS query to a known DNS server.
3. Send a broadcast query to any possible local subnet DNS server. (This step isn't widely supported.)

If the client doesn't obtain a DNS-to-IP resolution from any of these steps, the resolution fails and the communication can't be sent. There are many ways to attack or exploit DNS, most of which are used to return false results.

Rogue DNS Server

A *rogue DNS server* can listen in on network traffic for any DNS query or specific DNS queries related to a target site. Then the rogue DNS server sends a DNS response to the client with false IP information. Once the client receives the response from the rogue DNS server, the client closes the DNS query session, which causes the response from the real DNS server to be dropped and ignored as an out-of-session packet.

DNS queries are not authenticated, but they do contain a 16-bit value known as the *query ID (QID)*. The DNS response must include the same QID as the query to be accepted. Thus, a rogue DNS server must include the requesting QID in the false reply.

Performing DNS Cache Poisoning

DNS poisoning involves attacking DNS servers and placing incorrect information into its zone file or cache. Authorized DNS server attacks aim to alter the primary record of an FQDN in the zone file on the primary authoritative DNS server. This causes real DNS servers to send false data back to clients. However, an attack on an authoritative DNS server typically gets noticed very quickly, so it rarely results in widespread exploitation.

So, most attackers focus on caching DNS servers instead. A caching DNS server is any DNS system deployed to cache DNS information from other DNS servers. The content hosted on a caching DNS server is not being watched by the worldwide security community but just the local operators. Thus, an attack against a caching DNS server can potentially occur without

notice for a significant period of time. This variation can be called *DNS cache poisoning*.

Although both of these attacks focus on DNS servers, they ultimately affect clients. Once a client has performed a dynamic DNS resolution, the information received from an authoritative DNS server or a caching DNS server will be temporarily stored in the client's local DNS cache. If that information is false, then the client's DNS cache has been poisoned.

DNS Pharming

Another attack closely related to DNS poisoning and/or DNS spoofing is *DNS pharming*. Pharming is the malicious redirection of a valid website's URL or IP address to a fake website. Pharming typically occurs either by modifying the local `hosts` file on a system or by poisoning or spoofing DNS resolution.

Altering the Hosts File

Modifying the `hosts` file on the client by placing false DNS data into it redirects users to false locations. If an attacker is able to plant false information into the `hosts` file, then when the system boots, the contents of the `hosts` file will be read into memory, where they will take precedence. This attack is effective, but it is also highly targeted. It only affects the individual systems with a locally corrupted `hosts` file. If the attacker wishes to cause harm more broadly, any of the other methods would be more effective.

Corrupt the IP Configuration

Corrupting the IP configuration can result in a client having a false DNS server definition (i.e., DNS lookup address changing). The DNS server address is typically distributed to clients through DHCP, but it can also be assigned statically. Attacks to alter a client's DNS server lookup address can be performed by compromising DHCP or through a script.

DNS Query Spoofing

A *DNS query spoofing* attack occurs when the threat actor is able to eavesdrop on a client's query to a DNS server. The attacker then sends back

a reply with false information. In order for this to be successful, the false reply must include the correct QID cloned from the query.

Use Proxy Falsification

Although not strictly a DNS issue, a *proxy falsification* attack could be implemented via DNS if the proxy's domain name has to be resolved by the client to use the proxy. Attacks could modify the local configuration, the configuration script, or the routing table to redirect communications to a false proxy. This method works only against web communications (or other services or protocols that use a proxy). A rogue proxy server can modify traffic packets to reroute requests to whatever site the malicious actor wants.



An adversary in the middle (AitM) (also known as a man-in-the-middle attack [MitM] or on-path attack) can be performed using DNS abuses, such as DNS cache poisoning. Once a client receives a response from DNS, that response will be cached for future use. If false information can be fed into the DNS cache, then misdirecting communications is trivially easy. See Chapter 17, “Preventing and Responding to Incidents,” for more on this type of attack.

Defenses to DNS Poisoning

Organizations should use a *split-DNS* system (aka *split-horizon DNS*, *split-view DNS*, and *split-brain DNS*). A split-DNS is deploying a DNS server for public use and a separate DNS server for internal use. All data in the zone file on the public DNS server is accessible by the public via queries or probing. However, the internal DNS is for internal use only. Only internal systems are granted access to interact with the internal DNS server. Outsiders are prohibited from accessing the internal DNS server by blocking inbound port 53 for both TCP and UDP. TCP 53 is used for zone transfers (which includes most DNS server–to–DNS server communications), and UDP 53 is used for queries (which is any non-DNS system sending a query to a DNS server). Internal systems can be configured to interact only with the internal DNS servers, or they may be

allowed to send queries to external DNS servers (which does require the firewall to be a stateful inspection firewall configured to allow responses to return to the internal system from an approved outbound query).

Although there are many DNS poisoning methods, here are some basic security measures you can take that can greatly reduce their threat:

- Limit zone transfers from internal DNS servers to external DNS servers. This is accomplished by blocking inbound TCP port 53 (zone transfer requests) and UDP port 53 (queries).
- Require internal clients to resolve all domain names through the internal DNS. This will require that you block outbound UDP port 53 (for queries) while keeping open outbound TCP port 53 (for zone transfers).
- Limit the external DNS servers from which internal DNS servers pull zone transfers.
- Deploy a network intrusion detection system (NIDS) to watch for abnormal DNS traffic.
- Properly harden all DNS, server, and client systems in your private network.
- Use DNSSEC to secure your DNS infrastructure.
- Use DoH or OdoH on all clients where supported.

There is no easy patch or update that will prevent these exploits from being waged against a client. This is due to the fact that these attacks take advantage of the normal and proper mechanisms built into various protocols, services, and applications. Thus, the defense is more of a detection and preventive concern. Install both HIDS and NIDS tools to watch for abuses of these types. Regularly review the logs of your DNS and DHCP systems, as well as local client system logs and potentially firewall, switch, and router logs for entries indicating abnormal or questionable occurrences.

Another DNS defense mechanism is a DNS sinkhole. A *DNS sinkhole* is a specific example of a false telemetry system (aka sinkhole server, internet sinkhole, and blackhole DNS). This technique is effectively DNS spoofing

used as a defense. A DNS sinkhole attempts to provide false responses to DNS queries from malware, such as bots, to prevent access to command and control systems. It can also be used to protect users from visiting known malicious or phishing sites. Thus, DNS sinkholes can be used for both malicious and benign/investigative/defensive purposes.

Domain Hijacking

Domain hijacking, or *domain theft*, is the malicious action of changing the registration of a domain name without the authorization of the valid owner. This may be accomplished by stealing the owner's logon credentials, using XSRF, hijacking a session, using an AitM attack, or exploiting a flaw in the domain registrar's systems.

An example of a domain hijack is the theft of the `fox-it.com` domain; you can read about this attack at `http://blog.fox-it.com/2017/12/14/lessons-learned-from-a-man-in-the-middle-attack`.

Sometimes, when another person registers a domain name immediately after the original owner's registration expires, it is called domain hijacking, but it shouldn't be. This is a potentially unethical practice, but it is not an actual hack or attack. It is taking advantage of the oversight of the original owner's failure to manually extend their registration or configure auto-renewal. If an original owner loses their domain name by failing to maintain registration, there is often no recourse other than to contact the new owner and ask about reobtaining control.

When an organization loses its domain and someone else takes over control, this can be a devastating event both to the organization and its customers and visitors. The new FQDN owner might host completely different content or a false duplicate of the previous site. This later activity might result in fooling visitors, similar to a phishing attack, where personally identifiable information (PII) might be extracted and collected.

The best defense against domain hijacking is to use strong multifactor authentication when logging into your domain registrar. To defend against letting your domain registration lapse, set up auto-renew and double-check the payment method a week before the renewal date.

Homograph Attack

Another DNS, address, or hyperlink concern is that of the *homograph attack*. These attacks leverage similarities in character sets to register phony international domain names (IDNs) that to the naked eye appear legitimate. For example, in many fonts, some letters in Cyrillic look like Latin characters; for example, the l (i.e., lowercase L) in Latin looks like the Palochka Cyrillic letter. Thus, domain names of `apple.com` and `paypal.com` might look valid as Latin characters but could actually include Cyrillic characters that, when resolved, direct you to a different site than you intended. For a thorough discussion of the homograph attack, see `http://blog.malwarebytes.com/101/2017/10/out-of-character-homograph-attacks-explained`.



See Chapter 2, “Personnel Security and Risk Management Concepts,” for social engineering topics of typosquatting, URL hijacking, and clickjacking, which are also related to domain name poisoning or spoofing.

Internet Protocol (IP) Networking

Another important protocol in the TCP/IP protocol suite operates at the Network Layer of the OSI model, namely, *Internet Protocol (IP)*. IP provides route addressing for data packets. It is this route addressing that is the foundation of global internet communications because it provides a means of identity and prescribes transmission paths. Similar to UDP, IP is connectionless and is an unreliable communication service. IP does not offer guarantees that packets will be delivered or that packets will be delivered in the correct order, and it does not guarantee that packets will be delivered only once. However, it was designed to perform “best effort” in finding a path or route to a destination in spite of a damaged or corrupted network structure. Thus, you must employ TCP with IP to gain reliable and controlled communication sessions.

IPv4 vs. IPv6

IPv4 is the version of Internet Protocol that is most widely used around the world. However, *IPv6* is being rapidly adopted for both private and public network use. IPv4 uses a 32-bit addressing scheme, whereas IPv6 uses 128 bits for addressing. IPv6 offers many new features that are not available in IPv4. Some of IPv6's new features are scoped addresses, autoconfiguration, and quality of service (QoS) priority values. Scoped addresses give administrators the ability to group and then block or allow access to network services, such as file servers or printing. Autoconfiguration theoretically removes the need for traditional DHCP and network address translation (NAT). However, DHCPv6 and NAT66 exist (see Chapter 12 for NAT66). QoS priority values allow for traffic management based on prioritized content. Also, IPSec is native to IPv6, but it is an add-on for IPv4.

DHCPv6 has two modes of operation. In stateful mode, DHCPv6 assigns specific IPv6 addresses to devices and manages the allocation of network configuration parameters. This mode is similar to the operation of DHCP in IPv4 networks. In stateless mode, DHCPv6 provides network configuration parameters without assigning specific IPv6 addresses. Devices may use Stateless Address Autoconfiguration (SLAAC) to generate their own addresses.

SLAAC is based on routers periodically sending Router Advertisement (RA) messages to the local network segment. The RA messages include information about the network prefix that devices should use when forming their IPv6 addresses. The second element used to craft an IPv6 address is the interface identifier, which is typically derived from the Media Access Control (MAC) address of the network interface on the device. However, to enhance privacy, some implementations may use techniques like “privacy extensions” to generate random interface identifiers. The combination of the network prefix and the interface identifier results in a self-assigned unique IPv6 address for each device.



IPv4 has an equivalent concept to that of IPv6's QoS, which is named Type of Service (ToS). However, ToS seemed to go unused and was converted into the Differentiated Services (DS) by a later specification. The DS field offers a variety of definable characteristics that can be used to manage traffic flow. However, it still does not seem to have widespread use or support by network devices, which would perform such management. There is promise that IPv6 networks will include more common support and actually provide for traffic prioritization based on IPv6 header values.

IPv6 is supported by most operating systems released since 2000. For most OSs today, native IPv6 is standard. While the initial deployment of IPv6, both privately and publicly, has been slow, at the start of 2024, over 42 percent of Internet IP traffic was IPv6-based. For a glimpse into the status of IPv4 to IPv6 conversion on the Internet from Google's perspective, see the IPv6 statistics at `www.google.com/intl/en/ipv6/statistics.html`.

The transition or migration to IPv6 raises several security concerns. One issue is that with the larger 128-bit address space, there are many more addresses that attackers can use as source addresses; thus, IP filtering and block lists will be less effective as attackers can just use a different address to get past the filter.

A second issue is that secure deployment of IPv6 requires that all security filtering and monitoring products be upgraded to fully support IPv6 prior to enabling the protocol on the production network. Otherwise, IPv6 will serve as a covert channel, as it will be unmonitored and unfiltered.

The means by which IPv6 and IPv4 can coexist on the same network is to use one or more of three primary options: *dual stack*, tunneling, or NAT-PT. Dual stack means having systems operate both IPv4 and IPv6 and using the appropriate protocol for each conversation. Tunneling allows most systems to operate a single stack of either IPv4 or IPv6 and use an encapsulation tunnel to access systems of the other protocol. Network Address Translation-Protocol Translation (NAT-PT) (RFC-2766) can be used to convert between IPv4 and IPv6 network segments similar to how NAT

converts between internal and external addresses. See Chapter 12 on NAT66.

For those organizations still reluctant to deploy IPv6 internally, they might consider IPv6 at the edge. This is a configuration where IPv6 is only supported on the boundary devices connected directly to the Internet. This enables communications over the Internet using IPv6, while using IPv4 internally.



Both IPv4 and IPv6 have a header field that is used to control or limit infinite transmission. The *time to live (TTL)* field of IPv4 and the *hop limit* field of IPv6 are decremented by routers until it reaches zero (0). Once that occurs, the packet is discarded and an ICMP Type 11 Timeout Exceeded error message is sent back to the origin.

IP Classes

Basic knowledge of IPv4 addressing and IPv4 classes is a must for any security professional. If you are rusty on IPv4 addressing, subnetting, classes, and other related topics, take the time to refresh your knowledge.

Table 11.1 and Table 11.2 provide a quick overview of the key details of classes and default subnets. A full Class A subnet supports 16,777,214 hosts; a full Class B subnet supports 65,534 hosts; and a full Class C subnet supports 254 hosts. Class D is used for multicasting, whereas Class E is reserved for experimental and future use.

TABLE 11.1 IP classes

Class	First binary digits	Decimal range of first octet
A	0	1–126
B	10	128–191
C	110	192–223
D	1110	224–239
E	1111	240–255

TABLE 11.2 IP classes' default subnet masks

Class	Default subnet mask	CIDR equivalent
A	255.0.0.0	/8
B	255.255.0.0	/16
C	255.255.255.0	/24

Note that the entire Class A network of 127 was set aside for the *loopback address*, although only a single address is actually needed for that purpose. A Class A network of 0 is defined as the blackhole network where traffic is routed to be thrown away and discarded.



The loopback address for IPv4 is any address in the Class A subnet of 127.0.0.1–127.255.255.254, even though only the address of 127.0.0.1 is typically used. When an interface is configured for loopback, a subnet mask is not defined; it will use 255.255.255.255 by default, although some will document this as 127.0.0.0/8. Also note that under IPv4, the first address of a subnet is reserved as the network address (i.e., 127.0.0.0) and the last for the directed broadcast (i.e., 127.255.255.255) and, therefore, not directly usable as a host address (or in this case a loopback address). The IPv6 loopback is not a specific address—it is a notation: ::1/128 or ::1%128.

The original class-based grouping of IPv4 addresses is no longer strictly adhered to. Instead, a more flexible system has been adopted based on *variable length subnet masking (VLSM)* and *Classless Inter-Domain Routing (CIDR)*. CIDR provides for a subnet masking notation that uses mask bit counts rather than a full dotted-decimal notation subnet mask. Thus, instead of 255.255.0.0, a CIDR notation is added to the IP address after a slash, as in 172.16.1.1/16, for example. One significant benefit of CIDR over traditional subnet-masking techniques is the ability to combine multiple noncontiguous sets of addresses into a single subnet. For example, it is possible to combine several Class C subnets into a single larger subnet grouping. If CIDR piques your interest, see IETF's RFC for CIDR at `http://tools.ietf.org/html/rfc4632`.

ICMP

Internet Control Message Protocol (ICMP) is used to determine the health of a network or a specific link. ICMP is utilized by `ping`, `traceroute`, `tracert`, `pathping`, and other network management tools. The `ping` utility employs ICMP echo packets and bounces them off remote systems. Thus, you can use `ping` to determine whether the remote system is online, whether the remote system is responding promptly, whether the intermediary systems are supporting communications, and at what level of performance efficiency the intermediary systems are communicating. The `ping` utility includes a redirect function that allows the echo responses to be sent to a destination different from the system of origin.

Unfortunately, the features of ICMP were often exploited in various forms of bandwidth-based denial-of-service (DoS) attacks, such as ping of death, Smurf attacks, and ping floods. This fact has shaped how networks handle ICMP traffic today, resulting in many networks limiting the use of ICMP or at least limiting its throughput rates.

IGMP

Internet Group Management Protocol (IGMP) allows systems to support multicasting. *Multicasting* is the transmission of data to multiple specific recipients. RFC 1112 discusses the requirements to perform IGMP multicasting (`http://tools.ietf.org/html/rfc1112`). IGMP is used to manage a host's dynamic multicast group membership. With IGMP, a single initial signal is multiplied at the router if divergent pathways exist to the intended recipients. Multicasting can be assisted by a Trivial File Transfer Protocol (TFTP) system to host or cache content that is to be sent to multiple recipients.

ARP Concerns

Address Resolution Protocol (ARP) is used to resolve IP addresses (32-bit binary number for logical addressing) into MAC addresses (48-bit binary number for physical addressing). Traffic on a network segment (for example, from a client to a default gateway [i.e., a router] via a switch) is directed from its source system to its destination system using MAC

addresses. ARP is carried as the payload of an Ethernet frame and is a dependent Layer 2 protocol.

ARP uses caching and broadcasting to perform its operations. The first step is to check the local ARP cache. If the needed information is already present in the ARP cache, it is used. If not, then an ARP request in the form of a broadcast is transmitted. If the owner of the queried address is in the local subnet, it can respond with the necessary information in an ARP reply/response. If not, the system will default to using its default gateway's MAC address to transmit its communications. ARP can be abused using a technique called ARP cache poisoning, where an attacker inserts bogus information into the ARP cache.

ARP cache poisoning or *ARP spoofing* is caused by an attacker responding with falsified replies. ARP cache is updated each time an ARP reply is received. The dynamic content of ARP cache, whether poisoned or legitimate, will remain in cache until a timeout occurs (which is usually under 10 minutes). Once an IP-to-MAC mapping falls out of cache, then the attacker gains another opportunity to poison the ARP cache when the client re-performs the ARP broadcast query.

Another form of ARP poisoning uses *gratuitous ARP* or *unsolicited ARP* replies. This occurs when a system announces its MAC-to-IP mapping without being prompted by an ARP query. A gratuitous ARP broadcast may be sent as an announcement of a node's existence, to update an ARP mapping due to a change in IP address or MAC address, or when redundant devices are in use that share an IP address and may also share the same MAC address (regularly occurring gratuitous ARP announcements help to ensure reliable failover).

A third form of ARP cache poisoning is to create static ARP entries. This is done via the `ARP` command and must be done locally. Unfortunately, this is easily accomplished through a malicious script executed on the client. However, static ARP entries are not persistent across reboots.

The best defense against ARP-based attacks is port security on the switch. Switch port security can prohibit communications with unknown, unauthorized, rogue devices and may be able to determine which system is responding to all ARP queries and block ARP replies from the offending system. A local or software firewall, host intrusion detection and prevention

system (HIDPS), or special endpoint security products can also be used to block unrequested ARP replies/announcements. One popular tool used to detect ARP poisoning is `arpwatch`.

Another defense is to establish static ARP entries. Yes, this can be used as both an attack/abuse and a defense. However, this is not often recommended because it removes the flexibility of a system adapting to changing network conditions, such as other devices entering and leaving the network. Once a static ARP entry is defined, it is "permanent" in that it will not be overwritten by any ARP reply, but it will not be retained across a reboot (that feature would be called persistence). A boot or logon script would need to be crafted on each system to re-create the static entries each time the system rebooted.

Secure Communication Protocols

Protocols that provide security services for application-specific communication channels are called secure communication protocols. Examples include the following:

IPSec *Internet Protocol Security (IPSec)* uses public key cryptography to provide encryption, integrity, antireplay, access control, and message origin authentication, all using IP-based protocols. The primary use of IPSec is for virtual private networks (VPNs), so IPSec can operate in either transport or tunnel mode. IPSec is a standard of IP security extensions used as an add-on for IPv4 and integrated into IPv6. IPSec is discussed further in Chapter 12.

Kerberos *Kerberos* offers a single sign-on (SSO) solution for users and provides protection for logon credentials. Modern implementations of Kerberos use hybrid encryption to provide reliable authentication protection. Kerberos is discussed further in Chapter 14, "Controlling and Monitoring Access."

SSH *Secure Shell (SSH)* is a good example of an end-to-end encryption technique. This security tool can be used to encrypt numerous plaintext utilities (such as `rcp`, `rlogin`, and `rexec`), serve as a protocol encrypter (such as with SFTP), and function as a transport mode VPN (i.e., host-to-host and link encryption only). SSH is discussed further in Chapter 12.

Signal Protocol This is a cryptographic protocol that provides end-to-end encryption for voice communications, videoconferencing, and text message services. The *Signal Protocol* is a core element in the messaging app named Signal.

Secure Remote Procedure Call (S-RPC) *S-RPC* is an authentication service for cross-network service communications and is simply a means to prevent unauthorized execution of code on remote systems.

Transport Layer Security (TLS) This is an encryption protocol that operates at OSI Layer 4 (by encrypting the payload of TCP communications). Although it is primarily known to be used to encrypt web communications as HTTPS, it can encrypt any Application Layer protocol. *Transport Layer Security (TLS)* replaced *Secure Sockets Layer (SSL)*. Features of TLS include the following:

- Supports secure client-server communications across an insecure network while preventing tampering, spoofing, and eavesdropping.
- Supports one-way authentication.
- Supports two-way authentication using digital certificates.
- Often implemented as the initial payload of a TCP package, allowing it to encapsulate all higher-layer protocol payloads.
- Can be used to encrypt User Datagram Protocol (UDP) and Session Initiation Protocol (SIP) connections. (SIP is a protocol associated with Voice over IP [VoIP].)

Implications of Multilayer Protocols

TCP/IP is a *multilayer protocol*. TCP/IP derives several benefits from its multilayer design, specifically in relation to its mechanism of encapsulation. For example, when communicating between a web server and a web browser over a typical network connection, HTTP is encapsulated in TCP, which in turn is encapsulated in IP, which in turn is encapsulated in Ethernet. This could be presented as follows:

```
[ Ethernet [ IP [ TCP [ HTTP [Payload] ] ] ] ]
```

However, this is not the extent of TCP/IP's encapsulation support. It is also possible to add additional layers of encapsulation. For example, adding TLS encryption to the communication would insert a new encapsulation between HTTP and TCP (technically, this results in HTTPS, the TLS-encrypted form of HTTP):

```
[ Ethernet [ IP [ TCP [ TLS [ HTTP [Payload] ] ] ] ] ]
```

This in turn could be further encapsulated with a Network Layer encryption such as IPSec:

```
[ Ethernet [ IPSec [ IP [ TCP [ TLS [ HTTP [Payload] ] ] ] ] ]
]
```

This is an example of a VPN. VPNs use encapsulation to enclose or tunnel one protocol inside another protocol. Usually, the encapsulation protocol encrypts the original protocol. For more on VPNs, see Chapter 12.

However, encapsulation is not always implemented for benign purposes. Numerous covert channel communication mechanisms use encapsulation to hide or isolate an unauthorized protocol inside another authorized one. For example, if a network blocks the use of FTP but allows HTTP, then tools such as HTTPTunnel can be used to bypass this restriction. This could result in an encapsulation structure such as this:

```
[ Ethernet [ IP [ TCP [ HTTP [ FTP [Payload] ] ] ] ] ]
```

Normally, HTTP carries its own web-related payload, but with the HTTPTunnel tool, the standard payload is replaced with an alternative protocol.

This false encapsulation can even occur lower in the protocol stack. For example, ICMP is typically used for network health testing and not for general communication. However, with utilities such as Loki, ICMP is transformed into a tunnel protocol to support TCP communications. The encapsulation structure of Loki is as follows:

```
[ Ethernet [ IP [ ICMP [ TCP [ HTTP [Payload] ] ] ] ] ]
```

Another area of concern caused by unbounded encapsulation support is the ability to jump between virtual local area networks (VLANs). Please see Chapter 12 about VLANs.

Multilayer protocols provide the following benefits:

- A wide range of protocols can be used at higher layers.
- Encryption can be incorporated at various layers.
- Flexibility and resiliency in complex network structures is supported.

There are a few drawbacks of multilayer protocols:

- Covert channels are allowed.
- Filters can be bypassed.
- Logically imposed network segment boundaries can be overstepped.

DNP3

DNP3 (Distributed Network Protocol 3) is primarily used in the electric and water utility and management industries. It is used to support communications between data acquisition systems and the system control equipment. This includes substation computers, remote terminal units (RTUs) (i.e., devices controlled by an embedded microprocessor), intelligent electronic devices [IEDs], and SCADA primary stations (i.e., control centers). DNP3 is an open and public standard. It is a multilayer protocol that functions similarly to TCP/IP in that it has link, transport, and transportation layers. For more details on DNP3, please view the protocol primer at www.dnp.org/About/Overview-of-DNP3-Protocol.

Converged Protocols

Converged protocols are the merging of specialty or proprietary protocols with standard protocols, such as those from the TCP/IP suite. The primary benefit of converged protocols is the ability to use existing TCP/IP supporting network infrastructure to host special or proprietary services without the need for unique deployments of alternate networking hardware. Some common examples of converged protocols are described here:

Storage Area Network (SAN) A *storage area network (SAN)* is a secondary network (distinct from the primary communications network) used to consolidate and manage various storage devices into a single consolidated network-accessible storage container. SANs are often used to enhance networked storage devices such as hard drives, drive arrays, optical jukeboxes, and tape libraries so that they can be made to appear to servers as if they were local storage. SANs operate by encapsulating or converging data storage signals into TCP/IP communications to separate storage and proximity. A SAN can be a single point of failure, so redundancy needs to be integrated to provide protection of availability. In some instances, a SAN may implement deduplication to save space by not retaining multiple copies of the same file. However, this can sometimes result in data loss if the one retained original is corrupted.

Internet Small Computer Systems Interface (iSCSI) *Internet Small Computer Systems Interface (iSCSI)* is a networking storage standard based on IP that operates at Layer 5 (Session). This technology can be used to enable location-independent file storage, transmission, and retrieval over LAN, WAN, or public internet connections. iSCSI is often viewed as a low-cost alternative to Fibre Channel.

InfiniBand over Ethernet *InfiniBand over Ethernet (IBoE)* refers to the encapsulation of InfiniBand traffic within Ethernet frames, allowing InfiniBand protocols to run over Ethernet networks. InfiniBand is a high-performance and low-latency interconnect technology commonly used in high-performance computing (HPC) environments. IBoE provides a way to integrate InfiniBand technology into existing Ethernet infrastructures.

Compute Express Link Compute Express Link (CXL) is an advanced high-speed interconnect technology developed to address the increasing demands of data-intensive workloads in modern computing systems. It is designed to enhance the performance, efficiency, and scalability of data-centric applications in various domains such as artificial intelligence (AI), machine learning (ML), HPC, and more. As a converged protocol, CXL supports the communication and collaboration of various components, such as CPUs, GPUs, accelerators, memory, and other devices, over a single high-speed interconnect.

Other concepts that may be considered examples of converged technologies include VPN, SDN, cloud, virtualization, SOA, microservices, and serverless architecture.

Voice over Internet Protocol (VoIP)

Voice over IP (VoIP) is a tunneling mechanism that encapsulates audio, video, and other data into IP packets to support voice calls and multimedia collaboration. VoIP has become a popular and inexpensive telephony solution for companies and individuals worldwide. VoIP has the potential to replace or supplant *public switched telephone network (PSTN)* services because it's often less expensive and offers a wider variety of options and features. VoIP can be used as a direct telephone replacement on computer networks as well as mobile devices. VoIP is considered a converged protocol as it combines the audio (and video) encapsulation technology (operating as Application Layer protocols) with the established multilayer protocol stack of TCP/IP.

VoIP is available in both commercial and open source options. Some VoIP solutions require specialized hardware to either replace traditional telephone handsets/base stations or allow these to connect to and function over the VoIP system. Some VoIP solutions are software only, such as Skype, and allow the user's existing speakers, microphone, or headset to replace the traditional telephone handset. Others are hardware-based, such as magicJack, which allows the use of existing PSTN phone devices plugged into a USB adapter to take advantage of VoIP over the Internet. Commercial VoIP equipment typically looks and functions much like traditional PSTN equipment but simply replaces the prior *plain old telephone service (POTS)* line with VoIP connectivity. Often, VoIP-to-VoIP calls are free (assuming the same or compatible VoIP technology are in use on both ends), whereas VoIP-to-landline or VoIP-to-mobile calls are usually charged a per-minute fee.

It is important to keep security in mind when selecting a VoIP solution to ensure that it provides the privacy and security you expect. Some VoIP systems are essentially plain-form communications that are easily intercepted and eavesdropped; others are highly encrypted, and any attempt to interfere or wiretap is deterred and thwarted.

VoIP is not without its problems. Threat actors can wage a wide range of potential attacks against a VoIP solution:

- *Caller ID* can be falsified easily using any number of VoIP tools, so threat actors can perform vishing (VoIP phishing) or Spam over Internet Telephony (SPIT) attacks.
- The call manager systems and VoIP phones themselves might be vulnerable to host operating system attacks and DoS attacks. If a device's or software's host OS or firmware has vulnerabilities, there is increased risk of exploits.
- Threat actors might be able to perform AitM attacks by spoofing call managers or endpoint connection negotiations and/or responses.
- Depending on the deployment, there are also risks associated with deploying VoIP phones off the same switches as desktop and server systems. This could allow for 802.1X authentication falsification as well as VLAN and VoIP hopping (i.e., jumping across authenticated channels).
- Since VoIP traffic is just network traffic, it is often possible to listen in on VoIP communications by decoding the VoIP traffic when it isn't encrypted.

Secure Real-Time Transport Protocol or *Secure RTP (SRTP)* is a security improvement over the *Real-Time Transport Protocol (RTP)* that is used in many VoIP communications. SRTP aims to minimize the risk of DoS, AitM attacks, and other VoIP exploits through robust encryption and reliable authentication. RTP or SRTP takes over after *Session Initiation Protocol (SIP)* establishes the communication link between endpoints.

Software-Defined Networking

Software-defined networking (SDN) is a unique approach to network operation, design, and management. The concept is based on the theory that the complexities of a traditional network with on-device configuration (i.e., routers and switches) often force an organization to stick with a single device vendor and limit the flexibility of the network to adapt to changing physical and business conditions, as well as optimize costs of acquiring new

devices. SDN aims to separate the infrastructure layer (aka the data plane and the forwarding plane)—hardware and hardware-based settings—from the control layer—network services of data transmission management. The control plane uses protocols to decide where to send traffic, and the data plane includes rules that decide whether traffic will be forwarded. This form of traffic management also involves access control over what systems can communicate which protocols to whom. This type of access control is typically attribute-based access control (ABAC) based.

Instead of traditional networking equipment such as routers and switches, an SDN solution gives an organization the option to handle traffic routing using simpler network devices that accept instructions from the SDN controller. This eliminates some of the complexity related to traditional networking protocols. Furthermore, this also removes the traditional networking concepts of IP addressing, subnets, routing, and the like from needing to be programmed into or be deciphered by hosted applications.

SDN offers a new network design that is directly programmable from a central location, flexible, vendor-neutral, and open-standards-based. Using SDN frees an organization from having to purchase devices from a single vendor. It instead allows organizations to mix and match hardware as needed, such as to select the most cost-effective or highest throughput–rated devices regardless of vendor. The configuration and management of hardware are then controlled through a centralized management interface. APIs (application programming interfaces) play a crucial role in SDN by providing a standardized way for external software applications to interact with and manipulate the network. In addition, the settings applied to the hardware can be changed and adjusted dynamically as needed from a central console or control point.

In relation to SDN, a southbound interface is the communication path from the SDN controller to the network devices in the data plane, facilitating the control and management of network traffic. The northbound interface is the communication path from the SDN controller to the applications or services in the Application Layer, allowing applications to interact with the SDN controller and make use of the network's programmability. These terms describe the flow of information between different components within an SDN framework.



Sometimes the terms *eastbound* and *westbound* are used to describe communications within a layer of a network or services' lattice structure (i.e., between peer elements).

SDN and network functions virtualization (NFV) are two distinct but closely related concepts that have transformed the traditional networking landscape. While SDN focuses on the separation of the control plane and data plane, NFV is about virtualizing and abstracting network functions from dedicated hardware devices. When combined, SDN and NFV create a more flexible, scalable, and programmable network infrastructure. It is effectively network virtualization. It allows data transmission paths, communication decision trees, and flow control to be virtualized in the SDN/NFV control layer rather than being handled on the hardware on a per-device basis.

Another interesting development arising out of the concept of virtualized networks is that of a *virtual SAN (VSAN)*. A SAN is a network technology that combines multiple individual storage devices into a single consolidated network-accessible storage container. They are often used with multiple or clustered servers that need high-speed access to a single shared dataset. These have historically been expensive due to the complex hardware requirements of the SAN. VSANs bypass these complexities with virtualization. A virtual SAN or a software-defined shared storage system is a virtual re-creation of a SAN on top of a virtualized network or an SDN.

Software-defined storage (SDS) is another derivative of SDN. SDS is a SDN version of a SAN or NAS. SDS is a storage management and provisioning solution that is policy driven and is independent of the actual underlying storage hardware. It is effectively virtual storage.

Software-defined wide-area networks (SDWAN or SD-WAN) are an evolution of SDN that can be used to manage the connectivity and control services between distant data centers, remote locations, and cloud services over WAN links.

Segmentation

Networks are not typically configured as a single large collection of systems. Usually, networks are segmented or subdivided into smaller organizational units. These smaller units, groupings, segments, or subnetworks (i.e., subnets) can be used to improve various aspects of the network:

Boosting Network Performance *Network segmentation* can improve performance through an organizational scheme in which systems that often communicate are located in the same segment. Also, dividing broadcast domains can significantly improve performance for larger networks.

Reducing Communication Problems Network segmentation often reduces congestion and contains communication problems, such as broadcast storms.

Providing Security Network segmentation can also improve security by isolating traffic and user access to those segments where they are authorized.

Physical segmentation refers to the practice of physically separating different components or segments within a network or system to enhance security, isolate sensitive information, and control access. This segmentation is typically achieved through various means, and different types of physical segmentation include in-band, out-of-band, and air-gapped. In-band segmentation involves the use of the same communication path or network infrastructure for both data and control traffic. Components within the same in-band segment share the same network resources. Often this is synonymous with logical segmentation as it does not use physical divisions for segmentation. Out-of-band segmentation involves separating data and controlling traffic onto different communication paths or networks. Control signals and management traffic have a dedicated network that is distinct from the network used for regular data transmission. An out-of-band segment creates a separate and distinct network structure for traffic that would otherwise interfere with the production network or that may itself be put at risk if placed on the production network. Secondary (or additional) network paths or segments may be created to support data

storage traffic (such as with SANs), VoIP, backup data, patch distribution, and management operations.

Air-gapped segmentation is the most stringent form of physical segmentation, where there is a complete physical separation between two systems or networks. This isolation is typically achieved by having no direct physical connection, such as cables, between the systems. There is also a need to either avoid the use of wireless communications or to block them purposefully.

Logical segmentation can be created by using switch-based virtual local area networks (VLANs), virtual private networks (VPNs), routers, firewalls, virtual routing and forwarding (VRF), and virtual domains individually or in combination.

VLANs are a method of logically dividing a physical LAN into multiple isolated broadcast domains. Devices within the same VLAN can communicate with each other as if they were on the same physical network, even if they are located on different physical segments. VLANs help improve network performance, security, and flexibility by grouping devices logically instead of relying solely on physical network topology. VLANs are commonly used in enterprise networks to segregate traffic based on departments, functions, or security levels.

VPNs create secure and encrypted communication channels over a public or shared network (usually the Internet). They allow remote users or branch offices to securely connect to the main corporate network, creating a virtual private network. VPNs provide secure communication over untrusted networks, ensuring privacy and data integrity. They are widely used for remote access, site-to-site connectivity, and secure communication between different entities.

VRF is a technology that allows multiple instances of a routing table to coexist within a router. Each VRF instance operates as a separate and independent routing domain, enabling the isolation of routing information. VRF is often used in service provider networks to provide virtualization and isolation for different customers or departments. It allows the same physical router to maintain separate routing tables for different VRF instances, preventing the leakage of routing information between them.

Virtual domains, also known as virtual systems or virtual contexts, involve creating isolated instances of a network device, such as a firewall or switch, to operate independently. Each virtual domain has its own configuration and operates as a separate logical entity. Virtual domains allow multiple customers or departments to use the same physical network device while maintaining logical separation. This is common in firewall appliances where different organizations or business units require dedicated firewall policies.

A private LAN or intranet, a screened subnet, and an extranet are all types of network segments.

Another example of network segmentation is a VPC. A virtual private cloud (VPC) is a virtualized network infrastructure provided by a cloud computing service provider. It allows users to create and manage isolated, logically segmented networks within the public cloud environment. A VPC enables organizations to host their applications and resources in a secure and dedicated space in the cloud while maintaining control over network configuration.

An evolution of the concept of network segmentation is micro-segmentation. *Micro-segmentation* is a network security strategy that involves dividing a network into small, isolated segments to enhance security and minimize the potential impact of security breaches. Micro-segmentation may potentially create divisions as small as a single device, such as a high-value server or even a client or endpoint device. This approach focuses on applying fine-grained security controls to individual workloads, applications, or devices within the network. Any and all communications between zones are filtered, may require authentication, often require session encryption, and may be subjected to allow list and block list control (i.e., authorization) and be closely monitored and logged. In some cases, to communicate with entities external to the local segment, the communication must be encapsulated for egress. This is similar to using a VPN to access a remote network. Micro-segmentation is a key element in implementing zero trust (see Chapter 8, “Principles of Security Models, Design, and Capabilities”).

Several technologies and concepts contribute to the implementation of micro-segmentation, including network overlays/encapsulation, distributed

firewalls, distributed routers, and intrusion detection systems (IDSs)/intrusion prevention systems (IPSs):

- *Network overlays/encapsulation:* Network overlays involve creating logical networks on top of an existing physical network. Encapsulation is a technique where data packets are wrapped in an additional layer, providing a form of isolation. Overlays and encapsulation support micro-segmentation by creating isolated communication channels between different segments. This helps in preventing unauthorized access to data by encapsulating it within specific overlays.
- *Distributed firewalls:* Traditional firewalls are typically placed at the network perimeter. Distributed firewalls, on the other hand, are implemented at various points within the network, closer to individual workloads or devices. These firewalls are also called internal segmentation firewalls (ISFWs). Distributing firewall capabilities across the network means that security policies can be enforced at a more granular level. Each workload or segment can have its own firewall rules, allowing for customized security controls.
- *Distributed routers:* Similar to distributed firewalls, distributed routers are deployed at various points within the network rather than being centralized. Distributed routers enable localized routing decisions and help control the flow of traffic between different micro-segments. This approach enhances network efficiency and provides more control over routing at a finer granularity.
- *Intrusion detection system (IDS)/intrusion prevention system (IPS):* IDS monitors network and/or system activities for suspicious behavior or security policy violations. IPS goes a step further by actively preventing or blocking identified threats. Deploying IDS/IPS at various points within the network allows for real-time detection and prevention of security threats. These systems contribute to the proactive security posture of micro-segmented environments.

Micro-segmentation is a powerful security strategy, especially in the context of modern cybersecurity, where organizations seek to enhance their defenses against advanced threats and minimize the impact of security incidents.

Virtual Extensible LAN (VXLAN) is an encapsulation protocol that enables VLANs (see Chapter 12) to be stretched across subnets and geographic distances. VLANs are typically restricted to Layer 2 network areas and are not able to include members from other networks that are accessible only through a router portal. Additionally, VXLAN allows for up to 16 million virtual networks to be created, whereas traditional VLANs are limited to only 4,096. VXLAN can be used as a means to implement micro-segmentation without limiting segments to local entities only. VXLAN is defined in RFC 7348.

Edge Networks

An edge network is a carefully designed data architecture that strategically allocates computing resources to edge devices within a network. This design helps distribute processing power demands away from central servers, empowering the devices to handle a significant portion of the processing workload. Edge networks are designed to bring content, services, and applications closer to the users to reduce latency and improve performance.

Edge network ingress points are often strategically located at the edge of the network infrastructure to efficiently bring in data or content from external sources, ensuring a responsive and low-latency user experience.

Edge network egress points are strategically positioned to efficiently direct traffic from the network to external destinations. This helps optimize the flow of data and content to ensure a seamless and responsive user experience.

Edge network peering refers to the process of establishing direct interconnections between edge networks, allowing them to exchange traffic directly without relying on intermediaries. The objective is to optimize the exchange of data, content, or services between different edge networks, leading to improved performance, reduced latency, and enhanced efficiency.

Wireless Networks

Wireless networking is widely implemented because of the ease of deployment and relatively low cost. Wireless networks are subject to the same vulnerabilities, threats, and risks as any cabled network in addition to distance eavesdropping and new forms of DoS and intrusion.



A wireless network can be referred to as an unbounded network, while a cable-only network can be referred to as a bound network.

802.11 is the IEEE standard for wireless network communications. Various versions (technically called amendments) of the standard have been implemented, many of which offer better throughput, as described in Table 11.3. Any later amendments that use the same frequency as earlier ones maintain backward compatibility.

TABLE 11.3 802.11 wireless networking amendments

Amendments	Wi-Fi Alliance names	Theoretical data rates	Frequencies
802.11	Wi-Fi 0	2 Mbps	2.4 GHz
802.11a	Wi-Fi 2	54 Mbps	5 GHz
802.11b	Wi-Fi 1	11 Mbps	2.4 GHz
802.11g	Wi-Fi 3	54 Mbps	2.4 GHz
802.11n	Wi-Fi 4	600 Mbps	2.4 GHz or 5 GHz
802.11ac	Wi-Fi 5	3.5 Gbps	5 GHz
802.11ax	Wi-Fi 6/Wi-Fi 6E	9.6 Gbps	Between 1 GHz and 7.125 GHz
802.11be	Wi-Fi 7	40 Gbps	Between 1 GHz and 7.250 GHz; coexists with 2.4, 5, & 6 GHz



Wi-Fi 0, 1, 2, and 3 are named by retroactive inference. They do not exist in the official standards or Wi-Fi Alliance documentation.

802.11x is sometimes used to collectively refer to all of these specific implementations as a group; however, 802.11 is preferred because 802.11x is easily confused with 802.1X, which is an authentication technology independent of wireless.

Wi-Fi can be deployed in either ad hoc mode (aka peer-to-peer Wi-Fi) or infrastructure mode. *Ad hoc mode* means that any two wireless networking devices can communicate without a centralized control authority (i.e., base station or access point). *Wi-Fi Direct* is an upgraded version of ad hoc mode that can support WPA2 and WPA3 (ad hoc supported only WEP). *Infrastructure mode* means that a *wireless access point (WAP)* is required and restrictions for wireless network access are enforced.

Infrastructure mode includes several variations, including stand-alone, wired extension, enterprise extended, and bridge. A *stand-alone mode* deployment is when there is a WAP connecting wireless clients to one another but not to any wired resources (thus, the WAP is on its own). A *wired extension mode* deployment is when the WAP acts as a connection point to link the wireless clients to the wired network. An *enterprise extended mode* deployment is when multiple wireless access points (WAPs) are used to connect a large physical area to the same wired network. Each WAP will use the same *extended service set identifier (ESSID)* so that clients can roam the area while maintaining network connectivity, even while their wireless NICs change associations from one WAP to another. A *bridge mode* deployment is when a wireless connection is used to link two wired networks. This type of deployment often uses dedicated wireless bridges and is used when wired bridges are inconvenient, such as when linking networks between floors or buildings.

A *fat access point* is a base station that is a fully managed wireless system, which operates as a stand-alone wireless solution. A *thin access point* is little more than a wireless transmitter/receiver, which must be managed

from a separate external centralized management console called a *wireless controller*. The benefit of using thin access points is that management, security, routing, filtering, and more are centralized at a management console, whereas numerous thin access points simply handle the radio signals. Most fat access points require device-by-device configuration and thus are not as flexible for enterprise use. Controller-based WAPs are thin access points that are managed by a central controller. A stand-alone WAP is a fat access point that handles all management functions locally on the device.

Securing the SSID

Wireless networks are assigned a *service set identifier (SSID)* to differentiate one wireless network from another. This effectively defines the Wi-Fi network's name. An SSID is used when a single WAP is in use, while an *ESSID (extended service set identifier)* is used when there are multiple WAPs supporting the same network by name over a larger area. An *independent service set identifier (ISSID)* is used by Wi-Fi Direct or in ad hoc mode. The *basic service set identifier (BSSID)* is the MAC address of the base station (or initiating device in an ad hoc/Wi-Fi Direct network), which is used to differentiate multiple base stations supporting an ESSID.

If a wireless client knows the SSID, they can configure their wireless NIC to communicate with the associated WAP. Knowledge of the SSID does not always grant entry, though, because the WAP can use numerous security features to block unwanted access. SSIDs are defined by default by vendors and thus are well known. Standard security practice dictates that the SSID should be changed to something unique before deployment.

The SSID is broadcast by the WAP via a special transmission called a *beacon frame*. A beacon frame allows any wireless NIC within range to see the wireless network and make connecting as simple as possible. This default SSID broadcast can be disabled to attempt to keep the wireless network secret. However, attackers can still discover the SSID with a wireless sniffer since the SSID is still used in transmissions between connected wireless clients and the WAP. Thus, disabling SSID broadcasting is not a true mechanism of security. Instead, use WPA2 or WPA3 as a reliable authentication and encryption solution rather than trying to hide the existence of the wireless network.

Wireless Channels

Within the assigned frequency of the wireless signal are subdivisions of that frequency known as *channels*. Think of channels as lanes on the same highway. In the United States, there are 11 channels defined within the *2.4 GHz* frequency range, in Europe there are 13, and in Japan there are 14. The differences stem from local laws regulating frequency management—think international versions of the Federal Communications Commission (FCC).

When two or more 2.4 GHz access points are relatively close to one another physically, signals on one channel can interfere with signals on another channel. One way to avoid this is to set the channels of physically close access points as differently as possible to minimize channel overlap interference. For example, if a building has four access points arranged in a line along the length of the building, the channel settings could be 1, 11, 1, and 11. However, if the building is square and an access point is in each corner, the channel settings may need to be 1, 4, 8, and 11. If three WAPs are positioned in close proximity, they can be set to channels 1, 6, and 11 without interference since these three channels do not overlap with each other.

5 GHz wireless was designed to avoid this channel overlap and interference issue. While 2.4 GHz channels are 22 MHz wide and 5 MHz apart, 5 GHz channels are 20 MHz wide and 20 MHz apart. Therefore, adjacent 5 GHz channels do not interfere with one another. Furthermore, adjacent channels can be combined or bonded into a larger width channel for faster throughput.

Wi-Fi band/frequency selection should be based on the purpose or use of the wireless network as well as the level of existing interference. For external networks, 2.4 GHz is often preferred because it can provide good coverage over a distance but at slower speeds; 5 GHz is often preferred for internal networks because it provides higher throughput rates (but less coverage area), but it does not penetrate solid objects, like walls and furniture. Most of the mesh Wi-Fi (multiple WAP) options are based on 5 GHz and use three or more mini-WAP devices to provide ML-optimized (machine learning–capable) coverage throughout a home or office. The 6 GHz spectrum range supports up to seven 160 MHz–wide channels more than the 5 GHz spectrum. This is possible due to the fact that the 6 GHz

spectrum is a contiguous 1.2 GHz frequency range rather than the multiple noncontiguous ranges in the 5 GHz spectrum. This provides for more top-speed connections than earlier forms of Wi-Fi. Devices that support the1–6 GHz spectrum range are labeled Wi-Fi 6E (as the version without the E only supports 1–5 GHz). However, 6 GHz is even more restricted by obstacles and distance.

Conducting a Site Survey

Wireless cells are the areas within a physical environment where a wireless device can connect to a wireless access point. You should adjust the strength of the WAP to maximize authorized user access and minimize outside intruder access. Doing so may require unique placement of wireless access points, shielding, and noise transmission. Often WAP placement is determined by performing a site survey to generate a heat map. A site survey is useful for evaluating existing wireless network deployments, planning expansion of current deployments, and planning for future deployments.

A *site survey* is a formal assessment of wireless signal strength, quality, and interference using an RF signal detector. A site survey is performed by placing a wireless base station in a desired location and then collecting signal measurements from throughout the area. These measurements are evaluated to determine whether sufficient signal is present where needed while minimizing signals elsewhere. If the base station is adjusted, then the site survey should be repeated. The goal of a site survey is to maximize performance in the desired areas (such as within a home or office) while minimizing ease of unauthorized access in external areas.

A site survey is often used to produce a heat map. A *heat map* is a mapping of signal strength measurements over a building's blueprint. The heat map helps to locate hot spots (oversaturation of signal) and cold spots (lack of signal) to guide adjustments to WAP placement, antenna type, antenna orientation, and signal strength.

Wireless Security

Wi-Fi is not always encrypted, and even when it is, the encryption is only between the client device and the base station. For end-to-end encryption of

communications, use a VPN or an encrypted communications application to pre-encrypt communications before transmitting them over Wi-Fi. For foundational encryption concepts, see Chapter 6, “Cryptography and Symmetric Key Algorithms,” and Chapter 7, “PKI and Cryptographic Applications.”

The original IEEE 802.11 standard defined two methods that wireless clients can use to authenticate to WAPs before normal network communications can occur across the wireless link. These two methods are *open system authentication (OSA)* and *shared key authentication (SKA)*.

OSA means no real authentication is required. As long as a radio signal can be transmitted between the client and WAP, communications are allowed. It is also the case that wireless networks using OSA typically transmit everything in cleartext, thus providing no secrecy or security.

With SKA, some form of authentication must take place before network communications can occur. The 802.11 standard defines one optional technique for SKA known as Wired Equivalent Privacy (WEP). Later 802.11 amendments added WPA, WPA2, WPA3, and other technologies.

Wired Equivalent Privacy (WEP)

Wired Equivalent Privacy (WEP) is defined by the original IEEE 802.11 standard. WEP uses a predefined shared Rivest Cipher 4 (RC4) secret key for both authentication (i.e., SKA) and encryption. Unfortunately, the shared key is static and shared among the WAP(s) and clients. Due to flaws in its implementation of RC4, WEP is weak.

WEP was cracked almost as soon as it was released. Today, it is possible to crack WEP in less than a minute. Fortunately, there are alternatives to WEP that you should use instead.

Wi-Fi Protected Access (WPA)

Wi-Fi Protected Access (WPA) was designed as the replacement for WEP; it was a temporary fix until the new 802.11i amendment was completed. WPA is a significant improvement over WEP in that it does not use the same static key to encrypt all communications. Instead, it negotiates a unique key set with each host. Additionally, it separated authentication from

encryption. WPA borrowed the authentication options from the then-still-draft of 802.11i.

WPA uses the RC4 algorithm and employs the *Temporal Key Integrity Protocol (TKIP)* or the Cisco alternative, *Lightweight Extensible Authentication Protocol (LEAP)*. However, WPA is no longer secure. Attacks specific to WPA (i.e., coWPAtty and GPU-based cracking tools) have rendered WPA's security unreliable. WPA might still be deployed to support EOSL or legacy equipment (although this is a very poor security option).



Temporal Key Integrity Protocol (TKIP) was designed as a temporary measure to support WPA features without requiring the replacement of legacy wireless hardware. TKIP and WPA were officially replaced by WPA2 in 2004. In 2012, TKIP was officially deprecated and is no longer considered secure.

Wi-Fi Protected Access 2 (WPA2)

IEEE 802.11i or *Wi-Fi Protected Access 2 (WPA2)* replaced WEP and WPA. It implements AES-CCMP instead of RC4. To date, no attacks have been successful against AES-CCMP encryption. However, there have been exploitations of the WPA2 key exchange processes (research KRACK [Key Reinstallation AttaCKs], if interested).



Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP) (Counter-Mode/CBC-MAC Protocol) is the combination of two block cipher modes to enable streaming by a block algorithm. CCMP can be used on many block ciphers. The AES-CCMP implementation was defined as part of WPA2, which replaced WEP and WPA, and is also used in WPA3 as the preferred means of wireless encryption.

WPA2/802.11i defined two “new” authentication options known as *preshared key (PSK)* or *personal (PER)* and IEEE 802.1X or *enterprise (ENT)*. They were also supported in WPA, but they were borrowed from the draft of IEEE 802.11i before it was finalized. PSK is the use of a static fixed password for authentication. ENT enables the leveraging of an existing AAA service, such as RADIUS or TACACS+, to be used for authentication.



Don't forget about the ports related to common AAA services: UDP 1812 (authentication and authorization) and UDP 1813 (accounting) for RADIUS and TCP 49 for TACACS+.

Wi-Fi Protected Access 3 (WPA3)

Wi-Fi Protected Access 3 (WPA3) was finalized in January 2018. WPA3-ENT uses 192-bit AES CCMP encryption, and WPA3-PER remains at 128-bit AES CCMP. WPA3-PER replaces the preshared key authentication with Simultaneous Authentication of Equals (SAE). Some 802.11ac/Wi-Fi 5 devices were the first to support or adopt WPA3.

Simultaneous Authentication of Equals (SAE) still uses a password, but it no longer encrypts and sends that password across the connection to perform authentication. Instead, SAE performs a zero-knowledge proof process known as Dragonfly Key Exchange, which is itself a derivative of Diffie–Hellman. The process uses the preset password and the MAC addresses of the client and AP to perform authentication and session key exchange. (There have been attacks against SAE; research Dragonblood attack, if interested.)

WPA3 also implements IEEE 802.11w-2009 management frame protection so that a majority of network management operations have confidentiality, integrity, authentication of source, and replay protection.

802.1X/EAP

WPA, WPA2, and WPA3 support the enterprise (ENT) authentication known as *802.1X/EAP*, a standard port-based network access control that

ensures that clients cannot communicate with a resource until proper authentication has taken place. Effectively, 802.1X is a handoff system that allows network devices to leverage the existing network infrastructure's authentication services. You can almost consider 802.1X as an authentication proxy service. Through the use of 802.1X, other techniques and solutions such as Remote Authentication Dial-In User Service (RADIUS), Terminal Access Controller Access Control System (TACACS), certificates, smartcards, token devices, and biometrics can be integrated into wireless networks, providing techniques for both mutual and multifactor authentication.

Extensible Authentication Protocol (EAP) is not a specific mechanism of authentication; rather it is an authentication framework. Effectively, EAP allows for new authentication technologies to be compatible with existing wireless or point-to-point connection technologies. For more on EAP and 802.1X, see Chapter 12.

LEAP

Lightweight Extensible Authentication Protocol (LEAP) is a Cisco proprietary alternative to TKIP for WPA. This was developed to address deficiencies in TKIP before the 802.11i/WPA2 system was ratified as a standard.

An attack tool known as `asleap` was released in 2004 that could exploit the ultimately weak protection provided by LEAP. LEAP should be avoided when possible; use of EAP-TLS as an alternative is recommended, but if LEAP is used, a complex password is strongly recommended.

PEAP

Protected Extensible Authentication Protocol (PEAP) encapsulates EAP methods within a TLS tunnel that provides authentication and potentially encryption. Since EAP was originally designed for use over physically isolated channels and hence assumed secured pathways, EAP is usually not encrypted. So PEAP can provide encryption for EAP methods.

Wi-Fi Protected Setup (WPS)

Wi-Fi Protected Setup (WPS) is a security standard for wireless networks. It is intended to simplify the effort involved in adding new clients to a well-secured wireless network. It operates by auto-connecting and automatically authenticating the first new wireless client to initiate a connection to the network once WPS is triggered. WPS can be initiated by a button on the WAP or a code or PIN that can be sent to the base station remotely. This allows for a brute-force guessing attack that could enable a threat actor to guess the WPS code in less than six hours, which in turn would enable the threat actor to connect their own unauthorized system to the wireless network.



The PIN code is composed of two four-digit segments, which can be guessed one segment at a time, with confirmation from the base station of each segment.

WPS is a feature that is enabled by default on most WAPs because it is a requirement for device Wi-Fi Alliance certification. It's important to disable it as part of a security-focused predeployment process. If a device doesn't offer the ability to turn off WPS (or the configuration Off switch doesn't work), upgrade or replace the base station's firmware or replace the whole device.

Wireless MAC Filter

A *MAC filter* can be used on a WAP to limit or restrict access to only known and approved devices. The MAC filter is a list of authorized wireless client interface MAC addresses that is used by a WAP to block access to all nonauthorized devices. Though a potentially useful feature, it can be difficult to manage and tends to be used only in small, static environments. However, even with WPA2 or WPA3, the Ethernet header remains in cleartext, which enables threat actors to sniff and spoof authorized MAC addresses. Additionally, many modern mobile devices offer randomized Wi-Fi MAC addresses. Thus, MAC filtering is no longer a useful option to block unknown devices while allowing in known ones.

Wireless Antenna Management

A wide variety of antenna types can be used for wireless clients and base stations. Many devices can have their standard antennas replaced with stronger (i.e., signal-boosting) antennas.

The standard straight or pole antenna is an *omnidirectional antenna*. This is the antenna found on most base stations and client devices. This type of antenna is sometimes also called a base antenna or a rubber duck antenna (due to most being covered in a flexible rubber coating).

Most other types of antennas are directional, meaning they focus their sending and receiving capabilities in one primary direction. Some examples of *directional antennas* include Yagi, cantenna, panel, and parabolic. A Yagi antenna is similar in structure to that of traditional roof TV antennas, which are crafted from a straight bar with cross-sections. Cantennas are constructed from tubes with one sealed end. Panel antennas are flat devices that focus from only one side of the panel. Parabolic antennas are used to focus signals from very long distances or weak sources.

Consider the following guidelines when seeking optimal antenna placement:

- Use a central location.
- Avoid solid physical obstructions.
- Avoid reflective or other flat metal surfaces.
- Avoid electrical equipment.

If a base station has external omnidirectional antennas, typically, they should be positioned pointing straight up vertically. If a directional antenna is used, point the focus toward the area of desired use. Keep in mind that wireless signals are affected by interference, distance, and obstructions.

Some WAPs provide a physical or logical adjustment of the antenna power levels. Power level controls are typically set by the manufacturer to a setting that is suitable for most situations. After performing site surveys, if wireless signals are still not satisfactory, power level adjustment might be necessary. However, changing channels, avoiding reflective and signal-

scattering surfaces, and reducing interference can often be more significant in terms of improving connectivity reliability.

When adjusting power levels, make minor adjustments instead of attempting to maximize or minimize the setting. Also, take note of the initial/default setting so that you can return to that setting if desired. After each power level adjustment, reset/reboot the WAP before re-performing site survey and quality tests. Sometimes, lowering the power level can improve performance. Some WAPs are capable of providing higher power levels than are allowed by regulations in countries where they are available.

Using Captive Portals

A *captive portal* is an authentication technique that redirects a newly connected client to a web-based portal access control page. The portal page may require the user to input payment information, provide login credentials, or input an access code. A captive portal is also used to display an acceptable use policy, privacy policy, and tracking policy to the user, who must consent to the policies before being able to communicate across the network.

Captive portals are most often located on wireless networks implemented for public use, such as at hotels, restaurants, bars, airports, libraries, and so on. However, they can be used on cabled Ethernet connections as well. Captive portals can be used in any scenario where the owner or administrator of a connection wants to limit access to authorized entities (which might include paying customers, overnight guests, known visitors, or those who agree to a security policy and/or terms of service).

General Wi-Fi Security Procedure

Here is a general guide or procedure to follow when deploying a Wi-Fi network. These steps are in order of consideration and application/installation:

1. Update firmware.
2. Change the default administrator password to something unique and complex.

3. Enable WPA2 or WPA3 encryption.
4. Enable ENT authentication or PSK/SAE with long, complex passwords.
5. Change the SSID (the default is often the vendor name).
6. Change the wireless MAC address (to hide OUI and device make/model that may be encoded into the default MAC address).
7. Decide whether to disable the SSID broadcast based on your deployment requirements (even though this doesn't increase security).
8. Enable MAC filtering if the pool of wireless clients is relatively small (usually less than 20) and static.
9. Consider using static IP addresses or configure DHCP with reservations (applicable only for small deployments).
10. Treat wireless as external or remote access, and separate the WAP from the wired network using a firewall.
11. Treat wireless as an entry point for attackers and monitor all WAP-to-wired-network communications with an NIDS.
12. Deploy a wireless intrusion detection system (WIDS) and a wireless intrusion prevention system (WIPS).
13. Consider requiring the use of a VPN across a Wi-Fi link.
14. Implement a captive portal.
15. Track/log all wireless activities and events.

Wireless Communications

Wireless communication is a quickly expanding field of technologies for networking, connectivity, communication, and data exchange. As wireless technologies continue to proliferate, your organization's security efforts need to encompass wireless communications.

General Wireless Concepts

Wireless communications employ radio waves to transmit signals over a distance. The radio spectrum is differentiated using frequency. Frequency is

a measurement of the number of wave oscillations within a specific time and is identified using the unit *Hertz (Hz)* (i.e., oscillations per second). Radio waves have a frequency between 3 Hz and 300 GHz. Several spectrum-use techniques were developed to manage the simultaneous use of the limited radio frequencies, including spread spectrum, FHSS, DSSS, OFDM, MIMO, TDMA, and CDMA.



Most devices operate within a small subsection of frequencies rather than all available frequencies. This is because of frequency-use regulations (in other words, the FCC in the United States), power consumption, and the expectation of interference.

Spread spectrum means that communication occurs over multiple frequencies. Thus, a message is broken into pieces, and each piece is sent at the same time but using a different frequency. Effectively, this is a parallel communication rather than a serial communication.

Frequency Hopping Spread Spectrum (FHSS) was an early implementation of the spread spectrum concept. FHSS transmits data in series across a range of frequencies, but only one frequency at a time is used.

Direct Sequence Spread Spectrum (DSSS) employs frequencies simultaneously in parallel. DSSS uses a special encoding mechanism known as chipping code to allow a receiver to reconstruct data even if parts of the signal were distorted because of interference.

Orthogonal Frequency-Division Multiplexing (OFDM) employs a digital multicarrier modulation scheme that allows for a more tightly compacted transmission. The modulated signals are perpendicular (orthogonal) and thus do not cause interference with one another. Ultimately, OFDM requires a smaller frequency set (aka channel bands) but can offer greater data throughput.

Multiple Input, Multiple Output (MIMO) is a wireless communication technology that uses multiple antennas at both the transmitter and receiver ends to improve the performance of a communication link. MIMO is widely

used in modern wireless communication systems to enhance data rates, reliability, and overall spectral efficiency.

Time Division Multiple Access (TDMA) is a digital communication technology used in various wireless communication systems, including mobile and satellite communication. TDMA is a multiple access scheme that divides the available communication channel into time slots, allowing multiple users to share the same frequency without interfering with each other.

Code Division Multiple Access (CDMA) is a digital cellular technology that allows multiple users to share the same frequency band simultaneously. CDMA assigns a unique code to each entity. This unique code allows multiple signals to occupy the same frequency at the same time.

Bluetooth

Bluetooth was originally defined in *IEEE 802.15.1* but is currently managed by Bluetooth SIG (www.bluetooth.com). The Bluetooth SIG (special interest group) is an industry association that oversees the development and standardization of Bluetooth technology. The Bluetooth SIG is responsible for defining the specifications, promoting interoperability, and certifying Bluetooth products.

Bluetooth is a wireless communication technology that allows devices to exchange data over short distances using radio waves. It uses the 2.4 GHz frequency. Bluetooth is plaintext by default in most implementations, but it can be encrypted with specialty transmitters and peripherals. Bluetooth operates between devices that have been paired, which often use a default pair code, such as 0000 or 1234. Bluetooth is generally a short-distance communication method (used to create personal area networks [PANs]), but that distance is based on the relative strengths of the paired devices' antennas. Standard or official use of Bluetooth ranges up to 100 meters, and an estimated 350+ meters with the introduction of Bluetooth 5.

Bluetooth Low Energy (Bluetooth LE, BLE, Bluetooth Smart) is a low-power-consumption derivative of standard Bluetooth. BLE was designed for IoT, edge/fog devices, mobile equipment, medical devices, and fitness trackers. It uses less power while maintaining a similar transmission range

to that of standard Bluetooth. Standard Bluetooth and BLE are not compatible, but they can coexist on the same device.

iBeacon is a location-tracking technology developed by Apple based on BLE. iBeacon can be used by a store to track customers while they shop as well as by customers as an indoor positioning system to navigate to an interior location.

Zigbee is an IoT equipment communications concept that is based on Bluetooth. Zigbee has low power consumption and a low throughput rate, and requires close proximity of devices. Zigbee communications are encrypted using a 128-bit symmetric algorithm.

Bluetooth is vulnerable to a wide range of attacks:

- *Bluesniffing* is Bluetooth-focused network packet capturing.
- *Bluesmacking* is a DoS attack against a Bluetooth device that can be accomplished through the transmission of garbage traffic or signal jamming.
- *Bluejacking* involves sending unsolicited messages to Bluetooth-capable devices without the permission of the owner/user. These messages may appear on a device's screen automatically, but many modern devices prompt whether to display or discard such messages.
- *BLUFFS (Bluetooth Forward and Future Secrecy)* is a series of exploits targeting Bluetooth, aiming to break Bluetooth sessions' forward and future secrecy, compromising the confidentiality of past and future communications between devices.
- *Bluesnarfing* is the unauthorized access of data via a Bluetooth connection. Sometimes, the term *bluejacking* is mistakenly used to describe or label the activity of bluesnarfing. Bluesnarfing typically occurs over a paired link between the threat actor's system and the target device. However, bluesnarfing is also possible against non-discoverable devices if their Bluetooth MAC addresses are known, which could be gathered using bluesniffing.
- *Bluebugging* grants an attacker remote control over the hardware and software of your devices over a Bluetooth connection. The name is

derived from enabling the microphone on a compromised system to use it as a remote wireless bug.

All Bluetooth devices are vulnerable to bluesniffing, bluesmacking, bluejacking, and BLUFF attacks. Only a few devices have been discovered to be vulnerable to bluesnarfing or bluebugging.

The defenses for all of these Bluetooth threats are to minimize use of Bluetooth, especially in public locations, and to leave Bluetooth turned off completely when not in active use.

RFID

Radio Frequency Identification (RFID) is a tracking technology based on the ability to power a radio transmitter using current generated in an antenna (see Figure 11.7) when placed in a magnetic field. RFID can be triggered/powered and read from a considerable distance away (potentially hundreds of meters). RFID can be attached to devices and components or integrated into their structure. This can allow for quick inventory tracking without having to be in direct physical proximity to the device. Simply walking into a room with an RFID reader, a malicious actor can collect the information transmitted by the activated chips in the area.

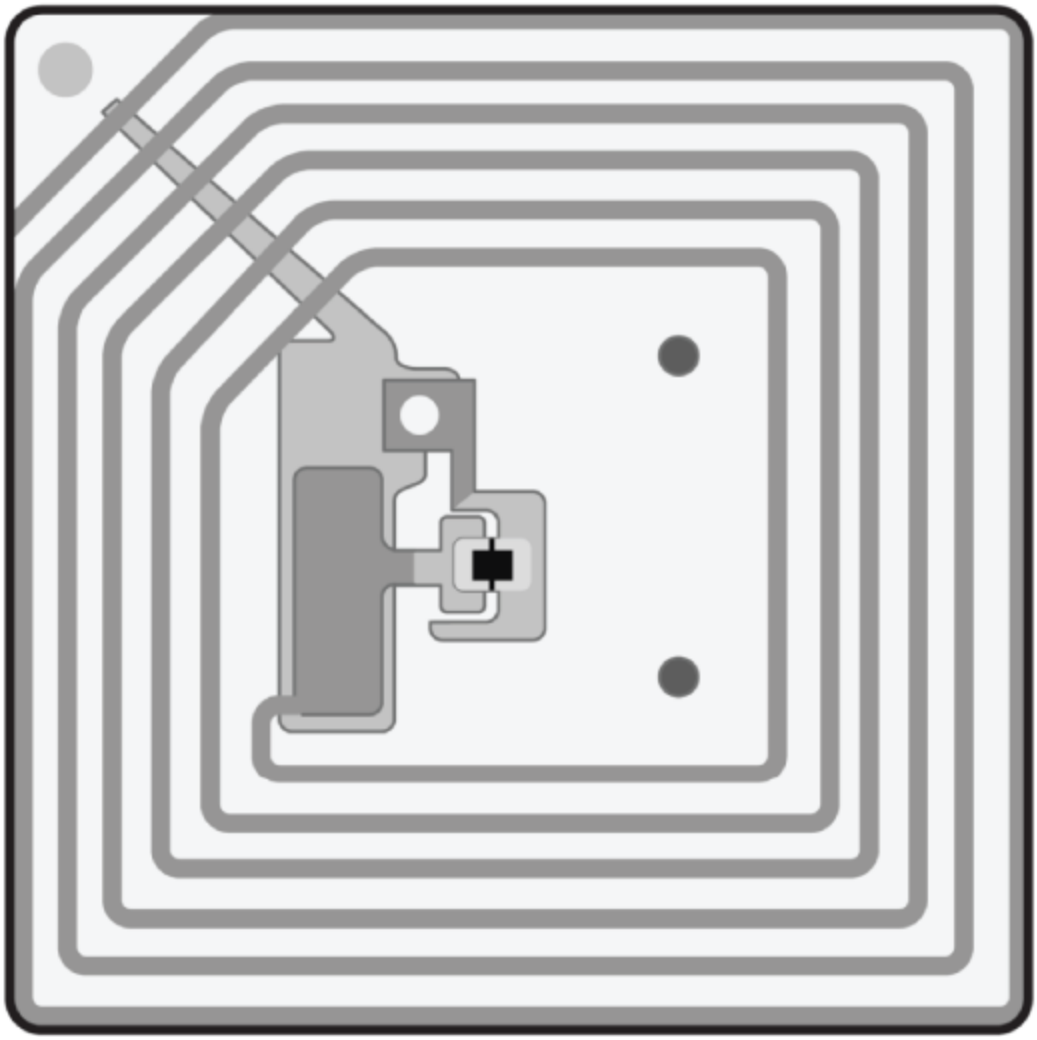


FIGURE 11.7 An RFID antenna

Adapted from http://electrosome.com/rfid-radio-frequency-identification

There is some concern that RFID can be a privacy-violating technology. If you are in possession of a device with an RFID chip, then anyone with an RFID reader can take note of the signal from your chip. When an RFID chip is awakened or responds to being near a reader, the chip (also called the RFID tag) transmits a unique code or serial number. That unique number is meaningless without the corresponding database that associates the number with the specific object (or person). However, if you are the only one around and someone detects your RFID chip code, then they can associate you and/or your device with that code for all future detections of the same code.

NFC

Near-field communication (NFC) is a standard that establishes radio communications between devices in close proximity (4 centimeters or less versus meters for passive RFID). It lets you perform a type of automatic synchronization and association between devices by touching them together or bringing them within centimeters of one another. NFC can be implemented as a field-powered or field-triggered device. NFC is a derivative technology from RFID and is a form of field-powered or manually triggered device.

NFC is commonly found on smartphones and many mobile device accessories. It's often used to perform device-to-device data exchanges, set up direct communications, or access more complex services such as WPA2/WPA3 wireless networks by linking with the WAP via NFC. Many contactless payment systems are based on NFC. NFC can function just like RFID (such as when using an NFC tile or sticker) or support more complex interactions. NFC chips can support challenge-response dialogs and even use public key infrastructure (PKI) encryption solutions.

NFC attacks can include AitM attacks, eavesdropping, data manipulation, and replay attacks. So, while some NFC implementations support reliable authentication and encryption, not all of them do. A best practice is to leave NFC features disabled until they need to be used.

Wireless Attacks

Wireless networking has become common on both corporate and home networks. Even with wireless security present, wireless attacks can still occur.

Wi-Fi Scanners

War driving is someone using a detection tool to look for wireless networking signals, often ones they aren't authorized to access. The name comes from the legacy attack concept of *war dialing*, which was used to discover active computer modems by dialing all the numbers in a prefix or an area code. War driving can be performed with a dedicated handheld detector, with a *mobile device* with Wi-Fi capabilities, with a notebook that has a wireless network card, or even with a drone (*war flying*). It can be

performed using native features of the OS or using specialized scanning and detecting tools (aka wireless scanners).

A *wireless scanner* is used to detect the presence of a wireless network. Any active wireless network that is not enclosed in a Faraday cage can be detected, since the base station will be transmitting radio waves, even those with SSID broadcast disabled.

A wireless scanner is able to determine whether there are wireless networks in the area, what frequency and channel they are using, the SSID, and what type of encryption is in use (if any). A wireless cracker can be used to break the encryption of WEP and WPA networks. WPA2 networks might be vulnerable to Key Reinstallation AttaCKs (KRACK) if devices have not been updated since 2017.

Rogue Access Points

A rogue WAP may be planted by an employee for convenience, installed internally by a physical intruder, or operated externally by an attacker. Such unauthorized access points usually aren't configured for security, or, if they are, they aren't configured properly or in line with the organization's approved access points. Rogue WAPs should be discovered and removed to eliminate an unregulated access path into your otherwise secured network.

A rogue WAP or false WAP can be deployed by an attacker externally to target your existing wireless clients or future visiting wireless clients. An attack against existing wireless clients requires that the rogue WAP be configured to duplicate the SSID, MAC address, and wireless channel of the valid WAP, although operating at a higher power rating. This may cause clients with saved wireless profiles to inadvertently select or prefer to connect to the rogue WAP instead of the valid original WAP.

A second method used by a rogue WAP focuses on attracting new visiting wireless clients. This type of rogue WAP is configured with a social engineering trick by setting the SSID to an alternate name that appears legitimate or even preferred over the original valid wireless network's SSID. The rogue WAP's MAC address and channel do not need to be clones of the original WAP.

The defense against rogue WAPs is to operate a wireless intrusion detection system (WIDS) to monitor the wireless signals for abuses, such as newly

appearing WAPs, especially those operating with mimicked or similar SSID and MAC values.

An administrator or security team member could attempt to locate rogue WAPs through the use of a wireless scanner and a directional antenna to perform triangulation. Once a rogue device is located, the investigation can turn to figuring out how it got there and who was responsible.

For clients, the best option is to connect a VPN across the wireless link, and only if the VPN connection is established successfully should the wireless link be used. VPNs can be set up in private networks for local wireless clients, or a public VPN provider can be used when connecting to public wireless networks.

Evil Twin

Evil twin is an attack in which a threat actor operates a false access point that will automatically clone or twin the identity of an access point based on a client device's request to connect. Each time a typical device successfully connects to a wireless network, it retains a wireless profile in its history. These wireless profiles are used to reconnect to a network automatically whenever the device is within range of the related base station. Each time the wireless adapter is enabled on a device, it sends out reconnection requests to each of the networks in its wireless profile history. These reconnect requests include the original base station's MAC address and the network's SSID. The evil twin attack system eavesdrops on the wireless signal for these reconnect requests. Once the evil twin sees a reconnect request, it spoofs its identity with those parameters and offers a plaintext connection to the client. The client accepts the request and establishes a connection with the false evil twin base station. This enables the malicious actor to eavesdrop on communications through an AitM attack, which could lead to session hijacking, data manipulation credential theft, and identity theft.

This attack works because authentication and encryption are managed by the base station, not enforced by the client. Thus, even though the client's wireless profile will include authentication credentials and encryption information, the client will accept whatever type of connection is offered by the base station, including plaintext.

To defend against evil twin attacks, pay attention to the wireless network your devices connect to. If you connect to a network that you know is not located nearby, it may be a sign that you are under attack. Disconnect and go elsewhere for internet access. You should also prune unnecessary and old wireless profiles from your history list to give attackers fewer options to target.

You can be easily fooled into thinking that you are connected to a proper and valid base station or connected to a false one. On most systems, you can check to see what if any communication security (i.e., encryption) is currently in use. If your network connection is not secure, you can either disconnect and go elsewhere or connect to a VPN. We always recommend attempting to connect to a VPN when using a wireless connection, even if your network properties show a valid security type.

Disassociation

Disassociation is one of the many types of wireless management frames. A disassociation frame is used to disconnect a client from one WAP as it is connecting to another WAP in the same ESSID network coverage area. If used maliciously, the client loses their wireless link.

A similar attack can be performed using a *deauthentication* packet. This packet is normally used immediately after a client initiates WAP authentication but fails to provide proper credentials. However, if sent at any time during a connected session, the client immediately disconnects as if its authentication did fail.

These management frames can be used in several forms of wireless attacks, including the following:

- For networks with hidden SSIDs, a disassociation packet with a MAC address spoofed as that of the WAP is sent to a connected client that causes the client to lose its connection and then send a Reassociation Request packet (in an attempt to reestablish a connection), which includes the SSID in the clear.
- An attack can send repeated disassociation frames to a client to prevent reassociation, thus causing a DoS.

- A session hijack event can be initiated by using disassociation frames to keep the client disconnected while the attacker impersonates the client and takes over their wireless session with the WAP.
- An AitM attack can be implemented by using a disassociation frame to disconnect a client. Then the attacker provides a stronger signal from their rogue/fake WAP using the same SSID and MAC as the original WAP; once the client connects to the false WAP, the attacker connects to the valid WAP.

The main defense against these attacks is to use WAP3 and/or operate a WIDS, which monitors for wireless abuses.

Jamming

Jamming is the transmission of radio signals to intentionally prevent or interfere with communications by decreasing the effective signal-to-noise ratio. To avoid or minimize interference and jamming, start by adjusting the physical location of devices. Next, check for devices using the same frequency and/or channel (i.e., signal configuration). If there are conflicts, change the frequency or channel in use on devices you control. If an interference attack is occurring, try to triangulate the source of the attack and take appropriate steps to address the concern—that is, contact law enforcement if the source of the problem is outside of your physical location.

Initialization Vector (IV) Abuse

An *initialization vector (IV)* is a mathematical and cryptographic term for a random number. Most modern crypto functions use IVs to increase their security by reducing predictability and repeatability. An IV becomes a point of weakness when it's too short, exchanged in plaintext, or selected improperly. One example of an IV attack is cracking WEP encryption using the `wesside-ng` tool from the Aircrack-ng suite at http://aircrack-ng.org.

Replay

A *replay attack* is the retransmission of captured communications in the hope of gaining access to the targeted system. Replay attacks attempt to

reestablish a communication session by replaying (i.e., retransmitting) captured traffic against a system. This may grant an adversary access into an account without the attacker possessing the account's actual credentials.

The replay attack concept is also used against cryptographic algorithms that don't incorporate temporal protections. In this attack, the malicious individual intercepts an encrypted message between two parties (often a request for authentication) and then later “replays” the captured message to open a new session.

Many wireless replay attack variants exist. They include capturing new connection requests of a typical client and then replaying that connect request to fool the base station into responding as if another new client connection request was initiated. Wireless replay attacks can also focus on DoS by retransmitting connection requests or resource requests of the base station to keep it busy focusing on managing new connections rather than maintaining and providing service for existing connections.

Wireless replay attacks can be mitigated by keeping the firmware of the base station updated. A WIDS will be able to detect such abuses and inform the administrators promptly about the situation. Additional defenses include using one-time authentication mechanisms, a timestamp and expiration period in each message, using challenge-response based authentication, and using sequenced session identification.

Satellite Communications

Satellite communications are primarily based on transmitting radio waves between terrestrial locations and an orbiting artificial satellite. Satellites are used to support telephone, television, radio, internet, and military communications. Satellites can be positioned in three primary orbits: *low Earth orbit (LEO)*, 160–2,000 km, *medium Earth orbit (MEO)*, 2,000–35,786 km, and *geostationary orbit (GEO)*, 35,786 km.

LEO satellites often have stronger signals than other orbits, but they do not remain in the same position over the earth, so multiple devices must be used to maintain coverage. Starlink (from SpaceX) is an example of a LEO satellite-based internet service. Starlink has plans to deploy a constellation

of over 40,000 satellites to provide global coverage of their internet from space service.

MEO satellites are in the sky above a terrestrial location for longer than a LEO satellite. Individual MEO satellites also usually have a larger transmission footprint (area of the earth covered by its transmitter/receiver) than that of LEO satellites. However, due to the higher orbit, there is additional delay and a weaker signal from MEO satellites.

GEO satellites appear motionless in the sky, as they are rotating around the earth at the same angular velocity as the earth rotates. Thus, GEO satellites maintain a fixed position above a terrestrial location. GEO satellites have a larger transmission footprint than MEO satellites but also a higher latency. But GEO satellites do not require that a ground station track the movement of the satellite across the sky, as is necessary with LEO and MEO satellites, so GEO ground stations can use fixed antennas.

Cellular Networks

A *cellular network*, *mobile network*, or wireless network is the primary communications technology that is used by many mobile devices, especially cell phones and smartphones. The network is organized around areas of access called cells, which are centered around a primary transceiver, known as a cell site, cell tower, or base station. The services provided over cellular networks are often referred to by a generational code, such as 2G, 3G, 4G, and 5G.

Generally, cellular service is encrypted, but only while the communication is being transmitted from the mobile device to a transmission tower. Communications are effectively plaintext once they are being transmitted over wires. So, avoid performing any task over cellular that is sensitive or confidential in nature. Use an encrypted communications application to pre-encrypt communications before transmitting them over a cellular connection, such as TLS or a VPN.

4G has been in use since the early 2000s and most cellular devices support 4G communications. The 4G standard allows for mobile devices to achieve 100 Mbps, whereas stationary devices can reach 1 Gbps. 4G is primarily using IP-based communications for both voice and data, rather than the

traditional circuit-switching telephony services of the past. 4G is provided by various transmission systems, the most common being LTE, followed by WiMAX.

5G is the latest mobile service technology that is available for use on some mobile phones, tablets, and other equipment. Many ICS, IoT, and specialty devices may have embedded 5G capabilities. 5G uses higher frequencies than previous cellular technologies, which has allowed for higher transmission speeds (up to 10 Gbps) but at a reduced distance. Organizations need to be aware of when and where 5G is available for use and enforce security requirements on such communications.

There are a few key issues to keep in mind with regard to cell phone wireless transmissions. First, communications over a cell phone provider's network, whether voice, text, or data, are not necessarily secure. Second, with specific wireless-sniffing equipment, your cell phone transmissions can be intercepted. In fact, your provider's towers can be simulated to conduct adversary-in-the-middle attacks. Third, using your cell phone connectivity to access the Internet or your office network provides attackers with yet another potential avenue of attack, access, and compromise. Many of these devices can potentially act as bridges, creating insecure access into a company network.

Content Distribution Networks (CDNs)

A *content distribution network (CDN)*, or *content delivery network*, is a collection of resource services deployed in numerous data centers across the Internet to provide low latency, high performance, and high availability of the hosted content. CDNs provide the desired multimedia performance quality demanded by customers through the concept of distributed data hosts. Rather than having media content stored in a single central location to be transmitted to all parts of the Internet, the media is distributed to numerous geographically distributed pre-staging internet locations that are closer to groups of customers. This results in a type of geographic and logical load balancing (see Chapter 12). No one server or cluster of servers will be strained under the load of all resource requests, and the hosting servers are located closer to the requesting customers. The overall result is lower-latency and higher-quality throughput. There are many CDN service

providers, including Cloudflare, Akamai, Amazon CloudFront, and CacheFly.

Although most CDNs focus on the physical distribution of servers, client-based CDN is also possible. This is often referred as *P2P (peer-to-peer)*. The most widely recognized P2P CDN is BitTorrent.

A service delivery platform (SDP) is a collection of components that provide the architecture for service delivery. SDP is often used in relation to telecommunications, but it can be used in many contexts, including VoIP, Internet TV, SaaS, and online gaming. An SDP is similar to a content delivery network (CDN), as both are designed for the support of and efficient delivery of a resource (such as services of an SDP and multimedia of a CDN). The goal of an SDP is to provide transparent communication services to other content or service providers. Both SDPs and CDNs can be implemented using microservices.

Secure Network Components

There are two basic types of private network segments: intranets and extranets. An *intranet* is a private network (i.e., LAN) that is often designed to host information services privately, similar to services found on the Internet. Networks that rely on external servers (in other words, ones positioned on the public internet) to provide information services for internal use are not considered intranets. Intranets provide users with access to the web, email, and other services on internal servers that are not accessible to anyone outside the private network.

An *extranet* is a cross between the Internet and an intranet. An extranet is a section of an organization's network that has been sectioned off so that it acts as an intranet for the private network but also serves information to authorized outsiders or external entities. An extranet is often reserved for use by specific partners, suppliers, distributors, remote salesforce, or select customers. An extranet for public consumption is typically labeled a screened subnet or perimeter network.

A *screened subnet* (previously known as a demilitarized zone [DMZ]) is a special-purpose extranet that is designed specifically for low-trust and unknown users to access specific systems, such as the public accessing a

web server. It can be implemented with two firewalls or one multihomed firewall. The two firewall deployment method positions one firewall between the screened subnet and the Internet and the second between the screened subnet and the intranet. This positions the subnet for outside access as a buffer between the Internet and the intranet, and the firewalls bounding the subnet effectively filter or screen all communications related to it. The multihomed firewall deployment method uses a single firewall with one interface connected to the Internet, a second interface to the screened subnet, and a third interface to the intranet.

A *screened host* is a firewall-protected system logically positioned just inside a network segment. All inbound traffic is routed to the screened host, which in turn acts as a proxy for all the trusted systems within the private network. It is responsible for filtering traffic coming into the private network as well as for protecting the identity of the internal system.



East-west traffic refers to the traffic flow that occurs within a specific network, data center, or cloud environment. *North-south traffic* refers to the traffic flow that occurs inbound or outbound between internal systems and external systems.

Secure Operation of Hardware

Strong familiarity with secure network components can assist you in designing an IT infrastructure that avoids single points of failure and provides strong support for availability. Part of operating hardware is to ensure that it is reliable and sufficient to support business operations. Some of the issues to consider in this regard include redundant power, warranty, and support.

Computer systems don't work without power. Providing reliable power is essential for a reliable IT/IS infrastructure. The concepts of surge protectors and UPSs were covered in Chapter 10, “Physical Security Requirements,” but another option you should consider is the deployment of redundant power supplies. Most deployments of failover power supplies are configured so that both provide half the power consumed by the system.

But in the event of a failure of one, the other can take over to provide 100 percent of the system's power needs. Some solutions offer hot swapping support so that failed supplies can be replaced or lower-capacity supplies can be swapped out with those with higher capacity.

The majority of equipment that is purchased and deployed today will likely operate without issue for years. However, it is still possible for devices to fail, causing excessive downtime or data loss. These problems can be minimized with planning and preparation, such as implementing redundancy and avoiding single-point-of-failure deployments (see Chapter 18, “Disaster Recovery Planning”). However, that doesn't resolve the issue that you have a failed device. That's when a warranty or a return policy can be helpful. When acquiring equipment, always inquire about the warranty coverage and return policy restrictions. You may be able to get a refund or a replacement if the device fails within a specific time frame.

Another aspect of hardware management that might be undervalued is support. Many of the hardware products in use today, such as VPN appliances, firewalls, switches, routers, and WAPs, are quite advanced. Some might even require specialized training or certification just to configure, set up, and deploy. If your organization does not have staff with expertise and experience with a specific hardware device, then you will need to rely on the support services provided by the vendor. Therefore, when obtaining new equipment, inquire about the technical support services available and whether they are included with the product purchase or if such services require an additional fee, subscription, or contract.

Common Network Equipment

These are some of the typical hardware devices in a network:

Repeaters, Concentrators, and Amplifiers *Repeaters, concentrators, and amplifiers (RCAs)* are used to strengthen the communication signal over a cable segment as well as connect network segments that use the same protocol. RCAs operate at OSI Layer 1. Systems on either side of an RCA are part of the same collision domain and broadcast domain. Aka *line driver*.

Multiplexer A multiplexer, often abbreviated as MUX, is a digital electronic device that combines multiple input signals into a single output

signal for transmission over a shared medium.

Collision Domains vs. Broadcast Domains

A collision occurs when two systems transmit data at the same time onto a connection medium that supports only a single transmission path. A *collision domain* is the group of networked systems that could cause a collision if any two (or more) systems in that group transmitted simultaneously. Collision domains are divided by using any Layer 2 or higher device.

A broadcast occurs when a single system transmits data to all possible recipients. A *broadcast domain* is the group of networked systems in which all other members receive a broadcast signal when one of the members of the group transmits it. Usually, the term *broadcast domain* is used to refer specifically to Ethernet broadcast domains (and not the broadcast features of IPv4). Ethernet broadcast domains are divided by using any Layer 3 or higher device.

Hubs *Hubs* are used to connect multiple systems and connect network segments that use the same protocol. A hub is a multiport repeater. Hubs operate at OSI Layer 1. Systems on either side of a hub are part of the same collision and broadcast domains.

Modems A traditional landline dial-up *modem* (modulator-demodulator) is a communications device that covers or modulates between an analog carrier signal and digital information to support computer communications of PSTN lines. From about 1960 until the mid-1990s, modems were a common means of WAN communications. Modems have generally been replaced by digital broadband technologies, including cable modems, fiber-optic modems, DSL modems, satellite modems, 802.11 wireless, and various forms of wireless modems.



The term *modem* is used incorrectly on any device that does not actually perform modulation. Most modern devices labeled as modems (cable, DSL, wireless, etc.) are routers, not modems. Integrated cable modem routers contain both functionalities of a modem and router in one device.

Bridges A *bridge* is used to connect two networks together—even networks of different topologies, cabling types, and speeds—to connect network segments that use the same protocol. A bridge forwards traffic from one network to another. Bridges that connect networks using different transmission speeds may have a buffer to store packets until they can be forwarded to the slower network. This is known as a *store-and-forward device*. Bridges operate at OSI Layer 2. Bridges were primarily used to connect hub networks together and thus have mostly been replaced by switches.

Switches *Switches* manage the transmission of frames via MAC address. Switches can also create separate broadcast domains when used to create VLANs (see Chapter 12). Switches operate primarily at OSI Layer 2. When switches have additional features, such as routing among VLANs, they can operate at OSI Layer 3 as well, known as Layer 3 switches.



MPLS (Multiprotocol Label Switching) is a high-throughput, high-performance network technology that directs data across a network based on short path labels rather than longer network addresses. This technique saves significant time over traditional IP-based routing processes, which can be quite complex. Furthermore, MPLS is designed to handle a wide range of protocols through encapsulation.

Routers *Routers* are used to control traffic flow on networks and are often used to connect similar networks and control traffic flow between the two. Routers manage traffic based on logical IP addressing. They can

function using statically defined routing tables, or they can employ a dynamic routing system. Routers operate at OSI Layer 3.

LAN Extenders A *LAN extender* is a remote access, multilayer switch used to connect distant networks over WAN links. Aka *WAN switch* or *WAN router*.

Jumpbox A *jump server* or *jumpbox* is a remote access system deployed to make accessing a specific system or network easier or more secure. A jump server is often deployed in extranets, screened subnets, or cloud networks where a standard direct link or private channel is not available or is not considered safe. A jump server can be deployed to receive an in-band VPN connection, but most are configured to accept out-of-band connections, such as direct dial-up or internet-origin broadband links. No matter what form of connection is used to access the jump server, it is important to ensure that only encrypted connections are employed.

Sensor A *sensor* collects information and then transmits it back to a central system for storage and analysis. Sensors are common elements of fog computing, ICS, IoT, IDS/IPS, and SIEM/security orchestration, automation, and response (SOAR) solutions. Many sensors are based on an SoC (system on a chip).

Collector A *security collector* is any system that gathers data into a log or record file. A collector's function is similar to the functions of auditing, logging, and monitoring. A collector watches for a specific activity, event, or traffic and then records the information into a record file.

Aggregators *Aggregators* are a type of multiplexor. Numerous inputs are received and directed or transmitted to a single destination. MPLS is an example of an aggregator. Some IDSs/IPSs use aggregators to collect or receive input from numerous sensors and collectors to integrate the data into a single data stream for analysis and processing.

System on a Chip (SoC)

A *system on a chip (SoC)* is an integrated circuit (IC) or chip that has all of the elements of a computer integrated into a single chip. This often includes the main CPU, RAM, GPU, Wi-Fi, wired networking, peripheral interfaces (such as USB), and power management. In most cases, the only item missing from an SoC compared to a full computer is bulk storage. Often, a bulk storage device must be attached or connected to the SoC to store its programs and other files since the SoC usually contains only enough memory to retain its own firmware or OS.

The security risks of an SoC include the fact that the firmware or OS of an SoC is often minimal, which leaves little room for most security features. An SoC may be able to filter input (such as by length or to escape metacharacters), reject unsigned code, provide basic firewall filtering, use communication encryption, and offer secure authentication. However, these features are not universally available on all SoC products. A few devices that use an SoC include the mini-computer Raspberry Pi, fitness trackers, smart watches, and some smartphones.

Network Access Control

Network access control (NAC) is the concept of controlling access to an environment through strict adherence to and enforcement of security policy. NAC is meant to be an automated detection and response system that can react in real time to ensure that all monitored systems are current on patches and updates and are in compliance with the latest security configurations, as well as keep unauthorized devices out of the network. The goals of NAC are as follows:

- Prevent/reduce known attacks directly and zero-day indirectly
- Enforce security policy throughout the network
- Use identities to perform access control

The goals of NAC can be achieved through the use of strong, detailed security policies that define all aspects of security control, filtering, prevention, detection, and response for every device from client to server and for every internal or external communication.

Originally, 802.1X (which provides port-based NAC) was thought to embody NAC, but most supporters believe that 802.1X is only a simple form of NAC or just one optional component in a complete NAC solution.

NAC can be implemented with a preadmission philosophy or a postadmission philosophy, or aspects of both:

- The preadmission philosophy requires a system to meet all current security requirements (such as patch application and malware scanner updates) before it is allowed to communicate with the network.
- The postadmission philosophy allows and denies access based on user activity, which is based on a predefined authorization matrix.

NAC options include using a host/system agent (*agent-based*) or performing overall network monitoring and assessment (*agentless*). A typical operation of an agent-based NAC system would be to install a NAC monitoring agent on each managed system. The NAC agent retrieves a configuration file on a regular basis, possibly daily or upon network connection, to check the current configuration baseline requirements against the local system. If the system is not compliant, it can be quarantined into a remediation subnet where it can communicate only with the NAC server. The NAC agent can download and apply updates and configuration files to bring the system into compliance. Once compliance is achieved, the NAC agent returns the system to the normal production network.

NAC agents can be either dissolvable or permanent. A dissolvable NAC agent is usually written in a web/mobile language and is downloaded and executed to each local machine when the specific management web page is accessed (such as a captive portal). A dissolvable NAC agent can be set to run once and then terminate. A permanent NAC agent is installed onto the monitored system as a persistent software background service.

An agentless or network monitoring and assessment NAC solution performs port scans, service queries, and vulnerability scans against networked

systems from the NAC server to determine whether devices are authorized and baseline compliant. An agentless system requires an administrator to manually resolve any discovered issues.

NAC systems can be implemented using both physical and virtual solutions. Physical NAC appliances are dedicated hardware devices that are deployed at key points in the network infrastructure. These devices actively monitor and control network access based on established policies. Virtual NAC solutions are software-based implementations that can run on virtual machines or as part of existing network infrastructure. They provide flexibility and scalability, allowing organizations to deploy NAC capabilities without dedicated hardware.

Other issues around NAC include out-of-band versus in-band monitoring, as well as resolving any remediation, quarantine, or captive portal strategies. You should evaluate these and other NAC concerns before implementation.

Firewalls

Firewalls are essential tools in managing, controlling, and filtering network traffic. A firewall can be a hardware or software component designed to protect one network segment from another. Firewalls are deployed between areas of higher and lower trust, like a private network and a public network (such as the Internet), or between two network segments that have different security levels/domains/classifications. Many commercial firewalls are hardware-based and can be called hardware firewalls, appliance firewalls, or network firewalls.



A *virtual firewall* is a firewall created for use in a virtualized or hypervisor environment or the cloud. A virtual firewall is a software re-creation of an appliance firewall or a standard host-based firewall installed into a guest OS in a VM.

Firewalls filter traffic based on a defined set of rules, also called filters or access control lists. They are basically a set of instructions that are used to

distinguish authorized traffic from unauthorized and/or malicious traffic. Only authorized traffic is allowed to cross the security barrier provided by the firewall. A typical firewall is based on the deny-by-default or implicit deny security stance. Only communications that meet an explicit allow exception are transmitted toward their destination. This concept is also known as allow listing.

The typical actions of a filter rule are allow, deny, drop, alert, and/or log. Some firewalls use a first-match mechanism when applying rules. Allow rules enable the packet to continue toward its destination. Deny rules block the packet from going any further (effectively discarding it). When first-match is used, the first rule that applies to the packet is followed, but no other rules are considered. Thus, rules need to be placed in a priority order. A final rule is the deny-all rule so that nothing is allowed to traverse the firewall unless it is granted an explicit exception. However, some firewalls perform a consolidated or accumulated result of all the rules that match a packet. Such amalgamation firewalls do not have a written or specific deny-all rule—instead they use *implicit deny*. This method also ensures that only traffic meeting explicit allow rules (which is not explicitly denied) is allowed to pass.



Sometimes a firewall's rule set is referred to by the term *tuple*. Tuple is a mathematical term meaning a collection of related data items. Tuple is also used with databases, where it references a record or row in a table.

Firewalls are most effective against unrequested traffic, initiations from outside the private network, and known malicious data, messages, or packets based on content, application, protocol, port, or source address. Most firewalls offer extensive logging, auditing, and monitoring capabilities as well as alarms and basic IDS functions.



A *bastion host* is a system specifically designed to withstand attacks, such as a firewall appliance or a jump server. The word *bastion* comes from medieval castle architecture. A bastion guardhouse was positioned in front of the main entrance (typically on the other side of the moat from the castle, where it controlled entrance onto the drawbridge) to serve as a first layer of protection. Using this term to describe a host indicates that the system is acting as a sacrificial host that will receive all inbound attacks.

Common ingress filters and egress filters can be used to block spoofed packets that often relate to malware, botnets, and other unwanted activities. Examples include the following:

- Blocking inbound packets claiming to have an internal source address
- Blocking outbound packets claiming to have an external source address
- Blocking packets with source or destination addresses listed on a block list (a list of known malicious IP addresses)
- Blocking packets that have source or destination addresses from the local area network (LAN) but that haven't been officially assigned to a host



Remotely triggered black hole (RTBH) is an edge filtering concept to discard unwanted traffic based on source or destination address long before it reaches the destination.

Firewalls, on their own, are typically unable to directly block viruses or malicious code transmitted through otherwise authorized communication channels, prevent unauthorized but accidental or intended disclosure of information by users, prevent attacks by malicious users already behind the

firewall, or protect data after it passes out of or into the private network. However, you can add these features through special add-in modules or companion products, such as antimalware scanners, DLP, and IDS tools. Firewall appliances are available that are preconfigured to perform all (or most) of these add-on functions natively. These types of firewall can be called a multifunction device (MFD), a unified threat management (UTM) device, or a next-generation firewall (NGFW).

In addition to logging network traffic activity, firewalls should log several other events:

- A reboot of the firewall
- Proxies or dependencies unable to start or not starting
- Proxies or other important services crashing or restarting
- Changes to the firewall configuration file
- A configuration or system error while the firewall is running

Firewalls are only one part of an overall security solution. With a firewall, many of the security mechanisms are concentrated in one place, and thus a firewall can be a single point of failure. Firewall failure is most commonly caused by human error and misconfiguration. Firewalls provide protection only against traffic that crosses the firewall.

There are several basic types of firewalls, which can be mixed to create hybrid or complex firewall solutions:

Static Packet-Filtering Firewalls A *static packet-filtering firewall* (aka *screening router)* filters traffic by examining data from a message header. Usually, the rules are concerned with source and destination IP address (Layer 3) and port numbers (Layer 4). This is also a type of stateless firewall since each packet is evaluated individually rather than in context (which is performed by a stateful firewall).



A *stateless firewall* analyzes packets on an individual basis against the filtering access control lists (ACLs) or rules. The context of the communication (that is, any previous packets) is not used to make an allow or deny decision on the current packet.

Application-Level Firewalls An *application-level firewall* filters traffic based on a single internet service, protocol, or application. Application-level firewalls operate at the Application Layer (Layer 7) of the OSI model. An example is the web application firewall (WAF). This firewall may be implemented stateless or stateful.



A *web application firewall (WAF)* is an appliance, server add-on, virtual service, or system filter that defines a strict set of communication rules for communications to and from a website. It's intended to prevent web application attacks.

A *next-generation secure web gateway (SWG, NGSWG, NG-SWG)* is a variation of and combination of the ideas of an NGFW and a WAF. An SWG is a cloud-based web gateway solution that is often tied to a subscription service that provides ongoing updates to filters and detection databases. This cloud-based firewall is designed to provide filtering services between CSP-based resources and on-premises systems. An SWG/NG-SWG often supports standard WAF functions; TLS decryption; cloud access security broker (CASB) functions; advanced threat protection (ATP), such as sandboxing and ML-based threat detection; DLP; rich metadata about traffic; and detailed logging and reporting.

Circuit-Level Firewalls *Circuit-level firewalls* (aka *circuit proxies)* are used to establish communication sessions between trusted partners. In theory, they operate at the Session Layer (Layer 5) of the OSI model (although in reality, they operate in relation to the establishment of TCP

sessions at the Transport Layer [Layer 4]). *SOCKS* (from Socket Secure, as in TCP/IP ports) is a common implementation of a circuit-level firewall. Circuit-level firewalls focus on the establishment of the circuit (or session), not the content of traffic, based on simple rules for IP and port, using captive portals, requiring port authentication via 802.1X, or more complex elements such as context- or attribute-based access control. This is also a type of stateless firewall.



A *TCP Wrapper* is an application that can serve as a basic firewall by restricting access to ports and resources based on user IDs or system IDs. Using TCP Wrappers is a form of port-based access control.

Stateful Inspection Firewalls *Stateful inspection firewalls* (aka *dynamic packet filtering firewalls*) evaluate the state, session, or context of network traffic. By examining source and destination addresses, application usage, source of origin (i.e., local or remote, physical port, or even routed path/vector), and the relationship between current packets and the previous packets of the same session, stateful inspection firewalls are able to grant a broader range of access for authorized users and activities and actively watch for and block unauthorized users and activities. Stateful inspection firewalls operate at OSI layers 3 and up.

A stateful inspection firewall is aware that any valid outbound communication (especially related to TCP) will trigger a corresponding response or reply from the external entity. Thus, this type of firewall automatically creates a temporary response rule for the request. But that rule exists only as long as the conversation is taking place.

Additionally, stateful inspection firewalls can retain knowledge of previous packets in a conversation to detect unwanted or malicious traffic that isn't noticeable or detectable when evaluating only individual packets. This is known as context analysis or contextual analysis. A stateful inspection firewall may also perform deep packet inspection (DPI), which is the analysis of the payload or content of a packet.



Deep packet inspection (DPI), *payload inspection*, or *content filtering* is the means to evaluate and filter the payload contents of a communication rather than only on the header values. DPI can also be known as complete packet inspection and information extraction. DPI filtering is able to block domain names, malware, spam, malicious scripts, abusive content, or other identifiable elements in the payload of a communication. DPI is often integrated with application-layer firewalls and/or stateful inspection firewalls.

Next-Generation Firewalls (NGFWs) A *next-generation firewall (NGFW)* is a *multifunction device (MFD)* or *unified threat management (UTM)* composed of several security features in addition to a firewall; integrated components can include application filtering, deep packet inspection, TLS offloading and/or inspection (aka TLS termination proxy), domain name and URL filtering, IDS, IPS, web content filtering, QoS management, bandwidth throttling/management, NAT, VPN anchoring, authentication services, identity management, and antivirus/antimalware scanning.



A *host-based firewall*, local, software, or personal firewall, is a security application that is installed on client systems. A host-based firewall provides protection for the local system from the activities of the user and from communications from the network or the Internet. It can often limit communications of installed applications and protocols and can block externally initiated connections. A host-based firewall can be a simple static filtering firewall, stateful inspection, or even an NGFW.

Internal Segmentation Firewall (ISFW) An *internal segmentation firewall (ISFW)* is a firewall deployed between internal network segments or company divisions. Its purpose is to prevent the further spread of malicious code or harmful protocols already within the private network. With an ISFW, network segments can be created without resorting to air

gaps, VLANs, or subnet divisions. An ISFW is commonly used in micro-segmentation architectures.

Proxy

A *proxy server* is a variation of an application-level firewall or circuit-level firewall. A proxy server is used to mediate between clients and servers. Proxies are most often used in the context of providing clients on a private network with internet access while protecting the identity of the clients. Often a proxy serves as a barrier against external threats to internal clients by accepting requests from clients, altering the source address of the requester, maintaining a mapping of requests to clients, and sending the altered request packets out. Once a reply is received, the proxy server determines which client it is destined for by reviewing its mappings and then sends the packets to the originally requesting client. This is effectively NAT (see Chapter 12). In addition to features such as NAT, proxy servers can provide caching and site or content filtering.

A *forward proxy* is a standard or common proxy that acts as an intermediary for queries of external resources. A forward proxy handles queries from internal clients when accessing outside services.

A *reverse proxy* provides the opposite function of a forward proxy; it handles inbound requests from external systems to internally located services. A reverse proxy is similar to the functions of port forwarding and static NAT. A reverse proxy is sometimes used on the border of a screened subnet to use private IP addresses on resource servers but allows for visitors from the public internet.

If a client is not configured (Figure 11.8, left) to send queries directly to a proxy, but the network routes outbound traffic to a proxy anyway, then a *transparent proxy* is in use. A *nontransparent proxy* is in use when a client is configured (Figure 11.8, right) to send outbound queries directly to a proxy. The settings for a nontransparent proxy can be set manually or using a *proxy auto-config (PAC)* file. PAC can be implemented with a script or via DHCP.

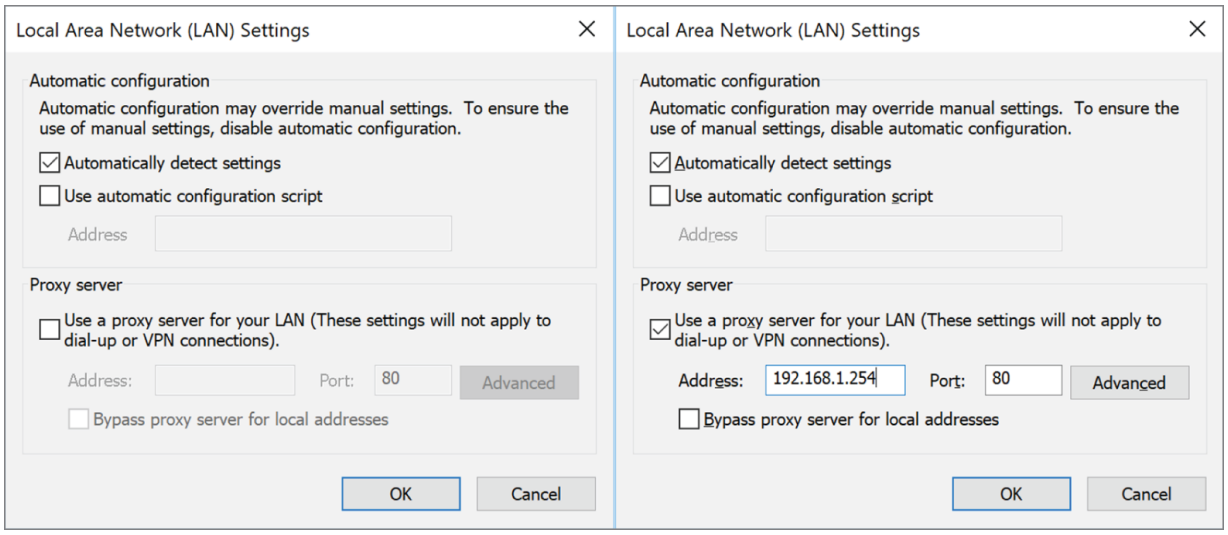


FIGURE 11.8 The configuration dialog boxes for a transparent (left) versus a nontransparent (right) proxy

Content/URL Filter

Content filtering or content inspection is the security-filtering function in which the contents of the application protocol payload are inspected. Often, such inspection is based on keyword matching. A primary block list of unwanted terms, addresses, or URLs is used to control what is or isn't allowed to reach a user. This is sometimes known as *deep packet inspection*. *Malware inspection* is the use of a malware scanner to detect unwanted software content in network traffic.

URL filtering, also known as *web filtering*, is the act of blocking access to a site based on all or part of the URL used to request access. URL filtering can focus on all or part of a fully qualified domain name (FQDN), specific pathnames, filenames, file extensions, or entire URLs. Many URL-filtering tools can obtain updated primary URL block lists from vendors as well as allow administrators to add or remove URLs from a custom list.

A *web security gateway* is a device that is a web-content filter (often URL and content keyword–based) that also supports malware scanning. Some web security gateways incorporate non-web features as well, including instant messaging (IM) filtering, email filtering, spam blocking, and spoofing detection. Thus, some are considered to be UTMs or NGFWs.

Endpoint Security

Managing network security with filtering devices such as firewalls and proxies is important, but you must not overlook the need for endpoint security. *Endpoint security* is the concept that each individual device must maintain local security, whether or not its network or telecommunications channels also provide security. Sometimes, this is expressed as "The end device is responsible for its own security" or host-based security. However, a clearer perspective is that any weakness in a network, whether on the border, on a server, or on a client, presents a risk to all elements within the organization.

As computing has evolved from a host/terminal model (where users could be physically distributed, but all functions, activity, data, and resources reside on a single centralized system) to a client/server model (where users operate independent, fully functional desktop computers but also access services and resources on networked servers), security controls and concepts have had to evolve to follow suit. This means that clients have computing and storage capabilities and, typically, multiple servers do likewise. The concept of a *client/server model* network is also known as a *distributed system* or a *distributed architecture*. Thus, security must be addressed everywhere instead of at a single centralized host. From a security standpoint, this means that because processing and storage are distributed on multiple clients and servers, all those computers must be properly secured and protected. It also means that the network links between clients and servers (and in some cases, these links may not be purely local) must also be secured and protected. When evaluating security architecture, be sure to include an assessment of the needs and risks related to distributed architectures.

Distributed architectures are prone to vulnerabilities that are unthinkable in monolithic host/terminal systems. Desktop systems can contain sensitive information that may be at some risk of being exposed and must, therefore, be protected. Individual users may lack general security savvy or awareness, and therefore, the underlying architecture has to compensate for those deficiencies. Desktop PCs, workstations, and laptops can provide avenues of access to critical information systems elsewhere in a distributed environment because users require access to networked servers and services to do their jobs. By permitting user machines to access a network and its distributed resources, organizations must also recognize that those user

machines can become threats if they are misused or compromised. Such software and system vulnerabilities and threats must be assessed and addressed properly.

Communications equipment can also provide unwanted points of entry into a distributed environment. For example, modems attached to a desktop machine that's also attached to an organization's network can make that network vulnerable to dial-in attacks. There is also a risk that wireless adapters on client systems can be used to create open networks. Likewise, users who download data from the Internet increase the risk of infecting their own and other systems with malicious code, Trojan horses, and so forth. Desktops, laptops, tablets, mobile phones, and workstations—and associated disks or other storage devices—may not be secure from physical intrusion or theft. Finally, when data resides only on client machines, it may not be secured with a proper backup (it's often the case that although servers are backed up routinely, the same is not true for client computers).

You should see that the foregoing litany of potential vulnerabilities in distributed architectures means that such environments require numerous safeguards to implement appropriate security and to ensure that such vulnerabilities are eliminated, mitigated, or remedied. Clients must be subjected to policies that impose safeguards on their contents and their users' activities.

These include the following:

- Email must be screened so that it cannot become a vector for infection by malicious software; email should also be subject to policies that govern appropriate use and limit potential liability.
- Download/upload policies must be created so that incoming and outgoing data is screened and suspect materials are blocked.
- Systems must be subject to robust access controls, which may include multifactor authentication and/or biometrics to restrict access to end-user devices and to prevent unauthorized access to servers and services.
- Restricted user-interface mechanisms and database management systems should be installed, and their use required, to restrict and

manage access to critical information so that users have minimal but necessary access to sensitive resources.

- File encryption may be appropriate for files and data stored on client machines (indeed, drive-level encryption is a good idea for laptops and other mobile computing gear that is subject to loss or theft outside an organization's premises).
- Enforce screen savers after a timeout. This will hide any confidential materials behind a screen saver, which should then require a valid login to regain access to the desktop, applications, storage devices, and so forth.
- It's essential to separate and isolate processes that run in user and supervisory modes so that unauthorized and unwanted access to high-privilege processes and capabilities is prevented.
- Protection domains or network segments should be created so that the compromise of a client won't automatically compromise an entire network.
- Disks and other sensitive materials should be clearly labeled according to their security classification or organizational sensitivity; procedural processes and system controls should combine to help protect sensitive materials from unwanted or unauthorized access.
- Files on desktop machines, as well as files on servers, should be backed up—ideally, using some form of centralized backup utility that works with client agent software to identify and capture files from clients stored in a secure backup storage archive.
- Desktop users need regular security awareness training to maintain proper security awareness; they also need to be notified about potential threats and instructed on how to deal with them appropriately.
- Desktop computers and their storage media require protection against environmental hazards (temperature, humidity, power loss/fluctuation, and so forth).
- Desktop computers should be included in your organization's disaster recovery and business continuity planning because they're potentially

as important as (if not more important than) other systems and services in getting users back to work on other systems.

- Developers of custom software built-in and for distributed environments also need to take security into account, including using formal methods for development and deployment, such as code libraries, change control mechanisms, configuration management, and patch and update deployment.

In general, safeguarding distributed environments means understanding the vulnerabilities to which they're subject and applying appropriate safeguards. These can (and do) range from technology solutions and controls to policies and procedures that manage risk and seek to limit or avoid losses, damage, unwanted disclosure, and so on. Configuring security on numerous endpoint devices can be complex, time-consuming, and tedious. The use of system imaging of a properly configured primary device will ensure the application of a consistent baseline across the upgraded endpoint devices.

Endpoint detection and response (EDR) is a security mechanism that is an evolution of traditional antimalware products, IDS, and firewall solutions. EDR seeks to detect, record, evaluate, and respond to suspicious activities and events, which may be caused by problematic software or by valid and invalid users. It is a natural extension of continuous monitoring focusing on both the endpoint device itself and network communications reaching the local interface. Some EDR solutions employ an on-device analysis engine, whereas others report events back to a central analysis server or to a cloud solution. The goal of EDR is to detect abuses that are potentially more advanced than what can be detected by traditional antivirus programs or HIDSs, while optimizing the response time of incident response, discarding false positives, implementing blocking for advanced threats, and protecting against multiple threats occurring simultaneously and via various threat vectors.

A few related concepts to EDR include managed detection and response (MDR), endpoint protection platform (EPP), and extended detection and response (XDR). MDR focuses on threat detection and mediation but is not limited to the scope of endpoints. MDR is a service that attempts to monitor an IT environment in real time to quickly detect and resolve threats. Often, an MDR solution is a combination and integration of numerous

technologies, including SIEM, network traffic analysis (NTA), EDR, and IDS.

EPP is a variation of EDR, much like IPS is a variation of IDS. The focus of EPP is on four main security functions: predict, prevent, detect, and respond. Thus, EPP is the more active prevent and predict variation of the more passive EDR concept.

XDR is not so much another tool as the collection and integration of several concepts into a single solution. XDR components can vary between vendors, but they often include EDR, MDR, and EPP elements. Also, XDR is not solely focused on endpoints, but often includes NTA, NIDS, and NIPS functions as well.

From there, we might as well mention that a managed security service provider (MSSP) can provide XDR solutions that are centrally controlled and managed. MSSP solutions can be deployed fully on-premises, fully in the cloud, or as a hybrid structure. MSSP solutions can be overseen through an SOC, which is itself local or remote. Typically, working with an MSSP to provide EDR, MDR, EPP, or XDR services can allow an organization to gain the benefits of these advanced security products and leverage the experience and expertise of the MSSP's staff of security management and response professionals.

Cabling, Topology, and Transmission Media Technology

Establishing security on a network involves more than just managing the operating system and software. You must also address physical issues, including cabling, topology, and transmission media technology.

LANs vs. WANs

There are two basic types of networks: LANs and WANs. A *local area network (LAN)* is a network in a limited geographical area, typically spanning a single floor or building. *Wide area network (WAN)* is the term usually assigned to the long-distance connections between geographically remote networks.

Transmission Media

Transmission media refers to the physical pathways or channels through which data is transmitted from one location to another. The characteristics of transmission media impact the quality and reliability of signal propagation. The type of connectivity media employed in a network is important to the network's design, layout, and capabilities. Without the right transmission media, a network may not be able to span your entire enterprise, or it may not support the necessary traffic volume. In fact, the most common causes of network failure (in other words, violations of availability) are cable failures or misconfigurations. It is important for you to understand that different types of network devices and technologies are used with different types of cabling. Each cable type has unique useful lengths, throughput rates, and connectivity requirements.

Physical protection measures, such as using secure conduits, enclosures, and access controls, help prevent unauthorized tampering with transmission media. Implementing encryption technologies ensures that even if the transmission medium is physically accessed, the data remains secure and unreadable without the proper decryption keys.

In the realm of transmission media signal propagation quality, several factors come into play:

- *Attenuation:* This refers to the loss of signal strength as it travels through the medium. Over longer distances, attenuation can significantly reduce the signal's strength, potentially leading to degradation.
- *Interference:* Unwanted signals that disrupt the transmission fall under interference. Sources such as electromagnetic interference (EMI) and radio-frequency interference (RFI) can disturb the signal and impact the overall quality of communication.
- *Noise:* Unwanted random variations in the signal constitute noise. Noise can distort the original signal, complicating the accurate interpretation of transmitted data.
- *Jitter:* The variation in latency between different packets.

- *Bandwidth:* The range of frequencies that a medium can support is defined as bandwidth. A higher bandwidth allows for the transmission of more data, contributing to improved signal quality and faster communication.
- *Propagation delay or latency:* The time taken for a signal to travel from the sender to the receiver is known as propagation delay. Propagation delay directly influences the speed of communication, and in certain applications, minimizing this delay is crucial.

Understanding and addressing these factors are essential components of designing reliable and efficient transmission systems, ensuring optimal performance and data integrity in communication networks.

Remember that many forms of transmission media are not cables. This includes wireless, Bluetooth, Zigbee, and satellites, which were all discussed earlier in this chapter.

Coaxial Cable

Coaxial cable, also called *coax*, was a popular networking cable type used throughout the 1970s and 1980s. In the early 1990s, its use quickly declined because of the popularity and capabilities of twisted-pair wiring (explained in more detail later). In the 2020s, you are unlikely to encounter coax being used as a LAN network cable but may still see some use of it as an audio/visual connection cable (such as between an over-the-air antenna and your television) or as an internet access media (such as from the wall to your cable modem).

Coaxial cable has a center core of copper wire surrounded by a layer of insulation, which is in turn surrounded by a conductive braided shielding and encased in a final insulation sheath. There are two legacy types of coaxial cable: thinnet and thicknet. Thinnet (10Base2) was commonly used to connect systems to backbone trunks of thicknet cabling. Thinnet can span distances of 185 meters and provide throughput up to 10 Mbps. Thicknet (10Base5) can span 500 meters and provide throughput up to 10 Mbps. A more modern coax format is RG6, which is commonly used for various applications in telecommunications, audio/video, and broadband communication systems.

The most common problems with coax cable are as follows:

- Bending the coax cable past its maximum arc radius and thus breaking the center conductor
- Deploying the coax cable in length greater than its maximum recommended length (which is 185 meters for 10Base2 or 500 meters for 10Base5)
- Not properly terminating the ends of the coax cable with a 50-ohm BNC resistor
- Not grounding at least one end of a terminated coax cable

Baseband and Broadband Cables

The naming convention used to label most network cable technologies follows the syntax *XXyyyyZZ*. *XX* represents the maximum speed the cable type offers, such as 10 Mbps for a 10Base2 cable. The next series of letters, *yyyy*, represents the baseband or broadband aspect of the cable, such as baseband for a 10Base2 cable. *Baseband* cables can transmit only a single signal at a time, and *broadband* cables can transmit multiple signals simultaneously. Most networking cables are baseband cables. However, when used in specific configurations, coaxial cable can be used as a broadband connection, such as RG6 coax used with cable modems. *ZZ* either represents the maximum distance the cable can be used or acts as shorthand to represent the technology of the cable, such as the approximately 200 meters for 10Base2 cable (actually 185 meters, but it's rounded up to 200) or T or TX for twisted-pair in 100BaseT or 100BaseTX.

Twisted-Pair

Twisted-pair cabling is extremely thin and flexible compared to coaxial cable. It consists of four pairs of wires that are twisted around each other and then sheathed in a PVC insulator. If there is a metal foil wrapper around the wires underneath the external sheath, the wire is known as *shielded twisted-pair (STP)*. The foil provides additional protection from external EMI. Twisted-pair cabling without the foil is known as *unshielded twisted-pair (UTP)*.

The wires that make up UTP and STP are small, thin copper wires that are twisted in pairs. The twisting of the wires provides protection from external radio frequencies and electric and magnetic interference and reduces crosstalk between pairs. Crosstalk occurs when data transmitted over one set of wires is picked up by another set of wires due to radiating electromagnetic fields produced by the electrical current. Each wire pair within the cable is twisted at a different rate (in other words, twists per foot); thus, the signals traveling over one pair of wires cannot cross over onto another pair of wires (at least within the same cable). The tighter the twist (the more twists per foot), the more resistant the cable is to internal and external interference and crosstalk, and thus, the capacity for throughput (that is, higher bandwidth) is greater.

There are several classes of UTP cabling. The various categories are created through the use of tighter twists of the wire pairs, variations in the quality of the conductor, and variations in the quality of the external shielding. Table 11.4 shows the original UTP categories.

TABLE 11.4 UTP categories

UTP category	Throughput	Notes
Cat 1	1 Mbps	Primarily used for voice. Not suitable for networks, but usable by modems.
Cat 2	4 Mbps	Original Token Ring networks and host-to-terminal connections on mainframes.
Cat 3	10 Mbps	Primarily used in Ethernet networks (10BaseT) and as telephone cables.
Cat 4	16 Mbps	Primarily used in Token Ring networks.
Cat 5	100 Mbps	Used in 100BaseTX, FDDI, and ATM networks.
Cat 5e	1 Gbps	Gigabit Ethernet (1000BaseT).
Cat 6	1 Gbps	Gigabit Ethernet (10G Ethernet with 55-meter distance limit).
Cat 6a	10 Gbps	Gigabit Ethernet, 10G Ethernet.
Cat 7	10 Gbps	Gigabit Ethernet, 10G Ethernet.
Cat 8	40 Gbps	10G+ Ethernet.

The following problems are the most common with twisted-pair cabling:

- Using the wrong category of twisted-pair cable for high-throughput networking
- Deploying a twisted-pair cable longer than its maximum recommended length (in other words, 100 meters)
- Using UTP in environments with significant interference

Conductors

The distance limitations of conductor-based network cabling stem from the resistance of the metal used as a conductor. Copper, the most popular conductor, is one of the best and least expensive room-temperature conductors available. However, it is still resistant to the flow of electrons. This resistance results in a degradation of signal strength and quality over the length of the cable.

The maximum length defined for each cable type indicates the point at which the level of degradation could begin to interfere with the efficient transmission of data. This degradation of the signal is known as *attenuation*. It is often possible to use a cable segment that is longer than the cable is rated for, but the number of errors and retransmissions will be increased over that cable segment, ultimately resulting in poor network performance. Attenuation is more pronounced as the speed of the transmission increases. We recommend that you use shorter cable lengths as the speed of the transmission increases.

Long cable lengths can often be supplemented through the use of repeaters or concentrators. A repeater is a signal amplification device, much like the amplifier for your car or home stereo. The repeater boosts the signal strength of an incoming data stream and rebroadcasts it through its second port. A concentrator does the same thing except it has more than two ports. However, using more than four repeaters (or hubs) in a row is discouraged (see the sidebar “5-4-3 Rule”).

5-4-3 Rule

The 5-4-3 rule is used whenever Ethernet or other IEEE 802.3 shared-access networks are deployed using hubs and repeaters as network connection devices in a tree topology (in other words, a central trunk with various splitting branches). This rule defines the number of repeaters/concentrators and segments that can be used in a network design. The rule states that between any two nodes (a node can be any type of processing entity, such as a server, client, or router), there can be a maximum of five segments connected by four repeaters/concentrators, and it states that only three of those five segments can be populated (in other words, have additional or other host or networking device connections).

The 5-4-3 rule does not apply to switched networks or the use of bridges or routers.

Fiber-Optic Cables

An alternative to conductor-based network cabling is fiber-optic cable. *Fiber-optic cables* transmit pulses of light rather than electricity. This gives fiber-optic cable the advantage of being extremely fast and nearly impervious to tapping and interference. Fiber will typically cost more to deploy than twisted pair, but its price premium has decreased to be more in line with other deployments and is often well worth the expense for its security, interference resilience, and performance. Fiber can be deployed as single-mode (supporting a single light signal) or multimode (supporting multiple light signals). Single-mode fiber has a thinner optical core, lower attenuation over distance, and potentially unlimited bandwidth. It uses a 1310 nm or 1550 nm wavelength laser, can be deployed in runs up to 10 km without repeaters, and is typically sheathed in yellow. Multimode fiber has a larger optical core, higher attenuation over distance, and bandwidth limitations (inversely related to distance), and it uses 850 nm or 1300 nm wavelength LEDs or lasers, has a maximum run length of 400m, and is typically sheathed in blue, aqua, or orange.

Dense Wavelength Division Multiplexing (DWDM) is an optical communication technology used in fiber-optic communication systems to increase the capacity and efficiency of the network. DWDM enables multiple data streams or channels to be simultaneously transmitted over a single optical fiber, each using a different wavelength of light. This allows for the simultaneous transmission of a large number of independent signals, significantly increasing the overall capacity of the fiber-optic infrastructure.

Transport Architecture

The transport architecture in networking encompasses several key aspects. First, network topology refers to the physical or logical layout of devices and connections, including configurations like bus, ring, star, mesh, tree, and hybrid. The chosen topology influences factors such as scalability, fault tolerance, and ease of management. Network topologies are discussed in the next section.

Second, transport architecture is often focused on the concept of planes. The concept of network architecture planes refers to the division of networking functionality and responsibilities into distinct layers or planes, each serving a specific purpose. This separation helps in organizing and managing the different aspects of network operations. The three primary

planes in network architecture are the data plane, control plane, and management plane. The data plane, or forwarding plane, is responsible for the transmission of user data between network devices, performing tasks like packet forwarding, switching, and routing. The control plane manages and maintains forwarding tables used by the data plane, handling activities such as routing protocols and decision-making on data forwarding. The management plane is in charge of overall network device administration, covering tasks like configuration, monitoring, performance analysis, and network maintenance. The separation of these planes allows for modular design, scalability, and the ability to upgrade or modify one plane without affecting the others.

A third critical consideration in transport architecture is the choice between cut-through and store-and-forward switching. Cut-through switching forwards a frame as soon as it reads the destination address, providing low latency and suitability for low-latency applications. On the other hand, store-and-forward switching receives and stores the entire frame in a buffer before forwarding, offering greater error checking and suitability for ensuring data integrity. The selection between cut-through and store-and-forward depends on factors such as network requirements, latency sensitivity, and the level of error checking needed.

Understanding and designing the transport architecture is essential for optimizing network performance, managing resources efficiently, and ensuring that the network meets its intended requirements.

Network Topologies

The physical layout and organization of computers and networking devices is known as the network topology. The *logical topology* is the grouping of networked systems into trusted collectives. The *physical topology* is not always the same as the logical topology. There are four basic topologies of the physical layout of a network:

Ring Topology A *ring topology* connects each system as points on a circle (see Figure 11.9). The connection medium acts as a unidirectional transmission loop. Only one system can transmit data at a time. Traffic management is performed by a token. A token is a digital hall pass that travels around the ring until a system grabs it. A system in possession of the

token can transmit data. Data and the token are transmitted to a specific destination. As the data travels around the loop, each system checks to see whether it is the intended recipient of the data. If not, it passes the token on. If so, it reads the data. Once the data is received, the token is released and returns to traveling around the loop until another system grabs it. If any one segment of the loop is broken, all communication around the loop ceases. Some implementations of ring topologies employ a fault tolerance mechanism, such as dual loops running in opposite directions, to prevent single points of failure.

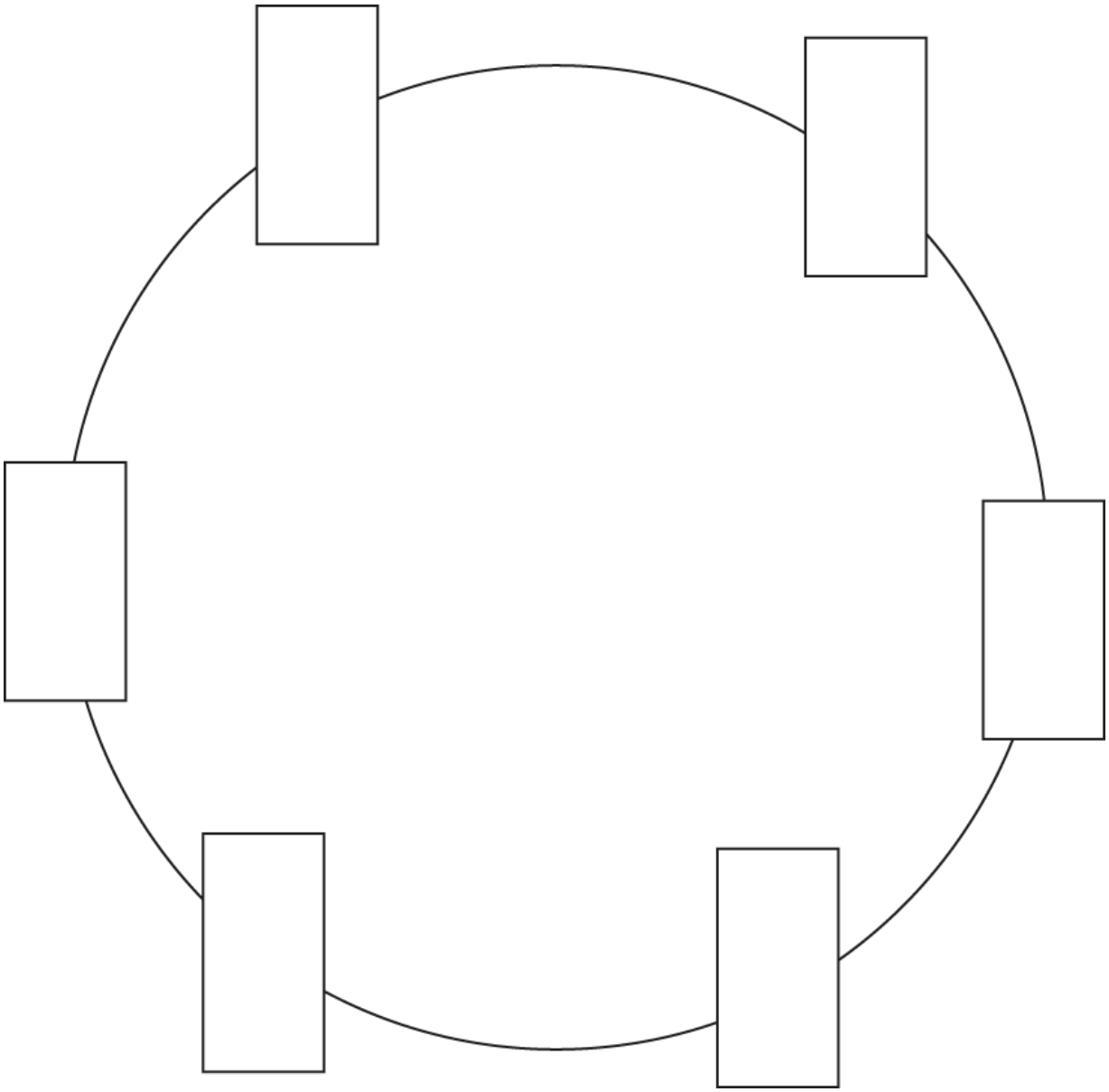


FIGURE 11.9 A ring topology

Bus Topology A *bus topology* connects each system to a trunk or backbone cable. All systems on the bus can transmit data simultaneously, which can result in collisions. A collision occurs when two systems transmit data at the same time; the signals interfere with each other. To avoid this, the systems employ a collision avoidance mechanism that basically "listens" for any other currently occurring traffic. If traffic is heard, the system waits a few moments and listens again. If no traffic is heard, the system transmits its data. When data is transmitted on a bus topology, all systems on the network hear the data. If the data is not addressed to a specific system, that system just ignores the data. The benefit of a bus topology is that if a single segment fails, communications on all other segments continue uninterrupted. However, the central trunk line remains a single point of failure.

There are two types of bus topologies: linear and tree. A linear bus topology employs a single trunk line with all systems directly connected to it. A tree topology employs a single trunk line with branches that can support multiple systems. Figure 11.10 illustrates both types. The primary reason a bus is rarely if ever used today is that it must be terminated at both ends and any disconnection can take down the entire network.

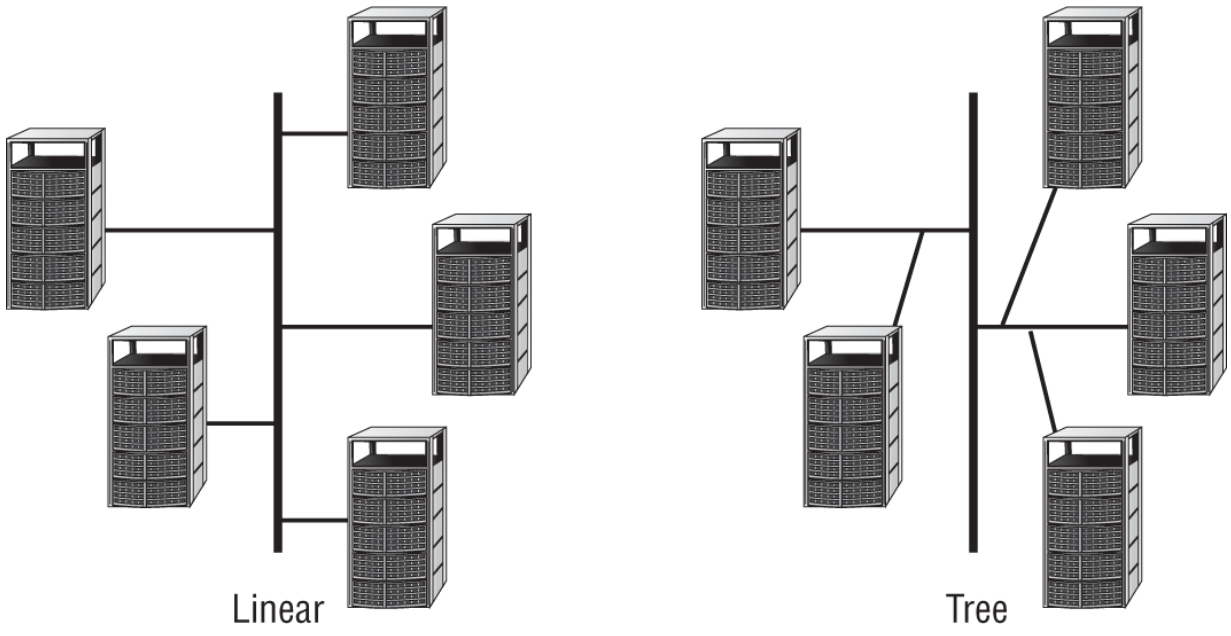


FIGURE 11.10 A linear bus topology and a tree bus topology

Star Topology A *star topology* employs a centralized connection device. This device can be a simple hub or switch. Each system is connected to the central hub by a dedicated segment (see Figure 11.11). If any one segment fails, the other segments can continue to function. However, the central hub is a single point of failure. Generally, the star topology uses less cabling than other topologies and makes the identification of damaged cables easier.

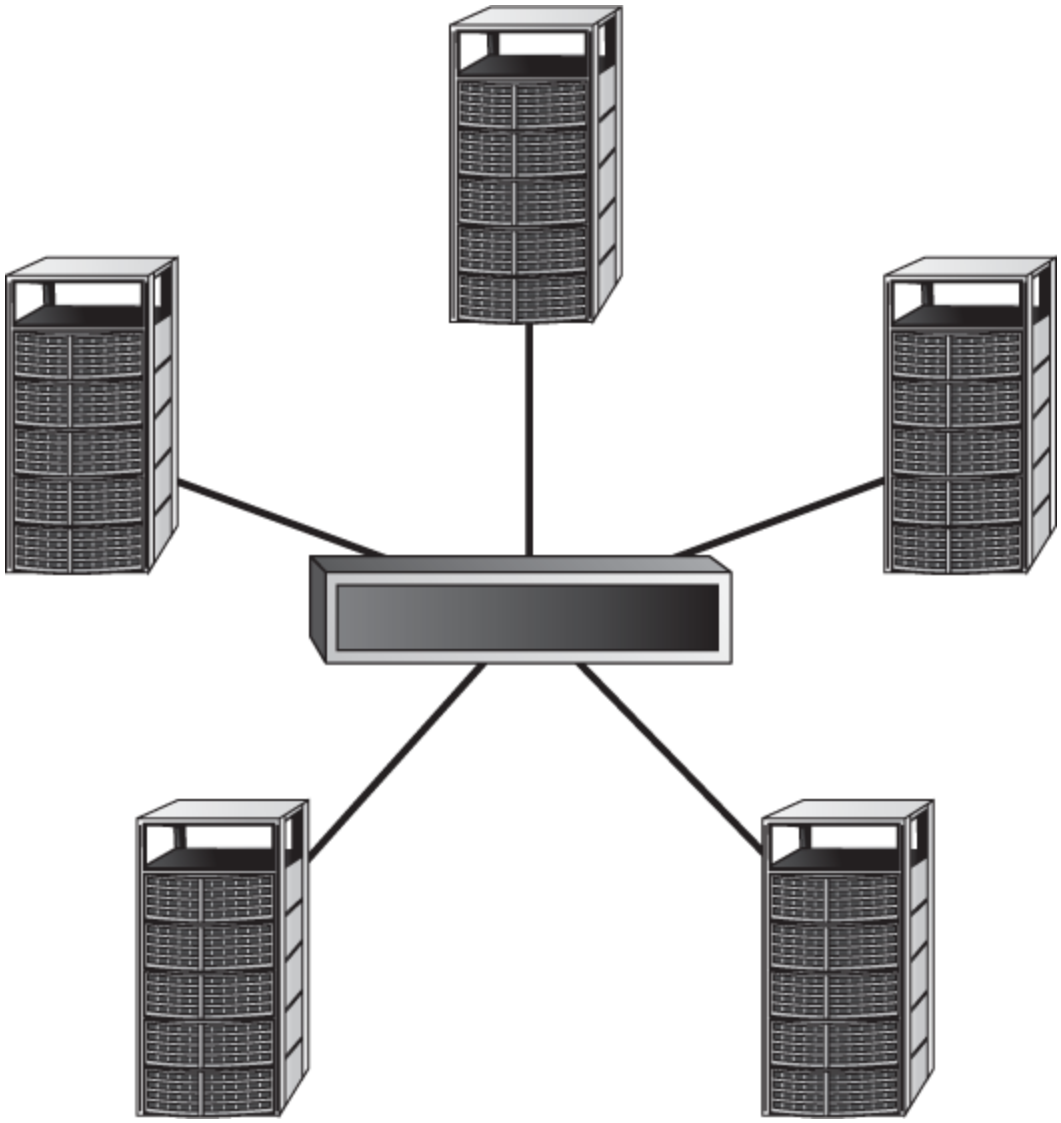


FIGURE 11.11 A star topology

A logical bus can be implemented as a physical star. Ethernet is a bus-based technology. It can be deployed as a physical star, but the hub or switch device is actually internally a logical bus connection device.

Mesh Topology A *mesh topology* connects systems to other systems using numerous paths (see Figure 11.12). A full-mesh topology connects each system to all other systems on the network. A partial-mesh topology connects many systems to many other systems. Mesh topologies provide redundant connections to systems, allowing multiple segment failures without seriously affecting connectivity.

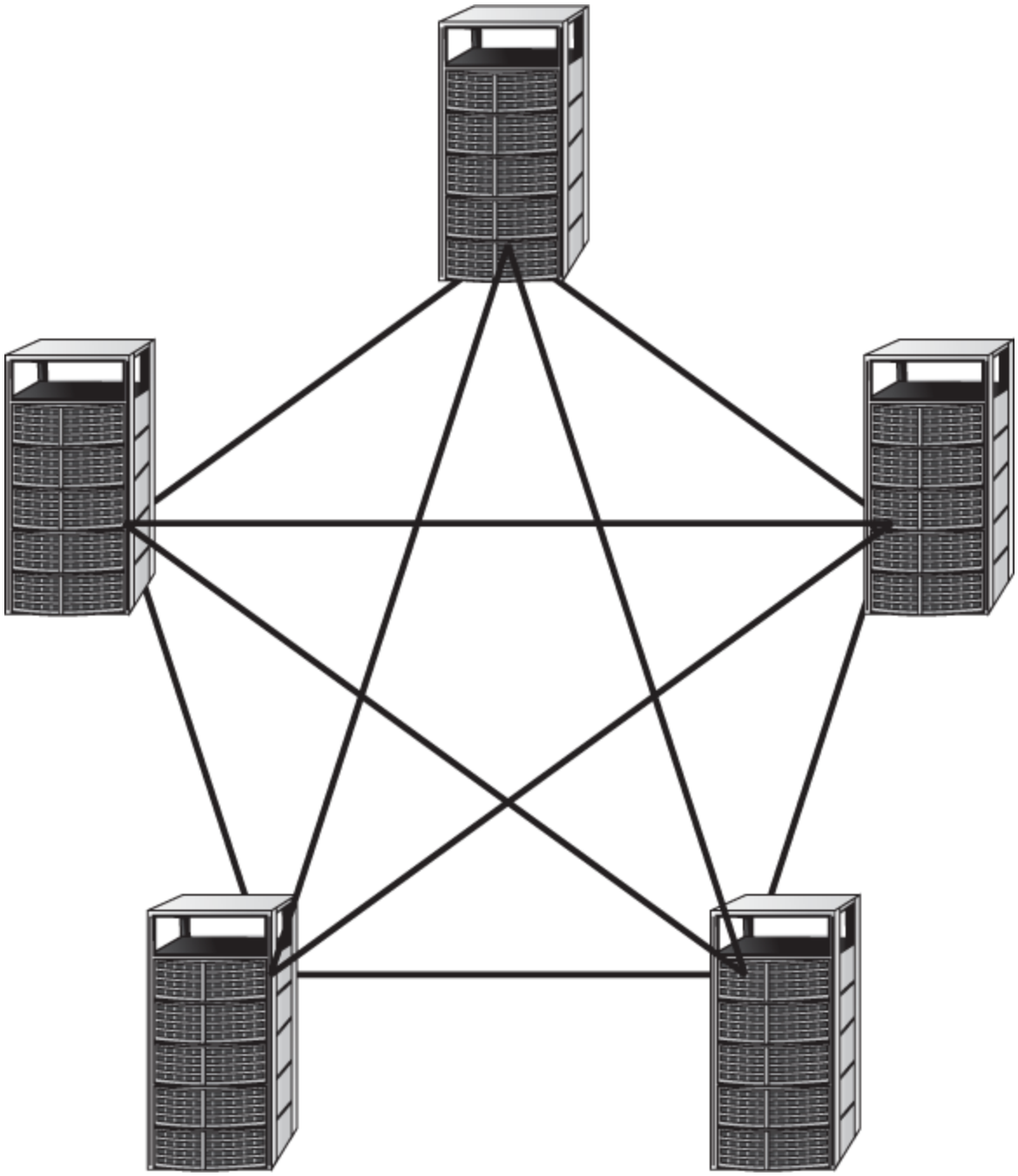


FIGURE 11.12 A mesh topology

Ethernet

Ethernet is a shared-media LAN technology (aka a broadcast technology). That means it allows numerous devices to communicate over the same medium but requires that the devices take turns communicating and performing collision detection and avoidance. Ethernet employs broadcast and collision domains (see the general feature “Collision Domains vs. Broadcast Domains”). Ethernet is an example of a media access methodology.

Ethernet can support full-duplex communications (in other words, full two-way) and usually employs twisted-pair cabling. (Coaxial cabling was originally used.) Ethernet is most often deployed on star or bus topologies. Ethernet is based on the IEEE 802.3 standard. Individual units of Ethernet data are called *frames*. Fast Ethernet supports 100 Mbps throughput. Gigabit Ethernet supports 1,000 Mbps (1 Gbps) throughput. 10 Gigabit Ethernet supports 10,000 Mbps (10 Gbps) throughput.

Sub-Technologies

Most networks comprise numerous technologies rather than a single technology. For example, Ethernet is not just a single technology but a superset of sub-technologies that support its common and expected activity and behavior. Ethernet includes the technologies of digital communications, synchronous communications, and baseband communications, and it supports broadcast, multicast, unicast, and anycast communications and Carrier-Sense Multiple Access with Collision Detection (CSMA/CD).

LAN technologies may include many of the sub-technologies described in the following sections.

Analog and Digital

One sub-technology common to many forms of network communications is the mechanism used to actually transmit signals over a physical medium, such as a cable. There are two types:

- *Analog communications* occur with a continuous signal that varies in frequency, amplitude, phase, voltage, and so on. The variances in the continuous signal produce a wave shape (as opposed to the square

shape of a digital signal). The actual communication occurs by variances in the constant signal.

- *Digital communications* occur through the use of a discontinuous electrical signal and a state change or on-off pulses.

Digital signals are more reliable than analog signals over long distances or when interference is present. This is because of a digital signal's definitive information storage method employing direct current voltage where voltage-on represents a value of 1 and voltage-off represents a value of 0. These on-off pulses create a stream of binary data. Analog signals become altered and corrupted because of attenuation over long distances and interference. Since an analog signal can have an infinite number of variations used for signal encoding as opposed to digital signals' two states, unwanted alterations to the signal make extraction of the data more difficult as the degradation increases.

Synchronous and Asynchronous

Some communications are synchronized with some sort of clock or timing activity. Communications are either synchronous or asynchronous:

- *Synchronous communications* rely on a timing or clocking mechanism based on either an independent clock or a timestamp embedded in the data stream. Synchronous communications are typically able to support very high rates of data transfer.
- *Asynchronous communications* rely on stop and start delimiters to manage the transmission of data. Because of the use of delimiters and the stop and start nature of its transmission, asynchronous communication is best suited for smaller amounts of data. PSTN modems are good examples of asynchronous communication devices.

Baseband and Broadband

How many communications can occur simultaneously over a cable segment depends on whether you use baseband technology or broadband technology:

- *Baseband technology* can support only a single communication channel. It uses a direct current applied to the cable. A current that is at

a higher level represents the binary signal of 1, and a current that is at a lower level represents the binary signal of 0. Baseband is a form of digital signal. Ethernet is a baseband technology.

- *Broadband technology* can support multiple simultaneous signals. Broadband uses frequency modulation to support numerous channels, each supporting a distinct communication session. Broadband is suitable for high throughput rates, especially when several channels are multiplexed. Broadband is a form of analog signal. Cable television and cable modems, fiber optics, satellite, DSL, T1, and T3 are examples of broadband technologies.

Casting Technologies

Casting technologies determine how many destinations a single transmission can reach:

- *Broadcast* technology supports communications to all possible recipients.
- *Multicast* technology supports communications to multiple specific recipients.
- *Unicast* technology supports only a single communication to a specific recipient.
- *Anycast* technology supports communications where a single sender transmits data to the nearest or best-suited node among a group of potential receivers. The goal is to deliver data to the “nearest” or “best” node in terms of network topology or routing metrics.
- *Geocast* technology supports communications where data is sent to all devices within a specific geographical area. It is a one-to-all communication paradigm limited to a particular geographic region, and devices outside that area do not receive the broadcast.

LAN Media Access

Media access protocols in LANs dictate how multiple devices within a network contend for the right to access the shared communication medium.

There are several methods used to manage and control access to the communication medium in LANs:

- *Arbitration* is a media access protocol where a central authority or a predefined set of rules determines which device has the right to access the communication medium at any given time. Devices interested in transmitting data request permission from the central authority or follow established rules to access the medium. This approach is often used in centralized network architectures. Time Division Multiple Access (TDMA), as an example, divides time into fixed slots, and a central authority, such as a base station or network controller, assigns specific time slots to each device. Devices are allowed to transmit only during their allocated time slots, avoiding collisions.
- *Deconfliction* is a media access protocol that aims to avoid collisions and conflicts by assigning specific time slots or frequency bands to different devices for communication. Each device is allocated a dedicated time slot or frequency range during which it can transmit data without interference from other devices. Time-division multiplexing (TDM) and frequency-division multiplexing (FDM) are common techniques used for deconfliction.
- *Contention-based* protocols allow devices to contend for access to the communication medium. Devices transmit data when they have information to send and contend with other devices for the right to transmit. When multiple devices attempt to transmit simultaneously, collisions may occur. Contention-based protocols often include mechanisms to detect and manage collisions, such as Carrier Sense Multiple Access with Collision Detection (CSMA/CD) or Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA).

The choice of a specific media access protocol depends on factors such as network architecture, traffic patterns, and the desired trade-off between simplicity and efficiency.

There are numerous LAN media access technologies that are used to avoid or prevent transmission collisions. These technologies define how multiple systems, all within the same collision domain, are to communicate. Some of

these technologies actively prevent collisions, whereas others respond to collisions.

Carrier-Sense Multiple Access (CSMA) This is an arbitration LAN media access technology that performs communications using the following steps:

1. The host listens to the LAN media to determine whether it is in use.
2. If the LAN media is not being used, the host transmits its communication.
3. The host waits for an acknowledgment.
4. If no acknowledgment is received after a timeout period, the host starts over at step 1.

CSMA does not directly address collisions. If a collision occurs, the communication would not have been successful, and thus an acknowledgment would not be received. This causes the sending system to retransmit the data and perform the CSMA process again.

Carrier-Sense Multiple Access with Collision Detection (CSMA/CD) This is a contention-based LAN media access technology that performs communications using the following steps:

1. The host listens to the LAN media to determine whether it is in use.
2. If the LAN media is not being used, the host transmits its communication.
3. While transmitting, the host listens for collisions (in other words, two or more hosts transmitting simultaneously).
4. If a collision is detected, the host transmits a jam signal.
5. If a jam signal is received, all hosts stop transmitting. Each host waits a random period of time and then starts over at step 1.

Ethernet networks employ the CSMA/CD technology. CSMA/CD responds to collisions by having each member of the collision domain wait for a short but random period of time before starting the process over. Unfortunately, allowing collisions to occur and then responding or reacting to collisions causes delays in transmissions as well as a required repetition

of transmissions. This results in about 40 percent loss in potential throughput.

Carrier-Sense Multiple Access with Collision Avoidance (CSMA/CA) This is a contention-based LAN media access technology that performs communications using the following steps:

1. The host has two connections to the LAN media: inbound and outbound. The host listens on the inbound connection to determine whether the LAN media is in use.
2. If the LAN media is not being used, the host requests permission to transmit.
3. If permission is not granted after a timeout period, the host starts over at step 1.
4. If permission is granted, the host transmits its communication over the outbound connection.
5. The host waits for an acknowledgment.
6. If no acknowledgment is received after a time-out period, the host starts over at step 1.

802.11 wireless networking is an example of a network that employs CSMA/CA technologies. CSMA/CA attempts to avoid collisions by granting only a single permission to communicate at any given time. This system requires the designation of a primary system, which responds to the requests and grants permission to send data transmissions.

Token Passing This is an arbitration LAN media access technology that performs communications using a digital token. Possession of the token allows a host to transmit data. Once its transmission is complete, it releases the token to the next system. Token passing was used by ring topology–based networks, such as legacy Token Ring and Fiber Distributed Data Interface (FDDI). Token passing prevents collisions since only the system possessing the token is allowed to transmit data.

Polling This is an arbitration LAN media access technology that performs communications using a primary-secondary configuration. One system is labeled as the primary system. All other systems are labeled as secondary.

The primary system polls or inquires of each secondary system in turn whether they have a need to transmit data. If a secondary system indicates a need, it is granted permission to transmit. Once its transmission is complete, the primary system moves on to poll the next secondary system. Mainframes often supported polling.

Polling addresses collisions by attempting to prevent them from using a permission system. Polling is an inverse of the CSMA/CA method. Both use primary and secondary systems, but although CSMA/CA allows the secondary system to request permissions, polling has the primary system offer permission. Polling can be configured to grant one system (or more) priority over other systems. For example, if the standard polling pattern was 1, 2, 3, 4, then to give system 1 priority, the polling pattern could be changed to 1, 2, 1, 3, 1, 4.

Summary

The tasks of designing, deploying, and maintaining security on a network require intimate knowledge of the technologies involved in networking. This includes protocols, services, communication mechanisms, topologies, cabling, endpoints, and networking devices.

The OSI model is a standard against which all protocols are evaluated. Understanding how the OSI model is used and how it applies to real-world protocols can help system designers and system administrators improve security. The TCP/IP model is derived directly from the TCP/IP protocol suite and roughly maps to the OSI model.

Most networks employ TCP/IP as the primary protocol. IP networking includes IPv4 and IPv6. IPv4 is the version of Internet Protocol that is most widely used around the world. IPv6 is being rapidly adopted for both private and public network use. DNS and ARP were developed to interchange or resolve between domain names and IP addresses or IP addresses and MAC addresses, respectively. TCP/IP supports many secure protocols, including IPSec, SSH, and protocols encrypted by TLS. TCP/IP is a multilayer protocol suite that allows for flexibility, resiliency, and encryption.

Converged protocols are common on modern networks, including VoIP and iSCSI. SDN and CDN have expanded the definition of network as well as expanded the use cases for it.

Micro-segmentation divides an internal network into numerous subzones to allow for greater security and control of communications, which in turn supports a zero-trust security policy.

Wireless communications occur in many forms, including cell phone, Bluetooth (802.15.1 and Bluetooth SIG), RFID, NFC, and Wi-Fi networking (802.11). Wireless communication is more vulnerable to interference, eavesdropping, denial of service, and AitM attacks.

Routers, hubs, switches, repeaters, gateways, proxies, NAC, and firewalls are an important part of a network's security. Firewalls are essential tools in managing, controlling, and filtering network traffic. Endpoint security is the concept that each individual device must maintain local security whether or not its network or telecommunications channels also provide security.

A wide range of hardware components can be used to construct a network, not the least of which is the cabling used to tie all the devices together. Understanding the strengths and weaknesses of each transmission media type is part of designing a secure network.

Study Essentials

Know the OSI model layers. The OSI layers are as follows: Application, Presentation, Session, Transport, Network, Data Link, and Physical.

Know the network container names. The network containers are: OSI layers 7–5 protocol data unit (PDU), Layer 4 segment (TCP) or a datagram (UDP), Layer 3 packet, Layer 2 frame, and Layer 1 bits.

Understand the MAC address. Media Access Control (MAC) address is a 6-byte (48-bit) binary address written in hexadecimal notation, aka hardware address, physical address, the NIC address, and the Ethernet address. The first 3 bytes (24 bits) of the address is the organizationally unique identifier (OUI), which denotes the vendor or manufacturer.

Understand the TCP/IP model. Also known as DARPA or the DOD model, the model has four layers: Application (also known as Process),

Transport (also known as Host-to-Host), Internet (sometimes known as Internetworking), and Link (although the terms Network Interface and sometimes Network Access are used).

Understand DNS. The Domain Name System (DNS) is the hierarchical naming scheme used in both public and private networks. DNS links human-friendly fully qualified domain names (FQDNs) and IP addresses together. DNSSEC and DoH are DNS security features.

Understand DNS poisoning. DNS poisoning is the act of falsifying the DNS information used by a client to reach a desired system. It can be accomplished through a rogue DNS server, pharming, altering a `hosts` file, corrupting IP configuration, DNS query spoofing, and proxy falsification.

Know about ARP. Address Resolution Protocol (ARP) is essential to the interoperability of logical and physical addressing schemes. ARP is used to resolve IP addresses into MAC addresses. Also, know about ARP poisoning.

Know about micro-segmentation. Micro-segmentation is dividing up an internal network into numerous subzones, potentially as small as a single device, such as a high-value server or even a client or endpoint device. Each zone is separated from the others by internal segmentation firewalls (ISFWs), subnets, or VLANs.

Know about edge networks. An edge network is a carefully designed data architecture that strategically allocates computing resources to edge devices within a network. This design helps distribute processing power demands away from central servers, empowering the devices to handle a significant portion of the processing workload.

Understand the various wireless technologies. Cell phones, Bluetooth (802.15.1 and Bluetooth SIG), and Wi-Fi wireless networking (802.11) are all called wireless technologies, even though they are all different. Be aware of their differences, strengths, and weaknesses. Understand the basics of securing 802.11 networking. Know about RFID, NFC, satellite, narrow-band, and Zigbee.

Understand site surveys. A site survey is a formal assessment of wireless signal strength, quality, and interference using an RF signal detector. A site survey is performed by placing a wireless base station in a

desired location and then collecting signal measurements from throughout the area.

Understand WPS attacks. Wi-Fi Protected Setup (WPS) is intended to simplify the effort involved in adding new clients to a secured wireless network. It operates by automatically connecting the first new wireless client to seek the network once WPS is triggered.

Understand captive portals. A captive portal is an authentication technique that redirects a newly connected client to a web-based portal access control page.

Know wireless attacks. Attacks include war driving, wireless scanners/crackers, rogue access points, evil twin, disassociation, jamming, IV abuse, and replay.

Be familiar with CDNs. A content distribution network (CDN), or content delivery network, is a collection of resource services deployed in numerous data centers across the Internet to provide low latency, high performance, and high availability of the hosted content.

Understand NAC. Network access control (NAC) is the concept of controlling access to an environment through strict adherence to and enforcement of security policy. Know about 802.1X, preadmission, postadmission, agent-based, and agentless.

Understand the various types of firewalls. There are several types of firewalls: static packet filtering, application-level, circuit-level, stateful inspection, NGFW, and ISFW. Also, know about virtual firewall, filters/rules/ACLs/tuples, bastion host, ingress, egress, RTBH, stateless versus stateful, WAF, SWG, TCP wrapper, DPI, and content and URL filtering.

Know about proxies. A proxy server is used to mediate between clients and servers. Proxies are most often used in the context of providing clients on a private network with internet access while protecting the identity of the clients. Know about forward, reverse, transparent, and nontransparent.

Understand endpoint security. Endpoint security is the concept that each individual device must maintain local security whether or not its network or telecommunications channels also provide security. Endpoint detection and response (EDR) is a combination of firewall, intrusion

detection system (IDS), and antimalware. Managed detection and response (MDR) combines EDR with Security information and event management (SIEM), network traffic analysis (NTA), and network IDS. Endpoint protection platform (EPP) is an intrusion prevention system (IPS) variant of EDR. Extended detection and response (XDR) is the combination of EDR, MDR, and EPP often with cloud-based remote monitoring and analysis.

Be familiar with the common LAN technologies. The most common LAN technology is Ethernet. Also, be familiar with analog versus digital communications; synchronous versus asynchronous communications; duplexing; baseband versus broadband communications; broadcast, multicast, unicast, anycast, and geocast communications; CSMA, CSMA/CD, and CSMA/CA; token passing; and polling.

Written Lab

1. Name the layers of the OSI model and their numbers from top to bottom.
2. Name three problems with cabling and the methods to counteract those issues.
3. What are the various technologies employed by wireless devices to maximize their use of the available radio frequencies?
4. Discuss methods used to secure 802.11 wireless networking.
5. Name eight Application-Layer protocols and their ports (indicate whether the ports are TCP or UDP).

Review Questions

1. Dorothy is using a network sniffer to evaluate network connections. She focuses on the initialization of a TCP session. What is the first phase of the TCP three-way handshake sequence?

   A. SYN flagged packet

   B. ACK flagged packet

   C. FIN flagged packet

D. SYN/ACK flagged packet

2. UDP is a connectionless protocol that operates at the Transport Layer of the OSI model and uses ports to manage simultaneous connections. Which of the following terms is also related to UDP?

   A. Bits

   B. Logical addressing

   C. Data reformatting

   D. Simplex

3. Which of the following is a means for IPv6 and IPv4 to be able to coexist on the same network? (Choose all that apply.)

   A. Dual stack

   B. Tunneling

   C. IPSec

   D. NAT-PT

   E. IP sideloading

4. Security configuration guidelines issued by your CISO require that all HTTP communications be secure when communicating with internal web services. Which of the following is true in regards to using TLS? (Choose all that apply.)

   A. Allows for use of TCP port 443

   B. Prevents tampering, spoofing, and eavesdropping

   C. Requires two-way authentication

   D. Is backward compatible with SSL sessions

   E. Can be used as a VPN solution

5. Your network supports TCP/IP. TCP/IP is a multilayer protocol. It is primarily based on IPv4, but the organization is planning on deploying IPv6 within the next year. What is both a benefit and a potentially harmful implication of multilayer protocols?

   A. Throughput

B. Encapsulation

C. Hash integrity checking

D. Logical addressing

6. A new VoIP system is being deployed at a government contractor organization. They require high availability of five nines of uptime for the voice communication system. They are also concerned about introducing new vulnerabilities into their existing data network structure. The IT infrastructure is based on fiber optics and supports over 1 Gbps to each device; the network often reaches near full saturation on a regular basis. What option will provide the best outcome of performance, availability, and security for the VoIP service?

   A. Create a new VLAN on the existing IT network for the VoIP service.

   B. Replace the current switches with routers and increase the interface speed to 1,000 Mbps.

   C. Implement a new, separate network for the VoIP system.

   D. Deploy flood guard protections on the IT network.

7. Micro-segmentation is dividing up an internal network in numerous subzones, potentially as small as a single device, such as a high-value server or even a client or endpoint device. Which of the following is true in regard to micro-segmentation? (Choose all that apply.)

   A. It is the assignment of the cores of a CPU to perform different tasks.

   B. It can be implemented using ISFWs.

   C. Transactions between zones are filtered.

   D. It supports edge and fog computing management.

   E. It can be implemented with virtual systems and virtual networks.

8. A new startup company is designing a sensor that needs to connect wirelessly to a PC or IoT hub to transmit its gathered data to a local application or cloud service for data analysis. The company wants to

ensure that all transferred data from the device cannot be disclosed to unauthorized entities. The device is also intended to be located within 1 meter of the PC or IoT hub it communicates with. Which of the following concepts is the best choice for this device?

A. Zigbee

B. Bluetooth

C. GEO

D. 5G

9. James has been hired to be a traveling repair technician. He will be visiting customers all over the country to provide support services. He has been issued a portable workstation with 4G and 5G data service. What are some concerns when using this capability? (Choose all that apply.)

A. Eavesdropping

B. Rogue towers

C. Data speed limitations

D. Reliability of establishing a connection

E. Compatibility with cloud services

F. Unable to perform duplex communications

10. A new startup company needs to optimize delivery of high-definition media content to its customers. They are planning the deployment of resource service hosts in numerous data centers across the world to provide low latency, high performance, and high availability of the hosted content. What technology is likely being implemented?

A. VPN

B. CDN

C. SDN

D. CCMP

11. Which of the following is a true statement about ARP poisoning or MAC spoofing?

A. MAC spoofing is used to overload the memory of a switch.

B. ARP poisoning is used to falsify the physical address of a system to impersonate that of another authorized device.

C. MAC spoofing relies on ICMP communications to traverse routers.

D. ARP poisoning can use unsolicited or gratuitous replies.

12. An organization stores group project data files on a central SAN. Many projects have numerous files in common but are organized into separate project containers. A member of the incident response team is attempting to recover files from the SAN after a malware infection. However, many files are unable to be recovered. What is the most likely cause of this issue?

    A. Using Fibre Channel

    B. Performing real-time backups

    C. Using file encryption

    D. Deduplication

13. Jim was tricked into clicking on a malicious link contained in a spam email message. This caused malware to be installed on his system. The malware initiated a MAC flooding attack. Soon, Jim's system and everyone else's in the same local network began to receive all transmissions from all other members of the network as well as communications from other parts of the next-to-local members. The malware took advantage of what condition in the network?

    A. Social engineering

    B. Network segmentation

    C. ARP queries

    D. Weak switch configuration

14. A _____ is an intelligent hub because it knows the hardware addresses of the systems connected on each outbound port. Instead of repeating traffic on every outbound port, it repeats traffic only out of the port on which the destination is known to exist.

A. Repeater

B. Switch

C. Bridge

D. Router

15. What type of security zone can be positioned so that it operates as a buffer between the secured private network and the Internet and can host publicly accessible services?

    A. Honeypot

    B. Screened subnet

    C. Extranet

    D. Intranet

16. An organization wants to use a wireless network internally, but they do not want any possibility of external access or detection. What security tool should be used?

    A. Air gap

    B. Faraday cage

    C. Biometric authentication

    D. Screen filters

17. Neo is the security manager for the southern division of the company. He thinks that deploying a NAC will assist in improving network security. However, he needs to convince the CISO of this at a presentation next week. Which of the following are goals of NAC that Neo should highlight? (Choose all that apply.)

    A. Reduce social engineering threats

    B. Detect rogue devices

    C. Map internal private addresses to external public addresses

    D. Distribute IP address configurations

    E. Reduce zero-day attacks

F. Confirm compliance with updates and security settings

18. The CISO wants to improve the organization's ability to manage and prevent malware infections. Some of her goals are to (1) detect, record, evaluate, and respond to suspicious activities and events, which may be caused by problematic software or by valid and invalid users, (2) collect event information and report it to a central ML analysis engine, and (3) detect abuses that are potentially more advanced than what can be detected by traditional antivirus or HIDSs. The solution needs to be able to reduce response and remediation time, reduce false positives, and manage multiple threats simultaneously. What solution is the CISO wanting to implement?

    A. EDR

    B. NGFW

    C. WAF

    D. XSRF

19. A(n) _____ firewall is able to make access control decisions based on the content of communications as well as the parameters of the associated protocol and software.

    A. Application-level

    B. Stateful inspection

    C. Circuit-level

    D. Static packet filtering

20. Which of the following is true regarding appliance firewalls? (Choose all that apply.)

    A. They are able to log traffic information.

    B. They are able to block new phishing scams.

    C. They are able to issue alarms based on suspected attacks.

    D. They are unable to prevent internal attacks.

Chapter 12
Secure Communications and Network Attacks

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 4.0: Communication and Network Security**
  - 4.1 Apply secure design principles in network architectures
    - 4.1.7 Performance metrics (e.g., bandwidth, latency, jitter, throughput, signal-to-noise ratio)
    - 4.1.18 Monitoring and management (e.g., network observability, traffic flow/shaping, capacity management, fault detection and handling)
  - 4.3 Implement secure communication channels according to design
    - 4.3.1 Voice, video, and collaboration (e.g., conferencing, Zoom rooms)
    - 4.3.2 Remote access (e.g., network administrative functions)
    - 4.3.3 Data communications (e.g., backhaul networks, satellite)
    - 4.3.4 Third-party connectivity (e.g., telecom providers, hardware support)

Communications security is designed to detect, prevent, and even correct data transportation errors (that is, it provides integrity protection as well as confidentiality). Communications security is used to sustain the security of networks while supporting the need to exchange and share data. This

chapter covers the many forms of communications security, vulnerabilities, and countermeasures.

The Communication and Network Security domain deals with topics related to network components (i.e., network devices and protocols), specifically how they function and how they are relevant to security. This domain is discussed in this chapter and in Chapter 11, “Secure Network Architecture and Components.” Be sure to read and study the material in both chapters to ensure complete coverage of the essential material.

Protocol Security Mechanisms

Transmission Control Protocol/Internet Protocol (TCP/IP) is the primary protocol suite used on most networks and on the Internet. It is a robust protocol suite, but it has numerous security deficiencies. In an effort to improve the security of TCP/IP, many subprotocols, mechanisms, or applications have been developed to protect the confidentiality, integrity, and availability of transmitted data. It is important to remember that even with the foundational protocol suite of TCP/IP, there are literally hundreds, if not thousands, of individual protocols, mechanisms, and applications in use across the Internet. Some of them are designed to provide security services. Some protect integrity, others protect confidentiality, and others provide authentication and access control. In the next sections, we'll discuss some common network and protocol security mechanisms.

Authentication Protocols

The *Point-to-Point Protocol (PPP)* is an encapsulation protocol designed to support the transmission of IP traffic over dial-up or point-to-point links. PPP is a Data Link Layer protocol that allows for multivendor interoperability of WAN devices supporting serial links. Although it is rarely found on typical Ethernet networks today, it is the foundation on which many modern communications are based, as well as the foundation of communication authentication. PPP includes a wide range of communication services, such as the assignment and management of IP addresses, management of synchronous communications, standardized encapsulation, multiplexing, link configuration, link quality testing, error detection, and feature or option negotiation (such as compression).

PPP is an Internet standard documented in RFC 1661. It replaced the *Serial Line Internet Protocol (SLIP)*. SLIP offered no authentication, supported only half-duplex communications, had no error-detection capabilities, and required manual link establishment and teardown. PPP supports automatic connection configuration, error detection, full-duplex communications, and options for authentication. The original PPP options for authentication were PAP, CHAP, and EAP.

Password Authentication Protocol (PAP) PAP transmits usernames and passwords in cleartext. It offers no form of encryption; it simply provides a means to transport the logon credentials from the client to the authentication server.

Challenge Handshake Authentication Protocol (CHAP) CHAP performs authentication using a challenge-response dialogue that cannot be replayed. A challenge is a random number issued by the server, which the client uses along with the password hash to compute the one-way function-derived response. CHAP also periodically reauthenticates the remote system throughout an established communication session to verify the persistent identity of the remote client. This activity is transparent to the user. However, since CHAP is based on MD5, it is no longer considered secure. A Microsoft customization named MS-CHAPv2 uses updated algorithms, adds optional session encryption, and is preferred over the original CHAP.

Extensible Authentication Protocol (EAP) This is a framework for authentication instead of an actual protocol. EAP allows customized authentication security solutions, such as supporting smartcards, tokens, and biometrics. EAP was originally designed for use over physically isolated channels and thus assumed secured pathways. Some EAP methods use encryption, but others do not. Over 40 EAP methods are defined, including LEAP, PEAP, EAP-SIM, EAP-FAST, EAP-MD5, EAP-POTP, EAP-TLS, and EAP-TTLS.

EAP Derivatives

Lightweight Extensible Authentication Protocol (LEAP) is a Cisco proprietary alternative to TKIP for WPA. It was developed to address deficiencies in TKIP before 802.11i/WPA2 was ratified as a standard. LEAP is now a legacy solution to be avoided.

Protected Extensible Authentication Protocol (PEAP) encapsulates EAP in a TLS tunnel. PEAP is preferred to EAP because PEAP imposes its own security. PEAP supports mutual authentication.

Subscriber Identity Module (EAP-SIM) is a means of authenticating mobile devices over the Global System for Mobile Communications (GSM) network. Each device/subscriber is issued a subscriber identity module (SIM) card, which is associated with the subscriber's account and service level.

Flexible Authentication via Secure Tunneling (EAP-FAST) is a Cisco protocol proposed to replace LEAP, which is now obsolete, thanks to the development of WPA2.

EAP-MD5 was one of the earliest EAP methods. It hashes passwords using MD5. It is now deprecated.

EAP Protected One-Time Password (EAP-POTP) supports the use of OTP tokens (which includes hardware devices and software solutions) in multifactor authentication for use in both one-way and mutual authentication.

EAP Transport Layer Security (EAP-TLS) is an open IETF standard that implements the TLS protocol for use in protecting authentication traffic. EAP-TLS is most effective when both client and server have a digital certificate (i.e., mutual certificate authentication).

EAP Tunneled Transport Layer Security (EAP-TTLS) is an extension of EAP-TLS that creates a VPN-like tunnel between endpoints prior to authentication. This ensures that even the client's username is never transmitted in cleartext.

EAP Internet Key Exchange v. 2 (EAP-IKEv2) is based on the IKEv2 (Internet Key Exchange) protocol from IPSec. It provides mutual

authentication, session key establishment, and supports authentication using passwords, symmetric keys, or asymmetric key pairs.

Nimble out-of-band authentication for EAP (EAP-NOOB) is a versatile bootstrapping solution for devices lacking preconfigured authentication credentials and those not yet registered on any server. It uses various OOB channels, such as QR codes, NFC tags, and audio. It is particularly useful for Internet-of-Things (IoT) gadgets and toys that arrive without any details regarding ownership, network affiliation, or server registration. For a more extensive list of EAP methods, see http://en.wikipedia.org/wiki/Extensible_Authentication_Protocol.

IEEE 802.1X defines the use of encapsulated EAP to support a wide range of authentication options for LAN connections. The *IEEE 802.1X* standard is formally named "Port-Based Network Access Control," where *port* refers to any network link, not just physical RJ-45 jacks. This technology ensures that clients can't communicate with a resource until proper authentication has taken place. It's based on Extensible Authentication Protocol (EAP) from PPP.

Many people encounter 802.1X in relation to wireless networking, where it serves as the basis for wireless enterprise authentication. In that implementation, 802.1X serves as an authentication proxy by forwarding wireless client authentication requests to a dedicated remote authentication server or AAA server (typically RADIUS or TACACS+; see Chapter 14, "Controlling and Monitoring Access").

Thus, it is important to remember that 802.1X isn't a wireless technology (i.e., IEEE 802.11)—it is an authentication technology that can be used anywhere authentication is needed, including WAPs, firewalls, routers, switches, proxies, VPN gateways, and remote access servers (RASs)/network access servers (NASs).

When 802.1X is in use, it makes a port-based decision about whether to allow or deny a connection based on the authentication of a user or service.

Like many technologies, 802.1X may be vulnerable to adversary-in-the-middle (AiTM) (aka MiTM or on-path) and hijacking attacks because the authentication mechanism occurs only when the connection is established.

Not all 802.1X or EAP authentication methods are secure; some only check for superficial IDs, such as a MAC address, before granting access. This issue can be addressed by using periodic mid-session reauthentication, as well as implementing session encryption in addition to any authentication protections provided by 802.1X/EAP.

For a discussion of 802.1X, LEAP, and PEAP in relation to wireless networking, see Chapter 11, “Secure Network Architecture and Components.”

Port Security

Port security in IT can mean several things. It can mean the physical control of all connection points, such as RJ-45 wall jacks or device ports (such as those on a switch, router, or patch panel), so that no unauthorized users or devices can attempt to connect to an open port. This control can be accomplished by locking down the wiring closet and server vaults and then disconnecting the workstation run from the patch panel (or punch-down block) that leads to a room's wall jack. Any unneeded or unused wall jacks can (and should) be physically disabled in this manner. Another option is to use a smart patch panel that can monitor the MAC address of any device connected to each wall port across a building and detect not just when a new device is connected to an empty port, but also when a valid device is disconnected or replaced by an invalid device.

Another meaning for port security is the management of TCP and User Datagram Protocol (UDP) ports. If a service is active and assigned to a port, then that port is open. All the other 65,535 ports (TCP or UDP) are closed if a service isn't actively using them. Threat actors can detect the presence of active services by performing a port scan. Firewalls, IDSs, IPSs, and other security tools can detect this activity and either block it or send back false/misleading information. This measure is a type of port security that makes port scanning less effective.

Port security can also refer to the need to authenticate to a port before being allowed to communicate through or across the port. This may be implemented on a switch, router, smart patch panel, or even a wireless network. This concept is often referred to as IEEE 802.1X. For the full discussion of network access control (NAC), see Chapter 11.

Quality of Service (QoS)

Quality of service (QoS) is the oversight and management of the efficiency and performance of network communications. QoS controls protect the availability of data networks under load. Many different factors contribute to the quality of the end-user experience, and QoS attempts to manage all of those factors to create an experience that meets business requirements.

Some of the performance metrics or factors contributing to QoS are as follows:

Bandwidth The network capacity available to carry communications.

Latency The time it takes a packet to travel from source to destination.

Jitter The variation in latency between different packets.

Packet Loss Some packets may be lost between source and destination, requiring retransmission.

Interference Electrical noise, faulty equipment, and other factors may corrupt the contents of packets.

Throughput The actual amount of data transmitted successfully over a network or communication channel within a given period. It is a measure of the effective data transfer rate and represents the real-world performance of the network.

Signal-to-Noise Ratio (SNR) A measure of the quality of a signal in a communication channel. It compares the strength of the desired signal to the level of background noise or interference present in the channel. The higher the SNR, the better the quality of the signal.

Based on the recorded/detected metrics in these areas, network traffic can be adjusted, throttled, or reshaped to account for unwanted conditions. QoS systems often prioritize certain traffic types that have a low tolerance for interference and/or have high business requirements. High-priority traffic or time-sensitive traffic (such as VoIP) can be prioritized, and other traffic can be held back as needed. Throttling or shaping can be implemented on a protocol or IP basis to set a maximum use or consumption limit. In some cases, using alternate transmission paths, time-shifting noncritical data transfers, or deploying more or higher-capacity connections may be necessary to maintain a desired QoS.

Most network administrators don't automatically consider QoS an aspect of security. However, availability is one of the elements of the CIA Triad. By monitoring and managing QoS, essential communications and their related business operations, processes, and tasks may have their availability sustained and protected. QoS may also include specific security requirements, such as requiring encryption for certain types of traffic.



Transparency is the characteristic of a service, security control, or access mechanism that ensures that it is unseen by users. The more transparent a security mechanism is, the less likely a user will be able to circumvent it or even be aware that it exists. With transparency, there is a lack of direct evidence that a feature, service, or restriction exists, and its impact on performance is minimal.

Secure Voice Communications

Telephony is the collection of methods by which telephone services are provided to an organization or the mechanisms by which an organization uses telephone services for either voice and/or data communications. Telephony includes *public switched telephone network (PSTN)* (aka plain old telephone service, or POTS), private branch exchange (PBX), mobile/cellular services (see Chapter 9, “Security Vulnerabilities, Threats, and Countermeasures”), and VoIP.

Public Switched Telephone Network

The vulnerability of voice communication is tangentially related to IT system security. However, most voice communication solutions have moved on to the network (i.e., technology convergence) by employing digital devices and VoIP; therefore, securing voice communications is an increasingly important issue. When voice communications occur over the IT infrastructure, it is important to implement mechanisms to provide for authentication and integrity. Confidentiality should be maintained by

employing an encryption service or protocol to protect voice communications while in transit.

PBX and PSTN voice communications are vulnerable to interception, eavesdropping, tapping, and other exploitations. Often, physical security is required to maintain control over voice communications within the confines of your organization's physical locations. Security of voice communications outside your organization is typically the responsibility of the phone company from which you lease services. If voice communication vulnerabilities are an important issue for sustaining your security policy, you should deploy an encrypted communication mechanism and use it exclusively.

PSTN connections were the only or primary remote network links for many businesses until high-speed, cost-effective, and ubiquitous access methods were available. POTS/PSTN also waned in use for home-user Internet connectivity once broadband and wireless services became more widely available. However, PSTN connections are sometimes still used as a backup option for remote connections when broadband solutions fail. PSTN may still be the only option for rural Internet and remote connections. PSTN is also used as standard voice lines when VoIP or broadband solutions are unavailable, interrupted, or not cost-effective.

Voice over Internet Protocol (VoIP)

Voice over Internet Protocol (VoIP) is a technology that encapsulates audio into IP packets to support telephone calls over TCP/IP network connections. VoIP is also the basis for many multimedia messaging services that combine audio, video, chat, file exchange, whiteboard, and application collaboration.

In Chapter 11, we discussed VoIP and mentioned that Secure Real-time Transport Protocol (SRTP) may be used to provide encryption. However, it is important to clarify when and if this encryption is of any use. VoIP encryption is widely available but rarely end-to-end. VoIP is not a single technology, even though it uses common standardized protocols—just as there are many different operating systems that communicate over the TCP/IP protocol suite. VoIP products from different vendors often do not

interoperate on anything other than the transmission of the audio communication itself.

For example, if you have VoIP phone service provided by your ISP, you may have a VoIP phone sitting on your desk that looks and acts like a traditional PSTN phone. The difference is that it is plugged into the LAN rather than a telephone line. The VoIP service provided by your ISP might not offer any form of encryption. Thus, it would be impossible to obtain end-to-end encryption using that service. However, even if your ISP provided encrypted VoIP services, it would only establish end-to-end encryption if you called someone using the same ISP-provided VoIP service. If you called someone using another VoIP solution, you likely would not end up with an end-to-end encrypted connection.

This is one of the most misunderstood aspects of VoIP services. It is often marketed as being an encrypted service. However, the advertisements fail to point out that the encryption is only established between compatible devices and service providers, which is usually limited to their own proprietary variation of VoIP. In order to communicate with another phone outside of the ISP's VoIP services, a VoIP-to-PSTN gateway must be present. This gateway supports calls from a VoIP phone to make their way to a traditional PSTN landline or mobile phone, and vice versa. If you are using ISP A's VoIP service to call someone using ISP B's VoIP service, your call will likely go through one or more gateways and likely traverse some portion of the PSTN network. Therefore, your call may be encrypted from your phone to the gateway, but it will have to be decrypted to traverse the gateway and the intermediary network. When the connection reaches the callee's service gateway, it may be encrypted again from the gateway to the destination phone.

There are likely some VoIP providers that have a direct gateway interface between their VoIP solution and another VoIP provider's network, but unless they happen to have compatible configurations, they still will have to decrypt and re-encrypt at the gateway. Therefore, unless you stay within the same VoIP provider's network, you cannot be assured that your connection is protected by end-to-end encryption.

However, even if your VoIP services somehow provide you with secured connections, a VoIP solution is still vulnerable to a number of other threats.

These include all of the standard network attacks, like AitM, hijacking, pharming, and denial-of-service (DoS). Plus, there are also the concerns of vishing, phreaking, fraud, and abuse.

Securing VoIP communications often involves specific application of many common security concepts:

- Use strong passwords and two-factor authentication.
- Record call logs and inspect for unusual activity.
- Block international calling.
- Outsource VoIP to a trusted SaaS.
- Update VoIP equipment firmware.
- Restrict physical access to VoIP-related networking equipment.
- Train users on VoIP security best practices.
- Prevent ghost or phantom calls on IP phones by blocking nonexistent or invalid-origin numbers.
- Implement NIPS with VoIP evaluation features.

Vishing and Phreaking

Malicious individuals can exploit voice communications through social engineering. Social engineering is a means by which an unknown, untrusted, or at least unauthorized person gains the trust of someone inside your organization in order to gain access to information or to a system. For more on social engineering in general, see Chapter 2, “Personnel Security and Risk Management Concepts.”

VoIP services are a favorite tool of social engineers because it allows them to call anyone with little to no expense. VoIP also allows the adversary to falsify their Caller ID in order to mask their identity or establish a pretext to fool the victim. Anyone who can receive a call, whether using a traditional PSTN landline, a PBX business line, a mobile phone, or a VoIP solution, can be the target of a VoIP-originated voice-based social engineering attack. This type of attack is known as *vishing*, which stands for voice-based phishing.

The only way to protect against vishing is to teach users how to respond and interact with any form of communication. Here are some guidelines:

- Always err on the side of caution whenever voice communications seem odd, out of place, or unexpected.
- Always request proof of identity before continuing a call related to anything sensitive, personal, financial, or confidential.
- Require callback authorizations on all voice-only requests for network alterations or activities. A callback authorization occurs when the initial client connection is disconnected, and a person or party calls the client on a predetermined number that will usually be stored in a corporate directory in order to verify the identity of the client.
- Classify information (usernames, passwords, IP addresses, manager names, dial-in numbers, and so on), and clearly indicate which information can be discussed or even confirmed using voice communications.
- If privileged information is requested over the phone by an individual who should know that giving out that particular information over the phone is against the company's security policy, ask why the information is needed and verify their identity again. This incident should also be reported to the security administrator.
- Never give out or change passwords via voice-only communications.
- Block numbers that are associated with vishing.
- Don't assume that the displayed Caller ID is valid. Caller ID should be used as an indicator of who you don't want to talk to, not a confirmation of who is calling.

Malicious attackers known as *phreakers* abuse phone systems in much the same way that attackers abuse computer networks (the “ph” represents “phone”). *Phreaking* is a specific type of attack directed toward the telephone system and voice services in general. Phreakers use various types of technology to circumvent the telephone system to make free long-distance calls, alter the function of telephone service, steal specialized services, and even cause service disruptions. Some phreaker tools are actual

devices, whereas others are just particular ways of using a regular telephone.

Although phreakers originally focused on PSTN phones and systems, they have evolved as voice technology has evolved. Phreakers can attack mobile devices, PBX systems, and VoIP solutions.

PBX Fraud and Abuse

Another voice communications threat is private branch exchange fraud and abuse. *Private branch exchange (PBX)* is a telephone switching or exchange system deployed in private organizations in order to enable multistation use of a small number of external PSTN lines. For example, a PBX may allow 150 phones in the office to have shared access to 20 leased PSTN lines. Many PBX systems allowed for interoffice calls without using external lines, assigned extension numbers to each handset, supported voice mail per extension, and remote calling. Remote calling, also known as hoteling, is the ability to be outside the offices, call into the office PBX system, type in a code to access a dial tone, and then dial another phone number. The original purpose of remote calling was to save money by having external personnel call the office on a toll-free number, and then make any long-distance calls on the office's long-distance calling plan.

Many PBX systems can be exploited by malicious individuals to avoid toll charges and hide their identity. Phreakers may be able to gain unauthorized access to personal voice mailboxes, redirect messages, block access, and redirect inbound and outbound calls.

Countermeasures to PBX fraud and abuse include many of the same precautions you would employ to protect a typical computer network: logical or technical controls, administrative controls, and physical controls. Here are several key points to keep in mind when designing a PBX security solution:

- Consider replacing remote access or long-distance calling through the PBX with a credit card or calling card system.
- Restrict dial-in and dial-out features to authorized individuals who require such functionality for their work tasks.

- If you still have dial-in modems, use unpublished phone numbers that are outside the prefix block range of your voice numbers.
- Protect administrative interfaces for the PBX.
- Block or disable any unassigned access codes or accounts.
- Define an acceptable use policy and train users on how to properly use the system.
- Log and audit all activities on the PBX and review the audit trails for security and use violations.
- Disable maintenance modems (i.e., remote access modems used by the vendor to remotely manage, update, and tune a deployed product) and/or any form of remote administrative access.
- Change all default configurations, especially passwords, and capabilities related to administrative or privileged features.
- Block remote dialing.
- Keep the system current with vendor/service provider updates.
- Deploy direct inward system access (DISA) technologies to reduce PBX fraud by external parties.

Direct inward system access (DISA), like any other security feature, must be properly installed, configured, and monitored in order to obtain the desired security improvement. DISA adds authentication requirements to all external connections to the PBX. Simply having DISA is not sufficient. Be sure to disable all features that are not required by the organization, craft user codes/passwords that are complex and difficult to guess, and then turn on auditing to keep watch on PBX activities.

Additionally, maintaining physical access control to all PBX connection centers, phone portals, and wiring closets prevents direct intrusion from on-site attackers. PBX systems of the past were primarily hardwar-based. Today, there are numerous PBX systems that are primarily software solutions, which may control and manage PSTN lines or VoIP connections. These software-based PBX systems are potentially vulnerable to the same application and network attacks that “standard” software and computers are subjected to, such as buffer overflows, malware, DoS, AitM attacks,

hijacking, and eavesdropping. Thus, if your network is not secure, then your PBX system is likely not being securely managed either.

Remote Access Security Management

Telecommuting, or working remotely, has become a common feature of business computing. Telecommuting usually requires remote access, the ability of a distant client to establish a communication session with a network. Remote access can take the following forms (among others):

- Connecting to a network over the Internet through a VPN
- Connecting to a WAP (which the local environment treats as remote access)
- Connecting to a terminal server system, mainframe, virtual private cloud (VPC) endpoint, virtual desktop interface (VDI), or virtual mobile interface (VMI) through a thin-client connection
- Connecting to an office-located PC using a remote desktop service
- Using cloud-based virtual desktop solutions
- Using a modem to dial up directly to a remote access server

The first three examples use fully capable clients. They establish connections just as if they were directly connected to the LAN. In the last three examples, all computing activities occur on the connected central system rather than on the remote client.

Remote Access and Telecommuting Techniques

Telecommuting is performing work at a remote location (i.e., other than the primary office). In fact, there is a good chance that you perform some form of telecommuting as part of your current job. Telecommuting clients use many remote access techniques to establish connectivity to the central office LAN. There are several types of remote access techniques:

Service-Specific *Service-specific remote access* gives users the ability to connect to and manipulate or interact with a single service, such as email, remotely.

Remote Control *Remote-control remote access* grants a remote user the ability to fully control another system that is physically distant from them. The monitor and keyboard act as if they are directly connected to the remote system.

Remote Node Operation *Remote node operation* is just another name for when a remote client establishes a direct connection to a LAN, such as with wireless, VPN, or dial-up connectivity. A remote system connects to a remote access server, which provides the remote client with network services and possible Internet access.

Remote Connection Security

When remote access capabilities are deployed in any environment, security must be considered and implemented to provide protection for your private network against remote access complications:

- Remote access users should be stringently authenticated before being granted access.
- Only those users who specifically need remote access for their assigned work tasks should be granted permission to establish remote connections.
- All remote communications should be protected from interception and eavesdropping. Doing so usually requires an encryption solution that provides strong protection for the authentication traffic as well as all data transmission.

It is important to establish secure communication channels before initiating the transmission of sensitive, valuable, or personal information. Remote connections can pose several potential security concerns if not protected and monitored sufficiently:

- If anyone with a remote connection can attempt to breach the security of your organization, the benefits of physical security are reduced.
- Telecommuters might use insecure or less secure remote systems to access sensitive data and thus expose it to a greater risk of loss, compromise, or disclosure.

- Remote systems might be exposed to malicious code and could be used as a carrier to bring malware into the private LAN.
- Remote systems might be less physically secure and thus at risk of being used by unauthorized entities or stolen.
- Remote systems might be more difficult to troubleshoot, especially if the issues revolve around a remote connection.
- Remote systems might not be as easy to upgrade or patch due to their potential infrequent connections or slow throughput links. However, this issue is lessened when high-speed, reliable broadband links are present.

These issues, and likely others, need to be considered, and a remote access security policy needs to be established.

Plan a Remote Access Security Policy

When outlining your remote access security management strategy, be sure to address the following issues in the policy:

Remote Connectivity Technology Each type of connection has its own unique security issues. Fully examine every aspect of your connection options. This can include cellular/mobile services, PSTN modems, cable TV Internet services, Digital Subscriber Line (DSL), fiber connections, wireless networking, and satellite.

Transmission Protection There are several forms of encrypted protocols, encrypted connection systems, and encrypted network services or applications. Use the appropriate combination of secured services for your remote connectivity needs. This can include VPNs and/or TLS.

Authentication Protection In addition to protecting data traffic, you must ensure that all logon credentials are properly secured. This requires the use of a secure authentication protocol, may mandate the use of a centralized remote access authentication system, and should require multifactor authentication.

Remote User Assistance Remote access users may periodically require technical assistance. You must have a means established to provide this as efficiently as possible. This can include, for example, addressing software

and hardware issues and user training issues. If an organization is unable to provide a reasonable solution for remote user technical support, it could result in a loss of productivity, compromise of the remote system, or an overall breach of organizational security.

If it is difficult or impossible to maintain a similar level of security on a remote system as is maintained in the private LAN, then remote access should be reconsidered in light of the security risks it represents. Network access control (NAC) can assist with this but may burden slower connections with large updates and patch transfers.

The ability to use remote access or establish a remote connection should be tightly controlled. You can control and restrict the use of remote connectivity by means of filters, rules, or access controls based on user identity, workstation identity, protocol, application, content, and time of day. (See attribute-based access control [ABAC] in Chapter 14.)

It should be a standard element in your security policy that no unauthorized modems be present on any system connected to the private network. You may need to specify this policy further by indicating that those with portable systems must either remove their modems before connecting to the network or boot with a hardware profile that disables the modem's device driver. This is the same prohibition concept that should be applied to secondary connection options of all types, including wireless and cellular.

Network Administrative Functions

Remote access does not need to focus exclusively on general workers for telecommuting. Remote access can also be an essential tool of administrators for the operation of network administrative functions. Network administrative functions encompass a range of tasks and capabilities that administrators perform to manage and control network resources from locations outside the physical infrastructure. These functions are vital for optimizing network performance, ensuring security, and responding to evolving requirements.

Configuration management is a key aspect, allowing administrators to remotely configure and modify network devices such as routers, switches, and firewalls. This involves adjusting settings, updating configurations, and implementing changes to meet the network's needs.

Monitoring and analysis are facilitated through remote access, enabling administrators to track network performance, analyze traffic patterns, and identify potential issues or security threats. Remote monitoring tools help monitor network metrics, analyze logs, and respond to alerts to ensure network health.

Troubleshooting and diagnostics benefit from remote access, as administrators can diagnose and address network issues from a remote location. This includes accessing devices, running diagnostic tests, analyzing logs, and implementing solutions to resolve connectivity or performance problems.

Security management involves configuring security policies, access controls, and authentication mechanisms remotely to protect the network. Administrators can manage firewalls, VPNs, and other security measures to safeguard the network infrastructure.

User account management is streamlined with remote access, allowing administrators to create, modify, or deactivate user accounts, reset passwords, and control access rights to network resources.

Software updates and patch management can be performed remotely, with administrators deploying updates, patches, and security fixes to network devices and servers. This helps address vulnerabilities and ensures that the network operates with the latest features and security enhancements.

Backup and recovery tasks are facilitated through remote access, enabling administrators to schedule remote backups, verify data integrity, and implement recovery procedures in case of data loss or system failures.

Policy enforcement involves administrators ensuring that network configurations align with organizational policies, security standards, and industry regulations. This is done remotely to enforce compliance and maintain a secure network environment.

Remote access network administrative functions are essential for maintaining operational efficiency, responding to issues promptly, and ensuring the overall health and security of network infrastructure, especially in scenarios where physical presence at the network site is not feasible. Any and all remote access-based administrative functions and their necessary security requirements should be defined in an organizational security policy.

Multimedia Collaboration

Multimedia collaboration is the use of various multimedia-supporting communication solutions to enhance distance collaboration (people working on a project together remotely). Often, collaboration allows workers to work simultaneously as well as across different time frames. Collaboration can also be used to track changes and include multimedia functions. Collaboration can incorporate email, chat, voice/VoIP, video/video conferencing, use of a whiteboard, online document editing, real-time file exchange, versioning control, and other tools. It is often a feature of advanced forms of remote meeting technology.

Whatever SaaS service is implemented to support multimedia collaboration, it is essential that it be thoroughly reviewed against the organization's security policy. Just because someone is working remotely does not mean that security should be relaxed. It is important to verify that connections are encrypted, that robust multifactor authentication is in use, and that tracking is available for the hosting organization to review.

One means to support remote access and collaboration activities is through the use of online conferencing solutions. One such product is Zoom, which gained extreme popularity in 2020 due to the pandemic lockdowns that resulted in many employees needing to continue working from home. Zoom is only one of several video conferencing products available.

Now that many employees have returned to the office, there is still a significant need for online remote video conferencing and collaboration. Some organizations have established Zoom rooms. A *Zoom room* refers to a dedicated physical space equipped with audio-visual and communication technology designed for hosting video meetings, conferences, and collaborations. A Zoom room is set up to enhance the quality and experience of online video collaboration meetings. They often include larger displays, high-quality camera and microphone equipment, a quality sound system, touchscreen system/room controls, lighting controls, and other smart technologies.

The concept of a Zoom room is part of the broader trend toward creating collaborative and technology-enabled meeting spaces within workplaces. It allows for a more immersive and effective collaboration experience during

virtual meetings, particularly in settings where teams or groups gather for discussions, presentations, or remote interactions.

Remote Meeting

Remote meeting technology is used for any product, hardware, or software that allows for interaction between remote parties. These technologies and solutions are known by many other terms: digital collaboration, virtual meetings, videoconferencing, software or application collaboration, shared whiteboard services, virtual training solutions, and so on. Any service that enables people to communicate, exchange data, collaborate on materials/data/documents, and otherwise perform work tasks together can be considered a remote meeting technology service.

No matter what form of multimedia collaboration is implemented, the attendant security implications must be evaluated. There are many questions about security that need to be asked and satisfactory answers uncovered prior to deployment or use:

- Does the service use strong authentication techniques?
- Does the communication occur across an open protocol or an encrypted tunnel?
- Is the encryption just from the endpoint to the central server, or is it end-to-end?
- Does the solution allow for true deletion of content?
- Are the activities of users audited and logged?
- Can unauthorized entities join in a private meeting?
- Can attendees interject into the meeting with voice, image, video, or file sharing?
- Does the platform integrate advertising/spam into the interface, and can it be disabled?
- What tracking mechanisms are used, can the tracking be disabled, and what is the data collected for?

- Are sessions recorded? Who has access to the recordings? Can they be exported and distributed?

Multimedia collaboration and other forms of remote meeting technology can improve the work environment and allow for input from a wider range of diverse workers across the globe, but this is a benefit only if the security of the communications solution can be ensured and personnel are trained to use it effectively and in compliance with company policy.

Instant Messaging and Chat

Instant messaging (IM), real-time messaging, or chat is a mechanism that allows for real-time text-based chat between two or more people located anywhere on the Internet. Some IM utilities allow for file transfer, multimedia, voice and videoconferencing, and more. Some forms of IM are based on a peer-to-peer service, whereas others use a centralized controlling server. Peer-to-peer-based IM and cloud-based IM systems are easy for end users to deploy and use, but it's difficult to manage from a corporate perspective because it may lack security or management controls. Messaging systems and chat services usually have numerous vulnerabilities, such as being susceptible to packet sniffing/eavesdropping, lacking native security capabilities such as multifactor authentication and encryption, and providing little or no protection for privacy.

Many stand-alone chat clients have been susceptible to malicious code deposits or infection through their file transfer capabilities. Also, chat users are often subject to numerous forms of social engineering attacks, such as impersonation or convincing a victim to reveal information that should remain confidential (such as passwords, PII, or intellectual property).

When selecting collaboration products, always consider the locus of control and the availability and effectiveness of security features, such as logging, multifactor authentication, and transmission encryption.

Monitoring and Management

Monitoring and management in the context of network operations involve various practices and tools aimed at ensuring the reliability, performance, and security of a network. Network observability, a key aspect, refers to the

ability to gain insights into the internal state of a network by collecting and analyzing relevant data. This involves monitoring various metrics, logs, and traces to understand how the network components are performing. The purpose is to enhance visibility into network behavior, detect issues, and gain actionable insights to optimize performance and troubleshoot problems.

Another critical area is traffic flow/shaping, which involves managing the flow of data within a network to optimize performance, allocate resources efficiently, and ensure a consistent user experience. By shaping traffic, administrators can prioritize certain types of data, manage bandwidth usage, and control the flow of information to prevent congestion or bottlenecks.

Capacity management is the practice of planning, monitoring, and optimizing the network's capacity to ensure it can handle current and future demands effectively. By monitoring resource usage and predicting future needs, administrators can allocate resources appropriately, prevent performance degradation, and ensure scalability.

Fault detection and handling involve identifying and addressing issues, errors, or failures within the network. The goal is to detect problems as soon as possible, minimize downtime, and implement strategies for fault tolerance and resilience. Automated alerts and notifications can aid in a prompt response to faults.

Monitoring and management in networking encompass practices related to network observability, traffic flow/shaping, capacity management, and fault detection and handling. These practices collectively contribute to maintaining a healthy and efficient network infrastructure, ensuring that it meets performance expectations, is resilient to faults, and can adapt to changing demands. Advanced tools and technologies, such as network monitoring software, traffic shaping mechanisms, and predictive analytics, play a crucial role in implementing effective monitoring and management strategies in modern network environments.

Load Balancing

The purpose of *load balancing* is to obtain more optimal infrastructure utilization, minimize response time, maximize throughput, reduce

overloading, and eliminate bottlenecks. A *load balancer* is used to spread or distribute network traffic load across several network links or network devices. Although load balancing can be used in a variety of situations, a common implementation is spreading a load across multiple members of a server farm or cluster. Scheduling or load-balancing methods are the means by which a load balancer distributes the work, requests, or loads among the devices behind it. A load balancer might use a variety of scheduling techniques to perform load distribution, as described in Table 12.1.

TABLE 12.1 Common load-balancing scheduling techniques

Technique	Description
Random choice	Each packet or connection is assigned a destination randomly.
Round robin	Each packet or connection is assigned the next destination in order, such as 1, 2, 3, 4, 5, 1, 2, 3, 4, 5, and so on.
Load monitoring	Each packet or connection is assigned a destination based on the current load or capacity of the targets. The device/path with the lowest current load receives the next packet or connection.
Preferencing or weighted	Each packet or connection is assigned a destination based on a subjective preference or known capacity difference. For example, suppose system 1 can handle twice the capacity of systems 2 and 3; in this case, preferencing would look like 1, 2, 1, 3, 1, 2, 1, 3, 1, and so on.
Least connections/traffic/latency	Each packet or connection is assigned a destination based on the least number of active connections, traffic load, or latency.
Locality based (geographic)	Each packet or connection is assigned a destination based on the destination's relative distance from the load balancer (used when cluster members are geographically separated or across numerous router hops).
Locality based (affinity)	Each packet or connection is assigned a destination based on previous connections from the same client, so subsequent requests go to the same destination to optimize continuity of service. Aka persistence.

Load balancing can be either a software service or a hardware appliance. Load balancing can also incorporate many other features, depending on the

protocol or application, including caching, TLS offloading, compression, buffering, error checking, filtering, and even firewall and IDS capabilities.



TLS offloading is the process of removing the TLS-based encryption from incoming traffic to relieve a web server of the processing burden of decrypting and/or encrypting traffic sent.

Virtual IP Addresses

In load-balancing scenarios, virtual IP addresses (VIP or VIPA) serve as key components to efficiently distribute incoming network traffic across multiple servers or resources. Unlike a physical IP address associated with a specific network interface, a VIP is mapped to a cluster or group of servers. When clients access the VIP, a load balancer directs their requests to one of the servers in the pool, ensuring that the overall load is evenly distributed and preventing any single server from becoming a bottleneck.

Load balancers utilize virtual IP addresses as the entry point for incoming traffic, making decisions on how to distribute the load based on defined algorithms or balancing schedules. The result is an optimized distribution of client requests across the backend servers, contributing to efficient resource utilization.

One significant advantage of virtual IP addresses in load balancing is their role in enhancing high availability. In the event of a server failure, the load balancer can seamlessly redirect traffic to other healthy servers, minimizing downtime and ensuring continuous service availability. This redundancy and failover capability are crucial for maintaining uninterrupted service in dynamic and scalable environments.

Virtual IP addresses also facilitate scalability by allowing the dynamic addition or removal of servers in response to changes in infrastructure size. As the system scales, load balancers adapt to the evolving server pool, and clients can access the service through the virtual IP address without being affected by modifications in the backend server configuration.

Load balancers often handle SSL/TLS termination at the virtual IP address, decrypting incoming encrypted traffic before distributing it to backend servers. Additionally, virtual IP addresses can be associated with content switching, enabling the load balancer to route traffic based on content types or specific application services.

In Global Server Load Balancing (GSLB) scenarios, virtual IP addresses play a crucial role in distributing traffic across multiple data centers or locations on a global scale. This approach considers factors such as proximity, server health, or other criteria to optimize both performance and reliability on a global level.

Virtual IP addresses are fundamental in load balancing architectures, providing a versatile and efficient mechanism for directing incoming traffic to backend servers. Their role spans from optimizing resource utilization and enhancing high availability to supporting scalability and global server load balancing.

Active-Active vs. Active-Passive

An *active-active system* is a form of load balancing that uses all available pathways or systems during normal operations. In the event of a failure of one or more of the pathways, the remaining active pathways must support the full load that was previously handled by all. This technique is used when the traffic levels or workload during normal operations need to be maximized (i.e., optimizing availability), but reduced capacity will be tolerated during adverse conditions (i.e., reducing availability).

An *active-passive system* is a form of load balancing that keeps some pathways or systems in an unused dormant state during normal operations. If one of the active elements fails, then a passive element is brought online and takes over the workload for the failed element. This technique is used when the level of throughput or workload needs to be consistent between normal states and adverse conditions (i.e., maintaining availability consistency).

Manage Email Security

Email is one of the most widely and commonly used Internet services. The email infrastructure employed on the Internet primarily consists of email servers using *Simple Mail Transfer Protocol (SMTP)* (TCP port 25) to accept messages from clients, transport those messages to other servers, and deposit them into a user's server-based inbox. In addition to email servers, the email infrastructure includes email clients. Clients retrieve email from their server-based inboxes using *Post Office Protocol version 3 (POP3)* (TCP port 110) or *Internet Message Access Protocol (IMAP)* (technically version 4) (TCP port 143). Internet-compatible email systems rely on the X.400 standard for addressing and message handling.

Postfix is the most common SMTP server for Unix systems (replacing the previously popular Sendmail product), and Exchange is the most common SMTP server for Microsoft systems. In addition to these popular products, numerous alternatives exist, but they all share the same basic functionality and compliance with Internet email standards.

If you deploy an SMTP server, it is imperative that you properly configure strong authentication for both inbound and outbound mail. SMTP is designed to be a mail relay system. This means it relays mail from the sender to the intended recipient. However, you want to avoid turning your SMTP server into an *open relay* (also known as an open relay agent or *relay agent*), which is an SMTP server that does not authenticate senders before accepting and relaying mail. Open relays are prime targets for spammers because they allow spammers to send out floods of emails by piggybacking on an insecure email infrastructure. As open relays are locked down—becoming *closed relays* or *authenticated relays*—adversaries are often resorting to hijacking authenticated user accounts through social engineering or credential stuffing/spraying/guessing attacks.

Another option to consider for corporate email is an SaaS email solution. Examples of cloud or hosted email include Gmail (Google Workspace) and Outlook/Exchange Online. SaaS email enables you to leverage the security experience and management expertise of some of the largest email service providers to support your company's communications. Benefits of SaaS email include high availability, distributed architecture, ease of access, standardized configuration, and physical location independence. However, there are some potential risks with using a hosted email solution, including

block listing issues, rate limiting, app/add-on restrictions, and what (if any) additional security mechanisms you can deploy.

Email Security Goals

The basic email mechanisms in use on the Internet offer efficient delivery of messages but lack controls to provide for confidentiality, integrity, or even availability. In other words, basic email is not secure. However, you can add security to email in many ways. Adding security to email may satisfy one or more of the following objectives:

- Restrict access to messages to their intended recipients (i.e., privacy and confidentiality).
- Maintain the integrity of messages.
- Authenticate and verify the source of messages.
- Provide for nonrepudiation.
- Verify the delivery of messages.
- Classify sensitive content within or attached to messages.

There is no real method to guarantee the availability of email, such as access to an inbox or assured delivery. However, these can be compensated for using verified delivery and maintaining several access vectors from clients to email servers (such as LAN, general Internet, and mobile data services).

As with any aspect of IT security, email security begins in a security policy approved by upper management. Within the security policy, you must address several issues:

- Acceptable use policies for email
- Access control and privacy
- Email management
- Email backup and retention policies

Acceptable use policies define what activities can and cannot be performed over an organization's email infrastructure. It is often stipulated that professional, business-oriented emails and a limited amount of personal emails can be sent and received through company-owned or provided email systems. Specific restrictions are usually placed on performing personal business (i.e., work for another organization, including self-employment) and sending or receiving illegal, immoral, or offensive communications as well as engaging in any other activities that would have a detrimental effect on productivity, profitability, or public relations.

Access control over email should be maintained so that users have access only to their specific inbox and email archive databases. An extension of this rule implies that no other user, authorized or not, can gain access to an individual's email. Access control should provide for both legitimate access and some level of privacy, at least from other employees and unauthorized intruders.

The mechanisms and processes used to implement, maintain, and administer email for an organization should be clarified. End users may not need to know the specifics of email management, but they do need to know whether email is considered private communication.

Email has recently been the focus of numerous court cases in which archived messages were used as evidence—often to the chagrin of the author or recipient of those messages. If email is to be retained (that is, backed up and stored in archives for future use), users need to be made aware of this. If email is to be reviewed for violations by an auditor, users need to be informed of this as well. Some companies have elected to retain only the last three months of email archives before they are destroyed, whereas others have opted to retain email for years. Depending on your country and industry, there are often regulations that dictate retention policies. But keep in mind, that although your organization may discard sent or received messages after only a few months, external entities may retain their copies of the conversations for years. The details of an email retention policy may need to be shared with affected subjects, which may include privacy implications, how long the messages are maintained, and for what purposes the messages can be used (such as auditing or violation investigations).

Understand Email Security Issues

The first step in deploying email security is to recognize the vulnerabilities specific to email. The standard protocols used to support email (i.e., SMTP, POP3, and IMAP) do not employ encryption natively. Thus, all messages are transmitted in the form in which they are submitted to the email server, which is often plaintext. This makes interception and eavesdropping easy.

Email is a common delivery mechanism for viruses, worms, Trojan horses, documents with destructive macros, and other malicious code. The proliferation of support for various scripting languages, auto-download capabilities, and auto-execute features has transformed hyperlinks within the content of email and attachments into a serious threat to every system. Many email clients now natively support HTML code (and thus JavaScript), which may be rendered automatically when a message is accessed.

Email offers little in the way of native source verification. Spoofing the source address of an email is a simple process for even a novice attacker. Email headers can be modified at their source or at any point during transit. Furthermore, it is also possible to deliver email directly to a user's inbox on an email server by directly connecting to the email server's SMTP port. And speaking of in-transit modification, there are no native integrity checks to ensure that a message was not altered between its source and destination.

In addition, email itself can be used as an attack mechanism. When sufficient numbers of messages are directed to a single user's inbox or through a specific SMTP server, a DoS attack can result. This attack is often called mail-bombing and is simply a DoS performed by inundating a system with messages. The DoS can be the result of storage capacity consumption or processing capability utilization. Either way, the result is the same: legitimate messages cannot be delivered.

A similar DoS issue is called a *mail storm*. This is when someone responds with a Reply All to a message that has a significant number of other recipients in the To: and CC: lines. As others receive these replies, they, in turn, use Reply All with their comments or demands to be removed from the conversation. This is further exacerbated if recipients have auto-responders set to Reply All for out-of-office notifications or other announcements.

Like email flooding and malicious code attachments, unwanted email can be considered an attack. Sending unwanted, inappropriate, or irrelevant messages is called spamming. Spamming is often little more than a nuisance, but it does waste system resources both locally and over the Internet. It is often difficult to stop spam because the source of the messages is usually spoofed.

Email Security Solutions

Imposing security on email is possible, but the efforts should be in tune with the value and confidentiality of the messages being exchanged. You can use several protocols, services, and solutions to add security to email without requiring a complete overhaul of the entire Internet-based SMTP infrastructure. Many of these email security improvements are forms of encryption; see Chapter 6, "Cryptography and Symmetric Key Algorithms," and Chapter 7, "PKI and Cryptographic Applications," for information on cryptography.

Secure Multipurpose Internet Mail Extensions (S/MIME) S/MIME is an email security standard that offers authentication and confidentiality to email through public key encryption, digital envelopes, and digital signatures. Authentication is provided through X.509 digital certificates issued by trusted third-party CAs. Privacy is provided through the use of Public Key Cryptography Standard (PKCS) standards-compliant encryption. Two types of messages can be formed using S/MIME: signed messages and secured enveloped messages. A signed message provides integrity, sender authentication, and nonrepudiation. An enveloped message provides recipient authentication and confidentiality.

Pretty Good Privacy (PGP) PGP is a peer-to-peer public-private key–based email system that uses a variety of encryption algorithms to encrypt files and email messages. PGP is not a standard but rather an independently developed product that has wide Internet grassroots support, which has elevated its proprietary certificates to de facto standard status.

DomainKeys Identified Mail (DKIM) DKIM is an email authentication method designed to verify the authenticity of the sender of an email message. It allows the recipient's mail server to check that an email claiming to come from a specific domain was indeed authorized by the

owner of that domain. DKIM helps combat email spoofing, phishing, and other forms of email fraud by providing a way to verify the integrity of the email's origin. See http://dkim.org.

Sender Policy Framework (SPF) To protect against spam and email spoofing, an organization can also configure its SMTP servers for Sender Policy Framework (SPF). SPF operates by checking that inbound messages originate from a host authorized to send messages by the owners of the SMTP origin domain. For example, if you receive a message from mark.nugget@abccorps.com, then SPF checks with the administrators of smtp.abccorps.com that `mark.nugget` is authorized to send messages through their system before the inbound message is accepted and sent into your recipient's inbox.

Domain-based Message Authentication Reporting and Conformance (DMARC) DMARC is a DNS-based email authentication system. It is an email authentication and policy framework that builds on the SPF (Sender Policy Framework) and DKIM (DomainKeys Identified Mail) protocols. DMARC allows domain owners to specify how their emails should be authenticated, how failed authentication should be handled, and how feedback about email activity should be provided. DMARC is intended to protect against business email compromise (BEC), phishing, and other email scams. Email servers can verify if a received message is valid by following the DNS-based instructions; if invalid, the email can be discarded, quarantined, or delivered anyway.

STARTTLS A lot of organizations are using Secure SMTP over TLS nowadays; however, it's not as widespread as it should be. STARTTLS (aka *explicit TLS* or *opportunistic TLS* for SMTP) will attempt to set up an encrypted connection with the target email server in the event that it is supported. STARTTLS is not a protocol but instead an SMTP command. Once the initial SMTP connection is made to the email server, the STARTTLS command will be used. If the target system supports TLS, then an encrypted channel will be negotiated. Otherwise, it will remain as plaintext. STARTTLS's secure session will take place on TCP port 587. STARTTLS can also be used with IMAP connections, whereas POP3 connections use the STLS command to perform a similar function.

Implicit SMTPS This is the TLS-encrypted form of SMTP, which assumes the target server supports TLS. If accurate, then an encrypted session is negotiated. If not, then the connection is terminated because plaintext is not accepted. SMTPS communications are initiated against TCP port 465.

Free PGP Solution

PGP started off as a free product for all to use, but it has since splintered into various divergent products. PGP is a commercial product, whereas OpenPGP is a developing standard that GnuPG is compliant with and that was independently developed by the Free Software Foundation. If you have not used PGP before, we recommend downloading the appropriate GnuPG version for your preferred email platform. This secure solution is sure to improve your email privacy and integrity. You can learn more about GnuPG at `http://gnupg.org`. You can learn more about PGP by visiting its pages on Wikipedia.

By using these and other security mechanisms for email and communication transmissions, you can reduce or eliminate many of the security vulnerabilities of email. Digital signatures can help eliminate impersonation. The encryption of messages reduces eavesdropping. And the use of email filters keeps spamming and mail-bombing to a minimum.

Blocking attachments at the email gateway system on your network can ease the threats from malicious attachments. You can have a 100 percent no-attachments policy or block only attachments that are known or suspected to be malicious, such as attachments with extensions that are used for executable and scripting files. If attachments are an essential part of your email communications, you'll need to train your users and use antimalware tools for protection. Training users to avoid contact with suspicious or unexpected attachments greatly reduces the risk of malicious code transference via email. Antimalware products are generally effective against known malicious code, but they offer little protection against new or unknown varieties.

Unwanted emails can be a hassle, a security risk, and a drain on resources. Whether spam, malicious email, or just bulk advertising, there are several

ways to reduce the impact on your infrastructure. Block list services offer a subscription system to a list of known email abuse sources. You can integrate the block list into your email server so that any messages originating from a known abusive domain or IP address are automatically discarded. Another option is to use a challenge/response filter. In these services, when an email is received from a new/unknown origin address, an autoresponder sends a request for a confirmation message. Spammers and auto-emailers will not respond to these requests, but valid humans will. Once they have confirmed that they are human and agree not to spam the destination address, their source address is added to an allow list for future communications.

Unwanted email can also be managed through the use of email *reputation filtering*. Several services maintain a grading system of email services in order to determine which are used for standard/normal communications and which are used for spam. These services include Sender Score, Cisco SenderBase Reputation Service, Broadcom's Symantec Email Security.cloud, Spamhaus ZEN, and Barracuda Reputation Block List (BRBL). These and other mechanisms are used as part of several spam filtering technologies, such as Apache SpamAssassin and spamd.

Fax Security

Fax communications are waning in popularity because of the widespread use of email. Even with declining use, faxes still represent a communications path that is vulnerable to attack. Like any other telephone communication, faxes can be intercepted and are susceptible to eavesdropping.

Some of the mechanisms that can be deployed to improve the security of faxes are fax encrypters, link encryption, activity logs, and exception reports. A fax encrypter gives a fax machine the capability to use an encryption protocol to scramble the outgoing fax signal. Link encryption is the use of an encrypted communication path, like a VPN link or a secured telephone link, to transmit the fax. Activity logs and exception reports can be used to detect anomalies in fax activity that could be symptoms of an attack.

In addition to the security of a fax transmission, it is important to consider the security of a received fax. Faxes that are automatically printed may sit in the out tray for a long period of time, therefore making them subject to viewing by unintended recipients. Studies have shown that adding banners of CONFIDENTIAL, PRIVATE, and so on spur the curiosity of passersby. So, disable automatic printing. Also, avoid fax machines that retain a copy of the fax in memory or on a local storage device. Consider integrating your fax system with your network so that you can email faxes to intended recipients instead of printing them to paper.

Virtual Private Network

A *virtual private network (VPN)* is a communication channel between two entities across an intermediary untrusted network. VPNs can provide several critical security functions, such as access control, authentication, confidentiality, and integrity. Most VPNs use encryption to protect the encapsulated traffic, but encryption is not necessary for the connection to be considered a VPN. A VPN is an example of a virtualized network.

VPNs are most commonly associated with establishing secure communication paths through the Internet between two distant networks. However, they can exist anywhere, including within private networks or between end-user systems connected to an ISP. The VPN can link two networks or two individual systems. They can link clients, servers, routers, firewalls, and switches. VPNs are also helpful in providing security for legacy applications that rely on risky or vulnerable communication protocols or methodologies, especially when communication is across a network.

Although VPNs can provide confidentiality and integrity over insecure or untrusted intermediary networks, they do not provide or guarantee availability. VPNs are also in relatively widespread use to get around location requirements for services like Netflix and Hulu and thus provide a (at times questionable) level of anonymity.

A *VPN concentrator* is a dedicated hardware device designed to support a large number of simultaneous VPN connections, often hundreds or thousands. It provides high availability, high scalability, and high performance for secure VPN connections. A VPN concentrator can also be called a *VPN server*, a *VPN gateway*, a *VPN firewall*, a *VPN remote access server (RAS)*, a *VPN device*, a *VPN proxy*, or a *VPN appliance*. The use of VPN devices is transparent to networked systems. Therefore, individual hosts do not need to support VPN capabilities locally if a VPN appliance is present.

Tunneling

Before you can truly understand VPNs, you must first grasp the concept of tunneling. *Tunneling* is the network communications process that protects the contents of protocol packets by encapsulating them in packets of another protocol. The encapsulation is what creates the logical illusion of a communications tunnel over the untrusted intermediary network. This virtual path exists between the encapsulation and the deencapsulation entities located at the ends of the communication.

As data is transmitted from one system to another across a VPN link, the normal LAN TCP/IP traffic is encapsulated (encased or enclosed) in the VPN protocol. The VPN protocol acts like a security envelope that provides

special delivery capabilities (for example, across the Internet) as well as security mechanisms (such as data encryption).

In fact, sending a snail mail letter to your grandmother involves the use of a tunneling system. You create the personal letter (the primary content protocol packet) and place it in an envelope (the tunneling protocol). The envelope is delivered through the postal service (the untrusted intermediary network) to its intended recipient. You can use tunneling in many situations, such as when you're bypassing firewalls, gateways, proxies, or other traffic control devices. The bypass is achieved by encapsulating the restricted content inside packets that are authorized for transmission. The tunneling process prevents the traffic control devices from blocking or dropping the communication because such devices don't know what the packets actually contain.

Tunneling is often used to enable communications between otherwise disconnected systems. If two systems are separated by a lack of network connectivity, a communication link can be established by a modem dial-up link or other remote access or wide area network (WAN) networking service. The actual LAN traffic is encapsulated in whatever communication protocol is used by the temporary connection, such as Point-to-Point Protocol in the case of modem dial-up. If two networks are connected by a network employing a different protocol, the protocol of the separated networks can often be encapsulated within the intermediary network's protocol to provide a communication pathway.

Regardless of the actual situation, tunneling protects the contents of the inner protocol and traffic packets by encasing, or wrapping, it in an authorized protocol used by the intermediary network or connection. Tunneling can be used if the primary protocol is not routable and to keep the total number of protocols supported on the network to a minimum.

If the act of encapsulating a protocol involves encryption, tunneling can provide a means to transport sensitive data across untrusted intermediary networks without fear of losing confidentiality and integrity.

Tunneling is not without its problems. It is generally an inefficient means of communicating because most protocols include their own error detection, error handling, acknowledgment, and session management features, so using more than one protocol at a time compounds the overhead required to

communicate a single message. Furthermore, tunneling creates either larger packets or additional packets that in turn consume additional network bandwidth. Tunneling can quickly saturate a network if sufficient bandwidth is not available. In addition, tunneling is a point-to-point communication mechanism and is not designed to handle broadcast traffic.

Tunneling also makes it difficult, if not impossible, to monitor the contents of the traffic in some circumstances, creating issues for security practitioners. When firewalls, intrusion detection systems, malware scanners, or other packet-filtering and packet-monitoring security mechanisms are used, you must realize that the data payload of VPN traffic won't be viewable, accessible, scannable, or filterable, because it's encrypted. Thus, for these security mechanisms to function against VPN-transported data, they must be placed outside of the VPN tunnel to act on the data after it has been decrypted and returned to normal LAN traffic.

How VPNs Work

A VPN link can be established over any other network communication connection. Examples include a typical LAN cable connection, a wireless LAN connection, a remote access dial-up connection, a WAN link, or even a client using an Internet connection for access to an office LAN. A VPN link acts just like a typical direct LAN cable connection; the only possible difference would be speed based on the intermediary network and on the connection types between the client system and the server system. Over a VPN link, a client can perform the same activities and access the same resources as if they were directly connected via a LAN cable. This remote access method is known as remote node operation.

VPNs can connect two individual systems or two entire networks. The only difference is that the transmitted data is protected only while it is within the VPN tunnel. Remote access servers or firewalls on the network's border act as the start points and endpoints for VPNs. Thus, traffic is unprotected within the source LAN, protected between the border VPN servers, and then unprotected again once it reaches the destination LAN.

VPN links through the Internet for connecting to distant networks are often inexpensive alternatives to direct links or leased lines. The cost of two high-

speed Internet links to local ISPs to support a VPN is often significantly less than the cost of any other connection means available.

VPNs can operate in two modes: *transport mode* and *tunnel mode*.

Transport mode links or VPNs are anchored or end at the individual hosts connected together. Let's use IPSec as an example (more on IPSec later in this chapter). In transport mode, IPSec provides encryption protection for just the payload and leaves the original message header intact (see Figure 12.1). This type of VPN is also known as a *host-to-host VPN* or an *end-to-end encrypted VPN*, since the communication remains encrypted while it is in transit between the connected hosts. Since transport mode VPNs do not encrypt a communication's header, this mode is best used only within a trusted network between individual systems. When needing to cross untrusted networks or link to and/or from multiple systems, then tunnel mode should be used.

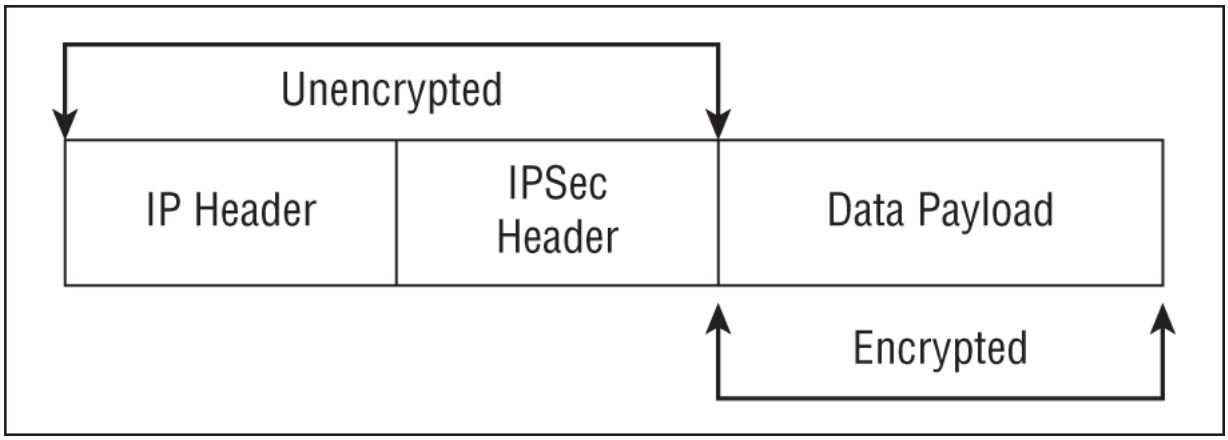


FIGURE 12.1 IPSec's encryption of a packet in transport mode

Tunnel mode links or VPNs terminate (i.e., are anchored or end) at VPN devices on the boundaries of the connected networks (or one remote device). In tunnel mode, IPSec provides encryption protection for both the payload and message header by encapsulating the entire original LAN protocol packet and adding its own temporary IPSec header (see Figure 12.2).

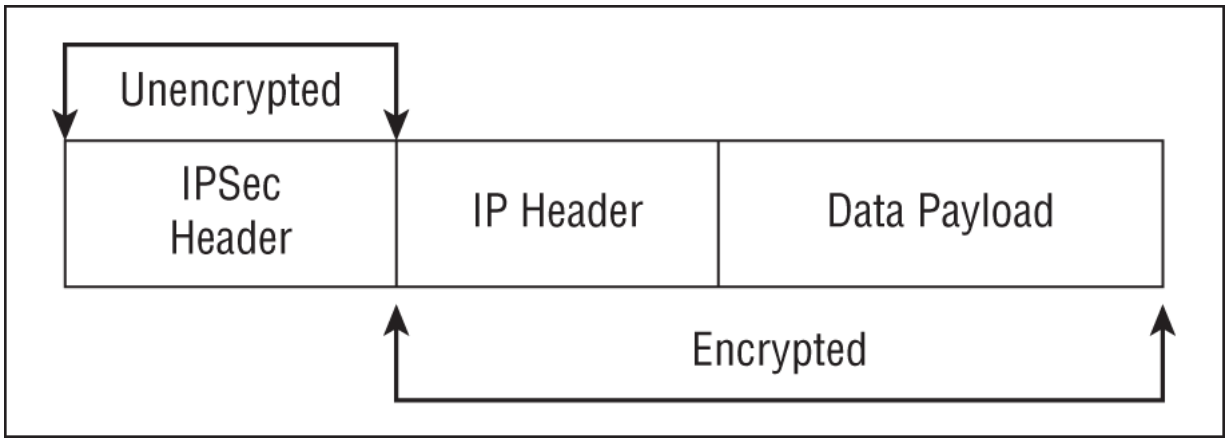


FIGURE 12.2 IPSec's encryption of a packet in tunnel mode

Numerous scenarios lend themselves to the deployment of tunnel mode VPNs; for example, VPNs can be used to connect two networks across the Internet (see Figure 12.3) (aka site-to-site VPN) or to allow distant clients to connect to an office local area network (LAN) across the Internet (see Figure 12.4) (aka remote access VPN). Once a VPN link is established, the network connectivity for the VPN client is the same as a local LAN connection. A *remote access VPN* is a variant of the *site-to-site VPN*. This type of VPN is also known as a *link encryption VPN*, since encryption is only provided when the communication is in the VPN link or portion of the communication. There may be network segments before and after the VPN, which are not secured by the VPN.

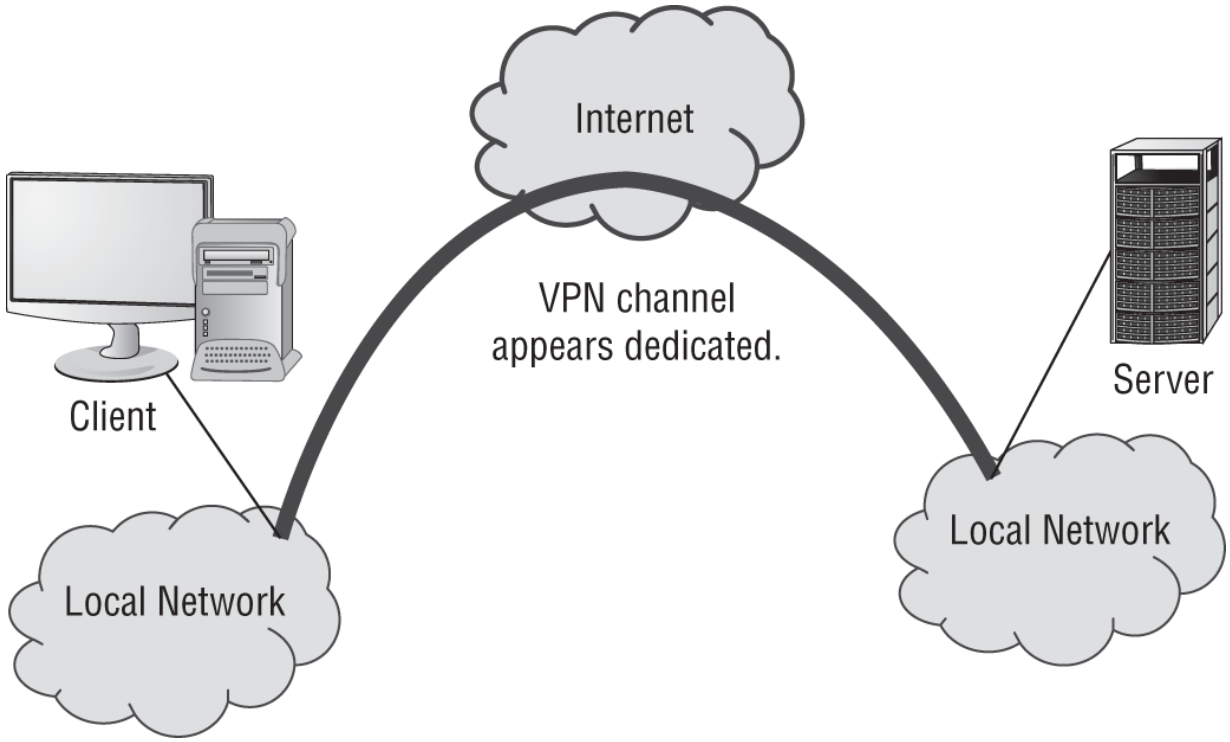


FIGURE 12.3 Two LANs being connected using a tunnel-mode VPN across the Internet



A *wide area network (WAN)* is a network over a long distance. A *metropolitan area network (MAN)* is a network within a town or city. A *campus area network (CAN)* is a network within a college campus or a business park. A VPN can be used over any type of network.

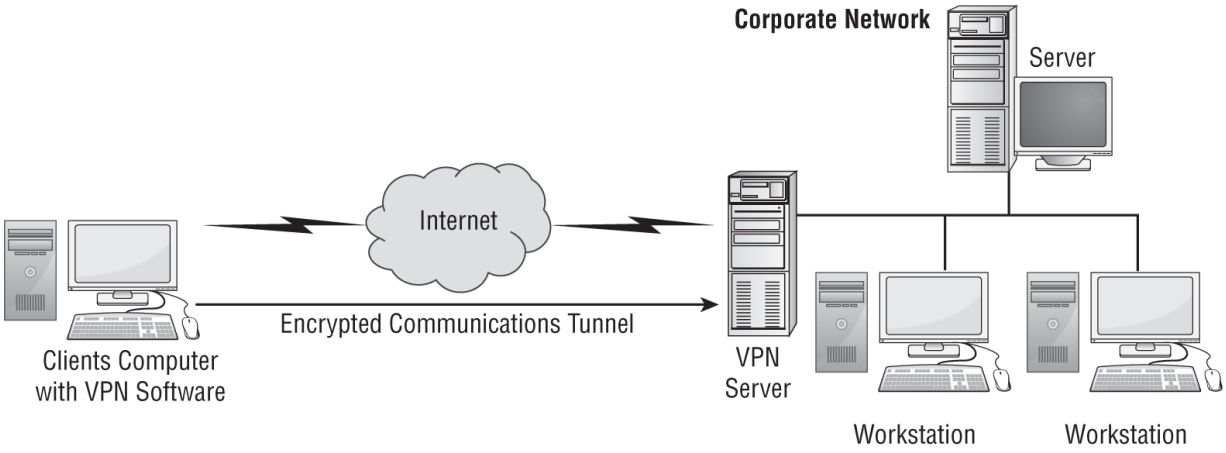


FIGURE 12.4 A client connecting to a network via a remote-access/tunnel VPN across the Internet

Always-On

An *always-on VPN* is one that attempts to auto-connect to the VPN service every time a network link becomes active. Always-on VPNs are mostly associated with mobile devices. Some always-on VPNs can be configured to engage only when an Internet link is established rather than a local network link or only when a Wi-Fi link is established rather than a wired link. Due to the risks of using an open public Internet link, whether wireless or wired, having an always-on VPN will ensure that a secure connection is established every time when attempting to use online resources.

Split Tunnel vs. Full Tunnel

A *split tunnel* is a VPN configuration that allows a VPN-connected client system (i.e., remote node) to access both the organizational network over the VPN and the Internet directly at the same time. The split tunnel thus simultaneously grants an open connection to the Internet and to the organizational network. This is usually considered a security risk for the organizational network since, when a split-tunnel VPN is established, an open pathway exists from the Internet through the client to the LAN. With a VPN connection to the LAN, the client is considered trusted, so filtering is not often used. Clients don't usually have the best filtering services themselves. So, this split tunnel pathway is an easier means for transference of malicious code, initiating intrusions, or exfiltrating confidential data than the direct LAN-to-Internet link, which is filtered by a firewall.

A *full tunnel* is a VPN configuration in which all of the client's traffic is sent to the organizational network over the VPN link, and then any Internet-destined traffic is routed out of the organizational network's proxy or firewall interface to the Internet. A full tunnel ensures that all traffic is filtered and managed by the organizational network's security infrastructure.

Common VPN Protocols

VPNs can be implemented using software or hardware solutions. In either case, there are several common VPN protocols: PPTP, L2TP, SSH, OpenVPN (i.e., TLS), and IPSec.

Point-to-Point Tunneling Protocol

Point-to-Point Tunneling Protocol (PPTP) is an obsolete encapsulation protocol developed from the dial-up Point-to-Point Protocol. It operates at the Data Link Layer (Layer 2) of the OSI model and is used on IP networks. PPTP uses TCP port 1723. PPTP offers protection for authentication traffic through the same authentication protocols supported by PPP:

- Password Authentication Protocol (PAP)
- Challenge Handshake Authentication Protocol (CHAP)
- Extensible Authentication Protocol (EAP)
- Microsoft Challenge Handshake Authentication Protocol (MS-CHAPv2)

The initial tunnel negotiation process used by PPTP is not encrypted. Thus, the session establishment packets that include the IP address of the sender and receiver—and can include usernames and hashed passwords—could be intercepted by a third party. Most modern uses of PPTP have adopted the Microsoft customized implementation (MS-CHAPv2), which supports session encryption using Microsoft Point-to-Point Encryption (MPPE) and supports various secure authentication options. Although PPTP is obsolete, many OSs and VPN services still support it.

Layer 2 Tunneling Protocol (L2TP)

Layer 2 Tunneling Protocol (L2TP) was developed by combining features of PPTP and Cisco's Layer 2 Forwarding (L2F) VPN protocol. Since its development, L2TP has become an Internet standard (RFC 2661). Obviously, L2TP operates at Layer 2 and thus can support just about any Layer 3 networking protocol. L2TP uses UDP port 1701.

L2TP can rely on PPP's supported authentication protocols, specifically IEEE 802.1X, which is a derivative of EAP from PPP. IEEE 802.1X enables L2TP to leverage or borrow authentication services from any available AAA server on the network, such as RADIUS or TACACS+. L2TP does not offer native encryption, but it supports the use of payload encryption protocols. Although it isn't required, L2TP is most often deployed using IPSec's ESP for payload encryption.



Generic Routing Encapsulation (GRE) is also a proprietary Cisco tunneling protocol that can be used to establish VPNs. GRE provides encapsulation but not encryption.

SSH

Secure Shell (SSH) is a secure replacement for Telnet (TCP port 23) and many of the Unix "r" tools, such as `rlogin`, `rsh`, `rexec`, and `rcp`. While Telnet provides plaintext remote access to a system, all SSH transmissions (both authentication and data exchange) are encrypted. SSH operates over TCP port 22. SSH is frequently used with a terminal emulator program such as Minicom or PuTTY. An example of SSH use would involve remotely connecting to a web server, firewall, switch, or router in order to make configuration changes.

SSH is a very flexible tool. It can be used as a secure Telnet replacement; it can be used to encrypt protocols (such as SFTP, SEXEC, SLOGIN, and SCP) similar to how TLS operates; and it can be used as a VPN protocol. However, as a VPN, SSH is limited to transport mode (i.e., end-to-end encryption between individual hosts, aka link encryption and host-to-host VPN). The tool OpenSSH is a means to implement SSH VPNs.



For most secure protocols, if the *S* in the name is a prefix, like with SFTP, then the encryption is provided by SSH (which has an *S* as its first letter). If the *S* in the name is a suffix, like with HTTPS, then the encryption is provided by TLS (which has S as its last letter).

OpenVPN

OpenVPN is based on TLS (formally SSL) and provides an easy-to-configure but robustly secured VPN option. OpenVPN is an open source implementation that can use either preshared passwords or certificates for authentication. Many WAPs support OpenVPN, which is a native VPN option for using a home or business WAP as a VPN gateway.

IP Security Protocol

Internet Protocol Security (IPSec) is a standard of IP security extensions used as an add-on for IPv4 and integrated into IPv6. The primary use of IPSec is for establishing VPN links between internal and/or external hosts or networks. IPSec works only on IP networks and provides for secured authentication as well as encrypted data transmission. IPSec is sometimes paired with L2TP as L2TP/IPSec.

IPSec isn't a single protocol but rather a collection of protocols, including AH, ESP, HMAC, IPComp, and IKE.

Authentication Header (AH) provides assurances of message integrity and nonrepudiation. AH also provides the primary authentication function for IPSec, implements session access control, and prevents replay attacks.

Encapsulating Security Payload (ESP) provides confidentiality and integrity of payload contents. It provides encryption, offers limited authentication, and prevents replay attacks. Modern IPSec ESP typically uses advanced encryption standard (AES) encryption. The limited authentication allows ESP to establish its own links without using AH and perform periodic mid-session reauthentication to detect and respond to session hijacking. ESP can operate in either transport mode or tunnel mode.

Hash-based Message Authentication Code (HMAC) is the primary hashing or integrity mechanism used by IPSec.

IP Payload Compression (IPComp) is a compression tool used by IPSec to compress data prior to ESP encrypting it in order to attempt to keep up with wire-speed transmission.

IPSec uses public-key cryptography and symmetric cryptography to provide encryption (aka hybrid cryptography), secure key exchange, access control, nonrepudiation, and message authentication, all using standard Internet protocols and algorithms. The mechanism of IPSec that manages cryptography keys is *Internet Key Exchange (IKE)*. IKE is composed of three elements: OAKLEY, SKEME, and ISAKMP. OAKLEY is a key generation and exchange protocol similar to Diffie–Hellman. *Secure Key Exchange Mechanism (SKEME)* is a means to exchange keys securely, similar to a digital envelope. Modern IKE implementations may also use ECDHE for key exchange. *Internet Security Association and Key Management Protocol (ISAKMP)* is used to organize and manage the encryption keys that have been generated and exchanged by OAKLEY and SKEME. A security association is the agreed-on method of authentication and encryption used by two entities (a bit like a digital keyring). ISAKMP is used to negotiate and provide authenticated keying material (a common method of authentication) for security associations in a secured manner. Each IPSec VPN uses two security associations, one for encrypted transmission and the other for encrypted reception. Thus, each IPSec VPN is composed of two simplex communication channels that are independently encrypted. ISAKMP's use of two security associations per VPN is what enables IPSec to support multiple simultaneous VPNs from each host.

Switching and Virtual LANs

Switches are the most common modern network management device. A switch operates primarily at Layer 2 but may be equipped to operate at Layer 3 (or higher) for specialty purposes. An unmanaged switch has no configuration options. A managed switch may offer numerous configuration options, such as VLANs and MAC limiting.

All switches operate around four primary functions: learning, forwarding, dropping, and flooding.

Learning or learning mode is how a switch becomes aware of its local network. Each received inbound Ethernet frame is evaluated. First, the source MAC address is checked against the content addressable memory (CAM) table. The CAM table is held in switch memory and contains a mapping between MAC address and port number. In this case, the port number is the physical RJ-45 jack rather than a Transport-layer protocol concern. If the Ethernet frame's source MAC address is not in the CAM table, it is added. Second, the destination MAC address is checked against the CAM table. If the address is present, then the exit port in the table is compared to the port that the current Ethernet frame was received on. If the port numbers are different, then the frame is forwarded out the exit port. If the port numbers are the same, then the frame is dropped (since it is already present on the correct network segment). If the destination MAC address is not present in the CAM table, then it is flooded or sent out all ports. This is done to hopefully allow the frame to reach its destination even if the destination is not known.

A *virtual local area network (VLAN)* is a hardware-imposed network segmentation created by switches. By default, all ports on a switch are part of VLAN 1. But as the switch administrator changes the VLAN assignment on a port-by-port basis, various ports can be grouped together and kept distinct from other VLAN port designations. VLANs can also be assigned or created based on the device's MAC address, IP subnetting, specified protocols, or authentication. VLAN management is most commonly used to distinguish between user traffic and management traffic. VLAN 1, the default VLAN, is typically designated as the VLAN for management traffic.

VLANs are used for traffic management because they are a form of network segmentation. Network segments exist to contain traffic within and block traffic attempting to exit or enter. Communications between members of the same VLAN occur without hindrance, but communications between VLANs require a routing function. VLAN routing can be provided either by an external router or by the switch's internal software (one reason for the terms *L3 switch* and *multilayer switch*). VLANs are treated like subnets but aren't subnets. VLANs are created by switches. Subnets are created by IP address and subnet mask assignments.

VLAN management is the use of VLANs to control traffic for security or performance reasons. VLANs can be used to isolate traffic between network segments. This can be accomplished by not defining a route between different VLANs or by specifying a deny filter between certain VLANs (or certain members of a VLAN). Any network segment that doesn't need to communicate with another in order to accomplish a work task/function shouldn't be able to do so. VLANs should be used to allow communications that are necessary and to block/deny anything that isn't necessary. Remember, “deny by default; allow by exception” isn't a guideline just for firewall rules but for security in general.

VLANs are used to segment a network logically without altering its physical topology. They are easy to implement, have little administrative overhead, and are a hardware-based solution (specifically a Layer 3 switch). As networks are being crafted in virtual environments or in the cloud, software switches or virtual switches are often used. In these situations, VLANs are not hardware-based but instead are switch software–based implementations or impositions. A VLAN is an example of a virtualized network.



In cloud and virtual environments, *distributed virtual switches* are becoming more common than stand-alone virtual switches because they help reduce the chance of introducing configuration errors. They are more easily centrally managed and can be managed using an infrastructure as code (IaC) architecture approach.

VLANs control and restrict broadcast traffic and reduce a network's vulnerability to sniffers because a switch treats each VLAN as a separate network division. It's the routing function between VLANs that blocks Ethernet broadcasts between subnets and VLANs, because a router (or any device performing Layer 3 routing functions such as a Layer 3 switch) doesn't forward Layer 2 Ethernet broadcasts. This feature of a switch blocks Ethernet broadcasts between VLANs and so helps protect against broadcast storms. A *broadcast storm* is a flood of unwanted Ethernet broadcast network traffic.

Another element of some VLAN deployments is that of *port isolation* or *private ports*. These are private VLANs that are configured to use a dedicated or reserved uplink port. The members of a private VLAN or a port-isolated VLAN can interact only with each other and over the predetermined exit port or uplink port. A common implementation of port isolation occurs in hotels. A hotel network can be configured so that the Ethernet ports in each room or suite are isolated on unique VLANs. This way, connections in the same unit can communicate but connections between units cannot. However, all of these private VLANs have a path out to the Internet (i.e., the uplink port).

Switch Eavesdropping

A port mirror is a common feature found on managed switches; it will duplicate traffic from one or more other ports out a specific port. A switch may have a hardwired *Switched Port Analyzer (SPAN)* port, which duplicates the traffic for all other ports, or any port can be configured as the mirror, audit, IDS, or monitoring port for one or more other ports. Port mirroring or port spanning takes place on the switch itself. Port mirroring and spanning is often used for network traffic analysis, packet capture, evidence collection, and intrusion detection.

A *port tap* is a means to eavesdrop on network communications, especially when a switch's SPAN function isn't available or doesn't meet the current interception needs. Modern inline taps have mostly replaced vampire taps. To install an inline tap, first, the original cable must be unplugged from the port and then plugged into the tap. Then, the tap is plugged into the vacated original port. A tap should be installed wherever traffic monitoring on a specific cable is required.

If there are more devices in an area than there are ports on a switch, additional switches can be deployed. Several switches can be linked together through their trunk ports. A trunk port is a dedicated port with higher bandwidth capacity than the other standard access ports. Switches are typically linked using a crossover cable, but if the ports are Auto-MDIX (medium-dependent interface crossover), then they will automatically configure themselves to adapt to whatever cable is used to link the devices.

The trunk link allows the switches to talk to each other directly, direct traffic between hosts, and stretch VLAN definitions across multiple physical switches. In this manner, VLAN3 on switch 2 can be part of the same VLAN as VLAN3 on switches 4 and 5. This is accomplished using special signaling defined in *IEEE 802.1q* (Dot1q) known as VLAN tagging. VLAN tags modify the standard construction of an Ethernet frame header to include a VLAN tag value. A standard Ethernet header is:

```
[Dst MAC | Src MAC | Ethertype]
```

A modified Ethernet header with a VLAN tag is structured like this:

```
[Dst MAC | Src MAC | VLAN | Ethertype]
```

Thus, a VLAN tag–modified Ethernet header cannot be interpreted by any host other than a switch, and then the switch is prepared to do so only on a trunk port.

However, there is the possibility of abuse of the VLAN tag system. An attacker could construct a header with multiple tags in order to perform *VLAN hopping*. The double-tagged Ethernet frame could start off in VLAN3 but then move into VLAN2. Early switches were not prepared for double tagging, so after reading the first VLAN tag into memory (such as VLAN3), the second VLAN tag (such as VLAN2) would overwrite the first in memory, thus only retaining the second value. When the switch then began to forward the frame, it would be placed into the second VLAN group.

The concept of OS virtualization has given rise to other virtualization topics, such as virtualized networks. A virtualized network or network virtualization is the combination of hardware and software networking components into a single integrated entity. The resulting system allows for software control over all network functions: management, traffic shaping, address assignment, and so on. A single management console or interface can be used to oversee every aspect of the network, a task requiring physical presence at each hardware component in the past. Virtualized networks have become a popular means of infrastructure deployment and management by corporations worldwide. They allow organizations to implement or adapt other interesting network solutions, including software-defined networks, VLANs, virtual switches, virtual SANs, guest operating

systems, port isolation, and more. Virtual networks are also discussed in Chapter 11, and software-defined networking (SDN) is discussed in Chapter 9.

MAC Flooding Attack

A *MAC flooding* attack is an intentional abuse of a switch's learning function to cause it to get stuck flooding. This is accomplished by flooding a switch with Ethernet frames with randomized source MAC addresses. The switch will attempt to add each newly discovered source MAC address to its content addressable memory (CAM) table. Once the CAM table is full, older entries will be dropped to make room for new entries (it is a first-in, first-out [FIFO] queue). Once the CAM is full of only false addresses, the switch is unable to properly forward traffic, so it reverts to flooding mode, where it acts like a hub or a multiport repeater and sends each received Ethernet frame out of every port.

MAC flooding is distinct from ARP poisoning and other types of AitM attacks in that the attacker does not get into the path of the communication between client and server; instead, the attacker (as well as everyone else on the local network) gets a copy of the communication. At this point, the attacker can eavesdrop on any communications taking place across the compromised switch.

A defense against MAC flooding is often present on managed switches. The feature, known as *MAC limiting*, restricts the number of MAC addresses that will be accepted into the CAM table from each jack/port. A network intrusion detection system (NIDS) may also be useful in identifying when a MAC flooding attack is attempted.

MAC Cloning

No two devices can have the same MAC address in the same local Ethernet broadcast domain; otherwise, an address conflict occurs. It is also good practice to verify that all MAC addresses across a private enterprise network are unique. This can be accomplished through manual NIC configuration checks as well as by remote queries performed by network discovery scanners. Although the design of MAC addresses should make them unique, vendor errors have produced duplicate MAC addresses. When

this happens, either the NIC hardware must be replaced, or the MAC address must be modified (i.e., spoofed) to a nonconflicting alternative address.

An adversary may eavesdrop on a network and take note of the MAC addresses in use. One of these addresses can then be spoofed into a system by altering the system's software copy of the NIC's MAC. This causes the Ethernet driver to operate based on the modified or spoofed MAC address instead of the original manufacturer's assigned MAC. Thus, it is quite simple to falsify, spoof, or clone a MAC address.

MAC spoofing is the changing of the default MAC address to some other value. *MAC cloning* is used to impersonate another system, often a valid or authorized network device, to bypass port security or MAC filtering limitations. *MAC filtering* is a security mechanism intended to limit or restrict network access to those devices with known specific MAC addresses. MAC filtering is commonly used on WAPs and switches.

Countermeasures to MAC spoofing/cloning include the following:

- Using intelligent switches that monitor for odd MAC address uses and abuses
- Using an NIDS that monitors for odd MAC address uses and abuses
- Maintaining an inventory of devices and their MAC addresses to confirm whether a device is authorized or unknown and rogue



To spoof a MAC address on *nix systems, you can use the utility `macchanger`. On Windows, use the free tools of Technitium from http://technitium.com/tmac or the SMAC Tool from http://smac-tool.com.

Network Address Translation

The goals of hiding the identity of internal clients, masking the design of your private network, and keeping public IPv4 address leasing costs to a minimum are all simple to achieve through the use of *network address translation (NAT)*. NAT hides the IPv4 configuration of internal clients and substitutes the IPv4 configuration of the proxy server's own public external NIC in outbound requests. This effectively prevents external hosts from learning the internal configuration of the network. This is an essential function when using RFC 1918 (Address Allocation for Private Internets) private IPv4 addresses internally while communicating with Internet resources.

NAT was developed to allow private networks to use any IPv4 address set without causing collisions or conflicts with public Internet hosts with the same IPv4 addresses. In effect, NAT translates the IPv4 addresses of your internal clients to leased addresses outside your environment. Functionally, NAT is a form of virtualized network; it hides or masks the real network configuration behind its own public identity.

NAT offers numerous benefits, including the following:

- You can connect an entire network to the Internet using only a single (or just a few) leased public IPv4 addresses.
- You can use the private IPv4 addresses defined in RFC 1918 in a private network and still be able to communicate with the Internet.
- NAT hides the IPv4 addressing scheme and network topography from the Internet.
- NAT restricts connections so that only traffic stemming from connections originating from the internal protected network is allowed back into the network from the Internet. Thus, most intrusion attacks are automatically repelled.
- NAT serves as a basic one-way firewall by only allowing incoming traffic that is in response to an internal system's request.

Are You Using NAT?

Most networks, whether at an office or at home, employ NAT. There are at least three ways to tell whether you are working within a “NATed” network:

- Check your client's IPv4 address. If it is one of the RFC 1918 addresses and you are still able to interact with the Internet, then you are on a NATed network.
- Check the configuration of your proxy, router, firewall, modem, or gateway device to see whether NAT is configured. (This action requires authority and access to the networking device.)
- If your client's IPv4 address is not an RFC 1918 address, then compare your address to what the Internet thinks your address is. You can do this by visiting any of the IP-checking websites; a popular one is `http://whatismyipaddress.com`. If your client's IPv4 address and the address that What Is My IP Address claims is your address are different, then you are working from a NATed network.

NAT is part of a number of hardware devices and software products, including firewalls, routers, gateways, WAPs, and proxies.

Strictly, NAT dynamically converts or maps the private IPv4 addresses of internal systems found in the header of network packets into public or external IPv4 addresses. NAT performs this operation on a one-to-one basis; thus, a single leased public IPv4 address can allow a single internal system to access the Internet. Closely related to NAT is *port address translation (PAT)*—also known as *overloaded NAT*, *network and port address translation (NPAT)*, and *network address port translation (NAPT)*—which allows a single public IPv4 address to host up to 65,536 simultaneous communications from internal clients (a theoretical maximum; in practice, you should limit the number to 4,000 or fewer in most cases due to hardware limitations). Instead of mapping IPv4 addresses on a one-to-one basis, PAT uses the Transport Layer port numbers to host multiple

simultaneous communications across each public IPv4 address by mapping internal sockets (i.e., the combination of an IPv4 address and a port number) to external sockets. PAT is effectively multiplexing numerous sessions from internal systems over a single external IPv4 address. So, with NAT, you must lease as many public IPv4 addresses as you want to have simultaneous communications, whereas with PAT you can lease significantly fewer IPv4 addresses.

The use of the term NAT in the IT industry has come to include the concept of PAT. Thus, when you hear or read about NAT, you can assume that the material is referring to PAT. This is true for most OSs, devices, and services. *Source Network Address Translation (SNAT)* is yet another term for NAT. NAT can also be called Stateful NAT or Dynamic NAT since the mapping and IPv4 address or socket allocation is created when a session is initiated and dissolved when the session is torn down (see the section "Stateful NAT," later in this chapter). From this point forward, our use of the term NAT is meant to imply the more likely use of PAT.

Another issue to be familiar with is that of *NAT traversal (NAT-T)* (RFC 3947). Traditional NAT doesn't support IPSec VPNs, because of the requirements of the IPSec protocol and the changes NAT makes to packet headers (which is perceived as corruption or violating integrity). However, NAT-T was designed specifically to support IPSec and other tunneling VPN protocols, such as Layer 2 Tunneling Protocol (L2TP), so that organizations can benefit from both NAT and VPNs across the same border device/interface.

Although NAT by default is a dynamic outbound mapping mechanism, it can be configured to perform inbound mapping as well. Known as *static NAT*, *reverse proxy*, *port forwarding*, or *destination network address translation (DNAT)*, this technique allows an external entity to initiate communication with an internal entity behind a NAT by using a public socket that is mapped to redirect to an internal system's private address. Though this is technically possible, it is generally to be avoided. Granting the easy ability for an external entity to initiate a connection with an internal system is not usually a secure solution. Static NAT may be useful for systems in a screened subnet or extranet, but definitely not for accessing systems in the internal private LAN.

NAT66, or Network Address Translation for IPv6, is a technique used to map multiple private IPv6 addresses to a smaller pool of public IPv6 addresses. The primary goal of NAT66 is similar to traditional NAT used in IPv4 networks, which is to enable multiple devices within a private network to share a single or a limited set of globally routable IPv6 addresses.

NAT66 allows multiple devices within a private IPv6 network to share the same public IPv6 address when communicating with external networks, such as the Internet. NAT66 provides a level of privacy and security by hiding internal network details from external entities. It assigns global IPv6 addresses to devices within the private network, and external entities see only the public IPv6 address. Although IPv6 has a vastly larger address space compared to IPv4, there may still be scenarios where organizations want to conserve public IPv6 addresses. NAT66 can be used to achieve this goal by allowing multiple internal devices to share a common public IPv6 address.

It's important to note that while NAT66 is an option, IPv6 was originally designed with the goal of providing globally unique addresses to all devices, promoting end-to-end connectivity without the need for address translation. The use of NAT in IPv6 has been a topic of debate, and some advocate for maintaining the original design principles of IPv6. However, in certain deployment scenarios or due to specific network requirements, organizations may choose to implement NAT66 for address conservation and security purposes.

Private IP Addresses

The world has simply deployed more devices using IPv4 than there are unique IPv4 addresses available. Fortunately, the early designers of the Internet and TCP/IP had good foresight and put aside a few blocks of addresses for private, unrestricted use. These IPv4 addresses, commonly called the *private IPv4 addresses*, are defined in *RFC 1918*. They are as follows:

- 10.0.0.0–10.255.255.255 (a full Class A range)
- 172.16.0.0–172.31.255.255 (16 Class B ranges)
- 192.168.0.0–192.168.255.255 (256 Class C ranges)

Can't NAT Again

On several occasions we've needed to “re-NAT” an already “NATed” network. This might occur in the following situations:

- You need to make an isolated subnet within a NATed network and attempt to do so by connecting a router to host your new subnet to the single port offered by the existing network.
- You have a DSL or cable modem that offers only a single connection but you have multiple computers or want to add wireless to your environment.

By connecting a NAT proxy router or a wireless access point, you are usually attempting to re-NAT what was NATed to you initially. One configuration setting that can either make or break this setup is the IPv4 address range in use. It is not possible to re-NAT the same subnet. For example, if your existing network is offering 192.168.1.x addresses, then you cannot use that same address range in your new NATed subnet. So change the configuration of your new router/WAP to perform NAT on a slightly different address range, such as 192.168.5.x, and you won't have the conflict. This seems obvious, but it is quite frustrating to troubleshoot the unwanted result without this insight.

All routers and traffic-directing devices are configured by default not to forward traffic to or from these private IPv4 addresses. In other words, the private IPv4 addresses are not routed by default. Thus, they cannot be directly used to communicate over the Internet. However, they can be easily used on private networks where routers are not employed or where slight modifications to router configurations are made. Using private IPv4 addresses in conjunction with NAT greatly reduces the cost of connecting to the Internet by allowing fewer public IPv4 addresses to be leased from an ISP.



Attempting to use the RFC 1918 private IPv4 addresses directly on the Internet is futile because all publicly accessible routers will drop data packets containing a source IPv4 address from these RFC 1918 ranges.

Stateful NAT

NAT operates by maintaining a mapping between requests made by internal clients, a client's internal IP address, and the IP address of the Internet service contacted. When a request packet is received by NAT from a client, it changes the source address in the packet from the client's to the NAT server's. This change is recorded in the NAT mapping database along with the destination address. Once a reply is received from the Internet server, NAT matches the reply's source address to an address stored in its mapping database and then uses the linked client address to redirect the response packet to its intended destination. This process is known as *stateful NAT* because it maintains information about the communication sessions between clients and external systems.

Automatic Private IP Addressing

Automatic Private IP Addressing (APIPA), also known as IPv4 link-local address assignment (defined in RFC 3927), assigns an IP address to a system in the event of a Dynamic Host Configuration Protocol (DHCP) assignment failure. APIPA is primarily a feature of Windows, since no other OS has adopted the standard. APIPA assigns each failed DHCP client an IP address from the range of 169.254.0.1 to 169.254.255.254, along with the default Class B subnet mask of 255.255.0.0. This allows the system to communicate only with other APIPA-configured clients within the same broadcast domain but not with any system across a router or with a correctly assigned IP address.



Don't confuse APIPA with the private IP address ranges defined in RFC 1918.

APIPA is not usually directly concerned with security. However, it is still an important issue to understand. If you notice that a system is assigned an APIPA address instead of a valid network address, that indicates a problem. It could be as mundane as a bad cable or power failure on the DHCP server, but it could also be a symptom of a malicious attack on the DHCP server. You might be asked to decipher issues in a scenario where IP addresses are presented. You should be able to discern whether an address is a public address, an RFC 1918 private address, an APIPA address, or a loopback address (see Chapter 11).

The Loopback Address

Another IP address range that you should be careful not to confuse with the private IP address ranges defined in RFC 1918 is the loopback address. The *loopback address* is purely a software entity. It is an IP address used to create a software interface that connects back to itself via TCP/IP. The loopback address allows for the testing of local network settings in spite of missing, damaged, or nonfunctional network hardware and related device drivers. Technically, the entire 127.x.x.x network is reserved for loopback use. However, only the 127.0.0.1 address is widely used.

Third-Party Connectivity

Third-party connectivity is a growing concern for almost every business. Very few organizations operate exclusively using internal resources—most organizations interact with outside third-party providers. Most of these external entities do not need to interact directly with an organization's IT/IS. However, for those few that do, it is important to consider the risks and ramifications. Any time an organizational network is connected directly to

another entity's network, their local threats and risks affect each other. A compromise of one organization can lead easily to the compromise of the other.

Any connection between IT environments should be planned out in detail well in advance of actually interconnecting the cabling (whether physical or virtual). Often, this process starts with an MOU and ends with an ISA:

- A *memorandum of understanding (MOU)* or memorandum of agreement (MOA) is an expression of agreement or aligned intent, will, or purpose between two entities. It is not typically a legal agreement or commitment, but rather a more formal form of a reciprocal agreement or handshake (neither of which is typically written down). An MOU can also be called a letter of intent. It is a means to document the specifics of an agreement or arrangement between two parties without necessarily legally binding them to the parameters of the document.
- An *interconnection security agreement (ISA)* is a formal declaration of the security stance, risks, and technical requirements of a link between two organizations' IT infrastructures. The goal of an ISA is to define the expectations and responsibilities of maintaining security over a communications path between two networks. Connecting networks can be mutually beneficial, but it also raises additional risks that need to be identified and addressed. An ISA is a means to accomplish that.

Additionally, a full risk assessment should be performed in order to predict issues and preemptively protect against adverse events as much as possible.

Keep in mind that direct linking of IT environments is not the only possible solution in most circumstances. Using an extranet to host servers to be accessed by the other party via a VPN is a reasonable alternative. Another option is to work with a cloud solution to establish a shared private cloud between the two entities so that only project-related content is ever shared between the two parties. A third option is to keep all datasets separate and use secure email, file sharing, and multimedia collaboration services.

Whatever approach you decide to use, don't let the rush or haste of establishing a new relationship with a third party or engaging in a new project cause security to be discarded or overlooked.

Similar care should be taken when electing to use a cloud service, since they are third parties. As an organization adopts cloud services, from SaaS to IaaS, the level of connectivity and direct interaction with on-premises equipment increases. Clear security guidelines and policies should be established, and when possible, technologies such as cloud access security brokers (CASBs) should be deployed to enforce those security requirements.

Yet another possible interpretation of third-party connectivity is a remote worker or telecommuter. As mentioned previously, there needs to be clear justification for allowing remote work, which requires a direct link or access to internal resources. When possible, limit telecommuters to extranet servers or only publicly facing systems (such as email and websites). It may also be important to provide company-owned and -controlled equipment to remote workers rather than depending on personal equipment, which may not be securable or may be used for nonwork purposes or by nonemployees.

Third-party connectivity is a risk that can be managed, but it requires focused and purposed attention. Remember that any means of data transmission or communication can be employed by benign actors for legitimate purposes as well as by adversaries for malicious purposes.

WAN technologies are critical for establishing connectivity over large geographical areas. In the context of third-party connectivity involving telecom providers and hardware support, several key considerations come into play. Telecom providers play a central role in WAN connectivity, offering services like leased lines, Multiprotocol Label Switching (MPLS), and virtual private network (VPN) services. Leased lines provide dedicated point-to-point connections, whereas MPLS and VPN services enable secure and efficient data transmission over shared infrastructure.

Hardware support is equally vital in WAN connectivity, involving components such as routers, switches, and WAN optimization appliances. These devices are essential for establishing and managing WAN connections. Hardware support ensures proper functioning, maintenance, and troubleshooting, either through internal IT teams or third-party vendors specializing in networking hardware.

MPLS services are commonly employed in WAN connectivity, providing businesses with the means to create private and secure networks across

multiple locations. MPLS incorporates quality of service (QoS) features, addressing specific performance requirements. With the rise of cloud services, WAN technologies play a pivotal role in connecting organizations to cloud providers. Telecom providers offer solutions facilitating direct connections to major cloud platforms, enhancing performance, security, and reliability for cloud-based applications.

Software-defined WAN (SD-WAN) is a modern approach to WAN connectivity that utilizes software-defined networking principles. It allows organizations to dynamically route traffic over various connections, optimizing performance and cost-effectiveness. Telecom providers may offer SD-WAN services to enhance network flexibility and efficiency.

Redundancy and failover mechanisms are crucial components of WAN technologies to ensure continuous connectivity. This may involve the use of multiple telecom providers or diverse network paths to minimize the risk of service disruption. Hardware support is essential for maintaining and configuring these redundant setups.

WAN technologies in the context of third-party connectivity involve collaboration with telecom providers and the utilization of hardware components. Organizations leverage a mix of technologies, including leased lines, MPLS, VPNs, SD-WAN, and others, to establish efficient and reliable connectivity across diverse locations. Hardware support, whether provided internally or by third-party vendors, is essential for maintaining the integrity and performance of the WAN infrastructure. The choice of WAN technologies and third-party partnerships depends on factors such as performance requirements, cost considerations, and the specific needs of the organization.

Switching Technologies

When two systems (individual computers or LANs) are connected over multiple intermediary networks, the task of transmitting data from one to the other is a complex process. Switching technologies were developed to simplify this task.

Circuit Switching

Circuit switching was originally developed to manage telephone calls over the public switched telephone network. In circuit switching, a dedicated physical pathway is created between the two communicating parties. Once a call is established, the links between the two parties remain the same throughout the conversation. Circuit switching provides for fixed or known transmission times, a uniform level of quality, and little or no loss of signal or communication interruptions. These systems employ permanent, physical connections. However, the term permanent applies only to each communication session. The path is permanent throughout a single conversation. Once the path is disconnected, if the two parties communicate again, a different path may be assembled. During a single conversation, the same physical or electronic path is used throughout the communication and is used only for that one communication. Circuit switching grants exclusive use of a communication path to the current communication partners. Only after a session has been closed can a pathway be reused by another communication.

Real-World Circuit Switching

There is very little actual circuit switching in the modern world (or at least in the past 20 to 25 years or so). Packet switching, discussed next, has become ubiquitous for data and voice transmissions. Decades ago, we could often point to the public switched telephone network (PSTN) as a prime example of circuit switching, but with the advent of digital switching and VoIP systems, those days are long gone. That's not to say that circuit switching is nonexistent in today's world; it is just not being used for data transmission. Instead, you can still find circuit switching in rail yards, irrigation systems, and even electrical distribution systems.

Packet Switching

Eventually, as computer communications increased as opposed to traditional voice communications, a new form of switching was developed. *Packet switching* occurs when the message or communication is broken up into small segments (fixed-length cell or variable-length packets, depending on the protocols and technologies employed) and sent across the

intermediary networks to the destination. Each segment of data has its own header that contains source and destination information. The header is read by each intermediary system and is used to route each packet to its intended destination. Each channel or communication path is reserved for use only while a packet is actually being transmitted over it. As soon as the packet is sent, the channel is made available for other communications.

Packet switching does not enforce the exclusivity of communication pathways. It can be seen as a logical transmission technology because addressing logic dictates how communications traverse intermediary networks between communication partners. Table 12.2 compares circuit switching to packet switching.

TABLE 12.2 Circuit switching vs. packet switching

Circuit switching	Packet switching
Constant traffic	Bursty traffic
Fixed known delays	Variable delays
Connection-oriented	Connectionless
Sensitive to connection loss	Sensitive to data loss
Used primarily for voice	Used for any type of traffic

In relation to security, you should consider a few potential issues. A packet-switching system places data from different sources on the same physical connection. This can lend itself to disclosure, corruption, or eavesdropping. Proper connection management, traffic isolation, and usually encryption are needed to protect against shared physical pathway concerns. A benefit of packet-switching networks is that they are not as dependent on specific physical connections as circuit switching is. Thus, when or if a physical pathway is damaged or goes offline, an alternate path can be used to continue the data/packet delivery. A circuit-switching network is often interrupted by physical path violations.

Virtual Circuits

A *virtual circuit* (also called a communication path) is a logical pathway or circuit created over a packet-switched network between two specific

endpoints. Within packet-switching systems are two types of virtual circuits:

- *Permanent virtual circuits (PVCs)*
- *Switched virtual circuits (SVCs)*

A PVC is like a dedicated leased line; the logical circuit always exists and is waiting for the customer to send data. A PVC is a predefined virtual circuit that is always available. The virtual circuit may be closed down when not in use, but it can be instantly reopened whenever needed. An SVC has to be created each time it is needed using the best paths currently available before it can be used and then disassembled after the transmission is complete. In either type of virtual circuit, when a data packet enters point A of a virtual circuit connection, that packet is sent directly to point B or the other end of the virtual circuit. However, the actual path of one packet may be different from the path of another packet from the same transmission. In other words, multiple paths may exist between point A and point B as the ends of the virtual circuit, but any packet entering at point A will end up at point B.

A PVC is like a two-way radio or walkie-talkie. Whenever communication is needed, you press the button and start talking; the radio reopens the predefined frequency automatically (that is, the virtual circuit). An SVC is more like a shortwave or ham radio. You must tune the transmitter and receiver to a new frequency every time you want to communicate with someone.

WAN Technologies

WAN technologies contribute to the efficiency, scalability, and reliability of long-distance communications. The selection of a specific technology hinges on factors such as geographical locations, bandwidth requirements, latency sensitivity, and the unique needs of the applications being supported. Each technology plays a role in addressing connectivity challenges across diverse and often remote environments.

Wide area network links are used to connect distant networks, nodes, or individual devices together. A WAN link can improve communications and

efficiency, but it can also place data at risk. Proper connection management and transmission encryption is needed for a secure connection, especially over public network links. WAN links and long-distance connection technologies can be divided into two primary categories: dedicated and nondedicated.

A *dedicated line* (also called a *leased line* or *point-to-point link*) is one that is continually reserved for use by a specific customer. A dedicated line is always on and waiting for traffic to be transmitted over it. The link between the customer's LAN and the dedicated WAN link is always open and established. A dedicated line connects two specific endpoints and only those two endpoints. This type of connection is often used between multiple business locations, so they can effectively communicate as a single entity.

There have been numerous types of dedicated lines over the years, ranging from the T1 (telephone line 1 with 1.54 Mbps capacity) to DS3 (Digital Service 3 with 44.7 Mbps capacity) (originally known as the T3). Other options included X.25, Asynchronous Transfer Mode (ATM), and Frame Relay. These technologies have mostly been replaced by fiber optic–based solutions.



Cable TV–based Internet service does not fit well into either the dedicated or the nondedicated classification. Cable Internet is an always-on system, but not between two client locations. Instead, it is a link from your premises to an Internet gateway. Thus, it can be labeled as a point-to-multipoint connection. Another wrinkle is that cable Internet service is also typically shared with the other subscribers in the neighborhood. Privacy is maintained through encryption, similar to a VPN, from the cable modem deployed at your location to an exit point in the cable company's network, typically immediately connected to the Internet gateway.

A *nondedicated line* is one that requires a connection to be established before data transmission can occur. A nondedicated line can be used to connect with any remote system that uses the same type of nondedicated line.

Fault Tolerance with Carrier Network Connections

To obtain fault tolerance with leased lines or with connections to carrier networks, you must deploy two redundant connections. For even greater redundancy, you should purchase the connections from two different telcos or service providers. However, when you're using two different service providers, be sure they don't connect to the same regional backbone or share any major pipeline. The physical location of multiple communication lines leading from your building is also of concern because a single disaster or human error (e.g., a misguided backhoe) could cause multiple lines to fail at once. If you cannot afford to deploy an exact duplicate of your primary dedicated leased line, consider a nondedicated connection. These less expensive options may still provide partial availability in the event of a primary leased line failure.

Standard classic dial-up modems and DSL modems are examples of nondedicated lines. Digital subscriber line (DSL) is a technology that exploits the upgraded telephone network to grant consumers speeds from 144 Kbps to 100 Mbps (or more). There are numerous formats of DSL (e.g., ASDL, VSDL, and SDSL), and each format varies according to the specific downstream and upstream bandwidth provided.

Backhaul networks constitute the segment linking smaller or local networks to a central hub or the broader Internet. Various WAN technologies are applied in this context to ensure effective connectivity. MPLS (Multiprotocol Label Switching) stands out as a prevalent choice, offering scalability and efficiency by utilizing labels for packet routing. This approach is conducive to connecting diverse locations and supporting quality of service (QoS) features. Additionally, Ethernet-based solutions, such as Metro Ethernet, are commonly employed in backhaul connections, providing high bandwidth, scalability, and flexibility for applications spanning cell towers, business locations, and data centers.

Satellite communications involve the transmission of data between Earth-based stations or satellite-enabled devices, relying on satellites orbiting Earth. In this domain, very-small-aperture terminal (VSAT) technology plays a pivotal role. VSAT enables small, remote terminals to communicate

with geostationary satellites, proving useful in locations where traditional wired or terrestrial connections pose challenges. Satellite broadband services leverage satellites to extend Internet connectivity to remote or underserved areas. Users with satellite dishes can establish connections, facilitating broadband access where terrestrial options may be limited. Moreover, the emergence of low Earth orbit (LEO) satellites, exemplified by projects like Starlink, offers a contemporary approach to satellite communications with reduced latency compared to traditional geostationary satellites. LEO satellites aim to provide high-speed Internet access globally.

Broadband over power lines (BPL) is a technology that enables high-speed data transmission over existing electrical power lines. In a BPL system, data signals are transmitted along the same infrastructure that is used to deliver electric power. This technology leverages the extensive network of power lines to provide broadband Internet access to homes, businesses, and other locations. While BPL has been explored as a potential solution for extending broadband access, its adoption has been limited compared to other technologies like DSL, cable, and fiber optics. Factors such as technical challenges, regulatory considerations, and the growth of alternative broadband technologies have influenced the deployment and acceptance of BPL in different regions.



Integrated Services Digital Network (ISDN) was the planned replacement for PSTN, but with the advent of DSL, cable Internet, and fiber options, it did not gain widespread adoption. Most ISDN services have been discontinued.

Fiber-Optic Links

Synchronous Digital Hierarchy (SDH) and *Synchronous Optical Network (SONET)* are fiber-optic high-speed networking standards. SDH was standardized by the International Telecommunications Union (ITU) and SONET by the American National Standards Institute (ANSI). SDH and SONET are mostly hardware or physical layer standards defining

infrastructure and line speed requirements. SDH and SONET use synchronous time-division multiplexing (TDM) for high-speed duplex communications with minimal need for control and management overhead.

While SDH and SONET have some regional differences in their standards, they are functionally similar and are often used interchangeably or in conjunction with each other, particularly in international networks. *STS (Synchronous Transport Signal)* (or *Optical Carrier (OC)*) is associated with SONET, while *STM (Synchronous Transport Module)* is associated with SDH. Both STS and STM represent standardized levels within their respective hierarchies for organizing and multiplexing digital signals. These optical transmission services support a foundational speed of 51.48 Mbps. The main bandwidth levels of SDH and SONET are shown in Table 12.3.

TABLE 12.3 Bandwidth levels of SDH and SONET

SONET	SDH	Data rate
STS-1/OC-1	STM-0	51.84 Mbps
STS-3/OC-3	STM-1	155.52 Mbps
STS-12/OC-12	STM-4	622.08 Mbps
STS-48/OC-48	STM-16	2.488 Gbps
STS-96/OC-96	STM-32	4.876 Gbps
STS-192/OC-192	STM-64	9.953 Gbps
STS-768/OC-768	STM-256	39.813 Gbps

Note: The SDH service numbers are 1/3 that of SONET's.

SDH and SONET both support mesh and ring topologies. These fiber solutions are often implemented as the backbone of a telco service, and divisions or fractions of the capacity are subscribed out to customers.

Prevent or Mitigate Network Attacks

Communication systems are vulnerable to attacks in much the same way any other aspect of the IT infrastructure is vulnerable. Understanding the threats and possible countermeasures is an important part of securing an environment. Any activity or condition that can cause harm to data,

resources, or personnel must be addressed and mitigated if possible. Keep in mind that harm includes more than just destruction or damage; it also includes disclosure, access delay, denial of access, fraud, resource waste, resource abuse, and loss. Common threats against communication system security include DoS (see Chapter 17, “Preventing and Responding to Incidents”), impersonation (see Chapter 2), replay (see Chapter 11), ARP poisoning (see Chapter 11), DNS poisoning (see Chapter 11), eavesdropping, and transmission modification.

Eavesdropping

As the name suggests, *eavesdropping* is listening to communication traffic for the purpose of duplicating it. The duplication can take the form of recording data to a storage device or using an extraction program that dynamically attempts to extract the original content from the traffic stream. Once a copy of traffic content is in the hands of an attacker, they can often extract many forms of confidential information, such as usernames, passwords, process procedures, and data.

Eavesdropping usually requires physical access to the IT infrastructure to connect a physical recording device to an open port or cable splice or to install a software-recording tool onto the system. Eavesdropping is often facilitated by the use of a network traffic capture or monitoring program or a protocol analyzer system (often called a sniffer). Eavesdropping devices and software are usually difficult to detect because they are used in passive attacks. When eavesdropping or wiretapping is transformed into altering or injecting communications, the attack is considered an active attack.

You can combat eavesdropping by maintaining physical access security to prevent unauthorized personnel from accessing your IT infrastructure. As for protecting communications that occur outside your network or for protecting against internal attackers, using encryption (such as IPSec or SSH) and onetime authentication methods (onetime pads or token devices) on communication traffic will greatly reduce the effectiveness and timeliness of eavesdropping. Application allow listings should also be considered as a means to prevent the execution of unauthorized software, such as sniffers.

Modification Attacks

In *modification attacks*, captured packets are altered and then played against a system. Modified packets are designed to bypass the restrictions of improved authentication mechanisms and session sequencing. Countermeasures to modification replay attacks include using digital signature verifications and packet checksum verification (i.e., integrity checking).

Summary

Transmission Control Protocol/Internet Protocol (TCP/IP) is the primary protocol suite used on most networks and on the Internet. It is a robust protocol suite, but it has numerous security deficiencies. Authentication and encryption need to be implemented to account for TCP/IP's deficiencies.

When securing communication channels, be sure to address voice, remote access, multimedia collaboration, data communications (such as email), and virtualized networks.

Secure voice communications can be achieved by evaluating and hardening PSTN, PBX, mobile, and VoIP solutions. VoIP security is often achieved through general network security practices and using Secure Real-time Transport Protocol (SRTP).

Remote access security management requires security system designers to address the hardware and software components of the implementation along with policy issues, work task issues, and encryption issues. This includes deployment of secure communication protocols. Secure authentication for both local and remote connections is an important foundational element of overall security.

Maintaining control over communication pathways is essential to supporting confidentiality, integrity, and availability for network, voice, and other forms of communication. Numerous attacks are focused on intercepting, blocking, or otherwise interfering with the transfer of data from one location to another. Fortunately, there are also reasonable countermeasures to reduce or even eliminate many of these threats.

VPNs are a common means to achieve data communications security. VPNs are based on encrypted tunneling. Tunneling, or encapsulation, is a means by which messages in one protocol can be transported over another network

or communications system using a second protocol. VPN solutions include IPSec, TLS, SSH, L2TP, and PPTP.

Telecommuting, or remote connectivity, has become a common feature of business computing. When remote access capabilities are deployed in any environment, security must be considered and implemented to provide protection for your private network against remote access complications. Remote access users should be stringently authenticated before being granted access. Remote access services include Voice over IP (VoIP), application streaming, VDI, multimedia collaboration, and instant messaging.

Email is insecure unless you take steps to secure it. To secure email, you should provide for nonrepudiation, restrict access to authorized users, make sure integrity is maintained, authenticate the message source, verify delivery, and classify sensitive content. These issues must be addressed in a security policy before they can be implemented in a solution. They often take the form of acceptable use policies, access controls, privacy declarations, email management procedures, and backup and retention policies.

Email is a common delivery mechanism for malicious code. Filtering attachments, using antivirus software, and educating users are effective countermeasures against that kind of attack. Email spamming or flooding is a form of denial of service that can be deterred through filters and IDSs. Email security can be improved using S/MIME and PGP.

Fax and voice security can be improved by using encryption to protect the transmission of documents and prevent eavesdropping. Training users effectively is a useful countermeasure against social engineering attacks.

Virtual networks are software or digital re-creations of physical concepts in order to achieve security or performance improvements. Examples of virtual networks include software-defined networks (SDNs), VPNs, VLANs, virtual switches, virtual SANs, guest operating systems, port isolation, and NAT.

A VLAN is a hardware-imposed network segmentation created by switches. VLANs are used to logically segment a network without altering its physical topology. VLANs are used for traffic management.

NAT is used to hide the internal structure of a private network as well as to enable multiple internal clients to gain Internet access through a few public IP addresses.

Third-party connectivity is a growing concern for businesses. Thus, it is important to consider the risks and ramifications. Any time an organizational network is connected directly to another entity's network, their local threats and risks affect each other. A compromise of one organization can lead easily to the compromise of the other. Any connection between IT environments should be planned out in detail well in advance of actually interconnecting the cabling (whether physical or virtual). Often, this process starts with an MOU and ends with an ISA.

WAN links, or long-distance connection technologies, can be divided into two primary categories: dedicated and nondedicated lines. A dedicated line connects two specific endpoints and only those two endpoints. A nondedicated line is one that requires a connection to be established before data transmission can occur.

Communication systems are vulnerable to many attacks, including distributed denial-of-service (DDoS), eavesdropping, impersonation, replay, modification, spoofing, and ARP and DNS attacks. Fortunately, effective countermeasures exist for each of these.

Study Essentials

Understand PPP. Point-to-Point Protocol (PPP) is an encapsulation protocol designed to support the transmission of IP traffic over dial-up or point-to-point links. The original PPP options for authentication were PAP, CHAP, and EAP.

Define PAP, CHAP, and EAP. Password Authentication Protocol (PAP) transmits usernames and passwords in cleartext. Challenge Handshake Authentication Protocol (CHAP) performs authentication using a challenge-response dialogue that cannot be replayed. Extensible Authentication Protocol (EAP) allows customized authentication security solutions.

Understand IEEE 802.1X. IEEE 802.1X defines the use of encapsulated EAP to support a wide range of authentication options for LAN

connections. The IEEE 802.1X standard is formally named “Port-Based Network Access Control.”

Know about port security. Port security can mean the physical control of all connection points, such as RJ-45 wall jacks or device ports. Port security is the management of TCP and User Datagram Protocol (UDP) ports. Port security can also refer to the need to authenticate to a port before being allowed to communicate through or across the port (i.e., IEEE 802.1X).

Understand voice communications security. Voice communications are vulnerable to many attacks, especially as voice communications become an important part of network services. You can obtain confidentiality by using encrypted communications. Countermeasures must be deployed to protect against interception, eavesdropping, tapping, and other types of exploitation.

Know the threats associated with PBX systems and the countermeasures to PBX fraud. Countermeasures to PBX fraud and abuse include many of the same precautions you would employ to protect a typical computer network: logical or technical controls, administrative controls, and physical controls.

Understand the security issues related to VoIP. VoIP is at risk for caller ID spoofing, vishing, call manager software/firmware attacks, phone hardware attacks, DoS, AitM/MitM/on-path attacks, spoofing, and switch hopping.

Recognize what phreaking is. Phreaking is a specific type of attack in which various types of technology are used to circumvent the telephone system to make free long-distance calls, to alter the function of telephone service, to steal specialized services, or to cause service disruptions. A phreaker is an attacker who performs phreaking.

Understand the issues of remote access security management. Remote access security management requires that security system designers address the hardware and software components of an implementation along with issues related to policy, work tasks, and encryption.

Know various issues related to remote access security. Be familiar with remote access, dial-up connections, screen scrapers, virtual applications/desktops, and general telecommuting security concerns.

Understand multimedia collaboration. Multimedia collaboration is the use of various multimedia-supporting communication solutions to enhance distance collaboration and communications.

Know the purpose of load balancers. The purpose of load balancing is to obtain more optimal infrastructure utilization, minimize response time, maximize throughput, reduce overloading, and eliminate bottlenecks. A load balancer is used to spread or distribute network traffic load across several network links or network devices.

Understand active/active. An active-active system is a form of load balancing that uses all available pathways or systems during normal operations but that has reduced capacity in adverse conditions.

Understand active/passive. An active-passive system is a form of load balancing that keeps some pathways or systems in an unused dormant state during normal operations. It is able to maintain consistent capacity during abnormal conditions.

Understand virtualized networks. A virtualized network or network virtualization is the combination of hardware and software networking components into a single integrated entity. Examples include software-defined networks (SDNs), VLANs, VPNs, virtual switches, virtual SANs, guest operating systems, port isolation, and NAT.

Define tunneling. Tunneling is the encapsulation of a protocol-deliverable message within a second protocol. The second protocol often performs encryption to protect the message contents.

Understand VPNs. VPNs are based on encrypted tunneling. They can offer authentication and data protection as a point-to-point solution. Common VPN protocols are PPTP, L2TP, SSH, TLS, and IPSec.

Understand split vs. full tunnel. A split tunnel is a VPN configuration that allows a VPN-connected client system (i.e., remote node) to access both the organizational network over the VPN and the Internet directly at the same time. A full tunnel is a VPN configuration in which all of the client's traffic is sent to the organizational network over the VPN link, and then any Internet-destined traffic is routed out of the organizational network's proxy or firewall interface to the Internet.

Be able to explain NAT. NAT protects the addressing scheme of a private network, allows the use of the private IP addresses, and enables multiple internal clients to obtain Internet access through a few public IP addresses. NAT is supported by many security border devices, such as firewalls, routers, gateways, WAPs, and proxies.

Know about third-party connectivity. Most organizations interact with outside third-party providers. Most of these external entities do not need to interact directly with an organization's IT/IS. However, for those few that do, it is important to consider the risks and ramifications. This includes partnerships, cloud services, and remote workers.

Understand the difference between packet switching and circuit switching. In circuit switching, a dedicated physical pathway is created between the two communicating parties. Packet switching occurs when the message or communication is broken up into small segments and sent across the intermediary networks to the destination. Within packet-switching systems are two types of communication paths, or virtual circuits: permanent virtual circuits (PVCs) and switched virtual circuits (SVCs).

Understand the various network attacks and countermeasures associated with communications security. Communication systems are vulnerable to many attacks, including distributed denial-of-service (DDoS), eavesdropping, impersonation, replay, modification, spoofing, and ARP and DNS attacks. Be able to supply effective countermeasures for each.

Written Lab

1. Describe the differences between transport mode and tunnel mode of VPNs.
2. Discuss the benefits of NAT.
3. What are the main differences between circuit switching and packet switching?
4. What are some security issues with email and options for safeguarding against them?

5. What are the private IP addresses, APIPA addresses, and loopback addresses?
6. Name at least six facts about VLANs.

Review Questions

1. Among the many aspects of a security solution, the most important is whether it addresses a specific need (i.e., a threat) for your assets. But there are many other aspects of security you should consider as well. A significant benefit of a security control is when it goes unnoticed by users. What is this called?

   A. Invisibility

   B. Transparency

   C. Diversion

   D. Hiding in plain sight

2. Extensible Authentication Protocol (EAP) is one of the three authentication options provided by Point-to-Point Protocol (PPP). EAP allows customized authentication security solutions. Which of the following are examples of actual EAP methods? (Choose all that apply.)

   A. LEAP

   B. EAP-VPN

   C. PEAP

   D. EAP-SIM

   E. EAP-FAST

   F. EAP-MBL

   G. EAP-MD5

   H. VEAP

   I. EAP-POTP

   J. EAP-TLS

K. EAP-TTLS

3. In addition to maintaining an updated system and controlling physical access, which of the following is the most effective countermeasure against PBX fraud and abuse?

   A. Encrypting communications

   B. Changing default passwords

   C. Using transmission logs

   D. Taping and archiving all conversations

4. A phreaker has been apprehended who had been exploiting the technology deployed in your office building. Several handcrafted tools and electronics were taken in as evidence that the phreaker had in their possession when they were arrested. What was this adversary likely focusing on with their attempts to compromise the organization?

   A. Accounting

   B. NAT

   C. PBX

   D. Wi-Fi

5. Multimedia collaboration is the use of various multimedia-supporting communication solutions to enhance distance collaboration (people working on a project together remotely). Often, collaboration allows workers to work simultaneously as well as across different time frames. Which of the following are important security mechanisms to impose on multimedia collaboration tools? (Choose all that apply.)

   A. Encryption of communications

   B. Multifactor authentication

   C. Customization of avatars and filters

   D. Logging of events and activities

6. Michael is configuring a new web server to offer instruction manuals and specification sheets to customers. The web server has been positioned in the screened subnet and assigned an IP address of

172.31.201.17, and the public side of the company's split-DNS has associated the documents.myexamplecompany.com domain name with the assigned IP. After verifying that the website is accessible from his management station (which accesses the screened subnet via a jumpbox) as well as from several worker desktop systems, he declares the project completed and heads home. A few hours later, Michael thinks of a few additional modifications to perform to improve site navigation. However, when he attempts to connect to the new website using the FQDN, he receives a connection error stating that the site cannot be reached. What is the reason for this issue?

A. The jumpbox was not rebooted.

B. Split-DNS does not support Internet domain name resolution.

C. The browser is not compatible with the site's coding.

D. A private IP address from RFC 1918 is assigned to the web server.

7. Mark is configuring the remote access server to receive inbound connections from remote workers. He is following a configuration checklist to ensure that the telecommuting links are compliant with company security policy. What authentication protocol offers no encryption or protection for logon credentials?

A. PAP

B. CHAP

C. EAP

D. RADIUS

8. Users have reported data loss and the inability to maintain connections throughout the workday. You suspect that something about the network structure has changed to cause this QoS reduction. Which of the following are aspects of networking you need to investigate to track down the issue? (Choose all that apply.)

A. Bandwidth

B. System uptime

C. Latency

D. Jitter

E. Application Layer protocol

F. Packet loss

G. Interference

H. Throughput

I. OS versions

J. Signal-to-noise ratio

9. While evaluating network traffic, you discover several addresses that you are not familiar with. Several of the addresses are in the range of addresses assigned to internal network segments. Which of the following IP addresses are private IPv4 addresses as defined by RFC 1918? (Choose all that apply.)

   A. 10.0.0.18

   B. 169.254.1:.119

   C. 172.31.8.204

   D. 192.168.6.43

10. The CISO has requested a report on the potential communication partners throughout the company. There is a plan to implement VPNs between all network segments in order to improve security against eavesdropping and data manipulation. Which of the following cannot be linked over a VPN?

    A. Two distant Internet-connected LANs

    B. Two systems on the same LAN

    C. A system connected to the Internet and a LAN connected to the Internet

    D. Two systems without an intermediary network connection

11. What networking device can be used to create digital virtual network segments that can be altered as needed by adjusting the settings

internal to the device?

A. Router

B. Switch

C. Proxy

D. Firewall

12. The CISO is concerned that the use of subnets as the only form of network segments is limiting growth and flexibility of the network. They are considering the implementation of switches to support VLANs but aren't sure VLANs are the best option. Which of the following is not a benefit of VLANs?

    A. Traffic isolation

    B. Data/traffic encryption

    C. Traffic management

    D. Reduced vulnerability to sniffers

13. The CISO has tasked you to design and implement an IT port security strategy. While researching the options, you realize there are several potential concepts that are labeled as port security. You prepare a report to present options to the CISO. Which of the following are port security concepts you should include on this report? (Choose all that apply.)

    A. Shipping container storage

    B. NAC

    C. Transport Layer

    D. RJ-45 jacks

14. _____ is the oversight and management of the efficiency and performance of network communications. Items to measure include throughput rate, bit rate, packet loss, latency, jitter, transmission delay, and availability.

    A. VPN

    B. QoS

C. SDN

D. Sniffing

15. You are configuring a VPN to provide secure communications between systems. You want to minimize the information left in plaintext by the encryption mechanism of the chosen solution. Which IPSec mode provides for encryption of complete packets, including header information?

    A. Transport

    B. Encapsulating Security Payload

    C. Authentication Header

    D. Tunnel

16. Internet Protocol Security (IPSec) is a standard of IP security extensions used as an add-on for IPv4 and integrated into IPv6. What IPSec component provides assurances of message integrity and identity verification?

    A. Authentication Header

    B. Encapsulating Security Payload

    C. IP Payload Compression protocol

    D. Internet Key Exchange

17. When you're designing a security system for Internet-delivered email, which of the following is least important?

    A. Nonrepudiation

    B. Data remanent destruction

    C. Message integrity

    D. Access restriction

18. You have been tasked with crafting the organization's email retention policy. Which of the following is typically not an element that must be discussed with end users in regard to email retention policies?

    A. Privacy

B. Auditor review

C. Length of retainer

D. Backup method

19. Modern networks are built on multilayer protocols, such as TCP/IP. This provides for flexibility and resiliency in complex network structures. All of the following are implications of multilayer protocols except which one?

    A. VLAN hopping

    B. Multiple encapsulation

    C. Filter evasion using tunneling

    D. Static IP addressing

20. Which of the following is a type of connection that can be described as a logical circuit that always exists and is waiting for the customer to send data?

    A. SDN

    B. PVC

    C. VPN

    D. SVC

Chapter 13
Managing Identity and Authentication

THE CISSP TOPICS COVERED IN THIS CHAPTER INCLUDE:

- ✓ **Domain 5.0: Identity and Access Management (IAM)**
  - 5.1 Control physical and logical access to assets
    - 5.1.1 Information
    - 5.1.2 Systems
    - 5.1.3 Devices
    - 5.1.4 Facilities
    - 5.1.5 Applications
    - 5.1.6 Services
  - 5.2 Design identification and authentication strategy (e.g., people, devices, and services)
    - 5.2.1 Groups and Roles
    - 5.2.2 Authentication, Authorization and Accounting (AAA) (e.g., multi-factor authentication (MFA), password-less authentication)
    - 5.2.3 Session management
    - 5.2.4 Registration, proofing, and establishment of identity
    - 5.2.5 Federated Identity Management (FIM)
    - 5.2.6 Credential management systems (e.g., Password vault)
    - 5.2.7 Single Sign On (SSO)
    - 5.2.8 Just-In-Time
  - 5.3 Federated identity with a third-party service
    - 5.3.1 On-premise
    - 5.3.2 Cloud
    - 5.3.3 Hybrid

- 5.5 Manage the identity and access provisioning lifecycle
  - 5.5.1 Account access review (e.g., user, system, service)
  - 5.5.2 Provisioning and deprovisioning (e.g., on/off boarding and transfers)
  - 5.5.3 Role definition and transition (e.g., people assigned to new roles)
  - 5.5.5 Service accounts management

The Identity and Access Management (IAM) domain focuses on issues related to granting and revoking privileges to access data or perform actions on systems. A primary focus is on identification, authentication, authorization, and accounting. In this chapter and Chapter 14, “Controlling and Monitoring Access,” we discuss all the objectives in the Identity and Access Management domain. Be sure to read and study the materials from both chapters to ensure complete coverage of this domain's essential material.

Controlling Access to Assets

Controlling access to assets is one of the central themes of security, and you'll find that many different security controls work together to provide access control. Note that assets can be tangible or intangible. Tangible assets refer to things you can touch, such as physical equipment, whereas intangible assets refer to information and data, such as intellectual property. In addition to personnel, technology assets can be information, systems, devices, facilities, applications, or services:

Information An organization's information includes all of its data. Data is stored in simple files on servers, computers, and smaller devices. It can also be stored in databases within a server farm or the cloud. It can even be paper records maintained in a file cabinet. Logical access controls attempt to prevent unauthorized access to information.

Systems An organization's systems include any IT systems that provide one or more services. For example, a simple file server that stores user files

is a system. Additionally, a web server working with a database server to provide an e-commerce service is a system. Permissions assigned to user and system accounts control system access.

Devices Devices refer to any computing equipment, including networking devices (routers and switches), storage devices (SAN and NAS), computing devices (servers, desktop computers, portable laptop computers, tablets, and smartphones), and external devices such as printers and scanners. Organizations have increasingly adopted policies allowing employees to connect their personally owned devices (such as smartphones or tablets) to an organization's network. Although the employees may own the devices, organizational data stored on the devices is still an asset of the organization.

Facilities An organization's facilities include any physical location that it owns or rents. This could be individual rooms, entire buildings, or whole complexes of several buildings. Physical security controls help protect facilities.

Applications Applications frequently provide access to an organization's data. Controlling access to applications provides an additional layer of control for the organization's data. Permissions are an easy way to restrict logical access to applications and be assigned to specific users or groups.

Services Services offered by an organization may include printing capabilities, network capacity, end-user support, and a variety of other offerings. Access control systems ensure that only authorized users gain access to these services.

Controlling Physical and Logical Access

In addition to understanding what assets need to be protected, you must know how to protect them. You can do so with physical security controls and logical access controls.

Chapter 10, “Physical Security Requirements,” discusses physical security controls in depth. In general, a physical security control is one you can touch, such as perimeter security controls (fences, gates, guards, and turnstiles) and environmental controls such as heating, ventilation, and air-conditioning (HVAC) systems and fire suppression.

Physical security controls protect systems, devices, and facilities by controlling access and controlling the environment. As an example, organizations often have a server room where servers are running, and it's common for server rooms to include routers and switches. The benefit is that server rooms have increased security, such as cipher locks controlling entry into the server room. Desktop computers typically aren't as valuable as servers, but regular physical security controls such as locks provide protection.

Servers store important information (data), and also many servers host applications accessed by employees throughout the organization. These applications and data enjoy the same benefits from the other physical security controls protecting these servers.

Logical access controls are the technical controls used to protect access to information, systems, devices, and applications. They include authentication, authorization, and permissions. Combined, they help prevent unauthorized access to data and configuration settings on systems and other devices. For example, only people who can authenticate on a system or network can access data. Permissions help ensure only authorized entities can access data. Similarly, logical access controls restrict access to configuration settings on systems and network devices to only authorized individuals. Many of these logical access controls can apply to resources on-site or in the cloud.

The CIA Triad and Access Controls

One of the primary reasons an organization implements access control mechanisms is to prevent losses. There are three categories of IT loss: loss of *confidentiality*, *integrity*, and *availability* (CIA). Protecting against these losses is so integral to IT security that they are frequently referred to as the *CIA Triad* (or sometimes the AIC Triad or Security Triad). Chapter 1, "Security Governance Through Principles and Policies," covers these in more depth. The following list identifies them in the context of access control:

Confidentiality Access controls help ensure that only authorized subjects can access objects. When unauthorized entities can access systems or data, it results in a loss of confidentiality.

Integrity Integrity ensures that data or system configurations are not modified without authorization, or if unauthorized changes occur, security controls detect the changes. If unauthorized or unwanted changes to objects occur, the result is a loss of integrity.

Availability Authorized requests for objects must be granted to subjects within a reasonable amount of time. In other words, systems and data should be available to users and other authorized subjects when they are needed. If the systems are not operational or the data is not accessible, the result is a loss of availability.

The AAA Model

The core functions of identity and access management systems are:

- *Authenticating* users, systems, services, and other subjects to confirm they are who they claim to be
- *Authorizing* actions attempted by those entities
- *Accounting* for activity by maintaining an audit trail

Together, these three core functions are described as the *AAA (or "Triple-A") model* of access control.

Identification and Authentication Strategy

Identification is the process of a subject claiming, or professing, an identity. A subject must provide an identity to a system to start the authentication, authorization, and accounting processes. Providing an identity might entail typing a username, swiping a smartcard, speaking a phrase, or positioning your face, hand, or finger in front of a camera or in proximity of a scanning device. A core identification principle is that all subjects must have unique identities.

Authentication verifies the subject's identity by comparing one or more factors against a database of valid identities, such as user accounts. The authentication information used to verify identity is private and needs to be protected. As an example, passwords are rarely stored in cleartext within a

database. Instead, authentication systems store hashes of passwords in the authentication database.



Chapter 6, “Cryptography and Symmetric Key Algorithms,” covers hashing in more depth.

Identification and authentication occur together as a single two-step process. Providing an identity is the first step, and providing the authentication information is the second step. Without both, a subject cannot gain access to a system.

In contrast, imagine a user claims an identity (such as with a username of john.doe@sybex.com) but doesn't prove the identity (with a password). This username is for the employee named John Doe. However, if a system accepts the username without the password, it has no proof that the user is John Doe. Anyone who knows John's username can impersonate him.

Each authentication technique or factor has benefits and drawbacks. Thus, it is important to evaluate each mechanism in the context of the environment where it is deployed. For example, a facility that processes Top Secret materials requires very strong authentication mechanisms. In contrast, authentication requirements for students within a classroom environment are significantly less.

While identification and authentication methods authenticate people, they also authenticate devices and services. The “Device Authentication” and “Service Authentication” sections, later in this chapter, explain devices and services in more depth.



You can simplify identification and authentication by thinking about a username and a password. Users identify themselves with usernames and authenticate (or prove their identity) with passwords. Of course, there are many more identification and authentication methods, but this simplification helps you keep the terms clear.

Comparing Subjects and Objects

Access control addresses more than just controlling which users can access which files or services. It is about the relationships between entities (subjects and objects). Access is the transfer of information from an object to a subject, which makes it important to understand the definition of both subject and object. Chapter 8, “Principles of Security Models, Design, and Capabilities,” covers subjects and objects in more depth. The following provides a short reminder:

Subject A *subject* is an active entity that accesses a passive object to receive information from, or data about, an object. Subjects can be users, programs, processes, services, computers, or anything else that can access a resource. When authorized, subjects can modify objects.

Object An *object* is a passive entity that provides information to active subjects. Examples of objects are files, databases, computers, programs, processes, services, printers, and storage media.



You can often simplify the access control topics by substituting the word *user* for *subject* and the word *file* for *object*. For example, instead of *a subject accesses an object*, you can think of it as *a user accesses a file*. However, it's also important to remember that subjects include more than users and that objects include more than just files.

You may have noticed that some examples, such as programs, services, and computers, are listed as both subjects and objects. This is because the roles of subject and object can switch back and forth. In many cases, when two entities interact, they perform different functions. Sometimes they may be requesting information and other times providing information. The key difference is that the subject is always the active entity that receives information about, or data from, the passive object. The object is always the passive entity that provides or hosts the information or data.

As an example, consider a common web application that provides dynamic web pages to users. Users query the web application to retrieve a web page, so the application starts as an object. The web application then switches to a subject role as it queries the user's computer to retrieve a cookie and then queries a database to retrieve information about the user based on the cookie. Finally, the application switches back to an object as it sends dynamic web pages back to the user.

Registration, Proofing, and Establishment of Identity

Within an organization, new employees prove their identity with appropriate documentation during the hiring process. Acceptable documentation for in-person identity proofing includes using physical documents such as a passport, driver's license, birth certificate, and more. This documentation establishes the identity of the new employee for the employer.

After verifying the documents are authentic, employees within a human resources (HR) department begin the registration process. This process can be as simple as creating an account for the new employee and having the new employee set a password. If the organization uses more secure authentication methods, such as biometrics, the registration process is more complex. For example, if the organization uses fingerprinting as a biometric method for authentication, registration includes capturing the new employee's fingerprints.

Online organizations often use knowledge-based authentication (KBA) for identity proofing of someone new, such as a new customer. For example, if you create an online savings account, the bank will ask you a series of